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Luiz A. B. San Martin

Lie Groups

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To the future generations,
represented by my grandchildren,
Pedro and João.

Preface

The purpose of this book is to offer an introductory text on Lie groups, by presenting the theory from its first principles.

The concept of group is by now one of the cornerstones of contemporary mathematics and its applications. This is due both to its simplicity as an algebraic structure and also to the fact that the idea of symmetry, in a broad sense, is formalized through the notion of invariants under transformation groups.

Lie groups form a special class of groups, which are studied using the methods of differential and integral calculus. As a mathematical object, a Lie group combines two structures of different flavor: an algebraic group structure on one side and that of a differentiable manifold on the other. Lie groups began to be investigated around 1870 as groups of symmetries of differential equations and of the several geometries that had appeared at the time. Since then, the theory of Lie groups, or what is more generally referred to as Lie theory, has undergone a substantial development. Through multiple ramifications and incarnations, it has established itself in most areas of mathematics, both pure and applied.

The techniques for studying Lie groups are based on the construction of their Lie algebras, which was started by Sophus Lie in the 1870s. (Incidentally, the name "Lie theory" itself has its roots in this construction.) Once the idea of Lie algebra of a Lie group has been set up, the core of the method is to establish local or global properties for Lie groups from the properties of the Lie algebras. This transfer process, so to speak, is very effective, for it allows to describe the typical nonlinearities of Lie groups by means of the linear algebra encapsulated by Lie algebras.

The book presents the body of results that set up the relationship between Lie groups and Lie algebras. It is divided into four parts, plus a fifth part consisting of appendices.

The heart of the theory of Lie groups—and the classification thereof based on Lie algebras—is developed in Parts II and III. Part II consists of four chapters, where the Lie algebra of a Lie group is defined, and incorporates the proofs of the fundamental relationships between the multiplication in the group and the bracket operation of the Lie algebra. The main ingredient in these formulas is the exponential map, which is based on solutions to ordinary differential equations (invariant vector fields).

This part also discusses Lie subgroups of a Lie group and their link to Lie subalgebras, besides other standard concepts of group theory such as homomorphisms, normal subgroups, and quotient spaces. The results of Part II build up to an existence and uniqueness theorem for Lie groups with the given Lie algebra. Uniqueness is reserved to Lie groups that satisfy the global topological property of being connected and simply connected.

Part II does not require a deep knowledge of Lie algebras, since the few concepts used are introduced when needed. In Part III, instead, Lie algebras come in full force and enable to portray the complete picture of how Lie algebras determine Lie groups. This includes, in particular, delicate issues such as the classification of semi-simple Lie algebras. Although the latter class of Lie algebras is not discussed exhaustively in this book, an effort was made to indicate the main argumentative steps, as well as references to full proofs in the existing literature.

Regarding the other parts of the book, Part I is dedicated to general topological groups. Here, only the first chapter, dealing with topological features groups may enjoy, is required for reading the rest of the book (Chapter 2). These properties are satisfied by Lie groups and are widely used throughout. The other two chapters (on Haar measures and representations of compact groups) rely on rather separate premises, for they require a different background (measure theory and basic functional analysis), and they appear in the sequel only tangentially.

Part IV is concerned with the actions of Lie group. The orbits of an action are investigated in detail after proving that they are differentiable submanifolds. Also introduced here are the notions of principal bundle and associated bundle, whose respective fibers are Lie groups and spaces acted on by Lie groups. The final chapter is an invitation to a broad and very active area of modern differential geometry, namely that of invariant geometric structures on homogeneous spaces.

Every Part unfolds with an initial detailed outline of the contents to follow. These summaries provide an overview of the main results in the various chapters and show how they become interconnected.

The introductory chapter opening the book is meant to survey in an informal and friendly manner the relationship between Lie groups and Lie algebras, including a review of the classification of Lie groups.

Readers approaching this matter for the first time are recommended to follow this chapter sequence: Chapters 2, 5, 6, 7, 8 (up to Section 8.1), 9, 10, 11 (up to Section 11.2), and 13, to become acquainted with the foundations of Lie group theory.

Finally, I would like to thank the people who variously contributed to the completion of this book, be it with suggestions, by pointing out several flaws in preliminary versions, or by expressing their support. I am particularly grateful to all the students who took my classes on Lie groups at the University of Campinas over the years. My deepest gratitude goes to my friends and colleagues Alexandre Santana, Adriano João da Silva, Carlos Braga Barros, Caio Negreiros, Elizabeth Gasparim, Lino Grama, Lonardo Rabelo, Lucas Seco, Luciana Alves, Mauro Patrão, Osvaldo do Rocio, Paolo Piccione, Paulo Ruffino, Pedro Catuogno, and Victor Ayala.

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Chapter 1
Introduction



This introductory chapter has an informal character. Its purpose is to provide the reader with a panoramic view of the theory developed in this book by discussing some of the main results through examples which are both concrete and illustrative and therefore central to the theory.

The formal definition of a Lie group will be presented later, in Chapter 5. For all effects, a Lie group G is a group whose product

$$(g, h) \in G \times G \longmapsto gh \in G$$

is a differentiable map. An example rich enough to cover much of the theory, and which should always be used as a guide, is the general linear group $\mathrm{Gl}\,(n, \mathbb{R})$. The elements of this group are the invertible $n \times n$ matrices with real entries, or, what is essentially the same, the invertible linear maps of a finite dimension real vector space.

In what follows, some aspects of the group $\mathrm{Gl}\,(n, \mathbb{R})$ are discussed. The first observation is that it is an open subset of the vector space of $n \times n$ matrices, that is, of $\mathbb{R}^{n^2}$. It has two connected components, determined by the sign of the determinant. One of them is

$$\mathrm{Gl}^{+}\,(n, \mathbb{R}) = \{g \in \mathrm{Gl}\,(n, \mathbb{R}) : \det g > 0\},$$

which is a subgroup of $\mathrm{Gl}\,(n, \mathbb{R})$. The other connected component is formed by matrices with determinant < 0, and is not a subgroup.

The group structure in $\mathrm{Gl}\,(n, \mathbb{R})$ is given by the usual matrix product. If $X = \left(x_{ij}\right)$ and $Y \in \left(y_{ij}\right)$ are $n \times n$ matrices, then $Z = XY = \left(z_{ij}\right)$ is given by

$$z_{ij} = \sum_{k=1}^{n} x_{ik} y_{kj},$$

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which is a polynomial map of degree two in the variables x_{ij}, y_{ij}. Therefore, the product is a differentiable map. For this reason, $\mathrm{Gl}(n, \mathbb{R})$ is a Lie group.

The great strength of Lie group theory is based on the existence of Lie algebras associated with the groups. Lie algebras make it possible to transfer methods of linear algebra to the study of nonlinear objects such as Lie groups. A Lie algebra is a quintessential algebraic structure. It is defined as a vector space $\mathfrak{g}$ endowed with a product (bracket) $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$ which satisfies the following properties:

1. Bilinearity, that is, $[\cdot, \cdot]$ is linear in both variables or, equivalently, the bracket product is distributive with respect to vector space operations.
2. Skew-symmetry, that is, $[X, Y] = -[Y, X]$, for $X, Y \in \mathfrak{g}$.
3. Jacobi Identity: For $X, Y, Z \in \mathfrak{g}$,

$$[X, [Y, Z]] = [[X, Y], Z] + [Y, [X, Z]].$$

The elements of the Lie algebra of a Lie group are ordinary differential equations (vector fields) in the group; they satisfy a symmetry property arising from the multiplicative structure of the group (translation invariant vector fields, see Chapter 5). In turn, the group elements are obtained through the solutions of these equations, given by their flows. Usually the vector space underlying the Lie algebra of a Lie group is identified with $T_1 G$, the space of tangent vectors at the identity $1 \in G$.

In other words, the Lie algebra is a linear object that approximates the group: A Lie algebra element is given by the derivative of a curve in the group. The inverse procedure consists in solving differential equations. For this reason, in the first decades of development of the theory the expression **infinitesimal group** was used instead of Lie algebra.

In the case of $\mathrm{Gl}(n, \mathbb{R})$, its Lie algebra is the vector space of $n \times n$ matrices with bracket given by the matrix commutator[1]

$$[A, B] = BA - AB.$$

This Lie algebra will be denoted $\mathfrak{gl}(n, \mathbb{R})$. In order to establish the relation between the algebra and the group, consider, for each matrix $A \in \mathfrak{gl}(n, \mathbb{R})$ the vector field

$$g \mapsto Ag$$

in the space of matrices. This vector field induces the linear differential equation

$$\frac{dg}{dt} = Ag. \tag{1.1}$$

[1]The reverse order in the definition of this commutator is due to the choice of right invariant vector fields which will soon be made.

This equation is just the linear system $\dfrac{dx}{dt} = Ax$, $x \in \mathbb{R}^n$, repeated n times, one time for each column of matrix g. The fundamental solution of the linear system in $\mathbb{R}^n$ is given by

$$e^{tA} = \sum_{n \geq 0} \frac{1}{n!} (tA)^n ,$$

showing that the solution of Equation (1.1) with initial condition $g(0) = 1$ (where 1 denotes the $n \times n$ identity matrix) is $g(t) = e^{tA}$. This solution is entirely contained in $\mathrm{Gl}(n, \mathbb{R})$ as exponentials are invertible matrices. Moreover, the curve

$$g : \mathbb{R} \to \mathrm{Gl}(n, \mathbb{R}) \qquad g(t) = e^{tA}$$

is a homomorphism when the additive group structure in $\mathbb{R}$ is considered, since the equality $e^{(t+s)A} = e^{tA} e^{sA}$ holds. The image of this homomorphism is what is called a **1-parameter subgroup** of the Lie group.

In short, there is a natural construction that associates to each Lie algebra element a subgroup of the Lie group. This construction defines the **exponential map** of the Lie group. It is fundamental for the development of the theory as the exponential map provides the link between the Lie algebra bracket and the group product, thus determining—almost completely—the structure of a Lie group from its Lie algebra. This link is accomplished by means of formulas involving $[\cdot, \cdot]$, exp and the product in the group.

An example is the **Baker–Campbell–Hausdorff formula** (see Chapter 8). For X and Y in the Lie algebra, this formula is written as

$$e^X e^Y = e^{c(X,Y)},$$

where $c(X, Y)$ is a series (similar to a Taylor series) containing only X, Y and its successive brackets. The first terms of the series are

$$c(X, Y) = X + Y + \frac{1}{2}[X, Y] + \frac{1}{12}[[X, Y], Y] - \frac{1}{12}[[X, Y], X] + \cdots \qquad (1.2)$$

The remaining terms involve brackets with four or more elements. The series $c(X, Y)$ converges if X and Y are small enough, showing that, for such values of X and Y, the product $e^X e^Y$ is completely determined by the Lie algebra, that is, by bracket combinations of its elements. Hence, the group product is fully determined locally, around the origin (that is the neutral element), by the Lie algebra bracket.

This kind of relation between bracket and group product can be extended to the entire group, allowing us to show that, up to global topological properties (such as, for example, that the underlying topological space of the group is connected and simply connected), there exists a unique Lie group associated with a given Lie algebra.

Another formula is the Taylor expansion of a commutator of exponentials given by the curve

$$\alpha(t) = e^{tB} e^{tA} e^{-tB} e^{-tA} \tag{1.3}$$

on the linear group $\mathrm{Gl}(n, \mathbb{R})$. Applying repeatedly the derivative

$$\frac{d}{dt} e^{tA} = A e^{tA} = e^{tA} A,$$

yields $\alpha'(0) = 0$ and

$$\alpha''(0) = 2[A, B].$$

As $\alpha(0) = 1$, this means that

$$\alpha(t) = 1 + t^2 [A, B] + \cdots,$$

where the relevant term is $[A, B]$. This allows us to see the bracket as the infinitesimal object associated with the group commutator. Derivatives of this kind can be extended to general vector fields. This Taylor expansion was at the origin of the concept of a Lie bracket of vector fields, as it is currently known. The concept was introduced by Sophus Lie and thus the entire theory got his name.

These formulas, although they are useful to illustrate the relation between Lie groups and Lie algebras, do not constitute the main technical tools of this theory. The bridge between Lie groups and Lie algebras is built with the adjoint representations defined in Chapter 5. Such representations provide formulas relating conjugations $C_g(x) = gxg^{-1}$ within the group, their differentials $\mathrm{Ad}(g)$, which are linear maps of the Lie algebra, and the differentials of $\mathrm{Ad}(g)$ which are given by the bracket defining the Lie algebra. In applying those formulas to go from Lie groups to Lie algebras it is necessary to differentiate twice (possibly different functions). The reverse process, from Lie algebra to Lie group, involves two integrals which are generally obtained with the help of theorems on the existence and uniqueness of solutions of ordinary differential equations. The second derivative in the Taylor expansion of the conjugation in (1.3) provides an heuristic idea that the passage from group to algebra takes place by means of two derivatives.

The following are other examples of Lie groups and their respective Lie algebras:

1. If G is an abelian Lie group, then its Lie algebra is abelian, that is, the bracket $[\cdot, \cdot]$ is identically zero (and vice versa, in the case of connected groups, as can be seen from the Campbell–Hausdorff formula). Connected abelian Lie groups are described in Chapter 7, Section 7.3.
2. Let

$$G = \mathrm{O}(n) = \{g \in \mathrm{Gl}(n, \mathbb{R}) : gg^T = g^T g = 1\}$$

be the group of orthogonal matrices. Its Lie algebra is the subalgebra of skew-symmetric matrices:

$$\mathfrak{so}(n) = \{A \in \mathfrak{gl}(n, \mathbb{R}) : A + A^T = 0\}.$$

The bracket in $\mathfrak{so}(n)$ is the matrix commutator. The reason for this is that A is a skew-symmetric matrix if and only if e^{tA} is an orthogonal matrix for every $t \in \mathbb{R}$. This means that the group $\mathrm{O}(n)$ is a submanifold of the space of matrices whose tangent space at the identity 1 coincides with the subspace of skew-symmetric matrices.

3. The group $\mathrm{Gl}(n, \mathbb{C})$ of invertible $n \times n$ complex matrices is a Lie group for the same reason as $\mathrm{Gl}(n, \mathbb{R})$ is. The Lie algebra of $\mathrm{Gl}(n, \mathbb{C})$ is $\mathfrak{gl}(n, \mathbb{C})$, which is the Lie algebra $n \times n$ complex matrices.

The program of Lie theory consists in studying Lie groups through their Lie algebras, making an effort to describe the geometric and algebraic properties of the group by means of the corresponding properties of its Lie algebra. Such a description should ultimately approach structural properties which would allow the classification of Lie groups in terms of Lie algebras. In this program, two concepts from abstract group theory play a central role, the concepts of subgroup and of group homomorphism. Such concepts are thoroughly worked with the help of Lie algebras and the results are the best possible:

1. **Subgroups:** If G is a Lie group with Lie algebra $\mathfrak{g}$, then the subgroups of G are in bijective correspondence with the Lie subalgebras of $\mathfrak{g}$, with two important restrictions about the subgroups entering such a bijection. The first one is that only **Lie subgroups** which are both subgroups and differentiable submanifolds are considered, so that their underlying structures make them Lie groups. The second restriction is that the bijection involves only connected Lie groups. This must be so because the exponential map sees only the connected component of the Lie group that contains the identity.

   In this bijection, a Lie subalgebra is associated with a Lie subgroup by taking derivatives (twice, as mentioned above). In the reverse process a Lie subgroup is obtained from a Lie subalgebra as an integral manifold of a distribution. (For the bijection, see Chapter 6. The theory of distributions is presented in Appendix B.)

   With regard to Lie subgroups, it must be mentioned the famous closed subgroup theorem by Cartan, which states that if a subgroup of G is a closed set, then it is automatically a Lie subgroup.
2. **Homomorphisms:** If $\phi : G \to H$ is a differentiable homomorphism between Lie groups, then its differential $d\phi_1 : T_1G \to T_1H$ is a linear map between the tangent spaces at their identities, which are the Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$ of G and H, respectively. This linear map turns out to be a Lie algebra homomorphism between $\mathfrak{g}$ and $\mathfrak{h}$. The reverse construction does not work in all generality due to global topological restrictions on G. It happens that a homomorphism $\theta : \mathfrak{g} \to \mathfrak{h}$ gives rise to a local homomorphism between groups G and H, defined around the identity of G and with values in a neighborhood of the identity of H. The only

obstacle for this application to extend to the whole group G is its fundamental group, so that if G is connected and simply connected, then $\theta : \mathfrak{g} \to \mathfrak{h}$ is the differential of a homomorphism $\phi : G \to H$ (see Chapter 7).

A good example of this phenomenon is given by groups $(\mathbb{R}, +)$ and $S^1 = \{z \in \mathbb{C} : |z| = 1\}$. Their Lie algebras are isomorphic (both have dimension 1), there are homomorphisms $\mathbb{R} \to S^1$ $(t \mapsto e^{ait})$, but there are no homomorphisms $S^1 \to \mathbb{R}$. Around their identities, the groups are isomorphic.

These comments on homomorphisms are in line with the above discussion about the Baker–Campbell–Hausdorff formula, in which it was concluded that the bracket in the Lie algebra determines the product in the Lie group around its identity.

This analysis of homomorphisms, especially the theorem on the extension to simply connected groups, gives rise to the description of all connected Lie groups on the basis of some classification of Lie algebras. This description comes down to two points (see Chapter 7):

1. Given a (real, finite dimensional) Lie algebra $\mathfrak{g}$ there exists a unique connected and simply connected Lie group $\widetilde{G}$ with Lie algebra $\mathfrak{g}$. Uniqueness stems from the extension theorem mentioned above: An isomorphism between the Lie algebras defines an isomorphism between connected and simply connected Lie groups. Existence is proved in two steps: (1) the construction of a Lie group G with Lie algebra isomorphic to g (in Chapter 7 this is done with the help of the theorem of Ado, which ensures that every Lie algebra is isomorphic to a matrix Lie subalgebra); (2) the formal construction of a Lie group structure in the universal covering space $\widetilde{G}$ of a Lie group G.
2. Every connected Lie group is equal to the quotient of a simply connected Lie group $\widetilde{G}$ by a discrete subgroup $\Gamma \subset \widetilde{G}$ contained in the center of $\widetilde{G}$.

This description works well for connected groups, as these are the groups that can be accessed by Lie algebras through the solution of differential equations.

An example are the groups of dimension 1. The additive group $(\mathbb{R}, +)$ is simply connected and its Lie algebra is the unique (up to isomorphisms) Lie algebra of dimension 1. Hence, any connected and simply connected Lie group of dimension 1 is isomorphic to $(\mathbb{R}, +)$. A discrete subgroup of $\mathbb{R}$ has the form $\omega\mathbb{Z}$, with $\omega > 0$. It thus follows that any group of dimension 1 is isomorphic to $\mathbb{R}$ or to $\mathbb{R}/\omega\mathbb{Z} \approx S^1$.

The classification of connected Lie groups is generally done in three steps: 1) classifying real Lie algebras; 2) determining, for each real Lie algebra $\mathfrak{g}$ (or rather for its Lie algebra isomorphism class), a simply connected Lie group $\widetilde{G}$ whose Lie algebra is $\mathfrak{g}$; 3) finding the center $Z\left(\widetilde{G}\right)$ of $\widetilde{G}$ and the discrete subgroups $\Gamma \subset Z(\widetilde{G}$.

From this point on, a deeper development of the theory of Lie algebras is required. Lie algebras are divided into two great classes, solvable algebras and semi-simple algebras. The Levi decomposition theorem combines these two types of algebra by means of the semi-direct product, in order to build all finite dimensional Lie algebras (see Chapter 9). This decomposition of Lie algebras extends to simply connected Lie groups, in such a way that all that is necessary to do is to determine

separately the simply connected groups for semi-simple and solvable Lie algebras, respectively.

In the case of solvable algebras, it is possible to prove that the manifolds underlying connected and simply connected groups are diffeomorphic to Euclidean spaces $\mathbb{R}^n$. As usual when one deals with solvable algebras, the proof is done by induction, starting from the group $(\mathbb{R}, +)$, of dimension 1, (see Chapter 10). A typical example is the group of upper triangular matrices

$$\begin{pmatrix} a_1 & \cdots & * \\ \vdots & \ddots & \vdots \\ 0 & \cdots & a_n \end{pmatrix} \qquad a_1, \ldots, a_n > 0,$$

whose manifold is diffeomorphic to $(\mathbb{R}_+)^n \times \mathbb{R}^{n(n-1)/2}$.

The semi-simple case presents a greater wealth of detail and a more engaging geometry. Unlike solvable algebras, real semi-simple Lie algebras are classified to the point where it is possible to distinguish them one by one. One of the first major results of Lie theory, already in late nineteenth century, is the classification by W. Killing and E. Cartan of complex simple Lie algebras (and thus of semi-simple algebras, which are direct sums of simple ones). This classification is described by the Dynkin diagrams, which are reproduced in Chapter 11 (in Section 11.3.2). The classification provides four series of matrix algebras (referred to as classical algebras), namely $\mathfrak{sl}(n, \mathbb{C})$ (complex traceless matrices), $\mathfrak{so}(n, \mathbb{C})$ (skew-symmetric complex matrices, comprehending two series, for even and odd n, respectively), and $\mathfrak{sp}(n, \mathbb{C})$ (complex symplectic matrices). In addition to the series, there are five specific Lie algebras (called exceptional), which go by their prototype names E_6, E_7, E_8, F_4 e G_2.

The classification of simple real algebras is obtained from the classification of complex algebras by means of a complexification process. Real algebras are divided into two types, compact and noncompact algebras.

As their name indicates, compact Lie algebras are linked to compact groups (although their definition is purely algebraic). Simple compact real algebras are in bijection with simple complex algebras through the so-called unitary Weyl trick, which states that a complex simple algebra is the complexification of a unique (up to isomorphisms) compact real algebra (see Chapter 11). The classical compact algebras are $\mathfrak{su}(n)$ (traceless skew-Hermitian matrices, which complexifies to $\mathfrak{sl}(n, \mathbb{C})$), $\mathfrak{so}(n)$ (skew-symmetric matrices, which complexifies to $\mathfrak{sl}(n, \mathbb{C})$), and $\mathfrak{sp}(n, \mathbb{C})$ (skew-Hermitian quaternionic matrices, which complexifies to $\mathfrak{sp}(n, \mathbb{C})$). A central result about these algebras is a theorem by H. Weyl on the finiteness of the fundamental group, which states that a Lie group is compact if its Lie algebra, is a semi-simple compact algebra. This theorem allows the classification of compact Lie groups, which are obtained as cartesian products of compact semi-simple Lie groups by tori (see Chapter 11). In addition to the group $\mathrm{O}(n)$ mentioned above (which is not connected), other examples of compact matrix groups are:

1. $\mathrm{SO}(n) = \{g \in \mathrm{O}(n) : \det g = 1\}$, with Lie algebra $\mathfrak{so}(n)$. These Lie algebras are simple if $n \neq 2$ and $n \neq 4$. The Lie algebra $\mathfrak{so}(2)$ is abelian, while $\mathfrak{so}(4)$ is semi-simple and decomposes into two simple components isomorphic to $\mathfrak{so}(3)$. The groups $\mathrm{SO}(n)$ are not simply connected.
2. $\mathrm{SU}(n) = \{g \in \mathrm{Gl}(n, \mathbb{C}) : g\overline{g}^T = \overline{g}^T g = 1, \det g = 1\}$, with Lie algebra $\mathfrak{su}(n)$, which is simple for $n \geq 2$. The groups $\mathrm{SU}(n)$, $n \geq 2$, are simply connected.
3. The group $\mathrm{U}(n)$ is defined as $\mathrm{SU}(n)$, but without the requirement that the determinant must be 1. Its Lie algebra is $\mathfrak{u}(n)$ (defined as $\mathfrak{su}(n)$, without the restriction on the trace). Lie algebras $\mathfrak{u}(n)$ are not semi-simple.
4. $\mathrm{Sp}(n)$, with Lie algebra $\mathfrak{sp}(n)$. These groups are simply connected. Their elements are given by quaternionic unitary matrices, that is, matrices with entries in $\mathbb{H}$ which satisfy $g\overline{g}^T = \mathrm{id}$.

Simple noncompact algebras are also classified (see the tables in Chapter 12). The corresponding Lie groups have the property that their underlying manifolds are diffeomorphic to the cartesian product of a compact group by a Euclidian space $\mathbb{R}^N$. A product of this kind is given by a Cartan decomposition or an Iwasawa decomposition (Chapter 12). Using this fact, the Levi decomposition theorem and the information about solvable simply connected groups, the conclusion is that every connected and simply connected Lie group is diffeomorphic to the cartesian product of a compact Lie group by a Euclidian space.

The following are examples of noncompact semi-simple groups:

1. $\mathrm{Sl}(n, \mathbb{R}) = \{g \in \mathrm{Gl}(n, \mathbb{R}) : \det g = 1\}$.
2. $\mathrm{Sl}(n, \mathbb{C}) = \{g \in \mathrm{Gl}(n, \mathbb{C}) : \det g = 1\}$.
3. $\mathrm{Sp}(n, \mathbb{R}) = \{g \in \mathrm{Gl}(2n, \mathbb{R}) : gJg^T = 1, \det g = 1\}$, where

$$J = \begin{pmatrix} 0 & -1_{n\times n} \\ 1_{n\times n} & 0 \end{pmatrix}.$$

4. $\mathrm{SO}(p, q) = \{g \in \mathrm{Gl}(p+q, \mathbb{R}) : gI_{p,q}g^T = 1, \det g = 1\}$, where

$$I_{p,q} = \begin{pmatrix} 1_{p\times p} & 0 \\ 0 & -1_{q\times q} \end{pmatrix}.$$

5. $\mathrm{SU}(p, q) = \{g \in \mathrm{Gl}(p+q, \mathbb{C}) : gI_{p,q}\overline{g}^T = 1, \det g = 1\}$.

1.1 Exercises

1. Find the first three terms of the Baker–Campbell–Hausdorff formula (1.2) for the linear group $\mathrm{Gl}(n, \mathbb{R})$, expanding the product $e^{tA}e^{tB}$ and grouping terms in t^k, $k = 0, 1, 2, 3$.

2. Let $\mathfrak{g}$ be a Lie algebra satisfying $[X,[Y,Z]]=0$ for every $X,Y,Z\in\mathfrak{g}$, so that the Baker–Campbell–Hausdorff series reduces to

$$c(X,Y)=X+Y+\frac{1}{2}[X,Y].$$

Show that the product $X*Y=c(X,Y)$ defines a group structure on $\mathfrak{g}$.

3. Let A be an $n\times n$ matrix. If $\exp A=\sum_{k\geq 0}\frac{1}{k!}A^k$, show that A is skew-symmetric ($A+A^T=0$) if and only if $\exp tA$ is an orthogonal matrix for every $t\in\mathbb{R}$. (Hint: consider the curve $\alpha(t)=\exp tA\,(\exp tA)^T$.)

4. Let $\mathrm{Sl}(n,\mathbb{R})=\{g\in\mathrm{Gl}(n,\mathbb{R}):\det g=1\}$ be the group of unimodular matrices. Assume that $\mathrm{Sl}(n,\mathbb{R})$ is a Lie subgroup and verify, using exponentials, that its Lie algebra is

$$\mathfrak{sl}(n,\mathbb{R})=\{A\in M_{n\times n}(\mathbb{R}):\mathrm{tr}A=0\}.$$

5. Let $\mathrm{SU}(2)$ be the group of 2×2 unitary matrices, that is,

$$\mathrm{SU}(2)=\{g\in M_{2\times 2}(\mathbb{C}):\overline{g}^T g=g\overline{g}^T=\mathrm{id},\det g=1\}.$$

Assume that $\mathrm{SU}(2)$ is a Lie subgroup of invertible matrices and verify, using exponentials, that its Lie algebra is the space of skew-Hermitian matrices

$$\mathfrak{su}(2)=\{A\in M_{2\times 2}(\mathbb{C}):A+\overline{A}^T=0,\mathrm{tr}A=0\}.$$

Verify that $\mathfrak{su}(2)$ is a real Lie algebra with $\dim\mathfrak{su}(2)=3$ (where the Lie bracket is given by the matrix commutator). Verify also that $\mathfrak{su}(2)$ is isomorphic to the following Lie algebras: (1) $\mathfrak{so}(3)=\{A\in M_{3\times 3}(\mathbb{R}):A+A^T=0\}$ (with the commutator); (2) $\mathbb{R}^3$ endowed with the vector product $\wedge$.

6. Let $\mathbb{H}=\{a+bi+cj+dk:a,b,c,d\in\mathbb{R}\}$ be the algebra of quaternions. Write $\xi=a+ib+jc+kd$ as $\xi=(a+ib)+j(c-id)$, that is, $\xi=z+jw$ with $z,w\in\mathbb{C}$. Left multiplication by ξ can be seen as a linear map on $\mathbb{C}^2$. Write the matrix of this map in basis $\{1,j\}$ and show that the map

$$\phi:a+bi+cj+dk=z+jw\longmapsto\begin{pmatrix}z&-\overline{w}\\ w&\overline{z}\end{pmatrix}\in M_{2\times 2}(\mathbb{C})$$

is an injective homomorphism. Show also that the restriction of ϕ to the sphere $\{\xi\in\mathbb{H}:|\xi|=1\}$ is a bijection on $\mathrm{SU}(2)$ and conclude that $\mathrm{SU}(2)$ é connected and simply connected. Determine the center of $\mathrm{SU}(2)$ and find all connected Lie groups with Lie algebra $\mathfrak{su}(2)\approx\mathfrak{so}(3)\approx\left(\mathbb{R}^3,\wedge\right)$.

7. Find the tangent space of the linear groups (subgroups of $\mathrm{Gl}(n,\mathbb{R})$) presented in the text and verify that these tangent spaces are Lie subalgebras of $\mathfrak{gl}(n,\mathbb{R})$ (with bracket given by the matrix commutator).

Part I
Topological Groups

Overview

This part consists of three chapters on topological groups. The only chapter whose reading is essential to the rest of the book is Chapter 2, which deals with the concepts of group theory from a topological point of view. The purpose of this chapter is to set up the language to be used throughout the entire theory of Lie groups. We present the concepts of subgroups (open, closed, etc.), connected components of topological groups, quotient spaces (which inherit the quotient topology in a natural way), and quotient topological groups. As quotient spaces, together with their respective topologies, are closely related to the orbits of continuous actions of topological groups, we include a discussion about homeomorphisms between orbits and quotient spaces. The topological concepts presented in this chapter will be revisited later, within the differentiable universe of Lie groups.

Chapter 3 constructs the Haar measures on locally compact Hausdorff topological groups. Their uniqueness, up to a scale factor, is demonstrated. The Haar measure is a central object in the theory of topological groups, particularly Lie groups, as it allows the use of methods of integral calculus in the study of those groups. The chapter is entirely developed within the framework of measure theory. It is independent of the rest of the book, particularly because the construction of the Haar measure on Lie groups can be done in a technically simpler way by means of invariant volume forms (as described in Section 5.6). The study of Chapter 3 can be postponed without loss, except for the information contained in the statement of Theorem 3.1.

Chapter 4 provides an introduction to the beautiful theory of representations of compact groups, which generalizes the theory of Fourier series for periodic functions and which was initially developed by Schur and Weyl at the beginning of twentieth century. The main results of this chapter are the Schur orthogonality relations and the Peter–Weyl theorem, which provide a clear description of the space L^2 of a compact group endowed with its Haar measure. In this chapter, we use some results of functional analysis. Like the chapter on Haar measures, it is not essential

for the rest of the book except for some initial results on the decomposition of representations, which will be applied in Chapter 11, a chapter dealing with compact Lie groups.

Chapter 2
Topological Groups



Several properties of Lie groups depend only on their topology and not on their differentiable manifold structure. Some of these properties, which are also valid for more general topological groups, are studied in this chapter. The aim here is not to present an exhaustive development of the theory of topological groups, but only to establish a language and to prove some results which are useful for the theory of Lie groups.

The identity element of a group G will be denoted by 1. Given a subset $A \subset X$ of a topological space, the interior, the closure, and the boundary of A are denoted by A°, $\overline{A}$, and ∂A, respectively.

2.1 Introduction

A **topological group** is a group whose underlying set is endowed with a topology compatible with the group product, in the sense that:

1. The product $p : G \times G \to G$, $p(g,h) = gh$, is a continuous map if $G \times G$ has the product topology.
2. The map $\iota : G \to G$, $\iota(g) = g^{-1}$, is continuous (hence, a homeomorphism, as $\iota^{-1} = \iota$).

These two properties can be combined by taking the map $q : G \times G \to G$, defined by $q(g,h) \to gh^{-1}$. Indeed, q is continuous if p and ι are continuous. Conversely, if q is continuous, then $g \to (1,g) \to g^{-1}$ is continuous. So, $p(g,h) = q\left(g, h^{-1}\right)$ is continuous.

Each element g of a group G naturally defines the following maps:

- **left translation** $L_g : G \to G$, $L_g(h) = gh$;
- **right translation** $R_g : G \to G$, $R_g(h) = hg$;
- **conjugation** (or inner automorphism) $C_g : G \to G$, $C_g(h) = ghg^{-1}$.

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From these definitions, it follows that $L_g \circ L_{g^{-1}} = R_g \circ R_{g^{-1}} = \mathrm{id}$. Moreover, $C_g = L_g \circ R_{g^{-1}}$, so that all these maps are bijections of G. In the case of topological groups, these maps are continuous since $L_g = p \circ s_{g,1}$ and $R_g = p \circ s_{g,2}$, where $s_{g,1}(h) = (g, h)$ and $s_{g,2}(h) = (h, g)$ are continuous maps $G \to G \times G$. The continuity of translations and the identities $\left(L_g\right)^{-1} = L_{g^{-1}}$, $\left(R_g\right)^{-1} = R_{g^{-1}}$ e $^{-1} = C_{g^{-1}}$ show that these maps are in fact homeomorphisms of G. The following formulas follow immediately from the definitions:

- $R_g \circ L_h = L_h \circ R_g$.
- $L_g \circ \iota = \iota \circ R_{g^{-1}}$.
- $R_g \circ \iota = \iota \circ L_{g^{-1}}$.

One should observe that the continuity of translations and conjugations depends on a property weaker than the continuity of p since, for instance, L_g is continuous if and only if the "partial map" $h \mapsto gh$ is continuous. Generally, maps defined on product spaces can be continuous in each variable without being continuous. This phenomenon leads to the definition of a **semitopological group**, a group whose product is partially continuous, that is, all translations are continuous. In what follows, some examples of topological groups are presented, together with some examples of semitopological groups which are not topological.

Examples

1. The linear groups mentioned in the Introduction (subgroups of $\mathrm{Gl}(n, \mathbb{R})$) are topological groups with the topology induced by the space of matrices; among them there are $\mathrm{Gl}(n, \mathbb{C})$, $\mathrm{O}(n)$, $\mathrm{Sl}(n, \mathbb{R})$, $\mathrm{Sl}(n, \mathbb{C})$, $\mathrm{Gl}(n, \mathbb{H})$.
2. $(\mathbb{R}^n, +)$ with its usual topology, which includes $(\mathbb{R}, +)$. The multiplicative group $(\mathbb{R}_\times, \cdot)$ is also a topological group, with the same topology.
3. More generally, on an ordered field $(\mathbb{K}, +, \cdot, \leq)$ it is possible to define the order topology, which is generated by open intervals. With respect to this topology, the $+$ operation defines a topological group, while the product defines a topological group on $\mathbb{K}_\times = \mathbb{K} \setminus \{0\}$.
4. Any group whose underlying set is endowed with the discrete topology (in which all sets are open).
5. The circle S^1 has a natural group structure given by the product of complex numbers with modulus 1: $S^1 = \{z \in \mathbb{C} : |z| = 1\}$. With the canonical topology, S^1 is a topological group. Alternatively, the product on S^1 is given by the quotient $S^1 = \mathbb{R}/\mathbb{Z}$, where the product is given by the sum modulo 1 of real numbers. (In the sequel, quotients of topological groups in general will be considered.)
6. Examples more general than the previous one are the cylinders

$$\mathbb{T}^k \times \mathbb{R}^m = \mathbb{R}^{m+k}/\mathbb{Z}^k = \left(\mathbb{R}^k/\mathbb{Z}^k\right) \times \mathbb{R}^m,$$

with canonical topologies. (See below products and quotients of topological groups.)

7. Let $(\mathbb{C} \setminus \{0\}, \cdot)$ be endowed with the topology generated by the basis of open sets that is formed by the open intervals of the vertical straight lines $r_a = \{a + ix \in \mathbb{C} : x \in \mathbb{R}\}$. This group is not a topological group with respect to this topology. Indeed, the left translation $L_{e^{i\theta}}$ is a rotation by an angle $\theta \in \mathbb{R}$. The image of the open set $r_1 = \{1 + ix \in \mathbb{C} : x \in \mathbb{R}\}$ is not an open set if, for instance, $\theta = -\pi/2$.
8. Let G be a topological group and X a topological space. Denote by $\mathcal{A}(X, G)$ the set of continuous maps $f : X \to G$. This set has a group structure with the product $(fg)(x) = f(x)\, g(x)$, whose inverse is $\iota(f)(x) = f(x)^{-1}$. Introduce in $\mathcal{A}(X, G)$ the compact-open topology, which has as subbasis of open sets the open sets of the type

$$A_{K,U} = \{f \in \mathcal{A}(X, G) : f(K) \subset U\},$$

where $K \subset X$ is compact and $U \subset G$ is open. Endowed with these structures, $\mathcal{A}(X, G)$ becomes a topological group. Indeed, the Cartesian product $\mathcal{A}(X, G) \times \mathcal{A}(X, G)$ is homeomorphic to $\mathcal{A}(X, G \times G)$ through $(f, g) \mapsto h$, where $h(x) = (f(x), g(x))$, with the compact-open topology on $\mathcal{A}(X, G \times G)$. Let $q_{\mathcal{A}}(f, g) = fg^{-1}$. Through the identification of these spaces, $q_{\mathcal{A}}^{-1}\left(A_{K,U}\right)$ is the set of functions $h : X \to G \times G$ such that $h(K) \subset q^{-1}(U)$. That is,

$$q_{\mathcal{A}}^{-1}\left(A_{K,U}\right) = A_{K,q^{-1}(U)},$$

where the neighborhood of the second member is seen on $\mathcal{A}(X, G \times G)$. Hence, the group is a topological group.
9. As a particular case of the previous example, let $\{G_i\}_{i \in I}$ be a family of groups indexed by a set I. The Cartesian product $G = \prod_{i \in I} G_i$ is the set formed by the maps $f : I \to \bigcup_{i \in I} G_i$ such that $f(i) \in G_i$ for every $i \in I$. The Cartesian product admits a group structure in which the product is defined component-wise: $(fg)(i) = f(i)\, g(i)$. The product topology on $\prod_{i \in I} G_i$ is generated by open sets of type $\prod_{i \in I} A_i$, with $A_i \subset G_i$ open, $i \in I$ and $A_i = G_i$, except for a finite number of indices (a compact-open topology in which I has discrete topology). As the product is performed component-wise and each G_i is a topological group, G is a topological group with the product topology.

   In particular, if I is a finite set, $\prod_{i \in I} G_i = G_1 \times \cdots \times G_n$, its elements are n-tuples $g = (g_1, \ldots, g_n)$, $g_i \in G_i$, and multiplication is given by

$$gh = (g_1 h_1, \ldots, g_n h_n)$$

with the product topology, generated by subsets of the type $A_1 \times \cdots \times A_n$ with $A_i \subset G_i$ open.
10. This example illustrates a group with a topology in which the product is a continuous map, but $\iota(g) = g^{-1}$ is not continuous. Consider the additive group $(\mathbb{R}, +)$ with $\mathbb{R}$ endowed with the topology (**Sorgenfrey topology**) generated

by the basis formed by the intervals $[a, b)$, $a < b$. The product is a continuous map. To see that, take x, y, such that $x + y \in [a, b)$. Then, for some $\varepsilon > 0$, $x + y + \varepsilon < b$, ensuring that $[a, b)$ contains $[x, x + \varepsilon/2) + [y, y + \varepsilon/2)$ (= $\{z + w : z \in [x, x + \varepsilon/2)$ and $w \in [x, x + \varepsilon/2)\}$). This means that the open set $[x, x + \varepsilon/2) \times [y, y + \varepsilon/2)$ is contained in $p^{-1}[a, b)$, showing that p is continuous. On the other hand, $\iota(x) = -x$ is not continuous. For instance, $(-2, -1] = \iota^{-1}[1, 2)$ is not open.

11. This example illustrates the case of a group G in which the inverse $\iota(g) = g^{-1}$ is continuous and p is partially continuous (that is, G is semitopological), but not continuous. Take the additive group $\left(\mathbb{R}^2, +\right)$ with $\mathbb{R}^2$ endowed with the topology generated by the **Siamese balls**, which are defined as follows: Take two balls of equal radius, with centers lying on the same vertical straight line and which are tangent to each other. The corresponding Siamese ball is equal to the union of the interior of the two balls together with their point of tangency. The set of Siamese balls constitutes a basis for the topology. Endowed with this topology, the inverse in $\mathbb{R}^2$ is continuous (by symmetry with respect to the origin), as well as translations. However, the product $p = +$ is not continuous. Indeed, $(1, 0) + (-1, 0) = (0, 0)$. Take a Siamese ball B with point of tangency at $(0, 0)$ and let B_1 and B_2 be Siamese balls with points of tangency respectively at $(1, 0)$ and $(-1, 0)$. Then, the set $B_1 + B_2$ is not contained in B, as can be geometrically verified. This means that $B_1 \times B_2$ is not contained in $p^{-1}(B)$. As B, B_1, and B_2 are arbitrary elements of the basis of the topology, it follows that p is not continuous on $((1, 0), (-1, 0))$.

□

If A is a subset of G and $g \in G$, then the translation $L_g(A)$ is denoted simply by $gA = \{gx : x \in A\}$. The fact that translations are homeomorphisms implies that gA is open or closed if A is open or closed, respectively. The same observation is valid for right translations Ag. More generally, let $B \subset G$ and write

$$A \cdot B = AB = \{xy \in G : x \in A, y \in B\}.$$

By definition, $AB = \bigcup_{x \in B} Ax = \bigcup_{x \in A} xB$. In this way, if A (or B) is open, then AB is open, as it is a union of open sets. It must be noted that the same statement is not valid for closed sets. For instance, take in $\left(\mathbb{R}^2, +\right)$ the closed sets $A = \left\{\left(x, \frac{1}{x}\right) : x > 0\right\}$, $B = \left\{\left(-x, \frac{1}{x}\right) : x > 0\right\}$. Then, the sum $A + B$ is contained in the half-plane $y > 0$ and, nevertheless, $(0, 0)$ is in the closure of $A + B$. In the case in which one of the sets is closed, the following statement is valid.

Proposition 2.1 *If $K \subset G$ is compact and $F \subset G$ is closed, then KF and FK are closed.*

Proof If $x \in \overline{KF}$, then x is the limit of a net $k_\alpha f_\alpha$ with $k_\alpha \in K$, $f_\alpha \in F$, and $\alpha \in \mathcal{D}$, where $\mathcal{D}$ is a directed set. As K is compact, there exists a subnet k_{α_j}

such that $k = \lim_{\alpha_j} k_{\alpha_j} \in K$, and this implies that $k^{-1} = \lim_{\alpha_j} k_{\alpha_j}^{-1}$ by continuity of the inverse. Using now the continuity of the product, it is clear that the subnet $f_{\alpha_j} = k_{\alpha_j}^{-1}\left(k_{\alpha_j} f_{\alpha_j}\right)$ converges to $f = k^{-1}x$, which belongs to the closed set F, as $f_{\alpha_j} \in F$. Hence, $x = kf$ with $k \in K$ and $f \in F$, that is, $x \in KF$. In the same way, it can be shown that FK is closed. □

Together with notation AB, there arise in a natural way the notations $A^2 = A \cdot A$, $A^3 = A^2 \cdot A = A \cdot A^2$, etc.

For $A \subset G$, $A^{-1} = \{x^{-1} \in G : x \in A\}$. As $\iota(g) = g^{-1}$ is a homeomorphism, $A^{-1} = \iota(A)$ is open or closed if and only if A is open or closed, respectively.

A neighborhood U of the identity is called **symmetric** if $U = U^{-1}$. It is not difficult to construct symmetric neighborhoods. Indeed, if V is any neighborhood of 1, then V^{-1} is a neighborhood too and $V \cap V^{-1}$ is a symmetric neighborhood.

2.2 Neighborhoods of Identity

Let $U \subset G$ be a nonempty open set and take $g \in U$. Then, $g^{-1}U$ and Ug^{-1} are neighborhoods of the identity of G. Conversely, if V is a neighborhood of 1, then, given $g \in G$, gV and Vg are neighborhoods of g. These observations have as a consequence that all information about the topology of G is concentrated in the set of open neighborhoods of the identity. The set of these neighborhoods is denoted $\mathcal{V}(1)$ or simply $\mathcal{V}$. The next proposition lists some properties of $\mathcal{V}$ which will be later used to describe the topology of G.

Proposition 2.2 *Let G be a topological group and denote by $\mathcal{V}$ the set of open neighborhoods of the identity* 1. *Then,*

(T1) *The identity* 1 *belongs to all subsets $U \in \mathcal{V}$.*
(T2) *Given two sets U, V in $\mathcal{V}$, $U \cap V$ is in $\mathcal{V}$.*
(GT1) *For every $U \in \mathcal{V}$, there exists $V \in \mathcal{V}$ such that $V^2 \subset U$.*
(GT2) *Given $U \in \mathcal{V}$, $U^{-1} \in \mathcal{V}$.*
(GT3) *For every $g \in G$ and $U \in \mathcal{V}$, $gUg^{-1} \in \mathcal{V}$.*

Proof Properties (T1) and (T2) hold for the neighborhoods of a point in any topological space. Property (GT1) is equivalent to saying that the product is continuous in the identity 1. Indeed, $p^{-1}(U) \subset G \times G$ is an open set containing $(1, 1)$. Hence, there exists an open subset V of G with $(1, 1) \in V \times V \subset p^{-1}(U)$. This means that $V^2 = p(V \times V) \subset U$. Property (GT2) has already been commented on above and is equivalent to the continuity in 1 of map ι. Finally, (GT3) follows from $g1g^{-1} = 1$ and the continuity of $C_g(x) = gxg^{-1}$. □

The properties stated in this proposition completely characterize the set of neighborhoods of the identity.

Definition 2.3 A **system of neighborhoods** of the identity (or neutral element) in a group G is a family of sets $\mathcal{V}$ satisfying the properties of Proposition 2.2.

It will be shown below that a system of neighborhoods of the identity defines in a unique way the topology of a topological group. To do that, it is necessary a lemma ensuring the continuity of maps from their continuity at a single point. Results analogous to this lemma are constantly used in the theory.

A topology $\mathcal{T}$ on a group G is called **left invariant** if gA is an open set of $\mathcal{T}$ for every $g \in G$ and every $A \in \mathcal{T}$. A topology is left invariant if and only if left translations are continuous (hence, homeomorphisms). **Right invariant** topologies are defined in the same way.

If $\mathcal{T}$ is a left invariant topology on G, then the product topology on $G \times G$ is left invariant, as $(g, h)(A \times B) = (gA) \times (hB)$ if $A, B \subset G$ and $(g, h) \in G \times G$. In the same way, the product topology is right invariant on $G \times G$ if it is right invariant on G.

Lemma 2.4 *Suppose that $\mathcal{T}$ is a left and right invariant topology on G. Then, G is a topological group if and only if*

1. *p is continuous at $(1, 1)$;*
2. *$\iota : G \to G$, $\iota(g) = g^{-1}$, is continuous at 1.*

Proof It is clear that the conditions are necessary. In order to prove that they are sufficient, let $L_{(g,h)}$ and $R_{(g,h)}$ be respectively the left and right translations in $G \times G$. Then, it follows immediately from the definitions that $p \circ L_{(g,1)} = L_g \circ p$ and $p \circ R_{(1,g)} = R_g \circ p$. Hence, by the associativity of the product, $p \circ L_{(g,1)} \circ R_{(1,h)} = L_g \circ R_h \circ p$ if $(g, h) \in G \times G$. The second member in this equality is a map that is continuous at $(1, 1)$, since $L_g \circ R_h$ is a homeomorphism. Thus, $p \circ L_{(g,1)} \circ R_{(1,h)}$ is continuous at $(1, 1)$. But $L_{(g,1)} \circ R_{(1,h)}$ is also a homeomorphism, whence it follows that p is continuous at $(g, h) = L_{(g,1)} \circ R_{(1,h)}(1, 1)$.

On the other hand, $R_{g^{-1}} \circ \iota$ is continuous at 1, hence $\iota \circ L_g$ is continuous at 1 and it follows that ι is continuous at $g = L_g(1)$. □

In order to characterize the topology of G on the basis of the systems of neighborhoods of the identity, recall that a **fundamental system of neighborhoods** for a point x in a topological space X is a family $\mathcal{F}$ of open sets of X, such that each element of $\mathcal{F}$ contains x, and if $A \subset X$ is an open set with $x \in A$, then there exists $B \in \mathcal{F}$ such that $B \subset A$.

Proposition 2.5 *Let G be a group and suppose that $\mathcal{V}$ is a system of neighborhoods of the identity in G, as in Definition 2.3. Then, there exists a unique topology $\mathcal{T}$ that makes G a topological group in such a way that $\mathcal{V}$ is a fundamental system of neighborhoods for the identity with respect to $\mathcal{T}$.*

Proof Define $\mathcal{T}$ as the family of subsets of G formed by the empty set $\emptyset$ and the subsets $A \subset G$, such that, for every $g \in A$, there exists $U \in \mathcal{V}$ with $gU \subset A$, together with $\emptyset$. To see that $\mathcal{T}$ is a topology, take $A, B \in \mathcal{T}$ and $x \in A \cap B$. Then, there exist $U, V \in \mathcal{V}$ such that $xU \subset A$ and $xV \subset B$. By property (T2), $U \cap V \in \mathcal{V}$. But

$$x(U \cap V) = xU \cap xV \subset A \cap B,$$

showing that $A \cap B \in \mathcal{T}$. The definition of $\mathcal{T}$ shows that any union of sets in $\mathcal{T}$ is an element of $\mathcal{T}$.

Now, the open neighborhoods of 1 with respect to $\mathcal{T}$ are the elements of $\mathcal{V}$. Indeed, the very definition of $\mathcal{T}$ shows that the elements of $\mathcal{V}$ are neighborhoods of 1. On the other hand, let U be a neighborhood of 1 with respect to $\mathcal{T}$. Then, there exists $V \in \mathcal{V}$, such that $1 \cdot V \subset U$. Hence, $\mathcal{V}$ is a fundamental system of neighborhoods of 1 with respect to $\mathcal{T}$.

The definition of $\mathcal{T}$ and property (GT3) ensure that $\mathcal{T}$ is left and right invariant. Indeed, a left translation gU, $u \in \mathcal{V}$, is also a right translation with the form $gU = \left(gUg^{-1}\right) g$. By (GT3), if $U \in \mathcal{V}$, then $gUg^{-1} \in \mathcal{V}$. Hence, using the previous lemma, in order to ensure that G endowed with $\mathcal{T}$ is a topological group, it suffices to verify that p and ι are continuous at (1, 1) and 1, respectively. But these continuities are equivalent to properties (GT1) and (GT2), respectively, and this concludes the proof that G is a topological group with topology $\mathcal{T}$.

Finally, suppose that $\mathcal{T}'$ is another topology satisfying the same conditions. Then $\mathcal{V}$ is a fundamental system of neighborhoods of 1 with respect to $\mathcal{T}'$. By performing left translations, it is clear that gV, with V varying in $\mathcal{V}$, is a fundamental system of neighborhoods of $g \in G$. Hence, for every $A \in \mathcal{T}'$ and $g \in A$, there exists $V \in \mathcal{V}$ such that $gV \subset A$. It follows that every open set of $\mathcal{T}'$ is an open set of $\mathcal{T}$, that is, $\mathcal{T}' \subset \mathcal{T}$. Exchanging the roles of $\mathcal{T}$ and $\mathcal{T}'$, it follows that $\mathcal{T} = \mathcal{T}'$, concluding the proof. □

Example A situation illustrating the construction made above appears when the elements of $\mathcal{V}$ are subgroups of G. In this case, the conditions on $\mathcal{V}$ are restricted to (T2) and (GT3), since $1 \in V \cap U$ and $V^2 = V^{-1} = V$ if U and V are subgroups. An example of a system $\mathcal{V}$ of this kind is constructed on group $\mathbb{Z}$. Given a prime number $p > 0$, denote by $\mathcal{V}_p$ the family of subgroups $V_n = p^n\mathbb{Z}$, $n \geq 1$. As $\mathbb{Z}$ is abelian, condition (GT3) is automatically satisfied. Condition (T2) also holds, as $p^n\mathbb{Z} \cap p^m\mathbb{Z} = p^{\max\{n,m\}}\mathbb{Z}$. Hence, $\mathcal{V}$ defines a topology on $\mathbb{Z}$, making it a topological group. This is the so-called p-adic topology on $\mathbb{Z}$. □

The above description of topology in terms of neighborhoods of identity establishes the principle that every topological description on G is made by means of such neighborhoods. The proposition below follows this principle in giving a criterion for a topology to be Hausdorff in terms of neighborhoods of identity.

Proposition 2.6 *Let G be a topological group. Then, the following conditions are equivalent:*

1. *The topology of G is Hausdorff.*
2. *$\{1\}$ is a closed set.*
3. $\bigcap_{U \in \mathcal{V}(1)} U = \{1\}$.

Proof In a Hausdorff topology, every unitary set is closed; in particular, $\{1\}$ is closed. Suppose that $\{1\}$ is closed. To show that the intersection of neighborhoods reduces to the identity, it is necessary to show that, for every $x \neq 1$, there exists $U \in \mathcal{V}$, such that $x \notin U$. As $\{1\}$ is closed, there exists $V \in \mathcal{V}$ such that $1 \notin x^{-1}V$,

that is, $x \notin V$. Finally, assume that the intersection in item (3) reduces to $\{1\}$ and take $x \neq 1$. Then, there exists $U \in \mathcal{V}$ such that $x \notin U$. From (GT1), there exists $V \in \mathcal{V}$ such that $V^2 \subset U$. Then, $V \cap xV^{-1} = \emptyset$, since $z \in V \cap xV^{-1}$ must satisfy $z = u = xv^{-1}$, $u, v \in V$, whence it follows that $x = uv \in V^2 \subset U$ and this contradicts the choice of U. As a consequence, the open sets V and xV^{-1} separate 1 from x. Now, take arbitrary $y \neq z$. Then, there exist open sets U_1 and U_2, with $y^{-1}z \in U_1$ and $1 \in U_2$ and $U_1 \cap U_2 = \emptyset$. Hence the open sets yU_1 and yU_2 separate z from y. □

2.3 Metrizable Groups

A distance $d : G \times G \to \mathbb{R}_+$ on a group G é said left invariant if $d(gx, gy) = d(x, y)$ for every $g, x, y \in G$. In other words, d is left invariant if the left translations L_g are isometries. Right invariant distances are defined analogously. A distance is called bi-invariant if it is both left- and right invariant.

A necessary condition for a topological space to be metrizable is that every point admits a denumerable fundamental system of neighborhoods. In the case of topological groups, this condition is also sufficient and, as before, it suffices to verify it for the identity.

Theorem 2.7 *Let G be a topological group and suppose that there exists a denumerable system of neighborhoods of the identity. Then, there exist distances d_L and d_R which are respectively right and left invariant and which are compatible with the topology of G.*

This theorem will not be proved here. When G is a Lie group, the enumerability condition is satisfied because G is locally homeomorphic to $\mathbb{R}^n$. So, Lie groups are metrizable. However, for the particular case of Lie groups, there is a construction simpler than the general proof of Theorem 2.7, using Riemannian metrics on differentiable manifolds. This proof will be presented later.

In any case, it is worth pointing out that the theorem ensures the existence of both a left invariant and a right invariant distance. Nevertheless, it is possible that there exists no bi-invariant distance on a metrizable group.

Examples Some examples of invariant distances are:

1. Let $|\cdot|$ be any norm on $\mathbb{R}^n$ and $d(x, y) = |x - y|$. Then, d is a bi-invariant distance on $(\mathbb{R}^n, +)$.

   Observe that a distance defined by a norm on the space of $n \times n$ matrices is not necessarity invariant when restricted to the group $\mathrm{Gl}(n, \mathbb{R})$.
2. Let G be a compact group that is metrizable by a distance d'. Define

$$d(x, y) = \sup_{g \in G} d'(gx, gy).$$

Then, d is a left invariant distance on G, compatible with its topology. The distance d can be also viewed in the following manner: denote by $\mathrm{Hom}(G)$ the group of homeomorphisms of G and let $\rho : G \to \mathrm{Hom}(G)$ be the map $\rho(g) = L_g$. Then, d is the restriction to $\rho(G)$ of the distance on $\mathrm{Hom}(G)$ that defines uniform convergence with respect to d'. □

2.4 Homomorphisms

Proposition 2.8 *Let G and H be topological groups and $\phi : G \to H$ a homeomorphism. Then, ϕ is continuous if and only if ϕ is continuous at the identity $1 \in G$.*

Proof It is sufficient to show that continuity at 1 entails continuity at every point. As ϕ is a homeomorphism, $\phi \circ L_g = L_{\phi(g)} \circ \phi$ for every $g \in G$. The second member is continuous at 1. Hence, $\phi \circ L_g$ is continuous at 1 and, as L_g is a homeomorphism, it follows that ϕ is continuous at $g = L_g(1)$. □

Given two groups G and H, the Cartesian product $G \times H$ is a group whose product is defined component-wise: $(g, x)(h, y) = (gh, xy)$, $g, h \in G$ and $x, y \in H$. Projections $\pi_1 : G \times H \to G$ and $\pi_2 : G \times H \to H$ are homeomorphisms.

The graph $\mathrm{gr}\phi$ of a map $\phi : G \to H$ is the set of elements of the form $(x, \phi(x))$ with $x \in G$. As $(x, \phi(x))(y, \phi(y)) = (xy, \phi(x)\phi(y))$, the map ϕ is a homeomorphism of groups if and only if its graph is a subgroup of $G \times H$.

When this happens, the groups G and $\mathrm{gr}\phi$ are isomorphic, since the map

$$l : x \in G \mapsto (x, \phi(x)) \in \mathrm{gr}\phi$$

is an isomorphism. The inverse of l is the projection $p : \mathrm{gr}\phi \to G$, $p(x, \phi(x)) = x$, which is the restriction to the graph of projection π_1 in the first coordinate.

Conversely, a subgroup $\Gamma \subset G \times H$ is the graph of a homeomorphism $\phi : G \to H$ if and only if the restriction of π_1 to Γ is an isomorphism. In this case, $\phi = \pi_2 \circ l$.

In topological contexts, the graphs of continuous homeomorphisms are characterized through closed subgroups.

Proposition 2.9 *Let G and H be topological groups, such that H is Hausdorff. A map $\phi : G \to H$ is a continuous homeomorphism if and only if its graph*

$$\mathrm{gr}\phi = \{(x, \phi(x)) \in G \times H : x \in G\}$$

is a closed subgroup of $G \times H$ homeomorphic to G by the projection $p(x, \phi(x)) = x$.

Proof By the comments above, it is sufficient to verify that ϕ is continuous if and only if its graph is closed and homeomorphic to G. But this property is true for maps in general. (If ϕ is continuous, then $l(x) = (x, \phi(x))$ is a homeomorphism whose inverse is é $p(x, \phi(x)) = x$. Moreover, let $\theta : G \times H \to H \times H$ the map given by $\theta(x, y) = (\phi(x), y)$. Then, $\mathrm{gr}\phi = \theta^{-1}(\Delta_H)$, where $\Delta_H = \{(y, y) \in H \times H : y \in H\}$ is the diagonal of $H \times H$. This diagonal is closed if and only if H is Hausdorff. Hence, $\mathrm{gr}\phi$ is closed. Conversely, if the graph is closed and homeomorphic to the domain, then $(F \times G) \cap \mathrm{gr}\phi$ is a closed subset of $\mathrm{gr}\phi$ for every $F \subset H$ closed. It follows that $\phi^{-1}(F) = \pi_1((F \times G) \cap \mathrm{gr}\phi)$ is closed in G and, hence, ϕ is continuous.) □

2.5 Subgroups

Let G be a topological group and H a subgroup of G. As H is a subset of G, it can be endowed with the induced topology, whose open sets have the form $A \cap H$, with A an open set in G. Then, H becomes a topological group. Indeed, denote by $p_H : H \times H \to H$ the product in H, which is the restriction to H of the product p of G. For every subset $A \subset G$, $p_H^{-1}(A \cap H) = p^{-1}(A) \cap (H \times H)$. It follows from this equality that the set $p_H^{-1}(A \cap H)$ is open in the topology induced on $H \times H$ by the product topology on $G \times G$, for every open set $A \subset G$. But this induced topology coincides with the product topology of H. Hence p_H is continuous. In the same way, it can be shown that $\iota_H(h) = h^{-1}$ is continuous in H.

A subgroup $H \subset G$ with the induced topology is called a **topological subgroup** of G.

In what follows, some results about the topological properties of the subgroups of G are presented. The following lemma, of a general character, is used in some proofs.

Lemma 2.10 *Let X be a topological space and $\phi : X \to X$ a homeomorphism. Suppose that $A \subset X$ is a subset invariant by ϕ, that is, $\phi(A) \subset A$. Then, $\overline{A}$ and A° are also invariant. Moreover, if $\phi(A) \subset \overline{A}$, then $\phi\left(\overline{A}\right) \subset \overline{A}$.*

Proof As ϕ is a homeomorphism, $\phi\left(\overline{A}\right) = \overline{\phi(A)}$. Then $\phi\left(\overline{A}\right) = \overline{\phi(A)} \subset \overline{A}$ if $\phi(A) \subset A$. In the same way, $\phi\left(\overline{A}\right) = \overline{\phi(A)} \subset \overline{\overline{A}} = \overline{A}$ if $\phi(A) \subset \overline{A}$. But $\phi(A^\circ) \subset \phi(A)$ and thus $\phi(A^\circ)$ is an open set contained in A if A is invariant. It then follows that $\phi(A^\circ) \subset A^\circ$. □

Proposition 2.11 *Let $H \subset G$ be a subgroup. Then its closure $\overline{H}$ is also a subgroup. Moreover, if H is normal, the same is true for $\overline{H}$.*

Proof It is necessary to show that $xy \in \overline{H}$ if $x, y \in \overline{H}$. To this end, suppose first that $x \in H$. Then $L_x(H) = H$ and the above lemma ensures that $L_x\left(\overline{H}\right) \subset \overline{H}$. But this means that $xy \in \overline{H}$ if $y \in \overline{H}$. Thus, given $x \in H$ and $y \in \overline{H}$, $xy \in \overline{H}$, and

this implies that $R_y(H) \subset \overline{H}$ for every $y \in \overline{H}$. But R_y is a homeomorphism, hence $R_y\left(\overline{H}\right) \subset \overline{H}$ for every $y \in \overline{H}$, and this means that $xy \in \overline{H}$ if $x, y \in \overline{H}$.

By means of a similar reasoning, it can be proved that the inverse ι leaves invariant $\overline{H}$, showing that $\overline{H}$ is a subgroup.

Finally, saying that H is normal is equivalent to saying that H is invariant by the conjugations C_g, $g \in G$. From Lemma 2.10, it follows that $\overline{H}$ is also invariant by C_g, that is, $\overline{H}$ is normal. □

Closed subgroups play a central role in the study of actions of topological (and Lie) groups because, in the case of continuous actions, subgroups that fix a point (isotropy subgroups) are closed. Proposition 2.11 shows that there are a great number of closed subgroups. On the other hand, the situation is much simpler for the interior H° of a subgroup H as it will be empty or it will be equal to H itself (in case H is connected), that is, H is open and, in this case, closed, as stated by the following propositions.

Proposition 2.12 *Let $H \subset G$ be a subgroup and suppose that $H^\circ \neq \emptyset$. Then, H is open.*

Proof Suppose that there exists $x \in H^\circ$. Then, for every $y \in H$, the set $yx^{-1}H^\circ$ is open, contains y, and is contained in H. This shows that $y \in H^\circ$ and, thus, $H \subset H^\circ$, that is, $H = H^\circ$. □

Proposition 2.13 *Suppose that H is an open subgroup of G. Then, H is closed.*

Proof A coset gH of H is obtained from H by means of a left translation. Hence, if H is open, the same happens to gH. But group G is the union of H with the cosets gH, $g \notin H$. This means that the complement of H in G is a union of open sets, whence it follows that H is closed. □

A subset A of a topological space X that is both open and closed is a union of connected components of X, that is, if a connected component $C \subset X$ satisfies $C \cap A \neq \emptyset$, then $C \subset A$. This observation, together with Proposition 2.13, shows that the open subgroups of G are unions of connected components of G. In particular, if the group is connected, it is its only open subgroup.

In any case, the connected components of G are related to the open subgroups. In the sequel, these components are described using the connected component G_0 containing the identity $1 \in G$. This connected component is called the **identity component** (or neutral element component).

Proposition 2.14 *Denote by G_0 the connected component of the identity element. Then, G_0 is a closed normal subgroup of G. Every other connected component is a coset $gG_0 = G_0g$ of G_0. Conversely, every coset $gG_0 = G_0g$ is a connected component of G.*

Proof A left translation L_g, $g \in G$, is a homeomorphism. Hence, the image by L_g of a connected component of G is also a connected component. In particular, if $g \in G_0$, then $L_g(G_0)$ is a connected component of G. However, $1 \in G_0$ and

$L_g(1) = g \in G_0$. This implies that $L_g(G_0) \subset G_0$. Then, taking $g, h \in G_0$, it is clear that $gh \in G_0$. Analogously, the set $\iota(G_0)$ is contained in a connected component of G which must be G_0, as $\iota(1) = 1$. This shows that G_0 is a subgroup. To see that it is normal, it suffices to repeat the same reasoning with conjugations C_g, $g \in G$, using the fact that $C_g(1) = 1$. Finally, because it is a connected component, G_0 is closed.

As G_0 is normal, $gG_0 = G_0 g$ for every $g \in G$. It is clear that $gG_0 = L_g(G_0)$ is connected, so gG_0 is contained in a connected component C of G. To obtain a contradiction, suppose that $gG_0 \neq C$. Then, $G_0 = L_{g^{-1}}(gG_0) \neq g^{-1}C$, contradicting the fact that G_0 is a connected component, since $g^{-1}C$ is connected. □

Generally, the identity component is not an open subgroup. For instance, consider in $(\mathbb{R}, +)$ the subgroup $\mathbb{Q} \subset \mathbb{R}$ endowed with the induced topology. Then, the identity component reduces to $\{0\}$, which is not an induced open set.

A condition for the connected component of identity G_0 to be open is that the group be locally connected, in the sense that every point has an open connected neighborhood. Because Lie groups are locally homeomorphic to $\mathbb{R}^n$, they are locally connected. Thus, the following proposition ensures that the connected components of such groups are open.

Proposition 2.15 *Suppose that G is locally connected. Then, G_0 is an open subgroup.*

Proof As G is locally connected, there exists a connected neighborhood U of the identity. It is clear that $U \subset G_0$. Hence, the interior of G_0 is nonempty and so it is open. □

Finally, the following result provides a way to generate connected groups which is very useful in the study of Lie groups.

Proposition 2.16 *Suppose that G is connected and consider a neighborhood U of the identity element. Then $G = \bigcup_{n \geq 1} U^n$.*

Proof Let $V = U \cap U^{-1}$ be a symmetric neighborhood contained in U. As $\bigcup_{n \geq 1} V^n \subset \bigcup_{n \geq 1} U^n$, it suffices to show that $G = \bigcup_{n \geq 1} V^n$. The union $\bigcup_{n \geq 1} V^n$ is closed by products. Besides, as V is symmetric, $(V^n)^{-1} = V^n$. This implies that $\bigcup_{n \geq 1} V^n$ is a subgroup of G that has a nonempty interior, as $V \subset \bigcup_{n \geq 1} V^n$. As subgroups with nonempty interiors are open, it follows that $\bigcup_{n \geq 1} V^n$ is an open subgroup and hence that $G = \bigcup_{n \geq 1} V^n$, as G is connected. □

2.6 Group Actions

2.6.1 Algebraic Description

A **left action** of a group G on a set X is a function that associates to $g \in G$ a map $a(g) : X \to X$ which satisfies the properties:

1. $a(1) = \mathrm{id}_X$, that is, $a(1)(x) = x$, for every $x \in X$;
2. $a(gh) = a(g) \circ a(h)$.

These properties ensure that each $a(g)$ is a bijection, as

$$a\left(g^{-1}\right) a(g) = a(1) = a(g)\, a\left(g^{-1}\right) = \mathrm{id}_X.$$

Viewed in a different way, a left action is a homomorphism $a : G \to \mathcal{B}(X)$, where $\mathcal{B}(X)$ is the group of bijections of X, with product given by the composition of two maps.

A **right action** is defined analogously, replacing the second property by $a(gh) = a(h) \circ a(g)$.

Alternatively, a left action is defined as a map $\phi : G \times X \to X$ satisfying

1. $\phi(1, x) = x$;
2. $\phi(gh, x) = \phi(g, \phi(h, x))$, $g, h \in G$ and $x \in X$.

The relation between ϕ and a is obvious: $\phi(g, x) = a(g)(x)$, that is, $a(g)$ is the partial map ϕ_g of ϕ when the first coordinate is fixed: $\phi_g(x) = \phi(g, x)$.

The other partial map associated with ϕ is obtained by fixing $x \in X$, that is, $\phi_x : G \to X$, $\phi_x(g) = \phi(g, x) = a(g)(x)$.

Symbols a or ϕ are usually suppressed in the notation for group actions. Thus, a left action is just written $g(x)$, $g \cdot x$ or gx instead of $a(g)(x)$. For right actions, it is more adequate to write the value of $a(g)$ on x as $(x)\, a(g)$, and then there appear the notations $(x)\, g$, $x \cdot g$ or xg. With these notations, a left action satisfies $1x = x$ and $g(hx) = (gh)\, x$, while a right action satisfies $x1 = x$ and $(xg)\, h = x(gh)$.

If a is a left action of G on X, then the map a' defined by $a'(g) = a\left(g^{-1}\right)$ is a right action and vice versa. In what follows, only left actions will be dealt with. The properties stated are automatically transferred to right actions by replacing $a(g)$ by $a\left(g^{-1}\right)$.

Given $x \in X$, its **orbit** by G, denoted by $G \cdot x$ or Gx, is defined as the set

$$G \cdot x = \{gx \in X : g \in G\}.$$

More generally, if $A \subset G$, then $Ax = \{gx : g \in A\}$, that is, $Ax = \phi_x(A)$. Each orbit is an equivalence class of the equivalence relation $x \sim y$ if there exists $g \in G$ such that $y = gx$. Therefore, two orbits are disjoint or coincident.

A subset $B \subset X$ is G-invariant if $gB \subset B$ for every $g \in G$. An invariant set is a union of orbits of G. If B is an invariant set, then the restriction of the group action to $G \times B$ defines an action $G \times B \to B$ of G on B.

In particular, the group G acts on its orbits.

The set G_x of elements of G that fix x is called **isotropy subgroup** or **stabilizer** of x:

$$G_x = \{g \in G : gx = x\}.$$

An isotropy subgroup is in fact a subgroup of G, since $(gh)\, x = g\,(hx)$, so that gh fixes x if $gx = hx = x$. Moreover, $g^{-1}x = x$ if $gx = x$, since $a\left(g^{-1}\right) = a\,(g)^{-1}$.

Isotropy subgroups are obtained from other isotropy subgroups through the following relation.

Proposition 2.17 *Given $x, y \in X$, suppose that $y = gx$, with $g \in G$. Then $G_y = gG_xg^{-1}$, where G_x and G_y denote the isotropy subgroups.*

Proof By definition, $h \in G_y$ if and only if $h\,(gx) = gx$. Applying g^{-1} to this equality, it follows that $\left(g^{-1}hg\right) x = x$, that is, $g^{-1}hg \in G_x$. Hence, $h \in G_y$ if and only if $h \in gG_xg^{-1}$. □

The actions of a group G are separated in classes according to the properties of their orbits and isotropy groups. The following definition distinguishes some types of actions which appear more frequently along the development of the theory.

Definition 2.18 Let a be an action of G on X.

1. The action is called **effective** if $\ker a = \{g \in G : a\,(g) = \mathrm{id}_X\} = \{1\}$. That is, if $gx = x$ for every $x \in X$, then $g = 1$.
2. The action is called **free** if the isotropy subgroups reduce to the identity element of G, that is, if $gx = x$ for some $x \in X$, then $g = 1$.
3. The action is called **transitive** if X is an orbit of G, that is, if for every two elements $x, y \in X$, there exists $g \in G$ such that $gx = y$.

It is clear from these definitions that all free actions are effective, though not every effective action is free. In an effective action, $\ker a = \{1\}$, therefore G is isomorphic to its image $a\,(G)$ by a. For this reason, an effective action is also called a **faithful action**.

It should be noted that the restriction of an action to an orbit is a transitive action. Therefore, every statement about transitive actions applies to restrictions of actions to an orbit.

A particular case of group action takes place in quotient spaces. Let $H \subset G$ be a subgroup and denote by G/H the set of cosets gH, $g \in G$. Then, the map $(g, g_1H) \mapsto g\,(g_1H) = (gg_1)\,H$ defines a natural left action of G on G/H. Denoting by $\pi : G \to G/H$ the canonical surjective map (projection) $\pi\,(g) = gH$, this action is written as $g\pi\,(g_1) = \pi\,(gg_1)$.

It is evident that the action of G on G/H is transitive. On the other hand, every transitive action can be identified (or rather, is in bijection) with a quotient space of G.

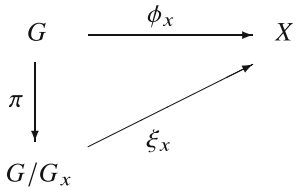
Proposition 2.19 *Suppose that the action of G on X is transitive and take $x \in X$. Then, the map $\xi_x : gG_x \in G/G_x \mapsto gx \in X$ is a bijection between G/G_x and X. The map ξ_x is **equivariant** in the sense that $g\xi_x(g_1G_x) = \xi_x((gg_1)G_x)$, $g, g_1 \in G$, that is, ξ_x commutes with the actions of G on G/G_x and X, respectively. Moreover, if $y = gx$, then $\xi_y = \xi_x \circ R_g$.*

Proof First, the map is well defined, because if g_1 and g_2 are in the same coset, that is, $g_1G_x = g_2G_x$, then $g_2^{-1}g_1 \in G_x$, and this means that $g_2^{-1}g_1x = x$, that is $g_1x = g_2x$. By definition, the map is surjective if and only if the action is transitive. Suppose now that $g_1x = g_2x$. Then $g_2^{-1}g_1x = x$, that is, $g_2^{-1}g_1 \in G_x$, and so $g_1G_x = g_2G_x$, showing the injectivity of the map.

Let $y = \xi_x(g_1G_x)$. Then $y = g_1x$ and therefore $gy = g(g_1x) = (gg_1)x$. It follows that $g\xi_x(g_1G_x) = \xi_x((gg_1)G_x)$.

Finally, if $y = gx$, then $\xi_y(h) = h(gx) = (hg)x = \xi_x(hg)$, showing that $\xi_y = \xi_x \circ R_g$. □

The map ξ_x in the proposition above is related to the partial map ϕ_x through the following commutative diagram:



Because of this identification, a quotient G/H is also called a **homogeneous space**, as are usually called the sets on which groups act transitively. The point x chosen to establish the identification between X and G/G_x is called **origin** or **basis** of the homogeneous space X. The identification of X with G/G_x depends on the choice of origin. However, changing x does not substantially modify the quotient space because, in a transitive action, isotropy groups are conjugate to each other, as shown by Proposition 2.17. Indeed, if $H \subset G$ is a subgroup then, for every $g \in G$, the map

$$hH \longmapsto g(hH)g^{-1} = \left(ghg^{-1}\right)\left(gHg^{-1}\right)$$

establishes a bijection between G/H and G/gHg^{-1}.

The facts described above for transitive actions apply immediately to the orbits of any action $G \times X \to X$. In this case, the restriction of the action to an orbit $G \cdot x$ is transitive, allowing the identification of $G \cdot x$ with G/G_x.

All the above discussion can be extended in an analogous way to right actions, where the homogeneous spaces are the quotients $H \setminus G$ formed by the cosets Hg, $g \in G$.

In a homogeneous space G/H, that is, in the presence of a transitive action, the free actions are those for which the isotropy subgroup H reduces to $\{1\}$. In this case, the homogeneous space is identified with G. In what follows, transitive and effective actions are described by the normal subgroups contained in the isotropy group.

Proposition 2.20 *Let G be a transitive action on $X = G/H$. Then, the action is effective if and only if H does not contain any normal subgroups of G other than* $\{1\}$.

Proof Suppose that $N \subset H$ is a normal subgroup of G, that is, $gNg^{-1} \subset N$ for every $g \in G$. It is clear that H is the isotropy group of the origin. But according to Proposition 2.17, isotropy subgroups are conjugate to each other. Hence, any $h \in N$ is contained in all isotropy subgroups. But this means that $hy = y$ for every $y \in X$, that is, $h = \mathrm{id}_X$. Therefore, if the action is effective, $N = \{1\}$.

Conversely, the normal subgroup $\ker a = \{g \in G : \forall y \in X, gy = y\}$ is contained in H. Thus, if H does not contain normal subgroups other than the trivial subgroup, then $\ker a = \{1\}$ and the action is effective. □

A way to obtain an action of a group G on a vector space V is through a **representation** of G in V, which is a homomorphism $\rho : G \to \mathrm{Gl}(V)$, where $\mathrm{Gl}(V)$ is the group of invertible linear transformation of V. The space V is called the **representation space** and $\dim V$ is its dimension. The representation defines the action $a : G \times V \to V$ given by $a(g, v) = \rho(g)\, v$.

2.6.2 Continuous Actions

In a topological context, actions should be considered continuous in the following sense.

Definition 2.21 Let G be a topological group and let X be a topological space. An action of G on X is continuous if the map $\phi : G \times X \to X$, $\phi(g, x) = gx$, is continuous.

If $H \subset G$ is a subgroup, the restriction to H of the action of G on X is an action of H. Taking in H the induced topology, the restriction of a continuous action is also continuous.

In the case of a continuous action, the objects previously introduced admit good topological properties.

Indeed, if ϕ is continuous, then the partial maps $\phi_x : G \to X$, $x \in X$, and $\phi_g : X \to X$, $g \in G$, are continuous. Moreover, since $a(g) = \phi_g$ and $a(g)^{-1} = a\left(g^{-1}\right)$, it follows that, for each $g \in G$, $a(g) : X \to X$ is a homeomorphism of X.

Proposition 2.22 *Suppose that the action of G on X is continuous and that X is a Hausdorff space. Then, any isotropy subgroup G_x, $x \in X$, is closed.*

Proof In terms of map ϕ, the isotropy subgroup is given by

$$G_x = \{g \in G : \phi(g, x) = x\} = \phi_x^{-1}\{x\}.$$

Since X is Hausdorff, it follows that G_x is closed. □

A variation of the concept of action of a group G on a topological space X are the local actions, in which the elements of an open subset of G are local homeomorphisms of X, that is, homeomorphisms between open subsets of X. Formally, a continuous (left) **local action** is a map $\phi : V \subset G \times X \to X$ satisfying the following conditions:

1. The domain $V \subset G \times X$ is open and such that
   (a) for every $x \in X$, the open set $V_x = V \cap (G \times \{x\})$ contains $(1, x)$;
   (b) given $g \in G$ and $x \in X$, $\left(g, \phi\left(g^{-1}, x\right)\right) \in V$ if $\left(g^{-1}, x\right) \in V$.
2. $\phi(1, x) = x$ for every $x \in X$.
3. $\phi(g, \phi(h, x)) = \phi(gh, x)$ if $(g, \phi(h, x))$, (h, x), and (gh, x) are elements of V (that is, $\phi(h, x) \in V_g$, $x \in V_h$ and $x \in V_{gh}$).

In a local action, the open set $V_g = V \cap (\{g\} \times X)$, $g \in G$, is the domain of the partial map $\phi_g : V_g \to X$. In principle, the set V_g may be empty. However, the set of elements $g \in G$ such that $V_g \neq \emptyset$ is open and nonempty. Indeed, $V_1 = X$ and if $(g, x) \in V$, then, elements of the type $(h, x) \in V$ if h is near g. That is, for h near g, if $V_g \neq \emptyset$, then $V_h \neq \emptyset$.

If $V_g \neq \emptyset$, then condition (1b) ensures that $V_{g^{-1}} \neq \emptyset$ and the partial map $\phi_g : X \to X$ is a homeomorphism between V_g and $V_{g^{-1}}$. Indeed, if $(g, x) \in V$, that is, $x \in V_g$, then, from condition (1b), $\left(g^{-1}, \phi(g, x)\right) \in V$. Hence, it is possible to write condition (3) to obtain $\phi\left(g^{-1}, \phi(g, x)\right) = \phi(1, x) = x$. That is, $\phi_g(x) \in V_{g^{-1}}$ and $\phi_{g^{-1}}\phi_g(x) = x$. Exchanging the roles of g and g^{-1}, it follows that $\phi_g : V_g \to V_{g^{-1}}$ is a homeomorphism.

2.7 Quotient Spaces

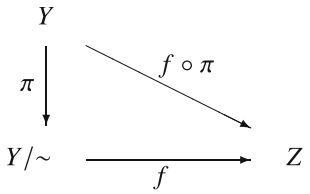
The tool to analyze the orbits of an action is the bijection $G/G_x \approx G \cdot x$ of Proposition 2.19. In order to consider this bijection from the point of view of continuity, it is necessary to introduce topologies in quotient spaces G/H; this is the subject of this section.

Usually, the following topology is used for the quotient of a topological space by an equivalence relation.

Definition 2.23 Let Y be a topological space and $\sim$ an equivalence relation in Y. Denote by $Y/\sim$ the set of equivalence classes of $\sim$ and by $\pi : Y \to Y/\sim$ the canonical surjective map, which associates to each $y \in Y$ its equivalence class. The **quotient topology** in $Y/\sim$ is that topology in which a subset $A \subset Y/\sim$ is open if and only if $\pi^{-1}(A)$ is open in Y. Equivalently, $F \subset Y/\sim$ is closed if and only if $\pi^{-1}(F)$ is closed in Y.

The quotient topology is the finest topology (the one containing the largest possible quantity of open sets) in which the canonical projection $\pi : Y \to Y/\sim$ is a continuous map. The continuity, with respect to the quotient topology, of functions defined in $Y/\sim$ is verified by means of the following property.

Proposition 2.24 *Let Y and Z be topological spaces, and let Y be endowed with an equivalence relation $\sim$. Then, a map $f : Y/\sim \to Z$ is continuous if and only if $f \circ \pi : Y \to Z$ is continuous.*



Proof If f is continuous, then $f \circ \pi$ is continuous since π is continuous. Conversely, suppose that $f \circ \pi$ is continuous and let $A \subset Z$ be an open set. Then, $(f \circ \pi)^{-1}(A) = \pi^{-1}\left(f^{-1}(A)\right)$ is open in Y. From the definition of quotient topology, it follows that $f^{-1}(A)$ is open in $Y/\sim$, concluding the proof. □

In the case in which G is a group and $H \subset G$ is a subgroup, the quotient G/H is the set of equivalence classes of the equivalence relation in G, in which $x \sim y$ if and only if $xH = yH$. Hence, G/H can be endowed with the quotient topology by this equivalence relation, when G is a topological group.

Proposition 2.25 *Let G be a topological group, $H \subset G$ a subgroup, and $\pi : G \to G/H$ the canonical projection. Then, π is an open map with respect to the quotient topology. If, moreover, H is compact, then π is a closed map.*

Proof Take an open set $A \subset G$. Then, $\pi^{-1}(\pi(A)) = AH = \bigcup_{h \in H} Ah$ is an open subset of G. It follows that $\pi(A)$ is open in the quotient topology.

Now, if H is compact and $F \subset G$ is closed, then $\pi^{-1}(\pi(F)) = FH$ is compact, according to Proposition 2.1, showing that π is a closed map. □

One should also observe that, in general, the projection is not a closed map. For instance, if $G = \mathbb{R}^2$, $H = \{0\} \times \mathbb{R}$, and $F = \{(x, y) \in \mathbb{R}^2 : -\pi/2 < x < \pi/2 \text{ and } y = \operatorname{tg}(x)\}$, then $\pi(F)$ is not closed.

The quotient topology presents a good behavior with respect to the Cartesian product of groups. Let G_1 and G_2 be topological groups and let $H_1 \subset G_1$, $H_2 \subset G_2$

be subgroups. The product $H_1 \times H_2$ is a subgroup of $G_1 \times G_2$ and the quotient $(G_1 \times G_2) / (H_1 \times H_2)$ is identified with $(G_1/H_1) \times (G_2/H_2)$ by means of the bijection

$$\phi : (g_1, g_2)(H_1 \times H_2) \longmapsto (g_1 H_1, g_2 H_2) .$$

This bijection is a homeomorphism with respect to quotient topologies in homogeneous spaces. This can be easily seen from the definition of quotient topology and the following commutative diagram:

$$\begin{array}{ccc} G_1 \times G_2 & \xrightarrow{\text{id}} & G_1 \times G_2 \\ \downarrow & & \downarrow \\ (G_1 \times G_2) / (H_1 \times H_2) & \xrightarrow{\phi} & (G_1/H_1) \times (G_2/H_2) \end{array}$$

Proposition 2.26 *The quotient topology in G/H is Hausdorff if and only if H is closed.*

Proof The map $\pi : G \to G/H$ is continuous and $H = \pi^{-1}\{x\}$ if x is the origin of G/H. Hence, if G/H is Hausdorff, then H is closed.

Conversely, suppose that H is closed. In order to prove that G/H is Hausdorff, it must be shown that the diagonal

$$\Delta = \{(x, x) \in G/H \times G/H : x \in G/H\}$$

is closed in the product topology in $G/H \times G/H$, which coincides with the quotient topology in $(G \times G) / (H \times H)$. By definition, Δ is closed if and only if $p^{-1}(\Delta)$ is a closed set in $G \times G$, where $p : G \times G \to G/H \times G/H$ is the canonical projection. But $p(g, h) \in \Delta$ if and only if $gH = hH$, that is, if $h^{-1}g \in H$. Hence,

$$p^{-1}(\Delta) = q^{-1}(H) ,$$

where q is the continuous map $q(x, y) = x^{-1}y$. It follows that, if H is closed, then $p^{-1}(\Delta)$ is closed and G/H is Hausdorff. □

The criterion of continuity for maps defined on the quotient allows us to prove that the natural action of G on G/H is continuous.

Proposition 2.27 *The action of G on G/H is continuous with respect to the quotient topology.*

Proof The map $\phi : G \times G/H \to G/H$ that defines the action is part of the following commutative diagram:

$$\begin{array}{ccc} G \times G & \xrightarrow{p} & G \\ \mathrm{id} \downarrow\downarrow \pi & & \downarrow \pi \\ G \times G/H & \xrightarrow{\phi} & G/H \end{array}$$

Let $A \subset G/H$ be an open set. Then, $p^{-1}\pi^{-1}(A)$ is open, whence it follows that $(\mathrm{id} \times \pi)^{-1} \phi^{-1}(A)$ is open in $G \times G$. But this means that $\phi^{-1}(A)$ is open in $G \times G/H$, by the definition of the quotient topology. □

2.7.1 Quotient Groups

A special situation occurs for the quotients considered above when the subgroup H is normal in G. In this case, the quotient G/H has a group structure defined by $(gH)(hH) = (gh)H$, and the canonical projection $\pi : G \to G/H$ is a homomorphism. With the quotient topology, this group becomes a topological group. To see this, it suffices to consider Proposition 2.24 and write the diagram

$$\begin{array}{ccc} G \times G & \xrightarrow{p} & G \\ \pi \downarrow\downarrow \pi & & \downarrow \pi \\ G/H \times G/H & \xrightarrow{\phi} & G/H \end{array}$$

where ϕ denotes the product in G/H. Then, in the same way as for Proposition 2.27, one can show that ϕ is continuous. On the other hand, the continuity of the inverse in G/H arises from the commutativity of the diagram

$$\begin{array}{ccc} G & \xrightarrow{\iota} & G \\ \downarrow \pi & & \downarrow \pi \\ G/H & \xrightarrow{\iota} & G/H \end{array}$$

together with Proposition 2.24.

With respect to the quotient topology, the projection $\pi : G \to G/H$ is a continuous homomorphism and an open map.

2.7.2 Compact and Connected Groups

The two results proved here can be used to verify, with the help of quotient spaces, whether certain topological groups are compact or connected.

The first result concerns compactness. In its proof, the following finite intersection property that characterizes compact spaces will be used: A topological space K is compact if and only if, for a family $\mathcal{F}$ of closed subsets of K, it is true that

$\bigcap_{F\in\mathcal{F}} F \neq \emptyset$ if it satisfies the finite intersection property, that is, if every finite intersection $F_1 \cap \cdots \cap F_k$ of elements of $\mathcal{F}$ is nonempty. In this case it can be assumed, without loss of generality, that $\mathcal{F}$ is complete, that is, it is closed under finite intersection of its elements, since the family of all finite intersections of elements of $\mathcal{F}$ also satisfies the finite intersection property.

Proposition 2.28 *Let G be a topological group and $H \subset G$ a subgroup. If H and G/H are compact, then G is compact.*

Proof Let $\mathcal{F}$ be a complete family of closed sets in G satisfying the finite intersection property. The compactness of H ensures that the projections $\pi(F)$, $F \in \mathcal{F}$, are closed in G/H (see Proposition 2.25). The family $\{\pi(F)\}_{F\in\mathcal{F}}$ also satisfies the finite intersection property, as

$$\pi(F_1 \cap \cdots \cap F_k) \subset \pi(F_1) \cap \cdots \cap \pi(F_k).$$

Now, the hypothesis that G/H is compact ensures the existence of $g \in G$ such that

$$gH \in \bigcap_{F\in\mathcal{F}} \pi(F).$$

This means that every $F \in \mathcal{F}$ intersects the coset gH. As $\mathcal{F}$ is a complete family, it follows that

$$(F_1 \cap gH) \cdots \cap (F_s \cap gH) = (F_1 \cap \cdots \cap F_s) \cap gH \neq \emptyset$$

for every $F_1, \ldots, F_s \in \mathcal{F}$. This means that the family of closed sets $F \cap gH$, $F \in \mathcal{F}$, satisfies the finite intersection hypothesis. Using again the compactness of H (and, therefore, of gH), it follows that

$$\left(\bigcap_{F\in\mathcal{F}} F\right) \cap gH = \bigcap_{F\in\mathcal{F}} (F \cap gH) \neq \emptyset.$$

It follows that $\bigcap_{F\in\mathcal{F}} F \neq \emptyset$, concluding the proof. □

It is clear that G/H is compact if G is compact, since the canonical projection $\pi : G \to G/H$ is continuous and surjective. On the other hand, if H is closed and G is compact, then H is also compact. Hence, the converse of the above theorem is true with the additional hypothesis that H is closed.

Proposition 2.29 *Suppose that H and G/H are connected. Then, G is connected.*

Proof Suppose by contradiction that $A, B \subset G$ are disjoint, nonempty open sets such that $A \cup B = G$. Then, $\pi(A)$ and $\pi(B)$ are nonempty open sets such that $\pi(A) \cup \pi(B) = G/H$. As G/H is connected, $\pi(A) \cap \pi(B) \neq \emptyset$. This means that

there exists a coset gH that intersects both sets A and B. Then, $A \cap gH$ and $B \cap gH$ are disjoint and nonempty open sets. But

$$gH = (A \cup B) \cap gH = (A \cap gH) \cup (B \cap gH),$$

and this contradicts the fact that H is connected, since gH is homeomorphic to H. □

As for the converse of the previous proposition, it is clear that G/H is connected if G is connected. Nevertheless, it may happen that both G and G/H are connected, but H is not connected.

Section 2.9 presents some examples that illustrate the application of the previous propositions to prove that certain topological groups are compact or connected.

2.8 Homeomorphism $G/G_x \to G \cdot x$

An orbit $G \cdot x$ of a continuous action $G \times X \to X$ admits two natural topologies. One of them is the induced topology from X. On the other hand, $G \cdot x$ is in bijection with the quotient G/G_x. By means of this bijection, it is possible to introduce a topology in $G \cdot x$ by stating that a subset $A \subset G \cdot x$ is open if the corresponding set in G/G_x is an open set in the quotient topology.

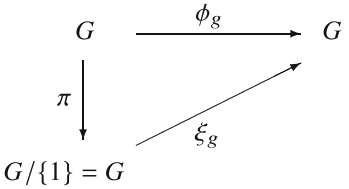
The aim of the following discussion is to compare these topologies, analyzing the homeomorphism property of the map $\xi_x : G/G_x \to G \cdot x$, in the case of a transitive action.

Proposition 2.30 *Let $G \times X \to X$ be a continuous and transitive action of G on X. Fix $x \in X$ and consider the bijection $\xi_x : G/G_x \to X$ given by $\xi_x(gG_x) = gx$. Then, ξ_x is continuous with respect to the quotient topology in G/G_x.*

Proof By Proposition 2.24, it suffices to show that $\xi_x \circ \pi$ is continuous. Now, $\xi_x \circ \pi(g) = \xi_x(gH) = gx$, that is, $\xi_x \circ \pi = \phi_x$, which is continuous if the action is continuous. □

The ideal situation would be to identify, as topological spaces, the space X on which a transitive action takes place with the quotient G/G_x. In general, this is not possible, as the map ξ_x may not be a homeomorphism for not being an open map, as shows the following example.

Example If G is a group, then the map $g \in G \mapsto L_g$ defines a left action of G on itself. This action is clearly transitive and the isotropy subgroup $G_g = \{1\}$ for every $g \in G$. Therefore, for each $g \in G$, there is a diagram



where $\xi_g(h) = hg$. In particular, $\xi_1(h) = h$ is the identity map. Thus, in order to have an example of a continuous action in which ξ_x is not an open map, it suffices to show the existence of a group endowed with two topologies $\mathcal{T}_1$ and $\mathcal{T}_2$, with $\mathcal{T}_2 \subset_{\neq} \mathcal{T}_1$. In this case,

$$\xi_1 = \mathrm{id} : (G, \mathcal{T}_1) \longrightarrow (G, \mathcal{T}_2)$$

is continuous but is not open. If both topologies make G a topological group, then the left action of G on G is continuous.

An example of such a group is provided by the real line $(\mathbb{R}, +)$. Let $\mathcal{T}_1$ be the usual topology. As for $\mathcal{T}_2$, consider an irrational flow on the thorus $\mathbb{T}^2$, that is, the image in $\mathbb{R}^2/\mathbb{Z}^2$ of a straight line $r \subset \mathbb{R}^2$, with irrational slope. This set is a subgroup of $\mathbb{T}^2$ isomorphic to $\mathbb{R}$, but the topology induced on the image is a topology $\mathcal{T}_2$ in $\mathbb{R}$ strictly contained in the usual topology. In both topologies, $\mathbb{R}$ is a topological group, but it is topology $\mathcal{T}_2$ that makes $\mathbb{R}$ a topological subgroup of $\mathbb{T}^2$. □

In what follows, a general result ensuring that ξ_x is an open map, within the context of the category theorem of Baire, is presented. Before doing that, it is convenient to reduce the problem to a single point.

Lemma 2.31 *Suppose that there exists $x_0 \in X$ such that, for every open neighborhood $U \in \mathcal{V}(1)$, the set $U \cdot x_0 = \xi_{x_0}(U)$ contains x_0 in its interior. Then, ξ_x is an open map for every $x \in X$ and hence is a homeomorphism.*

Proof Consider first ξ_{x_0}. In this case, given an open set $V \subset G$, it must be shown that $V \cdot x_0$ is open, that is, if $g \in V$, then gx_0 is an interior point of $V \cdot x_0$. By hypothesis, if $g \in V$, then $U = g^{-1}V \in \mathcal{V}(1)$ is such that $U \cdot x_0$ is a neighborhood of x_0. This implies that $V \cdot x_0 = (gU) \cdot x_0 = g(U \cdot x_0)$ is a neighborhood of gx_0, since g is a homeomorphism. This shows that ξ_{x_0} is an open map.

Now, if $x = hx_0$, then $\xi_x = \xi_{x_0} \circ R_h$. Hence, if ξ_{x_0} is an open map, the same occurs with ξ_x. □

The general result proved in the sequel, about homeomorphism $G/G_x \to X$, is valid when X is a Baire space, that is, a countable union of closed subsets of X with empty interior still has an empty interior. Examples of Baire spaces are the complete metric spaces or the topological spaces which are Hausdorff and locally compact. The following lemma will be used to prove this result.

Lemma 2.32 *Let G be a topological group, $D \subset G$ a dense subset, and $U \in \mathcal{V}(1)$ a neighborhood of identity. Then,*

$$G = \bigcup_{g \in D} gU.$$

Proof Take a symmetric neighborhood $W \subset U$. Then, given $x \in G$, there exists $g \in D$ such that $g \in xW$, that is, $x^{-1}g \in W$. The symmetry of W ensures that $g^{-1}x \in W$, and this means that $x \in gW \subset gU$, concluding the proof. □

Proposition 2.33 *Let $G \times X \to X$ be a continuous and transitive action. Suppose that G is locally compact and separable (that is, admits a countable dense set) and that X is Hausdorff and a Baire space. Then, the maps $\xi_x : G/G_x \to X$ are homeomorphisms.*

Proof Take $x_0 \in X$, $U \in \mathcal{V}(1)$ an open neighborhood and W a compact symmetric neighborhood of 1, such that $W^2 \subset U$. By Lemma 2.31, it suffices to show that $U \cdot x_0$ is a neighborhood of x_0. Let g_n be a dense sequence in G. By the previous lemma, the compact sets $g_n W$ cover G and, therefore, the compact sets $g_n W \cdot x_0$ cover X. However, X is a Baire space, and this ensures that, for some n_0, $g_{n_0} W \cdot x_0$ has a nonempty interior, that is, it contains $g_{n_0} g \cdot x_0$ in its interior for some $g \in W$. As $g_{n_0} g$ is a homeomorphism, it follows that x_0 is an interior point of $g^{-1} g_{n_0}^{-1} \left(g_{n_0} W \cdot x_0\right)$. But

$$g^{-1} g_{n_0}^{-1} \left(g_{n_0} W \cdot x_0\right) = g^{-1} W \cdot x_0 \subset U \cdot x_0,$$

concluding the proof. □

Finally, it must be noted that, in the case of differentiable actions of Lie groups, it will be shown later, with the help of differential calculus, that the maps ξ_x are homeomorphisms (indeed, diffeomorphisms).

2.9 Examples

In what follows, some examples are given of group actions that provide homeomorphisms between quotients and certain concrete spaces.

1. The group $G = \mathrm{Gl}(n, \mathbb{R})$ acts on $\mathbb{R}^n$ in the canonical way: $\phi(g, x) = gx$, $g \in \mathrm{Gl}(n, \mathbb{R})$, $x \in \mathbb{R}^n$. This action is continuous, as ϕ is the restriction of a (second degree) polynomial map $M_n(\mathbb{R}) \times \mathbb{R}^n \to \mathbb{R}^n$. There are exactly two orbits, the origin $\{0\}$ and its complement $\mathbb{R}^n \setminus \{0\}$. It is evident that the origin is an orbit. To see that its complement is also an orbit, take $e_1 = (1, 0, \ldots, 0) \in \mathbb{R}^n \setminus \{0\}$ and $x = (x_1, \ldots, x_n) \neq 0$. Then, there exists a matrix $g \in \mathrm{Gl}(n, \mathbb{R})$ such that $ge_1 = x$. Indeed, it is possible to extend x to a basis $\{x, v_2, \ldots, v_{n-1}\}$ of $\mathbb{R}^n$.

Denote by $\{e_1, \ldots, e_n\}$ the canonical basis of $\mathbb{R}^n$. Then, g defined by $ge_1 = x$ and $ge_i = v_i$, $i = 2, \ldots, n$ is an element of $\mathrm{Gl}(n, \mathbb{R})$ with the desired property.

The isotropy subgroup at 0 is the entire $\mathrm{Gl}(n, \mathbb{R})$. The isotropy subgroup G_{e_1} at e_1 is formed by the matrices of the form (with respect to the canonical basis)

$$\begin{pmatrix} 1 & b \\ 0 & C \end{pmatrix} \tag{2.1}$$

where b is a $1 \times (n-1)$ row matrix and $C \in \mathrm{Gl}(n-1, \mathbb{R})$, and hence is homeomorphic to $\mathrm{Gl}(n-1, \mathbb{R}) \times \mathbb{R}^{n-1}$. The isotropy groups at $x \neq 0$ are conjugate to G_{e_1}. The conditions of Proposition 2.33 are satisfied here. Therefore, the quotient $\mathrm{Gl}(n, \mathbb{R}) / G_{e_1}$ is homeomorphic to the cylinder $\mathbb{R}^n \setminus \{0\}$.

The same considerations apply to subgroup $\mathrm{Gl}^+(n, \mathbb{R}) = \{g \in \mathrm{Gl}(n, \mathbb{R}) : \det g > 0\}$, with two differences. One is, if $n = 1$, $\mathbb{R}_+$ must be used instead of $\mathbb{R}$. Another difference is that the matrix C in (2.1) must have a positive determinant. In this way, the isotropy group G_{e_1} is homeomorphic to $\mathrm{Gl}^+(n-1, \mathbb{R}) \times \mathbb{R}^{n-1}$.

Homeomorphisms $\mathrm{Gl}^+(n, \mathbb{R}) / G_{e_1} \approx \mathbb{R}^n \setminus \{0\}$ and $G_{e_1} \approx \mathrm{Gl}^+(n-1, \mathbb{R}) \times \mathbb{R}^{n-1}$, together with Proposition 2.29, allow us to show that $\mathrm{Gl}^+(n, \mathbb{R})$ is connected. Indeed, $\mathrm{Gl}^+(1, \mathbb{R}) = \mathbb{R}_+$ is connected. Since $\mathbb{R}^2 \setminus \{0\}$ is connected and, in the case $n = 2$, the isotropy group $G_{e_1} \approx \mathrm{Gl}^+(1, \mathbb{R}) \times \mathbb{R}$ is connected, it follows that $\mathrm{Gl}^+(2, \mathbb{R})$. Proceeding by induction, Proposition 2.29 ensures that $\mathrm{Gl}^+(n, \mathbb{R})$ is connected.

As a consequence, $\mathrm{Gl}(n, \mathbb{R})$ has two connected components, which are $\mathrm{Gl}^+(n, \mathbb{R})$ and $\mathrm{Gl}^-(n, \mathbb{R}) = \{g \in \mathrm{Gl}(n, \mathbb{R}) : \det g < 0\}$, which is a coset of $\mathrm{Gl}^+(n, \mathbb{R})$ in $\mathrm{Gl}(n, \mathbb{R})$.

The canonical action of $\mathrm{Gl}(n, \mathbb{R})$ on $\mathbb{R}^n$ induces, by restriction, actions of its subgroups. These actions are all continuous. However, the structure of orbits varies according to the subgroup. Some examples are presented in the following items.

2. Let $\mathrm{O}(n) \subset \mathrm{Gl}(n, \mathbb{R})$. The orbits are the spheres

$$S_r = \{x \in \mathbb{R}^n : |x| = r\} \qquad r \geq 0.$$

(The norm $|\cdot|$ used here is the one arising from the canonical inner product, recalling that this inner product is implicit in the definition of $\mathrm{O}(n)$.) The argument to show that the spheres are the orbits is similar to the one used above, extending non-null vectors to bases, now taking care not to choose orthonormal bases. The isotropy subgroup at e_1 (or at λe_1, $\lambda \neq 0$) is formed by orthogonal matrices of the form (2.1), that is, by matrices of the form

$$\begin{pmatrix} 1 & 0 \\ 0 & C \end{pmatrix}$$

with $C \in \mathrm{O}(n-1)$. This group is isomorphic to $\mathrm{O}(n-1)$. By Proposition 2.33, the quotient $\mathrm{O}(n)/\mathrm{O}(n-1)$ is homeomorphic to the sphere of dimension $n-1$. It follows from Proposition 2.28 that $\mathrm{O}(n)$ is compact, as the spheres are compact and $\mathrm{O}(1)$ reduces to two points.

3. The same arguments of the previous example allow us to show that the orbits of $\mathrm{SO}(n) = \{g \in \mathrm{O}(n) : \det g = 1\}$ are also spheres. In this case, the isotropy group at e_1 is isomorphic to $\mathrm{SO}(n-1)$, allowing us to use Proposition 2.29 to prove, by induction, that $\mathrm{SO}(n)$ is connected, since $\mathrm{SO}(2) \approx S^1$ is connected, as are the spheres S^n.

   It then follows that $\mathrm{O}(n)$ has two connected components: $\mathrm{SO}(n)$ and the coset formed by the elements of $\mathrm{O}(n)$ with determinant -1.
4. The group $\mathrm{Sl}(n,\mathbb{R}) = \{g \in \mathrm{Gl}(n,\mathbb{R}) : \det g = 1\}$ acts transitively on $\mathbb{R}^n \setminus \{0\}$, as can be verified through the argument of construction of bases. Thus, $\mathrm{Sl}(n,\mathbb{R})$ has exactly two orbits in its canonical action on $\mathbb{R}^n$. In this case, the isotropy groups are homeomorphic to $\mathrm{Sl}(n-1,\mathbb{R}) \times \mathbb{R}^{n-1}$. As in the previous cases, an application of Proposition 2.29 allows us to prove, by induction, that $\mathrm{Sl}(n,\mathbb{R})$ is connected.
5. Just as in previous examples, Proposition 2.29 can be applied to show, through their action on $\mathbb{C}^n$, that the groups $\mathrm{Gl}(n,\mathbb{C})$ and $\mathrm{Sl}(n,\mathbb{C})$ are connected. The difference from the real case is that, here, $\mathrm{Gl}(1,\mathbb{C}) \approx \mathbb{C} \setminus \{0\}$ is connected (contrary to $\mathbb{R} \setminus \{0\}$), allowing the beginning of induction.

   In the same way, the groups $\mathrm{U}(n)$ and $\mathrm{SU}(n)$ are compact and connected.
6. Again, let $G = \mathrm{Gl}(n,\mathbb{R})$ and let $X = \mathbb{P}^{n-1}$ be the projective space of the 1-dimensional subspaces of $\mathbb{R}^n$. If $V \in \mathbb{P}^{n-1}$ and $g \in \mathrm{Gl}(n,\mathbb{R})$, then $gV = \{gx : x \in V\}$ is a subspace of $\mathbb{R}^n$ of dimension 1, and therefore $gV \in \mathbb{P}^{n-1}$. The map $V \mapsto gV$ defines an action of $\mathrm{Gl}(n,\mathbb{R})$ on $\mathbb{P}^{n-1}$. This action is continuous with respect to the following quotient topology in $\mathbb{P}^{n-1}$: given $v \in \mathbb{R}^n$, denote by $[v]$ the subspace generated by v. If $v \neq 0$, $[v] \in \mathbb{P}^{n-1}$. Hence, there exists a surjective map $\pi : v \in \mathbb{R}^n \setminus \{0\} \mapsto [v] \in \mathbb{P}^{n-1}$. The open sets of $\mathbb{P}^{n-1}$ are the sets $A \subset \mathbb{P}^{n-1}$ such that $\pi^{-1}(A)$ is open, that is, the topology in $\mathbb{P}^{n-1}$ is the quotient topology by the equivalence relation $v \sim w$ if $v = aw$, $a \neq 0$, in $\mathbb{R}^n \setminus \{0\}$. With this topology, the action of $\mathrm{Gl}(n,\mathbb{R})$ is continuous. This action is transitive and the isotropy subgroup in $[e_1]$ is formed by matrices of the type

$$\begin{pmatrix} a & b \\ 0 & C \end{pmatrix}$$

   with $a \in \mathbb{R}$, b a $1 \times (n-1)$ row matrix and $C \in \mathrm{Gl}(n-1,\mathbb{R})$. The projection π is equivariant with respect to the actions of G on $\mathbb{R}^n \setminus \{0\}$ and $\mathbb{P}^{n-1}$. As in the case of the action on $\mathbb{R}^n$, this action induces actions of all linear groups, that is, of the subgroups of $\mathrm{Gl}(n,\mathbb{R})$. These actions are called **projective actions**.
7. Analogously to projective actions, the group $\mathrm{Gl}(n,\mathbb{R})$ acts on the Grassmannian $\mathrm{Gr}_k(n)$, formed by the subspaces of $\mathbb{R}^n$ of dimension k. The action is given by $(g, V) \mapsto gV$, where gV is the image of subspace V by the linear map g. This action of $\mathrm{Gl}(n,\mathbb{R})$ is also transitive and is continuous with respect to the

following topology in $\mathrm{Gr}_k(n)$: denote by $B_k(n)$ the set of $n \times k$ matrices of rank k, endowed with the topology induced by the topology of the vector space of all $n \times k$ matrices. There exists a surjective map $\pi : B_k(n) \to \mathrm{Gr}_k(n)$ which associates to a matrix $p \in B_k(n)$ the vector space generated by the columns of p. Now, define in $B_k(n)$ the equivalence relation $p \sim q$ if $p = qa$ for some $a \in \mathrm{Gl}(k, \mathbb{R})$. Then, $\mathrm{Gr}_k(n)$ is identified with the set of equivalence classes $B_k(n) / \sim$ and $\pi : B_k(n) \to \mathrm{Gr}_k(n)$, with the canonical projection $B_k(n) \to B_k(n) / \sim$. This defines the quotient topology in $\mathrm{Gr}_k(n)$, whose open sets are the sets $A \subset \mathrm{Gr}_k(n)$ such that $\pi^{-1}(A)$ is open in $B_k(n)$.

Let $V_0 \in \mathrm{Gr}_k(n)$ be the subspace generated by the first k vectors of the canonical basis. Then, the isotropy subgroup in V_0 is formed by matrices of the type

$$\begin{pmatrix} P & Q \\ 0 & R \end{pmatrix}$$

with $P \in \mathrm{Gl}(k, \mathbb{R})$ and $Q \in \mathrm{Gl}(n-k, \mathbb{R})$.

8. Let Z be a vector field on a differentiable manifold M of class C^1 and suppose that Z is complete, that is, the maximal solutions of Z extend to the interval $(-\infty, +\infty)$. Denote by Z_t, $t \in \mathbb{R}$, the flow of Z, that is, $t \mapsto Z_t(x)$ is the trajectory of Z that passes by x at $t = 0$. The flow satisfies the properties $Z_0(x) = x$ and $Z_{t+s}(x) = Z_t(Z_s(x))$. Therefore, $(t, x) \mapsto Z_t(x)$ defines an action of $\mathbb{R}$ on M. The theorems about the dependence of solutions on initial conditions ensure that this action is continuous. The orbits of this action are the trajectories of the vector field. The isotropy subgroup at x is described by: (1) $G_x = \mathbb{R}$ if x is a singularity of the vector field, that is, $Z(x) = 0$; (2) $G_x = \{0\}$ if the trajectory x is not a closed curve; (3) $G_x = \omega\mathbb{Z}$ if the trajectory passing by x is periodic with period ω. By Exercise 23 below, these are the only closed subgroups of $\mathbb{R}$ (compare with the examples at the end of Section 7.3).

2.10 Exercises

1. Let $G \times X \to X$ be a continuous action of a topological group G on a topological space X. Let $A \subset X$ be a G-invariant subset. Show that the restriction $G \times A \to A$ of the action to A is also continuous, with the topology induced on A.
2. Let G be a topological group such that $\{1\}$ is closed. Show that, if $H \subset G$ is an abelian subgroup, then the closure $\overline{H}$ is also abelian.
3. Let $H \subset G$ be a subgroup and denote by $\mathrm{N}(H) = \{g \in G : gHg^{-1} \subset H\}$ its normalizer. Show that $\mathrm{N}(H)$ is closed if H is closed.
4. Let G be a Hausdorff topological group. Show that the centralizer $\{g \in G : \forall x \in M, gx = xg\}$ of the set M is a closed subgroup.

5. Let G be a Hausdorff topological group and let $K_1, K_2 \subset G$ be compact subsets with $K_1 \cap K_2 = \emptyset$. Show that there exist neighborhoods of the identity V and W such that $K_1 V \cap K_2 V = \emptyset$ and $W K_1 \cap W K_2 = \emptyset$. (Hint: $K_1 K_2^{-1}$ and $K_2^{-1} K_1$ are compact.)
6. Let G be a topological group and M a metric space. A map $f : G \to M$ is uniformly continuous if, for every $\varepsilon > 0$, there exists $U \in \mathcal{V}(1)$ such that, if $xy^{-1} \in U$, then $d(f(x), f(y)) < \varepsilon$. Show that f is uniformly continuous if G is compact and $f : G \to M$ is continuous.
7. Let G be a connected, noncompact topological group. Also, let $V \subset G$ be a compact neighborhood of the identity. Verify that, for every $k \geq 1$, V^k is compact. Use this result to prove that, for every $k \geq 1$, V^{k+1} contains V^k properly.
8. A subgroup Γ of a topological group G is **discrete** if there exists a neighborhood V of the identity such that $V \cap \Gamma = \{1\}$. Show that, if Γ is discrete, with neighborhood V, then $gV \cap \Gamma = \{g\}$ for every $g \in \Gamma$. Show also that Γ is closed if G is Hausdorff.
9. Let G be a locally connected topological group. Show that, if $\Gamma \subset G$ is a discrete subgroup, then the projection $\pi : G \to G/\Gamma$ is a covering map (see definition in Section 7.4).
10. Let G be a topological group and $\Gamma \subset G$ a discrete subgroup. Show that, if G is connected and Γ is a normal subgroup, then Γ is contained in the center $Z(G)$ of G. (Hint: for $x \in \Gamma$, consider the map $g \in G \mapsto gxg^{-1} \in \Gamma$.)
11. Let G be a topological group and $H \subset G$ a subgroup. Suppose that $\Gamma \subset G$ is a discrete subgroup. Show that $\Gamma \cap H$ is a discrete subgroup of H.
12. Let G be a group (not necessarily topological) acting on topological space X, such that, for every $g \in G$, the induced map $g : X \to X$ is a homeomorphism. Define an equivalence relation on X by $x \sim y$ if x and y belong to the same G-orbit. Show that the canonical projection $X \to X/\sim$ is an open map in the quotient topology.
13. Let X be a topological space and $x \sim y$ an equivalence relation on X. Show that the space of equivalence classes $X/\sim$, endowed with the quotient topology, is Hausdorff if and only if the relation is a closed subset of $X \times X$.
14. Let G be a compact group and take $x \in G$. Show that the closure $\overline{\{x^n : n \geq 1\}}$ of the set of powers of x is a subgroup.
15. A subsemigroup S of a group is a set closed by the group product: if $x, y \in S$, then $xy \in S$ (not necessarily $x^{-1} \in S$). Show that a closed subsemigroup of a compact group is a group (use the previous exercise).
16. Given a continuous action $G \times X \to X$ of the topological group G on space X, let $F \subset X$ be a closed subset. Show that the semigroup $S_F = \{g \in G : g(F) \subset F\}$ is closed. Conclude that the subgroup $G_F = \{g \in G : g(F) = F\}$ is also closed.
17. Let G be a topological group and $H_1 \subset H_2 \subset G$ subgroups. Define $\pi : G/H_1 \to G/H_2$ by $\pi(gH_1) = gH_2$. Verify that this map is well defined and show that it is continuous and open (with respect to the quotient topologies). Show also that π is equivariant, that is, $g\pi(x) = \pi(gx)$, $x \in G/H_1$.

18. Let G be a topological group and let $H_1 \subset H_2$ be closed subgroups. Show that G/H_1 is compact if G/H_2 and H_2/H_1 are compact. Do the same replacing "compact" by "connected."
19. Let G be a locally connected topological group and $H \subset G$ a locally connected closed subgroup. Show that, if H is not connected, then G/H is not simply connected. (Hint: Consider the identity component H_0 of H.)
20. Let G be a *compact* topological group and let $\phi : G \to \mathbb{R}$ be a continuous homomorphism. Show that $\phi \equiv 0$.
21. Show that, if $G \subset \mathrm{Gl}(n, \mathbb{R})$ is a compact group, then, for every $g \in G$, $\det g = 1$ or -1.
22. Let G be a topological group and suppose that the derived group $[G, G]$ (that is, the subgroup of G generated by the commutators $xyx^{-1}y^{-1}$, $x, y \in G$) is dense. Show that, if H is an abelian topological group and $\phi : G \to H$ is a continuous homomorphism, then ϕ is trivial, that is, $\phi(x) = 1$ for every $x \in G$.
23. Show that the only closed subgroups of $(\mathbb{R}, +)$ are the trivial subgroups $\{0\}$, $\mathbb{R}$ and the subgroups with the form $\mathbb{Z}\omega$, $\omega > 0$.
24. Show that $\mathrm{O}(n)$ is compact and that $\mathrm{Sl}(n, \mathbb{R})$ is not compact.
25. Show that $\mathrm{SO}(n)$ is path-connected without using Proposition 2.29. (Hint: Write the Jordan canonical form of an orthogonal matrix.)
26. Consider the action of $\mathrm{Sl}(n, \mathbb{R})$ on the real projective space $\mathbb{P}^{n-1}$, given by $g[v] = [gv]$, where $[v]$ denotes the subspace generated by $0 \neq v \in \mathbb{R}^n$. Show that this action is transitive. Show that the restriction of this action to $\mathrm{SO}(n)$ is also transitive.
27. Give an example of a noncompact subgroup $G \subset \mathrm{Gl}(n, \mathbb{R})$ whose action on $\mathbb{P}^{n-1}$ is not transitive.
28. In the previous exercises, replace $\mathbb{P}^{n-1}$ by the Grassmannians $\mathrm{Gr}_k(n)$ of the subspaces of dimension k of $\mathbb{R}^n$.
29. Show that a compact group admits a bi-invariant distance.
30. Denote by $\overline{S}(\infty)$ the group of all bijections (permutations) of $\mathbb{N}$. For each $n \in \mathbb{N}$, let $\overline{S}_n(\infty)$ be the subgroup of $\overline{S}_n(\infty)$ formed by the elements that leave fixed each integer of $\{1, \ldots, n\}$. Show that the set $\overline{S}_n(\infty)$, $n \geq 1$, forms a system of neighborhoods of the identity of $\overline{S}(\infty)$, giving rise to a topology on $\overline{S}(\infty)$ that makes it a topological group. Show that this topology is totally disconnected.
31. A function $f : X \to \mathbb{R}$ on a topological space X is upper (respectively lower) semicontinuous if, given $\varepsilon > 0$, there exists an open set $U \ni x_0$ such that $f(U) \subset (-\infty, f(x_0) + \varepsilon)$ (respectively $f(U) \subset (f(x_0) - \varepsilon, +\infty)$). Let G be a topological group and $\phi : G \to (\mathbb{R}, +)$ a homomorphism. Show that ϕ is continuous if it is upper (or lower) semicontinuous at 1. Do the same for a homomorphism $\phi : G \to (\mathbb{R}_\times, \cdot)$.

Chapter 3
Haar Measure



A Haar measure on a topological group G is a measure on the σ-algebra of the Borel sets of G (that is, the σ-algebra generated by its open subsets), which is invariant under translations in the group. A Haar measure may be left or right invariant. In this chapter, the construction of Haar measures on locally compact topological groups is done. It will also be proved the uniqueness, up to multiplication by a positive constant, of the Haar measure.

Reading this chapter requires some previous knowledge of measure theory.

3.1 Introduction

Let $(X, \mathcal{F}, \mu)$ be a measure space, where $\mathcal{F}$ is a σ-algebra of subsets of X (σ-algebra of measurable sets) and μ is a σ-finite measure on $\mathcal{F}$. Given a measurable map $g : X \to X$ with respect to $\mathcal{F}$ (that is, $g^{-1}(A) \in \mathcal{F}$ if $A \in \mathcal{F}$), a new measure $g_*\mu$ on $\mathcal{F}$ is defined by

$$g_*\mu(A) = \mu\left(g^{-1}A\right).$$

Measure μ is invariant by g if $g_*\mu = \mu$, and this means that, for every measurable set $A \in \mathcal{F}$, $\mu\left(g^{-1}A\right) = \mu(A)$. In terms of integrals, the translated measure $g_*\mu$ satisfies the equality

$$\int f(x)\,(g_*\mu)\,(dx) = \int f \circ g(x)\,\mu(dx)$$

for every function $f : X \to \mathbb{R}$ integrable with respect to μ. This equality serves as a definition of $g_*\mu$ since, if $f = \chi_A$ is the characteristic function of set A ($\chi_A(x) = 1$

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if $x \in A$ and $\chi_A(x) = 0$ if $x \notin A$), then $g_*\mu(A) = \int \chi_A(x)\,(g_*\mu)(dx)$ and $\mu\left(g^{-1}A\right) = \int \chi_A \circ g(x)\,\mu(dx)$.

Moving on to Haar measures, let G be a topological group and denote by $\mathcal{F}$ the σ-algebra of the Borel sets of G (σ-algebra generated by the open subsets of G). The translations of G are measurable maps, as they are continuous. A **left invariant** Haar measure μ on G is a measure on $\mathcal{F}$ (Borel measure) such that, for every left translation L_g, it holds $\left(L_g\right)_*\mu = \mu$, that is, for a Borel set A, $\mu(gA) = \mu(A)$. The **right invariant** Haar measures are defined in an analogous way by the property $\mu(Ag) = \mu(A)$.

It should be noted that if μ is a Haar measure and $a > 0$, then measure $a\mu$ is also a Haar measure, as $\left(L_g\right)_*(a\mu) = a\left(\left(L_g\right)_*\mu\right)$ (or $\left(R_g\right)_*(a\mu) = a\left(\left(R_g\right)_*\mu\right)$ in the right invariant case).

Theorem 3.1 *Let G be a Hausdorff, locally compact topological group. Then G admits a unique (up to multiplication by a constant $a > 0$) left invariant (respectively right invariant) Haar measure $\mu \neq 0$. This measure satisfies the following properties:*

1. *If K is compact, then $\mu(K) < \infty$.*
2. *If $U \neq \emptyset$ is open, then $\mu(U) > 0$.*
3. *μ is **regular**, that is, if A is a Borel set, then*

   a. $\mu(A) = \inf \mu(U)$, with $A \subset U$ and U open (outer regularity):
   b. $\mu(A) = \sup \mu(K)$, with $K \subset A$ compact (inner regularity).

This theorem on Haar measures will be proved in subsequent sections of this chapter. In the proof, it may be considered only left invariant measures because, if μ is a left invariant Haar measure on G, then $\hat{\mu} = \iota_*\mu$ is right invariant if ι is the inverse of G. Indeed, if A is a Borel set, then

$$\hat{\mu}(Ag) = \mu\left(g^{-1}A^{-1}\right) = \mu\left(A^{-1}\right) = \hat{\mu}(A).$$

Conversely, if μ is right invariant, then $\hat{\mu}$ is left invariant. This means that left invariant and right invariant Haar measures are obtained from each other through the inverse map.

In general, left and right invariant measures do not coincide. A group G is called **unimodular** if the left and right invariant Haar measures coincide, that is if they are bi-invariant. For instance, abelian groups are unimodular, since left and right translations are equal.

Examples

1. The guiding example to Haar measures is the Lebesgue measure λ on $\mathbb{R}$ (or more generally on $\mathbb{R}^n$), which is invariant by left or right translations, as the group is abelian. Therefore, the Lebesgue measure is a Haar measure normalized by $\lambda([0, 1]^n) = 1$.

2. A group G endowed with the discrete topology is locally compact. A Haar measure μ is given by $\mu\{x\} = 1$ for every $x \in G$. In the case in which G is finite, it is possible to normalize the Haar measure by putting $\mu\{x\} = 1/|G|$, so that $\mu(G) = 1$.
3. Lie groups are locally compact and Hausdorff. Hence, they admit Haar measures. As will be seen later, in Chapter 5, the construction of Haar measures on a Lie group does not require the general theorem above, as in this case the measure can be defined by means of an invariant volume form. □

Before entering the proof of Theorem 3.1, it is worth observing that, on a Hausdorff compact group G, any Haar measure on G satisfies $\mu(G) < \infty$, since Haar measures are finite on compact sets. On the other hand, the next proposition shows that a Hausdorff, locally compact group G is compact if and only if the Haar measure of G is finite.

Proposition 3.2 *Let G be a Hausdorff, locally compact topological group with Haar measure μ. Then, $\mu(G) = \infty$ if G is not compact.*

Proof Let K be a compact and symmetric neighborhood of identity. The union

$$H = \bigcup_{n \geq 1} K^n$$

is a subgroup of G satisfying $\mu(H) > 0$, since $K \subset H$ and $\mu(K) > 0$. If H is compact, then there exist infinite cosets gH, $g \in G$, as G is not compact. As $\mu(gH) = \mu(H) > 0$ and two distinct cosets are disjoint, it follows that $\mu(G) > \infty$.

On the other hand, if H is not compact, then, for every $n \geq 1$, the inclusion $K^n \subset K^{n+1}$ is proper, for otherwise there would exist an n_0 such that $K^{n_0+k} = K^{n_0}$, $k \geq 1$, and $H = \bigcup K^n = K^{n_0}$, contradicting the hypothesis that H is not compact.

Now, if $g \in K^{n+1} \setminus K^n$, then $gK \cap K^{n-1} = \emptyset$. Indeed, if there is a $z \in gK \cap K^{n-1}$, then $z = gk = k_1 \cdots k_{n-1}$, with $k, k_i \in K$. But this implies that $g = k_1 \cdots k_{n-1} k^{-1} \in K^n$. On the other hand, $gK \subset K^{n+2}$, as $g \in K^{n+1}$. Choose a sequence $g_n \in K^{10n+1} \setminus K^{10n}$, $n \geq 1$. Then it holds $g_n K \cap g_m K = \emptyset$ if $n \neq m$. As $\mu(g_n K) = \mu(K) > 0$, σ-additivity ensures that

$$\mu\left(\bigcup_{n \geq 1} g_n K\right) = \sum_{n \geq 1} \mu(K) = \infty,$$

concluding the proof that $\mu(G) = \infty$. □

3.2 Construction of Haar Measure

It is constructed here a left invariant Haar measure on a locally compact group G. A right invariant measure is obtained by the application of the group inverse to the left invariant measure.

Denote by $\mathcal{K}$ the family of compact subsets of G and, as before, let $\mathcal{V}(1)$ be the set of open neighborhoods of 1.

The method for constructing a Haar measure follows what is called the Caratheodory procedure. It consists of the following steps:

1. Construct a pre-measure on $\mathcal{K}$, that is, a map $\lambda : \mathcal{K} \to \mathbb{R}_+$ that is monotonic, subadditive, and additive on disjoint compact sets (see Proposition 3.8 below).
2. Define, starting from λ, a function λ_* on the open subsets of G. This function is monotonic, σ-subadditive, and σ-additive (see Proposition 3.12 below).
3. Extend λ_* to an **outer measure** μ_e defined on all subsets of G. This outer measure is monotonic and σ-subadditive (see Proposition 3.13 below).
4. Measure theory then ensures that μ_e is σ-additive on the measurable sets of μ_e (see definition below).
5. Finally, prove that Borel sets are measurable (see Proposition 3.16 below).

In this procedure, the only step specific to topological groups is the first one. The remaining steps apply to general Hausdorff locally compact spaces.

To carry out the construction, it is fixed once and for all a compact set $K_0 \subset G$ with nonempty interior. The existence of K_0 stems from the hypothesis that G is locally compact. This compact set serves to normalize the Haar measure, just as the cube $[0, 1]^n$ normalizes the Lebesgue measure on $\mathbb{R}^n$.

The definition of the pre-measure on $\mathcal{K}$ requires the following concept:

- Let $\emptyset \neq K \subset G$ compact and $\emptyset \neq V \subset G$ open. The open sets xV, $x \in K$, cover K and, therefore, there exist finite subcovers.[1] The **index** of K with respect to V, denoted by $(K : V)$, is the least n such that there exists a finite set $\{x_1, \ldots, x_n\} \subset K$ with

$$K \subset x_1 V \cup \cdots \cup x_n V.$$

  Obviously, $(K : V) \geq 1$ as $K \neq \emptyset$.

  For a given open set $V \in \mathcal{V}(1)$, define $\lambda_V : \mathcal{K} \to \mathbb{R}_+$ by

$$\lambda_V(K) = \frac{(K : V)}{(K_0 : V)}.$$

Lemma 3.3 *The map $\lambda_V : \mathcal{K} \to \mathbb{R}_+$, $V \in \mathcal{V}(1)$, satisfies the following properties:*

[1] The choice of translations xV leads to a left invariant Haar measure. The same argument applies to translations Vx, yielding right invariance.

1. $\lambda_V(K_1) \leq \lambda_V(K_2)$ *if* $K_1 \subset K_2$.
2. $\lambda_V(K_1 \cup K_2) \leq \lambda_V(K_1) + \lambda_V(K_2)$.
3. $\lambda_V(K_1 \cup K_2) = \lambda_V(K_1) + \lambda_V(K_2)$ *se* $K_1 V^{-1} \cap K_2 V^{-1} = \emptyset$.

Proof These properties follow almost immediately from the definition of index:

1. If $\{x_1, \ldots, x_{(K_2:V)}\}$ is such that $K_2 \subset x_1 V \cup \cdots \cup x_{(K_2:V)} V$, then these open sets also cover $K_1 \subset K_2$. From this it follows that $(K_1 : V) \leq (K_2 : V)$ and, therefore, $\lambda_V(K_1) \leq \lambda_V(K_2)$.
2. Let $\{x_1, \ldots, x_{(K_1:V)}\}$ and $\{y_1, \ldots, y_{(K_2:V)}\}$ be such that $K_1 \subset x_1 V \cup \cdots \cup x_{(K_1:V)} V$ and $K_2 \subset y_1 V \cup \cdots \cup y_{(K_2:V)} V$. Then, the union of these $(K_1 : V) + (K_2 : V)$ open sets covers $K_1 \cup K_2$. Hence, $(K_1 \cup K_2 : V) \leq (K_1 : V) + (K_2 : V)$ and, therefore, $\lambda_V(K_1 \cup K_2) \leq \lambda_V(K_1) + \lambda_V(K_2)$.
3. Take $F = \{x_1, \ldots, x_{(K_1 \cup K_2:V)}\} \subset K_1 \cup K_2$ such that $K_1 \cup K_2 \subset x_1 V \cup \cdots \cup x_{(K_1 \cup K_2:V)} V$, and take $x \in F \cap K_1$. Then, $xV \cap K_2 = \emptyset$, because if $z \in xV \cap K_2$, then $z \in K_2$ and $z = xv$, with $v \in V$. This means that $x = zv^{-1} \in K_2 V^{-1} \cap K_1$, contradicting the hypothesis. In the same way, if $y \in F \cap K_2$, then $yV \cap K_1 = \emptyset$. Hence,

$$K_1 \subset \bigcup_{x \in F \cap K_1} xV \qquad \text{and} \qquad K_2 \subset \bigcup_{y \in F \cap K_2} yV.$$

   It follows that if n (respectively m) is the number of elements in $F \cap K_1$ (respectively in $F \cap K_2$), then $n \geq (K_1 : V)$ and $m \geq (K_2 : V)$. As $n + m = (K_1 \cup K_2 : V)$, it follows that $(K_1 \cup K_2 : V) \geq (K_1 : V) + (K_2 : V)$. Hence, $\lambda_V(K_1 \cup K_2) \geq \lambda_V(K_1) + \lambda_V(K_2)$ and the equality follow from the previous item. □

The properties stated in the lemma above are of a measure theoretic nature. The next lemma, also of immediate proof, considers the invariance of λ_V that is related to the group structure.

Lemma 3.4 *If* $g \in G$, *then* $\lambda_V(gK) = \lambda_V(K)$.[2]

Proof As in the proof of the previous lemma, everything reduces to a property of the index. Let $\{x_1, \ldots, x_n\} \subset K$, such that $K \subset x_1 V \cup \cdots \cup x_n V$. Then, $gK \subset gx_1 V \cup \cdots \cup gx_n V$ and $\{gx_1, \ldots, gx_n\} \subset gK$. Taking specifically $n = (K : V)$, it follows that the least number of V translates that cover gK is $\leq (K : V)$. That is, $(gK : V) \leq (K : V)$. Replacing K by gK and g by g^{-1}, the converse inequality is obtained, and it follows that $(gK : V) = (K : V)$. From the definition of λ_V, this means that $\lambda_V(gK) = \lambda_V(K)$. □

In order to define λ independently of V, the following additional property of indices is used.

[2]The left invariance of λ_V stated in this lemma is a consequence of the choice of covers of type xV. The choice of open sets Vx would lead to right invariance.

Lemma 3.5 $(K : V) \leq \left(K : K_0^\circ\right)(K_0 : V)$, *where* K_0° *is the interior of* K_0.

Proof Take $\{x_1, \ldots, x_n\} \subset K$ and $\{y_1, \ldots, y_m\} \subset K_0$, such that $K \subset x_1 K_0^\circ \cup \cdots \cup x_n K_0^\circ$ and $K_0 \subset y_1 V \cup \cdots \cup y_m V$, where $n = \left(K : K_0^\circ\right)$ and $m = (K_0 : V)$. Then

$$K \subset \bigcup_{i=1}^{n} x_i K_0^\circ \subset \bigcup_{i=1}^{n} \bigcup_{j=1}^{m} x_i y_j V.$$

Hence, K is covered by $x_i y_j V$, $i = 1, \ldots, n$, $j = 1 \ldots, n$. It follows that $(K : V) \leq mn = \left(K : K_0^\circ\right)(K_0 : V)$. □

The inequalities of the following corollary are immediate consequences of the above lemma and the definition of λ_V.

Corollary 3.6 $0 \leq \lambda_V(K) \leq \left(K : K_0^\circ\right)$.

The inequalities in this corollary show that, for any open set V, the map $\lambda_V : \mathcal{K} \to \mathbb{R}_+$ can be seen as an element of the Cartesian product

$$\mathcal{P} = \prod_{K \in \mathcal{K}} [0, \left(K : K_0^\circ\right)]$$

of the compact intervals $[0, \left(K : K_0^\circ\right)] \subset \mathbb{R}$, which are independent of V. This product is also compact.

Viewing a map λ_V as an element of $\mathcal{P}$, define for each open set $V \neq \emptyset$ the set

$$M(V) = \{\lambda_U \in \mathcal{P} : U \in \mathcal{V}(1),\ U \subset V\}.$$

Denote by $C(V)$ the closure of $M(V)$ in $\mathcal{P}$.

Lemma 3.7 $\bigcap_{V \in \mathcal{V}(1)} C(V) \neq \emptyset$.

Proof Since $\mathcal{P}$ is compact, it suffices to verify that the family of closed sets $C(V) \subset \mathcal{P}$ satisfies the property of finite intersection. Take $V_1, \ldots, V_k \in \mathcal{V}(1)$ and write $V = V_1 \cap \cdots \cap V_k$. By definition, $\lambda_V \in M(V_i)$, $i = 1, \ldots, k$. So,

$$\lambda_V \in C(V_1) \cap \cdots \cap C(V_k),$$

showing that this intersection is nonempty. □

It is now possible to define the desired pre-measure λ on compact sets.

Definition of Pre-measure λ on $\mathcal{K}$ Choose any element $\lambda \in \bigcap_{V \in \mathcal{V}(1)} C(V)$, which is interpreted as a map $\lambda : \mathcal{K} \to \mathbb{R}$.

The following propositions show that $\lambda \in \bigcap_{V \in \mathcal{V}(1)} C(V)$ is in fact an invariant pre-measure on $\mathcal{K}$. To prove that, the properties stated in Lemma 3.3 for the maps

λ_V will be extended to λ by continuity. Continuity is used in the following way: Take a compact set $K \in \mathcal{K}$ and denote by $p_K : \mathcal{P} \to [0, \left(K : K_0^\circ\right)]$ the projection of the Cartesian product $\mathcal{P} = \prod_{K \in \mathcal{K}} [0, \left(K : K_0^\circ\right)]$ on its K-component. This projection is continuous by the definition of the product topology. Moreover, if $\xi \in \mathcal{P}$ is viewed as a map $\xi : \mathcal{K} \to \mathbb{R}$, then $\xi(K) = p_K(\xi)$.

Proposition 3.8 *The map $\lambda : \mathcal{K} \to \mathbb{R}_+$ ($\lambda \in \mathcal{P}$) is a finitely additive pre-measure, in the sense that $\lambda \geq 0$, $\lambda(\emptyset) = 0$ and satisfies the following properties:*

1. $\lambda(K_1) \leq \lambda(K_2)$ *if* $K_1 \subset K_2$.
2. $\lambda(K_1 \cup K_2) \leq \lambda(K_1) + \lambda(K_2)$.
3. $\lambda(K_1 \cup K_2) = \lambda(K_1) + \lambda(K_2)$ *if* $K_1 \cap K_2 = \emptyset$.
4. $\lambda(K_1 \cup \cdots \cup K_n) \leq \sum_{i=1}^n \lambda(K_i)$.

Moreover, $\lambda(K) \leq \left(K : K_0^\circ\right)$.

Proof First, $\lambda \geq 0$. Indeed, take a compact set K and $U \in \mathcal{V}(1)$. Then, by definition, $\lambda_U(K) \geq 0$. Looking at λ_U as an element of the Cartesian product $\mathcal{P}$, the equality $\lambda_U(K) = p_K(\lambda_U)$ is immediate. So, the projection p_K is ≥ 0 in any set $M(V)$. By continuity, $p_K \geq 0$ in $C(V)$. Since $\lambda \in C(V)$ for every $V \in \mathcal{V}(1)$, it follows that $\lambda(K) = p_K(\lambda) \geq 0$. The proof that $\lambda(\emptyset) = 0$ is similar.

The same continuity argument works in the proof of the properties stated:

1. Let $V \in \mathcal{V}(1)$. By definition, an element of $M(V)$ has the form λ_U, with $U \subset V$. By Lemma 3.3, the inequality $\lambda_U(K_1) \leq \lambda_U(K_2)$ is true if $K_1 \subset K_2$; in terms of projections p_K, this means that $p_{K_1}(\lambda_U) \leq p_{K_2}(\lambda_U)$. Hence, $p_{K_1} \leq p_{K_2}$ in $M(V)$. By continuity, it follows that $p_{K_1}(\xi) \leq p_{K_2}(\xi)$ for every ξ in the closure $C(V)$ of $M(V)$. But λ belongs to $C(V)$ for every V, whence it follows that $p_{K_1}(\lambda) \leq p_{K_2}(\lambda)$, that is, $\lambda(K_1) \leq \lambda(K_2)$.
2. The proof is similar to item (1), now taking the projections $p_{K_1 \cup K_2}$, p_{K_1}, and p_{K_2}.
3. Since G is Hausdorff, there exists $V \in \mathcal{V}(1)$, such that $K_1 V^{-1} \cap K_2 V^{-1} = \emptyset$ (see Chapter 2, Exercise 5).[3] The same property holds for every open set U, with $1 \in U \subset V$, that is, $K_1 U^{-1} \cap K_2 U^{-1} = \emptyset$. Hence, by Lemma 3.3 (3), the equality $\lambda_U(K_1 \cup K_2) = \lambda_U(K_1) + \lambda_U(K_2)$ holds for every $\lambda_U \in M(V)$. By the continuity of projections $p_{K_1 \cup K_2}$, p_{K_1}, and p_{K_2}, it follows that $\lambda(K_1 \cup K_2) = \lambda(K_1) + \lambda(K_2)$, as $\lambda \in C(V)$.
4. Finite subadditivity follows from property (2) by induction.

Finally, the last statement follows from Corollary 3.6 by the same continuity argument. □

Parallel to the properties mentioned in the proposition above, related to measure theory, the invariance of λ also holds.

[3] In this argument, the hypothesis that G is Hausdorff is essential to separate the compact sets K_1 and K_2.

Proposition 3.9 *If $g \in G$, then $\lambda(gK) = \lambda(K)$ for every compact K.*

Proof The proof follows from Lemma 3.4 and the continuity of projections p_K, $K \in \mathcal{K}$. Fix $g \in G$ and $K \in \mathcal{K}$. By Lemma 3.4, $\lambda_U(gK) = \lambda_U(K)$ for every $\lambda_U \in M(V)$, that is, $U \subset V$. This means that $p_{gK}(\lambda_U) = p_K(\lambda_U)$ for every $U \subset V$. By the continuity of projection p_K, it follows that $p_{gK}(\xi) = p_K(\xi)$ for every $\xi \in C(V)$. As $\lambda \in C(V)$, it follows that $p_{gK}(\lambda) = p_K(\lambda)$, that is, $\lambda(gK) = \lambda(K)$, concluding the proof. □

The next statement concludes the discussion about the pre-measure λ, showing that the compact set K_0, chosen at the beginning, normalizes λ.

Proposition 3.10 $\lambda(K_0) = 1$.

Proof Indeed, if $U \in \mathcal{V}(1)$, then $\lambda_U(K_0) = 1$ by definition, which means that $p_{K_0}(\lambda_U) = 1$. By continuity, $p_{K_0}(\xi) = 1$ for every $\xi \in C(V)$ and so $p_{K_0}(\lambda) = 1$, that is, $\lambda(K_0) = 1$. □

Once the pre-measure λ on compact sets is defined, the construction of the Haar measure follows a procedure valid for locally compact spaces in general. First define a map λ_* on open sets, by stating that, for an open set U,

$$\lambda_*(U) = \sup\{\lambda(K) : K \in \mathcal{K},\ K \subset U\} \in [0, +\infty).$$

Next, a map μ_e is defined on arbitrary sets $A \subset G$ by

$$\mu_e(A) = \inf\{\lambda_*(U) : A \subset U \subset G,\ U \text{ open}\} \in [0, +\infty).$$

Just as λ, these maps are left invariant. Indeed, if $g \in G$ and U is open, then $K \subset gU$ if and only if $g^{-1}K \subset U$ and K is compact if and only if $g^{-1}K$ is compact. Thus,

$$\begin{aligned}\lambda_*(gU) &= \sup\{\lambda(gK) : K \in \mathcal{K},\ K \subset U\} \\ &= \sup\{\lambda(K) : K \in \mathcal{K},\ K \subset U\} \\ &= \lambda_*(U).\end{aligned}$$

Analogously, $\mu_e(gA)$ is the infimum of $\lambda_*(gU)$, with $A \subset U$, and so $\mu_e(gA) = \mu(A)$.

The Haar measure μ will be given by the restriction of μ_e to Borel sets. The theorem to be proved states that this restriction is in fact a σ-additive measure; this will be done in what follows. The arguments for this involve only measure theory and are not specific to topological groups.

The first step is the proof of the σ-subadditivity and the additivity of λ_*. To show that, the following topological lemma is needed.

Lemma 3.11 *Let X be a Hausdorff topological space, $U, V \subset X$ open and $K \subset U \cup V$ compact. Then, there exist compact sets $K_1 \subset U$ and $K_2 \subset V$, such that $K = K_1 \cup K_2$.*[4]

Proof The sets $L_1 = K \setminus U$ and $L_2 = K \setminus V$ are compact. From the inclusion $K \subset U \cup V$, it follows that $L_1 \subset V$, $L_2 \subset U$ and $L_1 \cap L_2 = \emptyset$. The hypothesis that X is Hausdorff ensures the existence of open sets V_1 and V_2 with $V_1 \cap V_2 = \emptyset$, $L_1 \subset V_1$ and $L_2 \subset V_2$.

Define the compacts $K_1 = K \setminus V_1$ and $K_2 = K \setminus V_2$. Then, $K_1 \cup K_2 = K \setminus (V_1 \cap V_2) = K$, as $V_1 \cap V_2 = \emptyset$. Moreover, $K_1 = K \cap V_1^c \subset K \cap L_1^c$ and since $L_1^c = K^c \cup U$, it follows that

$$K_1 \subset K \cap \left(K^c \cup U\right) = \emptyset \cup (K \cap U) \subset U.$$

In the same way, $K_2 \subset V$, concluding the proof. □

Proposition 3.12 *λ_* is a σ-additive pre-measure*[5] *in the sense that $\lambda_* \geq 0$, $\lambda_*(\emptyset) = 0$ and satisfies the following properties:*

1. *λ_* is monotonic, that is, $\lambda_*(U_1) \leq \lambda_*(U_2)$ if $U_1 \subset U_2$.*
2. *λ_* is σ-subadditive, that is, $\lambda_*\left(\bigcup_{n\geq 1} U_n\right) \leq \sum_{n\geq 1} \lambda_*(U_n)$.*
3. *λ_* is σ-additive, that is, $\lambda_*\left(\bigcup_{n\geq 1} U_n\right) = \sum_{n\geq 1} \lambda_*(U_n)$ if the open sets are pairwise disjoint.*

Proof It follows directly from the definition that $\lambda_* \geq 0$ and that $\lambda_*(\emptyset) = 0$. On the other hand, if U_1 and U_2 are open sets with $U_1 \subset U_2$ and if $K \subset U_1$ is compact, then $K \subset U_2$. Therefore, $\sup_{K\subset U_1} \lambda(K) \leq \sup_{K\subset U_2} \lambda(K)$, that is, $\lambda_*(U_1) \leq \lambda_*(U_2)$.

The proof of σ-subadditivity and additivity of items (2) and (3), is done first for finite unions and sums.

Take open sets U and V. If K is compact with $K \subset U \cup V$, then, by the above lemma, there exist compact sets $K_1 \subset U$ and $K_2 \subset V$ such that $K = K_1 \cup K_2$. By definition, $\lambda_*(U) \geq \lambda(K_1)$ and $\lambda_*(V) \geq \lambda(K_2)$, and therefore

$$\lambda_*(U) + \lambda_*(V) \geq \lambda(K_1) + \lambda(K_2) \geq \lambda(K_1 \cup K_2) = \lambda(K)$$

by the additivity of λ. It follows that $\sup_{K\subset U\cup V} \lambda(K) \leq \lambda_*(U) + \lambda_*(V)$, that is, $\lambda_*(U \cup V) \leq \lambda_*(U) + \lambda_*(V)$.

Suppose, moreover, that $U \cap V = \emptyset$ and take compacts $C_1 \subset U$ and $C_2 \subset V$. Then, $C_1 \cap C_2 = \emptyset$ and

[4] In this lemma, there appears again the need to work with Hausdorff spaces.

[5] The use of prefix "pre" is due to the fact that the set of open sets is not a σ-algebra.

$$\lambda(C_1) + \lambda(C_2) = \lambda(C_1 \cup C_2) \leq \lambda_*(U \cup V).$$

Taking the supremum it follows $\lambda_*(U) + \lambda_*(V) \leq \lambda_*(U \cup V)$, which, together with the previous inequality provides $\lambda_*(U \cup V) = \lambda_*(U) + \lambda_*(V)$.

By induction,

$$\lambda_*\left(\bigcup_{i=1}^{n} U_i\right) \leq \sum_{i=1}^{n} \lambda_*(U_i),$$

where the equality holds if the open sets U_i are pairwise disjoint.

Now, take a sequence U_n of open sets and a compact set $K \subset \bigcup_{n\geq 1} U_n$. Then, there exists n such that $K \subset \bigcup_{i=1}^{n} U_i$. By subadditivity,

$$\lambda(K) \leq \lambda_*\left(\bigcup_{i=1}^{n} U_i\right) \leq \sum_{i=1}^{n} \lambda_*(U_i) \leq \sum_{i\geq 1} \lambda_*(U_i).$$

Taking the supremum, it follows that $\lambda_*\left(\bigcup_{n\geq 1} U_n\right) \leq \sum_{n\geq 1} \lambda_*(U_n)$.

If the open sets in the sequence are pairwise disjoint, then, for every integer m,

$$\lambda_*\left(\bigcup_{n\geq 1} U_n\right) \geq \lambda_*\left(\bigcup_{i=n}^{m} U_n\right) = \sum_{n=1}^{m} \lambda_*(U_n).$$

And since m is arbitrary,

$$\sum_{n\geq 1} \lambda_*(U_n) \leq \lambda_*\left(\bigcup_{n\geq 1} U_n\right),$$

showing the σ-additivity of λ_*. □

It is now possible to show that map μ_e satisfies similar properties.

Proposition 3.13 *μ_e is an outer measure in the sense that $\mu_e \geq 0$, $\mu_e(\emptyset) = 0$ and satisfies the following properties:*

1. $\mu_e(A_1) \leq \mu_e(A_2)$ *if* $A_1 \subset A_2$.
2. $\mu_e\left(\bigcup_{n\geq 1} A_n\right) \leq \sum_{n\geq 1} \mu_e(A_n)$.

Proof It follows directly from its definition that $\mu_e \geq 0$ and $\mu_e(\emptyset) = 0$. Monotonicity, too, is almost immediate: If $A_2 \subset U$ with U open, then $A_1 \subset U$, and

hence $\mu_e(A_1) \leq \lambda_*(U)$. Taking the infimum with respect to $U \supset A_2$, it follows that $\mu_e(A_1) \leq \mu_e(A_2)$.

The σ-subadditivity is immediate if $\mu_e(A_n) = \infty$ for some n. Suppose, then, that for every $n \geq 1$, $\mu_e(A_n) < \infty$. In this case, given $\varepsilon > 0$ and $n \geq 1$, there exists an open set $U_n \supset A_n$, such that

$$\lambda_*(U_n) < \mu_e(A_n) + \frac{\varepsilon}{2^n}.$$

Then, $\bigcup_{n\geq 1} A_n \subset \bigcup_{n\geq 1} U_n$ and so, by the σ-subadditivity of λ_*, it holds

$$\begin{aligned}\mu_e\left(\bigcup_{n\geq 1} A_n\right) &\leq \lambda_*\left(\bigcup_{n\geq 1} A_n\right) \leq \sum_{n\geq 1} \lambda_*(U_n) \\ &< \sum_{n\geq 1} \mu_e(A_n) + \sum_{n\geq 1} \frac{\varepsilon}{2^n} \\ &= \sum_{n\geq 1} \mu_e(A_n) + \varepsilon.\end{aligned}$$

Since $\varepsilon > 0$ is arbitrary, the σ-subadditivity $\mu_e\left(\bigcup_{n\geq 1} A_n\right) \leq \sum_{n\geq 1} \mu_e(A_n)$ holds. □

The next proposition establishes relations between maps λ, λ_*, and μ_e.

Proposition 3.14 *If $U \subset G$ is open, then $\mu_e(U) = \lambda_*(U)$. On the other hand, if K is compact, then $\mu_e(K^\circ) \leq \lambda(K) \leq \mu_e(K)$.*

Proof If V is open, with $U \subset V$, then $\lambda_*(U) \leq \lambda_*(V)$, since λ_* is monotonic. Hence, $\mu_e(U) = \inf_{V\supset U} \lambda_*(V) = \lambda_*(U)$.

If K is compact and V is open, with $K \subset V$, then $\lambda(K) \leq \lambda_*(V)$ by the definition of λ_*. Therefore,

$$\lambda(K) \leq \inf_{V\supset K} \lambda_*(V) = \mu_e(K).$$

On the other hand, if $L \subset K^\circ$ is compact, then $L \subset K$ and hence $\lambda(L) \leq \lambda(K)$ by the monotonicity of λ. Hence,

$$\mu_e(K^\circ) = \sup_{L\subset K^\circ} \lambda(L) \leq \lambda(K).$$ □

Corollary 3.15 *If K is compact, then $\mu_e(K) < \infty$.*

Proof Indeed, let K_0 be the basic compact set with nonempty interior. Then, $\lambda(K_0) = 1$ by Proposition 3.10. Take $\{x_1, \ldots, x_n\} \subset K$, such that $K \subset x_1 K_0^\circ \cup \cdots \cup x_n K_0^\circ$. Then,

$$\mu_e(K) \leq \sum_{i=1}^{n} \mu_e\left(x_i K_0^\circ\right) \leq \sum_{i=1}^{n} \lambda(x_i K_0) \leq n. \qquad \square$$

As already mentioned, the Haar measure μ is given by the restriction of μ_e to the σ-algebra of Borel sets. For this restriction to be really a measure, it is still needed to ensure that it is σ-additive, complementing the subadditivity of Proposition 3.13. In order to do that, the following general facts of measure theory will be used.

A subset A is μ_e-**measurable** if, for every subset X,

$$\mu_e(X) = \mu_e(X \cap A) + \mu_e\left(X \cap A^c\right),$$

which is equivalent to $\mu_e(X) \geq \mu_e(X \cap A) + \mu_e(X \cap A^c)$, since μ_e is subadditive. Denote by $\mathcal{M}$ the family of sets μ_e-measurable. The following results, which will not be proved here,[6] hold:

1. $\mathcal{M}$ is a σ-algebra ($\emptyset \in \mathcal{M}$, if $A \in \mathcal{M}$, then $A^c \in \mathcal{M}$ and $\mathcal{M}$ is closed by enumerable unions).
2. μ_e is σ-additive on $\mathcal{M}$, that is, $\mu_e\left(\bigcup_{n\geq 1} A_n\right) = \sum_{n\geq 1} \mu_e(A_n)$ if the sets $A_n \in \mathcal{M}$ are pairwise disjoint.

These results ensure that the restriction of μ_e to $\mathcal{M}$ is a measure. Moreover, this measure is complete in the sense that if $\mu_e(A) = 0$ and $B \subset A$, then $B \in \mathcal{M}$ and $\mu_e(B) = 0$. This is so because A is μ_e-measurable if $\mu_e(A) = 0$ since, given a set X, the monotonicity of μ_e ensures that

$$\mu_e(X) = \mu_e(A) + \mu_e(X) \geq \mu_e(A \cap X) + \mu_e\left(X \cap A^c\right).$$

Moreover, if $B \subset A$ and $\mu_e(A) = 0$, then $\mu_e(B) = 0$ again because μ_e is monotonic.

In view of these results, it suffices, to conclude the construction of the Haar measure, to verify that Borel subsets are μ_e-measurable, which is done in the sequel.

Proposition 3.16 *Borel sets are μ_e-measurable.*

Proof It suffices to show that the open sets are measurable, since $\mathcal{M}$ is a σ-algebra. Let U be an open set and take an arbitrary subset X. If $\mu_e(X) = \infty$, then the inequality $\mu_e(X) \geq \mu_e(X \cap U) + \mu_e(X \cap U^c)$ is trivially satisfied.

Hence, assume that $\mu_e(X) < \infty$ and take $\varepsilon > 0$. Then, there exists an open set $V \supset X$ such that $\mu_e(X) \leq \lambda_*(V) < \mu_e(X) + \varepsilon$. Take a compact set $C \subset V \cap U$ such that $\lambda_*(V \cap U) - \varepsilon < \lambda(C) \leq \lambda_*(V \cap U)$. Take also a compact set $D \subset V \cap C^c$ such that $\lambda_*(V \cap C^c) - \varepsilon < \lambda(D) \leq \lambda_*(V \cap C^c)$. Then, $V \cap U^c \subset V \cap C^c$, as $C \subset U$, and hence

[6] See, for instance, Section 11 of Halmos.

$$\mu_e\left(V \cap U^c\right) - \varepsilon \leq \mu_e\left(V \cap C^c\right) - \varepsilon < \lambda(D),$$

as $\mu_e(V \cap C^c) = \lambda_*(V \cap C^c)$. Therefore,

$$\begin{aligned}\mu_e(X \cap U) + \mu_e\left(X \cap U^c\right) - 2\varepsilon &\leq \mu_e(V \cap U) + \mu_e\left(V \cap U^c\right) - 2\varepsilon \\ &\leq \lambda(C) + \lambda(D) = \lambda(C \cup D),\end{aligned}$$

since $C \cap D = \emptyset$ because $D \subset D \subset V \cap C^c$. But $C \cup D$ is a compact set contained in the open set $(V \cap U) \cup (V \cap C^c) \subset V$. Therefore it follows, by the definition of λ_*, that

$$\lambda(C \cup D) \leq \lambda_*(V \cap U) \cup \left(V \cap C^c\right) \leq \lambda_*(V).$$

But due to the choice of V, $\lambda_*(V) < \mu_e(X) + \varepsilon$. These inequalities show that

$$\mu_e(X \cap U) + \mu_e\left(X \cap U^c\right) < \mu_e(X) + 3\varepsilon,$$

and as $\varepsilon > 0$ is arbitrary, there follows the desired inequality $\mu_e(X \cap U) + \mu_e(X \cap U^c) \leq \mu_e(X)$, showing that the open set U is μ_e-measurable. □

Finally, the properties stated in Theorem 3.1 have been verified along the construction. They are:

1. $\mu(K) < \infty$ if K is compact, by Corollary 3.15.
2. If $U \neq \emptyset$ is open, then $\mu(U) > 0$. Indeed, let K_0 be the compact set chosen to normalize the Haar measure. Then, $\mu(K_0) \geq \lambda(K_0) = 1$ by Proposition 3.14. Given an open set $U \neq \emptyset$, there exist $\{x_1, \ldots, x_n\} \subset K_0$ such that $K_0 \subset x_1 U \cup \cdots \cup x_n U$, so that

$$\mu(K_0) \leq \sum_{i=1}^{n} \mu(x_1 U) = \sum_{i=1}^{n} \mu(U) = n\mu(U),$$

   showing that $\mu(U) > 0$.
3. Outer regularity: for a Borel set A, $\mu(A) = \inf \mu(U)$ with $A \subset U$ and U open. This is the very definition of $\mu_e = \mu$.
4. Inner regularity: if U is open, then $\mu(U) = \sup \mu(K)$ with $K \subset U$ compact. Indeed, $\mu(U) = \lambda_*(U) = \sup_{K \subset U} \lambda(K)$. But by Proposition 3.14, the inequality $\mu_e(K^\circ) \leq \lambda(K) \leq \mu_e(K) = \mu(K)$ holds, whence it follows that

$$\mu(U) = \sup_{K \subset U} \lambda(K) \leq \sup_{K \subset U} \mu(K) \leq \mu(U),$$

as $\mu(K) \leq \mu(U)$ if $K \subset U$. This shows that the open sets are inner regular. But a general theorem in measure theory ensures that a measure is regular if compact sets are outer regular and open sets are inner regular.[7] Therefore, μ is indeed a regular measure.

3.3 Uniqueness

Let μ_1 and μ_2 be left invariant Haar measures. Then

$$\mu = \mu_1 + \mu_2$$

is also a left invariant Haar measure. Measures μ_1 and μ_2 are absolutely continuous with respect to μ as, if A is a Borel set with $\mu(A) = 0$, then $\mu_1(A) + \mu_2(A) = 0$, which implies that $\mu_1(A) = \mu_2(A) = 0$.

Hence, by the Radon–Nikodym theorem, there exists a measurable function (the Radon–Nikodym derivative) $f : G \to \mathbb{R}_+$ such that

$$\mu_1(A) = \int_A f(x)\, \mu(dx)$$

for every measurable set A.

This said, the idea of the proof consists in verifying that f is (almost always) constant, as this entails that μ_1 is a multiple of μ. The same argument applies to μ_2, ensuring that the measures are multiples of each other.

To prove that f is almost always constant, let $F(x, y) = f\left(y^{-1}x\right)$. If $A, B \subset G$ are measurable sets, then

$$\begin{aligned}\int_{A\times B} F(x, y)\,(\mu \times \mu)\,(d(x, y)) &= \int_B \left(\int_A f\left(y^{-1}x\right)\mu(dx)\right)\mu(dy)\\ &= \int_B \left(\int_A f(x)\,\mu(dx)\right)\mu(dy)\\ &= \int_{A\times B} f(x)\,(\mu \times \mu)\,(d(x, y)).\end{aligned}$$

As A and B are arbitrary, this equality implies that the set $N = \{(x, y) : F(x, y) \neq f(x)\}$ satisfies $\mu \times \mu(N) = 0$, that is,

[7] See, for instance, Section 52 of Halmos [18] and more specifically the F theorem.

$$
\begin{aligned}
0 = \mu \times \mu (N) &= \int_N (\mu \times \mu)(d(x, y)) \\
&= \int_G \left(\int_G \chi_N (x, y)\, \mu (dy) \right) \mu (dx) \\
&= \int_G \mu (N_x)\, \mu (dx),
\end{aligned}
$$

where $N_x = (\{x\} \times G) \cap N$. This implies that, for μ-almost all x, it holds that $\mu (N_x) = 0$. But $(x, y) \in N_x$ if and only if $f\left(y^{-1}x\right) \neq f(x)$. Hence, if $\mu\left(N_{x_0}\right) = 0$, then $f\left(y^{-1}x_0\right) = f(x_0)$ for every y, except for a set of measure zero (N_x itself). It then follows that $f(x)$ is almost always constant, and this shows that $\mu_1 = a_1\mu$ for $a_1 > 0$.

Finally, the same arguments show that $\mu_2 = a_2\mu$, $a_2 > 0$, concluding that $\mu_1 = a\mu_2$, $a = a_1a_2^{-1}$.

3.4 Modular Function

The modular function (actually a homomorphism) is a real-valued function on G which compares left invariant and right invariant Haar measures.

To define it, let μ be a left invariant Haar measure on a Hausdorff, locally compact group G. If $g \in G$, then $\left(R_g\right)_* \mu$ ($\left(\left(R_g\right)_* \mu (A) = \mu\left(Ag^{-1}\right)\right)$ is also a left invariant Haar measure, as left translations commute with right translations. By the uniqueness of the Haar measure, there exists a real $\Delta (g) > 0$, such that

$$
\left(R_g\right)_* \mu = \Delta (g)\, \mu.
$$

By definition, $\Delta (g)$ is the **modular function** of G. This definition does not depend on the choice of μ, since $\left(R_g\right)_* (a\mu) = a\left(R_g\right)_* (\mu) = \Delta (g)\, \nu$ if $\nu = a\mu$ with $a > 0$.

If the group G is unimodular, then the left invariant Haar measure μ is also right invariant and thus $\Delta (g) = 1$ for every g. Conversely, if Δ is constant and equal to 1, then μ is right invariant and the group is unimodular.

The modular function can be computed using a single measurable Borel set A such that $0 < \mu (A) < \infty$ (for instance, A can be compact with nonempty interior). This is so because, by definition, $\mu\left(Ag^{-1}\right) = \Delta (g)\, \mu (A)$ and so

$$
\Delta (g) = \mu\left(Ag^{-1}\right) / \mu (A).
$$

This equality also shows that $\mu\left(Ag^{-1}\right) > 0$ if $0 < \mu (A) < \infty$, since $\Delta (g) > 0$.

Using this form for writing Δ, it is possible to obtain the following homomorphism property of Δ.

Proposition 3.17 *Δ is a homomorphism with values in the multiplicative group of positive real numbers.*

Proof If $g, h \in G$ and A is measurable, such that $0 < \mu(A) < \infty$, then

$$\Delta(gh) = \frac{\mu\left(Ah^{-1}g^{-1}\right)}{\mu(A)} = \frac{\mu\left(Ah^{-1}g^{-1}\right)}{\mu\left(Ah^{-1}\right)} \frac{\mu\left(Ah^{-1}\right)}{\mu(A)} = \Delta(g)\,\Delta(h). \qquad \square$$

In what follows, it will be proved that Δ is continuous. To do it, the following two lemmas of topological character are required.

Lemma 3.18 *In the topological group G, let K be a compact set and U an open set, such that $K \subset U$. Then, there exists an open set V with $1 \in V$, such that $KV \subset U$.*

Proof If $x \in K$, then $1 \in x^{-1}U$ and hence there exists an open set V_x with $1 \in V_x$, such that $V_x \subset V_x^2 \subset x^{-1}U$. Since the open sets xV_x cover K, there exists $\{x_1, \ldots, x_n\} \subset K$ such that $K \subset x_1V_{x_1} \cup \cdots \cup x_nV_{x_n}$. The open set $V = V_{x_1} \cap \cdots \cap V_{x_n}$, which contains 1, satisfies $KV \subset U$. To see this, take $y \in K$. Then, there exists $i = 1, \ldots, n$ such that $y \in x_iV_{x_i}$. Hence,

$$yV \subset x_iV_{x_i}V \subset x_iV_{x_i}^2 \subset U,$$

because $V_{x_i}^2 \subset x_i^{-1}U$. Since $y \in K$ is arbitrary, this shows that $KV \subset U$. $\square$

Lemma 3.19 *Let G be a topological group and $\psi : G \to \mathbb{R}_\times$ a homomorphism. Then, ψ is continuous if and only if ψ is upper semicontinuous at 1, that is, for every $\delta > 0$, there exists a neighborhood $V \in \mathcal{V}(1)$ such that $\Delta(g) < 1 + \delta$ if $g \in V$.*

Proof It is clear that, if ψ is continuous, then it is upper semicontinuous at 1. For the converse, it suffices to verify its continuity at the identity 1. Now, given $\delta > 0$, let V be as in the statement and take a symmetric neighborhood $W = V \cap V^{-1}$, in such a way that $\psi(g) < 1 + \delta$ for every $g \in W$. Since $g^{-1} \in W$, it is also true that $\psi(g)^{-1} = \psi\left(g^{-1}\right) < 1 + \delta$, that is, $\psi(g) > 1/(1 + \delta)$. It thus follows that, given $\varepsilon > 0$, if $0 < \delta < \min\{\varepsilon, \varepsilon/(1 - \varepsilon)\}$ and $g \in W$, then $1 - \varepsilon < \psi(g) < 1 + \varepsilon$, showing that ψ is continuous at 1. $\square$

Proposition 3.20 *The modular function Δ is continuous and, therefore, measurable.*

Proof By the previous lemma, it suffices to show that Δ is upper semicontinuous at 1. To do it, take a compact $K \subset G$ with nonempty interior. As $0 < \mu(K) < \infty$, the modular function is given by $\Delta(g) = \mu\left(Kg^{-1}\right)/\mu(K)$ and the upper semicontinuity at 1 translates into the statement that, for every $\varepsilon > 0$, there exists a neighborhood $V \in \mathcal{V}(1)$ such that $\mu\left(Kg^{-1}\right) < \mu(K) + \varepsilon$ if $g \in V$.

Now the outer regularity of μ is applied. Given $\varepsilon > 0$, let U be an open set with $K \subset U$, such that $\mu(U) < \mu(K) + \varepsilon$. By Lemma 3.18, there exists $V \in \mathcal{V}(1)$ such that $KV^{-1} \subset U$. Then, $Kg^{-1} \subset U$ for every $g \in V$ and so it follows that

$$\mu\left(Kg^{-1}\right) \leq \mu(U) < \mu(K) + \varepsilon$$

if $g \in V$, showing the upper semicontinuity and concluding the proof. □

Another way to view the modular function Δ is through the Radon–Nikodym derivative of the right invariant Haar measure $\widehat{\mu}$ with respect to the left invariant measure μ.

Proposition 3.21 *Let μ be a left invariant Haar measure. Then, Δ is the Radon–Nikodym derivative of $\widehat{\mu} = \iota_* \mu$ with respect to μ.*

Proof Fix a measurable set A such that $0 < \mu(A) < \infty$ and $\Delta(g) = \mu\left(Ag^{-1}\right)/\mu(A)$. This equality shows, in particular, that the function $g \mapsto \mu\left(Ag^{-1}\right)$ is continuous, since Δ is continuous. By this continuity, if $K \subset G$ is compact, then there exists the integral

$$\int_G \chi_K(x)\, \mu\left(Ax^{-1}\right) \mu(dx).$$

This integral is determined by the following sequence of equalities:

$$\begin{aligned}\int_G \chi_K(x)\, \mu\left(Ax^{-1}\right) \mu(dx) &= \int_G \chi_K(x) \int_G \chi_{Ax^{-1}}(y)\, \mu(dy)\, \mu(dx) \\ &= \int_G \chi_K(x) \int_G \chi_A(yx)\, \mu(dy)\, \mu(dx) \\ &= \int_G \int_G \chi_K(x)\, \chi_A(yx)\, \mu(dx)\, \mu(dy),\end{aligned}$$

by the Theorem of Fubini. Using the left invariance of μ and changing again the order of integration, the last integral becomes

$$\begin{aligned}\int_G \chi_A(x) \int_G \chi_K\left(y^{-1}x\right) \mu(dy)\, \mu(dx) &= \int_G \chi_A(x) \int_G \chi_K(yx)\, \widehat{\mu}(dy)\, \mu(dx) \\ &= \int_G \chi_A(x) \int_G f(y)\, \widehat{\mu}(dy)\, \mu(dx),\end{aligned}$$

by the right invariance of $\widehat{\mu}$. Hence,

$$\int_G \chi_K(x)\, \mu\left(Ax^{-1}\right) \mu(dx) = \mu(A)\, \widehat{\mu}(K),$$

proving that

$$\begin{aligned}\widehat{\mu}(K) &= \frac{1}{\mu(A)} \int_G \chi_K(x)\, \mu\left(Ax^{-1}\right) \mu(dx) \\ &= \int_G \chi_K(x)\, \Delta(x)\, \mu(dx).\end{aligned}$$

Let μ' be the measure defined by

$$\mu'(B) = \int_B \Delta(x)\,\mu(dx).$$

Then, μ' is regular because Δ is continuous and coincides with $\widehat{\mu}$ on compact sets. The fact that $\widehat{\mu}$ is also regular entails $\widehat{\mu} = \mu'$, concluding the proof. □

Finally, a comment on Lie groups: For these groups, the presence of a differentiable structure allows a much simpler construction of Haar measures by means of integration with respect to differential forms. In this case, modular functions are defined by determinants of linear maps given by the adjoint representation (see Chapter 5, Section 5.6). The construction via volume forms will provide concrete examples of Haar measures.

3.5 Exercises

1. Let G and H be locally compact groups with Haar measures μ_G and μ_H, respectively. Show that $\mu_G \times \mu_H$ is a Haar measure on $G \times H$. Generalize for a finite product of topological groups.
2. Let G be a Hausdorff, locally compact group with left invariant Haar measure μ. Given a compact subgroup $K \subset G$, denote by $\pi : G \to G/K$ the canonical projection and define $\pi_*\mu(A) = \mu\left(\pi^{-1}(A)\right)$ for a Borel set $A \subset G/K$ (considering the quotient topology). Show that $\pi_*\mu$ is a well defined measure on the Borel sets of G/K, invariant by the action of G. In particular, if K is normal, then $\pi_*\mu$ is a Haar measure on G/K.
3. Let K be a Hausdorff, compact group with Haar measure μ and let $\rho : K \to \mathrm{Gl}(V)$ be a continuous representation of K on a finite dimensional vector space V. Take $v \in V$ and let $w \in V$ be given by

$$w = \int_K (\rho(k)\,v)\,\mu(dk).$$

   Show that w is a fixed point of K, that is, $\rho(k)\,w = w$ for every $k \in K$.

   Use this information to show that, if the representation is irreducible (that is, the only invariant subspaces are $\{0\}$ and V), then, for every $v \in V$,

$$\int_K (\rho(k)\,v)\,\mu(dk) = 0.$$

4. Let μ and ν be two probability measures on the Borel sets of a topological group G. The **convolution product** $\mu * \nu$ between μ and ν is defined by $\mu * \nu = p_*(\mu \times \nu)$, where $p : G \times G \to G$ is the product in G and $\mu \times \nu$ is the product measure in $G \times G$. Show that $\mu * \nu(A) = \int_G \nu\left(g^{-1}A\right)\mu(dg)$ for every Borel

set $A \subset G$. Show also that on a compact group G, a Haar measure μ satisfies $\mu * \mu = \mu$.

5. Let K be a Hausdorff, compact group with Haar measure μ. This exercise points to a proof that the set $C(K)$ of continuous functions on K is dense in the Hilbert space $L^2(K, \mu)$. (This fact is not restricted to the Haar measure, but holds for spaces with regular metrics.) Suppose, for contradiction, that $\overline{C(K)} \neq L^2(K, \mu)$ and verify that there exists $f \in L^2(K, \mu)$ such that $\int_K f(x) g(x) \mu(dx) = 0$ for every $g \in C(K)$.

   (a) Let $C \subset K$ be a compact set and A_n, $n \in \mathbb{N}$, a sequence of open sets in K such that $C = \bigcap_n A_n$. Use the Lemma of Urysohn to construct a sequence of continuous functions f_n that converges pointwise to the characteristic function χ_C of C.
   (b) Use the previous item (and the regularity of μ) to show that $\int_C f(x) \mu(dx) = 0$ for every compact set C.
   (c) Use the inner regularity of μ to show that $\int_A f(x) \mu(dx) = 0$ for a Borel set A.
   (d) Show that $f = 0$ almost everywhere, using the equalities $\int_{\{f<0\}} f(x) \mu(dx) = 0 = \int_{\{f>0\}} f(x) \mu(dx)$.

Chapter 4
Representations of Compact Groups



In this chapter, some results about representations of compact groups will be proved. Most proofs involve integration with respect to the Haar measure, which is finite. For this reason, it is assumed once and for all that all the groups are Hausdorff.

The main result is the Peter–Weyl theorem, which, together with Schur orthogonality relations, generalizes the construction of Fourier series on S^1.

4.1 Representations

A (Hausdorff) compact group K is unimodular, as the modular function $\Delta : K \to \mathbb{R}_+$ is a continuous homomorphism. So, its image $\Delta(K)$ is a compact subgroup of the multiplicative group $\mathbb{R}_+$. Since $\{1\}$ is the only compact subgroup of $\mathbb{R}_+$, it follows that $\Delta(K) = \{1\}$ and hence that Δ is constant $= 1$. For this reason, the Haar measure μ is bi-invariant and will always be chosen with normalization $\mu(K) = 1$.

The use of techniques of integration on compact groups is greatly favored by the fact that continuous functions are integrable. This is so because, if $f : K \to \mathbb{R}$ is continuous, then f is bounded, and hence the integrals of its positive part $f^+(x) = \max\{f(x), 0\}$ and of its negative part $f^-(x) = -\min\{f(x), 0\}$ are finite. It follows from this that

$$\int f(x)\,\mu(dx) = \int f^+(x)\,\mu(dx) - \int f^-(x)\,\mu(dx)$$

is well defined. In the same way, continuous functions with values in finite dimensional real vector spaces are integrable.

In many situations, the integration with respect to the Haar measure on a compact group is used to obtain objects that are invariant by the action of the group. For instance, let K be a compact group, and let $\rho : K \to \mathrm{Gl}(V)$ be a **representation** of K on a finite dimensional real vector space V. If ρ is continuous, then, for every

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$v_0 \in V$, the map $f : K \to V$ defined by $f(k) = k \cdot v_0$ is continuous and hence integrable. This means that

$$v_{\text{inv}} = \int_K f(k)\, \mu(dk) = \int_K \rho(k)\, v_0 \mu(dk)$$

is a well defined vector in V. This vector is fixed for every $g \in K$. Indeed,

$$\rho(g) \cdot v_{\text{inv}} = \rho(g) \int_K \rho(k)\, v_0 \mu(dk) = \int_K \rho(gk)\, v_0 \mu(dk)$$

by the linearity of the integral. Hence,

$$\rho(g) \cdot v_{\text{inv}} = \int_K \rho(k)\, v_0 \left(\left(L_g\right)_* \mu\right)(dk) = v_{\text{inv}},$$

as $\left(L_g\right)_* \mu = \mu$. This invariance means that v_{inv} is an eigenvector associated with eigenvalue 1 for every $\rho(g)$, $g \in K$. From this, it follows, for instance, that if some $\rho(g)$ has no eigenvalue 1, then $\int_K \rho(k)\, v_0 \mu(dk) = 0$ for every $v_0 \in V$.

The same integration method is used in the following proposition, which is fundamental in the theory of representations of compact groups.

Proposition 4.1 *Let K be a Hausdorff compact group, and let $\rho : K \to \mathrm{Gl}(V)$ be a continuous representation of K on a finite dimensional real (respectively, complex) vector space V. Then, there exists an inner (respectively, Hermitian) product $(\cdot, \cdot)$ in V that is K-invariant, that is, $(\rho(k)\, u, \rho(k)\, v) = (u, v)$ for every $k \in K$, $u, v \in V$.*[1]

Proof Let μ be the Haar measure of K normalized by $\mu(K) = 1$. Take an arbitrary inner product $B(\cdot, \cdot)$ on V, and define the map $(\cdot, \cdot) : V \times V \to \mathbb{R}$ by

$$(u, v) = \int_K B(\rho(k)\, u, \rho(k)\, v)\, \mu(dk). \tag{4.1}$$

This integral is well defined because μ is a finite measure and the function $k \in K \mapsto B(\rho(k)\, u, \rho(k)\, v) \in \mathbb{R}$ (with fixed u and v) is continuous and, therefore, integrable. As B is bilinear and symmetric, the same is true for $(\cdot, \cdot)$. If $u = v$, then the integrand of (4.1), $B(\rho(k)\, u, \rho(k)\, u) > 0$ for every $k \in K$. This implies that $(u, u) \geq 0$, and if $u \neq 0$, then $(u, u) > 0$, as $B(\rho(k)\, u, \rho(k)\, u)$ is continuous as a function of k. Hence, $(\cdot, \cdot)$ is indeed an inner product on V. To see that it is K-invariant, take $g \in K$. Since $\rho(kg) = \rho(k)\, \rho(g)$, it follows that

[1]The inner product or the Hermitian product are fixed points in the space of quadratic forms or sesquilinear forms.

$$(\rho(g)u, \rho(g)v) = \int_K B(\rho(kg)u, \rho(kg)v)\,\mu(dk)$$
$$= \int_K B(\rho(k)u, \rho(k)v)\left(\left(R_g\right)_*\mu\right)(dk).$$

But as μ is invariant by right translations, the last integral reduces to the right hand side of (4.1), and this shows that $(\rho(g)u, \rho(g)v) = (u, v)$, concluding the proof in the real case. The complex case is similar. □

In other words, the proposition above ensures that any finite dimensional representation of a compact Lie group assumes values in the group $\mathrm{O}(n)$ or $\mathrm{U}(n)$ of isometries of an inner product or a Hermitian product.

The fact that the elements $\rho(g)$ are isometries ensures that $W^\perp$ is a K-invariant subspace if W is K-invariant, that is, $\rho(k)W \subset W$ for every $k \in K$. Indeed, if $u \in W$, $v \in W^\perp$ and $k \in K$, then

$$(\rho(k)v, u) = \left(v, \rho\left(k^{-1}\right)u\right) = 0,$$

since $\rho\left(k^{-1}\right)u \in W$. As $u \in W$ is arbitrary, it follows that $\rho(k)v \in W^\perp$. This observation provides the following decomposition of the representation space.

Proposition 4.2 *Let K be a Hausdorff compact group and $\rho : K \to \mathrm{Gl}(V)$ a continuous representation of K on a real or complex vector space V. Then, V decomposes as a direct sum*

$$V = V_1 \oplus \cdots \oplus V_n,$$

where each subspace V_i is invariant by K (that is, $\rho(k)V_i \subset V_i$ if $k \in K$) and ***irreducible*** *(that is, it does not admit invariant subspaces other than the trivial subspaces $\{0\}$ and V_i).*

Proof Consider the real case and take a K-invariant inner product $(\cdot, \cdot)$ on V. If V is irreducible, there is nothing to prove (the decomposition contains one single term). If V is not irreducible, let $\{0\} \neq W \subset V$ be a K-invariant subspace. Then, $W^\perp$ is also invariant.

If W and $W^\perp$ are irreducible, then the desired decomposition is $V = W \oplus W^\perp$. Otherwise, just like V, W (or $W^\perp$) is written as a direct sum of invariant subspaces orthogonal to each other. Decomposing the invariant subspaces successively in this way, the decomposition of V in invariant subspaces is attained, since in each step, the dimensions of subspaces decrease. □

A useful tool in representation theory is the **Lemma of Schur**. It is a lemma about the centralizer of subsets of linear maps of a vector space V and applies, in particular, to representations of groups.

Let A and B be linear maps on $\mathrm{End}(V)$, which commute with each other. Then, $A(Bv) = BAv = 0$ if $Av = 0$ and $Bw = B(Av) = A(Bv)$ if $w = Av$. This means that $\ker A$ and $\mathrm{im}\, A$ are invariant by B.

This said, take a subset $\Gamma \subset \mathrm{End}(V)$, such that the only subspaces invariant by Γ are the trivial subspaces $\{0\}$ and V. If $L \in \mathrm{End}(V)$ commutes with the elements of Γ, then $\ker L$ and $\mathrm{im}\, L$ are subspaces invariant by Γ. As Γ is irreducible, it follows that the possibilities for $\ker L$ and $\mathrm{im}\, L$ are $\{0\}$ and V, and this means that $L = 0$ or L is bijective.

Suppose, moreover, that L has an eigenvector in V, associated with an eigenvalue λ in the scalar field of V. Then, $L - \lambda \cdot \mathrm{id}$ also commutes with the elements of Γ. This implies, in the irreducible case, that $L - \lambda \cdot \mathrm{id}$ is 0 or bijective. However, $L - \lambda \cdot \mathrm{id}$ cannot be bijective, since L has eigenvectors. Hence, it follows that $L - \lambda \cdot \mathrm{id} = 0$, that is, $L = \lambda \cdot \mathrm{id}$ is a scalar map. This is the result of the Lemma of Schur:

Proposition 4.3 *Let V be a vector space on $\mathbb{K}$ and $\Gamma \subset \mathrm{End}(V)$ an irreducible set of linear maps of V. Let $L \in \mathrm{End}(V)$, which commutes with all the elements of Γ. Suppose that L has an eigenvector in V associated with eigenvalue $\lambda \in \mathbb{K}$. Then, $L = \lambda \cdot \mathrm{id}$. In particular, if $\mathbb{K}$ is algebraically closed and $\dim V < \infty$, then the centralizer of Γ in $\mathrm{End}(V)$ is the subspace of scalar maps.*

An immediate consequence of the Lemma of Schur is that any irreducible representation ρ of an abelian group G (not necessarily compact) on a complex vector space has at most dimension 1. This is so because $\rho(g) = \lambda(g) \cdot \mathrm{id}$ for every $g \in G$, so that any subspace is invariant, and thus the only possibility of being irreducible is to have dimension 1 (or 0).

In the case of representations in real spaces in general, the elements of the centralizer are not scalars, as is shown by the example of the canonical representation of S^1 on $\mathbb{R}^2$ given by

$$\theta \mapsto \begin{pmatrix} \cos\theta & -\mathrm{sen}\theta \\ \mathrm{sen}\theta & \cos\theta \end{pmatrix}. \tag{4.2}$$

With the Lemma of Schur, it is possible to obtain the following results about representations of compact groups.

Proposition 4.4 *Let K be a Hausdorff compact group and $\rho : K \to \mathrm{Gl}(V)$ an irreducible and continuous representation of K on a real (respectively, complex) vector space V. If $(\cdot,\cdot)_1$ and $(\cdot,\cdot)_2$ are invariant inner products (respectively, Hermitian products), then $(\cdot,\cdot)_1 = \lambda(\cdot,\cdot)_2$ for some $\lambda > 0$.*

Proof In both cases, there exists $P \in \mathrm{End}(V)$ such that $(u, v)_1 = (Pu, v)_2$, with P symmetric in the real case and Hermitian in the complex case. In each case, the eigenvalues of P are real. The following inequalities hold for every $k \in K$ and $u, v \in V$, since both products are invariant:

$$\begin{aligned}(Pu, v)_2 &= (u, v)_1 = (\rho(k)u, \rho(k)v)_1 \\ &= (P\rho(k)u, \rho(k)v)_2 = \left(\rho\left(k^{-1}\right) P\rho(k)u, v\right)_2.\end{aligned}$$

Hence, $P = \rho\left(k^{-1}\right) P \rho(k)$, that is, P commutes with $\rho(k)$, $k \in K$. By the Lemma of Schur, $P = \lambda \cdot \mathrm{id}$, that is, $(\cdot,\cdot)_1 = \lambda\,(\cdot,\cdot)_2$, concluding the proof. □

As a consequence, the set of irreducible inner or Hermitian products in an irreducible representation is parameterized by $\mathbb{R}_+$. On the other hand, in an arbitrary finite dimensional representation of a compact group, this set is parameterized by $\mathbb{R}_+^n$, where n is the number of irreducible components, since the invariant inner or Hermitian products are obtained by direct sums of their invariant restrictions in the irreducible components.

Let now ρ_1 and ρ_2 be representations of K in spaces V_1 and V_2, respectively. Take a linear map $L : V_1 \to V_2$, which intertwines the representations, that is, $L\rho_1(k) = \rho_2(k) L$ for every $k \in K$. Then, as in the Lemma of Schur, $\ker L$ and $\mathrm{im}\, L$ are invariant by K (or rather, by representations ρ_1 and ρ_2, respectively). So, if ρ_1 is irreducible, $L = 0$ or L is injective and, if ρ_2 is irreducible, then $L = 0$ or L is surjective. If both representations are irreducible, then $L = 0$ or L is an isomorphism. In the latter case, it is said that the **representations** are **equivalent**.

The method of integration with respect to the Haar measure provides maps that intertwine representations.

Proposition 4.5 *Let ρ_1 and ρ_2 be representations of a compact group K on spaces V_1 and V_2 on $\mathbb{R}$ or $\mathbb{C}$, of finite dimension. For an arbitrary linear map $L_0 : V_1 \to V_2$, define $L_{\mathrm{inv}} : V_1 \to V_2$ by*

$$L_{\mathrm{inv}} = \int_K \rho_2\left(x^{-1}\right) L_0 \rho_1(x)\, \mu(dx). \tag{4.3}$$

Then, L_{inv} is linear and satisfies $L_{\mathrm{inv}}\rho_1(k) = \rho_2(k) L_{\mathrm{inv}}$ for every $k \in K$. Moreover, if $\rho_1 = \rho_2 = \rho$ and $V_1 = V_2 = V$, then $\mathrm{tr} L_{\mathrm{inv}} = \mathrm{tr} L_0$.

Proof The integral in (4.3) is well defined, as the integrand is a continuous map with values in the space $L(V_1, V_2)$ of linear maps $V_1 \to V_2$. If $k \in K$, then

$$\begin{aligned} L_{\mathrm{inv}}\rho_1(k) &= \int_K \rho_2\left(x^{-1}\right) L_0 \rho_1(xk)\, \mu(dx) \\ &= \int_K \rho_2(k)\, \rho_2\left((xk)^{-1}\right) L_0 \rho_1(xk)\, \mu(dx) \\ &= \rho_2(k)\, L_{\mathrm{inv}} \end{aligned}$$

by the right invariance of μ, showing that L_{inv} intertwines the representations.

To see the equality of traces, let $\{e_1, \ldots, e_n\}$ be an orthonormal basis of V with respect to the invariant inner (or Hermitian) product $(\cdot,\cdot)$. Then,

$$\begin{aligned} \mathrm{tr} L_{\mathrm{inv}} &= \sum_i \left(L_{\mathrm{inv}} e_i, e_i\right) = \sum_i \int_K \left(\rho\left(x^{-1}\right) L_0 \rho(x) e_i, e_i\right) \mu(dx) \\ &= \int_K \sum_i \left(L_0 \rho(x) e_i, \rho(x) e_i\right) \mu(dx) = \int_K (\mathrm{tr} L_0)\, \mu(dx), \end{aligned}$$

as $\{\rho(x)e_1, \ldots, \rho(x)e_n\}$ is also an orthonormal basis for any $x \in K$. The last integral is $\mathrm{tr}L_0$, concluding the proof. □

This proposition, together with the Lemma of Schur, shows that if ρ_1 and ρ_2 are nonequivalent irreducible representations, then the integral of (4.3) is zero. On the other hand, if $\rho_1 = \rho_2 = \rho$ is irreducible and V is a complex space, then the integral is a scalar $L_{\text{inv}} = \lambda \cdot \text{id}$. In the real case, it may be that the integral is not a scalar, as shown in the following example.

Example Take the canonical representation of S^1 on $\mathbb{R}^2$ given in (4.2) and the conjugation

$$\begin{pmatrix} \cos\theta & \text{sen}\theta \\ -\text{sen}\theta & \cos\theta \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} \cos\theta & -\text{sen}\theta \\ \text{sen}\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} \text{sen}\theta\cos\theta & \cos^2\theta \\ -\text{sen}^2\theta & -\text{sen}\theta\cos\theta \end{pmatrix}.$$

Its integral with respect to the normalized Haar measure is

$$\frac{1}{2\pi}\int_0^{2\pi} \begin{pmatrix} \text{sen}\theta\cos\theta & \cos^2\theta \\ -\text{sen}^2\theta & -\text{sen}\theta\cos\theta \end{pmatrix} d\theta = \begin{pmatrix} 0 & 1/2 \\ -1/2 & 0 \end{pmatrix}, \tag{4.4}$$

which is not a scalar. □

4.2 Schur Orthogonality Relations

Let $L^2(K, \mu)$ be the space of functions $f : K \to \mathbb{C}$ that are measurable and quadratically integrable with respect to the Haar measure, which is a Hilbert space with the Hermitian product

$$(f, g) = \int_K f(x)\overline{g(x)}\mu(dx).$$

In what follows, Proposition 4.5 will be used to show what are known as the Schur orthogonality relations, which give the values of the Hermitian product of $L^2(K, \mu)$ for certain functions defined by representations of K.

Let $\rho : K \to \text{Gl}(V)$ be a finite dimensional continuous representation. A function of the type

$$f_{u,v}(k) = (\rho(k)u, v),$$

where $(\cdot, \cdot)$ is the (inner or Hermitian) product invariant by K and $u, v \in V$, is called the **matrix entry** of ρ. The function takes real or complex values, depending on whether V is real or complex. The reason for this name is that if $\{e_1, \ldots, e_n\}$ is an orthonormal basis of V, then the entries of the matrix of $\rho(k)$ with respect to this basis are given by $f_{e_i,e_j}(k)$.

The **Schur orthogonality relations** determine integrals of the type

$$\left(f_{u_1,u_2},\, f_{v_1,v_2}\right) = \int_K f_{u_1,u_2}(k)\, \overline{f_{v_1,v_2}(k)}\mu(dk),$$

where u_1, u_2 and v_1, v_2 are elements of spaces of irreducible representations of the compact group K. These orthogonality relations are stated in the following result.

Proposition 4.6 *Let ρ_1 and ρ_2 be irreducible representations of K on **complex** vector spaces V_1 and V_2, respectively.*

1. *If ρ_1 and ρ_2 are not equivalent, then*

$$\int_K f_{v_1,v_2}(k)\, \overline{f_{u_1,u_2}(k)}\mu(dk) = 0$$

for every $u_1, u_2 \in V_1$ and $v_1, v_2 \in V_2$.

2. *If $\rho_1 = \rho_2 = \rho$ and $V_1 = V_2 = V$, then*

$$\int_K f_{v_1,v_2}(k)\, \overline{f_{u_1,u_2}(k)}\mu(dk) = \frac{(v_1, u_1)\, \overline{(v_2, u_2)}}{\dim V} \tag{4.5}$$

for $u_1, u_2, v_1, v_2 \in V$.

Proof To get the first integral, take $L_0 : V_1 \to V_2$ as $L_0(w) = (w, u_2)\, v_2$, where $(\cdot,\cdot)$ is the invariant Hermitian product on V_1. Then, for $k \in K$ and $w = u_1 \in V_1$, it holds

$$\begin{aligned}\rho_2\left(k^{-1}\right) L_0\rho_1(k)\, u_1 &= (\rho_1(k)\, u_1, u_2)\, \rho_2\left(k^{-1}\right) v_2 \\ &= f_{u_1,u_2}(k)\, \rho_2\left(k^{-1}\right) v_2.\end{aligned}$$

Taking the Hermitian product with $v_1 \in V_2$ yields

$$\begin{aligned}\left(v_1, \rho_2\left(k^{-1}\right) L_0\rho_1(k)\, u_1\right) &= \overline{f_{u_1,u_2}(k)}\left(v_1, \rho_2\left(k^{-1}\right) v_2\right) \\ &= \overline{f_{u_1,u_2}(k)}\,(\rho_2(k)\, v_1, v_2),\end{aligned}$$

from where it follows that

$$\left(v_1, \rho_2\left(k^{-1}\right) L_0\rho_1(k)\, u_1\right) = f_{v_1,v_2}(k)\, \overline{f_{u_1,u_2}(k)}. \tag{4.6}$$

By Proposition 4.5 and the Lemma of Schur,

$$\int_K \rho_2\left(k^{-1}\right) L_0\rho_1(k)\, u_1\mu(dk) = 0,$$

so that

$$\int_K f_{v_1,v_2}(k)\,\overline{f_{u_1,u_2}(k)}\mu(dk) = \int_K \left(v_1, \rho_2\left(k^{-1}\right) L_0 \rho_1(k)\, u_1\right) \mu(dk) = 0,$$

showing the first Schur orthogonality relation.

In the case of a single irreducible representation, the same L_0 yields the same expression (4.6) for $f_{v_1,v_2}(k)\,\overline{f_{u_1,u_2}(k)}$. The difference now is that

$$L_{\text{inv}} = \int_K \rho_2\left(k^{-1}\right) L_0 \rho_1(k)\, u_1 \mu(dk) = \lambda \cdot \text{id}$$

by the Lemma of Schur. Hence, $\text{tr} L_{\text{inv}} = \lambda \dim V$. But, by Proposition 4.5, $\text{tr} L_{\text{inv}} = \text{tr} L_0 = (v_2, u_2)$, and it follows that

$$L_{\text{inv}} = \lambda \cdot \text{id} = \frac{(v_2, u_2)}{\dim V}\text{id}.$$

Therefore,

$$\begin{aligned}\int_K f_{v_1,v_2}(k)\,\overline{f_{u_1,u_2}(k)}\mu(dk) &= \int_K \left(v_1, \rho_2\left(k^{-1}\right) L_0 \rho_1(k)\, u_1\right) \mu(dk) \\ &= (v_1, L_{\text{inv}} u_1) \\ &= \frac{(v_1, u_1)\,\overline{(v_2, u_2)}}{\dim V},\end{aligned}$$

concluding the proof. □

In the case of real representations, the first relation of Schur (distinct representations) continues to hold, with the same proof. This is so because the map $L : V_1 \to V_2$, obtained by integrating the conjugation by L_0, is zero in the case where the representations are not equivalent. However, the second relation, which uses the fact that $L = \lambda \cdot \text{id}$, may not be valid for real representations. The following example illustrates this.

Example Take the canonical representation of S^1 on $\mathbb{R}^2$ and $L_0(w) = (w, e_2)\, e_1$, where $(\cdot, \cdot)$ and $\{e_1, e_2\}$ are the canonical inner product and the canonical basis of $\mathbb{R}^2$. The matrix of L_0 is given by

$$L_0 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

and the calculation in (4.4) shows that

$$L = \begin{pmatrix} 0 & 1/2 \\ -1/2 & 0 \end{pmatrix}$$

is not a scalar matrix. Just like in the proof of Proposition 4.6, it holds

$$\left(v, \rho_2\left(k^{-1}\right) L_0 \rho_1(k)\, u\right) = f_{v,e_1}(k)\, f_{u,e_2}(k)$$

for every $u, v \in \mathbb{R}^2$. Hence, if $v = (x_1, x_2)$ and $u = (y_1, y_2)$, then

$$\begin{aligned}\int_K f_{v,e_1}(k)\, f_{u,e_2}(k)\, \mu(dk) &= (v, Lu) \\ &= \frac{1}{2}(x_1 y_2 - x_2 y_1),\end{aligned}$$

which is not identically zero. However, the second member of (4.5) is given in this case by

$$\frac{(v,u)(e_1,e_2)}{\dim V} = 0.$$

□

It is now possible to define the **character** of a finite dimensional representation $\rho : G \to \mathrm{Gl}(V)$, which is the function χ_ρ defined on G with values in the field of scalars of V given by

$$\chi_\rho(g) = \mathrm{tr}(\rho(g)).$$

The characters of group representations are useful to distinguish whether different representations are equivalent or not. This is so because $\chi_{\rho_1} = \chi_{\rho_2}$ if $\rho_i : G \to \mathrm{Gl}(V_i)$, $i = 1, 2$, are equivalent representations. Indeed, if $P : V_1 \to V_2$ is an isomorphism that intertwines the representations, then

$$\chi_{\rho_2}(g) = \mathrm{tr}\left(P \rho_1(g) P^{-1}\right) = \mathrm{tr}(\rho_1(g)) = \chi_{\rho_1}(g).$$

One consequence of Schur orthogonality relations is that the characters of complex representations of compact groups can distinguish these representations completely. To discuss this issue, take first an irreducible complex representation $\rho : K \to \mathrm{Gl}(V)$ of a compact group K. If $(\cdot, \cdot)$ is an invariant Hermitian product and $\{e_1, \ldots, e_n\}$ is an orthonormal basis, then

$$\chi_\rho(k) = \sum_{i=1}^{n} (\rho(k) e_i, e_i) = \sum_{i=1}^{n} f_{e_i,e_i}(k),$$

that is, χ_ρ is a sum of matrix entries. From relation (4.5), it follows that

$$\int_K |\chi_\rho(k)|^2 \mu(dk) = \sum_{i,j=1}^{n} \int_K f_{e_i,e_i}(k)\, \overline{f_{e_j,e_j}(k)}\, \mu(dk) = \frac{1}{\dim V} \sum_{i=1}^{n} |(e_i, e_i)|^2$$

and hence

$$\int_K |\chi_\rho(k)|^2 \mu(dk) = 1. \tag{4.7}$$

In particular, $\chi_\rho \neq 0$ if $\dim V > 0$. On the other hand, if $\rho_1 : K \to \mathrm{Gl}(V_1)$ is another irreducible representation of K, which is not equivalent to ρ, then the first Schur relation shows that

$$\int_K \chi_\rho(k)\, \overline{\chi_{\rho_1}(k)} \mu(dk) = 0, \tag{4.8}$$

and this ensures that $\chi_{\rho_1} \neq \chi_\rho$.

Therefore, two complex irreducible representations of a compact group K are equivalent if and only if their characters coincide.

The same can be said about representations that are not necessarily irreducible. To see that, take a representation $\rho : K \to \mathrm{Gl}(V)$ and a decomposition

$$V = W_1 \oplus \cdots \oplus W_n$$

in irreducible subspaces invariant by K. Denote by σ_i the restriction of ρ to W_i. Then, by definition,

$$\chi_\rho = \chi_{\sigma_1} + \cdots + \chi_{\sigma_n}.$$

Equations (4.7) and (4.8) show that, for every $i = 1, \ldots, n$, it holds

$$\int_K \chi_\rho(k)\, \overline{\chi_{\sigma_i}(k)} \mu(dk) = N_i,$$

where N_i is the number of irreducible components of V equivalent to W_i (that is, the number of times σ_i "appears" in ρ). In the same way, if $\rho_1 : K \to \mathrm{Gl}(V_1)$ is another representation of K, then $\int_K \chi_\rho(k)\, \overline{\chi_{\sigma_i}(k)} \mu(dk)$ is the number of times σ_i appears in ρ_1. Hence, if $\chi_{\rho_1} = \chi_\rho$, the number of times an irreducible representation appears in both representations is the same. This implies that the representations are equivalent.

In short, the following criterion for two representations to be equivalent, which is widely used in the theory of representations of compact groups, is obtained.

Proposition 4.7 *Let K be a Hausdorff compact group. Then, two finite dimensional complex representations ρ_1 and ρ_2 of K are equivalent if and only if their characters χ_{ρ_1} and χ_{ρ_2} are equal.*

Example The group $K = S^1 = \{z \in \mathbb{C} : |z| = 1\}$ is compact and abelian. Because it is abelian, the Lemma of Schur ensures that its irreducible representations have dimension 1 and are given by homomorphisms with values in the multiplicative group $\mathbb{C}_\times = \mathbb{C} \setminus \{0\}$. If ρ is an irreducible Representation, then its character coincides with ρ, as the representation space has dimension 1.

A continuous homomorphism $\rho : S^1 \to \mathbb{C}_\times$ has values in S^1, as its image is a compact subgroup. A homomorphism of this kind has the form $\rho\left(e^{it}\right) = e^{i\theta(t)}$, where $\theta : \mathbb{R} \to \mathbb{R}$ is a linear map, such that $\theta(\mathbb{Z}) \subset \mathbb{Z}$, that is, $\theta(t) = nt$ with $n \in \mathbb{Z}$. For this reason, the representations, as well as the characters, have the form $\rho(z) = z^n$, $z \in S^1$. □

4.3 Regular Representations

It will be considered here representations, by left or right translations, of a group K on the function spaces $C(K)$ of continuous functions $f : K \to \mathbb{C}$, and $L^2(K) = L^2(K, \mu)$, of functions $f : K \to \mathbb{C}$ that are measurable and quadratically integrable with respect to the Haar measure μ. As in the entire chapter, K is a Hausdorff compact group.

The space $C(K)$ is a Banach space with norm

$$\|f\|_\infty = \sup_{k \in K} |f(k)|,$$

while $L^2(K)$ is a Hilbert space with Hermitian product

$$(f, g) = \int_K f(x)\,\overline{g(x)}\mu(dx),$$

whose norm is denoted by $\|\cdot\|_2$. As K is compact, every continuous function is integrable, and this implies that $C(K) \subset L^2(K)$. Actually, $C(K)$ is dense in $L^2(K)$ (in the topology of this latter space). This is due to the fact that the Haar measure μ is regular and that, in compact spaces with regular measures, the set of continuous functions is dense in L^2 (see Chapter 3, Exercise 5).

The left and right translations of $f : K \to \mathbb{C}$ by $k \in K$ are defined by

$$(L_k f)(x) = f(kx) \qquad \text{and} \qquad (R_k f)(x) = f(xk).$$

If f is continuous, then $L_k f$ and $R_k f$ are continuous, in the same way $L_k f, R_k f \in L^2(K)$ if $f \in L^2(K)$, and this entails that L_k and R_k are linear maps of both $C(K)$ and $L^2(K)$. These linear maps are isometries, as $\|L_k f\|_\infty = \|R_k f\|_\infty = \|f\|_\infty$ and $\|L_k f\|_2 = \|R_k f\|_2 = \|f\|_2$ (as the Haar measure is bi-invariant).

For $k_1, k_2 \in K$ and a function f, it holds

$$\left(L_{k_1}\left(L_{k_2} f\right)\right)(x) = f\left(k_2 k_1 x\right) = \left(L_{k_2 k_1} f\right)(x),$$
$$\left(R_{k_1}\left(R_{k_2} f\right)\right)(x) = f\left(x k_1 k_2\right) = \left(R_{k_1 k_2} f\right)(x).$$

Hence, the maps $k \in K \to L_{k^{-1}}$ and $k \in K \to R_k$ define representations of K on $C(K)$ and on $L^2(K)$. These are called **regular representations** of K.

These representations satisfy the following continuity property.

Proposition 4.8 *If $V = C(K)$ or $V = L^2(K)$, then the maps $(k, f) \in K \times V \to L_k f \in V$ and $(k, f) \in K \times V \to R_k g \in V$ are continuous.*

Proof Consider first the case $C(K)$. If $f \in C(K)$, then f is uniformly continuous (see Chapter 2, Exercise 6). This means that, given $\varepsilon > 0$, there exists a neighborhood U of the identity such that $|f(k_1) - f(k_2)| < \varepsilon$ if $k_1 k_2^{-1} \in U$. If $x \in K$ and $k_1 k_2^{-1} \in U$, then $(k_1 x)(k_2 x)^{-1} = k_1 k_2^{-1} \in U$, and so $|f(k_1 x) - f(k_2 x)| < \varepsilon$, showing that $\left\|L_{k_1} f - L_{k_2} f\right\|_\infty < \varepsilon$. Now, if $\|g - f\|_\infty < \varepsilon$ and $k_1 k_2^{-1} \in U$, then

$$\begin{aligned}\left\|L_{k_1} f - L_{k_2} g\right\|_\infty &\leq \left\|L_{k_1} f - L_{k_2} f\right\|_\infty + \left\|L_{k_2} f - L_{k_2} g\right\|_\infty \\ &\leq \varepsilon + \|f - g\|_\infty < 2\varepsilon,\end{aligned}$$

and this shows the continuity of $(k, f) \to L_k f$. For $R_k f$, the proof is similar.

Now, given $f \in L^2(K)$ and $\varepsilon > 0$, choose $h \in C(K)$ such that $\|f - h\|_2 < \varepsilon$. By the case $C(K)$, there exists a neighborhood of the identity $U \subset K$ such that $\left\|L_{k_1} h - L_{k_2} h\right\|_\infty < \varepsilon$ if $k_1 k_2^{-1} \in U$. Therefore, if $k_1 k_2^{-1} \in U$, then

$$\begin{aligned}\left\|L_{k_1} f - L_{k_2} f\right\|_2 &\leq \left\|L_{k_1} f - L_{k_1} h\right\|_2 + \left\|L_{k_1} h - L_{k_2} h\right\|_2 + \left\|L_{k_2} h - L_{k_2} f\right\|_2 \\ &< 3\varepsilon,\end{aligned}$$

as L_{k_1} and L_{k_2} are isometries and $\left\|L_{k_1} h - L_{k_2} h\right\|_2 \leq \left\|L_{k_1} h - L_{k_2} h\right\|_\infty$.

Finally, if $\|f - g\|_2 < \varepsilon$, then

$$\begin{aligned}\left\|L_{k_1} f - L_{k_2} g\right\|_2 &\leq \left\|L_{k_1} f - L_{k_2} f\right\|_2 + \left\|L_{k_2} f - L_{k_2} g\right\|_2 \\ &\leq 3\varepsilon + \|f - g\|_2 < 4\varepsilon.\end{aligned}$$

□

An interesting feature of the regular representation (on $C(K)$ or on $L^2(K)$) is that it contains, up to equivalence, every finite dimensional irreducible representation. These "inclusions" are given by the matrix entries of the representations.

Let $\rho : K \to \mathrm{Gl}(V)$ be a continuous representation of K on a finite dimensional **complex** vector space V. Denote by $C(K)_\rho$ the subspace spanned by the matrix entries $f_{u,v}$ of ρ, which is a subspace of $C(K)$. This space has finite dimension since, if $\{v_1, \ldots, v_n\}$ is a basis of V, then the matrix entries $f_{v_i, v_j}(x) = \left(\rho(x) v_i, v_j\right)$ span $C(K)_\rho$. Moreover, $C(K)_\rho$ is invariant by left and right translations, since

$$f_{u,v}(kx) = (\rho(kx)u, v) = f_{u,\rho(k^{-1})v}(x) \tag{4.9}$$

and

$$f_{u,v}(xk) = (\rho(xk)u, v) = f_{\rho(k)u,v}(x). \tag{4.10}$$

From these relations, the following equivalence relations between V and subspaces of $C(K)_\rho$ are obtained.

Proposition 4.9 *Suppose that the representation on V is irreducible. Given $0 \neq v \in V$, define the map $P_v : V \to C(K)_\rho$ by $P_v(u) = f_{u,v}$. Then, P_v is linear, injective and intertwines representation ρ on V with the representation by right translations on $C(K)_\rho$. Hence, the image of P_v is the space of an irreducible representation equivalent to ρ.*

Proof P_v is linear, and by (4.10), it intertwines representations. To see that it is injective, observe that $f_{u,v}$ is identically zero if and only if $\rho(x)u$ is orthogonal to v for every $x \in K$. But the subspace spanned by $\{\rho(x)u : x \in K\}$ is K-invariant. If $f_{u,v} = 0$, then this subspace is contained in $v^\perp$ and, as the representation is irreducible, it must be zero. This means that $u = 0$ if $f_{u,v} = 0$, showing the injectivity. □

The same statements are valid for maps of the type $v \mapsto f_{u,v}$ with u fixed, which intertwine ρ with left translations as a consequence of (4.9). The difference is that these maps are antilinear, as $(\cdot, \cdot)$ is a Hermitian product.

The following proposition, still related to equivalent representations, shows that the spaces $C(K)_\rho$ are the same for equivalent representations.

Proposition 4.10 *Let $\rho_1 : K \to \mathrm{Gl}(V_1)$ and $\rho_2 : K \to \mathrm{Gl}(V_2)$ be equivalent representations. Then, $C(K)_{\rho_1} = C(K)_{\rho_2}$.*

Proof Let $P : V_1 \to V_2$ be an isomorphism such that $P\rho_1(k) = \rho_2(k)P$ for every $k \in K$. Choose an invariant Hermitian product $(\cdot, \cdot)_2$ on V_2 and define $(u, v)_1 = (Pu, Pv)_2$, which is an invariant Hermitian product on V_1. If $u, v \in V_1$, then

$$\begin{aligned} f_{Pu,Pv}(x) &= (\rho_2(x)Pu, Pv)_2 = (P\rho_2(x)u, Pv)_2 \\ &= (\rho_1(x)u, v)_1 = f_{u,v}(x), \end{aligned}$$

showing that spaces $C(K)_{\rho_1}$ and $C(K)_{\rho_2}$ are equal. □

On the other hand, if the irreducible representations $\rho_1 : K \to \mathrm{Gl}(V_1)$ and $\rho_2 : K \to \mathrm{Gl}(V_2)$ are not equivalent, then the Schur orthogonality relations show that $C(K)_{\rho_1}$ is orthogonal to $C(K)_{\rho_2}$ on $L^2(K)$. In particular, $C(K)_{\rho_1} \cap C(K)_{\rho_2} = \emptyset$.

In the case of a nonirreducible representation in which V decomposes in invariant subspaces,

$$V = V_1 \oplus \cdots \oplus V_s,$$

it holds the sum $C(K)_\rho = \sum_{i=1}^{s} C(K)_{\rho_i}$, where ρ_i is the representation on V_i. In general, this sum is not direct, as $C(K)_{\rho_i} = C(K)_{\rho_j}$ if ρ_i and ρ_j are equivalent. However, if representations ρ_i are irreducible and are not equivalent to each other, then

$$C(K)_\rho = C(K)_{\rho_1} \oplus \cdots \oplus C(K)_{\rho_s}.$$

A function F on a space $C(K)_\rho$ is called a **representative function**, since by item (1), its left and right translates span a finite dimensional subspace of $C(K)$ in which K is represented by translations.

The space of representative functions will be denoted by $\mathcal{R}(K)$. This space is the sum of subspaces $C(K)_\rho$ with ρ running through finite dimensional continuous representations. This sum can be taken only on representations ρ that are irreducible.

The space $\mathcal{R}(K)$ is a sum of finite dimensional representation spaces of K. On the other hand, Proposition 4.9 shows that $\mathcal{R}(K)$ contains all irreducible representations of K with finite dimension. The Peter–Weyl theorem, which will be proved in the next section, ensures that $\mathcal{R}(K)$ is dense in $L^2(K)$ and thus, in a certain way, finite dimensional representations exhaust $L^2(K)$.

Example If $K = S^1 = \{z \in \mathbb{C} : |z| = 1\}$, then the irreducible representations have dimension 1. This implies that the representative functions associated with those representations are multiples of the representations themselves (that is, of their characters), which are homomorphisms of S^1 with values in $\mathbb{C}_\times$. □

4.4 Peter–Weyl Theorem

Theorem 4.11 *Let K be a Hausdorff compact group. Then, the space $\mathcal{R}(K)$ of representative functions is dense in $L^2(K)$.*

The Peter–Weyl theorem will be proved along this section. To begin the proof, suppose by contradiction that the closure E of $\mathcal{R}(K)$ is different from $L^2(K)$, so that its orthogonal complement $E^\perp \neq \{0\}$. The following lemma is proved under the assumption that the contradiction hypothesis holds.

Lemma 4.12 *There exists a continuous function $f \in E^\perp$, $f \neq 0$, that satisfies the following properties:*

1. *$f\left(kxk^{-1}\right) = f(x)$ for every $x, k \in K$, that is, f is constant on the conjugacy classes of K.*
2. *$\overline{f\left(x^{-1}\right)} = f(x)$.*

The proof of this lemma will be done later.

For the moment, the idea to arrive at a contradiction and, hence, at the Peter–Weyl theorem consists in constructing a finite dimensional continuous representation of ρ such that the function f of the lemma is not orthogonal to $C(K)_\rho$. This

representation ρ will be defined on a subspace V of $L^2(K)$ that is a finite dimensional eigenspace of a linear operator T of $L^2(K)$.

To do that, the following results from functional analysis, more specifically, from the theory of operators on Hilbert spaces, will be used:

1. Let H be a complex Hilbert space with Hermitian product $(\cdot, \cdot)$. If $T : H \to H$ is a compact operator (that is, the image of the unit ball is relatively compact) and self-adjoint (that is, $(Tu, v) = (u, Tv)$), then T admits an eigenvalue λ whose eigenspace V_λ has finite dimension. (As T is self-adjoint, its eigenvalues are real, though this information will not be used below.)
2. Let $N \in C(K \times K)$ be a continuous Map, and define the integral linear operator $T : L^2(K) \to L^2(K)$ by

$$T(g)(x) = \int_K N(x, y)\, g(y)\, \mu(dy) \qquad g \in L^2(K, \mu). \tag{4.11}$$

Then, T is compact.

Now, let f be as in Lemma 4.12, and define $N(x, y) = f\left(x^{-1}y\right)$, which in turn defines the linear operator T as in (4.11). By item (2) above, T is compact. To apply item (1), it is necessary to verify that T is self-adjoint, and this is obtained from the properties of function f. Indeed, given $g, h \in L^2(K)$, it holds

$$\begin{aligned}(Tg, h) &= \int_K \int_K N(x, y)\, g(y)\, \overline{h(x)}\mu(dy)\, \mu(dx) \\ &= \int_K g(y) \int_K N(x, y)\, \overline{h(x)}\mu(dx)\, \mu(dy).\end{aligned}$$

But, by property (2) of Lemma 4.12, $N(y, x) = f\left(y^{-1}x\right) = \overline{f\left(x^{-1}y\right)} = N(x, y)$. Hence, the above integral with respect to x is written as

$$\int_K N(x, y)\, \overline{h(x)}\mu(dx) = \int_K \overline{N(y, x)h(x)}\mu(dx).$$

Therefore,

$$\begin{aligned}(Tg, h) &= \int_K g(y) \overline{\int_K N(y, x)\, h(x)\, \mu(dx)}\mu(dy) \\ &= (g, Th),\end{aligned}$$

that is, T is self-adjoint.

With these preliminaries out of the way, let λ be the eigenvalue of T whose eigenspace V_λ is finite dimensional, as ensured by the two results stated above. (It is implicit that $\dim V_\lambda > 0$.)

The desired representation is given by the restriction to V_λ of the left translation $(L_k g)(x) = g(kx)$ with $g \in L^2(K, \mu)$ and $k, x \in K$.

Lemma 4.13 *If $g \in V_\lambda$, then $L_k g \in V_\lambda$.*

Proof If $g \in V_\lambda$ and $k \in K$, then

$$\begin{aligned}(TL_k g)(x) &= \int_K N(x, y)\, g(ky)\, \mu(dy) \\ &= \int_K N\left(x, k^{-1}y\right) g(y)\, \mu(dy)\end{aligned}$$

by the invariance of the Haar measure. But $N\left(x, k^{-1}y\right) = f\left(x^{-1}k^{-1}y\right) = N(kx, y)$. It follows that

$$\begin{aligned}(TL_k g)(x) &= \int_K N(kx, y)\, g(y)\, \mu(dy) \\ &= (Tg)(kx) = \lambda g(kx).\end{aligned}$$

This equation says that $TL_k g = \lambda L_k g$, and this means that V_λ is invariant by left translations, proving the lemma. □

By this lemma, the restriction of $k \mapsto L_{k^{-1}}$ defines a representation of K on V_λ. This representation is continuous. Indeed, by Proposition 4.8, if $u, v \in V_\lambda$, then the function $k \mapsto \left(L_{k^{-1}}u, v\right)$ is continuous. Taking an orthonormal basis $\{v_1, \ldots, v_n\}$ of V_λ, it is seen that the entries $\left(L_{k^{-1}}v_i, v_j\right)$ of the matrices of $L_{k^{-1}}$ with respect to this basis are continuous. Thus, the representation is continuous.

Hence, the representation on V_λ admits matrix entry functions. They are given, for $g, h \in V_\lambda$, by

$$f_{g,h}(y) = \left(L_{y^{-1}}g, h\right) = \int_K g\left(y^{-1}x\right)\overline{h(x)}\mu(dx).$$

The following calculation provides a simple expression for the Hermitian product between matrix entries $f_{g,g}$ and function f of Lemma 4.12. If $g \in V_\lambda$, then

$$\begin{aligned}\left(f, f_{g,g}\right) &= \int_K f(y)\,\overline{f_{g,g}(y)}\mu(dy) \\ &= \int_K \int_K f(y)\,\overline{g\left(y^{-1}x\right)}g(x)\,\mu(dx)\,\mu(dy).\end{aligned}$$

Changing the order of integration and using the invariance of $\mu(dy)$ by $y \mapsto y^{-1}$ and by left multiplication by x^{-1}, the last integral becomes

$$\begin{aligned}\left(f, f_{g,g}\right) &= \int_K \int_K f\left(xy^{-1}\right) \overline{g\left(y^{-1}\right)} g(x)\, \mu(dy)\, \mu(dx) \\ &= \int_K \int_K f\left(y^{-1}x\right) \overline{g(y)} g(x)\, \mu(dy)\, \mu(dx),\end{aligned}$$

since $f\left(x^{-1}y\right) = f\left(x^{-1}yx^{-1}x\right) = f\left(yx^{-1}\right)$. Changing once more the order of integrations, it is finally found that

$$\begin{aligned}\left(f, f_{g,g}\right) &= \int_K \left(\int_K f\left(y^{-1}x\right) g(x)\, \mu(dx)\right) \overline{g(y)} \mu(dy) \\ &= \int_K Tg(y)\, \overline{g(y)} \mu(dy) = (Tg, g) \\ &= \lambda \|g\|_2^2.\end{aligned}$$

This equation leads immediately to a contradiction with the hypothesis that the closure E of $\mathcal{R}(K)$ is proper. Indeed, by construction, $f \in E^{\perp}$, and thus $\left(f, f_{g,g}\right) = 0$ if $g \in V_\lambda$. But this implies that $\|g\|_2 = 0$, contradicting the fact that $V_\lambda \neq \{0\}$.

It remains to prove Lemma 4.12. To do that, two other lemmas will be proved.

Lemma 4.14 *Let $\mathcal{H} \subset L^2(K)$ be a closed subspace, invariant by left and right translations. Given $h \in \mathcal{H}$ and $g \in L^2(K)$, the functions $G(x) = \int_K g(y)\, h\left(y^{-1}x\right) \mu(dy)$ and $C(x) = \int_K h\left(yxy^{-1}\right) \mu(dy)$ belong to $\mathcal{H}$.*

Proof Take $u \in \mathcal{H}^{\perp}$. Then,

$$\begin{aligned}(G, u) &= \int_K \int_K g(y)\, h\left(y^{-1}x\right) \overline{u(x)} \mu(dy)\, \mu(dx) \\ &= \int_K \int_K g(y)\, h\left(y^{-1}x\right) \overline{u(x)} \mu(dx)\, \mu(dy) \\ &= \int_K g(y) \left(L_{y^{-1}} h, u\right) \mu(dy) = 0,\end{aligned}$$

since $L_{y^{-1}} h \in \mathcal{H}$. This shows that $G \in \left(\mathcal{H}^{\perp}\right)^{\perp}$, which coincides with $\mathcal{H}$, since this subspace is closed. The proof for C is analogous. □

Lemma 4.15 *Given $h \in L^2(K)$ and an open set $U \subset K$ with $1 \in U$, define*

$$h_U = \int_U h\left(y^{-1}x\right) \mu(dy) = \int_K \chi_U(y)\, h\left(y^{-1}x\right) \mu(dy),$$

where χ_U is the characteristic function of U. Then,

$$\|h_U - h\|_2 \leq \int_K \chi_U(y) \left\|L_{y^{-1}} h - h\right\|_2 \mu(dy).$$

Proof By definition,

$$\begin{aligned}\|h_U - h\|_2^2 &= \int_K |h_U(x) - h(x)|^2 \mu(dx)\\ &= \int_K |\int_K \chi_U(y)\left(h\left(y^{-1}x\right) - h(x)\right)\mu(dy)|^2 \mu(dx)\\ &= \int_K |\int_K \chi_U(y)\left(L_{y^{-1}}h(x) - h(x)\right)\mu(dy)|^2 \mu(dx).\end{aligned}$$

To simplify the notation, write $F(x,y) = \chi_U(y)\left(L_{y^{-1}}h(x) - h(x)\right)$. Then,

$$\begin{aligned}\|h_U - h\|_2^2 &= \int_K \int_K F(x,y)\,\mu(dy)\overline{\int_K F(x,z)\,\mu(dz)}\mu(dx)\\ &= \int_K \int_K \int_K F(x,y)\,F(x,z)\,\mu(dy)\,\mu(dz)\,\mu(dx)\\ &= \int_K \int_K \left(\int_K F(x,y)\,\overline{F(x,z)}\mu(dx)\right)\mu(dy)\,\mu(dz).\end{aligned}$$

For $y \in K$, define $\phi_y(x) = F(x,y)$, so that

$$\|h_U - h\|_2^2 = \int_K \int_K \left(\int_K \left(\phi_y, \phi_z\right)\mu(dx)\right)\mu(dy)\,\mu(dz).$$

This integral is real, and therefore

$$\begin{aligned}\|h_U - h\|_2^2 &= \int_K \int_K \left(\int_K \mathrm{Re}\left(\phi_y, \phi_z\right)\mu(dx)\right)\mu(dy)\,\mu(dz)\\ &\le \int_K \int_K \left(\int_K |\left(\phi_y, \phi_z\right)|\mu(dx)\right)\mu(dy)\,\mu(dz)\\ &\le \int_K \int_K \|\phi_y\|_2 \|\phi_z\|_2\,\mu(dy)\,\mu(dz)\\ &= \left(\int_K \|\phi_y\|_2\,\mu(dy)\right)^2.\end{aligned}$$

That is, $\|h_U - h\|_2 \le \int_K \|\phi_y\|_2\,\mu(dy)$. But, by the definition of ϕ_y, it follows that

$$\begin{aligned}\|h_U - h\|_2 &\le \int_K \left(\int_K |\chi_U(y)\left(L_{y^{-1}}h(x) - h(x)\right)|^2 \mu(dx)\right)^{1/2}\mu(dy)\\ &= \int_K \chi_U(y)\left(\int_K |\left(L_{y^{-1}}h(x) - h(x)\right)|^2 \mu(dx)\right)^{1/2}\mu(dy)\\ &= \int_K \chi_U(y)\left\|L_{y^{-1}}h - h\right\|\mu(dy),\end{aligned}$$

which is the desired inequality. □

Before beginning the proof of Lemma 4.12, it must be observed that Equations (4.9) and (4.10) show that space $\mathcal{R}(K)$ is invariant by left and right translations. By continuity (Proposition 4.8), the closure E of $\mathcal{R}(K)$ is also invariant by translations. As translations are isometries, the orthogonal space $E^{\perp}$ is also invariant.

The same invariance holds for the map that associates a function $f \in L^2(K)$ with the function $\overline{f\left(x^{-1}\right)}$. This is so because, if $f \in C(K)_{\rho}$, then function $\overline{f\left(x^{-1}\right)}$ is also in $C(K)_{\rho}$, since

$$\overline{f_{u,v}\left(x^{-1}\right)} = \overline{\left(\rho\left(x^{-1}\right)u, v\right)} = f_{v,u}(x).$$

Proof of Lemma 4.12 Take an arbitrary $h \in E^{\perp}$, $h \neq 0$. Given a neighborhood U of 1 in K, define

$$\begin{aligned} h_U(x) &= \int_U h\left(y^{-1}x\right)\mu(dy) \\ &= \int_K \chi_U(y)\, h\left(y^{-1}x\right)\mu(dy), \end{aligned}$$

where χ_U is the characteristic function of U. By Lemma 4.14, $h_U \in E^{\perp}$, because this subspace is closed and invariant by translations.

For each U, the function h_U is continuous. Indeed, if $x_1, x_2 \in K$, then, by the Cauchy–Schwarz inequality,

$$|h_U(x_1) - h_U(x_2)| \leq \left\| R_{x_1}h - R_{x_2}h \right\|.$$

Hence, the continuity of h_U follows from the continuity of $x \in K \mapsto R_x h$ (Proposition 4.8). Moreover, the neighborhood U can be chosen in such a way that h_U is close enough to h so as to make $h_U \neq 0$. This is so because $\|h_U - h\|_2 \leq \int_U \left\| L_{y^{-1}}h - h \right\|_2 \mu(dy)$ by Lemma 4.15, and map $y \mapsto L_{y^{-1}}h$ is continuous.

Once $h_U \neq 0$ is chosen, take $x_0 \in K$ such that $h_U(x_0) \neq 0$ and define $h_1 = L_{x_0}h_U$ and $h_2 = \overline{h_1(1)}h_1$. These functions also belong to $E^{\perp}$. They are continuous and not zero. Now, define

$$h_3(x) = \int_K h_2\left(yxy^{-1}\right)\mu(dy),$$

which by Lemma 4.14 is in $E^{\perp}$. This function is continuous, as it can be verified in the same way as for h_U. Moreover, $h_3\left(xyx^{-1}\right) = h_3(y)$ for every $y \in K$.

Finally, define $f(x) = h_3(x) + \overline{h_3\left(x^{-1}\right)}$. Then, f satisfies the conditions required by Lemma 4.12, concluding the proof. □

4.5 Exercises

1. Given a quaternion $q \neq 1$ such that $q^n = 1$, show that $1 + q + \cdots + q^{n-1} = 0$. Show also that

$$\int_{S^3} (p \cdot q)\, dp = 0$$

for any quaternion q, where dp is the Haar measure on S^3.

2. Let V be the space of an irreducible representation of a (Hausdorff) compact group K. Show that the set $\bigcup_{k,g \in K} f(kxg)$ spans $C(K)_\rho$ for any $f \in C(K)_\rho$, $f \neq 0$.
3. Let K be a compact Hausdorff group and $\rho : K \to \mathrm{Gl}(V)$ be a finite dimensional representation of K. Let $\rho(K) \cdot v$ be an orbit of this representation, and denote by $\mathrm{co}(\rho(K) \cdot v)$ the convex closure of this orbit. Show that there exists $w \in \mathrm{co}(\rho(K) \cdot v)$ that is a fixed point of K, that is, $\rho(k) w = w$ for every $k \in K$.
4. Let K be a compact Lie group and $\rho : K \to \mathrm{Gl}(V)$ an irreducible representation of K on a real, finite dimensional vector space V. Show that, if there exists a proper convex cone $C \subset V$ invariant by K (that is, $\rho(k) C \subset C$ for every $k \in K$), then $\dim V = 1$.
5. Find the characters of the irreducible representations of SU (2).

Part II
Lie Groups and Algebras

Overview

In this part of the book, the foundations of Lie group theory are established.

Chapter 5 introduces the Lie algebra of a Lie group. The concepts that relate the two structures of Lie group and Lie algebra are the exponential map and the adjoint representations of the Lie group and its Lie algebra. Both representations are realized by linear maps in the Lie algebra. The properties of the Lie groups are obtained from their Lie algebras, and vice versa, by mutual interaction of these three objects, and materialize concretely in formulas (5.8) and (5.9). The first of these formulas relates, through the exponential map, group conjugation with the adjoint representation of the group, while the second one couples the adjoint representations of the Lie group and the Lie algebra. The proofs in Chapter 5 freely use the existence and uniqueness theory of ordinary differential equations and Lie brackets of vector fields. This background material can be found in Appendix A. Also in Chapter 5 is a section on time-dependent ODEs on Lie groups, plus the construction of the Haar measure on Lie groups, which is carried out using invariant volume forms.

Chapter 6 deals with Lie subgroups of a Lie group and their quotients. The definition of Lie subgroup is the obvious one, namely a subgroup that is at the same time a differentiable submanifold such that the group multiplication becomes differentiable. There is a subtlety in this definition: the product should be differentiable with respect to the intrinsic differentiable structure of the submanifold rather than with respect to the ambient differentiable structure. This observation is addressed in detail with the help of the concept of quasi-regular (or quasi-embedded) submanifold, as defined in Appendix B. (This attention to such nuances at the beginning will become clear later on, when it will be explained that every subgroup that is at the same time a separable immersed submanifold is in fact a Lie subgroup.) The Lie algebra of a Lie subgroup is a Lie subalgebra. Conversely, the integrability theory of distributions allows to construct a unique connected Lie subgroup with a

given Lie subalgebra. This result establishes a one-to-one correspondence between connected Lie subgroups and Lie subalgebras.

One of the central results of Lie group theory is the celebrated closed subgroup theorem proved by Cartan. It ensures that if a subgroup is closed as a subset, then it is a Lie subgroup (with differentiable manifold structure built a posteriori.)

Another result in the same direction is a theorem due to Kuranishi and Yamabe, which shows that if a subgroup is path connected as a subset, then it is a Lie subgroup. These theorems are proved in Chapter 6, the last one with the stronger hypothesis that the paths are differentiable. Finally, the technique developed in the proof of Cartan's theorem allows to define a structure of differentiable manifold on a quotient space, which leads in particular to the definition of quotient Lie group. The proofs of this chapter make extensive use of the theory of distributions described in Appendix B.

Chapter 7 is of global nature and culminates in Theorem 7.15, which establishes the one-to-one correspondence between isomorphism classes of connected and simply connected Lie groups and isomorphism classes of Lie algebras. This result is complemented by Theorem 7.15, realizing any connected Lie group as a quotient of a connected and simply connected one. In such quotient, the kernel is a central discrete group (hence abelian, as contained in the center). The latter is also the result that provides us with classifications of Lie groups based on possible classifications of Lie algebras.

The proof of the theorem on simply connected groups requires the study of differentiable homomorphisms between Lie groups and how they are determined by the corresponding infinitesimal homomorphisms (the differentials at the origin).

A crucial result is the extension theorem ensuring that any Lie algebra homomorphism is the differential of a homomorphism between Lie groups, as long as the domain is a connected and simply connected Lie group. To prove this, one needs to construct a subgroup of the Cartesian product as candidate for the graph of the group homomorphism. In general, this subgroup will not be a graph. The projection on the first factor, however, is a covering map over the domain. For this reason, the assumption that the domain is simply connected guarantees that the subgroup of the Cartesian product is indeed the graph of a homomorphism. Regardless of this hypothesis on the domain, this method builds local homomorphisms between Lie groups, which then allows to show that Lie groups with isomorphic Lie algebras are locally isomorphic.

The reasoning in the proof of the extension theorem is also employed in an interesting application of Cartan's closed subgroup theorem, whereby a continuous homomorphism between Lie groups is differentiable.

The extension theorem ensures the uniqueness (up to isomorphism) of the connected and simply connected Lie group with a given Lie algebra. The proof of existence is carried out in two steps.

First of all, one proves that a given Lie algebra (real and finite dimensional) is the Lie algebra of one Lie group at least. This is shown here in an indirect way as an application of Ado's theorem, which proves that every finite dimensional Lie algebra is isomorphic to a matrix Lie algebra.

The second step in the existence proof applies the theory of covering spaces to manufacture a Lie group structure on the universal covering space of a Lie group. A summary of the theory of covering spaces is included at the end of the chapter.

Chapter 8 is devoted to the proof of two formulas of local character: a formula for the differential of the exponential map and the Baker–Campbell–Hausdorff formula. These expressions are given by power series involving the bracket of Lie algebra. The former will be used at a later stage to decide whether the exponential map on a Lie group is a diffeomorphism (or at least a local diffeomorphism). The Baker–Campbell–Hausdorff series, on the other hand, allows to construct a structure of analytic manifold on a Lie group in such a way that the product map, and other maps derived from it, becomes analytic. These relations are proved first for linear groups, where the terms in the series are given by matrix products. The proof for arbitrary Lie groups relies on Ado's theorem, which ensures the local isomorphism with a linear group.

Chapter 5
Lie Groups and Lie Algebras



The objective of this chapter is to introduce the concepts of Lie groups and their Lie algebras. The Lie algebra $\mathfrak{g}$ of a Lie group G is defined as the space of invariant vector fields (left or right, depending on choice), with bracket given by the Lie bracket of vector fields. The flows of invariant vector fields establish the exponential map $\exp : \mathfrak{g} \to G$, which is the main link between $\mathfrak{g}$ and G. These constructions make exhaustive use of results about vector fields on manifolds and their Lie brackets. A short collection of such results can be found in Appendix A. Adjoint representations are another tool for linking Lie groups and their Lie algebras. The formulas involving such representations are developed in this chapter. They are used along the whole theory.

5.1 Definition

A Lie group is a group whose underlying set has a differentiable manifold structure, in such a way that the product map

$$p : (g, h) \in G \times G \longmapsto gh \in G$$

is differentiable.

Both the differentiable manifold structure of G and the differentiability of p presuppose a degree of differentiability C^k, $1 \leq k \leq \omega$. For the development of most part of the theory, it is necessary to take first order derivatives in G and in the tangent bundle TG. So that it must be assumed that both G and p are of class C^2. However, there is no loss of generality in assuming that G and p are analytic (C^ω) because it is possible to show that, if p is of class C^2, then p is analytic with respect to an analytic manifold structure contained in the C^k structure, $2 \leq k \leq \infty$ (see

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Chapter 8).[1] Anyway, it is assumed that G is of class C^∞, as well as the product p, so that there is no harm in taking freely derivatives of any order.

Regarding the topology of the manifold underlying G, it is assumed that G is **paracompact** and is therefore metrizable by Riemannian metrics. This hypothesis is necessary when submanifolds and Lie subgroups of G are considered. (See Appendix B, Section B.5, about paracompact manifolds.)

Given $g \in G$, the left and right translations $L_g : G \to G$ and $R_g : G \to G$ are defined by $L_g(h) = gh$ and $R_g(h) = hg$, respectively. These maps are differentiable because $L_g = p \circ s_{g,1}$ and $R_g = p \circ s_{g,2}$, where $s_{g,1}(h) = (g, h)$ and $s_{g,2}(h) = (h, g)$ are differentiable maps $G \to G \times G$. Actually, both left and right translations are diffeomorphisms since $L_g \circ L_{g^{-1}} = R_g \circ R_{g^{-1}} = \mathrm{id}$. In the same way, the inner automorphisms $C_g = L_g \circ R_{g^{-1}}$, $g \in G$ are diffeomorphisms.

Unlike topological groups, the definition of Lie groups does not require a priori that the inverse $\iota(g) = g^{-1}$ is differentiable or continuous. The reason for this is that the differentiability of p implies that of ι through the implicit function theorem, as will be proved in the following proposition.

In what follows, the differential of a map f at a point x is denoted by df_x.

Proposition 5.1 *On a Lie group G, the map $\iota : g \in G \mapsto g^{-1} \in G$ is a diffeomorphism. The differential of ι is given by*

$$d\iota_g = -\left(dL_{g^{-1}}\right)_1 \circ \left(dR_{g^{-1}}\right)_g .$$

In particular, $(d\iota)_1 = -\mathrm{id}$.

Proof Given $(g, h) \in G \times G$, the partial differential of the product p with respect to the second variable is

$$\partial_2 p_{(g,h)} = d\left(L_g\right)_h .$$

As L_g is a diffeomorphism, it follows that $d\left(L_g\right)_h$ is bijective and, in particular, surjective. The implicit function theorem then ensures that, for a fixed $c \in G$, the equation $p(g, h) = c$ has a local differentiable solution $h = \phi_c(g)$, writing h as a function of g, that is, $p(g, \phi_c(g)) = c$. When $c = 1$, $\phi_1 = \iota$, showing that ι is differentiable. It then follows that ι is a diffeomorphism because its inverse ι^{-1} coincides with ι, that is, $\iota \circ \iota = \mathrm{id}$.

Also by the implicit function theorem, the differential $d\iota_g$ is given by

$$d\iota_g = -\left(\partial_2 p\right)^{-1}_{(g,g^{-1})} \circ \left(\partial_1 p\right)_{(g,g^{-1})} ,$$

[1] **Hilbert's fifth problem** (of 23 problems publicized in 1900) asks which topological groups are differentiable. A solution to this problem shows that a topological group is a Lie group if it is a (locally Euclidean) topological manifold. More generally, a locally compact group is a Lie group if it does not admit "small subgroups" (any neighborhood of the identity contains only the trivial subgroup). See Montgomery and Zippin [40] and Yang [60].

where $\left(\partial_j p\right)_{(x,y)}$ denotes the differential of p with respect to variable $j = 1, 2$ at point (x, y). These partial differentials are given by $(\partial_2 p)_{(x,y)} = d\left(L_x\right)_y$ and $(\partial_1 p)_{(x,y)} = d\left(R_y\right)_x$. Therefore, $(\partial_1 p)_{(g,g^{-1})} = d\left(R_{g^{-1}}\right)_g$ and $(\partial_2 p)^{-1}_{(g,g^{-1})} = \left(d\left(L_g\right)_{g^{-1}}\right)^{-1} = d\left(L_{g^{-1}}\right)_1$ and from these results the formula in the statement follows.

Finally, in case $g = 1$, $R_1(g) = g$ is the identity map and hence $(dR_1)_1$ is the identity map on tangent space T_1G. In the same way, $(dL_1)_1 = \mathrm{id}$, and it then follows that $d\iota_1 : T_1G \to T_1G$ is $-\mathrm{id}$. □

The proposition above shows that every Lie group is a topological group, as defined in Chapter 2.

In many situations, it is convenient to use a simplified notation for the differentials of translations on a Lie group G. Let $t \mapsto g_t$ be a differentiable curve on G and take $h \in G$. Using the following notations:

$$hg'_t = d\left(L_h\right)_{g_t}\left(g'_t\right), \qquad g'_t h = d\left(R_h\right)_{g_t}\left(g'_t\right),$$

the computation of derivatives on G can be carried out as if they were done on a matrix group. For instance, $\left(g_t^2\right)' = g'_t g_t + g_t g'_t$ and from $g_t g_t^{-1} = 1$ it follows that $\left(g_t g_t^{-1}\right)' = g'_t g_t^{-1} + g_t\left(g_t^{-1}\right)' = 0$. Hence,

$$\left(g_t^{-1}\right)' = -g_t^{-1} g'_t g_t^{-1},$$

which is the formula for $d\iota_g$ in the last proposition.

Let G and H be Lie groups. Then, the Cartesian product $G \times H$ admits the product manifold structure and the product group structure $(g_1, h_1)(g_2, h_2) = (g_1 g_2, h_1 h_2)$, making $G \times H$ a Lie group. Indeed, the differentiability of the product is a consequence of the fact that each coordinate is differentiable. More generally, if G_i, $i = 1, \ldots, k$, is a finite number of Lie groups, then the direct product $G_1 \times \cdots \times G_k$ is a Lie group where the group and differentiable manifold structures are the product ones.

Other constructions with Lie groups, such as the quotient of a group by a subgroup and the semi-direct product, will be made later.

Examples

1. Let G be any group endowed with the discrete topology. With this topology, G has a differentiable manifold structure of dimension 0 in which the product is differentiable. It may be an infinite group of any cardinality since the hypothesis of paracompactness does not restrict the cardinality of connected components.

   Therefore, every group G can be viewed as a Lie group. A Lie group with discrete topology is called a **discrete Lie group**. (This is a purely formal example since the discrete differentiable structure does not add any information to the algebraic structure of group G.)

2. If G is a Lie group, then its connected components are open submanifolds. In particular, the connected component of the identity, G_0, is an open and closed normal subgroup. The restriction to G_0 of the product on G is differentiable (as G_0 is open) and this makes G_0 a Lie group.
3. Any finite dimensional vector space V over $\mathbb{R}$ is an abelian Lie group with the operation $+$ on V. In particular, $(\mathbb{R}, +)$ is a Lie group. The multiplicative groups $\mathbb{R}_\times = \mathbb{R} \setminus \{0\}$ and $\mathbb{C}_\times = \mathbb{C} \setminus \{0\}$ are also Lie groups.
4. Let $\mathrm{Gl}\,(n, \mathbb{R})$ be the group of invertible linear maps of $\mathbb{R}^n$ or, what is the same, the group of $n \times n$ invertible matrices. This group is an open subset of the vector space $M_n\,(\mathbb{R})$ of $n \times n$ matrices and is therefore a differentiable manifold. The group product in $\mathrm{Gl}(n, \mathbb{R})$ comes from the usual matrix product. If $X = \left(x_{ij}\right)$ and $Y = \left(y_{ij}\right)$ are $n \times n$ matrices, then

$$z_{ij} = \sum_{k=1}^{n} x_{ik} y_{kj},$$

which is a polynomial of degree two in variables x_{ij}, y_{ij} and hence a differentiable map. For this reason, $\mathrm{Gl}\,(n, \mathbb{R})$ is a Lie group. If V is a finite dimensional real vector space, denote by $\mathrm{Gl}\,(V)$ the group of invertible linear maps of V. Taking a basis of V, an isomorphism between $\mathrm{Gl}\,(V)$ and $\mathrm{Gl}\,(n, \mathbb{R})$ is defined by $h \in \mathrm{Gl}\,(V) \mapsto [h] \in \mathrm{Gl}\,(n, \mathbb{R})$, where $[h]$ denotes the matrix for h with respect to the fixed basis. By this isomorphism, $\mathrm{Gl}\,(V)$ is a Lie group.
5. Let $\mathcal{A}$ be an **associative algebra** over $\mathbb{R}$, that is, $\mathcal{A}$ is a real vector space endowed with a product $\cdot : \mathcal{A} \times \mathcal{A} \to \mathcal{A}$, which is bilinear (distributive) and associative. Suppose that $\dim \mathcal{A} < \infty$ and that $\mathcal{A}$ has a multiplicative identity 1. An element $x \in \mathcal{A}$ admits a bilateral inverse if there exists $y \in \mathcal{A}$ such that $xy = yx = 1$. In this case, $y = x^{-1}$ is unique. The set $G\,(\mathcal{A})$ of invertible elements of $\mathcal{A}$,

$$G\,(\mathcal{A}) = \{x \in \mathcal{A} : \exists\, x^{-1}\},$$

is a group with the product of $\mathcal{A}$. On the other hand, $G\,(\mathcal{A})$ is a (nonempty) open subset of $\mathcal{A}$ (with real vector space topology). Indeed, consider the map $E : \mathcal{A} \to L\,(\mathcal{A})$ which associates with $x \in \mathcal{A}$ the left translation $L_x : \mathcal{A} \to \mathcal{A}$, $L_x\,(y) = x \cdot y$, which is a linear map of $\mathcal{A}$. The map E is a homomorphism of associative algebras: It is linear and $L_{xy} = L_x \circ L_y$. In particular, this implies that $L_{x^{-1}} = (L_x)^{-1}$ when $x \in G\,(\mathcal{A})$. Moreover, the existence of a multiplicative identity ensures that E is injective, since $L_x = 0$ implies that $0 = L_x\,(1) = x \cdot 1 = x$. Therefore, the function $\det L_x$ is a nonzero polynomial on $\mathcal{A}$. As $\det L_x \neq 0$ if and only if $x \in G\,(\mathcal{A})$, it follows that $G\,(\mathcal{A})$ is a nonempty open subset (which, moreover, is dense). Hence, $G\,(\mathcal{A})$ is a Lie group since the product, being a bilinear map of a finite dimensional space, is differentiable.

It is clear that $\mathrm{Gl}\,(n, \mathbb{R})$ is the particular case in which $\mathcal{A}$ is the associative algebra of $n \times n$ matrices.

6. A particular case of the previous example is the algebra $\mathbb{H}$ of **quaternions** with real coefficients, which has dimension 4 and is spanned by $\{1, i, j, k\}$, where 1 is the multiplicative identity and the remaining generators satisfy

$$ij = -ji = k,\ jk = -kj = i,\ ki = -ik = j,\ i^2 = j^2 = k^2 = -1.$$

This is an associative algebra and every nonzero element in $\mathbb{H}$ admits an inverse. Indeed, the conjugate of $q = a + bi + cj + dk$ is defined by $\overline{q} = a - bi - cj - dk$. Then, it holds that

$$q\overline{q} = |q|^2 = a^2 + b^2 + c^2 + d^2,$$

from where it follows that $q\left(\overline{q}/|q|^2\right) = 1$, showing that the inverse of $q \neq 0$ is given by $q^{-1} = \overline{q}/|q|^2$. Therefore, $\mathbb{H}_\times = \mathbb{H} \setminus \{0\}$ is a Lie group, since the product is a polynomial map.

The algebra of quaternions generalizes into the (associative) Clifford algebras, which are not constructed here. The same construction of Lie groups applies to such algebras.

7. Let G be the group of $n \times n$ upper triangular matrices with diagonal entries equal to 1:

$$G = \left\{ \begin{pmatrix} 1 & \cdots & * \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \end{pmatrix} \right\}.$$

The set G is in bijection with the vector space $\mathbb{R}^{n(n-1)/2}$. Hence, G has a differentiable manifold structure. With respect to this structure, the product in G (matrix product) is differentiable, making G a Lie group. □

Later, several results will be proved which ensure that certain subgroups of Lie groups are also Lie groups. On the basis of such results, it will be easy to produce a wide range of examples of Lie groups.

The tangent bundle TG and the cotangent bundle T^*G of a Lie group G are easily described by the left and right translations in G. Indeed, given $g \in G$, the differential of the left translation $d\left(L_g\right)_1$ is an isomorphism between T_1G and T_gG, since L_g is a diffeomorphism. For this reason, the map

$$(g, v) \in G \times T_1G \longmapsto d\left(L_g\right)_1(v) \in TG$$

is a bijection. This map can be rewritten as $\partial_2 p\,(g, 1)\,(v)$, so it can be seen that it is differentiable, as well as p. Its inverse is given by

$$v \in TG \mapsto \left(\pi(v), dL_{\pi(v)^{-1}} v\right) \in G \times T_1 G,$$

where $\pi : TG \to G$ is the canonical projection. This inverse is also differentiable, showing that TG is diffeomorphic to $G \times T_1 G$.

In the same way,

$$(g, v) \in G \times T_1 G \longmapsto d\left(R_g\right)_1 (v) \in TG$$

defines a diffeomorphism between $G \times T_1 G$ and TG, identifying TG with $G \times T_1 G$ through right translations.

A similar identification occurs with the cotangent bundle T^*G. Given $g \in G$, the pullbacks $d\left(L_{g^{-1}}\right)_1^* : T_1^*G \to T_g^*G$ and $d\left(R_{g^{-1}}\right)_g^* : T_1^*G \to T_g^*G$ are isomorphisms and define the diffeomorphisms

- $(g, \alpha) \in G \times T_1^*G \longmapsto d\left(L_{g^{-1}}\right)_1^* (\alpha) \in T^*G$,
- $(g, \alpha) \in G \times T_1^*G \longmapsto d\left(R_{g^{-1}}\right)_1^* (\alpha) \in T^*G$.

In other words, the tangent and cotangent bundles of Lie groups are trivial. This triviality allows the definition of **Maurer–Cartan forms**, which are differential 1-forms on G with values in $T_1 G$. They are defined by right or left translations by

$$\omega_g^r (v) = d\left(R_{g^{-1}}\right)_g (v) \qquad \text{and} \qquad \omega_g^l (v) = d\left(L_{g^{-1}}\right)_g (v)$$

for $g \in G$ and $v \in T_g G$.

A differentiable manifold M whose tangent bundle TM is trivial is called **parallelizable**. The differentiable manifolds underlying Lie groups are parallelizable, and this shows that not every manifold admits a Lie group structure. For instance, the sphere S^2 is not a parallelizable manifold, therefore there does not exist on S^2 a differentiable product that satisfies the group axioms. Along the development of the theory, other topological restrictions to the underlying manifold of a Lie group will appear. One of them is that the fundamental group $\pi_1(G)$ must be abelian (see Chapter 7).

5.2 Lie Algebra of a Lie Group

The first step in the study of Lie is the construction of the associated Lie algebras. A Lie algebra is a vector space $\mathfrak{g}$ endowed with a product (bracket) $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$ that satisfies the properties:

1. The bracket $[\cdot, \cdot]$ is bilinear, that is, it is linear in each of its variables.
2. Skew-symmetry, that is $[X, Y] = -[Y, X]$, for $X, Y \in \mathfrak{g}$.
3. Jacobi identity: for $X, Y, Z \in \mathfrak{g}$,

$$[X, [Y, Z]] = [[X, Y], Z] + [Y, [X, Z]].$$

A subspace $\mathfrak{h} \subset \mathfrak{g}$ of a Lie algebra $\mathfrak{g}$ is a **Lie subalgebra** if it is closed under the bracket. In this case, $\mathfrak{h}$ is also a Lie algebra.

An example of a Lie algebra is given by the vector space of vector fields on a differentiable (C^∞) manifold endowed with the Lie bracket of vector fields. Another example is the algebra $\mathfrak{gl}(n, \mathbb{R})$ formed by real $n \times n$ matrices with bracket given by the matrix commutator.

$$[A, B] = AB - BA.$$

In what follows, the Lie algebra of a Lie group G will be defined as a subalgebra of the Lie algebra of vector fields on G, formed by invariant vector fields on G.

5.2.1 Invariant Vector Fields

Definition 5.2 Let G be a Lie group. A vector field X on G is said to be

- **right invariant** if, for every $g \in G$, $\left(R_g\right)_* X = X$. In detail,

$$d\left(R_g\right)_h (X(h)) = X(hg)$$

  for every $g, h \in G$;
- **left invariant** if, for every $g \in G$, $\left(L_g\right)_* X = X$, that is,

$$d\left(L_g\right)_h (X(h)) = X(gh).$$

Right or left invariant vector fields are completely determined by their values at the identity $1 \in G$ because, for every $g \in G$, the condition for right invariance, for instance, implies that $X(g) = d\left(R_g\right)_1 (X(1))$. Therefore, each element in tangent space $T_1 G$ determines a unique right invariant vector field and a unique left invariant vector field.

Given $X \in T_1 G$, the notation X^r denotes the right invariant vector field such that $X^r(1) = X$, while X^l denotes the corresponding left invariant vector field. Explicitly,

$$X^r(g) = d\left(R_g\right)_1 (X) \qquad X^l(g) = d\left(L_g\right)_1 (X).$$

Denote by Inv^r the set of right invariant vector fields. This set is a vector subspace (over $\mathbb{R}$) of the space of all vector fields on G, since $\left(R_g\right)_*$ is a linear map on the vector fields. Analogously, the set Inv^l of left invariant vector fields is also a vector subspace (in general, different from the subspace of right invariant vector fields). The maps $X \in T_1 G \mapsto X^r \in \mathrm{Inv}^r$ and $X \in T_1 G \mapsto X^l \in \mathrm{Inv}^l$ are

isomorphisms between the corresponding vector spaces, whose inverses are given by $X^{r,l} \in \mathrm{Inv}^{r,l} \mapsto X^{r,l}(1) \in T_1 G$.

Examples

1. Let $G = \mathrm{Gl}(n, \mathbb{R})$ be the general linear group, which is an open set in the vector space of $n \times n$ matrices $M_n(\mathbb{R})$. Fixing $g \in G$, the left and right translations $L_g(h) = gh$ and $R_g(h) = hg$ are restrictions to $\mathrm{Gl}(n, \mathbb{R})$ of linear maps of $M_n(\mathbb{R}) = \mathbb{R}^{n^2}$. The tangent bundle of G is identified with $G \times M_n(\mathbb{R})$. It thus follows that a vector field X in G is a map $X : G \to M_n(\mathbb{R})$. Moreover, through this identification, the linear maps L_g and R_g satisfy $d\left(L_g\right)_h = L_g$ and $d\left(R_g\right)_h = R_g$ for every $g, h \in G$.

   With these observations, it is possible to describe the invariant vector fields on $\mathrm{Gl}(n, \mathbb{R})$. Suppose that $X : G \to M_n(\mathbb{R})$ is right invariant. Then, for every $g \in G$,

$$X(g) = d\left(R_g\right)_1 (X(1)) = R_g(X(1)) = X(1)\, g.$$

   Hence, all right invariant vector fields have the form $X(g) = Ag$, with A a matrix in $T_1 G$. The differential equation defined by X is the linear system

$$\frac{dg}{dt} = Ag$$

   in the space of matrices. The flow of X is given by $X_t(g) = e^{tA} g$, where $e^A = \sum_{k \geq 0} \frac{1}{k!} A^k$ is the matrix exponential.

   Analogously, left invariant fields have the form $X(g) = gA$, and are associated with linear systems $\dfrac{dg}{dt} = gA$. Their flows have the form $X_t(g) = ge^{tA}$.

   In $\mathrm{Gl}(n, \mathbb{R})$ there exist left invariant vector fields which are not right invariant and vice versa. Indeed, suppose that the vector field $X(g) = Ag$ coincides with the vector field $Y(g) = gB$, that is, $Ag = gB$ for every $g \in G$. In particular, if $g = 1$, then $A = B$. It then follows that $Ag = gA$ and, therefore, A commutes with all matrices in $\mathrm{Gl}(n, \mathbb{R})$. But this occurs if and only if $A = a \cdot 1$, $a \in \mathbb{R}$, that is, A is a scalar matrix. Hence, the right invariant vector field $X(g) = Ag$ is not left invariant if A is not a scalar matrix.
2. The group $G(\mathcal{A})$ of invertible elements of an associative algebra $\mathcal{A}$ is a Lie group (see Example 5 of the previous section). In the same way as with $\mathrm{Gl}(n, \mathbb{R})$, left and right translations are linear and their flows are determined by the exponential $e^A = \sum_{k \geq 0} \frac{1}{k!} A^k$, where the powers are given by the product in $\mathcal{A}$. This is so because the series $e^{tA} = \sum_{k \geq 0} \frac{1}{k!} t^k A^k$ is a power series in $t \in \mathbb{R}$ with radius of convergence ∞, whose derivative is given by the sum of the derivatives of its terms.

3. As a particular case of the previous item, the right translations for quaternions $\mathbb{H}_\times = \mathbb{H} \setminus \{0\}$ are restrictions of linear maps and, for this reason, right invariant vector fields have the form $X_q(x) = qx$, with $q \in \mathbb{H}$, and the exponential is given by $e^q = \sum_{k\geq 0} \frac{1}{k!} q^k$.
4. Let $G = (\mathbb{R}^n, +)$. Fixing $v \in \mathbb{R}^n$, the left and right translations coincide and are given by

$$L_v(x) = R_v(x) = x + v.$$

Hence, $d(L_v)_y = d(R_v)_y = \mathrm{id}$ for every $y \in \mathbb{R}^n$. This means that invariant vector fields are constant, that is, $X(x) = v$, with $v \in \mathbb{R}^n$ fixed. The corresponding differential equation is $\dot{x} = v$, and its flow is the translation $X_t(x) = x + tv$. □

The Lie algebra of a Lie group is defined on both invariant vector field spaces Inv^r and Inv^l endowed with the Lie bracket. The following lemma states this in precise terms.

Lemma 5.3 *Let X and Y be right invariant vector fields on a Lie group G. Then the Lie bracket $[X, Y]$ is right invariant. The same holds for left invariant vector fields.*

Proof It is a consequence of the following general formula: Let M be a manifold, X, Y vector fields on M and ϕ a diffeomorphism of M. Then, $\phi_*[X, Y] = [\phi_* X, \phi_* Y]$ (see Appendix A). Applying this formula to $\phi = R_g$ (or L_g) and invariant vector fields X, Y, the invariance of the bracket follows. □

In other words, the spaces Inv^r and Inv^l are Lie subalgebras of the Lie algebra of all vector fields on G. In particular, both vector spaces admit Lie algebra structures. The Lie algebra of the Lie group G is any of the Lie algebras Inv^r or Inv^l.

The following arguments show that these Lie algebras are essentially the same, that is, they are isomorphic, so that there is no ambiguity in this terminology.

The tangent space $T_1 G$ is isomorphic to both Inv^r and Inv^l. Through isomorphisms, the Lie bracket, restricted to subspaces of invariant vector fields, induces brackets $[\cdot, \cdot]_r$ and $[\cdot, \cdot]_l$ on $T_1 G$. These brackets are given, for $A, B \in T_1 G$, by

- $[A, B]_r = [A^r, B^r](1)$,
- $[A, B]_l = [A^l, B^l](1)$.

The following lemma allows to relate them.

Lemma 5.4 *Let $A \in T_1 G$ and $\iota(g) = g^{-1}$ the inverse in G. Then,*

$$(\iota)_*\left(A^r\right) = (-A)^l \qquad \text{and} \qquad (\iota)_*\left(A^l\right) = (-A)^r.$$

In detail: $(d\iota)_{g^{-1}}\left(A^r\left(g^{-1}\right)\right) = -A^l(g)$ and $(d\iota)_{g^{-1}}\left(A^l\left(g^{-1}\right)\right) = -A^r(g)$.

Proof Write $Y = (\iota)_* (A^r)$. Then,

$$\left(L_g\right)_* (Y) = \left(L_g\right)_* (\iota)_* \left(A^r\right).$$

Using the chain rule and the equality $L_g \circ \iota = \iota \circ R_{g^{-1}}$, it follows that $\left(L_g\right)_* (Y) = \left(\iota \circ R_{g^{-1}}\right)_* (A^r)$. The right hand side of this equality is $(\iota)_* (A^r) = Y$ by the chain rule and the fact that A^r is right invariant. Therefore, $\left(L_g\right)_* (Y) = Y$, and hence Y is left invariant. Now, $Y(1) = (d\iota)_1 (A^r(1))$. But $A^r(1) = A$ and $(d\iota)_1 = -\mathrm{id}$, so that $Y(1) = -A$, showing that $Y = (-A)^l$. The other equality is proved in the same way, or applying ι_* to the equality already proved. □

Proposition 5.5 *Let $A, B \in T_1 G$. Then, $[A, B]_r = -[A, B]_l$.*

Proof By definition, $[A, B]_r = [A^r, B^r](1)$. Applying $d\iota_1 = -\mathrm{id}$ to this equality yields

$$-[A, B]_r = (d\iota)_1 [A, B]_r = (d\iota)_1 \left([A^r, B^r](1)\right).$$

But by the previous lemma (and by the homomorphism property of ι_*), $\iota_*[A^r, B^r] = [\iota_* A^r, \iota_* B^r] = [A^l, B^l]$. Hence,

$$-[A, B]_r = [A^l, B^l](1) = [A, B]_l,$$

concluding the proof. □

Changing viewpoint a little, this proposition shows that the Lie algebra structures on $T_1 G$ induced by Inv^r and Inv^l are isomorphic in the sense that there exists an invertible linear map $L : T_1 G \to T_1 G$ such that $L[A, B]_r = [LA, LB]_l$. Indeed, take $L = -\mathrm{id}$. Then, $L[A, B]_r = -[A, B]_r$, while $[LA, LB]_l = [A, B]_l$, so $-\mathrm{id}$ is an isomorphism. Viewed in a different way, the previous proposition shows that the map ι_* defines an isomorphism between Inv^r and Inv^l endowed with the Lie bracket of vector fields.

Definition 5.6 The **Lie algebra** of G, denoted by $\mathfrak{g}$ or $\mathcal{L}(G)$ is any of the isomorphic Lie algebras Inv^r, Inv^l, $(T_1 G, [\cdot, \cdot]_r)$ or still $(T_1 G, [\cdot, \cdot]_l)$.

The isomorphism between $\mathrm{Inv}^{r,l}$ and $T_1 G$ reflects in the notation used, in which $X \in \mathfrak{g}$ means both an element of $T_1 G$ and an invariant vector field.

For the development of the theory, it is possible to choose any of these Lie algebras to represent $\mathfrak{g}$. The main difference lies, of course, in the choice between right and left invariance.

Due to these choices there are two parallel sets of formulas. They are obtained from each other symmetrically by applying the minus sign that relates the right and left brackets.

A criterion for the choice between left or right invariant vector fields arises in the study of group actions. If left actions are considered (this happens when the value of a function f is written as $f(x)$), then right invariant vector fields should

be considered. On the other hand, left invariant vector fields are the correct setting for right actions (when a function f is written as $(x)\, f$).

Examples

1. As shown in the examples of the previous section, right invariant vector fields in $\mathrm{Gl}(n, \mathbb{R})$ have the form $X_A(g) = Ag$, with A an $n \times n$ matrix, while left invariant fields have the form $Y_A(g) = gA$.

   In local coordinates, the Lie bracket of two vector fields is given by

   $$[X, Y] = dY(X) - dX(Y)$$

   (see Appendix A, Proposition A.5). For a matrix A, the vector field X_A extends to a linear map on the space of matrices. Therefore, $dX_A = X_A$. Thus, applying this bracket formula to X_A and X_B, one obtains

   $$[X_A, X_B](g) = B(Ag) - A(Bg),$$

   that is, $[X_A, X_B] = X_{BA-AB}$. On the other hand, the Lie bracket of left invariant vector fields is given by $[Y_A, Y_B] = X_{AB-BA}$. In this way the Lie algebras Inv^r and Inv^l are identified with the space of $n \times n$ matrices. On Inv^r, the bracket is given by $[A, B] = BA - AB$, while on Inv^l the bracket is given by $[A, B] = AB - BA$.
2. If $\mathcal{A}$ is an associative Lie algebra, then the Lie algebra of the Lie group $G(\mathcal{A})$ is given by commutators on $\mathcal{A}$, in the same way as on $\mathrm{Gl}(n, \mathbb{R})$.
3. For the group of quaternions $\mathbb{H}_\times$, its Lie algebra is $\mathbb{H}$ itself, with bracket given by the commutator

   $$[p, q] = qp - pq.$$

4. The invariant vector fields on $(\mathbb{R}^n, +)$ are the constant vector fields. As the Lie bracket of constant vector fields is zero (as follows from the bracket formula in coordinates, see Proposition A.5), the Lie algebra of the abelian group $(\mathbb{R}^n, +)$ is abelian, that is, it satisfies $[\cdot, \cdot] \equiv 0$.
5. If G and H are Lie groups with Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$, respectively, then the Lie algebra of $G \times H$ is $\mathfrak{g} \times \mathfrak{h}$, where the bracket is given by

   $$[(X_1, Y_1), (X_2, Y_2)] = ([X_1, X_2], [Y_1, Y_2]).$$

   More generally, the Lie algebra of a direct product $G_1 \times \cdots \times G_k$ is the direct product $\mathfrak{g}_1 \times \cdots \times \mathfrak{g}_k$ of its Lie algebras, where the bracket is given componentwise.
6. If G is a discrete Lie group, $\dim G = 0$ and hence $\mathfrak{g} = \{0\}$. □

Other examples of Lie algebras of Lie groups will be presented in the next chapter, on Lie subgroups of Lie groups.

5.3 Exponential Map

The exponential map $\exp : \mathfrak{g} \to G$ is the central object used to transfer to a Lie group G the properties of its Lie algebra $\mathfrak{g}$. The basic idea in its construction is that, by definition, the elements of $\mathfrak{g}$ are ordinary differential equations on G (invariant vector fields) that have flows, which are formed by local diffeomorphisms of G. The elements forming these flows are naturally identified with elements of G, allowing to construct, from $X \in \mathfrak{g}$, a subgroup of G parameterized by $t \in \mathbb{R}$ (1-parameter subgroup). The exponential map is constructed from these subgroups.

To put these comments precisely, let X be a left or right invariant vector field on G. Denote by X_t its flow. In principle, X_t is a local flow, that is, for a fixed t, the domain $\mathrm{dom}X_t$ of X_t is the open subset of G of initial conditions whose solutions can be extended to t.

The invariance of X entails the following symmetry of the flow X_t: Suppose, for instance, that $X \in \mathrm{Inv}^r$, take $g, h \in G$ with $h \in \mathrm{dom}X_t$ and consider the curve $\alpha(t) = X_t(h)\, g$. Its domain is an open interval of $\mathbb{R}$ containing 0, with $\alpha(0) = hg$, since $X_0(h) = h$. Moreover, by the chain rule, $\alpha'(t) = d\left(R_g\right)_{X_t(h)}(X(X_t(h)))$ and, as X is right invariant, it follows that

$$\alpha'(t) = X(X_t(h)\, g) = X(\alpha(t)).$$

Hence, α is a solution of $dg/dt = X(g)$ with initial condition $\alpha(0) = hg$, that is, $\alpha(t) = X_t(hg)$. This means that

$$X_t(hg) = X_t(h)\, g \qquad X \in \mathrm{Inv}^r. \tag{5.1}$$

In particular, taking $h = 1$ yields $X_t(g) = X_t(1)\, g$. That is, the trajectory through g is obtained by right translation of the trajectory starting at the identity.

Analogously, it is shown that

$$gY_t(h) = Y_t(gh) \qquad Y \in \mathrm{Inv}^l \tag{5.2}$$

and $Y_t(g) = gY_t(1)$ if Y is a left invariant vector field.

Since trajectories are obtained from each other by translations, they are extended to the same interval of $\mathbb{R}$, that is, all maximal solutions of invariant vector fields have the same intervals of definition. This allows to show that invariant vector fields are complete, that is, their trajectories are extended to $(-\infty, +\infty)$.

Proposition 5.7 *A (left or right) invariant vector field is complete.*

Proof Take a right invariant vector field X whose flow is denoted by X_t. Let (α, ω), with $\alpha < 0$ and $\omega > 0$, be the common interval of maximal solutions $t \mapsto X_t(g)$, $g \in G$. Suppose by contradiction that $\omega < \infty$. Define the curves

$$\begin{cases} x(t) = X_t(1) & t \in (\alpha, \omega). \\ y(t) = X_{t-\omega/2}\left(X_{\omega/2}(1)\right) & t \in (\alpha + \omega/2, 3\omega/2). \end{cases}$$

Both are solutions of the differential equation $\dot{g} = X(g)$. As $x(\omega/2) = y(\omega/2) = X_{\omega/2}(1)$, the uniqueness of solutions ensures that $x(t) = y(t)$ on the interval $(\alpha + \omega/2, \omega)$, which is the intersection of the domains of definition of the curves. Hence, both curves define a solution $z(t)$, whose domain of definition is the union $(\alpha, 3\omega/2)$ of the intervals of definition of x and y. As $z(0) = 1$, this contradicts the fact that the interval of the maximal solution is (α, ω). Therefore, $\omega = \infty$. In the same way, $\alpha = -\infty$, concluding the proof. □

Another consequence of invariance properties (5.1) and (5.2) are the following equalities:

- If $X \in \mathrm{Inv}^r$, then $X_{t+s}(1) = X_t(X_s(1)) = X_t(1) X_s(1) = X_s(1) X_t(1)$.
- If $Y \in \mathrm{Inv}^l$, then $Y_{t+s}(1) = Y_t(Y_s(1)) = Y_t(1) Y_s(1) = Y_s(1) Y_t(1)$.

These equalities imply that $X_{-t}(1) = (X_t(1))^{-1}$ and $Y_{-t}(1) = (Y_t(1))^{-1}$. Hence, if $X \in \mathrm{Inv}^r$ and $Y \in \mathrm{Inv}^l$, then their trajectories through the origin,

$$\{X_t(1) : t \in \mathbb{R}\} \qquad \text{and} \qquad \{Y_t(1) : t \in \mathbb{R}\}$$

are subgroups of G. Indeed, these subgroups coincide if $X(1) = Y(1)$.

Proposition 5.8 *Let X and Y be, respectively, right and left invariant vector fields which coincide at the identity, that is, $X(1) = Y(1)$. Then their trajectories $X_t(1)$ and $Y_t(1)$ through the identity coincide for every $t \in \mathbb{R}$.*

Proof It suffices to verify that the curve $\alpha(t) = X_t(1)$ satisfies the differential equation $\dot{g} = Y(g)$, as can be seen by computing its derivative:

$$\begin{aligned} \alpha'(t) &= \frac{d}{ds} X_{t+s}(1)_{|s=0} = \frac{d}{ds} X_t(1) X_s(1)_{|s=0} \\ &= \left(dL_{\alpha(t)}\right)_1 (X(1)) = Y(\alpha(t)). \end{aligned}$$ □

Once this discussion of invariant vector fields is done, the **exponential map** on a Lie group can be defined.

Definition 5.9 Let $X \in T_1 G$. Then $\exp X = (X^r)_{t=1}(1) = \left(X^l\right)_{t=1}(1)$. As usual, $\exp X$ is also written as e^X. This defines a map $\exp : \mathfrak{g} \to G$, where $\mathfrak{g} = T_1 G$ is the Lie algebra of G.

The exponential map is well defined because invariant vector fields are complete, so that the solution of $\dot{g} = X(g)$ through the identity when $t = 0$ extends to $t = 1$.

If Z is a vector field and $a \in \mathbb{R}$, then the trajectories of Z and aZ coincide and their flows satisfy $(aZ)_t = Z_{at}$. Applying this observation to the vector fields X^r and X^l, it is seen that their trajectories by the identity are given by

$$\left(X^r\right)_t(1) = \left(X^l\right)_t(1) = \exp tX.$$

By the properties of these trajectories stated above, it follows that the exponential map $t \mapsto \exp(tX)$, $X \in \mathfrak{g}$, is a homomorphism, that is,

$$\exp(t+s)X = \exp(tX)\exp(sX) = \exp(sX)\exp(tX)$$

and its image $\{\exp(tX) : t \in \mathbb{R}\}$ is a subgroup of G. This subgroup is called the 1-parameter subgroup generated by $X \in \mathfrak{g}$.

The following proposition collects properties of the exponential map and the flows of invariant vector fields discussed above.

Proposition 5.10 *The following statements hold:*

1. *If X is a right invariant vector field, then $X_t = L_{\exp(tX)}$, that is, $X_t(g) = e^{tX}g$.*[2]
2. *If X is a left invariant vector field, then $X_t = R_{\exp(tX)}$, that is, $X_t(g) = ge^{tX}$.*
3. *$e^0 = 1$.*
4. *If $n \in \mathbb{Z}$, then $\left(e^X\right)^n = e^{nX}$ for every $n \in \mathbb{Z}$. In particular, $\left(e^X\right)^{-1} = e^{-X}$.*

These properties generalize well known properties of the exponential in concrete situations.

Examples

1. As already seen the right invariant vector fields on $\mathrm{Gl}(n, \mathbb{R})$ have the form $X(g) = Ag$, with A an $n \times n$ matrix. The differential equation associated with X is the linear system

$$\frac{dg}{dt} = Ag$$

   in matrix space. Its fundamental solution is given by the matrix exponential $\exp A = \sum_{k\geq 0} \frac{1}{k!}A^k$, which coincides with the exponential map on $\mathrm{Gl}(n, \mathbb{R})$.
2. If $\mathcal{A}$ is an associative algebra, then the exponential map of the Lie group $G(\mathcal{A})$ is given by $\exp A = \sum_{k\geq 0} \frac{1}{k!}A^k$, in the same way as on $\mathrm{Gl}(n, \mathbb{R})$.
3. In $(\mathbb{R}^n, +)$, invariant vector fields are constant: $X(x) = v$. The flow of such a vector field is given by translations $X_t(x) = x + tv$. Taking $x = 0$, it is seen that

$$e^{tv} = tv.$$

   In particular, $e^v = v$ and $\exp = \mathrm{id}$.

[2] In this statement, X denotes both an invariant vector field and an element of T_1G. This apparent ambiguity arises from the isomorphism between the spaces of invariant vector fields and the tangent space T_1G.

4. If G and H are Lie groups with Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$, respectively, then $\exp(X, Y) = (\exp X, \exp Y)$ if $(X, Y) \in \mathfrak{g} \times \mathfrak{h}$, the Lie algebra of $G \times H$. This is due to the fact that the right invariant vector fields on the direct product $G \times H$ have the form (X, Y), with X a right invariant vector field on G and Y a right invariant vector field on H. Therefore, their solutions can be found component-wise.

   The same observation holds for a direct product $G_1 \times \cdots \times G_k$, where the exponential is given by the Cartesian product of the exponentials on each group. □

In addition to the algebraic properties of the exponential map, stated in Proposition 5.10, its differentiability is also relevant. The exponential map was defined through the solution of the differential equation $dg/dt = X(g)$, with X an invariant vector field. The set of equations defined (for instance) by right invariant vector fields can be put in a single equation, depending on a parameter $A \in T_1 G$, writing

$$\frac{dg}{dt} = f(A, g), \tag{5.3}$$

where $f : T_1 G \times G \to TG$ is given by $f(A, g) = \left(dR_g\right)_1 (A)$. In case when G and p are of class at least C^2, f is of class ≥ 1. Therefore, the solutions of (5.3) depend differentiably on parameter A. This means that the exponential map $\exp : \mathfrak{g} \to G$ is differentiable.

The differential of exp is widely used in the development of the theory. There exists a formula for this differential in terms of a series whose successive terms are brackets of elements in $\mathfrak{g}$. This formula will be proved in Chapter 8. One of its particular cases is the expression stated below for the differential of the exponential map at the origin $0 \in \mathfrak{g}$. To read the expression that will be written, it must be taken into account that $\exp 0 = 1$, so that $(d \exp)_0$ is a linear map $\mathfrak{g} \to T_1 G = \mathfrak{g}$.

Proposition 5.11 $(d \exp)_0 = \mathrm{id}$.

Proof Given $X \in \mathfrak{g}$, $(d \exp)_0 (X) = \dfrac{d}{dt} \exp(0 + tX)_{|t=0}$. But this derivate is precisely X, since the curve e^{tX} is a solution of $dg/dt = X^r(g)$. Hence, $(d \exp)_0 (X) = X$, concluding the proof. □

Corollary 5.12 *There exists a neighborhood U of $0 \in \mathfrak{g}$ and a neighborhood V of 1 in G, such that $\exp_{|U} : U \to V$ is a diffeomorphism.*

Proof It follows from the inverse function theorem and the fact that $(d \exp)_0 = \mathrm{id}$ is invertible. □

Corollary 5.13 *Let G be a connected Lie group and take $g \in G$. Then, there exist $X_1, \ldots, X_s \in \mathfrak{g}$, such that*

$$g = \exp(X_1) \cdots \exp(X_s).$$

Proof Since G is connected, the neighborhood V of the previous corollary generates G, that is,

$$G = \bigcup_{n \geq 1} V^n.$$

An element of V^n has the form $g_1 \cdots g_n$ with $g_i = \exp X_i \in V$. Hence, an element of V^n is a product of exponentials, and the same occurs to an arbitrary $g \in G$. □

By the previous corollary, every element of a connected Lie group G is a product of exponentials which, generally, involves more than one factor since not every element of G has the form $\exp X$, that is, the exponential map is not always surjective. Exercise 14, at the end of this chapter, shows this nonsurjectivity on the group $\mathrm{Gl}(2, \mathbb{R})$. In turn, Exercise 34 of Chapter 6 indicates the proof that G is the product of exponentials of elements in a generator of $\mathfrak{g}$, generalizing the above corollary.

According to Corollary 5.12, the map $\log = \exp^{-1} : V \to U$ is a diffeomorphism between an open subset of G and an open subset of a vector space. Hence, log can be considered a chart, or local coordinate system, of G. This chart is called a **coordinate system of the first kind**.

Another type of coordinate system on the neighborhood of identity, obtained from exponentials, is given by the following map: Take a basis $\{X_1, \ldots, X_N\}$ of $\mathfrak{g}$ and consider the map

$$\psi : (t_1, \ldots, t_N) \in \mathbb{R}^N \longmapsto e^{t_1 X_1} \cdots e^{t_N X_N} \in G. \tag{5.4}$$

It satisfies $\psi(0) = 1$ and $d\psi_0 = \mathrm{id}$ as for each element e_i of the canonical basis of $\mathbb{R}^n$ it holds

$$d\psi_0(e_i) = \frac{\partial \psi}{\partial t_i}(0) = \frac{d}{dt_i} \psi(0, \ldots, t_i, \ldots, 0)_{|t_i = 0} = X_i.$$

Hence, $d\psi_0$ is an isomorphism and this entails that, on some neighborhood of $0 \in \mathbb{R}^N$, ψ is a diffeomorphism. A map of this type is called a **coordinate system of the second kind**.

5.4 Homomorphisms

Let G and H be Lie groups. A differentiable homomorphism $\phi : G \to H$ between G and H is called a **homomorphism of Lie groups**. The same terminology applies to isomorphisms and automorphisms of Lie groups.

The condition of being differentiable is part of the definition of homomorphism of Lie groups. To verify whether a homomorphism $\phi : G \to H$ between Lie groups is differentiable, it suffices to verify its differentiability at a single point. Indeed, consider the equalities

$$\phi \circ R_g = R_{\phi(g)} \circ \phi, \qquad \phi \circ L_g = L_{\phi(g)} \circ \phi.$$

The first one implies that $\phi = R_{\phi(g)} \circ \phi \circ R_{g^{-1}}$. Applying the chain rule one sees that, if ϕ is differentiable at the identity 1, then ϕ is differentiable also at $g \in G$.

Taking into account the principle that Lie groups must be studied through Lie algebras, the homomorphisms between Lie groups will be described through homomorphisms between their Lie algebras.

A homomorphism between Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$ is a linear map $\theta : \mathfrak{g} \to \mathfrak{h}$ satisfying $\theta[X, Y] = [\theta X, \theta Y]$ for every $X, Y \in \mathfrak{g}$. The relation between homomorphisms of Lie groups and those of Lie algebras is given by the differential at the identity. This relation will be proved in the sequel using some formulas involving exponentials and homomorphisms of groups.

Lemma 5.14 *Let G and H be Lie groups with Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$, respectively. Let $\phi : G \to H$ be a differentiable homomorphism and take $X \in \mathfrak{g}$. Then, for every $g \in G$, it holds*

$$d\phi_g \left(X^r(g)\right) = Y^r(\phi(g)) \qquad d\phi_g \left(X^l(g)\right) = Y^l(\phi(g)),$$

where $Y = d\phi_1(X)$.

Proof Since X^r is a right invariant vector field,

$$d\phi_g \left(X^r(g)\right) = d\phi_g \circ d\left(R_g\right)_1 (X) = d\left(\phi \circ R_g\right)_1 (X).$$

But the last term coincides with

$$d\left(\phi \circ R_g\right)_1 (X) = d\left(R_{\phi(g)} \circ \phi\right)_1 (X) = d\left(R_{\phi(g)}\right)_1 \circ d\phi_1 (X),$$

and so it follows that $d\phi_g (X^r(g)) = d\left(R_{\phi(g)}\right) \circ d\phi_1 (X) = Y^r(\phi(g))$, which is the equality stated. The proof for left invariant vector fields is similar. □

Two vector fields X and Y are called ϕ-related if $d\phi_x (X(x)) = Y(\phi(x))$. In this case, the trajectories of Y are the images by ϕ of the trajectories of X (see Appendix A). The previous lemma ensures that right (or left) invariant vector fields defined by $X \in T_1 G$ and $Y = d\phi_1 (X)$ are ϕ-related. Since the trajectories of invariant vector fields are given by the corresponding exponentials, it follows from the above lemma the following fundamental formula for homomorphisms.

Proposition 5.15 *Let G and H be Lie groups with Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$, respectively. Let $\phi : G \to H$ be a differentiable homomorphism and take $X \in \mathfrak{g}$. Then,*

$$\phi(\exp(X)) = \exp(d\phi_1 (X)). \tag{5.5}$$

Another property of ϕ-related vector fields is that their Lie brackets are also ϕ-related (see Appendix A, Proposition A.2). It then follows, from Lemma 5.14, the homomorphism property of the differential $d\phi_1$.

Proposition 5.16 *Let G and H be Lie groups with Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$, respectively. Let $\phi : G \to H$ be a differentiable homomorphism. Then, $d\phi_1 : \mathfrak{g} \to \mathfrak{h}$ is a homomorphism, that is,*

$$d\phi_1[X, Y] = [d\phi_1 X, d\phi_1 Y], \tag{5.6}$$

with left or right invariant brackets.

Proof If $Z = d\phi_1(X)$ and $W = d\phi_1(Y)$, then X^r and Z^r, as well as Y^r and W^r, are ϕ-related. Hence, $[X, Y]^r$ and $[Z, W]^r$ are also ϕ-related by Proposition A.2 of Appendix A. It then follows that

$$[Z, W]_r = [Z, W]^r(1) = d\phi_1[X, Y]_r.$$

The same holds for left invariant vector fields. □

The homomorphism $d\phi_1$ between the Lie algebras is sometimes called **infinitesimal homomorphism** associated with ϕ.

The last proposition above states that homomorphisms of Lie groups induce, through the differential, homomorphisms of Lie algebras. The converse procedure, that is, the construction of homomorphisms of Lie groups which "extend" homomorphisms of Lie algebras is not always possible. For instance, if $G = S^1$ and $H = \mathbb{R}$, then $\dim \mathfrak{g} = \dim \mathfrak{h} = 1$ and so there exist isomorphisms between $\mathfrak{g}$ and $\mathfrak{h}$. But no isomorphism has the form $d\phi_1$, because the only homomorphism $G \to H$ is constant, since S^1 is compact and $\{0\}$ is the only subgroup of $\mathbb{R}$ contained in a compact (see Chapter 2, Exercise 20). What is at stake here are global topological properties of domain G (its fundamental group). This issue will be extensively discussed in Chapter 7; it is relevant for the study of classes of Lie groups isomorphisms from Lie algebras.

Example The determinant $\det : \mathrm{Gl}(n, \mathbb{R}) \to \mathbb{R} \setminus \{0\}$ is a homomorphism into the real multiplicative group. This homomorphism is differentiable. To get $d(\det)_1$, take a matrix A and a curve $g_t = \left(a_{ij}(t)\right)_{i,j=1}^{n} \in \mathrm{Gl}(n, \mathbb{R})$, such that $g_0 = 1$ and $a'_{ij}(0) = A$. The derivative of the product

$$\det(g_t) = \sum_{\sigma} (-1)^{\sigma} a_{1\sigma(1)}(t) \cdots a_{n\sigma(n)}(t)$$

at $t = 0$ is equal to $\mathrm{tr}A$. Hence, $d(\det)_1(A) = \mathrm{tr}A$. As can be directly verified, the map $A \in \mathfrak{gl}(n, \mathbb{R}) \mapsto \mathrm{tr}A \in \mathbb{R}$ is a homomorphism of Lie algebras. Formula (5.5) says that $\det e^A = e^{\mathrm{tr}A}$. □

5.4.1 Representations

A particular case of homomorphism between Lie groups arises when the range is a linear group $\mathrm{Gl}(V)$. In this case the homomorphism is called a **representation** of G on the vector space V. The space V is called **representation space** and $\dim V$ is its dimension. In what follows, V is a real vector space. (See Chapter 4 for more information on group representations.)

Let ρ be a (differentiable) finite dimensional representation of G on V. The Lie algebra of the group $\mathrm{Gl}(V)$ is denoted $\mathfrak{gl}(V)$; it coincides with the vector space of linear maps $V \to V$ with bracket given by the commutator. The differential of ρ at the identity $d\rho_1 : \mathfrak{g} \to \mathfrak{gl}(V)$ is a homomorphism of Lie algebras and as such, a representation on V of the Lie algebra $\mathfrak{g}$. This representation is called **infinitesimal representation** associated with ρ. It is usual to denote the infinitesimal representation with the same notation (that is, $\rho = d\rho_1$). The formula relating both representations is given by Proposition 5.15:

$$\rho(\exp X) = \exp(d\rho_1(X)). \tag{5.7}$$

The exponential in the right hand side is that of the linear group and it can therefore be written as the sum of a power series.

Examples

1. Let $G = \mathrm{Gl}(n, \mathbb{R})$. Its canonical representation on $\mathbb{R}^n$ is the identity map. The corresponding infinitesimal representation is also the identity, that is, it associates with an element of $\mathfrak{gl}(n, \mathbb{R})$ the corresponding linear map of $\mathbb{R}^n$. This statement follows from

$$\frac{d}{dt}\left(e^{tA}\right)_{|t=0} = A.$$

2. Again, let $G = \mathrm{Gl}(n, \mathbb{R})$ and consider the tensor product

$$T_k = \bigotimes^k \mathbb{R}^n = \mathbb{R}^n \otimes \cdots \otimes \mathbb{R}^n.$$

For $g \in G$, define the linear map $\rho_k(g) : T_k \to T_k$ in such a way that, for the tensor products $v_1 \otimes \cdots \otimes v_k$, $v_1, \ldots, v_k \in \mathbb{R}^n$, it holds

$$\rho_k(g)(v_1 \otimes \cdots \otimes v_k) = gv_1 \otimes \cdots \otimes gv_k.$$

Map ρ_k is a representation of $\mathrm{Gl}(n, \mathbb{R})$. Its infinitesimal representation is computed with the derivative

$$\frac{d}{dt}\left(e^{tA}v_1 \otimes \cdots \otimes e^{tA}v_k\right)_{|t=0} = \sum_{i=1}^{k} v_1 \otimes \cdots \otimes Av_i \otimes \cdots \otimes v_k.$$

The right hand side in this equality defines the linear map $(d\rho_k)_1(A)$. The tensor representation can be restricted to any linear group $G \subset \mathrm{Gl}(n, \mathbb{R})$.

Analogous representations are obtained for the k-th exterior product $\wedge^k \mathbb{R}^n$. The expressions for $\rho_k(g)$ and $(d\rho_k)_1$ are the same, replacing the tensor product $\otimes$ by the exterior product $\wedge$. □

If ρ is a representation of G on V, then the **dual representation** of ρ, denoted ρ^*, is the representation of G on the dual vector space V^* of V defined by

$$\rho^*(g)(\alpha) = \alpha \circ \rho(g)^{-1} \qquad g \in G, \alpha \in V^*.$$

If $\mathfrak{g}$ is the Lie algebra of G, then the corresponding infinitesimal dual representation is given by

$$\rho^*(X)(\alpha) = -\alpha \circ \rho(X) \qquad X \in \mathfrak{g}, \alpha \in V^*.$$

5.4.2 Adjoint Representations

There is a natural representation of a Lie group G on its Lie algebra $\mathfrak{g}$. This representation is constructed in the following way: An element $g \in G$ defines the inner automorphism $C_g(x) = gxg^{-1}$. It is clear that $C_g(1) = 1$, hence $d\left(C_g\right)_1$ is a linear map $\mathfrak{g} \to \mathfrak{g}$. Given $g, h \in G$,

$$C_g \circ C_h(x) = g\left(hxh^{-1}\right)g^{-1} = C_{gh}(x),$$

and this implies that $d\left(C_g\right)_1 \circ d(C_h)_1 = d\left(C_{gh}\right)_1$. This means that the map $g \mapsto d\left(C_g\right)_1$ is a representation of G on $\mathfrak{g}$, that is, a homomorphism of G into $\mathrm{Gl}(\mathfrak{g})$.

Definition 5.17 The **adjoint representation** $\mathrm{Ad} : G \to \mathrm{Gl}(\mathfrak{g})$, of G on its Lie algebra $\mathfrak{g}$ is defined by

$$\begin{aligned}\mathrm{Ad}(g) &= d\left(C_g\right)_1 = d\left(L_g \circ R_{g^{-1}}\right)_1 = d\left(R_{g^{-1}} \circ L_g\right)_1 \\ &= \left(dL_g\right)_{g^{-1}} \circ \left(dR_{g^{-1}}\right)_1 = \left(dR_{g^{-1}}\right)_g \circ \left(dL_g\right)_1.\end{aligned}$$

The representation Ad is differentiable.

According to Proposition 5.16, for any $g \in G$, $\mathrm{Ad}(g) = d\left(C_g\right)_1$ is a homomorphism of $\mathfrak{g}$. In fact, an automorphism, since $\mathrm{Ad}(g)^{-1} = \mathrm{Ad}\left(g^{-1}\right)$. This

means that the image of Ad is contained in the group of automorphisms Aut $(\mathfrak{g})$ of $\mathfrak{g}$ (which is a Lie group, as will be verified in the next chapter).

A formula used very frequently in relations involving the adjoint representation is obtained by applying Proposition 5.15 to $\phi = C_g$. From this proposition, it is found that $C_g(\exp X) = \exp\left(\left(dC_g\right)_1(X)\right)$, that is,

$$g \exp(X) g^{-1} = \exp(\mathrm{Ad}(g) X). \tag{5.8}$$

As Ad is a differentiable representation, it is possible to consider its infinitesimal representation, which is a representation of the Lie algebra $\mathfrak{g}$ on itself, that is, a homomorphism of Lie algebras $\mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$. As will be shown below, the infinitesimal representation is nothing more, nothing less than the adjoint representation of $\mathfrak{g}$, which is now defined.

Definition 5.18 Let $\mathfrak{g}$ be a Lie algebra. Its **adjoint representation** is the map $\mathrm{ad} : \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$ defined by

$$\mathrm{ad}(X)(Y) = [X, Y].$$

The Jacobi identity ensures that the map ad is in fact a homomorphism of Lie algebras, in which the bracket in $\mathfrak{gl}(\mathfrak{g})$ is given by the commutator.

A linear map $D : \mathfrak{g} \to \mathfrak{g}$ is called a **derivation** if

$$D[X, Y] = [DX, Y] + [X, DY].$$

The Jacobi identity for brackets on Lie algebras ensures that the maps $\mathrm{ad}(X)$, $X \in \mathfrak{g}$, are derivations of $\mathfrak{g}$. They are called **inner derivations** of $\mathfrak{g}$.

Proposition 5.19 *Let G be a Lie group with Lie algebra $\mathfrak{g}$, with bracket given by left invariant vector fields. Then, $d(\mathrm{Ad})_1(X) = \mathrm{ad}_l(X)$ for every $X \in \mathfrak{g}$ and*

$$\mathrm{Ad}(\exp X) = \exp(\mathrm{ad}_l(X)). \tag{5.9}$$

(Sub-index "l" was inserted to emphasize that the bracket is given by the left invariant vector fields.)

Proof Let X be a left invariant vector field. Then, $d(\mathrm{Ad})_1(X)$ is a linear map $\mathfrak{g} \to \mathfrak{g}$. In order to compute it, let Y be another left invariant vector field. If $t \in \mathbb{R}$, then

$$\begin{aligned}\mathrm{Ad}\left(e^{tX}\right)(Y) &= d\left(L_{e^{tX}} \circ R_{e^{-tX}}\right)_1(Y) \\ &= d\left(R_{e^{-tX}}\right)_{e^{tX}}\left(d\left(L_{e^{tX}}\right)_1(Y)\right).\end{aligned}$$

Since Y is invariant by left translations,

$$d\left(L_{e^{tX}}\right)_1(Y) = Y\left(e^{tX}\right).$$

Now, the flow X_t of X is given by $X_t = R_{\exp(tX)}$. Using this flow, the equality above is rewritten as

$$\mathrm{Ad}\left(e^{tX}\right)(Y) = d\left(X_{-t}\right)_{X_t(1)}\left(Y\left(X_t(1)\right)\right).$$

Differentiating this equality with respect to t and using the formula that defines the Lie bracket of vector fields, it is found that

$$\frac{d}{dt}\left(\mathrm{Ad}\left(e^{tX}\right)(Y)\right)_{|t=0} = [X, Y](1).$$

Since X and Y are left invariant vector fields, the last equality means that

$$\frac{d}{dt}\mathrm{Ad}\left(e^{tX}\right)_{|t=0} = \mathrm{ad}_e(X),$$

showing that ad is the infinitesimal representation associated with Ad. The second formula in the statement of the proposition is a particular case of (5.7), which holds for representations in general. □

The equality $[X, Y]_l = -[X, Y]_r$ implies that $\mathrm{ad}_l(X) = -\mathrm{ad}_r(X)$, $X \in \mathfrak{g}$, and this adds a sign to the formula of the previous proposition, for the case of right invariant vector fields.

Proposition 5.20 *If right invariant vector fields are used in the previous proposition, then* $d(\mathrm{Ad})_1(X) = -\mathrm{ad}_r(X)$ *for every* $X \in \mathfrak{g}$ *and*

$$\mathrm{Ad}(\exp X) = \exp\left(-\mathrm{ad}_r(X)\right). \tag{5.10}$$

Formulas (5.8) and (5.9) (or (5.10)) form the basis for establishing relations between the properties of a Lie group G and its Lie algebra $\mathfrak{g}$. The left hand side of (5.8) involves the product in G, while the right hand side of (5.9) depends only on the bracket in $\mathfrak{g}$. Both are linked to an intermediary term involving $\mathrm{Ad}(g)$, $g \in G$. In a typical application of (5.8) and (5.9), a property of G entails a property of $\mathrm{Ad}(g)$, $g \in G$, differentiating (5.8). A new derivative, now of (5.9), leads to a property of $\mathrm{ad}(X)$, $X \in \mathfrak{g}$. The inverse process is carried out by means of two "integrals." This process involving two derivatives is in the spirit of Proposition A.6 of Appendix A, in which the Lie bracket is interpreted as the second derivative of a commutator.

The case of abelian groups in the following example illustrates the method for applying formulas (5.8) and (5.9).

Example Let G be an abelian group. Then its Lie algebra is abelian. Indeed, by (5.7),

$$e^{t\,\mathrm{Ad}(g)X} = ge^{tX}g^{-1} = e^{tX}$$

for every $g \in G$, $X \in \mathfrak{g}$ and $t \in \mathbb{R}$. The derivative of this equality, at $t = 0$, furnishes $\mathrm{Ad}\,(g)\,X = X$ for every $g \in G$, $X \in \mathfrak{g}$, that is, $\mathrm{Ad}\,(g) = \mathrm{id}$ for every $g \in G$. Therefore, by (5.9), if $Y \in \mathfrak{g}$, then $\mathrm{id} = \mathrm{Ad}\,(\exp tY) = \exp\,(t\,\mathrm{ad}_l\,(Y))$. Differentiating this last term at $t = 0$ one obtains $\mathrm{ad}_l\,(Y) = 0$ for every $Y \in \mathfrak{g}$, and this means that the Lie algebra is abelian.

Conversely, G is abelian if it is connected and $\mathfrak{g}$ is abelian. In this case, one must begin applying (5.9), to conclude that

$$\mathrm{Ad}\left(e^{tY}\right) = e^{t\,\mathrm{ad}_l(Y)} = 1$$

if $Y \in \mathfrak{g}$. Hence, by (5.8), $e^Y e^X e^{-Y} = e^X$, that is, $e^Y e^X = e^X e^Y$ for every $X, Y \in \mathfrak{g}$. But G is generated by exponentials, since it is connected. Therefore, any two elements $g = e^{X_1} \cdots e^{X_n}$ and $h = e^{Y_1} \cdots e^{Y_m}$ of G commute. □

As observed above, each $\mathrm{Ad}\,(g)$, $g \in G$, is an automorphism of $\mathfrak{g}$. From this it follows that Ad can be considered a homomorphism of groups $\mathrm{Ad} : G \to \mathrm{Aut}\,(\mathfrak{g})$, where $\mathrm{Aut}\,(\mathfrak{g})$ denotes the group of automorphisms of $\mathfrak{g}$. The image of Ad is a subgroup of $\mathrm{Aut}\,(\mathfrak{g})$. In the next chapter it will be proved that this image is a Lie subgroup.

The kernel $\ker \mathrm{Ad}$ is a closed subgroup of G because Ad is a differentiable and, in particular, continuous map with values in group $\mathrm{Gl}\,(\mathfrak{g})$. The following proposition describes this kernel.

Proposition 5.21 *Let*

$$Z\,(G_0) = \{g \in G : \forall h \in G_0,\ gh = hg\}$$

be the ***centralizer*** *of the connected component of the identity G_0. Then,* $\ker \mathrm{Ad} = Z\,(G_0)$.

Proof It is a consequence of formula

$$e^{\mathrm{Ad}(g)X} = ge^X g^{-1}$$

and of Corollary 5.13. In the first place, if $g \in \mathrm{Z}\,(G_0)$, then

$$e^{\mathrm{Ad}(g)X} = e^X$$

for every $X \in \mathfrak{g}$ since $\exp X \in G_0$. But exp is a diffeomorphism around the origin, that is, there exists a neighborhood $V \subset \mathfrak{g}$ of 0 such that exp is injective in this neighborhood. Hence, $\mathrm{Ad}\,(g)\,X = X$ for every $X \in V$. This implies that $\mathrm{Ad}\,(g) =$

id because, for every $Y \in \mathfrak{g}$, there exists $r \in \mathbb{R}$ such that $rY \in V$ and, as $\mathrm{Ad}(g)$ is linear, it follows that $\mathrm{Ad}(g)\, Y = Y$. This shows that $\mathrm{Z}(G_0) \subset \ker \mathrm{Ad}$.

On the other hand, if $g \in \ker \mathrm{Ad}$, then $\mathrm{Ad}(g)\, X = X$ for every $X \in \mathfrak{g}$, and hence

$$e^X = g e^X g^{-1}.$$

This means that g commutes with all the exponentials $\exp X$, $X \in \mathfrak{g}$. Therefore, g commutes with products of exponentials $\exp(X_1) \cdots \exp(X_s)$, that is, g commutes with all the elements of G_0. □

Corollary 5.22 *The* ***center*** $Z(G) = \{g \in G : \forall h \in G,\ gh = hg\}$ *is contained in* $\ker \mathrm{Ad}$ *and both coincide if* G *is connected.*

Proof Indeed, $Z(G) \subset \mathrm{Z}(G_0)$ and the equality holds if $G = G_0$. □

The center of a group is an abelian subgroup. Then, it follows from the above corollary that $\ker \mathrm{Ad}$ is abelian if G is connected. This statement is not generally true. For instance, if G is a discrete group, then $G = \ker \mathrm{Ad}$, which does not need to be abelian.

The above propositions (especially the corollary) are in accordance with the fact that the kernel $\ker \mathrm{ad}$ of the adjoint representation of $\mathfrak{g}$ is its center

$$\mathfrak{z}(\mathfrak{g}) = \{X \in \mathfrak{g} : \forall Y \in \mathfrak{g},\ [X, Y] = 0\}.$$

It will be shown later that the center $Z(G)$ is a Lie subgroup of G. Its Lie algebra is the center $\mathfrak{z}(\mathfrak{g})$ of $\mathfrak{g}$.

Examples

1. On $\mathrm{Gl}(n, \mathbb{R})$, $\mathrm{Ad}(g)$ coincides with the conjugation C_g, for C_g extends to a linear map on the space of matrices, hence it coincides with $\mathrm{Ad}(g)$, which is its differential at the identity. In other words, if $A \in \mathfrak{gl}(n, \mathbb{R})$ and $g \in \mathrm{Gl}(n, \mathbb{R})$, then

$$\mathrm{Ad}(g)\, A = g A g^{-1}.$$

   The center of $\mathrm{Gl}(n, \mathbb{R})$ is the subgroup of scalar matrices $a \cdot 1$, $0 \neq a \in \mathbb{R}$. Even though $\mathrm{Gl}(n, \mathbb{R})$ has two connected components, the expression for $\mathrm{Ad}(g)$ immediately confirms that $Z(\mathrm{Gl}(n, \mathbb{R})) = \ker \mathrm{Ad}$.
2. In an abelian group G, the adjoint representation is trivial: $\mathrm{Ad}(g) = \mathrm{id}$ for every $g \in G$, since $C_g = \mathrm{id}$. □

The dual representation Ad^* of the adjoint representation Ad is called the **coadjoint representation**.

5.5 Ordinary Differential Equations

Let G be a Lie group with Lie algebra $\mathfrak{g}$. Given a curve $A : (a, b) \to \mathfrak{g}$ defined on an interval $(a, b) \subset \mathbb{R}$, it is possible to define on G a time-dependent ordinary differential equation by means of a right translation as

$$\frac{dg}{dt} = dR_g\left(A\left(t\right)\right). \tag{5.11}$$

In compact notation, this equation can be written as $\dot{g} = A\left(t\right) g$. In the same way, the equation obtained by left translation can be written as

$$\frac{dg}{dt} = dL_g\left(A\left(t\right)\right) = gA\left(t\right). \tag{5.12}$$

The theorems on the existence and uniqueness of solutions apply to these equations under quite general conditions for A. This is so because the equations depend differentiably on g. As for its dependence on t, which comes from A, it must be assumed that A is measurable and locally integrable (with respect to the Lebesgue measure restricted to interval (a, b)), in the sense that, for every $t \in (a, b)$, there exists $\varepsilon > 0$ such that $A\left(\cdot\right)$, restricted to $(t - \varepsilon, t + \varepsilon)$, is integrable. This condition is satisfied, for instance, when A is continuous or piecewise continuous.

Under these conditions, the theory of existence and uniqueness of solutions of ordinary differential equations ensures that, given an initial condition $(t_0, g_0) \subset (a, b) \times G$, there exists $\delta > 0$ and a unique solution $\phi : (t_0 - \delta, t_0 + \delta) \to G$ with $\phi\left(t_0\right) = g_0$. This solution is an absolutely continuous function that has derivatives at almost all points of $(t_0 - \delta, t_0 + \delta)$ and at those points the equation is satisfied. Moreover, by its continuous dependence on initial conditions, δ can be chosen in such a way that, for every (t, g) in a neighborhood of (t_0, g_0), the solution with initial condition (t, g) is defined on the whole interval $(t_0 - \delta, t_0 + \delta)$.

Invariant differential equations generalize the equations defined by left and right invariant vector fields and have similar properties. For instance, a right translation of a solution of (5.11) is also a solution. Indeed, given $\alpha\left(t\right)$ with $\alpha'\left(t\right) = A\left(t\right)\alpha\left(t\right)$ and $g \in G$, define $\beta\left(t\right) = R_g\left(\alpha\left(t\right)\right)$. Then,

$$\beta'\left(t\right) = dR_g\left(\alpha'\left(t\right)\right) = dR_g dR_{\alpha(t)}\left(A\left(t\right)\right) = dR_{\beta(t)}\left(A\left(t\right)\right),$$

that is, β is also a solution of 5.11) In the same way, Equation (5.12) is left invariant.

In particular, the solutions of both equations are obtained respectively by right or left translations of the solutions through the identity.

So, for each $s \in (a, b)$, denote by $\phi^r\left(s, t\right)$ the solution of the right invariant equation (5.11) with initial condition $\phi^r\left(s, s\right) = 1$, defined on an open interval $I \subset (a, b)$ containing s. Then, the solution with initial condition (s, g), $g \in G$, is the right translation by g: $\phi^r\left(s, t\right) g$. In particular, if $g = \phi^r\left(u, s\right)$ then, as a function of t, the product $\phi^r\left(s, t\right)\phi^r\left(u, s\right)$ is a solution with initial condition $(u, 1)$. Therefore, there is the formula

$$\phi^r(u,t) = \phi^r(s,t)\,\phi^r(u,s)\,. \tag{5.13}$$

The situation is the same for the left invariant equation (5.12): If $\phi^l(s,t)$ is the solution with initial condition $(s,1)$, then $g\phi^l(s,t)$ is the solution with initial condition (s,g) and there is the equality

$$\phi^l(u,t) = \phi^l(u,s)\,\phi^l(s,t)\,. \tag{5.14}$$

Notice that, in particular, any solution with initial condition (s,g), $g \in G$, has the same maximal interval of definition. The objective now is to prove that this interval coincides with the domain of definition (a,b) of A. This result generalizes Proposition 5.7.

Proposition 5.23 *Let $A : (a,b) \to \mathfrak{g}$ be a measurable, locally integrable curve. Then, for every $t_0 \in (a,b)$ and $g_0 \in G$, there exists a unique solution $\psi : (a,b) \to G$ of (5.11), defined on the whole interval (a,b), such that $\psi(t_0) = g_0$. The same result holds for (5.12).*

Proof Take $c \in (a,b)$ and suppose, to fix ideas, that $t_0 < c$. It must be shown that the solution $t \mapsto \phi^r(t_0,t)$ extends to c. To do that, observe that for each $s \in [t_0,c]$, there exists $\delta_s > 0$ such that the solution with initial condition $(s,1)$ is defined on the interval $(s-\delta_s, s+\delta_s)$. By compactness, there exist finite elements $s_1 < \cdots < s_k$ with $t_0 = s_1$, $c = s_k$ and for each i the solution $\phi^r(s_i,t)$ extends to s_{i+1}. By repeatedly applying formula (5.13), it is then obtained that

$$\phi^r(t_0,c) = \phi^r(s_{k-1},c)\cdots\phi^r(s_1,s_2)\,\phi^r(t_0,s_1)$$

is well defined, concluding the proof. □

5.6 Haar Measure

The general construction of Haar measures on locally compact groups carried out in Chapter 3 can be very simplified in the case of Lie groups. This is so because on differentiable manifolds it is possible to define measures by means of volume forms, making it easy to construct invariant measures on Lie groups.

Let G be a Lie group with Lie algebra $\mathfrak{g}$ and denote by $\mathfrak{g}^*$ the dual of $\mathfrak{g}$. A volume form ν on $\mathfrak{g}$ is a nonzero n-form, where $n = \dim \mathfrak{g}$. For example, if $\alpha_1, \ldots, \alpha_n$ is a basis of $\mathfrak{g}^*$, then $\nu = \alpha_1 \wedge \cdots \wedge \alpha_n$ is an n-form on $\mathfrak{g}$. The space of n-forms $\wedge^n\mathfrak{g}^*$ has dimension 1.

A volume form ν on $\mathfrak{g}$ defines on G an invariant volume form (also denoted by ν) by translation:

$$\nu_g = \left(\left(dL_{g^{-1}}\right)_1\right)^* \nu \in \wedge^n T_g^* G.$$

This form is left invariant, that is, $\left(L_g\right)^* \nu = \nu$ for every $g \in G$. Conversely, a left invariant volume form is completely determined by its value at 1, which is an element of $\wedge^n \mathfrak{g}^*$. Right invariant volume forms are defined in the same way. (For more information about invariant differential forms, see Chapter 14.)

An invariant volume form on G is identically zero or is different from zero at every point of G. Moreover, if $\nu \neq 0$ is an invariant volume form, than any other invariant volume form is a multiple $\nu' = a\nu$ of ν with $a \in \mathbb{R}$.

Roughly speaking, the integral of a function $f : M \to \mathbb{R}$ on a manifold M with respect to a volume form ν is defined, in a coordinate system $\psi : U \subset \mathbb{R}^n \to M$, as

$$\int_{\psi(U)} f d\nu = \int f_\psi(x) \left| w_\psi(x) \right| dx,$$

where $dx = dx_1 \wedge \cdots \wedge dx_n$ is the canonical volume form on $\mathbb{R}^n$, $f_\psi = f \circ \psi$ and the function $w_\psi : U \to \mathbb{R}$ is defined by the equality $\psi^* \nu = w dx$. The integral in the right side does not vary with $\psi : U \to M$, and this allows to define intrinsically the integral $\int f d\nu$ of function f with respect to ν. Modulo some technical issues, these integrals give rise to a (regular) Borel measure μ_ν on M that satisfies:

1. If $\nu_1 = h \cdot \nu$, where $h : M \to \mathbb{R}$ is a function, then $\mu_{\nu_1} = |h| \mu_\nu$, as follows from the local definition in coordinate systems. That is, μ_{ν_1} is absolutely continuous with respect to μ_ν with Radon–Nikodym derivative $|h|$.
2. If $\phi : M \to M$ is a diffeomorphism and ν is different from zero at every point of M, then there exists a function $h : M \to \mathbb{R}$ such that $\phi^* \nu = h\nu$. Hence, $\mu_{\phi^* \nu} = |h| \cdot \mu_\nu$. Moreover, $\phi_* \mu_{\phi^* \nu} = \mu_\nu$.

Let now μ_ν be the Borel measure associated with the left invariant volume form ν on Lie group G. Then, μ_ν is invariant by left translations since $\left(L_g\right)^* \nu = \nu$ implies that $\mu_{L_g^* \nu} = \mu_\nu$, that is, $\left(L_g\right)_* \mu_\nu = \mu_\nu$. Therefore, it is a (left invariant) Haar measure on G. Two nonzero Haar measures constructed in this way are obtained from each other by scaling since, if $\nu_1 = a\nu$, $a \neq 0$, then $\mu_{\nu_1} = |a| \mu_\nu$. These are all the Haar measures on a Lie group, by the theorem of uniqueness of Haar measures on locally compact groups.

The right invariant Haar measures are obtained in the same way, by means of right invariant volume forms.

As discussed in Chapter 3, left invariant measures are not necessarily right invariant, and the relation between them is given by the modular function Δ. Let ν^r be a right invariant volume form. Its left translation is $\left(L_g\right)^* \nu^r = (\mathrm{Ad}(g)\, \nu)^r$. However, $\mathrm{Ad}(g)\, \nu = \det \mathrm{Ad}(g)\, \nu$ and consequently $L_g \nu^r = \det \mathrm{Ad}(g) \cdot \nu^r$.

In particular, the left invariant volume form ν^l, which is given by $\nu_g^l = \left(\left(dL_g\right)_1\right)^* \nu$ (where $\nu = \nu_1^r$), satisfies

$$\nu^l = \det \mathrm{Ad}(g) \cdot \nu^r. \tag{5.15}$$

Turning to Haar measures, let μ_{ν^l} and μ_{ν^r} be the measures defined by ν^l and ν^r, respectively. From the equality above it follows that

$$\mu_{\nu^l} = |\det \mathrm{Ad}\,(g)|\, \mu_{\nu^r}.$$

This means that $|\det \mathrm{Ad}\,(g)|$ is the Radon–Nikodym derivative of μ_{ν^l} with respect to μ_{ν^r}. By Section 3.4 of Chapter 3, it follows that the **modular function** of G is

$$\Delta\,(g) = |\det \mathrm{Ad}\,(g)|\,.$$

In particular, a Lie group is unimodular if $|\det \mathrm{Ad}\,(g)| = 1$ for every $g \in G$.

5.7 Exercises

1. Show that a right invariant vector field X on a Lie group G is also left invariant if and only if $\mathrm{Ad}\,(g)\,X = X$ for every $g \in G$. Show that this happens if and only if $e^{tX} \in Z\,(G)$ for every $t \in \mathbb{R}$.
2. On a Lie group G, consider a new product $g * h = hg$. Denote by G^* the group with this product. Show that G^* is still a Lie group, isomorphic to G. What is the relation between invariant vector fields on G and G^*?
3. Give an example of a Lie group whose underlying manifold is diffeomorphic to some $\mathbb{R}^n$ but whose product is not abelian.
4. An affine map of a real vector space V is a map of the form $g\,(x) = Px + v$ with $P : V \to V$ linear and $v \in V$. Verify that g is invertible if and only if P is invertible. Show that the group $\mathrm{Aff}\,(V)$ of invertible affine maps is a Lie group if $\dim V < \infty$. Describe the invariant vector fields on $\mathrm{Aff}\,(V)$ and the Lie algebra $\mathfrak{aff}\,(V)$ of $\mathrm{Aff}\,(V)$.
5. Describe the conjugates and adjoints on affine group $\mathrm{Aff}\,(n) = \mathrm{Aff}\,(\mathbb{R}^n)$ and on the corresponding Lie algebra $\mathfrak{aff}\,(n)$.
6. Show that, on a Lie group, right invariant vector fields commute with left invariant vector fields. Show that, if G is connected, then a vector field X is right invariant if and only if $[X, Y] = 0$ for every left invariant vector field Y. (Use the fact that every element of a connected Lie group is a product of exponentials.)
7. Let G be a Lie group with Lie algebra $\mathfrak{g}$ such that the center $Z\,(G)$ of G is a discrete subgroup. Given $g \in G$ and $X \in \mathfrak{g}$, suppose that $\mathrm{Ad}\,(g)$ commutes with $\mathrm{Ad}\left(e^{tX}\right)$ for every $t \in \mathbb{R}$ and show that g commutes with e^{tX}, $t \in \mathbb{R}$.
8. Let $\mathfrak{g}$ be a finite dimensional real Lie algebra. A derivation of $\mathfrak{g}$ is a linear map $D : \mathfrak{g} \to \mathfrak{g}$ that satisfies $D[X, Y] = [DX, Y] + [X, DY]$ for any $X, Y \in \mathfrak{g}$. Show that D is a derivation if and only if $\exp\,(tD)$ is an automorphism of $\mathfrak{g}$ for every $t \in \mathbb{R}$. (Hint: Consider the differential equations satisfied by $e^{tD}[X, Y]$ and $[e^{tD}X, e^{tD}Y]$.)

9. Let G be a Lie group. Show that there exists a neighborhood U of the identity which does not contain any subgroup of G except the trivial subgroup $\{1\}$.
10. Let G be a Lie group. Show that, for every $n \in \mathbb{N}$, there exists a neighborhood U of identity such that the order of x is greater than n if $x \in U$ and $x \neq 1$.
11. Let G be a Lie group with Lie algebra $\mathfrak{g}$. Given $X, Y \in \mathfrak{g}$, use formulas (5.8) and (5.9) to show that $[X, Y] = 0$ if and only if $e^{tX} e^{sY} = e^{sY} e^{tX}$ for every $s, t \in \mathbb{R}$. Show also that, in this case, $e^{X+Y} = e^X e^Y$.
12. Find the differentiable homomorphisms $\mathrm{Gl}(n, \mathbb{R}) \to \mathbb{R}$. (Hint: Find the infinitesimal homomorphisms $\theta : \mathfrak{gl}(n, \mathbb{R}) \to \mathbb{R}$.)
13. Show that every 1-parameter subgroup of $\mathrm{O}(3)$ is closed. Is this statement true for $\mathrm{O}(n)$, $n > 3$?
14. Show that $\exp : \mathfrak{gl}(2, \mathbb{R}) \to \mathrm{Gl}^+(2, \mathbb{R})$ is not surjective. (Hint: Use the Jordan canonical form to show that the real parts of the eigenvalues of $g = \exp A$ are equal if they are negative.)
15. Show that every element of $\mathrm{Sl}(2, \mathbb{R})$ can be written as a product $e^X e^Y$, $X, Y \in \mathfrak{sl}(2, \mathbb{R})$. (Hint: Write a matrix g as $g = kt$, with k an orthogonal matrix and t a triangular matrix, applying to the columns of matrix g the Gram–Schmidt orthonormalization procedure.)
16. Show that on $\mathrm{Gl}(n, \mathbb{C})$ the exponential map is surjective. (Hint: Reduce the problem to a Jordan block.)
17. Let G be a Lie group with Lie algebra $\mathfrak{g}$. Show that if $\phi : \mathbb{R} \to G$ is a differentiable homomorphism, then $\phi(t) = \exp(tX)$ for some $X \in \mathfrak{g}$.
18. The **Cartan–Killing form** of a Lie algebra $\mathfrak{g}$ is the bilinear symmetric form $\langle \cdot, \cdot \rangle$ defined by $\langle X, Y \rangle = \mathrm{tr}(\mathrm{ad}(X) \mathrm{ad}(Y))$, $X, Y \in \mathfrak{g}$. Show that every derivation D of $\mathfrak{g}$ is skew-symmetric with respect to the Cartan–Killing form, that is, $\langle DX, Y \rangle + \langle X, DY \rangle = 0$ for every $X, Y \in \mathfrak{g}$. Show also that an automorphism ϕ of $\mathfrak{g}$ is an "isometry" of the Cartan–Killing form, that is, $\langle \phi X, \phi Y \rangle = \langle X, Y \rangle$ for every $X, Y \in \mathfrak{g}$.
19. Let G be a compact Lie group with Lie algebra $\mathfrak{g}$. Show that the eigenvalues of $\mathrm{ad}(X)$, $X \in \mathfrak{g}$, are pure imaginary and conclude that the Cartan–Killing form of $\mathfrak{g}$ is negative semidefinite ($\langle X, X \rangle \leq 0$ for every $X \in \mathfrak{g}$).
20. Let G be a connected Lie group and $\rho : G \to \mathrm{Gl}(V)$ a representation of G on a vector space V, with $\dim V < \infty$. Let β be a bilinear form on V. Show that the elements of $\rho(G)$ are isometries of β ($\beta(\rho(g)u, \rho(g)v) = \beta(u, v)$) if and only if the elements of the infinitesimal representation are skew-symmetric linear maps with respect to β.
21. Given a connected Lie group G with Lie algebra $\mathfrak{g}$, let $\mathfrak{z}(\mathfrak{g}) = \{X \in \mathfrak{g} : \forall Y \in \mathfrak{g},\ [X, Y] = 0\}$ be the center of $\mathfrak{g}$ and $Z(G) = \{g \in G : \forall h \in G,\ gh = hg\}$ the center of G. Show that, for every $X \in \mathfrak{z}(\mathfrak{g})$, $\exp X \in Z(G)$. Conversely, $X \in \mathfrak{z}(\mathfrak{g})$ if, for every $t \in \mathbb{R}$, $\exp(tX) \in Z(G)$.
22. Let G be a connected Lie group such that $Z(G)$ is a discrete subgroup. Let $H = \mathrm{Ad}(G)$ be the image of the adjoint representation. Show that $Z(H) = \{1\}$. (Take $\mathrm{Ad}(g) \in Z(H)$ and show that $g e^{tX} g^{-1} = e^{tX}$ for $t \in \mathbb{R}$ and X in the Lie algebra of G.) (Alternative: Show that $\mathrm{Ad}^{-1}(Z(H))$ is a discrete normal subgroup and, therefore, is contained in the center of G.)

23. Let $\mathfrak{g}$ be a Lie algebra such that $[X, [Y, Z]] = 0$ for every $X, Y, Z \in \mathfrak{g}$. Show that the product $*$, given by

$$X * Y = X + Y + \frac{1}{2}[X, Y],$$

defines on $\mathfrak{g}$ a group structure. Show also that this group is a Lie group if $\mathfrak{g}$ is a finite dimensional Lie algebra over $\mathbb{R}$, in such a way that its Lie algebra coincides with $\mathfrak{g}$.

24. In the previous exercise, suppose that $\mathfrak{g}$ is finite dimensional. Given $X, Y \in \mathfrak{g}$, consider the curve $\alpha(t) = e^{tX} e^{tY} e^{-tX} e^{-tY}$ and compute $\alpha'(0)$ and $\alpha''(0)$.
25. Let G and H be Lie groups with Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$, respectively, and $\phi : G \to H$ a differentiable homomorphism such that $d\phi_1$ is an isomorphism. Show that $\ker \phi$ is a discrete subgroup. Show also that ϕ is a covering map if G and H are connected. Conclude that if H is simply connected, then ϕ is an isomorphism.
26. Let G and H be Lie groups with Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$, with G connected. Show that, if $\phi, \psi : G \to H$ are differentiable homomorphisms such that $d\phi_1 = d\psi_1$, then $\phi = \psi$. Give examples to show that this result does not hold if G is not connected.
27. Let G be a Lie group and $\dot{g} = A(t)\, g$ a right invariant ordinary differential equation on G. Denote by g_t a solution of this equation. Show that $h_t = \mathrm{Ad}(g_t)$ satisfies the differential equation $\dot{h} = \mathrm{ad}(A(t))\, h$.
28. Let G be a Lie group, M a differentiable manifold, and $f : M \times G \to \mathbb{R}$ a differentiable function with compact support. Show that the function $F(x) = \int_G f(x, g)\, \mu(dg)$ is differentiable if μ is the Haar measure on G.
29. Show that, if ϕ is an automorphism of the Lie group G and μ is a Haar measure of G, then $\phi_* \mu$ is the Haar measure $(1/\det \theta)\, \mu$, where $\theta = d\phi_1$.
30. Let G be a connected Lie group. Show that G is unimodular if and only if $\mathrm{tr}(\mathrm{ad}(X)) = 0$ for every $X \in \mathfrak{g}$.
31. Let G be a connected Lie group with Lie algebra $\mathfrak{g}$. Show that, if the derived algebra $\mathfrak{g}' = [\mathfrak{g}, \mathfrak{g}]$ coincides with $\mathfrak{g}$, then G is unimodular. (The derived algebra is the vector space spanned by the brackets $[X, Y]$ with $X, Y \in \mathfrak{g}$.)
32. Consider the Heisenberg group G, formed by matrices

$$\begin{pmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix}.$$

Write the expression of the Haar measure of G in coordinates (x, y, z), that is, in the form $f(x, y, z)\, dxdydz$.

Chapter 6
Lie Subgroups



In this chapter, the subgroups of a Lie group are studied from the viewpoint of differential calculus. This means that the subgroups considered are Lie groups with a differentiable submanifold structure. The Lie algebra of a Lie subgroup is a subalgebra of the Lie algebra of the ambient group (a subspace of the tangent space at the identity). One of the objectives is to establish the bijection between Lie subalgebras and Lie subgroups; this is done with the help of theorems on the integrability of distributions. (An overview of the theory of integrability of distributions is found in Appendix B, as well as several concepts and results about submanifolds used in this chapter.)

6.1 Definition and Examples

Definition 6.1 Let G be a Lie group and $H \subset G$ a subgroup. Then, H is a **Lie subgroup** of G if H is an immersed submanifold of G, such that the product $H \times H \to H$ is differentiable with respect to the intrinsic structure of H.

(As defined in Appendix B, an immersed submanifold of a manifold M is a subset $N \subset M$ that admits a manifold structure such that the inclusion $i : N \hookrightarrow M$ is an immersion. The differentiable structure on N is the intrinsic structure mentioned in the definition.)

In the definition of Lie subgroup, attention must be paid to the requirement that the product on H is differentiable with respect to the intrinsic structure. The point is that, if H is a subgroup and an immersed submanifold, then the restriction to $H \times H$ of the product on G provides a differentiable map $H \times H \to G$ that assumes values in H. This does not automatically ensure that the corresponding map $H \times H \to H$ (product on H) is differentiable or even continuous with respect to the intrinsic structure, the condition for H to be a Lie group.

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A quite general case in which the product $H \times H \to H$ is differentiable is when H is a **quasi-regular** (or **quasi-embedded**—see definition in Appendix B) immersed submanifold. In this case, the differentiable map $H \times H \to G$ assumes values in H and hence is differentiable with respect to the intrinsic topology (by Proposition B.3). In particular, subgroups that are embedded submanifolds are automatically Lie subgroups because embedded submanifolds are quasi-regular. Also, if a subgroup H is the preimage of a regular value of a differentiable map $f : G \to M$, then H is a Lie subgroup because these level manifolds are embedded submanifolds. (Some examples of subgroups of this kind will be presented below.)

It should be noted that the main method to build Lie subgroups is through integrability of distributions. By this method the connected submanifolds produced are quasi-regular (see Proposition B.24). Therefore for the development of the theory of connected Lie subgroups there is no loss of generality in assuming in the definition that a Lie subgroup is a quasi-regular submanifold.[1]

Another way to look at Lie subgroups arises from the observation that, if $H \subset G$ is a Lie subgroup of G, then the inclusion $j : H \hookrightarrow G$ is a homomorphism, which is an injective immersion. On the other hand, if $\phi : L \to G$ is a differentiable injective homomorphism from Lie group L, then its image is a Lie subgroup. Indeed, by the chain rule, the equality $\phi \circ L_g = L_{\phi(g)} \circ \phi$ implies that

$$d\phi_g = \left(dL_{\phi(g)}\right)_1 \circ d\phi_1 \circ \left(dL_{g^{-1}}\right)_g$$

for every $g \in G$. It thus follows that ϕ has constant rank. The rank theorem then ensures that ϕ is an immersion if ϕ is injective and hence its image is a Lie subgroup (see Chapter 7, Proposition 7.1). Thus, an alternative definition is that a Lie subgroup of G is the image by an injective differentiable homomorphism $\phi : L \to G$, where L is a Lie group. (Later, it will be proved in Chapter 7 that the image of any differentiable homomorphism $\phi : L \to G$ is a Lie subgroup of G, without the hypothesis that ϕ is injective.)

Examples

1. If G is a Lie group, then the connected component of the identity G_0 is an open subgroup and, therefore, a Lie subgroup, since open submanifolds are embeddings.
2. If G is a Lie group, then any 1-parameter subgroup

$$\{\exp(tX) : X \in \mathfrak{g},\ t \in \mathbb{R}\}$$

is a Lie subgroup. Indeed, if the curve $t \mapsto \exp(tX)$ is closed, one obtains an injective immersion $S^1 \to G$. Otherwise, the 1-parameter group defines an injective immersion $\mathbb{R} \to G$. In both cases, $t \mapsto \exp(tX)$ is an injective differentiable homomorphism. Hence, its image is a Lie subgroup.

[1] Some authors as, for instance, Varadarajan [53], adopt this definition.

3. If H is a Lie subgroup of G and L is a Lie subgroup of H, then L is also a Lie subgroup of G, as follows directly from the definition. In particular, the connected component of the identity H_0 of H, which is a normal subgroup of H, is also a Lie subgroup of G.
4. Let $\phi : G \to H$ be a differentiable homomorphism. Then ϕ has constant rank and it then follows that $\ker \phi = \phi^{-1}\{1\}$ is an embedded submanifold. Hence, $\ker \phi$ is a Lie subgroup of G.
5. The two dimensional torus $\mathbb{T}^2 \approx S^1 \times S^1$ is a Lie group. It is isomorphic to the quotient $\mathbb{R}^2/\mathbb{Z}^2$. Let $\pi : \mathbb{R}^2 \to \mathbb{R}^2/\mathbb{Z}^2 \approx \mathbb{T}^2$ be the homomorphism defined by the quotient. The straight lines $\pi(r_\alpha)$, where $r_\alpha = \{x(1, \alpha) \in \mathbb{R}^2 : x \in \mathbb{R}\}$ are subgroups of $\mathbb{T}^2$ and, at the same time, quasi-regular submanifolds of dimension 1 (see the example at the beginning of Appendix B). Hence, $\pi(r_\alpha)$ is a Lie subgroup. If α is rational, then the subgroup is closed (and compact); if α is irrational, the subgroup is dense.

   In the same way, let $\pi_n : \mathbb{R}^n \to \mathbb{R}^n/\mathbb{Z}^n$ be the canonical projection on the torus $\mathbb{T}^n$. If $V \subset \mathbb{R}^n$ is a vector subspace, then $\pi_n(V)$ is a Lie subgroup of $\mathbb{T}^n$. This example can be extended to cylinders $\mathbb{R}^n/\mathbb{Z}^k$, $0 \leq k \leq n$.
6. Let $\mathrm{O}(n)$ be the subgroup of orthogonal $n \times n$ matrices. To verify that $\mathrm{O}(n)$ is a Lie subgroup of $\mathrm{Gl}(n, \mathbb{R})$, consider the map $\tau : \mathrm{Gl}(n, \mathbb{R}) \to \mathrm{M}_{n\times n}(\mathbb{R})$ given by $\tau(g) = g^T g$. It is clear that $\mathrm{O}(n) = \tau^{-1}\{1\}$. On the other hand, if A is a matrix, then $d\tau_g(A) = A^T g + g^T A = \left(g^T A\right)^T + g^T A$. It then follows that the kernel of $d\tau_g$ is given by

$$\ker d\tau_g = \{\left(g^T\right)^{-1} B : B^T + B = 0\},$$

   which is the left translation by $\left(g^T\right)^{-1}$ of the space of skew-symmetric matrices. Therefore, τ has constant rank at every point of $\mathrm{Gl}(n, \mathbb{R})$. In particular, $\mathrm{O}(n) = \tau^{-1}\{1\}$ is an embedded submanifold of $\mathrm{Gl}(n, \mathbb{R})$, and this shows that the orthogonal group is a Lie subgroup.

   The connected component of the identity of $\mathrm{O}(n)$ is $\mathrm{SO}(n) = \{g \in \mathrm{O}(n) : \det g = 1\}$, which is also a Lie subgroup.
7. Other linear Lie groups, that is, Lie subgroups of $\mathrm{Gl}(n, \mathbb{R})$, are:

   (a) Special linear group: $\mathrm{Sl}(n, \mathbb{R}) = \{g \in \mathrm{Gl}(n, \mathbb{R}) : \det g = 1\} = \ker(\det)$.
   (b) Unitary group: $\mathrm{U}(n)$, of complex $n \times n$ matrices such that $\overline{g}^T g = g\overline{g}^T = 1$. This group, analogous to $\mathrm{O}(n)$, is an embedded submanifold, the preimage of a regular value of $\tau(g) = g\overline{g}^T$.
   (c) Real symplectic group: $\mathrm{Sp}(n, \mathbb{R})$, formed by real $n \times n$ matrices such that $g^T J g = g J g^T = J$, where

$$J = \begin{pmatrix} 0 & -\mathrm{id}_{n\times n} \\ \mathrm{id}_{n\times n} & 0 \end{pmatrix}.$$

   This group is the preimage of a regular value of $\tau(g) = g J g^T$.

8. The example that follows is rather artificial (and the submanifold that emerges is not paracompact), but it illustrates that differentiability with respect to the intrinsic structure must be treated carefully. The group $G = \mathbb{C}_\times$ has two differentiable manifold structures. The canonical one and another structure in which the open sets of the intrinsic topology are open subsets of vertical straight lines (see Chapter 2, Example 7). The second structure can be viewed as an immersed submanifold of the first one. In this immersed submanifold, the product is not continuous because a $\pi/2$ rotation of a vertical interval is not an intrinsic open set. □

6.2 Lie Subalgebras and Lie Subgroups

The principle that guides the construction of the theory of Lie groups is to obtain information about the structure of Lie groups from the structure of Lie algebras. Following this principle, the Lie subgroups of a group G are studied by relating them to the Lie subalgebras of the Lie algebra $\mathfrak{g}$ of G.

The inclusion $j : H \hookrightarrow G$ of a Lie subgroup H in a Lie group G is a differentiable homomorphism. As seen in the previous chapter, the differential dj_1 at the identity is a homomorphism of Lie algebras. But dj_1 is the inclusion of the tangent space $T_1 H$ into the tangent space $T_1 G$. This inclusion is a homomorphism. Hence, $T_1 H$ is a Lie subalgebra of $\mathfrak{g}$ and this means that the Lie algebra of a Lie subgroup can be identified with a subalgebra of the Lie algebra of the group. These observations are true for the Lie algebras of both right and left invariant vector fields. Indeed, the equality $[\cdot,\cdot]_r = -[\cdot,\cdot]_l$ shows that a subspace of $T_1 G$ is a subalgebra of $[\cdot,\cdot]_r$ if and only if it is a subalgebra of $[\cdot,\cdot]_l$.

Looking at the Lie algebra $\mathfrak{h}$ of H as a subalgebra of $\mathfrak{g}$, the exponential map on H turns out to be the restriction of the exponential map on G. This follows from Proposition 5.15. Indeed, write $\exp_H$ and $\exp_G$ for the exponential maps on H and G, respectively. As the inclusion j is a differentiable homomorphism, Proposition 5.15 shows that $j\left(\exp_H X\right) = \exp_G\left(dj_1\left(X\right)\right)$. Identifying $\mathfrak{h}$ with a subset of $\mathfrak{g}$, this equality means that the exponential map on H coincides with the restriction to $\mathfrak{h}$ of the exponential map on G.

The objective now is to obtain the Lie subgroups of G from the Lie subalgebras of $\mathfrak{g}$. The technique to do this comes from the theory of distributions on differentiable manifolds (see Appendix B), which allows to establish a bijection between the connected Lie subgroups and the Lie subalgebras. (This bijection does not work in the nonconnected case because a Lie subgroup and its identity component have the same Lie algebra. For instance, $\mathrm{O}(n)$ and its identity component $\mathrm{SO}(n)$.)

The idea of constructing subgroups from subalgebras arises from the following observations. Let $H \subset G$ be a Lie subgroup whose Lie algebra is $\mathfrak{h}$. For every $g \in H$, the right translation R_g leaves H invariant, that is, $R_g H \subset H$ and the restriction to H of R_g is a diffeomorphism of H. This implies that, for every $h \in H$,

the image by $d\left(R_g\right)_h$ of the tangent space $T_hH \subset T_hG$ is the subspace $T_{hg}H$. In particular, the tangent space to H on g is $d\left(R_g\right)_1 \mathfrak{h}$. In the same way, R_g is a diffeomorphism between H and the coset Hg, hence the tangent space to Hg at g is $d\left(R_g\right)_1 \mathfrak{h}$.

The expressions for the tangent spaces show that the connected subgroup H, as well as its cosets Hg, are connected integral submanifolds of the distribution $\Delta^r_{\mathfrak{h}}$ on G, defined by

$$\Delta^r_{\mathfrak{h}}(g) = d\left(R_g\right)_1 \mathfrak{h}. \tag{6.1}$$

This distribution depends only on $\mathfrak{h}$. The same comments apply to the left invariant distribution $\Delta^l_{\mathfrak{h}}(g) = d\left(L_g\right)_1 \mathfrak{h}$, defined by H. In this case, the integral manifolds are the cosets gH.

The idea is to reverse these arguments. Given a Lie subalgebra $\mathfrak{h}$, it is possible to define a distribution $\Delta^r_{\mathfrak{h}}$ as in (6.1) (or by means of left translations $\Delta^l_{\mathfrak{h}} = d\left(L_g\right)_1 \mathfrak{h}$). If $\Delta^r_{\mathfrak{h}}$ is shown to be integrable, then the candidate to be the connected Lie subgroup with Lie algebra $\mathfrak{h}$ is the maximal connected integral manifold of $\Delta^r_{\mathfrak{h}}$ (or of $\Delta^l_{\mathfrak{h}}$) through the identity 1. This strategy reproduces in greater dimensions the construction of 1-parameter subgroups and their cosets.

The integrability of distributions $\Delta^r_{\mathfrak{h}}$ and $\Delta^l_{\mathfrak{h}}$ follows by a straightforward application of the Frobenius theorem, which ensures that a distribution with constant dimension is integrable if the Lie brackets of tangent vector fields are tangent to the distribution (see Appendix B, Theorem B.11). More precisely, the version of Frobenius theorem stated in Corollary B.15 is directly applicable to a basis of invariant vector fields defined by a basis of $\mathfrak{h}$.

In what follows, a direct proof of the integrability of distributions $\Delta^r_{\mathfrak{h}}$ and $\Delta^l_{\mathfrak{h}}$ will be presented, a proof that explicitly shows the integral manifolds. This proof is based on the idea of characteristic distributions, which are integrable (see Theorem B.9). The point is that the distribution $\Delta^r_{\mathfrak{h}}$ is both right invariant and invariant by left translations L_{e^Y} if $Y \in \mathfrak{h}$. The same holds for $\Delta^l_{\mathfrak{h}}$, reversing the roles of right and left translations. The following lemma allows to prove this invariance.

Lemma 6.2 *Let $\mathfrak{h}$ be a Lie subalgebra and let $X, Y \in \mathfrak{h}$. Then,* $\mathrm{Ad}\left(e^Y\right) X \in \mathfrak{h}$.

Proof Indeed,

$$\mathrm{Ad}\left(e^Y\right) X = e^{\mathrm{ad}(Y)} X = \sum_{k \geq 0} \frac{1}{k!} \mathrm{ad}(Y)^k X.$$

Since $\mathfrak{h}$ is a subalgebra, each term in the series belongs to $\mathfrak{h}$. Therefore, the sum of the series is in $\mathfrak{h}$, which is a closed vector subspace. □

From this lemma there follows the following invariance by translations.

Lemma 6.3 *If $Y \in \mathfrak{h}$, then $\left(dL_{e^Y}\right)_x \left(\Delta^r_{\mathfrak{h}}(x)\right) = \Delta^r_{\mathfrak{h}}\left(e^Y x\right)$ for any $x \in G$. The same result holds for right translations by e^Y, $Y \in \mathfrak{h}$, of the distribution $\Delta^l_{\mathfrak{h}}$.*

Proof For $g, x \in G$ and $X \in \mathfrak{g}$,

$$\begin{aligned}\left(dL_g\right)_x \left(X^r(x)\right) &= \left(dL_g\right)_x \circ (dR_x)_1 (X) = d\left(R_x \circ L_g\right)_1 (X) \\ &= \left(dR_{gx}\right)_1 \circ d\left(R_{g^{-1}} \circ L_g\right)_1 (X) \\ &= (\mathrm{Ad}(g)\, X)^r (gx)\,.\end{aligned}$$

In particular, if $X \in \mathfrak{h}$ and $g = e^Y$, $Y \in \mathfrak{h}$, then by the previous lemma, $\mathrm{Ad}\left(e^Y\right) X \in \mathfrak{h}$, and therefore $\left(\mathrm{Ad}\left(e^Y\right) X\right)^r \left(e^Y x\right) \in \Delta\left(e^Y x\right)$, that is, $\left(dL_{e^Y}\right)_x \left(X^r(x)\right) \in \Delta\left(e^Y x\right)$. Since $\Delta^r_{\mathfrak{h}}(x) = \{X^r(x) : X \in \mathfrak{h}\}$, it follows that $\left(dL_{e^Y}\right)_x \left(\Delta^r_{\mathfrak{h}}(x)\right) \subset \Delta^r\left(e^Y x\right)$ and the equality holds because the dimensions coincide. □

It is now possible to construct the integral manifolds of the invariant distributions.

Theorem 6.4 *Let G be a Lie group with Lie algebra $\mathfrak{g}$ and $\mathfrak{h} \subset \mathfrak{g}$ a Lie subalgebra. Then the distributions $\Delta^r_{\mathfrak{h}}(g) = d\left(R_g\right)_1 \mathfrak{h}$ and $\Delta^l_{\mathfrak{h}}(g) = d\left(L_g\right)_1 \mathfrak{h}$ are integrable.*

Proof Consider the right invariant distribution $\Delta^r_{\mathfrak{h}}$. Take a basis $\{X_1, \ldots, X_k\}$ of $\mathfrak{h}$ and, for $g \in G$, define the map $\rho : \mathbb{R}^k \to G$ by

$$\rho(t_1, \ldots, t_k) = e^{t_1 X_1} \cdots e^{t_k X_k} g.$$

Its partial derivatives are given by

$$\frac{d\rho}{dt_i}(t_1, \ldots, t_k) = d\left(L_{e^{t_1 X_1} \ldots e^{t_{i-1} X_{i-1}}}\right)_{z_i} \left(X^r_i(z_i)\right),$$

where $z_i = e^{t_i X_i} \cdots e^{t_k X_x} g$. By means of successive applications of the previous lemma, it follows that these partial derivatives belong to $\Delta^r_{\mathfrak{h}}(z_i)$. This means the image of $d\rho_t$ is contained in $\Delta^r_{\mathfrak{h}}(\rho(t))$ for every $t \in \mathbb{R}^k$. On the other hand,

$$\frac{d\rho}{dt_i}(0) = X^r_i(g)\,,$$

ensuring that the differential $d\rho_0$ is injective at the origin. Therefore, $d\rho_t$ is injective on a neighborhood U of 0 and, since the dimensions coincide, the conclusion is that the restriction of ρ to U is an integral manifold of $\Delta^r_{\mathfrak{h}}$.

The left invariant distribution $\Delta^l_{\mathfrak{h}}$ is treated in the same way, multiplying the exponentials by the right side: $g e^{t_1 X_1} \cdots e^{t_k X_x}$. □

As mentioned above, this theorem is a consequence of the Frobenius theorem. In addition to these proofs, in Exercise 7 of Chapter 8, another construction of

integral manifolds is pointed out. It relies on the formula for the differential of the exponential map. Still on the above theorem, it should be noticed that the map ρ, which is globally defined on $\mathbb{R}^k$, is not, in general, an immersion on every $\mathbb{R}^k$. An example is suggested in Exercise 3, at the end of this chapter.

Once the integrability of distributions $\Delta^r_{\mathfrak{h}}$ and $\Delta^l_{\mathfrak{h}}$ is ensured, it is possible to consider their maximal connected integral manifolds. According to Appendix B, the following statements hold:

1. For each $g \in G$, there exists a unique maximal connected integral manifold through g of $\Delta^r_{\mathfrak{h}}$ and $\Delta^l_{\mathfrak{h}}$. These manifolds are denoted by $I^r_{\mathfrak{h}}(g)$ and $I^l_{\mathfrak{h}}(g)$, respectively. (See Theorem B.19.)
2. If $N \subset G$ is a connected integral manifold of $\Delta^r_{\mathfrak{h}}$ (respectively of $\Delta^l_{\mathfrak{h}}$) with $g \in N$, then N is an open submanifold of $I^r_{\mathfrak{h}}(g)$ (respectively of $I^l_{\mathfrak{h}}(g)$). (See Theorem B.19.)
3. The maximal connected integral manifolds are quasi-regular submanifolds. (See Proposition B.24. Here the hypothesis that G is a paracompact manifold is required.)

The integral manifolds $I^r_{\mathfrak{h}}(g)$ are invariant by right translations. Indeed, the right invariance of $\Delta^r_{\mathfrak{h}}$ implies that $I^r_{\mathfrak{h}}(g)\,h = R_h I^r_{\mathfrak{h}}(g)$ is a connected integral manifold of $\Delta^r_{\mathfrak{h}}$. Since $gh \in I^r_{\mathfrak{h}}(g)\,h$, it follows that $I^r_{\mathfrak{h}}(g)\,h \subset I^r_{\mathfrak{h}}(gh)$. This inclusion is an equality because $I^r_{\mathfrak{h}}(g) = R_{h^{-1}}\left(I^r_{\mathfrak{h}}(g)\,h\right) \subset R_{h^{-1}}\left(I^r_{\mathfrak{h}}(gh)\right)$ and this last term is a connected integral manifold. Therefore, the maximal connected integral manifolds are given by

- $I^r_{\mathfrak{h}}(g) = I^r_{\mathfrak{h}}(1)\,g$. In the same way, $I^l_{\mathfrak{h}}(g) = g I^l_{\mathfrak{h}}(1)$.

It is now possible to prove the theorem on the existence of a Lie subgroup with a given Lie subalgebra.

Theorem 6.5 *Given a Lie group G with Lie algebra $\mathfrak{g}$ let $\mathfrak{h} \subset \mathfrak{g}$ be a subalgebra. Then, the maximal connected integral submanifolds $I^r_{\mathfrak{h}}(1)$ and $I^l_{\mathfrak{h}}(1)$ are given by*

$$I^r_{\mathfrak{h}}(1) = I^l_{\mathfrak{h}}(1) = \{e^{Y_1} \cdots e^{Y_s} : s \geq 0,\ Y_i \in \mathfrak{h}\}. \tag{6.2}$$

This set is a Lie subgroup with Lie algebra $\mathfrak{h}$. Its cosets are the maximal integral manifolds $I^r_{\mathfrak{h}}(1)\,g = I^r_{\mathfrak{h}}(g)$ and $g I^l_{\mathfrak{h}}(1) = I^l_{\mathfrak{h}}(g)$.

Proof It all comes down to proving the equality (6.2). To do that, define in $I^r_{\mathfrak{h}}(1)$ the relation $g \sim h$ if there exist $Y_1, \ldots, Y_s \in \mathfrak{h}$ such that

$$g = e^{Y_1} \cdots e^{Y_s} h,$$

which is obviously an equivalence relation. Claim: The equivalence classes of this equivalence relation are open sets of $I^r_{\mathfrak{h}}(1)$. Indeed, take $g \in I^r_{\mathfrak{h}}(1)$ and a basis $\{X_1, \ldots, X_k\}$ of $\mathfrak{h}$. Then, as in the proof of Theorem 6.4, the map

$$\rho(t_1, \ldots, t_k) = e^{t_1 X_1} \cdots e^{t_k X_k} g$$

defines an integral manifold N with $g \in N$. By definition, if $h \in N$, then $h \sim g$ and, since $N \subset I^r_{\mathfrak{h}}(1)$ is open, it follows that g belongs to the interior of its equivalence class. This shows that any equivalence class is open. As the complement of an equivalence class is the union of the remaining classes, it follows that the equivalence classes are both open and closed sets. It follows that $g \sim h$ if $g, h \in I^r_{\mathfrak{h}}(1)$, since $I^r_{\mathfrak{h}}(1)$ is connected. Hence $I^r_{\mathfrak{h}}(1) = \{e^{Y_1} \cdots e^{Y_s} : s \geq 0,\ Y_i \in \mathfrak{h}\}$, as the right hand side is the equivalence class of 1.

In the same way, the integral submanifolds $(t_1, \ldots, t_k) \mapsto g e^{t_1 X_1} \cdots e^{t_k X_k}$, $g \in I^l_{\mathfrak{h}}(1)$, prove the equality of the statement for $I^l_{\mathfrak{h}}(1)$.

Now, the expression (6.2) shows immediately that $I^r_{\mathfrak{h}}(1) = I^l_{\mathfrak{h}}(1)$ is a subgroup. Finally, this subgroup is a Lie subgroup, because a maximal connected integral manifold is quasi-regular and this ensures the differentiability of the product with respect to the intrinsic structure. □

It should be stressed that $I^r_{\mathfrak{h}}(1) = I^l_{\mathfrak{h}}(1)$, as happens with the 1-parameter subgroups.

As a complement to the previous theorem, the uniqueness of the connected Lie subgroup with the given Lie subalgebra must be verified.

Proposition 6.6 *Let G be a Lie group with Lie algebra $\mathfrak{g}$. Then, for any Lie subalgebra $\mathfrak{h} \subset \mathfrak{g}$, there exists a unique connected subgroup $H \subset G$ whose Lie algebra is $\mathfrak{h}$.*

Proof Indeed, if H is a connected Lie subgroup with Lie algebra $\mathfrak{h}$, then H is generated by exponentials $\exp_H X$, $X \in \mathfrak{h}$. Since the exponential map on H is the restriction of the exponential map on G, then H is given by (6.2), showing the uniqueness of H. The existence is ensured by the theorem above. □

Notation The only connected subgroup H with Lie algebra $\mathfrak{h}$ is denoted by $\langle \exp \mathfrak{h} \rangle$. This notation is consistent with formula (6.2), which says that H is generated by the set $\exp \mathfrak{h}$.

The above results show that every connected Lie subgroup is an integral manifold of a distribution and, consequently, is a quasi-regular submanifold.

In general, integrable distributions admit adapted charts (see Appendix B, Section B.4). In the particular case of the distribution $\Delta^r_{\mathfrak{h}}$ (or $\Delta^l_{\mathfrak{h}}$), whose integral manifolds are the cosets of $\langle \exp \mathfrak{h} \rangle$, an adapted chart around the identity is given by the exponential on G in the following way.

Proposition 6.7 *Let G be a Lie group with Lie algebra $\mathfrak{g}$ and $\mathfrak{h} \subset \mathfrak{g}$ a subalgebra. Suppose that $\mathfrak{e} \subset \mathfrak{g}$ is a vector subspace that complements $\mathfrak{h}$ in $\mathfrak{g} = \mathfrak{e} \oplus \mathfrak{h}$. Then, there exist open sets $0 \in V \subset \mathfrak{e}$, $0 \in U \subset \mathfrak{h}$ and $1 \in W \subset G$ such that the map $\psi : V \times U \to W$ defined by $\psi(X, Y) = e^X e^Y$ is a diffeomorphism and therefore an adapted chart for $\Delta_{\mathfrak{h}}$. In other words, $W = e^V e^U$.*

Proof The map $\psi : \mathfrak{h} \times \mathfrak{e} \to G$, given by $\psi(X, Y) = e^X e^Y$, is well defined. Its differential at $(0, 0)$ is the identity map, that is, the inclusion of $V \times \mathfrak{h}$ into $\mathfrak{g}$. Hence, ψ is a local diffeomorphism on a neighborhood of the origin, ensuring the existence of the neighborhoods U and W of the statement. For each $Y \in \mathfrak{e}$, $\psi(\{Y\} \times \mathfrak{h})$ is contained in the coset $(\exp Y)\langle \exp \mathfrak{h} \rangle$, which is an integral manifold of $\Delta_{\mathfrak{h}}$. Hence, ψ is an adapted chart. □

Examples

1. Let $\mathfrak{g}$ be a finite dimensional real Lie algebra. The image of its adjoint representation $\mathrm{ad} : \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$ is the Lie algebra of linear maps

$$\mathrm{ad}(\mathfrak{g}) = \{\mathrm{ad}(X) \in \mathfrak{gl}(\mathfrak{g}) : X \in \mathfrak{g}\}.$$

   The only connected subgroup $\langle \exp(\mathrm{ad}(\mathfrak{g})) \rangle$ of $\mathrm{Gl}(\mathfrak{g})$ whose Lie algebra is $\mathrm{ad}(\mathfrak{g})$ is denoted by $\mathrm{Int}(\mathfrak{g})$. As $\exp \mathrm{ad}(X) = \mathrm{Ad}(\exp X)$, the elements of $\mathrm{Int}(\mathfrak{g})$ are automorphisms of $\mathfrak{g}$. They are called **inner automorphisms** of $\mathfrak{g}$. If G is connected, then $\mathrm{Int}(\mathfrak{g})$ is the image of the adjoint representation of G, since both G and $\langle \exp(\mathrm{ad}(\mathfrak{g})) \rangle$ are generated by exponentials.
2. The theorem of Ado[2] ensures that every finite dimensional Lie algebra $\mathfrak{g}$ admits a faithful (that is, injective) representation $\rho : \mathfrak{g} \to \mathfrak{gl}(V)$ of finite dimension $\dim V = n$. In this case, $\mathfrak{g}$ is isomorphic to its image $\rho(\mathfrak{g})$. Hence, if $\mathfrak{g}$ is a real Lie algebra, it is isomorphic to a Lie subalgebra of $\mathfrak{gl}(n, \mathbb{R})$ of the Lie group $\mathrm{Gl}(n, \mathbb{R})$. Therefore the subgroup $G = \langle \exp \rho(\mathfrak{g}) \rangle$ is a Lie group with Lie algebra isomorphic to $\mathfrak{g}$. In short, every Lie algebra on $\mathbb{R}$ is isomorphic to the Lie algebra of some Lie group. This is the content of what is known as the **third Lie theorem**. □

6.3 Ideals and Normal Subgroups

The objective of this section is to particularize for normal subgroups the relations between Lie subgroups and Lie subalgebras established in the previous section.

Let $\mathfrak{g}$ be a Lie algebra and $\mathfrak{v}$ a subspace of $\mathfrak{g}$. The **normalizer** of $\mathfrak{v}$ on $\mathfrak{g}$, denoted by $\mathfrak{n}(\mathfrak{v})$, is defined by

$$\mathfrak{n}(\mathfrak{v}) = \{X \in \mathfrak{g} : \mathrm{ad}(X)\,\mathfrak{v} \subset \mathfrak{v}\}.$$

It follows by the Jacobi identity that $\mathfrak{n}(\mathfrak{v})$ is a subalgebra of $\mathfrak{g}$. Clearly, $\mathfrak{v}$ is a subalgebra if and only if $\mathfrak{v} \subset \mathfrak{n}(\mathfrak{v})$ and $\mathfrak{v}$ is an **ideal** of $\mathfrak{g}$ if $\mathfrak{n}(\mathfrak{v}) = \mathfrak{g}$, that is, if $[X, \mathfrak{v}] \subset \mathfrak{v}$ for every $X \in \mathfrak{g}$.

[2] See *Álgebras de Lie* [47, Chapter 10] and Varadarajan [53, Section 3.17].

Let G be a Lie group with Lie algebra $\mathfrak{g}$ and take a subspace $\mathfrak{v} \subset \mathfrak{g}$. The **normalizer** of $\mathfrak{v}$ on G, denoted by $N(\mathfrak{v})$, is defined by

$$N(\mathfrak{v}) = \{g \in G : \mathrm{Ad}(g)\,\mathfrak{v} = \mathfrak{v}\}.$$

Since Ad is a homomorphism, the normalizer $N(\mathfrak{v})$ is a subgroup of G.

Proposition 6.8 *Let H be a Lie subgroup of G and denote by $\mathfrak{h}$ its Lie algebra. Suppose that $g \in G$ normalizes H, that is, $gHg^{-1} \subset H$. Then g normalizes $\mathfrak{h}$, that is,* $\mathrm{Ad}(g)\,\mathfrak{h} = \mathfrak{h}$.

Proof For $X \in \mathfrak{h}$, it holds

$$ge^{tX}g^{-1} = e^{t\mathrm{Ad}(g)X}$$

for every $t \in \mathbb{R}$. The fact that g normalizes H implies that $e^{t\mathrm{Ad}(g)X}$ is a curve on H. It is differentiable with respect to the intrinsic structure of H and its derivative at $t = 0$ is $\mathrm{Ad}(g)\,X$. Hence, $\mathrm{Ad}(g)\,X$ is in $T_1H \subset T_1G$, that is, in $\mathfrak{h}$. This shows the inclusion $\mathrm{Ad}(g)\,\mathfrak{h} \subset \mathfrak{h}$ and therefore the equality $\mathrm{Ad}(g)\,\mathfrak{h} = \mathfrak{h}$. □

Corollary 6.9 *If H is a normal Lie subgroup of G, then its Lie algebra $\mathfrak{h}$ is an ideal of the Lie algebra $\mathfrak{g}$ of G.*

Proof Given $X \in \mathfrak{g}$, the previous proposition ensures that $\mathrm{Ad}\left(e^{tX}\right)\mathfrak{h} = \mathfrak{h}$ for every $t \in \mathbb{R}$. In particular, if $Y \in \mathfrak{h}$, the curve $\mathrm{Ad}\left(e^{tX}\right)Y \subset \mathfrak{h}$ and this implies that its derivative at $t = 0$ is in $\mathfrak{h}$. But

$$\frac{d}{dt}\mathrm{Ad}\left(e^{tX}\right)Y_{|t=0} = [X, Y]_l,$$

concluding the proof. □

The converse of this corollary states that if $\mathfrak{h}$ is an ideal of $\mathfrak{g}$, then $\langle \exp \mathfrak{h} \rangle$ is a normal subgroup. This converse result does not hold if G is not connected (see an example below). The proof for the connected case is presented in the sequel.

Proposition 6.10 *Let G be a connected Lie group with Lie algebra $\mathfrak{g}$ and $\mathfrak{h} \subset \mathfrak{g}$ an ideal. Then, $H = \langle \exp \mathfrak{h} \rangle$ is a normal subgroup of G.*

Proof Since H is a product of exponentials of elements of $\mathfrak{h}$, it suffices to show that $g\left(e^X\right)g^{-1} \in H$ if $g \in G$ and $X \in \mathfrak{h}$. But

$$g\left(e^X\right)g^{-1} = e^{\mathrm{Ad}(g)X}.$$

Hence it is enough to show that $\mathrm{Ad}(g)\,\mathfrak{h} = \mathfrak{h}$ for every $g \in G$. To do it, take $Y \in \mathfrak{g}$. Then, $\mathrm{ad}(Y)\,\mathfrak{h} \subset \mathfrak{h}$, since $\mathfrak{h}$ is an ideal. As the linear map $\mathrm{ad}(Y)$ leaves $\mathfrak{h}$ invariant, the same occurs with its exponential. Hence, $\mathrm{Ad}\left(e^Y\right)\mathfrak{h} = e^{\mathrm{ad}(Y)}\mathfrak{h} = \mathfrak{h}$ for every

$Y \in \mathfrak{g}$. Now, using the hypothesis that G is connected, every element $g \in G$ is a product of exponentials and thus $\mathrm{Ad}(g)\,\mathfrak{h} = \mathfrak{h}$, concluding the proof. □

Normal subgroups appear in group theory as kernels of homomorphisms. In the case of a differentiable homomorphism ϕ, it was mentioned in Example 4 of Section 6.1 that its kernel is a Lie subgroup and hence a normal Lie subgroup whose Lie algebra is the ideal $\ker (d\phi)_1$.

Example In Proposition 6.10, it was proved that in a connected Lie group G the subgroup $\langle \exp \mathfrak{h} \rangle$ is normal if $\mathfrak{h}$ is an ideal. The hypothesis that G is connected is essential, as shown by the following example: Let R_θ be the finite subgroup of rotations on $\mathbb{R}^2$ generated by the angle $\theta = \dfrac{2\pi}{q}$, where q is an integer > 2 (for instance, if $\theta = \pi/2$, then R_θ has four elements). The set $G = R_\theta \times \mathbb{R}^2$ is a Lie subgroup of the group of affine maps of $\mathbb{R}^2$. The connected component of the identity of G is $\{1\} \times \mathbb{R}^2$ and the number of connected components of G is equal to the order of R_θ. The Lie algebra $\mathfrak{g}$ of G is the abelian 2-dimensional algebra $\mathbb{R}^2$. Any subspace of $\mathfrak{g}$ is an ideal and the associated connected subgroup is identified with the subspace. However, these subgroups are not normal if they have dimension 1 because the rotations in R_θ do not leave invariant any subspace with dimension 1. □

6.4 Limits of Products of Exponentials

In this section, two limits involving products of exponentials are computed. These limits will be used later in the proof of the closed subgroup theorem of Cartan.

Take G a Lie group with Lie algebra $\mathfrak{g}$. Through this section, U and W will be fixed neighborhoods of $0 \in \mathfrak{g}$ and $1 \in G$, respectively, such that $\exp : U \to W$ is a diffeomorphism.

Given $X, Y \in \mathfrak{g}$, consider the curve $\alpha(t) = e^{tX} e^{tY}$, with $\alpha(0) = 1$. If t is small enough, $\alpha(t) \in W$ and hence defines the curve $\beta(t) \in U$ by $\exp \beta(t) = \alpha(t)$. An immediate computation shows that

$$\alpha'(0) = X + Y.$$

Since $d(\exp)_1 = \mathrm{id}$, it follows that $\beta'(0) = \alpha'(0) = X + Y$. This ensures that, on some interval containing $0 \in \mathbb{R}$,

$$\beta(t) = t(X + Y) + o(t)$$

with $\lim\limits_{t \to 0} \dfrac{o(t)}{t} = 0$. In other words,

$$\exp(tX) \exp(tY) = \exp(t(X + Y) + o(t)). \tag{6.3}$$

This expression allows to show the limit below, known as the **Lie product formula**.

Proposition 6.11 *Given $X, Y \in \mathfrak{g}$, it holds*

$$\exp(X+Y) = \lim_{n\to\infty} \left(\exp\left(\frac{X}{n}\right)\exp\left(\frac{Y}{n}\right)\right)^n. \tag{6.4}$$

Proof Plugging $t = \dfrac{1}{n}$ into (6.3) yields

$$\exp\left(\frac{X}{n}\right)\exp\left(\frac{Y}{n}\right) = \exp\left(\frac{1}{n}(X+Y) + o\left(\frac{1}{n}\right)\right).$$

Since $\lim\limits_{n\to\infty} \dfrac{o(1/n)}{1/n} = 0$, it follows that

$$\lim_{n\to\infty} \exp\left(\frac{1}{n}(X+Y) + o\left(\frac{1}{n}\right)\right)^n = \exp(X+Y),$$

concluding the proof. □

This proposition will be used below to ensure that the set of elements of $\mathfrak{g}$ whose exponentials belong to a closed subgroup H is a vector subspace. In the sequel, another lemma will be proved which will ensure that this set is a subalgebra.

Given $X, Y \in \mathfrak{g}$, consider the curve

$$\alpha(t) = e^{tX}e^{tY}e^{-tX}e^{-tY},$$

which satisfies $\alpha(0) = 0$. If t is small enough $\alpha(t)$ is in the neighborhood W, where $\exp : U \to W$ is a diffeomorphism. For these values of t,

$$\alpha(t) = e^{\beta(t)}$$

with $\beta(t) \in U \subset \mathfrak{g}$, a differentiable curve. This curve β can be written in terms of flows of vector fields on U. Indeed, denote by $\log : W \to U$ the inverse of exp and let $\widehat{X} = \log_* X^r$ be the vector field on U which is log-related to X. Then, the images of the trajectories of $\widehat{X}$ by exp are trajectories of X. In this way, defining $\widehat{Y}$ in the same way and taking into account that the flow of the right invariant vector field X^r is given by $X^r_t(g) = \exp(tX)\,g$, the curve β is defined by

$$\beta(t) = \widehat{X}_t \circ \widehat{Y}_t \circ \widehat{X}_{-t} \circ \widehat{Y}_{-t}(0).$$

The first two derivatives at $t = 0$ of a curve defined by composition of flows of vector fields in a vector space can be computed using the properties of flows (see Appendix A, Proposition A.6). By these computations it follows that

$$\beta'(0) = 0 \qquad \text{and} \qquad \beta''(0) = -2[\widehat{X}, \widehat{Y}](0).$$

But $[X, Y]_r = [\widehat{X}, \widehat{Y}](0)$. Hence, in a neighborhood of $0 \in \mathbb{R}$, $\beta(t) = -t^2[X, Y]_r + o(t)$ with $\lim_{t\to 0} \frac{o(t)}{t^2} = 0$. In terms of exponentials, this expression for β is written

$$e^{tX} e^{tY} e^{-tX} e^{-tY} = \exp\left(-t^2[X, Y]_r + o(t)\right). \tag{6.5}$$

Proposition 6.12 *Given $X, Y \in \mathfrak{g}$, it holds*

$$e^{-[X,Y]_r} = \lim_{n\to\infty} \left(e^{\frac{X}{n}} e^{\frac{Y}{n}} e^{-\frac{X}{n}} e^{-\frac{Y}{n}}\right)^{n^2}.$$

Proof Plugging $t = \dfrac{1}{n}$ into (6.5) yields

$$e^{\frac{X}{n}} e^{\frac{Y}{n}} e^{-\frac{X}{n}} e^{-\frac{Y}{n}} = \exp\left(-\frac{1}{n^2}[X, Y]_r + o\left(\frac{1}{n}\right)\right).$$

Since $\lim_{n\to\infty} \dfrac{o(1/n)}{1/n^2} = 0$, it follows, taking the power n^2, that

$$\lim_{n\to\infty} \exp\left(-\frac{1}{n^2}[X, Y]_r + o\left(\frac{1}{n}\right)\right)^{n^2} = \exp\left(-[X, Y]_r\right),$$

concluding the proof. □

6.5 Closed Subgroups

The closed subgroup theorem of Cartan ensures that any closed subgroup H of a Lie group G is in fact a Lie subgroup, that is, admits a differentiable manifold structure that makes it a Lie subgroup. This is one of the classical results of Lie group theory and is widely used in many different situations.

Let $H \subset G$ be a closed subgroup. The strategy to prove that H is a Lie subgroup consists in defining, from H, a Lie subalgebra $\mathfrak{h} \subset \mathfrak{g}$ and then to construct the differentiable structure on H from the structure of $\langle \exp \mathfrak{h} \rangle$. So, define

$$\mathfrak{h}_H = \{X \in \mathfrak{g} : \forall t \in \mathbb{R},\ \exp tX \in H\}. \tag{6.6}$$

With the limits proved in the previous section, it is easy to verify that $\mathfrak{h}_H$ is a Lie subalgebra of $\mathfrak{g}$.

Proposition 6.13 *Given a Lie group G with Lie algebra $\mathfrak{g}$, let $H \subset G$ be a closed subgroup. Then $\mathfrak{h}_H$ is a Lie subalgebra of $\mathfrak{g}$.*

Proof The set $\mathfrak{h}_H$ is nonempty because $0 \in \mathfrak{h}_H$. Take $X, Y \in \mathfrak{h}_H$. The 1-parameter subgroups generated by X and aX coincide if $0 \neq a \in \mathbb{R}$. Hence, $aX \in \mathfrak{h}_H$, $a \in \mathbb{R}$. The Lie product formula (6.4) ensures that $X + Y \in \mathfrak{h}_H$, as H is closed. In the same way, Proposition 6.12 ensures that $\mathfrak{h}_H$ is closed by the bracket, concluding the proof. □

The main step in the proof of the closed subgroup theorem is the next lemma. This lemma is a useful tool and is applied in several other contexts like, for instance, in the construction of the differentiable structures on quotient spaces to be done later.

Lemma 6.14 *Given a closed subgroup H, define, as in (6.6), the subalgebra $\mathfrak{h}_H$. Choose a subspace $\mathfrak{e}$ such that $\mathfrak{g} = \mathfrak{h}_H \oplus \mathfrak{e}$. Then, there exist $U \subset \mathfrak{h}_H$ and $V \subset \mathfrak{e}$ (domains of an adapted chart as in Proposition 6.7) such that, if $W = e^U e^V$, then*

$$H \cap W = e^U. \tag{6.7}$$

Proof By definition, $\langle \exp \mathfrak{h}_H \rangle \subset H$, hence $e^U \subset H \cap W$. Moreover,

$$H \cap W = e^U \left(H \cap e^V \right),$$

since for $X \in U$ and $Y \in V$, $e^X e^Y \in H$ if and only if $e^Y = e^{-X} e^X e^Y \in H$. Thus, one must find V such that $H \cap e^V = \{1\}$.

Supposing by contradiction that there does not exist such an open set V. Then there exists a sequence $Y_n \in \mathfrak{e}$ with $Y_n \neq 0$, such that $y_n = e^{Y_n} \in H$ and $\lim Y_n = 0$, that is, $\lim y_n = 1$.

Take compact neighborhoods $0 \in V'' \subset V' \subset \mathfrak{e}$ such that $2V'' = V'' + V'' \subset V'$ (for instance, balls $V' = B[0, \delta]$ and $V'' = B[0, \delta/10]$, with respect to some norm). If n is big enough, then $Y_n \in V''$ and, by compactness, there is an integer $N(n) \geq 1$ such that $N(n) Y_n \notin V''$.

Let k_n be the least positive integer such that $k_n Y_n \notin V''$. Then, $(k_n - 1) Y_n \in V''$, and it follows that

$$k_n Y_n = (k_n - 1) Y_n + Y_n \in V'' + V'' \subset V'.$$

By compactness, it can be assumed that $k_n Y_n$ converges to $Y \in V'$. As $k_n Y_n \notin V''$, it follows that Y does not belong to the interior of V'', and hence that $Y \neq 0$. Taking exponentials, one obtains $y_n^{k_n} = e^{k_n Y_n} \in H$ and $y_n^{k_n} \to y = e^Y \neq 1$. Therefore, $y \in H$, since H is closed.

This yields a contradiction since it implies that $e^{tY} \in H$ for every $t \in \mathbb{R}$. Indeed, suppose first that $t = p/q$ is rational ($p, q \in \mathbb{Z}$, $q > 0$). If a_n is the quotient of the division of pk_n by q, then $pk_n = a_n q + b_n$ with $0 \leq b_n < q$. Dividing this equality by q and multiplying by Y_n, one obtains

$$tk_n Y_n = a_n Y_n + (b_n/q)\, Y_n.$$

Taking limits, the left hand side converges to tY, while $(b_n/q)\, Y_n \to 0$, because $b_n < q$ and $Y_n \to 0$. Therefore, $\lim_{n\to+\infty} a_n Y_n = tY$. Hence

$$e^{tY} = \lim_{n\to\infty} e^{a_n Y_n} = \lim_{n\to\infty} \left(e^{Y_n}\right)^{a_n} \in H,$$

since $\left(e^{Y_n}\right)^{a_n} \in H$ and H is closed. This proves that $e^{tY} \in H$ if t is rational. Using again the fact that H is closed, the continuity of the exponential map ensures that $e^{tY} \in H$ for every $t \in \mathbb{R}$. But this contradicts the definition of $\mathfrak{h}_H$, since Y is in the subspace $\mathfrak{e}$ complementary of $\mathfrak{h}_H$ and $Y \neq 0$, concluding the proof. □

It is now possible to state and conclude the proof of the closed subgroup theorem of Cartan.

Theorem 6.15 *Every closed subgroup H of a Lie group G is a Lie subgroup. More precisely: A closed subgroup H admits an **embedded** manifold structure that makes it a Lie subgroup. Its Lie algebra is $\mathfrak{h}_H$ (defined in (6.6)).*

Proof Let $\psi : V \times U \to W$, $\psi(X, Y) = e^X e^Y$ be the adapted chart around the identity from Lemma 6.14, with $H \cap W = e^U$. Then, for every $h \in H$, the set $Wh = e^V e^U h$ is a neighborhood of h in G. This neighborhood satisfies

$$H \cap Wh = \left(Hh^{-1} \cap W\right) h = (H \cap W)\, h = e^U h.$$

Hence, the diffeomorphism $R_h \circ \psi : V \times U \to W$ satisfies $R_h \circ \psi(\{0\} \times U) = H \cap Wh$. As $h \in H$ is arbitrary, this shows that H is an embedded submanifold of G (see Proposition B.1). In particular, H is a quasi-regular submanifold and hence H is a Lie group. Applying again Lemma 6.14, it is found that the tangent space to H at the identity is $\mathfrak{h}_H$ and therefore this is the Lie algebra of H. □

The closed subgroup theorem admits the following converse.

Proposition 6.16 *If $H \subset G$ is an embedded Lie subgroup, then H is closed.*

Proof As H is an embedded submanifold, there is a chart

$$\psi : U \times V \subset \mathbb{R}^k \times \mathbb{R}^{n-k} \to W \subset G$$

with $(0, 0) \in U \times V$ and $1 \in W$, such that $\psi(0, 0) = 1$ and $H \cap W = \psi(U \times \{0\})$. In this case, $H \cap W$ is closed in W because $U \times \{0\}$ is closed in $U \times V$. This implies that $\overline{H} \cap W = H \cap W$. But $\overline{H}$ is also a submanifold and this equality shows that $T_1 H = T_1 \overline{H}$, so that H is an open subgroup and, hence, a closed subgroup of $\overline{H}$. Consequently, H is closed in G. □

The results of this section show that a Lie subgroup $H \subset G$ is closed if and only if the submanifold H is an embedded manifold. But the proof of the closed subgroup

theorem goes further. It shows that H is an embedded Lie subgroup if H is **locally closed**, in the sense that every $h \in H$ admits a neighborhood W such that $H \cap W$ is closed in W. This is so because Lemma 6.14 continues to hold with the hypothesis that H is locally closed around 1. Hence there is the following consequence.

Corollary 6.17 *If H is a locally closed subgroup of a Lie group G, then H is closed.*

A case in which a subgroup $\Gamma \subset G$ is locally closed is when it is a **discrete subgroup** in the sense that there exists a neighborhood U of the identity such that $U \cap \Gamma = \{1\}$. Then the subgroup Γ is closed and $\dim \Gamma = 0$. Conversely, if Γ is a closed subgroup with $\dim \Gamma = 0$, then Γ is discrete because in this case the subalgebra $\mathfrak{h}_\Gamma = \{0\}$ and the open set $W = e^V e^U$ of Lemma 6.14 reduces to e^V and satisfies $\Gamma \cap e^V = \{1\}$. This fact is highlighted in the following proposition for future reference, because discrete subgroups play a central role in the description of connected Lie groups.

Proposition 6.18 *Let $\Gamma \subset G$ be a closed subgroup with $\dim \Gamma = 0$. Then Γ is discrete.*

Examples The following are some examples of closed Lie subgroups.

1. For $x \in G$, let

$$Z(x) = \{y \in G : yx = xy\}$$

be the centralizer of x in G. Then $Z(x)$ is a closed subgroup if G is a Lie group (or even if G is a Hausdorff topological group). Indeed, $y \in Z(x)$ if and only if $C_x(y) = xyx^{-1} = y$, that is, $Z(x)$ is the set of points at which the continuous maps C_x and id coincide. As G is a Hausdorff topological space, this set is closed.
2. Let

$$Z(G) = \{x \in G : \forall y \in G,\ xy = yx\}$$

be the center of group G. Then $Z(G)$ is closed if G is a Lie group (or even if G is a Hausdorff topological group). Indeed, $Z(G) = \bigcap_{x \in G} Z(x)$ and each $Z(x)$ is closed, according to the previous example.
3. Let $\mathfrak{g}$ be a finite dimensional real Lie algebra. Denote by $\mathrm{Aut}(\mathfrak{g})$ the group of automorphisms of $\mathfrak{g}$, that is $g \in \mathrm{Aut}(\mathfrak{g})$ if $g : \mathfrak{g} \to \mathfrak{g}$ is an invertible linear map and $g[X, Y] = [gX, gY]$ for every $X, Y \in \mathfrak{g}$. It is clear that $\mathrm{Aut}(\mathfrak{g})$ is a subgroup of $\mathrm{Gl}(\mathfrak{g})$. Let g_n be a sequence in $\mathrm{Aut}(\mathfrak{g})$ such that $g = \lim g_n$ is in $\mathrm{Gl}(\mathfrak{g})$. Since bilinear forms between finite dimensional vector spaces are continuous, the equalities $g_n[X, Y] = [g_n X, g_n Y]$, $n \geq 1$, are maintained in the limit, showing that $g \in \mathrm{Aut}(\mathfrak{g})$. Hence, $\mathrm{Aut}(\mathfrak{g})$ is a closed subgroup of $\mathrm{Gl}(\mathfrak{g})$ and, as such, is a Lie group. The Lie algebra of $\mathrm{Aut}(\mathfrak{g})$ is the algebra of derivations $\mathrm{Der}(\mathfrak{g})$ of $\mathfrak{g}$ because, if $D : \mathfrak{g} \to \mathfrak{g}$ is a linear map, then $e^{tD} \in \mathrm{Aut}(\mathfrak{g})$ if and only if $D \in \mathrm{Der}(\mathfrak{g})$.

4. The group $\mathrm{Gl}(n, \mathbb{C})$ of invertible complex-linear maps is a subgroup of $\mathrm{Gl}(2n, \mathbb{R})$. A linear map g of $\mathbb{R}^{2n}$ is complex if and only if it commutes with the linear map $J : \mathbb{R}^{2n} \to \mathbb{R}^{2n}$, which corresponds to the multiplication by i in $\mathbb{C}^n$. From this, it is easy to verify that $\mathrm{Gl}(n, \mathbb{C})$ is a closed subgroup of $\mathrm{Gl}(2n, \mathbb{R})$ and, hence, a Lie group. This can be seen in terms of matrices, since the left multiplication of a complex matrix $Z = A + iB$ by elements of $\mathbb{C}^n$ defines a linear map of $\mathbb{R}^{2n}$ with matrix

$$\begin{pmatrix} A & -B \\ B & A \end{pmatrix}.$$

   Thus, $\mathrm{Gl}(n, \mathbb{C})$ can be viewed as the subgroup of $\mathrm{Gl}(2n, \mathbb{R})$ of matrices of this form. This subgroup is closed.
5. A construction similar to the construction of the previous item can be done with $\mathbb{H}$. Write a quaternion as

$$q = a + ib + jc + kd = (a + ib) + j(c - id) = z + jw$$

   with $z, w \in \mathbb{C}$. Then the product of quaternions becomes

$$(z + jw)(z_1 + jw_1) = (zz_1 - \overline{w}z_1) + j(\overline{z}w_1 + wz_1),$$

   since $zj = j\overline{z}$ if $z \in \mathbb{C}$. In this way, the left multiplication in $\mathbb{H}$ by $q = z + jw$ defines a linear map on $\mathbb{C}^2$ with matrix

$$\begin{pmatrix} z & -\overline{w} \\ w & \overline{z} \end{pmatrix}.$$

   This extends to $n \times n$ quaternionic matrices. Such a matrix can be written as $Q = A + jB$, with A and B complex $n \times n$ matrices. The left multiplication by Q in the space of $n \times 1$ column matrices in $\mathbb{H}^n$ defines a linear map of $\mathbb{C}^{2n}$ with matrix

$$\begin{pmatrix} A & -\overline{B} \\ B & \overline{A} \end{pmatrix}. \tag{6.8}$$

   The subgroup of $\mathrm{Gl}(2n, \mathbb{C})$, of matrices of this form, is a closed subgroup and, hence, a Lie group. This subgroup is denoted by $\mathrm{Gl}(n, \mathbb{H})$.
6. All the linear groups presented in the introduction are closed and, therefore, are Lie subgroups. They are:

   - $\mathrm{O}(n)$, $\mathrm{SO}(n)$, $\mathrm{U}(n)$, $\mathrm{SU}(n)$, $\mathrm{Sp}(n)$, $\mathrm{Sl}(n, \mathbb{R})$, $\mathrm{Sl}(n, \mathbb{C})$, $\mathrm{Sp}(n, \mathbb{R})$, $\mathrm{SO}(p, q)$, $\mathrm{SU}(p, q)$.

   To these groups other linear groups can be added which are also closed groups, all called classical groups.

(a) $\mathrm{Sl}(n, \mathbb{H})$, the group of $n \times n$ quaternionic matrices, whose complex form (6.8) has determinant 1. This group is also denoted by $\mathrm{SU}^*(2n)$.
(b) $\mathrm{Sp}(n, \mathbb{C}) = \{g \in \mathrm{Gl}(2n, \mathbb{C}) : gJg^T = 1\}$, where

$$J = \begin{pmatrix} 0 & -1_{n\times n} \\ 1_{n\times n} & 0 \end{pmatrix}.$$

(c) $\mathrm{O}(p, q), = \{g \in \mathrm{Gl}(p+q, \mathbb{R}) : gI_{p,q}g^T = 1\}$, where

$$I_{p,q} = \begin{pmatrix} 1_{p\times p} & 0 \\ 0 & -1_{q\times q} \end{pmatrix}.$$

(d) $\mathrm{U}(p, q) = \{g \in \mathrm{Gl}(p+q, \mathbb{C}) : gI_{p,q}\overline{g}^T = 1, \det g = 1\}$. □

6.6 Path Connected Subgroups

Theorem 6.19 *Let G be a Lie group and $H \subset G$ a subgroup. Suppose that, for every $h \in H$, there exists a C^1 curve connecting 1 to h. Then H is a Lie subgroup and a quasi-regular submanifold.*

The condition of the theorem is equivalent to saying that H is connected by differentiable paths, since a path between arbitrary g and h can be obtained by translating a path between 1 and $g^{-1}h$.

The strategy to prove this theorem is similar to that used in the proof of the theorem of Cartan. Consider first the subset $\mathfrak{h}_H \subset \mathfrak{g} = T_1G$ formed by the derivatives $\dot{x}_0$ of the curves $x_t \in H$, of class C^1, with $x_0 = 1$.

Lemma 6.20 *The set $\mathfrak{h}_H$ defined above is a Lie subalgebra.*

Proof The constant curve $x_t = 1$ is in H, so $0 \in \mathfrak{h}_H$. Let x_t and y_t be two C^1 curves contained in H with $x_0 = y_0 = 1$ and denote by $X = \dot{x}_0$ and $Y = \dot{y}_0$ their derivatives at the origin. If $r \in \mathbb{R}$, then the derivative at $t = 0$ of x_{rt} is equal to rX, showing that $\mathfrak{h}_H$ is closed under scalar multiplication. On the other hand, the derivative at $t = 0$ of the curve $x_t y_t \in H$ is equal to $X + Y$. Hence, $\mathfrak{h}_H$ is a vector subspace.

To obtain the bracket, consider, for each t, the curve $s \mapsto C_{x_t}(y_s) = x_t y_s x_t^{-1} \in H$. Its derivative at $s = 0$ is

$$\left(dC_{x_t}\right)_1 (Y) = \mathrm{Ad}(x_t)(Y),$$

hence the curve $t \mapsto z_t = \mathrm{Ad}(x_t)(Y)$ is in $\mathfrak{h}_H$, as well as its derivative

$$\dot{z}_0 = d(\mathrm{Ad})_1 (\dot{x}_0)(Y) = \mathrm{ad}(\dot{x}_0)(Y) = [X, Y],$$

showing that $[X, Y] \in \mathfrak{h}_H$. □

Now, to prove Theorem 6.19, it suffices to verify that $H = \langle \exp \mathfrak{h}_H \rangle$. First, take $h \in H$ and a curve $x_t \in H$ connecting the identity to h. Then, $x_{t+s}x_t^{-1} \in H$ and therefore

$$\dot{x}_t x_t^{-1} = \frac{d}{ds}\left(x_{t+s}x_t^{-1}\right)_{|s=0} \in \mathfrak{h}.$$

This means that the curve x_t is tangent to the distribution $\Delta^r_{\mathfrak{h}_H}(x) = (dR_x)_1 \mathfrak{h}_H$. Hence, x_t is entirely contained in an integral manifold I of $\Delta^r_{\mathfrak{h}_H}$ (see Proposition B.22). Since $x_0 = 1$, the integral manifold I containing x_t can only be $\langle \exp \mathfrak{h} \rangle$, showing that $H \subset \langle \exp \mathfrak{h}_H \rangle$.

For the converse inclusion, it is proved that H has nonempty interior in $\langle \exp \mathfrak{h}_H \rangle$: Let $\{X_1, \ldots, X_n\}$ be a basis of $\mathfrak{h}_H$ and take curves $x_t^1, \ldots, x_t^n$ in H with $\dot{x}_0^i = X_i$. Define the map

$$\psi : (t_1, \ldots, t_n) \longmapsto x_{t_1}^1 \cdots x_{t_n}^n \in H \subset \langle \exp \mathfrak{h}_H \rangle,$$

whose domain is an open set of $\mathbb{R}^n$ containing the origin. This is a C^1 map and its partial derivatives at the origin are given by

$$\frac{\partial \psi}{\partial t_i}(0) = X_i.$$

By the inverse function theorem, the image of ψ has a nonempty interior in $\langle \exp \mathfrak{h}_H \rangle$. As this image is contained in H, it follows that H is an open subgroup of $\langle \exp \mathfrak{h}_H \rangle$ and, therefore, $H = \langle \exp \mathfrak{h}_H \rangle$, since $\langle \exp \mathfrak{h}_H \rangle$ is connected, concluding the proof of Theorem 6.19.

An immediate corollary of the above theorem is that, if a subgroup $H \subset G$ is also a connected submanifold, then this submanifold is quasi-regular and H is a Lie subgroup. Actually, the following more general result holds.

Corollary 6.21 *Let G be a Lie group and $H \subset G$ a subgroup. Suppose that H is a submanifold with at most an enumerable quantity of connected components. Suppose also that the connected component H_0 containing the identity is a subgroup. Then H is a Lie subgroup and a quasi-regular submanifold.*

Proof Indeed, H_0 is a connected submanifold and, hence, is path connected. So, the theorem ensures that H_0 is a Lie subgroup and a quasi-regular submanifold. On the other hand, let g be an element of a connected component H_1. Then $gH_0 = L_g(H_0)$ is a connected set containing g, hence $gH_0 \subset H_1$. It follows that $H_0 = g^{-1}(gH_0) \subset g^{-1}H_1$, showing that $H_1 = gH_0$. Consequently, the connected components of H are cosets of H_0 and maximal integral manifolds of $\Delta_{\mathfrak{h}}$. By Corollary B.25, H is a quasi-regular submanifold and is a Lie group. □

In this corollary, the hypothesis that H_0 is a subgroup is necessary because it is not known in advance whether H, with the intrinsic topology, is a topological group.

6.7 Manifold Structure on G/H, H Closed

Let G be a Lie group and H a closed subgroup of G. The objective of this section is to construct a differentiable manifold structure on G/H compatible with the quotient topology. As seen in Chapter 2, the quotient topology on G/H is Hausdorff if H is closed. The next theorem states the properties of the quotient differentiable structure.

Theorem 6.22 *Let G be a Lie group and $H \subset G$ a closed subgroup. Then, there exists a differentiable structure on G/H, compatible with the quotient topology, which satisfies:*

1. $\dim G/H = \dim G - \dim H$.
2. *The canonical projection $\pi : G \to G/H$ is a submersion.*
3. *A function $f : G/H \to M$ is differentiable if and only if $f \circ \pi$ is differentiable.*
4. *The natural action $a : G \times G/H \to G/H$ is differentiable. (This action is given by $a(g, xH) = (gx)H$, that is, $g(xH) = (gx)H$.)*
5. *For each $g \in G$, the induced map $g : G/H \to G/H$, $xH \mapsto gxH$ is a diffeomorphism.*

The differentiable structure defined in this theorem is called, of course, **quotient differentiable structure**.

The construction of an atlas for the differentiable manifold structure on G/H is based on the good adapted chart ensured by Lemma 6.14 (improved in Lemma 6.23 below).

Denote by $\mathfrak{g}$ and $\mathfrak{h}$ the Lie algebras of G and H, respectively, and let $\mathfrak{e} \subset \mathfrak{g}$ be a vector subspace of $\mathfrak{g}$, such that $\mathfrak{g} = \mathfrak{e} \oplus \mathfrak{h}$. The charts on G/H will be diffeomorphisms defined on open sets of $\mathfrak{e}$ with values on open sets of G/H.

By Lemma 6.14, there exist V, U and W with $0 \in V \subset \mathfrak{e}$, $0 \in U \subset \mathfrak{h}$ and $1 \in W \subset G$, such that the map $\psi : V \times U \to W$, defined by

$$\psi(Y, X) = e^Y e^X,$$

is a diffeomorphism. The open set $W = e^V e^U$ satisfies $W \cap H = e^U$, which is the same as $e^V \cap H = \{1\}$. (Here, the order of factors $e^Y e^X$ was changed with respect to Lemma 6.14, to have $\psi(Y, X) \in e^Y H$.)

In what follows, it will be convenient to suppose that W is contained in a similar open set $W_1 = e^{V_1} e^{U_1}$ with $0 \in V \subset V_1 \subset \mathfrak{e}$, $0 \in U \subset U_1 \subset \mathfrak{h}$ and $W_1 \cap H = e^{U_1}$, and satisfying the conditions

- $W^2 \subset W_1$ and $W^{-1} W \subset W_1$.

In this case, the adapted coordinate system $\psi : V \times U \to W$ is the restriction of $\psi_1 : V_1 \times U_1 \to W_1$. This situation can be achieved by adequately shrinking the size of open sets V_1 and U_1 and by restricting ψ_1.

The construction of the atlas on G/H will be done with the help of extension of ψ to $V \times H$, defined by $\Psi : V \times H \to G$,

$$\Psi(Y, h) = e^Y h.$$

This map is differentiable because it is a composition of differentiable maps. Its differential, computed at $A \in \mathfrak{e}$ and $B^r(h)$, $B \in \mathfrak{h}$, is given by

$$\begin{aligned} d\Psi_{(Y,h)}\left(A, B^r(h)\right) &= \frac{d}{dt}\Psi\left(Y + tA, e^{tB}h\right)_{|t=0} = \frac{d}{dt}\left(e^{Y+tA}e^{tB}h\right)_{|t=0} \\ &= \frac{d}{dt}R_h\left(e^{Y+tA}e^{tB}\right)_{|t=0} \\ &= (dR_h)_{e^Y}\left(d\psi_{(Y,0)}((A, B))\right), \end{aligned}$$

that is,

$$d\Psi_{(Y,h)}\left(A, B^r(h)\right) = d\left(R_h \circ \psi\right)_{(Y,0)}(A, B). \tag{6.9}$$

Lemma 6.23 *Using the notations above, $\Psi(Y, h) = e^Y h$ is a diffeomorphism onto $e^V H$, which is an open set of G.*

Proof First, Ψ is injective. Indeed, suppose that $e^{Y_1}h_1 = e^{Y_2}h_2$. Then $e^{-Y_2}e^{Y_1} = h_2h_1^{-1}$. The left hand side of this equality is in W_1, since $W^{-1}W \subset W_1$. As the right hand side is in H, it follows that

$$e^{-Y_2}e^{Y_1} \in W_1 \cap H = e^{U_1},$$

that is, $e^{Y_1} = e^{Y_2}e^{X_1}$ for some $X_1 \in U_1$. This means that $\psi_1(Y_1, 0) = \psi_1(Y_2, X_1)$, hence $X_1 = 0$ and $Y_1 = Y_2$ since $\psi_1 : V_1 \times U_1 \to W_1$ is a diffeomorphism. Therefore, $h_1 = h_2$ and Ψ is injective. This allows to conclude that Ψ is a diffeomorphism since it is one-to-one and onto.

Expression (6.9) shows that $d\Psi_{(Y,h)}$ is an isomorphism, and this ensures that Ψ is a local diffeomorphism and, hence, its image $e^V H$ is an open set. The bijectivity of Ψ then allows to conclude that Ψ is a diffeomorphism. □

Now it is possible to define the atlas that provides the differentiable structure on G/H.

Definition 6.24 Given $g \in G$, define $\sigma_g : V \to G/H$ by

$$\sigma_g(Y) = \pi(g \exp Y) = g\pi(\exp Y),$$

that is, $\sigma_g = g \circ \pi \circ \Psi_{|V\times\{1\}}$. (In the right hand side of the latter expression, g is interpreted as a homeomorphism of G/H.)

The set of maps σ_g, $g \in G$, forms an atlas of a differentiable structure on G/H. The proof is carried out in the following items:

1. $\sigma_g = g \circ \sigma_1$, as follows immediately from the definition.

2. The image $\sigma_g(V)$ is an open set of G/H with respect to the quotient topology.
   Indeed, $\sigma_1(V) = \pi\left(e^V\right) = \pi\left(e^V H\right)$, which is open, because $e^V H$ is open and π is an open map. Hence $\sigma_g(V) = g(\sigma_1(V))$ is also open, since $g : G/H \to G/H$ is a homeomorphism.
3. $\sigma_g : V \to \sigma_g(V)$ is bijective.
   It suffices to verify the injectivity: If $\sigma_g(Y_1) = \sigma_g(Y_2)$, then there exist $h_1, h_2 \in H$ such that $ge^{Y_1}h_1 = ge^{Y_2}h_2$, that is, $e^{Y_1}h_1 = e^{Y_2}h_2$, and this means that $\Psi(Y_1, h_1) = \Psi(Y_2, h_2)$ and, hence, that $Y_1 = Y_2$ by the injectivity of Ψ.
4. $\sigma_g : V \to \sigma_g(V)$ is a homeomorphism.
   By construction, σ_g is continuous. To verify that it is an open map, take $A \subset V$. Then $\sigma_g(A) = g\pi\left(e^A\right) = g\pi\left(e^A H\right)$. If A is open, then $e^A H = \Psi(A \times H)$ is open and this implies that $\sigma_g(A)$ is open in G/H.
5. For $g_1, g_2 \in G$, the transition function $\sigma_{g_2}^{-1} \circ \sigma_{g_1}$ is given by

$$\begin{aligned} \sigma_{g_2}^{-1} \circ \sigma_{g_1}(Y) &= p\left(\Psi^{-1}\left(g_2^{-1} g_1 \Psi(Y, 1)\right)\right) \\ &= p\left(\Psi^{-1}\left(g_2^{-1} g_1 e^Y\right)\right), \end{aligned} \tag{6.10}$$

   where $p : V \times H \to V$ is the projection.
   Indeed, take $Y, Z \in V$ such that $\sigma_{g_1}(Y) = \sigma_{g_2}(Z)$. This means that $g_1 e^Y$ is in the same coset of $g_2 e^Z$, that is, there exists $h \in H$ such that

$$g_1 \Psi(Y, 1) = g_1 e^Y = g_2 e^Z h = g_2 \Psi(Z, h).$$

   This equality is rewritten as $\Psi(Z, h) = g_2^{-1} g_1 \Psi(Y, 1) = g_2^{-1} g_1 e^Y$. Using the fact that Ψ is bijective, it is seen that Z is the first coordinate of $\Psi^{-1}\left(g_2^{-1} g_1 e^Y\right)$, as stated.
   (It should be noticed that the domain of definition of $\sigma_{g_2}^{-1} \circ \sigma_{g_1}$ is the open set

$$\left\{Y \in V : g_2^{-1} g_1 e^Y \in e^V H\right\} = g_1^{-1} g_2\left(e^V H\right) \cap e^V,$$

   which is not empty if $g_2\left(e^V H\right) \cap g_1 e^V \neq \emptyset$, that is, if $\sigma_{g_1}(V) \cap \sigma_{g_2}(V) \neq \emptyset$. Moreover, if Y is in this domain of definition, then $\Psi^{-1}\left(g_2^{-1} g_1 e^Y\right)$ is well defined and hence the formula (6.10) makes sense.)

These statements show that the maps σ_g are the charts of a differentiable atlas on $G/H = \bigcup_{g \in G} \sigma_g(V)$. Item (4) ensures that σ_g is a coordinate system for an open set around gH, while item (5) shows that the transition functions are differentiable because they are compositions of differentiable maps, that is $\sigma_{g_2}^{-1} \circ \sigma_{g_1} = p \circ \Psi^{-1} \circ L_{g_2^{-1} g_1} \circ \exp$.

This concludes the construction of the differentiable manifold structure on G/H, proving the first part of Theorem 6.22. The remaining properties stated in Theorem 6.22 are obtained in the following way:

1. $\dim G/H = \dim G - \dim H$, because $\dim G/H = \dim V = \dim \mathfrak{e} = \dim \mathfrak{g} - \dim \mathfrak{h} = \dim G - \dim H$.
2. The canonical projection $\pi : G \to G/H$ is a submersion. Indeed, given $g \in G$, the maps $\psi_g = L_g \circ \psi$ and σ_g are charts around g and gH, respectively. On these charts, the projection π is read as

$$\sigma_g^{-1} \circ \pi \circ \psi_g (X, Y) = \sigma_g^{-1} \left(\pi \left(g e^Y e^X \right) \right).$$

   But $e^X \in H$, hence $\pi \left(g e^Y e^X \right) = \pi \left(g e^Y \right) = \sigma_g (Y)$. Thus $\sigma_g^{-1} \circ \pi \circ \psi_g (X, Y) = Y$ is the projection onto the second component. This also shows that π is differentiable and is a submersion.
3. The criterion of differentiability for a function $f : G/H \to M$ is an immediate consequence of the fact that $\pi : G \to G/H$ is a surjective submersion. In any case, for the charts of the previous item it holds

$$f \circ \pi \circ \psi_g (X, Y) = f \circ \sigma_g (Y),$$

   which shows that f is differentiable if and only if $f \circ \pi$ is differentiable.
4. The canonical action $a : G \times G/H \to G/H$ enters into the following commutative diagram:

$$\begin{array}{ccc} G \times G & \xrightarrow{p} & G \\ \mathrm{id} \downarrow\downarrow \pi & \searrow & \downarrow \pi \\ G \times G/H & \xrightarrow{a} & G/H \end{array}$$

   From this diagram and the criterion of differentiability for functions defined on G/H, it immediately follows that a is differentiable since $\pi \circ p$ is differentiable.
5. Given $g \in G$, the map $xH \mapsto gxH$ is differentiable because it is a partial map of action a. Its inverse is given by $xH \mapsto g^{-1}xH$, which is also differentiable, therefore, both maps are diffeomorphisms.

Finally, if H is a closed normal subgroup, then G/H is a group whose product $\overline{p} : G/H \times G/H \to G/H$ is differentiable, since it is defined by the diagram

$$\begin{array}{ccc} G \times G & \xrightarrow{p} & G \\ \pi \downarrow\downarrow \pi & \searrow & \downarrow \pi \\ G/H \times G/H & \xrightarrow{\overline{p}} & G/H \end{array}$$

(compare with Proposition 2.27). In this case, π is a differentiable homomorphism. Its differential $d\pi_1$ is a surjective homomorphism of Lie algebras whose kernel is $\mathfrak{h}$, which is an ideal of $\mathfrak{g}$. Hence, the Lie algebra of G/H, being the image of $d\pi_1$, is isomorphic to $\mathfrak{g}/\ker d\pi_1 = \mathfrak{g}/\mathfrak{h}$.

Proposition 6.25 *Let G be a Lie group and $H \subset G$ a closed normal subgroup. Then, G/H is a Lie group with the quotient structure. Its Lie algebra is isomorphic to the quotient Lie algebra $\mathfrak{g}/\mathfrak{h}$.*

6.8 Exercises

1. Show that every connected Lie subgroup of $(\mathbb{R}^n, +)$ is closed.
2. Describe the connected Lie subgroups of the Heisenberg group, that is, the Lie group of matrices of the form

$$\begin{pmatrix} 1 & x & y \\ 0 & 1 & z \\ 0 & 0 & 1 \end{pmatrix} \qquad x, y, z \in \mathbb{R}.$$

   Show that all Lie subgroups are closed. Is there any compact subgroup among them?
3. The objective of this exercise is to give an example that the map ρ defined in the proof of Theorem 6.4 is not an immersion in the whole of $\mathbb{R}^k$. Take $\mathfrak{g} = \mathfrak{gl}(n, \mathbb{R})$ and let $\mathfrak{h}$ be the subalgebra of upper triangular matrices with zeros in the diagonal. Choose the basis of $\mathfrak{h}$ given by $X_1 = E_{23} + E_{13}$, $X_2 = E_{12}$ and $X_3 = E_{23}$, where E_{ij} denotes the matrix with nonzero entry $= 1$ only in position ij. Show that the map $(t_1, t_2, t_3) \mapsto e^{t_1 X_1} e^{t_2 X_2} e^{t_3 X_3}$ is not an immersion.
4. Let $G \subset \mathrm{Gl}(n, \mathbb{R})$ be a Lie subgroup with Lie algebra $\mathfrak{g} \subset \mathfrak{gl}(n, \mathbb{R})$. Show that if G is compact, then the eigenvalues of every matrix $X \in \mathfrak{g}$ are pure imaginary.
5. Let G_1 and G_2 be Lie groups and $H_1 \subset G_1$ and $H_2 \subset G_2$ Lie subgroups. Show that $H_1 \times H_2$ is a Lie subgroup of $G_1 \times G_2$.
6. Let G be a Lie group with Lie algebra $\mathfrak{g}$ and $V \subset \mathfrak{g}$ a vector subspace. Define the distribution $\Delta_V(g) = \left(dR_g\right)_1(V)$. Show that Δ_V is not integrable if V is not a Lie subalgebra.
7. Let $H \subset G$ be a Lie subgroup with $\dim H < \dim G$ and at most an enumerable quantity of connected components. Show that H has empty interior in G.
8. Given a Lie group G, let $H_1, H_2 \subset G$ be Lie subgroups with H_1 connected. Show that, if $H_1 \cap H_2$ contains an intrinsic open subset of H_1, then $H_1 \subset H_2$.
9. Show that the following subgroups of the linear group are Lie groups: $\mathrm{O}(n)$; $\mathrm{SO}(n)$; $\mathrm{Sl}(n, \mathbb{R}) = \{g \in \mathrm{Gl}(n, \mathbb{R}) : \det g = 1\}$; $\mathrm{U}(n)$; $\mathrm{SU}(n)$; $\mathrm{Sp}(n, \mathbb{R}) = \{g \in \mathrm{Gl}(2n, \mathbb{R}) : g J g^{\mathrm{T}} = J\}$, where J is the matrix written in $n \times n$ blocks as

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix};$$

the subgroup of upper triangular matrices ($a_{ij} = 0$ if $i > j$). Describe their Lie algebras.

10. Let Γ be a discrete subgroup and H a connected subgroup of a Lie group G. Suppose that H normalizes Γ, that is, for every $h \in H$, $h\Gamma h^{-1} \subset \Gamma$, and show that h centralizes Γ, that is, $h\gamma = \gamma h$, $\gamma \in \Gamma$, $h \in H$. Conclude that a discrete normal subgroup of a connected Lie group is contained in its center.
11. Given a Lie group G, show that its center $Z(G)$ is a Lie subgroup whose Lie algebra is the center $\mathfrak{z}(\mathfrak{g}) = \{X \in \mathfrak{g} : \forall Y \in \mathfrak{g},\ [X, Y] = 0\}$ of the Lie algebra $\mathfrak{g}$ of G. Conclude that $Z(G)$ is a discrete subgroup if and only if $\mathfrak{z}(\mathfrak{g}) = \{0\}$.
12. Give an example of a connected Lie group G such that $\mathfrak{z}(\mathfrak{g}) = \{0\}$ but $Z(G)$ is infinite.
13. A finite dimensional Lie algebra $\mathfrak{g}$ is **simple** if $\dim \mathfrak{g} > 1$ and the only ideals of $\mathfrak{g}$ are the trivial ideals $\{0\}$ and $\mathfrak{g}$. Let G be a Lie group whose Lie algebra $\mathfrak{g}$ is simple. Show that its center $Z(G)$ is discrete.
14. Given a Lie subgroup H of G and $g \in G$, show that gHg^{-1} is a Lie subgroup whose Lie algebra is $\mathrm{Ad}(g)\,\mathfrak{h}$, where $\mathfrak{h}$ is the Lie algebra of H.
15. Let G be a Lie group with Lie algebra $\mathfrak{g}$ and $\mathfrak{a}, \mathfrak{b} \subset \mathfrak{g}$ Lie subalgebras such that, for every $X \in \mathfrak{a}$ and every $Y \in \mathfrak{b}$, $[X, Y] = 0$. Define the connected subgroups $A = \langle \exp \mathfrak{a} \rangle$ and $B = \langle \exp \mathfrak{b} \rangle$. Show that every $a \in A$ commutes with every $b \in B$.
16. Let $H \subset G$ be a connected Lie subgroup. Show that H is normal in its closure $\overline{H}$. Give an example of a nonconnected Lie subgroup H which is not normal in its closure.
17. Let H be a connected and not closed subgroup of G. Show that the Lie algebra of H is properly contained in the Lie algebra of its closure $\overline{H}$.
18. Suppose that $H \subset G$ is a connected and not closed Lie subgroup. Show that there exists a sequence $h_n \in H$ such that $h_n \to \infty$ (that is, for every compact $K \subset H$, there exists $n \in \mathbb{N}$ such that $h_n \notin K$) with respect to the intrinsic topology and, nevertheless, $h_n \to 1$ in the topology of G.
19. Given a Lie group G with Lie algebra $\mathfrak{g}$, take neighborhoods $V \subset \mathfrak{g}$ and $U \subset G$ of the origins, such that $\exp : V \to U$ is a diffeomorphism. Denote by $\log : U \to V$ the inverse of exp. Let $H \subset G$ be a connected Lie subgroup with Lie algebra $\mathfrak{h}$. Show that H is closed if $\log(U \cap H) \subset \mathfrak{h}$.
20. Suppose that $H \subset G$ is a connected Lie subgroup with Lie algebra $\mathfrak{h} \subset \mathfrak{g}$. Denote by $N(H) = \{g \in G : gHg^{-1} \subset H\}$ the normalizer of H and $\mathfrak{n}(\mathfrak{h}) = \{X \in \mathfrak{g} : \mathrm{ad}(X)\,\mathfrak{h} \subset \mathfrak{h}\}$ the normalizer of $\mathfrak{h}$ in $\mathfrak{g}$. Show that the normalizer $N(H)$ of H is a closed subgroup. Suppose also that H is connected and show that its Lie algebra is $\mathfrak{n}(\mathfrak{h})$. Give an example of a nonconnected Lie subgroup H such that the Lie algebra of $N(H)$ is not $\mathfrak{n}(\mathfrak{h})$.
21. Let G be a connected Lie group and $\rho : G \to \mathrm{Gl}(V)$ a differentiable finite dimensional representation of G. Show that $\rho(G)$ is a Lie subgroup of $\mathrm{Gl}(V)$. What is the Lie algebra of $\rho(G)$? Generalize for a differentiable homomorphism $\rho : G \to H$.

22. Let $\mathfrak{g}$ be a Lie algebra (over $\mathbb{R}$, with $\dim \mathfrak{g} < \infty$). Denote by $\mathrm{Aut}(\mathfrak{g})$ the group of automorphisms of $\mathfrak{g}$. Show that the Lie algebra of $\mathrm{Aut}(\mathfrak{g})$ is the algebra of derivations of $\mathfrak{g}$ (see the example at the end of Section 6.5).
23. A group of matrices G is called algebraic if there exist polynomials P_i, $i = 1, \ldots, r$ in the space of matrices such that $G = \{g : P_i(g) = 0\}$. Show that an algebraic group of matrices over $\mathbb{R}$ or $\mathbb{C}$ is a Lie group and describe its Lie algebra in terms of the polynomials. Give examples of algebraic groups.
24. Let G be a connected Lie group with Lie algebra $\mathfrak{g}$. Take elements $X, Y \in \mathfrak{g}$ which generate $\mathfrak{g}$ (that is, X and Y are not contained in any proper subalgebra of $\mathfrak{g}$ or, equivalently, the successive brackets between of X and Y generate $\mathfrak{g}$). Show that the 1-parameter groups $\exp tX$ and $\exp tY$ generate G.
25. Let G be a Lie group and $S \subset G$ a **subsemigroup** of G, that is, if $g, h \in S$, then $gh \in S$ (and g^{-1} may not belong to S). Let $\mathfrak{g}$ be the Lie algebra of G and consider the set

$$\mathcal{L}(S) = \{X \in \mathfrak{g} : \forall t \geq 0,\ \exp tX \in \overline{S}\}.$$

Suppose that $1 \in \overline{S}$ and use the formulas of Section 6.4 to show that $\mathcal{L}(S)$ satisfies the following properties: (1) $\mathcal{L}(S)$ is a convex cone, that is, given $X, Y \in \mathcal{L}(S)$ and a, b real > 0, then $aX + bY \in \mathcal{L}(S)$; (2) $\mathcal{L}(S)$ is a **Lie cone**, that is, if $\pm X \in \mathcal{L}(S)$, then $\exp t\,\mathrm{ad}(X)(\mathcal{L}(S)) = \mathcal{L}(S)$ for every $t \in \mathbb{R}$.
26. Show the uniqueness of the quotient structure, that is, two differentiable manifold structures on G/H satisfying the properties of Theorem 6.22 are diffeomorphic.
27. Let G be a connected Lie group with Lie algebra $\mathfrak{g}$ and $H, N \subset G$ Lie subgroups, with Lie algebras $\mathfrak{h}$ and $\mathfrak{n}$, respectively. Suppose that $\mathfrak{g} = \mathfrak{h} + \mathfrak{n}$ and that N is a normal subgroup. Show that $G = HN$.
28. Let G be a Lie group and $H \subset G$ a closed subgroup. Suppose that H contains a closed subgroup L which is normal in G. Show that there exists a diffeomorphism $\phi : G/H \to (G/L)/(H/L)$ such that, for every $g \in G$ and $x \in G/H$, $\phi(gx) = \pi(g)\phi(x)$, where $\pi : G \to G/L$ is the canonical projection.
29. Given a Lie group G and a closed subgroup $H \subset G$, let $\phi : G \to G$ be a differentiable homomorphism such that $\phi(H) \subset H$. Define $\overline{\phi}(gH) = \phi(g)H$. Show that $\overline{\phi} : G/H \to G/H$ is a well defined map and is differentiable. Show also that, if ϕ is an automorphism and $\phi(H) = H$, then $\overline{\phi}$ is a diffeomorphism.
30. Let G be a Lie group, H a closed subgroup and $M \subset G$ a submanifold such that the map $\phi : M \times H \to G$ given by $\phi(x, y) = xy$ is a diffeomorphism. Show that G/H is diffeomorphic to M.
31. This exercise presents a case where the decomposition of Lemma 6.23 is global. Let $G = \mathrm{Gl}(n, \mathbb{R})$ and $K = \mathrm{O}(n)$. Denote by $\mathfrak{e}$ the space of symmetric $n \times n$ matrices. Show that the map $\psi : \mathfrak{e} \times K \to G$ given by $\psi(X, k) = e^X k$ is a diffeomorphism. Do the same with $G = \mathrm{Sl}(n, \mathbb{R})$ and $K = \mathrm{SO}(n)$.
32. Let G be a Lie group with Lie algebra $\mathfrak{g}$. Given a subgroup $H \subset G$, consider the set $\mathfrak{h}_H \subset \mathfrak{g}$ formed by the derivatives $\alpha'(0)$ of the differentiable curves

$\alpha(t) \in H$ with $\alpha(0) = 1$. Show that $\mathfrak{h}_H$ is a Lie subalgebra. Suppose that G is connected, H is normal and $\mathfrak{h}_H = \{0\}$ and show that $H \subset Z(G)$.

33. The theorem of Chow, particularized for invariant vector fields on a Lie group G with Lie algebra $\mathfrak{g}$, has the following statement: Let $D \subset \mathfrak{g}$ be a subset that generates $\mathfrak{g}$ as a Lie algebra. Then, there exist $X_1, \ldots, X_n \in D$ (possibly with repetitions) such that if $\rho : \mathbb{R}^n \to G$ is defined by

$$\rho(t_1, \ldots, t_n) = e^{t_1 X_1} \cdots e^{t_n X_n} i,$$

then the image of ρ has a nonempty interior in G. The following items point out the proof of this theorem.

(a) Let $M \subset G$ be an embedded submanifold and suppose that for every $X \in D$ the right invariant vector field X^r is tangent to M (that is, $X^r(x) \in T_x M$ for every $x \in M$). Show that $\dim M = \dim G$, that is, M is an open submanifold. (Use the fact that if Z, W are tangent vector fields to an embedded submanifold, then the Lie bracket $[Z, W]$ is also tangent.)

(b) Verify that if $\dim G > 0$, then there exists $X_1 \in D$, $X_1 \neq 0$, and show that if $\dim G > 1$, then there exist $X_2 \in D$ and $t_0 \in \mathbb{R}$ such that $\{X_1^r(t_0), X_2^r(t_0)\}$ is linearly independent. Show that the map $(t, s) \mapsto e^{tX_1} e^{sX_2}$ has rank 2 in a neighborhood of $(t_0, 0)$.

(c) Repeat the arguments of the previous item to show by induction that there exist $X_1, \ldots, X_n \in D$ and $\tau_0 = \left(t_1^0, \ldots, t_n^0\right) \in \mathbb{R}^n$ such that ρ, as defined above, has maximum rank at τ_0. Conclude the proof of the theorem of Chow.

(d) With the same notations, show that there exist $Y_1, \ldots, Y_m \in D$ and $\sigma_0 = \left(s_1^0, \ldots, s_m^0\right) \in \mathbb{R}^m$ such that $e^{s_1^0 Y_1} \cdots e^{s_m^0 Y_m} = 1$ and the map

$$(s_1, \ldots, s_m) \in \mathbb{R}^m \mapsto e^{s_1 Y_1} \cdots e^{s_m Y_m} \in G$$

has rank equal to $\dim G$ at σ_0. (Hint: Take $(Y_1, \ldots, Y_m) = (-X_n, \ldots, X_1, X_1, \ldots, X_n)$ where $(X_1, \ldots, X_n)$ is as in the previous item.)

34. Let G be a connected Lie group with Lie algebra $\mathfrak{g}$ and $D \subset \mathfrak{g}$ a subset generating $\mathfrak{g}$. Use the previous exercise to show that G is a product of exponentials on $\mathbb{R}D$, that is, for every $g \in G$ there exist $X_1, \ldots, X_n \in D$ and $t_1, \ldots, t_n \in \mathbb{R}$ such that $g = e^{t_1 X_1} \cdots e^{t_n X_n}$.

35. Let G be a connected Lie group with Lie algebra $\mathfrak{g}$ and H a Lie subgroup with subalgebra $\mathfrak{h} \subset \mathfrak{g}$. Suppose that, for an element $g \in G$, it holds

$$\mathfrak{g} = \sum_{k \geq 0} \mathrm{Ad}(g)^k \mathfrak{h}.$$

Show that there exists an integer $k \geq 1$ such that the product $(gH)^k = gH \cdots gH$ (k times) has a nonempty interior in G. Use this to conclude that every element of G is a product of elements of gH or of Hg^{-1}.

Chapter 7
Homomorphisms and Coverings



The results about Lie subgroups proved so far allow to obtain information about homomorphisms between Lie groups. The idea is that the graph of a homomorphism $\phi : G \to H$ is a subgroup of the product group $G \times H$ isomorphic to G by the projection $\pi_1 : G \times H \to G$, $\pi_1(x, y) = x$ and, conversely, if the graph of a map ϕ is a subgroup, then ϕ is a homomorphism. In case the homomorphism ϕ is continuous or differentiable, its graph has topological or differentiable properties. One of the applications obtained is the proof that any homomorphism between the Lie algebras "extends" to the groups when the domain is simply connected. Bringing together these extensions and the construction of a Lie group structure on the universal covering of a given group, one obtains a description of connected Lie groups from simply connected Lie groups. The isomorphism classes of connected and simply connected groups are in bijection with the isomorphism classes of finite dimensional Lie algebras.

7.1 Homomorphisms

7.1.1 Immersions and Submersions

Let $\phi : G \to H$ be a differentiable homomorphism. Then ϕ has constant rank because, for every $g \in G$, $\phi \circ L_g = L_{\phi(g)} \circ \phi$ and hence

$$d\phi_g = d\left(L_{\phi(g)}\right)_1 \circ d\phi_1 \circ d\left(L_{g^{-1}}\right)_g .$$

So that the rank is constant and equal to the rank of the infinitesimal homomorphism $d\phi_1$ since translations are diffeomorphisms. Therefore, ϕ is an immersion if and only if $d\phi_1$ is injective. In the same way, ϕ is a submersion if and only if $d\phi_1$ is surjective.

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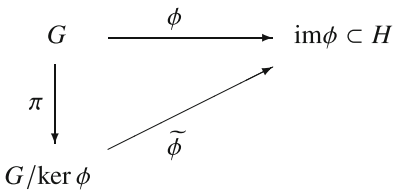
L. A. B. San Martin, *Lie Groups*, Latin American Mathematics Series,
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For a differentiable homomorphism, the properties of immersion and injectivity, as well as the properties of submersion and surjectivity, are closely related.

Consider first injectivity. The kernel $\ker\phi$ of ϕ is a closed Lie subgroup. Its Lie algebra is the ideal $\ker d\phi_1$, since $\phi\left(e^{tX}\right) = e^{td\phi_1(X)} = 1$, $t \in \mathbb{R}$, if and only if $d\phi_1(X) = 0$. Therefore, if ϕ is an immersion, that is, if $d\phi_1$ is injective, then, by Proposition 6.18, $\ker\phi$ is a discrete subgroup. Conversely, if $\ker\phi$ is discrete, then $\ker d\phi_1 = \{0\}$ and ϕ is an immersion. In particular, if ϕ is injective, that is, if $\ker\phi = \{1\}$, then ϕ is an immersion. Hence, in the case in which ϕ is injective, its image is a Lie subgroup of H by one of the definitions of Lie subgroup presented at the beginning of Chapter 6.

In general, for homomorphisms not necessarily injective there is the following **isomorphism theorem**.

Proposition 7.1 *Let G and H be Lie groups and $\phi : G \to H$ a differentiable homomorphism. Then $G/\ker\phi$ is a Lie group. Let $\widetilde{\phi}(g\ker\phi) = \phi(g)$ be the homomorphism that makes the diagram*



commutative. Then $\widetilde{\phi}$ is an injective immersion into H and this implies that $\mathrm{im}\phi$ *is a Lie subgroup of H isomorphic to $G/\ker\phi$ and, hence,* $\dim(\mathrm{im}\phi) = \dim G - \dim(\ker\phi)$.

Moreover, if G is connected, then $\mathrm{im}\phi = \langle\exp(\mathrm{im}d\phi_1)\rangle$.

Proof The Lie group construction on $G/\ker\phi$ was carried out in Chapter 6 (see Proposition 6.25). The differentiability of $\widetilde{\phi}$ (with respect to the quotient structure) is a consequence of item (3) of Theorem 6.22, since $\widetilde{\phi}\circ\pi = \phi$ is differentiable. Finally, by definition, $\widetilde{\phi}$ is an injective map and this implies that its image is a Lie subgroup isomorphic to $G/\ker\phi$.

The last statement is a consequence of the fact that if G is connected, then $\mathrm{im}\phi$ is a connected Lie subgroup whose tangent space at the identity is $\mathrm{im}d\phi_1$. Hence, $\mathrm{im}\phi$ is the only connected subgroup with this Lie algebra, that is, $\mathrm{im}\phi = \langle\exp(\mathrm{im}d\phi_1)\rangle$. □

Now, consider surjectivity. If $d\phi_1$ is surjective, then ϕ is a submersion and hence an open map. It follows that $\mathrm{im}\phi$ is a subgroup of H with nonempty interior and as such contains the connected component of the identity H_0. In particular, if H is connected, then ϕ is surjective.

The converse holds without further restrictions in case the domain G is connected. In this case, $\mathrm{im}\phi$ is a connected subgroup with Lie algebra $\mathrm{im}d\phi_1$. Therefore,

if $\mathrm{im}\phi = H$, then $\mathrm{im} d\phi_1 = \mathfrak{h}$ by the uniqueness of a connected Lie group with a given Lie algebra.

On the other hand, in the nonconnected case, pathologies may arise due to the arbitrariness of the quantity of connected components. For instance, a Lie group G with $\dim G \geq 1$ can also be seen as a Lie group G_{d} of dimension 0, with the discrete topology. The identity $\mathrm{id} : G_{\mathrm{d}} \to G$ is a surjective homomorphism but its differential $d\phi_1 = 0$ is not surjective. This does not occur with groups with an enumerable quantity of connected components, as the following application of the theorem of Baire shows.

Proposition 7.2 *Let $\phi : G \to H$ be a differentiable homomorphism and suppose that the identity component H_0 of H is in the image of ϕ. Suppose also that G has at most an enumerable quantity of connected components. Then $d\phi_1$ is surjective, that is, ϕ is a submersion.*

Proof Let G_0 be the identity component of G. Then

$$H_0 \subset \bigcup_{g \in G} \phi\left(g G_0\right) = \bigcup_{g \in G} \phi(g)\, \phi\left(G_0\right).$$

By the theorem of Baire, at least one of the sets $\phi(g)\, \phi\left(G_0\right) \cap H_0$ has a nonempty interior. As these components are diffeomorphic, all the components that intersect H_0 have a nonempty interior. In particular, $\phi\left(G_0\right)$ is a connected subgroup with nonempty interior in H_0 and, hence, $\phi\left(G_0\right) = H_0$. This implies that $d\phi_1$ is surjective, as two connected Lie groups coincide if and only if their Lie algebras are equal. □

This information about differentiable homomorphisms will be later applied to the case in which G and H are connected.

If $d\phi_1$ is an isomorphism, then ϕ is a local diffeomorphism. Also if G and H are connected, then ϕ is surjective and H is isomorphic to $G/\ker\phi$, where $\ker\phi$ is a discrete subgroup. In this case, besides being a local diffeomorphism, ϕ is a covering map (see definition in Section 7.4 below). This follows from the statement of Exercise 8 of Chapter 2. As this fact will be crucial for the description of connected Lie groups, it is proved here.

Proposition 7.3 *Let G be a locally connected topological group and $\Gamma \subset G$ a discrete subgroup. Then the canonical projection $\pi : G \to G/\Gamma$ is a covering map.*

Proof Take an open set U with $U \cap \Gamma = \{1\}$ and let $V \ni 1$ be a connected open set with $V^{-1} = V$ and $V^2 \subset U$. For $g \in G$, the projection $\pi\left(gV\right)$ is a connected neighborhood of $x = gH$ and it holds

$$\pi^{-1}\left(\pi\left(gV\right)\right) = \bigcup_{\gamma \in \Gamma} gV\gamma.$$

The open sets $gV\gamma$ in this union are connected and satisfy $gV\gamma_1 \cap gV\gamma_2 \neq \emptyset$ if and only if $\gamma_1 = \gamma_2$. Indeed, if $y \in gV\gamma_1 \cap gV\gamma_2$, then $g^{-1}y = v_1\gamma_1 = v_2\gamma_2$, with $v_1, v_2 \in V$. Hence, $v_2^{-1}v_1 = \gamma_2\gamma_1^{-1} \in U \cap \Gamma$ and this implies that $v_1 = v_2$ and $\gamma_1 = \gamma_2$. Therefore, $\pi^{-1}(\pi(gV))$ is a union of connected components homeomorphic to $\pi(gV)$, showing that π is a covering map. □

From this fact about topological groups there follows the following statement about homomorphisms whose differentials are isomorphisms.

Proposition 7.4 *Let G and H be Lie groups with Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$, respectively. Let $\phi : G \to H$ be a surjective homomorphism and suppose that $d\phi_1 : \mathfrak{g} \to \mathfrak{h}$ is an isomorphism. Then ϕ is a covering map.*

Proof The isomorphism theorem, combined with the hypothesis that ϕ is surjective, ensures that H is isomorphic to $G/\ker\phi$ by the isomorphism $\widetilde{\phi}$ defined in Proposition 7.1. Moreover, $\ker\phi$ is a discrete subgroup because $(d\phi)_1$ is an isomorphism. The previous proposition then ensures that ϕ is a covering map. □

Corollary 7.5 *Suppose that the differentiable homomorphism $\phi : G \to H$ is an immersion. Then, $\phi : G \to \mathrm{im}\phi$ is a covering map.*

7.1.2 Graphs and Differentiability

Given the groups G and H, a map $\phi : G \to H$ is a homomorphism of groups if and only if its graph $\mathrm{gr}\phi = \{(x, \phi(x)) : x \in G\}$ is a subgroup of $G \times H$. When this happens, the groups G and $\mathrm{gr}\phi$ are isomorphic by the projection $p : \mathrm{gr}\phi \to G$, $p(x, \phi(x)) = x$, whose inverse is

$$l : x \in G \mapsto (x, \phi(x)) \in \mathrm{gr}\phi.$$

The homomorphism ϕ is recovered by l with the formula $\phi = \pi_2 \circ l$, where π_2 is the projection on H.

In the topological context, a homomorphism $\phi : G \to H$ is continuous if and only if its graph is a closed subgroup when H is a Hausdorff space (see Proposition 2.9).

A similar criterion holds for differentiable homomorphisms.

Proposition 7.6 *Let G and H be Lie groups. A map $\phi : G \to H$ is a differentiable homomorphism if and only if its graph*

$$\mathrm{gr}\phi = \{(x, \phi(x)) \in G \times H : x \in G\}$$

is a closed Lie subgroup of $G \times H$ diffeomorphic to G by the projection $p(x, \phi(x)) = x$.

Proof If the graph is a closed Lie subgroup diffeomorphic to G by the projection, then the equality $\phi = \pi_2 \circ l$ shows that ϕ is differentiable.

The converse is a consequence of a general result about maps between manifolds: if ϕ is differentiable, then its graph is an embedded closed submanifold diffeomorphic to the domain. For this reason, if ϕ is differentiable, then $\mathrm{gr}\phi$ is a Lie subgroup. □

The interpretation of differentiability in terms of the graph, combined with the closed subgroup theorem, allows to show that the continuous homomorphisms between Lie groups are in fact differentiable homomorphisms.

Theorem 7.7 *Let G and H be Lie groups and $\phi : G \to H$ a continuous homomorphism. Then ϕ is differentiable.*

Proof It can be assumed without loss of generality that G is connected, since the homomorphism ϕ is differentiable if and only if it is differentiable at the identity. Moreover, if $\mathrm{gr}\phi \subset G \times H$ is closed, then $\mathrm{gr}\left(\phi_{|G_0}\right) \subset G_0 \times H$ is also closed because G_0 is closed in G.

Now, the continuity of ϕ implies that $\mathrm{gr}\phi$ is closed. Hence it is a Lie subgroup of $G \times H$ homeomorphic to G by the projection $p : \mathrm{gr}\phi \to G$. This projection is a differentiable homomorphism. Its injectivity implies that $\ker dp_{(1,1)}$ is injective and, thus, p is an immersion. The surjectivity of p implies that the infinitesimal homomorphism $dp_{(1,1)}$ is surjective and it then follows that p is a submersion.

Therefore, p is a diffeomorphism and this ensures that $\phi = \pi_2 \circ p^{-1}$ is differentiable. □

Example The above theorem is a wide generalization of the fact that the continuous homomorphisms of the additive group $\mathbb{R}$ are differentiable. For this case, it is possible to give the following elementary proof: If $\phi : \mathbb{R} \to \mathbb{R}$ is a homomorphism, then for every integer $n \in \mathbb{Z}$, $\phi(n) = n\phi(1)$ and $\phi(1) = n\phi(1/n)$, that is, $\phi(1/n) = 1/n\phi(1)$. This implies that ϕ is linear when $\mathbb{R}$ is seen as a vector space over $\mathbb{Q}$, that is, $\phi(p/q) = p/q\phi(1)$. If, moreover, ϕ is continuous, then it is linear also over $\mathbb{R}$. In particular, ϕ is differentiable. □

7.2 Extensions of Homomorphisms

Theorem 7.7 exploited the subgroup property of the graph of a homomorphism, together with the closed subgroup theorem. The next step is to exploit the same property, now taking into account the construction of connected subgroups on the cartesian product $G \times H$ from the Lie subalgebras of the Lie algebra $\mathfrak{g} \times \mathfrak{h}$ of $G \times H$. What is obtained here are "extensions" of homomorphisms $\theta : \mathfrak{g} \to \mathfrak{h}$ of Lie algebras to homomorphisms of Lie groups $\phi : G \to H$, in the sense that $d\phi_1 = \theta$.

In general, these extensions are not possible, though they hold locally. The global case works only with the hypothesis that the domain is simply connected.

Let $\mathfrak{g}$ and $\mathfrak{h}$ be Lie algebras. In the same way as with maps between groups, a map $\theta : \mathfrak{g} \to \mathfrak{h}$ is a homomorphism if and only if its graph is a subalgebra of the product algebra $\mathfrak{g} \times \mathfrak{h}$. Conversely, a subspace $\mathfrak{v}$ of $\mathfrak{g} \times \mathfrak{h}$ is the graph of a homomorphism $\theta : \mathfrak{g} \to \mathfrak{h}$ if and only if $\mathfrak{v}$ is a subalgebra isomorphic to $\mathfrak{g}$ by the projection $\mathfrak{g} \times \mathfrak{h}$, restricted to $\mathfrak{v}$. The graph of homomorphism θ will be denoted by $\mathfrak{g}(\theta)$.

Suppose now that $\mathfrak{g}$ and $\mathfrak{h}$ are the Lie algebras of the Lie groups G and H, respectively. Then, $\mathfrak{g} \times \mathfrak{h}$ is a Lie algebra of $G \times H$. Let $G(\theta) = \langle \exp \mathfrak{g}(\theta) \rangle$ be the only connected Lie subgroup of $G \times H$ whose Lie algebra is $\mathfrak{g}(\theta)$. It would be desirable that $G(\theta)$ is the graph of a homomorphism $G \to H$ that extends θ, that is, whose differential is θ. However, $G(\theta)$ is not always the graph of a map between G and H, as shows the following example.

Example Let $G = S^1$ and $H = \mathbb{R}$. Their Lie algebras coincide with $\mathbb{R}$, the unidimensional abelian algebra. The homomorphisms $\theta : \mathbb{R} \to \mathbb{R}$ have the form $\theta_a(X) = aX$, $a \in \mathbb{R}$. The graph of θ_a is the straight line r_a with equation $u = av$, $(u, v) \in \mathbb{R}^2$ and the subgroup $G(\theta_a) \subset S^1 \times \mathbb{R}$ is the image of r_a by the canonical projection $\mathbb{R} \times \mathbb{R} \to S^1 \times \mathbb{R}$. Geometrically, the groups $G(\theta_a)$ are spirals if $a \neq 0$, and the circle $S^1 \times \{0\}$ if $a = 0$. Only in the case $a = 0$, $G(\theta_a)$ is the graph of a map $S^1 \to \mathbb{R}$ (the trivial homomorphism, identically zero). If $a \neq 0$, then the projection of $G(\theta_a)$ on S^1 is not injective and, therefore, is not the graph of a function. □

Before looking at the existence of extensions of homomorphisms of Lie algebras, the following proposition ensures the uniqueness of such extensions in case the domain is connected.

Proposition 7.8 *Let G and H be Lie groups with Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$, respectively. Let also $\phi, \psi : G \to H$ be differentiable homomorphisms such that $d\phi_1 = d\psi_1$. Suppose that G is connected. Then, $\phi = \psi$.*

Proof If $X \in \mathfrak{g}$, then

$$\phi\left(e^X\right) = e^{d\phi_1(X)} = e^{d\psi_1(X)} = \psi\left(e^X\right),$$

so that ϕ and ψ coincide in a product $e^{X_1} \cdots e^{X_k}$, $X_i \in \mathfrak{g}$. Therefore, $\phi = \psi$ since G is connected. □

In case G is not connected, the extensions are not unique. For instance, if G is a discrete group, then $(d\phi)_1 = 0$ for any homomorphism. However, if ϕ is a homomorphism, then $C_h \circ \phi$ is also a homomorphism for every $h \in H$. In general (if H is not abelian), $C_h \circ \phi \neq \phi$.

Even though $G(\theta)$ is not in general the graph of a map, it is locally the graph of a local homomorphism, in the following sense.

Definition 7.9 A **local homomorphism** between the Lie groups G and H is a map $\phi : U \to H$ with $1 \in U \subset G$ a neighborhood of the identity, such that $\phi(g_1 g_2) = \phi(g_1)\phi(g_2)$ whenever g_1, g_2 and $g_1 g_2$ are in U. This implies that $\phi\left(g^{-1}\right) = \phi(g)^{-1}$ if $g, g^{-1} \in U$. If, moreover, $\phi : U \to V \subset H$ is a

diffeomorphism, then ϕ is a **local isomorphism**. In this case, ϕ^{-1} is also a local homomorphism.

To verify that θ extends to a local homomorphism, denote by $p : G(\theta) \to G$ the restriction to $G(\theta) \subset G \times H$ of the projection $G \times H \to G$. It is clear that p is a differentiable homomorphism $G(\theta) \to G$. The differential dp_1 of p at the identity is the restriction to $\mathfrak{g}(\theta)$ of the projection $\mathfrak{g} \times \mathfrak{h} \to \mathfrak{g}$, and it follows that $dp_1 : \mathfrak{g}(\theta) \to \mathfrak{g}$ is an isomorphism of Lie algebras, since $\mathfrak{g}(\theta)$ is the graph of a homomorphism.

Proposition 7.10 *Let $\theta : \mathfrak{g} \to \mathfrak{h}$ be a homomorphism. Then, there exists a differentiable local homomorphism $\phi : U \subset G \to H$ such that $d\phi_1 = \theta$.*

Proof Since dp_1 is an isomorphism, there exist neighborhoods of the identity $V_1 \subset G(\theta)$ and $U_1 \subset G$ such that the restriction $p_V : V_1 \to U_1$ of p to V_1 is a diffeomorphism. Let $\overline{\phi}$ be the inverse of p_V, that is, $p\left(\overline{\phi}(x)\right) = x$ for every $x \in U_1$. By definition, $p(x, y) = x$ if $y \in H$, hence $\overline{\phi}$ has the form $\overline{\phi}(x) = (x, \phi(x))$ with $\phi(x) \in H$, for every $x \in U_1$. This defines the map $\phi : U_1 \to H$ and, as $\overline{\phi}(x)$ runs through V_1 when x varies in U_1, the graph of ϕ is V_1.

To obtain the local homomorphism, let $V \subset V_1$ be a neighborhood of 1 such that $V^2 \subset V_1$ and define $U = p(V)$. Restricting the maps to V and U, ϕ is a local homomorphism. Indeed, if g_1, g_2 and $g_1 g_2$ are in U, then $\overline{\phi}(g_1) = (g_1, \phi(g_1))$, $\overline{\phi}(g_2) = (g_2, \phi(g_2))$ and $\overline{\phi}(g_1 g_2) = (g_1 g_2, \phi(g_1 g_2))$ belong to V. But $G(\theta)$ is a subgroup of $G \times H$ and this ensures the

$$(g_1, \phi(g_1))(g_2, \phi(g_2)) = (g_1 g_2, \phi(g_1)\phi(g_2)) \in G(\theta).$$

Then, $(g_1 g_2, \phi(g_1)\phi(g_2)) \in V_1$, since $V^2 \subset V_1$. As p is injective and $p(g_1 g_2, \phi(g_1)\phi(g_2))$ coincides with $p(g_1 g_2, \phi(g_1 g_2))$, it follows that $\phi(g_1 g_2) = \phi(g_1)\phi(g_2)$.

Finally, ϕ is differentiable since its graph is a submanifold V of $G \times H$. □

Corollary 7.11 *Two Lie groups are locally isomorphic if and only if their Lie algebras are isomorphic.*

Proof By the above proposition, an isomorphism between the Lie algebras extends to a local isomorphism between the groups. Conversely, the differential $d\phi_1$ is an isomorphism of Lie algebras if ϕ is a local isomorphism. □

The objective now is to extend the local homomorphism to the whole group G. Taking into account global topological properties of G conditions are obtained in order that $G(\theta)$ is the graph of a map $G \to H$.

The following proposition ensures that, in the connected case, $G(\theta)$ meets one of the requirements to be the graph of a map, namely that for every $g \in G$ there exists $h \in H$ such that $(g, h) \in G(\theta)$.

Proposition 7.12 *With the above notations, the image of the projection* $p : G(\theta) \to G$ *is the connected component* G_0 *of the identity of* G*. In particular,* p *is surjective if* G *is connected.*

Proof Since dp_1 is an isomorphism, its image has a nonempty interior in G and, therefore, is an open subgroup containing the connected component of the identity. On the other hand, the image of p is connected because $G(\theta)$ is connected. □

It is now possible to prove the main theorem about extensions of homomorphisms. This theorem is known as the **monodromy principle**.

Theorem 7.13 *Let* G *and* H *be Lie groups with Lie algebras* $\mathfrak{g}$ *and* $\mathfrak{h}$*, respectively. Suppose that* G *is connected and simply connected. Then, for every homomorphism* $\theta : \mathfrak{g} \to \mathfrak{h}$*, there exists a unique homomorphism* $\phi : G \to H$ *such that* $\theta = d\phi_1$*.*

Proof As before, let $\mathfrak{g}(\theta) \subset \mathfrak{g} \times \mathfrak{h}$ be the graph of θ and $G(\theta) \subset G \times H$ the connected Lie subgroup whose Lie algebra is $\mathfrak{g}(\theta)$. By assumption, G is connected, hence the Proposition 7.12 ensures that the projection $p : G(\theta) \to G$ is a surjective differentiable homomorphism. Its differential dp_1 is an isomorphism. Therefore, by Proposition 7.4, p is a covering map. But, by hypothesis, G is simply connected. Hence, p is a homeomorphism. In particular, p is injective, and this ensures that $G(\theta)$ is the graph of a map $\phi : G \to H$. A posteriori, ϕ is a differentiable homomorphism, since $G(\theta)$ is a Lie subgroup. Finally, $d\phi_1 = \theta$, since the tangent space at the identity of $G(\theta)$ is the graph of θ. The uniqueness follows from Proposition 7.8. □

As a consequence of this theorem it follows that up to isomorphisms there exists at most one connected and simply connected Lie group with Lie algebra $\mathfrak{g}$.

Corollary 7.14 *Let* G_1 *and* G_2 *be connected and simply connected Lie groups with isomorphic Lie algebras. Then,* G_1 *and* G_2 *are isomorphic.*

Proof Let $\mathfrak{g}_1$ and $\mathfrak{g}_2$ be the Lie algebras of G_1 and G_2, respectively, and $\theta : \mathfrak{g}_1 \to \mathfrak{g}_2$ an isomorphism. Denote by $\phi : G_1 \to G_2$ and $\psi : G_2 \to G_1$ the homomorphisms with $d\phi_1 = \theta$ and $d\psi_1 = \theta^{-1}$. By the uniqueness of extensions, it follows that $\phi \circ \psi = \mathrm{id}_{G_2}$ and $\psi \circ \phi = \mathrm{id}_{G_1}$, that is, $\psi^{-1} = \phi$ is an isomorphism between G_1 and G_2. □

7.3 Universal Covering

The objective of this section is to conclude one of the programs of the theory of Lie groups, namely to describe the connected Lie groups from their Lie algebras. This description reduces to the following theorem, whose proof will be carried out along this section.

Theorem 7.15 *Let* $\mathfrak{g}$ *be a real Lie algebra with* $\dim \mathfrak{g} < \infty$*. Then,*

1. *there exists a unique (up to isomorphism) connected and simply connected Lie group* $\widetilde{G}(\mathfrak{g})$ *with Lie algebra* $\mathfrak{g}$*;*
2. *if* G *is a connected Lie group with Lie algebra* $\mathfrak{g}$*, then* $G \approx \widetilde{G}(\mathfrak{g})/\Gamma$*, where* $\Gamma \subset \widetilde{G}(\mathfrak{g})$ *is a* ***discrete central subgroup****, that is,* Γ *is contained in the center* $Z\left(\widetilde{G}(\mathfrak{g})\right)$ *of* $\widetilde{G}(\mathfrak{g})$*. In this case,* Γ *is isomorphic to the fundamental group* $\pi_1(G)$*.*

By this theorem, the connected Lie groups are classified on the basis of a classification (up to isomorphism) of the real Lie algebras and a description of the centers $Z\left(\widetilde{G}(\mathfrak{g})\right)$ of the simply connected groups $\widetilde{G}(\mathfrak{g})$.

The uniqueness of $\widetilde{G}(\mathfrak{g})$ was ensured by the monodromy principle (Theorem 7.13) and its Corollary 7.14.

The existence of the simply connected group $\widetilde{G}(\mathfrak{g})$ still has to be proved. The part of the proof involving the quotient $G = \widetilde{G}(\mathfrak{g})/\Gamma$ will be easily obtained by a new application of the monodromy principle and will be left to the end.

The existence of $\widetilde{G}(\mathfrak{g})$ is ensured in two steps:

1. If $\mathfrak{g}$ is a finite dimensional real Lie algebra, then there exists a connected Lie group G with Lie algebra (isomorphic to) $\mathfrak{g}$. This existence result is known as the **third Lie theorem**.
2. If G is a connected Lie group with Lie algebra $\mathfrak{g}$, then its simply connected covering manifold $\widetilde{G}$ admits a Lie group structure whose Lie algebra is (isomorphic to) $\mathfrak{g}$.

As mentioned in the examples of Section 6.2, a way to show the existence of some Lie group with a given Lie algebra $\mathfrak{g}$ is by applying the following result about Lie algebras.

Theorem of Ado *Every finite dimensional real Lie algebra admits a faithful (that is, injective) representation, also of finite dimension.*[1]

The image of a faithful representation of $\mathfrak{g}$ is a matrix Lie algebra isomorphic to $\mathfrak{g}$ and, hence, is the Lie algebra of some connected group.

It is now possible to fix a connected Lie group G with Lie algebra $\mathfrak{g}$ and apply the theory of covering spaces to construct a Lie group structure on the space $\widetilde{G}$, the simply connected universal covering of G. (See an overview of this theory in the appendix to this chapter.) The existence of space $\widetilde{G}$ is due to the fact that G is locally connected. Moreover, $\widetilde{G}$ is a differentiable manifold such that the covering map $\pi : \widetilde{G} \to G$ is a local diffeomorphism.

Theorem 7.16 *Let* G *be a connected Lie group. Denote by* $\pi : \widetilde{G} \to G$ *the universal covering of the underlying manifold and choose an element* $\widetilde{1} \in \widetilde{G}$ *with* $\pi\left(\widetilde{1}\right) = 1$*. Then, there exists a unique product on* $\widetilde{G}$ *that makes it a Lie group in*

[1] See *Álgebras de Lie* [47, Chapter 10] and Varadarajan [53, Section 3.17]. The proof of the theorem of Ado is quite engaging from the algebraic point of view. However, the proof is trivial for Lie algebras with center $\{0\}$, as is the case with semi-simple algebras.

such a way that $\widetilde{1}$ is the identity and π is a homomorphism. The Lie algebras of G and $\widetilde{G}$ are isomorphic.

Proof The product on $\widetilde{G}$ is obtained by lifting the product on $p : G \times G \to G$ in the following manner: Let $q : \widetilde{G} \times \widetilde{G} \to G$ be the differentiable map given by

$$q(x, y) = p(\pi(x), \pi(y)).$$

As $\widetilde{G} \times \widetilde{G}$ is simply connected, there exists a unique differentiable map $\widetilde{p} : \widetilde{G} \times \widetilde{G} \to \widetilde{G}$, lifting of q, such that $\widetilde{p}\left(\widetilde{1}, \widetilde{1}\right) = \widetilde{1}$:

$$\begin{array}{ccc} \widetilde{G} \times \widetilde{G} & \xrightarrow{\widetilde{p}} & \widetilde{G} \\ \downarrow \pi & q \searrow & \downarrow \pi \\ G \times G & \xrightarrow{p} & G \end{array}$$

(See item (4) of Section 7.4.)

The product on $\widetilde{G}$ defined by $\widetilde{p}$ satisfies the group axioms. This is proved by exploiting repeatedly the existence and uniqueness of liftings, as follows:

1. The base point $\widetilde{1}$ is the identity because the map $x \in \widetilde{G} \mapsto \widetilde{p}\left(\widetilde{1}, x\right) \in \widetilde{G}$ is a lifting of $x \in \widetilde{G} \mapsto q\left(\widetilde{1}, x\right) \in G$. This map is just the projection $\pi : \widetilde{G} \to G$. Since, by definition, $\widetilde{p}\left(\widetilde{1}, \widetilde{1}\right) = \widetilde{1}$, it follows that $\widetilde{p}\left(\widetilde{1}, x\right) = x$, as the only lifting that fixes a point is the identity. The same way $\widetilde{p}\left(x, \widetilde{1}\right) = x$.
2. The only lifting $\widetilde{\iota}$ of the map $x \in \widetilde{G} \mapsto \pi(x)^{-1} \in G$, satisfying $\widetilde{\iota}\left(\widetilde{1}\right) = \widetilde{1}$, defines the inverse in $\widetilde{G}$. Indeed, the map $x \in \widetilde{G} \mapsto \widetilde{p}(x, \widetilde{\iota}(x)) \in \widetilde{G}$ is a lifting of the constant map $x \in \widetilde{G} \mapsto 1 \in G$. As $x \in \widetilde{G} \mapsto \widetilde{1} \in \widetilde{G}$ is also a lifting and both coincide at $\widetilde{1}$, it follows that $\widetilde{p}(x, \widetilde{\iota}(x))$ is constant and equal to $\widetilde{1}$, showing that $\widetilde{\iota}(x) = x^{-1}$.
3. Associativity is a consequence of the fact that the maps $\widetilde{G} \times \widetilde{G} \times \widetilde{G} \to \widetilde{G}$ determined by products $x(yz)$ and $(xy)z$ are liftings of the map $\widetilde{G}^3 \to G$ given by $\pi(x)\pi(y)\pi(z)$. Both liftings coincide at $\left(\widetilde{1}, \widetilde{1}, \widetilde{1}\right)$ and therefore they coincide.

The constructions of q and $\widetilde{p}$ show that

$$\pi \widetilde{p}(x, y) = q(x, y) = p(\pi x, \pi y),$$

and hence π is a homomorphism. Conversely, these equalities show that any lifting of q satisfies the homomorphism property. It then follows, from the uniqueness of liftings, the uniqueness of the product on $\widetilde{G}$.

Finally, $d\pi_{\widetilde{1}}$ is an isomorphism between the Lie algebras of $\widetilde{G}$ and G. □

In the construction of the above theorem, the starting point was an arbitrary connected Lie group G. The fact that π is a surjective homomorphism then ensures that G is isomorphic to the quotient $\widetilde{G}/\ker\pi$. Having in mind this remark,

the following observations conclude the proof of the first part of item (2) of Theorem 7.15.

1. $\Gamma = \ker \pi$ is a discrete subgroup, as π is a local isomorphism.
2. If $\Gamma \subset L$ is a discrete normal subgroup of the connected group L, then $\Gamma \subset Z(L)$ (see Chapter 2, Exercise 10). Indeed, take $\gamma \in \Gamma$ and let $U \subset L$ be an open set such that $\gamma U \cap \Gamma = \{\gamma\}$. The continuity of the map $l \in L \mapsto l\gamma l^{-1} \in G$ implies that $l\gamma l^{-1} \in \gamma U$ for l in a neighborhood V of the identity. As Γ is normal, it follows that $l\gamma l^{-1} = \gamma$, that is, l commutes with γ if $l \in V$. Therefore, γ commutes with any $x \in L$, which is a product of elements of V, that is, $\gamma \in Z(L)$.

The only assertion in Theorem 7.15 that remains to be checked is the isomorphism between the discrete subgroup and the fundamental group, which follows from the following general fact.

Proposition 7.17 *Let $\widetilde{G}$ be a connected and simply connected group and $\Gamma \subset \widetilde{G}$ a normal and discrete subgroup (that is, $\Gamma \subset Z\left(\widetilde{G}\right)$). Then, Γ is isomorphic to $\pi_1(G)$ if $G = \widetilde{G}/\Gamma$.*

Proof The universal covering of G is $\pi : \widetilde{G} \to G$. Hence, the fundamental group $\pi_1(G)$ is isomorphic to the group Λ of continuous liftings of π, that is, the group of continuous maps $\theta : \widetilde{G} \to \widetilde{G}$ such that $\pi \circ \theta = \pi$ (see item (8) of Section 7.4). If $g \in \Gamma$, then the right translation $R_g \in \Lambda$. So it makes sense to define the homomorphism

$$g \in \Gamma \mapsto R_g \in \Lambda,$$

which is injective, since $g = 1$ if $R_g = \mathrm{id}$. To verify that it is surjective, take $\theta \in \Lambda$. Then $\pi\theta(1) = \pi(1)$, which means that $\theta(1) \in \Gamma$. Now, $R_{\theta(1)}$ and θ are two liftings of π with the same initial condition $R_{\theta(1)}(1) = \theta(1)$. As the lifting is unique, it follows that $R_{\theta(1)} = \theta$, showing the isomorphism between Γ and $\Lambda \approx \pi_1(G)$. □

Corollary 7.18 *The fundamental group of a connected Lie group G is abelian.*

Proof If $G = \widetilde{G}/\Gamma$, then $\Gamma \approx \pi_1(G)$ is central and, therefore, abelian. □

Corollary 7.19 *Two connected Lie groups are **locally isomorphic** if and only if their universal coverings are isomorphic.*

The following are some concrete examples of simply connected Lie groups and their centers. Later, a general analysis will be carried out for the different classes of Lie algebras.

Examples

1. The additive group $(\mathbb{R}, +)$ is the only simply connected Lie group of dimension 1, since all 1-dimensional Lie algebras are isomorphic. Let $\Gamma \subset \mathbb{R}$ be a discrete subgroup with $\Gamma \neq \{0\}$. Then,

$$\omega = \inf\{x \in \Gamma : x > 0\}$$

exists and is necessarily > 0. As Γ is closed, $\omega \in \Gamma$, and it follows that $\mathbb{Z}\omega \subset \Gamma$. The converse inclusion is true because, if $x \in \Gamma$, then it is possible to write $x = n\omega + q$ with $n \in \mathbb{Z}$ and $0 \leq q < \omega$. In this case, $q = x - n\omega \in \Gamma$, which requires that $q = 0$ as the converse would contradict the definition of ω.

In short, $\Gamma = \mathbb{Z}\omega$ and $\mathbb{R}/\Gamma \approx S^1$, showing that $\mathbb{R}$ and S^1 are the only connected Lie groups of dimension 1. (Observe that a closed subgroup of $\mathbb{R}$ has dimension 0 (and, thus, is discrete) or has dimension 1 (and hence is the whole $\mathbb{R}$). This shows that the arguments above determine the closed subgroups of $\mathbb{R}$.)

2. A connected Lie group is **abelian** if and only if its Lie algebra is abelian. Thus, to determine those groups, it suffices to present an abelian simply connected group $\widetilde{G}$ and determine its discrete subgroups (as they are all contained in the center of $\widetilde{G}$). For $\widetilde{G}$ take the additive group $\mathbb{R}^n$. The discrete subgroups of $\mathbb{R}^n$ are isomorphic to $\mathbb{Z}^k$, $k = 1, \ldots, n$. Indeed, the following result holds: Let V be a real vector space of dimension n and $H \subset V$ a discrete subgroup of the additive group of V such that $H \neq \{1\}$. Then, there exists a linearly independent set $\{v_1, \ldots, v_k\}$, $1 \leq k \leq n$, such that

$$H = \{n_1 v_1 + \cdots + n_k v_k : n_i \in \mathbb{Z}\}.$$

The proof of this statement is done by induction on n. First, for $n = 1$, the discrete subgroups of the real line $\mathbb{R}$ have the form $\mathbb{Z}\omega$ with $\omega \in \mathbb{R}$, which is the desired form (see the previous example).

For $n \geq 2$, suppose that V is endowed with an inner product $\langle \cdot, \cdot \rangle$. The fact that H is discrete ensures that the infimum

$$\inf\{|v| \in \mathbb{R} : v \in H,\ v \neq 0\}$$

is attained, that is, there exists $v_1 \in H$, such that $|v_1|$ is minimum among the lengths of the nonzero elements of H. Let $\langle v_1 \rangle$ be the subspace spanned by v_1. The subgroup $\langle v_1 \rangle \cap H = \mathbb{Z}v_1$, since v_1 has minimum length in H. Moreover, the choice of v_1 ensures that the ball $B(0, |v_1|/3)$ with center at 0 and radius $|v_1|/3$ intersects H only at the origin.

Denote by $p : V \to V/\langle v_1 \rangle$ the canonical projection onto the quotient space $V/\langle v_1 \rangle$ of dimension $n - 1$ and consider the subgroup $p(H)$. This subgroup is discrete in $V/\langle v_1 \rangle$. This is checked by verifying that the ball $U = B\left(0, \frac{1}{3}|v_1|\right)$ satisfies $p(U) \cap p(H) = \{0\}$. Consider the set

$$p^{-1}(p(U)) = U + \langle v_1 \rangle.$$

An element x of this set has the form $x = av_1 + u$, $a \in \mathbb{R}$, $u \in U$. Suppose that $av_1 + u \in H$. If n is the integer part of a, then $(a - n)v_1 + u$ and $(a - (n + 1))v_1 + u$ are in H. It then follows that there exists b with $|b| \leq \frac{1}{2}$,

such that $bv_1 + u \in H$. But $bv_1 + u \subset B(0, |v_1|)$, since $|bv_1| \leq \frac{1}{2}|v_1|$ and $|u| \leq \frac{1}{3}|v_1|$. It follows, from the choice of v_1, that $bv_1 + u = 0$, which implies that $u \in \langle v_1 \rangle$. This shows that

$$p^{-1}(p(U)) \cap H = \langle v_1 \rangle \cap H.$$

As this equality is equivalent to $p(U) \cap p(H) = \{0\}$, the subgroup $p(H)$ is discrete in $V/\langle v_1 \rangle$. By the induction hypothesis, there exist linearly independent elements $w_2, \ldots, w_k$ in $V/\langle v_1 \rangle$, such that

$$p(H) = \mathbb{Z} \cdot w_2 + \cdots + \mathbb{Z} \cdot w_k.$$

Taking representatives in V, these elements are written as $w_i = [v_i]$, $i = 2, \ldots, k$. The set $\{v_1, v_2, \ldots, v_k\}$ is linearly independent in V. By construction, every element of $x \in H$ is written as

$$x = av_1 + n_2 v_2 + \cdots + n_k v_k$$

with $n_i \in \mathbb{Z}$. To conclude the proof it remains to check that $a \in \mathbb{Z}$. But $n_2 v_2 + \cdots + n_k v_k \in H$ and, hence, $av_1 \in H$, so that $a \in \mathbb{Z}$.

This description of the discrete subgroups shows that, up to conjugation (choice of basis), the discrete subgroups of $\mathbb{R}^n$ have the form

$$\mathbb{Z}^k = \{(x_1, \ldots, x_k, 0, \ldots, 0) : x_i \in \mathbb{Z}\}.$$

Hence, the connected abelian Lie groups have the form $\mathbb{R}^n/\mathbb{Z}^k$, $n \geq 0$, $k = 0, \ldots, n$. In case $k = n$, $\mathbb{R}^n/\mathbb{Z}^n$ is the torus $\mathbb{T}^n$, while $\mathbb{R}^n/\mathbb{Z}^k \approx \mathbb{R}^{n-k} \times \mathbb{T}^k$. In particular, the only abelian connected Lie groups that are compact are the tori $\mathbb{T}^n$, $n \geq 0$.

3. The affine group $\mathrm{Aff}(1)$ has dimension 2 and two connected components. Its Lie algebra $\mathfrak{aff}(1)$ is the only 2-dimensional Lie algebra that is not abelian. The connected component of the identity $\mathrm{Aff}(1)_0$ is diffeomorphic to $\mathbb{R}_+ \times \mathbb{R}$, which is simply connected. On the other hand, the center of $\mathrm{Aff}(1)$ is trivial because, if $(a, v) \in \mathrm{Aff}(1)$ commutes with (b, w), then $aw + v = bv + w$. If this occurs for every (b, w), then $v = 0$ (taking $w = 0$), and so $a = 1$. Consequently, $\mathrm{Aff}(1)_0$ is the only connected Lie group of dimension 2 that is not abelian. (There exist, therefore, four connected Lie groups of dimension 2: the abelian groups $\mathbb{R}^2$, $\mathbb{T}^1 \times \mathbb{R}$ and $\mathbb{T}^2$ together with $\mathrm{Aff}(1)_0$, which is not abelian.)
4. The connected group $\mathrm{Gl}^+(2, \mathbb{R})$ has the following geometric structure: Let g be a 2×2 invertible matrix. The columns of g form a basis of $\mathbb{R}^2$. The Gram–Schmidt orthonormalization procedure applied to this basis consists in multiplying g on the right by an upper triangular matrix of the form

$$t = \begin{pmatrix} a & x \\ 0 & b \end{pmatrix}$$

with $a, b > 0$. The procedure yields the matrix $gt = u$, whose columns form an orthonormal basis, that is, u is an orthogonal matrix. As $\det g > 0$ and $\det t = ab$, it follows that $\det u > 0$, that is, $u \in \mathrm{SO}(2)$. Hence, $\mathrm{Gl}^+(2, \mathbb{R}) = \mathrm{SO}(2)\, T$, where T is the group of upper triangular matrices with positive entries in the diagonal. The groups $\mathrm{SO}(2)$ and T are connected with $\mathrm{SO}(2)$, diffeomorphic to the circle S^1, and T is diffeomorphic to $\mathbb{R}^3$.

The map $\phi : \mathrm{SO}(2) \times T \to \mathrm{Gl}^+(2, \mathbb{R})$, given by $\phi(u, t) = ut$, is a diffeomorphism. It is surjective by the orthonormalization procedure. On the other hand, ϕ is injective, as $\mathrm{SO}(2) \cap T = \{1\}$ (and thus $u_1 t_1 = u_2 t_2$ implies that $u_1^{-1} u_2 = t_1 t_2^{-1} = 1$). Moreover, $d\phi_{(u,t)}$ is an isomorphism for each (u, t) (the verification uses the fact that the only skew-symmetric matrix which is also upper triangular is the zero matrix). Therefore, $\mathrm{Gl}^+(2, \mathbb{R})$ is diffeomorphic to the cylinder $S^1 \times \mathbb{R}^3$ and its universal covering is diffeomorphic to $\mathbb{R}^4$.

The same arguments apply to $\mathrm{Sl}(2, \mathbb{R})$ that has the decomposition $\mathrm{Sl}(2, \mathbb{R}) = \mathrm{SO}(2)\, T_1$, where T_1 is the group of upper triangular matrices with determinant 1 and positive entries in the diagonal. This group is diffeomorphic to $\mathbb{R}^2$ and thus $\mathrm{Sl}(2, \mathbb{R})$ is diffeomorphic to $\mathbb{S}^1 \times \mathbb{R}^2$ and its simply connected covering is diffeomorphic to $\mathbb{R}^3$. This construction is a particular case of the Iwasawa decomposition, which will be considered in Chapter 12.

5. Let

$$\mathrm{Sp}(1) = \{q \in \mathbb{H} : |q| = 1\}$$

be the unit sphere of quaternions $\mathbb{H}$. The Lie algebra of $\mathrm{Sp}(1)$ is the tangent space at the identity, which is the algebra of imaginary quaternions. This Lie algebra is isomorphic to $\mathfrak{so}(3)$. Therefore, $\mathrm{Sp}(1)$ is the only simply connected group with Lie algebra $\mathfrak{so}(3)$.

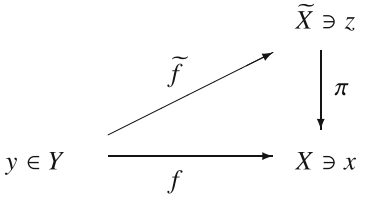
Since the Lie algebra of $\mathrm{SO}(3)$ is also $\mathfrak{so}(3)$, the general theory ensures that there exists a surjective homomorphism $\phi : \mathrm{Sp}(1) \to \mathrm{SO}(3)$, whose kernel is a discrete central subgroup of $\mathrm{Sp}(1)$. This homomorphism is concretely given by the adjoint representation of $\mathrm{Sp}(1)$ in its Lie algebra. In terms of the product of quaternions, $\mathrm{Ad}(z)(w) = zwz^{-1} = zw\bar{z}$, $z \in \mathrm{Sp}(1)$ and $\overline{w} = -w$. For every $z \in \mathrm{Sp}(3)$, $\mathrm{Ad}(z)$ is an isometry. Therefore, the image of Ad is a connected subgroup of dimension 3 of $\mathrm{SO}(3)$, and so $\mathrm{Ad}(\mathrm{Sp}(1)) = \mathrm{SO}(3)$, that is, $\mathrm{Ad} : \mathrm{Sp}(1) \to \mathrm{SO}(3)$ is a surjective homomorphism. The kernel of Ad is the center of $\mathrm{Sp}(1)$, which is $Z(\mathrm{Sp}(1)) = \{\pm 1\}$. Hence, $\mathrm{Sp}(1) \to \mathrm{SO}(3)$ is a double covering, and it follows that the fundamental group of $\mathrm{SO}(3)$ is isomorphic to $\mathbb{Z}_2 = \ker \mathrm{Ad}$.

□

7.4 Appendix: Covering Spaces (Overview)

The following is a list of several results about covering spaces which were used in the construction of the universal covering of a Lie group.

1. A map between locally connected topological spaces $f : A \to B$ is a covering map if for every $x \in B$ there exists a neighborhood $V \ni x$ such that the restriction of f to each connected component C of $f^{-1}(V)$ is a homeomorphism between C and V.
2. Let X be a path connected and locally path connected topological space. Suppose also that X is semilocally simply connected, that is, every $x \in X$ admits a neighborhood U such that the closed curves in U are homotopic (in X) to a point. Then, there exists a covering $\pi : \widetilde{X} \to X$ which is simply connected, that is, the fundamental group $\pi_1\left(\widetilde{X}\right)$ is trivial. This simply connected covering is unique (up to homeomorphism) and is called universal covering of X.
3. In particular a connected differentiable manifold M admits a universal covering $\pi : \widetilde{M} \to M$. In this case, $\widetilde{M}$ is also a differentiable manifold and π is a local diffeomorphism. (The charts of $\widetilde{M}$ are defined by composing the charts of M with the projection π, on suitable chosen domains.)
4. Continuous maps can be lifted to the universal covering, according to the following statement: Let X be a path connected and locally path connected topological space and $\pi : \widetilde{X} \to X$ its universal covering. Let Y be a simply connected space and $f : Y \to X$ a continuous map. Take $x \in X$, $z \in \widetilde{X}$ and $y \in Y$, such that $\pi(z) = x$ and $f(y) = x$. Then, there exists a unique lifting $\widetilde{f} = \widetilde{f}_{x,y,z} : Y \to \widetilde{X}$, such that $\widetilde{f}(y) = z$ and $\pi \circ \widetilde{f} = f$, that is, such that the following diagram is commutative:



5. In the previous item, if X and Y are manifolds and f is differentiable, then $\widetilde{f}$ is also differentiable.
6. A particular case of lifting of continuous (or differentiable) maps is when $Y = \widetilde{X}$ and $f = \pi$. Then, a lifting $\widetilde{\pi}_{x,y,z}$ satisfies $\pi \circ \widetilde{\pi}_{x,y,z} = \pi$ and is a homeomorphism of $\widetilde{X}$ (the inverse of $\widetilde{\pi}_{x,y,z}$ is $\widetilde{\pi}_{x,z,y}$).

   Every map $\theta : \widetilde{X} \to \widetilde{X}$ satisfying $\pi \circ \theta = \pi$ is a lifting of π. Moreover, the set Λ of continuous liftings of π forms a group with the multiplication given by composition of maps.
7. If $x \in X$ and $y \in \pi^{-1}\{x\}$ are fixed, then the liftings $\widetilde{\pi}_{x,y,z}$, $z \in \pi^{-1}\{x\}$ exhaust the group Λ of the previous item. Hence, Λ is in bijection with $\pi^{-1}\{x\}$.

8. The group Λ is isomorphic to the fundamental group $\pi_1(X)$. Isomorphisms are obtained by fixing $x \in X$ and $y \in \pi^{-1}\{x\}$: A continuous closed curve $\alpha : [0,1] \to X$ with $\alpha(0) = \alpha(1) = x$ is lifted in a unique way to a curve $\widetilde{\alpha} : [0,1] \to \widetilde{X}$ com $\widetilde{\alpha}(0) = y$. If $[\alpha] \in \pi_1(X,x)$ denotes the homotopy class of α, then the map $[\alpha] \mapsto \widetilde{\pi}_{x,y,\widetilde{\alpha}(1)} \in \Lambda$ is well defined. This map defines an isomorphism between the fundamental group $\pi_1(X,x)$ and Λ.

7.5 Exercises

1. Let $\phi : G \to H$ be a continuous invertible homomorphism between Lie groups. Show that ϕ is an isomorphism, that is, ϕ^{-1} is a differentiable homomorphism. Do the same for the case of a local isomorphism.
2. Let G and H be connected Lie groups and denote by $\widetilde{G}$ and $\widetilde{H}$ their simply connected coverings. Show that $\widetilde{G} \times \widetilde{H}$ is the universal covering of $G \times H$. Generalize for a product with more than two factors.
3. Use the Gram–Schmidt orthonormalization procedure to generalize the example of $\mathrm{Gl}^+(2,\mathbb{R})$ given in the text for the groups $\mathrm{Gl}^+(n,\mathbb{R})$, $\mathrm{Sl}(n,\mathbb{R})$, $\mathrm{Gl}(n,\mathbb{C})$ and $\mathrm{Sl}(n,\mathbb{C})$.
4. Show that the fundamental groups of $\mathrm{Sl}(n,\mathbb{R})$ and $\mathrm{Gl}^+(n,\mathbb{R})$ coincide with the fundamental group of $\mathrm{SO}(n)$. What can be said about the fundamental groups of $\mathrm{Sl}(n,\mathbb{C})$ and $\mathrm{Gl}(n,\mathbb{C})$?
5. Let $\mathfrak{g}$ be a finite dimensional Lie algebra such that $[X,[Y,Z]] = 0$ for every $X, Y, Z \in \mathfrak{g}$. Find the connected and simply connected Lie group associated with $\mathfrak{g}$. (See Chapter 5, Exercise 23.)
6. Let G be a connected Lie group with $\dim G = 2$ and $\exp : \mathfrak{g} \to G$ its exponential map. Show that exp is a covering map.
7. Let K be a compact abelian Lie group. Show that the set of elements $x \in K$ of finite order (that is, $x^k = 1$, for some $k \in \mathbb{N}$) is dense in K.
8. Show that $\mathrm{Sl}(2,\mathbb{R})$ is diffeomorphic to $\mathbb{S}^1 \times \mathbb{R}^2$, has fundamental group $\mathbb{Z}$ and its universal covering $\widetilde{\mathrm{Sl}(2,\mathbb{R})}$ is diffeomorphic to $\mathbb{R}^3$. Show also that the center of $\widetilde{\mathrm{Sl}(2,\mathbb{R})}$ is isomorphic to $\mathbb{Z}$.
9. Describe all the connected Lie groups whose Lie algebra is $\mathfrak{sl}(2,\mathbb{R})$.
10. Let $G = \mathbb{R}^n \times T^m$ be a connected abelian group and denote by $\widehat{G}$ the set of differentiable homomorphisms $\phi : G \to S^1$. Show that there exists a bijection between $\widehat{G}$ and a differentiable submanifold of the dual $\left(\mathbb{R}^{n+m}\right)^*$ diffeomorphic to $\mathbb{R}^n \times \mathbb{Z}^m$. In $\widehat{G}$, define the product pointwise by $(\phi\psi)(g) = \phi(g)\,\psi(g)$. Verify that this product is well defined in $\widehat{G}$ (using the fact that S^1 is abelian). Verify also that this product defines a Lie group structure on $\widehat{G}$.
11. Let G be the group of matrices

$$\begin{pmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix},$$

with $x, y, z \in \mathbb{R}$, and $\Gamma \subset G$ the subgroup of matrices with entries in $\mathbb{Z}$. Show that the manifold G/Γ does not admit a group structure that makes it a Lie group.

12. Denote by $\mathrm{Sl}(2, \mathbb{Z})$ the set of 2×2 matrices with integer entries and determinant 1. Verify that $\mathrm{Sl}(2, \mathbb{Z})$ is a closed subgroup of $\mathrm{Sl}(2, \mathbb{R})$. Show that there does not exist any group structure on the manifold $\mathrm{Sl}(2, \mathbb{R})/\mathrm{Sl}(2, \mathbb{Z})$ that makes it a Lie group.
13. Let G be a connected Lie group with Lie algebra $\mathfrak{g}$. Suppose that there exist ideals $\mathfrak{a}, \mathfrak{b} \subset \mathfrak{g}$ such that $\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{b}$. Define the connected subgroups $A = \langle \exp \mathfrak{a} \rangle$ and $B = \langle \exp \mathfrak{b} \rangle$. Show that $G = AB = BA$, that is, the map $A \times B \to G$, $(a, b) \mapsto ab$ is surjective.

    Show also that if the exponential maps of A and B are surjective, then the exponential on G is also surjective. (Hint: Go to the universal covering. Compare with Exercise 15 of Chapter 6.)
14. Given the connected Lie groups G and H, denote by $\pi_G : \widetilde{G} \to G$ and $\pi_H : \widetilde{H} \to H$ the universal coverings. Let $\phi : G \to H$ be a differentiable homomorphism. Show that there exists a unique homomorphism $\widetilde{\phi} : \widetilde{G} \to \widetilde{H}$ such that $\pi_H \circ \widetilde{\phi} = \phi \circ \pi_G$. Show also that a homomorphism $\theta : \mathfrak{g} \to \mathfrak{h}$, between the Lie algebras of G and H, extends to a homomorphism between G and H if and only if $\ker \phi \subset \ker \pi_G$, where $\widetilde{\phi} : \widetilde{G} \to \widetilde{H}$ is such that $d\widetilde{\phi}_1 = \theta$.
15. Let $\widetilde{G}$ be a connected and simply connected Lie group and $\Gamma_1, \Gamma_2 \subset \widetilde{G}$ discrete central subgroups. Show that $G_1 = \widetilde{G}/\Gamma_1$ is isomorphic to $G_2 = \widetilde{G}/\Gamma_2$ if and only if there exists an automorphism $\phi \in \mathrm{Aut}\left(\widetilde{G}\right)$, such that $\phi(\Gamma_1) = \Gamma_2$.

    Give examples in which G_1 is not isomorphic to G_2 and, nevertheless, Γ_1 is isomorphic to Γ_2, that is, $\pi_1(G_1) \approx \pi_1(G_2)$.
16. Let G be a connected Lie group and H a connected closed subgroup. Show that G is simply connected if H and G/H are simply connected. (Hint: Write $G = \widetilde{G}/\Gamma$, check that $G/H \approx \widetilde{G}/H'$, where $H' = \pi^{-1}(H)$ and $\pi : \widetilde{G} \to G$ is the canonical projection. Verify that $H = H'/\left(H' \cap \Gamma\right)$ and show that H' is not connected. Finally, consider the covering $\widetilde{G}/H'_0 \to \widetilde{G}/H' \approx G/H$.) (Another hint: Use the long homotopy exact sequence for fibrations.)
17. Show that the Lie group $\mathrm{SU}(n)$, $n \geq 1$, is simply connected. Do the same for the group $\mathrm{Sp}(n)$ of unitary quaternionic matrices. (Hint: In both cases, there exist spheres of suitable dimensions which are homogeneous spaces of the groups.)
18. Show that $\mathrm{SO}(n)$, $n \geq 2$, is not simply connected and that the fundamental group $\pi_1(\mathrm{SO}(n)) \approx \mathbb{Z}_2$ if $n \geq 3$.
19. The objective of this exercise is to show, using directly homotopy curves, that the fundamental group of a connected Lie group G is abelian. Denote by $[\alpha]$ the homotopy class of the curve α. Let $\gamma : [0, 1] \to G$ be a continuous curve with $\gamma(0) = \gamma(1) = 1$ and define the curves $\widetilde{\gamma}, \widehat{\gamma} : [0, 1] \to G$ by

$$\widetilde{\gamma}(t) = \begin{cases} 1 & t \in \left[0, \frac{1}{2}\right] \\ \gamma(2t-1) & t \in \left[\frac{1}{2}, 1\right] \end{cases} \quad \text{and} \quad \widehat{\gamma}(t) = \begin{cases} \gamma(2t) & t \in \left[0, \frac{1}{2}\right] \\ 1 & t \in \left[\frac{1}{2}, 1\right] \end{cases},$$

which run through γ, respectively, on the second and first halves of $[0, 1]$.

(a) Show that $[\gamma] = [\widetilde{\gamma}] = [\widehat{\gamma}]$. Hint: Define the homotopies

$$\widetilde{H}(t, s) = \begin{cases} 1 & t \in \left[0, \frac{s}{2}\right] \\ \gamma\left(\frac{2t-s}{2-s}\right) & t \in \left[\frac{s}{2}, 1\right] \end{cases} \quad \widehat{\gamma}(t) = \begin{cases} \gamma\left(\frac{2t}{2-s}\right) & t \in \left[0, \frac{2-s}{2}\right] \\ 1 & t \in \left[\frac{2-s}{2}, 1\right] \end{cases}.$$

(b) Given the closed curves $\gamma_1, \gamma_2 : [0, 1] \to G$ with initial and end points at $1 \in G$, consider the curves $\widehat{\gamma}_1(t)\,\widetilde{\gamma}_2(t)$ and $\widetilde{\gamma}_1(t)\,\widehat{\gamma}_2(t)$ (product in G). Show that

$$\widehat{\gamma}_1(t)\,\widetilde{\gamma}_2(t) = \begin{cases} \gamma_1(2t) & t \in \left[0, \frac{1}{2}\right] \\ \gamma_2(2t-1) & t \in \left[\frac{1}{2}, 1\right] \end{cases}$$

$$\widetilde{\gamma}_1(t)\,\widehat{\gamma}_2(t) = \begin{cases} \gamma_2(2t) & t \in \left[0, \frac{1}{2}\right] \\ \gamma_1(2t-1) & t \in \left[\frac{1}{2}, 1\right] \end{cases}$$

(c) Use the previous item to show that $[\widehat{\gamma}_1\widetilde{\gamma}_2] = [\gamma_1][\gamma_2]$ and $[\widetilde{\gamma}_1\widehat{\gamma}_2] = [\gamma_2][\gamma_1]$ and conclude that $[\gamma_1][\gamma_2] = [\gamma_2][\gamma_1]$.

Chapter 8
Series Expansions



8.1 The Differential of the Exponential Map

The exponential map $\exp : \mathfrak{g} \to G$ on a Lie group G is a differentiable map. Its differential at the origin $d\,(\exp)_0 : \mathfrak{g} \to \mathfrak{g}$ is the identity map of $\mathfrak{g} = T_1 G$.

On the other hand at $X \in \mathfrak{g}$ the differential $d\,(\exp)_X$ is a linear map between $\mathfrak{g} = T_1 G$ and the tangent space $T_{e^X} G$. The left translation of a linear map $T_X : \mathfrak{g} \to \mathfrak{g}$ is defined by

$$T_X = dL_{e^{-X}} \circ d\,(\exp)_X \ . \tag{8.1}$$

The purpose of this section is to write the linear map T_X as a power series in $\mathrm{ad}\,(X)$.

Theorem 8.1 *Given an element $X \in \mathfrak{g}$, let T_X be the left translation of $d\,(\exp)_X$ as defined in (8.1). Then*

$$T_X = \sum_{k \geq 0} \frac{1}{(k+1)!} \left(\mathrm{ad}\,(X)\right)^k . \tag{8.2}$$

In this expression for T_X the Lie-algebra structure on $\mathfrak{g} = T_1 G$ is assumed to be given by the bracket between right invariant vector fields. If one takes left invariant vector fields, then the sign of ad must be changed. Thus let $\mathrm{ad}_r\,(X)$ and ad_l denote the adjoints given by right and left invariant vector fields, respectively. Then $\mathrm{ad}_l\,(X) = -\mathrm{ad}_r\,(X)$, and with this notation the formula for the differential of the exponential reads

$$T_X = \sum_{k \geq 0} \frac{1}{(k+1)!} \left(\mathrm{ad}_r\,(X)\right)^k = \sum_{k \geq 0} \frac{(-1)^k}{(k+1)!} \left(\mathrm{ad}_l\,(X)\right)^k .$$

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The series can be written more concisely by taking into account that the power series

$$\sum_{k\geq 0} \frac{1}{(k+1)!} z^k$$

in the variable z represents the (real or complex) function $f(z) = \dfrac{e^z - 1}{z}$. Hence $T_X = f(\operatorname{ad}(X))$, that is,

$$T_X = \frac{e^{\operatorname{ad}_r(X)} - 1}{\operatorname{ad}_r(X)} = \frac{1 - e^{-\operatorname{ad}_l(X)}}{\operatorname{ad}_l(X)}.$$

The proof of Theorem 8.2 is divided into two steps. The first step is to prove the formula for the differential of exp for linear groups, that is, subgroups of $\mathrm{Gl}(n, \mathbb{R})$. Afterwards the result is extended to general Lie groups with the aid of Ado's theorem, which ensures that any Lie group is locally isomorphic to a linear group.

Therefore let $\mathfrak{g}$ be a Lie algebra of matrices and take $X, Y \in \mathfrak{g}$. By definition $T_X(Y) = dL_{\exp(-X)} \cdot d(\exp)_X(Y)$, which for matrices is given by

$$\frac{d}{dt}\left(e^{-X} e^{X+tY}\right)_{|t=0} = e^{-X} \frac{d}{dt}\left(e^{X+tY}\right)_{|t=0}.$$

On a linear group e^X is the power series $\sum_{k\geq 0} \dfrac{1}{k!} X^k$, which permits to compute the derivative $\dfrac{d}{dt}\left(e^{X+tY}\right)_{|t=0}$ explicitly as a power series in ad. In what follows this derivative will be computed through manipulations of the series, basically by rearranging the order of terms and using the associative property. All this is justified by the convergence in norm of the series involved. (In this case, it is convenient to take the standard operator norm, which satisfies $\|XY\| \leq \|X\| \cdot \|Y\|$.)

The exponential series converges normally, so

$$\frac{d}{dt}\left(e^{X+tY}\right)_{|t=0} = \sum_{k\geq 1} \frac{1}{k!} \frac{d}{dt} (X + tY)^k_{|t=0},$$

and the derivative of a product of matrices yields

$$\frac{d}{dt} (X + tY)^k_{|t=0} = \sum_{i=0}^{k-1} X^{k-i-1} Y X^i.$$

Hence

$$\frac{d}{dt}\left(e^{X+tY}\right)_{|t=0} = \sum_{k\geq 1} \sum_{i=0}^{k-1} \frac{1}{k!} X^{k-i+1} Y X^i. \tag{8.3}$$

The terms in this sum can be rewritten with the aid of the following commutation formula, holding on arbitrary associative algebras.

Lemma 8.2 *Let $\mathcal{A}$ be an associative algebra and take $x, y \in \mathcal{A}$. If $\mathrm{ad}_r(x)y = yx - xy$, then for every $n \geq 1$*

$$yx^n = \sum_{p=0}^{n} \binom{n}{p} x^{n-p} (\mathrm{ad}_r(x)^p y). \tag{8.4}$$

This lemma is proved by simple induction.

Using (8.4) in (8.3), the derivative $\frac{d}{dt}\left(e^{X+tY}\right)_{|t=0}$ becomes

$$\sum_{k\geq 1} \sum_{i=0}^{k-1} \sum_{j=0}^{i} \frac{1}{k!} \binom{i}{j} X^{k-i-1} X^{i-j} \mathrm{ad}\,(X)^j\,(Y)$$

$$= \sum_{k\geq 1} \sum_{i=0}^{k-1} \sum_{j=0}^{i} \frac{1}{k!} \binom{i}{j} X^{k-j-1} \mathrm{ad}\,(X)^j\,(Y)\,,$$

where $\mathrm{ad} = \mathrm{ad}_r$. The idea is now to write this sum as a series in $\mathrm{ad}\,(X)^j\,(Y)$. To this end its terms are rearranged as follows:

$$\sum_{k\geq 1} \sum_{i=0}^{k-1} \sum_{j=0}^{i} = \sum_{k\geq 1} \sum_{j=0}^{k-1} \sum_{i=j}^{k-1} = \sum_{j\geq 0} \sum_{k\geq j+1} \sum_{i=j}^{k-1},$$

giving for the differential the expression

$$\sum_{j\geq 0} \sum_{k\geq j+1} \frac{1}{k!} X^{k-j-1} \mathrm{ad}\,(X)^j\,(Y) \left(\sum_{i=j}^{k-1} \binom{i}{j} \right). \tag{8.5}$$

The sum inside the parentheses can be evaluated using the following lemma on binomial coefficients.

Lemma 8.3 $\sum_{i=j}^{k-1} \binom{i}{j} = \binom{k}{j+1}$.

Proof The argument goes by induction and uses the relation

$$\binom{n}{j} + \binom{n}{j+1} = \binom{n+1}{j+1}.$$

Induction starts with $\binom{j}{j} + \binom{j+1}{j} = j+2 = \binom{j+2}{j+1}$. □

By the lemma, sum (8.5) reads

$$\sum_{j\geq 0}\sum_{k\geq j+1}\frac{1}{k!}\begin{pmatrix}k\\ j+1\end{pmatrix}X^{k-j-1}\mathrm{ad}\,(X)^j\,(Y)$$
$$=\sum_{j\geq 0}\sum_{k\geq j+1}\frac{1}{(j+1)!\,(k-j-1)!}X^{k-j-1}\mathrm{ad}\,(X)^j\,(Y)\,.$$

Putting $r=k-j-1$ the last sum may be rewritten as

$$\sum_{j\geq 0}\frac{1}{(j+1)!}\left(\sum_{r\cdot\geq 0}\frac{1}{r!}X^r\right)\mathrm{ad}\,(X)^j\,(Y)\,,$$

that is,

$$\frac{d}{dt}\left(e^{X+tY}\right)_{|t=0}=e^X\sum_{j\geq 0}\frac{1}{(j+1)!}\mathrm{ad}\,(X)^j\,(Y)\,.$$

Multiplying on the left by e^{-X} produces

$$T_X=\sum_{j\geq 0}\frac{1}{(j+1)!}\mathrm{ad}\,(X)^j\,(Y)\,,$$

where $\mathrm{ad}=\mathrm{ad}_r$, since the commutation formula (8.4) was applied. This concludes the proof of Theorem 8.2 for linear groups.

The general case follows from the representation theorem of Ado and the following lemma.

Lemma 8.4 *Let $\phi : G\to H$ be a homomorphism of Lie groups, and denote by $\exp_G$ and $\exp_H$ the exponential maps on G and H, respectively. Take left-translates T_X^G and T_Y^H on G and H as of (8.1), with $Y=d\phi_1\,(X)$. Then*

$$d\phi_1\left(T_X^G\right)=T^H_{d\phi_1(X)}.$$

Proof Since ϕ is a homomorphism, $\phi\circ\exp_G=\exp_H\circ d\phi_1$. Hence if $Y=d\phi_1\,(X)$, then

$$d\phi_{\exp_G X}\circ d\left(\exp_G\right)_X=d\left(\exp_H\right)_Y\circ d\phi_1\,(X)\;.$$

Applying to this relation the left-translation $dL_{\exp_H(-Y)}$ yields

$$d\left(L_{\exp_H(-Y)}\right)_{\exp_H Y}\circ d\phi_{\exp_G X}\circ d\left(\exp_G\right)_X=T_Y^H\circ d\phi_1\,(X)\;. \tag{8.6}$$

However, $\phi \circ L_{\exp_G(-X)} = L_{\phi(\exp_G(-X))} \circ \phi = L_{\exp_H(-Y)} \circ \phi$. Substituting the latter in (8.6) and applying the chain rule, it follows that

$$\begin{aligned} d\phi_1 \left(T_X^G\right) &= d\left(\phi \circ L_{\exp_G(-X)}\right)_{\exp_G X} \circ d\left(\exp_G\right)_X \\ &= T_Y^H \circ d\phi_1(X) \ , \end{aligned}$$

i.e. the claimed formula. □

To conclude the proof of Theorem 8.1 the previous lemma need to be applied twice.

Let G be a Lie group with Lie algebra $\mathfrak{g}$. Taking a faithful representation ρ of $\mathfrak{g}$ (whose existence is warranted by Ado's theorem), define $H = \langle \exp \rho(\mathfrak{g}) \rangle$. The Lie algebras $\mathfrak{g}$ and $\mathfrak{h} = \rho(\mathfrak{g})$ of G and H are isomorphic. Hence the universal covering $\widetilde{G}$ of both groups is the same, and there exist homomorphisms $\phi : \widetilde{G} \to G$ and $\psi : \widetilde{G} \to H$ such that $d\phi_1$ and $d\psi_1$ are isomorphisms. Therefore, retaining the notation of the lemma for $X \in \mathfrak{g}$, one has

$$T_X^G = d\phi_1 \left(T^{\widetilde{G}}_{(d\phi_1)^{-1}(X)}\right) = d\phi_1 \circ (d\psi_1)^{-1} T_Y^H,$$

where $Y = \theta(X)$ and $\theta = d\psi_1 \circ (d\phi_1)^{-1}$. Since the theorem is true on the linear group H, it follows that

$$\begin{aligned} T_X^G &= \theta^{-1} \sum_{k \geq 0} \frac{1}{(k+1)!} (\mathrm{ad}(\theta X))^k \\ &= \sum_{k \geq 0} \frac{1}{(k+1)!} (\mathrm{ad}(X))^k \ , \end{aligned}$$

since $\theta^{-1} \mathrm{ad}(\theta X) = \mathrm{ad}(X)$ and θ is an isomorphism. This concludes the proof of Theorem 8.1 in general.

8.2 The Baker–Campbell–Hausdorff Series

Let G be a Lie group with Lie algebra $\mathfrak{g}$. The exponential map on G is a diffeomorphism $\exp : V \to U$ around the origin, and $0 \in V \subset \mathfrak{g}$ and $1 \in U \subset G$ are open sets. If $X, Y \in V$ are small enough, then the product $e^X e^Y$ still belongs to U, allowing one to write

$$e^X e^Y = e^{c(X,Y)}.$$

The map c expresses the product of G in a local coordinate system of the first kind.

The **Baker–Campbell–Hausdorff series** (BCH) gives an expression for $c(X, Y)$ as a sum of a series whose terms are iterated brackets of X, Y. This series is written as

$$c(X, Y) = X + Y + \sum_{n \geq 2} c_n(X, Y),$$

where the terms $c_n(X, Y)$, $n \geq 2$, are homogeneous of order n, meaning sums of n-fold brackets (in X and Y). For example, the first terms (written for right invariant vector fields) read

$$c_2(X, Y) = -\frac{1}{2}[X, Y],$$
$$c_3(X, Y) = \frac{1}{12}[[X, Y], Y] - \frac{1}{12}[[X, Y], X].$$

As for the differential of the exponential map, the formula for the BCH series is universal, too, in the sense that the expressions for the homogeneous terms are the same regardless of the Lie group (Lie algebra, as a matter of fact).

In what follows a proof of the convergence of the BCH series (for small X and Y) will be presented, as well as an inductive formula to compute $c_n(X, Y)$ from lower-degree terms.

The BCH series is defined locally (around the identity) and is expressed in terms of Lie brackets. Therefore if $\mathfrak{g}$ is a fixed Lie algebra, the series is the same for any Lie group with Lie algebra $\mathfrak{g}$ since any two are locally isomorphic. In view of this, the same strategy used for the formula of the exponential differential will be adopted here to study the BCH series. The idea is to use Ado's theorem and make computations with matrices, which facilitates the arguments, especially with regard to the series' convergence.

With this in place, the left-hand side of equality $e^X e^Y = e^{c(X,Y)}$ is written as a sum of a power series

$$\begin{aligned} e^X e^Y &= \left(\sum_{n \geq 0} \frac{1}{n!} X^n\right)\left(\sum_{n \geq 0} \frac{1}{n!} Y^n\right) \\ &= \sum_{n \geq 0} \left(\sum_{j=0}^{n} \frac{1}{j!} \frac{1}{(n-j)!} X^j Y^{n-j}\right) \\ &= \sum_{n \geq 0} e_n(X, Y). \end{aligned}$$

The last series converges normally for any X, Y since this happens to the exponential series. (Convergence is with respect to a norm given beforehand; for instance, the operator norm, which satisfies $\|XY\| \leq \|X\| \cdot \|Y\|$.)

Now consider the series of the logarithm

$$\log(1+x) = \sum_{k\geq 1} \frac{(-1)^{k+1}}{k} x^k,$$

which converges absolutely if $|x| < 1$. This series inverts the exponential in the sense that

$$\begin{aligned} \log(\exp x) &= \log(1 + (\exp x - 1)) \\ &= \sum_{k\geq 1} \frac{(-1)^{k+1}}{k} (\exp x)^k = x \end{aligned}$$

whenever $|\exp x - 1| < 1$. So, substituting the series for $e^X e^Y$ in the logarithm gives

$$\begin{aligned} c(X,Y) &= \sum_{k\geq 1} \frac{(-1)^{k+1}}{k} \left(\sum_{n\geq 0} e_n(X,Y) \right)^k \\ &= \sum_{k\geq 1} \frac{(-1)^{k+1}}{k} \left(\sum_{n\geq 0} \sum_{j=0}^{n} \frac{1}{j!} \frac{1}{(n-j)!} X^j Y^{n-j} \right)^k. \end{aligned}$$

This converges normally for X, Y small enough and satisfying $\left\| e^X e^Y - 1 \right\| < 1$. Hence the terms of the series can be rearranged (permuted and associated) to obtain new series that still converge normally. The above expression contains monomials in X and Y of the form $X^{i_1} Y^{j_1} \cdots X^{i_s} Y^{j_s}$, of degree $n = i_1 + j_1 + \cdots + i_s + j_s$ (coming from the kth power of the series in parentheses). By gathering monomials of the same degree n in the term $c_n(X,Y)$ produces the convergent series

$$c(X,Y) = \sum_{n\geq 1} c_n(X,Y), \tag{8.7}$$

as in the BCH formula. These arguments demonstrate the following statement.

Proposition 8.5 *There exists a number $\rho > 0$ such that if $\|X\|, \|Y\| < \rho$, then $c(X,Y)$ is given by the convergent series (8.7,) where the term $c_n(X,Y)$ is a homogeneous polynomial in X, Y of the type*

$$c_n(X,Y) = \sum a_{I,J} X^{i_1} Y^{j_1} \cdots X^{i_s} Y^{j_s}$$

with $n = i_1 + j_1 + \cdots + i_s + j_s$ and $I = (i_1, \ldots, i_s)$, $J = (j_1, \ldots, j_s)$.

The goal is now to achieve a recursive formula for c_n. It will be clear from the formula that $c_n(X,Y)$ is a **Lie element** , i.e. a sum of iterated brackets of X and Y.

The idea is to write the series for $c(tX, tY)$, with $|t| < 1$, and take into account that $c_n(tX, tY) = t^n c_n(X, Y)$ since $c_n(X, Y)$ is a homogeneous polynomial of degree n in X and Y. Therefore

$$c(tX, tY) = \sum_{n \geq 1} c_n(X, Y)\, t^n$$

is a power series in t, furthermore absolutely convergent if $|t| < 1$. Its derivative is $c(tX, tY)' = \sum_{n \geq 0} (n+1)\, c_{n+1}(X, Y)\, t^n$. If the derivative is properly computed, then $c_n(X, Y)$ can be found by comparing the terms of two series.

The derivative $c(tX, tY)'$ is computed through the equality $e^{tX} e^{tY} = e^{c(tX, tY)}$ and the formula for the differential of the exponential map.

The computations will be made for the bracket of right invariant fields, given by $[X, Y] = YX - XY$ in the case of linear groups (so that XY means matrix multiplication). For the right invariant bracket one has $d(\exp)_Z = e^Z T_Z$, where $T_Z = \phi(\operatorname{ad}(Z))$. In other words T_Z is the power series of the function

$$\phi(z) = \frac{e^z - 1}{z}$$

evaluated at $z = \operatorname{ad}(Z)$.

The computation of $c(tX, tY)'$ uses the following functions:

1. $\eta(z) = \phi(z)^{-1} = \frac{z}{e^z - 1}$. This function satisfies $\eta(-z) = \eta(z) + z$, because

$$\begin{aligned} \eta(-z) - z &= \frac{z}{1 - e^{-z}} - z = \frac{z e^{-z}}{1 - e^{-z}} \\ &= \frac{z}{e^z - 1} = \eta(z). \end{aligned}$$

2. $\psi(z) = \eta(z) + \frac{z}{2}$, which is an even function because $\psi(-z) = \eta(-z) - \frac{z}{2} = \psi(z)$. (Note that $\psi(z) = \frac{z}{e^z - 1} + \frac{z}{2} = \frac{z}{2} \frac{e^z + 1}{e^z - 1} = \frac{z}{2} \coth \frac{z}{2}$.) The power series of ψ has only terms of even degree. Since $\psi(0) = \eta(0) = 1$ the series can be written as

$$\psi(z) = 1 + \sum_{k \geq 1} a_{2k} z^{2k}. \tag{8.8}$$

The latter can be obtained by the following formal computations:

$$\begin{aligned} \psi(z) &= \frac{z}{e^z - 1} + \frac{z}{2} \\ &= \frac{z}{2} + \frac{1}{1 + \left(\frac{z}{2!} + \frac{z^2}{3!} + \cdots \right)} \end{aligned}$$

$$= \frac{z}{2} + 1 - \left(\frac{z}{2!} + \frac{z^2}{3!} + \cdots \right) + \left(\frac{z}{2!} + \frac{z^2}{3!} + \cdots \right)^2 + \cdots$$

Proposition 8.6 *Write* $f(t) = c(tX, tY)$. *Then*

$$c(tX, tY)' = f'(t) = \psi(\operatorname{ad}(f(t)))(X + Y) + \frac{1}{2}[f(t), X - Y]. \tag{8.9}$$

Proof The derivative in t of $c(tX, tY)$ is computed by considering the two t's as separate variables. To this purpose put $F(u, v) = c(uX, vY)$ so that $e^{uX}e^{vY} = e^{F(u,v)}$ and

$$c(tX, tY)' = \frac{\partial F}{\partial u}(t, t) + \frac{\partial F}{\partial v}(t, t).$$

First, the derivative in v equals

$$\frac{\partial}{\partial v} e^{uX} e^{vY} = e^{uX} e^{vY} Y$$

(recall X and Y are matrices). On the other hand,

$$\begin{aligned} \frac{\partial}{\partial v} e^{c(uX,vY)} &= d(\exp)_{c(uX,vY)} \left(\frac{\partial F}{\partial v}(u, v) \right) \\ &= e^{c(uX,vY)} T_{c(uX,vY)} \left(\frac{\partial F}{\partial v}(u, v) \right), \end{aligned}$$

where $T_{c(tX,tY)}$ comes from the differential of the exponential map. By comparing the two derivatives it follows that

$$e^{uX} e^{vY} Y = e^{c(uX,vY)} T_{c(uX,vY)} \left(\frac{\partial F}{\partial v}(u, v) \right).$$

Now, if X and Y are small enough the linear map $T_{c(tX,tY)}$ has an inverse so that

$$\frac{\partial F}{\partial v}(u, v) = T^{-1}_{c(uX,vY)}(Y).$$

But $T^{-1}_{c(uX,vY)} = \eta(\operatorname{ad}(c(uX, vY)))$ since $\eta(z)\phi(z) = 1$. Hence

$$\begin{aligned} \frac{\partial F}{\partial v}(u, v) &= \eta(\operatorname{ad}(c(uX, vY)))(Y) \\ &= \psi(\operatorname{ad}(c(uX, vY)))(X) - \frac{1}{2}\operatorname{ad}(c(uX, vY))(X). \end{aligned}$$

The derivative in u is obtained from the v-derivative combined with identity $e^{-vY}e^{-uX} = e^{-c(uX,vY)}$. Hence

$$e^{-vY}e^{-uX}X = e^{-c(uX,vY)}T_{-c(uX,vY)}\left(\frac{\partial F}{\partial u}(u,v)\right),$$

that is,

$$\begin{aligned}\frac{\partial F}{\partial u}(u,v) &= \eta\left(-\mathrm{ad}\left(c\left(uX,vY\right)\right)\right)(X)\\ &= \psi\left(-\mathrm{ad}\left(c\left(uX,vY\right)\right)\right)(X) + \frac{1}{2}\mathrm{ad}\left(c\left(uX,vY\right)\right)(X)\\ &= \psi\left(\mathrm{ad}\left(c\left(uX,vY\right)\right)\right)(X) + \frac{1}{2}\mathrm{ad}\left(c\left(uX,vY\right)\right)(X),\end{aligned}$$

since the function ψ is even.

Finally, summing the partial derivatives produces

$$c\left(tX,tY\right)' = \psi\left(\mathrm{ad}\left(c\left(tX,tY\right)\right)\right)(X+Y) + \frac{1}{2}\mathrm{ad}\left(c\left(tX,tY\right)\right)(X-Y),$$

as claimed. □

It is now possible to obtain the recursive formula for $c_n(X,Y)$ by comparing the coefficients of the power series in equality (8.9). To begin with, the first term $c_1(X,Y)$ equals

$$c_1(X,Y) = \frac{1}{2}c\left(tX,tY\right)'_{t=0} = X+Y,$$

because $\psi(0) = 1$.

Theorem 8.7 *The recursive formula for $c_n = c_n(X,Y)$ is given by $c_1 = X+Y$ and*

$$\begin{aligned}(n+1)\,c_{n+1} &= \frac{1}{2}\left[c_n, X-Y\right]\\ &\quad + \sum_{2\le 2k\le n} a_{2k}\sum_{J_{k,n}}\mathrm{ad}\left(c_{j_1}\right)\cdots\mathrm{ad}\left(c_{j_{2k}}\right)(X+Y),\end{aligned}$$

where the second sum extends over the multi-indices $J_{k,n} = (j_1,\ldots,j_{2k})$, with $2k$ elements $j_i \ge 1$ such that $j_1+\cdots+j_{2k} = n$.

Proof The left-hand side of (8.9) reads

$$c\left(tX,tY\right)' = \sum_{n\ge 0}(n+1)\,c_{n+1}t^n. \tag{8.10}$$

In the right side the series of the last term is

$$\frac{1}{2}\mathrm{ad}\left(c\left(tX,tY\right)\right)\left(X-Y\right)=\frac{1}{2}\sum_{n\geq 1}\mathrm{ad}\left(c_n\right)\left(X-Y\right)t^n, \tag{8.11}$$

while the series in the first term is given by

$$\begin{aligned}\psi\left(\mathrm{ad}\left(c\left(tX,tY\right)\right)\right)\left(X+Y\right)&=X+Y+\sum_{k\geq 1}a_{2k}\left(\mathrm{ad}\left(c\left(tX,tY\right)\right)\right)^{2k}\left(X+Y\right)\\&=X+Y+\sum_{k\geq 1}a_{2k}\left(\sum_{j\geq 1}\mathrm{ad}\left(c_j\right)t^j\right)^{2k}\left(X+Y\right).\end{aligned}$$

In the last series the coefficient of t^n, $n\geq 1$ is

$$\sum_{2\leq 2k\leq n}a_{2k}\sum_{j_1+\cdots+j_{2k}=n}\mathrm{ad}\left(c_{j_1}\right)\cdots\mathrm{ad}\left(c_{j_{2k}}\right)\left(X+Y\right). \tag{8.12}$$

To finish, the relation of the statement is obtained by comparing the nth term of (8.10) with the sum of the nth terms in (8.11) and (8.12). □

The theorem above allows to compute, in principle, the terms of the BCH series, although the inductive process quickly becomes complicated. The first terms of the series go as follows:

1. $c_1\left(X,Y\right)=X+Y$.
2. $2c_2\left(X,Y\right)=\frac{1}{2}\left[c_1,X-Y\right]=\frac{1}{2}\left[X+Y,X-Y\right]=-\left[X,Y\right]$, that is,

$$c_2\left(X,Y\right)=-\frac{1}{2}\left[X,Y\right].$$

(The minus sign is due to the fact that the bracket is between right invariant vector fields.)

3. For $n=3$ the recursive formula shows

$$\begin{aligned}3c_3&=\frac{1}{2}\left[c_2,X-Y\right]+a_2\sum_{j_1+j_2=2}\mathrm{ad}\left(c_{j_1}\right)\mathrm{ad}\left(c_{j_2}\right)\left(X+Y\right)\\&=\frac{1}{2}\left[c_2,X-Y\right]+a_2\mathrm{ad}\left(c_1\right)\mathrm{ad}\left(c_1\right)\left(X+Y\right).\end{aligned}$$

Since $c_1=X+Y$ the last term is zero, so

$$c_3=\frac{1}{12}\left[\left[X,Y\right],Y\right]-\frac{1}{12}\left[\left[X,Y\right],X\right].$$

8.3 Analytic-Manifold Structure

The convergence of the Baker–Campbell–Hausdorff series shows that the product map on a Lie group G becomes analytic when read in a small enough coordinate system of the first kind. Using the identity $e^X e^Y = e^{c(X,Y)}$, in fact, the definition of $c(X, Y)$ means that $c : V_0 \times V_0 \to V_0$ is the expression of the product $p : G \times G \to G$ in local coordinates given by a chart $\exp : V_0 \to U_0$ around the identity. That is, $c = \log \circ p \circ (\exp \times \exp)$, where $\log = \exp^{-1} : U_0 \to V_0$.

The BCH series $c(X, Y) = \sum_{n \geq 0} c_n(X, Y)$ is a power series expansion in the sense that the terms c_n are homogeneous polynomials of degree n in the variables X, Y. As proved in the previous section, this series converges normally on a neighborhood $V \times V \subset V_0 \times V_0$ of the origin. Hence the product map is analytic in a chart of G around the identity, and defined by $\exp : V \to U = \exp(V)$. In the sequel the translations of this chart will be shown to define on G an analytic atlas for which the product map is analytic.

- **Definition of the analytic atlas:** start with a coordinate system $\exp : V_0 \to U_0$ such that $V_0 \times V_0$ is contained in the domain of convergence of the BCH series. Choose $V \subset V_0$ with $-V = V$ and such that if $U = e^V$, then $U^2 \subset U_0$.

  Now define, for each $g \in G$, the chart $\varphi_g : V \to gU$ by $\varphi_g(X) = ge^X$. The collection of charts

$$\mathcal{A} = \{\varphi_g : V \to gU, g \in G\}$$

  is an atlas of G since the open sets gU, $g \in G$, cover G.

Proposition 8.8 *The atlas $\varphi_g : V \to gU$, $g \in G$, defined by $\varphi_g(X) = ge^X$ is analytic. The product $p : G \times G \to G$ is an analytic map with respect to this atlas.*

Proof Let $g, h \in G$ be such that $gU \cap hU \neq \emptyset$. The transition function is $\varphi_h^{-1} \circ \varphi_g : V_g \to V_h$, where $V_g = \varphi_g^{-1}(gU \cap hU)$. If $x \in gU \cap hU$, then $x = ge^X = he^Y$ with $X \in V_g$, $Y \in V_h$ and

$$Y = \varphi_h^{-1} \circ \varphi_g(X).$$

Since $ge^X = he^Y$ it follows that $h^{-1}g = e^Y e^{-X} \in U^2 \subset U_0$. Hence there exists $Z \in V_0$ such that $h^{-1}g = e^Z$, so that

$$e^Y = h^{-1}ge^X = e^Z e^X.$$

Now, both Z and X belong to the convergence domain of c, hence $Y = c(Z, X)$. Consequently $\varphi_h^{-1} \circ \varphi_g = c(Z, \cdot)$, where Z is determined by $h^{-1}g = e^Z$. This proves that the maps $\varphi_h^{-1} \circ \varphi_g$, $g, h \in G$ are analytic.

To show the analyticity of the product, take $g, h \in G$ and the charts $\varphi_g \times \varphi_h : V \times V \to gU \times hU$ and $\varphi_{gh} : V \to ghU$. Then

$$p\left(ge^X, he^Y\right) = ge^X he^Y = ghe^{\mathrm{Ad}(h^{-1})X} e^Y,$$

which shows that p, in the chosen charts, reads

$$(X, Y) \mapsto c\left(\mathrm{Ad}\left(h^{-1}\right) X, Y\right).$$

The above is the composite of an analytic map and a linear map, and hence is analytic. □

8.4 Exercises

1. Let G be a compact, nonabelian Lie group. Show that the exponential map on G is not a local diffeomorphism.
2. Let $\mathfrak{s}$ be the vector space of $n \times n$ symmetric matrices. Show that the restriction to $\mathfrak{s}$ of the exponential map is an injective immersion, whose image is the set S of positive-definite symmetric matrices. (Hint: if X symmetric, then there exists an orthogonal matrix g such that gXg^{-1} is diagonal. Combine this fact with the formula for d (exp).)
3. Let G be the group of real, upper triangular $n \times n$ matrices whose diagonal elements are positive. Show that the exponential map on G is diffeomorphism.
4. Use the BCH formula to show that if x_t and y_t are C^1 curves on a Lie group G, then there exists a curve w_t, also of class C^1, such that $w_{t^2} = x_t y_t x_t^{-1} y_t^{-1}$ for all t.
5. Let G be a Lie group with Lie algebra $\mathfrak{g}$. Given $X, Y \in \mathfrak{g}$ show that $[X, Y] = 0$ if $e^{t_n X} e^{t_n Y} e^{-t_n X} e^{-t_n Y} = 1$ for every element $t_n \in \mathbb{R}$ of a convergent sequence such that $t_n \neq t_m$ for $n \neq m$. (Hint: recall that the exponential map is analytic.)
6. Take the following matrices in the Lie algebra $\mathfrak{g} = \mathfrak{gl}(2, \mathbb{R})$:

$$X = \begin{pmatrix} 0 & -\pi \\ \pi & 0 \end{pmatrix} \quad Y = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

   Show that the BCH series for $c(X, Y)$ does not converge. (Hint: show that $e^X e^Y$ is not an exponential.)
7. Let G be a Lie group with Lie algebra $\mathfrak{g}$ and $\mathfrak{h} \subset \mathfrak{g}$ a subalgebra. Given $g \in G$ define maps $\theta^r, \theta^l : \mathfrak{h} \to G$ by $\theta^r(X) = e^X g$ and $\theta^l(X) = ge^X$. Apply the formula for the differential of the exponential map to show that for every $X, Y \in \mathfrak{h}$, $(d\theta^r)_X(Y) \in \Delta^r_{\mathfrak{h}}(\theta(X))$ and $\left(d\theta^l\right)_X(Y) \in \Delta^l_{\mathfrak{h}}(\theta(X))$, where $\Delta^r_{\mathfrak{h}}(g) = d\left(R_g\right)_1 \mathfrak{h}$ and $\Delta^l_{\mathfrak{h}}(g) = d\left(L_g\right)_1 \mathfrak{h}$. Use this to write an alternative proof of the integrability of the distributions $\Delta^r_{\mathfrak{h}}$ and $\Delta^l_{\mathfrak{h}}$ (see Theorem 6.4).
8. Show that the exponential map on the group $\mathrm{SU}(2) \approx \mathrm{Sp}(1) \approx S^3$ is not an open map.

9. Let G be a Lie group with Lie algebra $\mathfrak{g}$. Take a sufficiently small element $X \in \mathfrak{g}$. Show that 1 is an eigenvalue of $\mathrm{Ad}\left(e^X\right)$ with the same multiplicity of the eigenvalue 0 of $\mathrm{ad}(X)$.
10. Let G be a connected Lie group. For $g \in G$ let P_g denote the characteristic polynomial of $\mathrm{Ad}(g) - \mathrm{id}$, and write it as

$$P_g(\lambda) = \lambda^n + p_{n-1}(g)\lambda^{n-1} + \cdots + p_1(g)\lambda + p_0(g),$$

where the coefficients $p_i(g)$ are analytic functions (polynomials composed with Ad). Prove that there exists an integer $r > 0$ such that p_i is identically zero if $i < r$. The integer r is called the **rank** of G.
11. Let G be a connected Lie group. An element $g \in G$ is called a **regular element** if the multiplicity of 1 as an eigenvalue of $\mathrm{Ad}(g)$ equals the rank of G. Use analyticity to prove that if G is connected, then the set of regular elements of G is dense.
12. Let G be a Lie group with Lie algebra $\mathfrak{g}$. Prove that the rank of G coincides with the rank of $\mathfrak{g}$. (The **rank** of $\mathfrak{g}$ is the smallest dimension of $\ker \mathrm{ad}(X)$ as $X \in \mathfrak{g}$ varies.)

Part III
Lie Algebras and Simply Connected Groups

Overview

This part addresses the construction of connected and simply connected Lie groups. Lie algebras are divided into two major classes, namely semi-simple Lie algebras and the solvable ones (the latter class includes nilpotent and abelian algebras). The decomposition theorem of Levi ensures that a finite dimensional Lie algebra can be written as the semi-direct product of a semi-simple Lie algebra by a solvable Lie algebra. In other words, any Lie algebra $\mathfrak{g}$ is the direct sum of a semi-simple subalgebra plus a solvable ideal. Connected and simply connected Lie groups will be constructed separately for each of these classes of Lie algebras.

The central concept in the construction of simply connected groups is the semi-direct product of Lie groups and Lie algebras. This product will be used both for groups associated with solvable Lie algebras, through successive decompositions, and to combine the constructions for the two major classes of Lie algebras, the solvable and semi-simple ones.

The semi-direct product of Lie groups is constructed and studied in Chapter 9. The treatment given here to semi-direct products involves the affine group of a Lie group G, whose elements are maps of G given by composing automorphisms and translations. The first step of the procedure is to define a Lie group structure on the automorphism group $\mathrm{Aut}G$.

In case, G is connected and simply connected said structure comes directly from the fact that $\mathrm{Aut}G$ is isomorphic to the automorphism group $\mathrm{Aut}\mathfrak{g}$ of the Lie algebra $\mathfrak{g}$. For general connected groups, it will be proved that $\mathrm{Aut}G$ is isomorphic to a closed subgroup of the automorphism group $\mathrm{Aut}\widetilde{G}$ of the universal covering $\widetilde{G}$ of G. The affine group $\mathrm{Aff}G$ then acquires a Lie group structure after one proves that the action of $\mathrm{Aut}G$ on G is differentiable. As application of the semi-direct product, the groups of the composition series (derived series and lower central series) of connected and simply connected Lie groups become closed, connected, and simply connected subgroups.

In Chapter 10, connected and simply connected solvable Lie groups are shown to be diffeomorphic to Euclidean space ($\mathbb{R}^n$ for some n). The proof relies on successive semi-direct products, which furnish these groups with global coordinates of the second kind. For nilpotent groups, the result can be improved, since the exponential map is a diffeomorphism (for connected and simply connected Lie groups), and hence defines global coordinates of the first kind. It follows from this that a connected and simply connected nilpotent Lie group has its Lie algebra as underlying manifold, and the latter is endowed with the product given by the Baker–Campbell–Hausdorff formula.

Semi-simple Lie algebras are in turn divided into two categories, compact and noncompact.

Chapter 11 handles the compact case. What happens is that if a Lie group G is compact, then its Lie algebra $\mathfrak{g}$ is compact, in the sense that there exists an inner product that is invariant under the adjoint representation (this is an algebraic property of $\mathfrak{g}$). The converse is almost true: a compact Lie algebra decomposes as direct sum of its center and a compact semi-simple algebra. A Lie group with compact Lie algebra may not be compact due to the presence of the center. However, if G is a Lie group whose Lie algebra $\mathfrak{g}$ is compact and semi-simple, then Weyl's theorem on the finite center (or finite fundamental group, depending on the perspective) guarantees G is compact. In particular, the connected and simply connected Lie group with (compact and semi-simple) Lie algebra $\mathfrak{g}$ is compact. Chapter 11 provides two proofs of Weyl's theorem. The first one has an analytic flavor and requires few Lie algebra prerequisites. The second proof freely uses the properties of compact (and complex) semi-simple Lie algebras. It has the advantage of exhibiting the fundamental group of the adjoint group (or, which is the same thing, the center of the simply connected group) as the quotient of two lattices (discrete subgroups of $\mathbb{R}^n$). In preparation for the second demonstration of Weyl's theorem, some structural properties of compact groups will be examined. The main one is the existence of maximal tori and the theorem whereby every element in the group is conjugated to an element in a fixed maximal torus. This fact implies (and is equivalent to) that the exponential map on a compact and connected Lie group is onto.

Compact semi-simple Lie algebras are in one-to-one correspondence with complex semi-simple Lie algebras by virtue of Weyl's so-called unitary trick. Complex semi-simple Lie algebras were classified by Cartan and Killing at the end of the nineteenth century. This classification is codified by the Dynkin diagrams found in Chapter 11.

Chapter 11 closes with a number of remarks showing how to tackle results about compact Lie groups by way of Riemannian geometry.

Noncompact semi-simple Lie algebras are investigated in Chapter 12. For these Lie algebras and their Lie groups, the Cartan and Iwasawa decompositions are constructed. These splitting results show that the manifold underlying a noncompact semi-simple Lie group is diffeomorphic to Euclidean space ($\mathbb{R}^n$) times a compact Lie group.

Combining the results on solvable and semi-simple groups then explains why the underlying manifold of a connected and simply connected Lie group is diffeomorphic to the product of a Euclidean space and a compact Lie group.

This part of the book requires several non-basic results from the theory of Lie algebras. These facts are explained without providing detailed proofs. Further material regarding these results may be found in the Portuguese textbook *Álgebras de Lie* [47], or in classics such as Helgason [20], Knapp [33], Varadarajan [53], and Warner [57].

Chapter 9
The Affine Group and Semi-Direct Products



The present chapter is devoted to the study of automorphism groups of connected Lie groups. If $\widetilde{G}$ is connected and simply connected the extension Theorem 7.13 shows that the automorphism group $\mathrm{Aut}\widetilde{G}$ is isomorphic to the group of automorphisms $\mathrm{Aut}\mathfrak{g}$ of the Lie algebra $\mathfrak{g}$, which is a Lie group. In general, the automorphism group of a connected group G is isomorphic to a closed subgroup of the automorphism group $\mathrm{Aut}\widetilde{G}$ of the universal covering $\widetilde{G}$. Therefore, $\mathrm{Aut}G$ is a Lie group as well.

The differentiable structure on $\mathrm{Aut}G$ allows to define semi-direct products in the context of Lie groups, by means of the affine group $\mathrm{Aff}G$, which is also a Lie group. The multiplication on a semi-direct product $H \times G$ arises from the multiplication on $\mathrm{Aff}G$.

9.1 Automorphisms of Lie Groups

Automorphism groups of Lie groups are studied using the automorphisms groups of the Lie algebras. Let $\mathfrak{g}$ be a real Lie algebra of finite dimension. As was seen in Chapter 6 the automorphism group $\mathrm{Aut}\mathfrak{g}$ of $\mathfrak{g}$ is a closed subgroup of the linear group $\mathrm{Gl}(\mathfrak{g})$ (see one of the examples at the end of Section 6.5).

Hence $\mathrm{Aut}\mathfrak{g}$ is a Lie group. Its Lie algebra consists of the derivations of $\mathfrak{g}$. Recall that a **derivation** on a Lie algebra $\mathfrak{g}$ is a linear map $D : \mathfrak{g} \to \mathfrak{g}$ satisfying

$$D[X, Y] = [DX, Y] + [X, DY]$$

for all $X, Y \in \mathfrak{g}$. The set of derivations of $\mathfrak{g}$ is denoted by $\mathrm{Der}\mathfrak{g}$.

It is not hard to verify that $\mathrm{Der}\mathfrak{g}$ is a Lie algebra (a subalgebra of the Lie algebra of linear maps on $\mathfrak{g}$). To see that $\mathrm{Der}\mathfrak{g}$ is the Lie algebra of $\mathrm{Aut}\mathfrak{g}$ it suffices to check that D is a derivation of $\mathfrak{g}$ if and only if e^{tD} is an automorphism of $\mathfrak{g}$ for every

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$t \in \mathbb{R}$. Now, taking $X, Y \in \mathfrak{g}$, by differentiating at $t = 0$ the relation $e^{tD}[X, Y] = [e^{tD}X, e^{tD}Y]$ shows that D is a derivation if e^{tD} is an automorphism for every t. Conversely, if D is a derivation, then the curves

$$\alpha(t) = e^{tD}[X, Y] \qquad \text{and} \qquad \beta(t) = [e^{tD}X, e^{tD}Y]$$

satisfy the linear differential equation $\gamma' = D\gamma$, $\gamma \in \mathfrak{g}$, and have the same initial condition $\alpha(0) = [X, Y] = \beta(0)$. Hence $\alpha = \beta$, showing that e^{tD} is an automorphism for every $t \in \mathbb{R}$.

Given $X \in \mathfrak{g}$, the adjoint $\mathrm{ad}\,(X)$ is a derivation of $\mathfrak{g}$, as follows from the Jacobi identity. Derivations of type $\mathrm{ad}\,(X)$ are called **inner derivations** of $\mathfrak{g}$. The set of inner derivations is the image $\mathrm{ad}\,(\mathfrak{g})$ of the adjoint representation of $\mathfrak{g}$. Hence $\mathrm{ad}\,(\mathfrak{g})$ is a subalgebra of $\mathrm{Der}\mathfrak{g}$. If D is a derivation and one takes $X \in \mathfrak{g}$, the definition of derivation says that

$$[D, \mathrm{ad}\,(X)] = \mathrm{ad}\,(DX)\,.$$

This property shows that $\mathrm{ad}\,(\mathfrak{g})$ is actually an ideal of $\mathrm{Der}\mathfrak{g}$. In general the inclusion $\mathrm{ad}\,(\mathfrak{g}) \subset \mathrm{Der}\mathfrak{g}$ is proper, as shown by abelian algebras, for which every linear map is a derivation while $\mathrm{ad}\,(\mathfrak{g}) = \{0\}$.

Since the Lie algebra of $\mathrm{Aut}\mathfrak{g}$ is $\mathrm{Der}\mathfrak{g}$ it is clear that the subalgebra $\mathrm{ad}\,(\mathfrak{g})$ integrates to a connected subgroup of $\mathrm{Aut}\mathfrak{g}$. This subgroup is denoted by $\mathrm{Int}\mathfrak{g}$ and its elements are called **inner automorphisms** of $\mathfrak{g}$. The reason for this name is that $\mathrm{Int}\mathfrak{g}$ is related to the group of inner automorphisms of a Lie group G with Lie algebra $\mathfrak{g}$ (see Proposition 9.6 below). The elements of $\mathrm{Int}\mathfrak{g}$ are products of exponentials of the Lie algebra $\mathrm{ad}\,(\mathfrak{g})$. That is to say, if $g \in \mathrm{Int}\mathfrak{g}$ then

$$g = e^{\mathrm{ad}(X_1)} \cdots e^{\mathrm{ad}(X_n)}$$

with $X_i \in \mathfrak{g}$.

Moving now to Lie groups, only continuous (and hence differentiable) automorphisms will be considered. This is subsumed in all the following discussion. The group of differentiable automorphisms of G is denoted by $\mathrm{Aut}G$.

If τ is an automorphism of G, by Proposition 5.16 of Chapter 5 its differential at the origin $d\tau_1$ is an automorphism of the Lie algebra $\mathfrak{g}$ of G. This defines a map $\delta : \mathrm{Aut}G \to \mathrm{Aut}\mathfrak{g}$ by $\delta(\tau) = (d\tau)_1$. The chain rule ensures that this map is a group homomorphism.

The homomorphism δ is one-to-one, since Proposition 7.8 implies $\phi = \psi$ if $\phi, \psi : G \to H$ are Lie group homomorphisms whose differentials $d\phi_1$ and $d\psi_1$ are equal and the domain G is connected.

(It should be noted that δ may not be injective if G is not connected. For example, if G is a discrete group, then δ is constant, equal to the identity, but in general there will exist automorphisms of G different from the identity.)

The injectivity of δ shows that the automorphism group of a connected Lie group is isomorphic to a subgroup of the group of automorphisms in its Lie algebra.

On the other hand, δ may not be onto, as will become clear in a short while, after finishing on automorphism groups.

However, Theorem 7.13 ensures that if G is connected and simply connected, then every automorphism of $\mathfrak{g}$ extends to an automorphism of G, which means that δ is surjective. Therefore for such groups automorphisms can be described as follows.

Proposition 9.1 *If G is a connected and simply connected Lie group, then* $\mathrm{Aut}G$ *is isomorphic to* $\mathrm{Aut}\mathfrak{g}$ *under the identification* $\delta : \mathrm{Aut}G \to \mathrm{Aut}\mathfrak{g}$, $\delta(\tau) = d\tau_1$.

The isomorphism $\delta : \mathrm{Aut}G \to \mathrm{Aut}\mathfrak{g}$ allows the differentiable structure of $\mathrm{Aut}\mathfrak{g}$ to be transferred to $\mathrm{Aut}G$, making the latter a Lie group. This purely formal construction of a Lie group structure on $\mathrm{Aut}G$ fits well with the differentiable structure of G, in the sense that the natural action of $\mathrm{Aut}G$ on G given by $\alpha : \mathrm{Aut}G \times G \to G$, $\alpha(\tau, x) = \tau(x)$, becomes differentiable.

Differentiability will be proved below. Before that, consider the partial map (evaluation map) $\alpha_x : \mathrm{Aut}G \to G$, $x \in G$, given by $\alpha_x(\tau) = \alpha(\tau, x) = \tau(x)$. By the definition of differentiable structure on $\mathrm{Aut}G$, α_x is differentiable if and only if the composition $\alpha_x \circ \delta^{-1}$ is differentiable in $\mathrm{Aut}\mathfrak{g}$.

Hence given $\theta \in \mathrm{Aut}\mathfrak{g}$, let $\tau = \delta^{-1}(\theta) \in \mathrm{Aut}G$ be its extension to G. Write $x = e^{X_1} \cdots e^{X_k}$ with $X_1, \ldots, X_k \in \mathfrak{g}$. Then

$$\begin{aligned}\alpha_x \circ \delta^{-1}(\theta) = \alpha_x(\tau) &= \tau\left(e^{X_1}\right) \cdots \tau\left(e^{X_k}\right) \\ &= e^{\theta(X_1)} \cdots e^{\theta(X_k)},\end{aligned}$$

that is, $\left(\alpha_x \circ \delta^{-1}\right)(\theta) = e^{\theta(X_1)} \cdots e^{\theta(X_k)}$. This map is a composite of differentiable maps (product in G, exponential and restrictions of linear maps $\theta \in \mathrm{Aut}\mathfrak{g} \mapsto \theta(X_i) \in \mathfrak{g}$). Consequently $\alpha_x \circ \delta^{-1} : \mathrm{Aut}\mathfrak{g} \to G$ is differentiable, making α_x a differentiable map for any $x \in G$.

It is now possible to prove the differentiability of the action of $\mathrm{Aut}G$ on G.

Proposition 9.2 *If G is connected and simply connected the action* $\alpha : \mathrm{Aut}(G) \times G \to G$, $\alpha(\tau, x) = \tau(x)$ *is differentiable.*

Proof Take a local coordinate system in G of the form $\exp : V \subset \mathfrak{g} \to U \subset G$, $0 \in V \subset \mathfrak{g}$ and $1 \in U \subset G$. Then for each $g \in G$ the map $X \in V \mapsto ge^X \in gU$ is a frame system at g. The map $f_g : \mathrm{Aut}\mathfrak{g} \times V \mapsto \mathrm{Aut}G \times gU$, $f_g(\theta, X) = \left(\tau, ge^X\right)$ (where $d\tau_1 = \theta$, that is, $\tau = \delta^{-1}(\theta)$) should be viewed as a chart in $\mathrm{Aut}(G) \times G$ (which is global in the first component).

The composition $\alpha \circ f_g : \mathrm{Aut}\mathfrak{g} \times V \to G$ is given by

$$\begin{aligned}\alpha \circ f_g(\theta, X) = \tau\left(ge^X\right) &= \tau(g)\,\tau\left(e^X\right) \\ &= \tau(g)\, e^{\theta(X)}.\end{aligned}$$

This map is the composite of differentiable maps (the product in G, the evaluation α_g, the exponential and the linear map $\theta \mapsto \theta(X)$). Therefore, $\alpha \circ f_g$ is differentiable, which proves that the action is differentiable. □

The differentiability (or rather, continuity) of the action in the simply connected case will be used below to prove that $\mathrm{Aut}G$ is a Lie group provided G is connected.

Now let $G = \widetilde{G}/\Gamma$ be a connected Lie group with $\widetilde{G}$ simply connected and Γ a discrete central subgroup. In this general situation the automorphism group can be quite different from $\mathrm{Aut}\mathfrak{g}$. At any rate, as δ is injective $\mathrm{Aut}G$ identifies with a subgroup of $\mathrm{Aut}\mathfrak{g}$, which is the same as $\mathrm{Aut}\widetilde{G}$. By the definition of δ this subgroup is the image of the 1-1 homomorphism $\mathrm{Aut}G \to \mathrm{Aut}\widetilde{G}$ that associates to $\tau \in \mathrm{Aut}G$ the only automorphism $\widetilde{\tau} \in \mathrm{Aut}\widetilde{G}$ satisfying $d\widetilde{\tau}_1 = d\tau_1$.

To describe the image of this homomorphism let $\pi : \widetilde{G} \to G = \widetilde{G}/\Gamma$ indicate the canonical projection, whose differential $d\pi_1$ is an isomorphism.

Lemma 9.3 $\pi \circ \widetilde{\tau} = \tau \circ \pi$.

Proof The maps $\pi \circ \widetilde{\tau}$ and $\tau \circ \pi$ are homomorphisms $\widetilde{G} \to G$ satisfying

$$d(\pi \circ \widetilde{\tau})_1 = d\pi_1 \circ d\widetilde{\tau}_1 = d\tau_1 = d(\tau \circ \pi)_1 .$$

Hence the claim is a consequence of Proposition 7.8. □

Now define

$$\mathrm{Aut}_\Gamma \widetilde{G} = \{\sigma \in \mathrm{Aut}\widetilde{G} : \sigma(\Gamma) = \Gamma\}, \tag{9.1}$$

which is evidently a subgroup of $\mathrm{Aut}\widetilde{G}$. The claim is that it is closed. In fact, consider a sequence $\sigma_n \to \sigma$ with $\sigma_n \in \mathrm{Aut}_\Gamma \widetilde{G}$ and take $\gamma \in \Gamma$. Then $\sigma_n(\gamma) \to \sigma(\gamma)$ because the action of $\mathrm{Aut}\widetilde{G}$ on $\widetilde{G}$ is continuous. Since $\sigma_n(\gamma) \in \Gamma$ it follows that $\sigma(\gamma) \in \Gamma$ and hence $\sigma \in \mathrm{Aut}_\Gamma \widetilde{G}$. (Compare this argument to Exercise 16 of Chapter 2.) All-in-all, $\mathrm{Aut}_\Gamma \widetilde{G}$ is a Lie subgroup of $\mathrm{Aut}\widetilde{G}$.

Proposition 9.4 *Let G be a connected Lie group. Then* $\mathrm{Aut}G$ *is isomorphic to* $\mathrm{Aut}_\Gamma \widetilde{G}$ *where* $G = \widetilde{G}/\Gamma$. *One possible isomorphism is* $\ell : \tau \in \mathrm{Aut}G \mapsto \widetilde{\tau} \in \mathrm{Aut}\widetilde{G}$ *where $\widetilde{\tau}$ is the only automorphism of $\widetilde{G}$ fulfilling* $d\widetilde{\tau}_1 = d\tau_1$.

Proof First of all, noteℓ is in fact a homomorphism of groups, as composite of the homomorphism $\delta : \mathrm{Aut}G \to \mathrm{Aut}\mathfrak{g}$, given by the differential, with the extension $\mathrm{Aut}\mathfrak{g} \to \mathrm{Aut}\widetilde{G}$.

By Proposition 7.8 the homomorphism δ is 1-1, and since $\mathrm{Aut}\mathfrak{g}$ and $\mathrm{Aut}\widetilde{G}$ are isomorphic it follows that ℓ is injective. This implies that $\mathrm{Aut}G$ is isomorphic to the image of ℓ inside $\mathrm{Aut}\widetilde{G}$. So one must prove that $\mathrm{Aut}_\Gamma \widetilde{G}$ equals the image of ℓ.

To do this take $\gamma \in \Gamma$. By Lemma 9.3, $\pi \circ \widetilde{\tau}(\gamma) = \tau \circ \pi(\gamma) = \tau(1) = 1$, which means $\widetilde{\tau}(\gamma) \in \Gamma$. Since $\gamma \in \Gamma$ is arbitrary it follows that $\widetilde{\tau}(\Gamma) \subset \Gamma$. Applying the same reasoning to $\widetilde{\tau}^{-1} = \widetilde{\tau^{-1}}$ gives the inclusion $\widetilde{\tau}^{-1}(\Gamma) \subset \Gamma$, so $\widetilde{\tau}(\Gamma) = \Gamma$, that is $\widetilde{\tau} \in \mathrm{Aut}_\Gamma \widetilde{G}$. Hence the image of ℓ is contained in $\mathrm{Aut}_\Gamma \widetilde{G}$.

Conversely, take $\sigma \in \mathrm{Aut}_\Gamma \widetilde{G}$. Then σ factorizes through a homomorphism $\sigma' : G \to G$ such that $\sigma' \circ \pi = \pi \circ \sigma$. This homomorphism is $\sigma'(x\Gamma) = \sigma(x)\Gamma$, which is well defined because $\sigma(\Gamma) = \Gamma$. As the differential $d\pi_1$ of $\pi : \widetilde{G} \to G = \widetilde{G}/\Gamma$ is an isomorphism and $\sigma' \circ \pi = \pi \circ \sigma$ it follows that $d\sigma_1 = d\sigma'_1$. So $\sigma = \ell(\sigma')$, showing that $\mathrm{Aut}_\Gamma \widetilde{G}$ is contained in the image of ℓ, and thus concluding the proof. □

Corollary 9.5 *If G is connected, then* Aut$G$ *is a Lie group and its action on G is differentiable.*

Proof The group Aut$G$ is isomorphic to $\mathrm{Aut}_\Gamma \widetilde{G}$, which is a closed Lie subgroup of Aut$\widetilde{G}$. The isomorphism endows Aut$G$ with a Lie group structure. The proof of the differentiability of the action is the same as the proof of Proposition 9.2. As in that case, take "charts"

$$f_g : \mathrm{Aut}_\Gamma \widetilde{G} \times V \to \mathrm{Aut}G \times gU,$$

where $\mathrm{Aut}_\Gamma \widetilde{G}$ is seen as a subgroup of Aut$\mathfrak{g}$. In these charts the action α reads

$$\begin{aligned} \alpha \circ f_g(\theta, X) &= \tau\left(ge^X\right) = \tau(g)\,\tau\left(e^X\right) \\ &= \tau(g)\, e^{\theta(X)}, \end{aligned}$$

where $\theta = d\tau_1$. The map $\alpha \circ f_g$ is differentiable, ensuring that the action α is differentiable as well. □

An automorphism $\sigma \in \mathrm{Aut}_\Gamma \widetilde{G}$ satisfies the condition in (9.1), namely $\sigma(\Gamma) = \Gamma$. This means both σ and σ^{-1} leave Γ invariant: $\sigma(\Gamma) \subset \Gamma$ and $\sigma^{-1}(\Gamma) \subset \Gamma$. Neither inclusions implies the other except in special cases, like, for instance, when Γ is finite. If so, if $\sigma(\Gamma) \subset \Gamma$, then the map $\sigma_{|\Gamma} : \Gamma \to \Gamma$ is 1-1 because σ is. An injective map on a finite set is also onto, so $\sigma_{|\Gamma}$ is onto, which means $\sigma^{-1}(\Gamma) \subset \Gamma$.

In general, though, there exist automorphisms σ of $\widetilde{G}$ for which $\sigma(\Gamma) \subset \Gamma$ but Γ is not σ^{-1}-invariant. When this occurs, σ factors through a homomorphism ψ of $\widetilde{G}/\Gamma$ by putting $\psi(x\Gamma) = \sigma(x)\Gamma$. This homomorphism is no longer injective. In fact, if $\sigma^{-1}(\Gamma)$ is not contained in Γ there exist elements $x \in \widetilde{G}$ and $\gamma \in \Gamma$, with $x = \sigma^{-1}(\gamma)$, such that $x \notin \Gamma$. Then $\sigma(x) = \gamma$ implies that the cosets $\sigma(x)\Gamma$ and Γ coincide, but $x\Gamma \neq 1\Gamma$, i.e. $\psi(x\Gamma) = \psi(1\Gamma)$.

Example Consider the torus $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$. The automorphism group of $\mathbb{R}^n$ is $\mathrm{Gl}(n, \mathbb{R})$. Therefore by Proposition 9.4 the automorphism group of $\mathbb{T}^n$ equals

$$\mathrm{Aut}_{\mathbb{Z}^n}\mathbb{R}^n = \{g \in \mathrm{Gl}(n, \mathbb{R}) : g\left(\mathbb{Z}^n\right) = \mathbb{Z}^n\}.$$

An invertible linear map $g \in \mathrm{Gl}(n, \mathbb{R})$ leaves $\mathbb{Z}^n$ invariant, that is, it satisfies $g(\mathbb{Z}^n) \subset \mathbb{Z}^n$ if and only if the matrix representing g in the canonical basis has integer entries. Hence the condition to have $g \in \mathrm{Aut}_{\mathbb{Z}^n}\mathbb{R}^n$ is that both g and g^{-1} be

matrices with integer entries. This forces (by Cramer's rule) $\det g = 1$. Conversely, if $g \in \mathrm{Gl}(n, \mathbb{R})$ has integer entries and $\det g = 1$, then its inverse has integer entries too. So, the automorphism group of $\mathbb{T}^n$ is isomorphic to the discrete group

$$\mathrm{Sl}(n, \mathbb{Z}) = \{g = \left(x_{ij}\right) \in \mathrm{Gl}(n, \mathbb{R}) : \det g = 1,\ x_{ij} \in \mathbb{Z}\}.$$

□

Denote by $\mathfrak{aut}G$ the Lie algebra of $\mathrm{Aut}G$ (for connected G). If $\widetilde{G}$ is simply connected, then $\mathfrak{aut}\widetilde{G}$ is isomorphic to the algebra of derivations $\mathrm{Der}\mathfrak{g}$. On the other hand, if $G = \widetilde{G}/\Gamma$, then $\mathfrak{aut}G$ is the subalgebra of elements of $\mathfrak{aut}\widetilde{G}$ whose exponentials belong in $\mathrm{Aut}_\Gamma \widetilde{G}$. This means that the elements of $\mathfrak{aut}G$ are the derivations $D \in \mathrm{Der}\mathfrak{g}$ such that $\phi_t \in \mathrm{Aut}_\Gamma \widetilde{G}$, $t \in \mathbb{R}$, where ϕ_t is the automorphism of $\widetilde{G}$ extending e^{tD}. In other words, $\mathfrak{aut}G$ is the set of derivations $D \in \mathrm{Der}\mathfrak{g}$ such that, for every $t \in \mathbb{R}$, the automorphism e^{tD} extends to G. In the above example, $\mathfrak{aut}\mathbb{T}^n = \{0\}$ because the automorphism group is discrete.

Another way of viewing $\mathfrak{aut}G$ is as a Lie subalgebra of vector fields on G. The differentiable action of $\mathrm{Aut}G$ on G induces an "infinitesimal action," i.e. a homomorphism mapping $Z \in \mathfrak{aut}G$ to the vector field $\overrightarrow{Z}$ on G defined by

$$\overrightarrow{Z}(x) = \frac{d}{dt}\left(\exp(tZ)\, x\right)_{|t=0} \qquad x \in G,$$

where $\exp(tZ)\, x$ denotes the action of $\exp tZ \in \mathrm{Aut}G$ on $x \in G$ (see Chapter 13 for details on the infinitesimal action). The vector field $\overrightarrow{Z}$ is an **infinitesimal automorphism** of G. The infinitesimal action $Z \to \overrightarrow{Z}$ is 1-1. In fact if $\overrightarrow{Z} = 0$, then $e^{tZ} = \mathrm{id} \in \mathrm{Aut}G$ for all $t \in \mathbb{R}$, which implies that $Z = 0$. Hence $\mathfrak{aut}G$ is isomorphic to the Lie algebra of infinitesimal automorphisms of G. (This space of vector fields is indeed a Lie algebra, see Chapter 13.)

In the case of a connected and simply connected group $\widetilde{G}$ each derivation $D \in \mathrm{Der}\mathfrak{g}$ defines an infinitesimal automorphism $\overrightarrow{D}$ on $\widetilde{G}$. It is easy to evaluate the vector field $\overrightarrow{D}$ on exponentials of the type $x = e^X \in \widetilde{G}$. In fact, denote by ϕ_t the one-parameter group of automorphisms of $\widetilde{G}$ such that $d(\phi_t)_1 = e^{tD}$. If $X \in \mathfrak{g}$, then $\phi_t\left(e^X\right) = \exp\left(e^{tD}X\right)$. Hence

$$\begin{aligned} \overrightarrow{D}\left(e^X\right) &= (d\exp)_X (DX) \qquad (9.2) \\ &= \sum_{k\geq 0} \frac{1}{(k+1)!} dL_{e^X}\left((\mathrm{ad}(X))^k (DX)\right) \end{aligned}$$

by the formula expressing the differential of the exponential (see Chapter 8).

Turning to $G = \widetilde{G}/\Gamma$, the Lie algebra $\mathfrak{aut}G$ identifies with the algebra of infinitesimal automorphisms of $\widetilde{G}$ that annihilate every $\gamma \in \Gamma$:

$$\mathfrak{aut}G = \{Z \in \mathfrak{aut}\widetilde{G} : \forall \gamma \in \Gamma,\ Z(\gamma) = 0\}.$$

The vector fields in the right hand side are precisely those that are projectable under $\pi : \widetilde{G} \to G$.

Example Let $\mathfrak{g}$ and $\widetilde{G}$ be the Lie algebra and Lie group of matrices of the type

$$X = \begin{pmatrix} 0 & x & z \\ 0 & 0 & y \\ 0 & 0 & 0 \end{pmatrix} \qquad \text{and} \qquad g = \begin{pmatrix} 1 & u & w \\ 0 & 1 & v \\ 0 & 0 & 1 \end{pmatrix},$$

respectively. The fact that

$$e^X = \begin{pmatrix} 1 & x & z + xy/2 \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix}$$

shows that the exponential map is a diffeomorphism. Hence if $D \in \mathrm{Der}\mathfrak{g}$, then by (9.2) it follows that

$$\overrightarrow{D}(g) = g\left(DX + \frac{1}{2}[X, DX]\right),$$

where $X = \log g$. □

This section is finished with some remarks on the group $\mathrm{Int}G$ of **inner automorphisms** of the group G, whose elements are the conjugation maps $C_x : G \to G$, $C_x(z) = xzx^{-1}$, with $x \in G$. If τ is an automorphism of G, then $\tau \circ C_x \circ \tau^{-1} = C_{\tau(x)}$, which goes to show that $\mathrm{Int}G$ is a normal subgroup in $\mathrm{Aut}G$.

The group structure of $\mathrm{Int}G$ is clarified by observing that the map $x \in G \mapsto C_x \in \mathrm{Int}G$ is a group homomorphism. The kernel is the center $Z(G)$ and its image is $\mathrm{Int}G$ by definition. Therefore $\mathrm{Int}(G)$ is isomorphic to $G/Z(G)$. This group is also isomorphic to the image $\mathrm{Ad}(G)$ of the adjoint representation of G, which in turn is a subgroup of $\mathrm{Aut}\mathfrak{g}$.

When G is connected its elements are products of exponentials. The formula

$$\mathrm{Ad}\left(e^X\right) = e^{\mathrm{ad}(X)}$$

then shows that $\mathrm{Ad}(G)$ consists of products of exponentials of type $e^{\mathrm{ad}(X)}$. In other words, $\mathrm{Ad}(G)$ is a subgroup in the group $\mathrm{Int}\mathfrak{g}$ of inner automorphisms of $\mathfrak{g}$. At the same time the above formula shows that any inner automorphism of $\mathfrak{g}$ extends to an inner automorphism of G (even if G is not simply connected). Therefore the following characterization of $\mathrm{Int}\mathfrak{g}$ follows.

Proposition 9.6 *If G is a connected Lie group,* $\mathrm{Int}G$ *is isomorphic to* $\mathrm{Int}\mathfrak{g}$. *It follows that* $\mathrm{Int}G_1$ *is isomorphic to* $\mathrm{Int}G_2$ *if the connected Lie groups G_1 and G_2 are locally isomorphic, that is, they have isomorphic Lie algebras.*

9.2 The Affine Group

On a group G a **left-affine map** is the composite of a left translation with a homomorphism of G. Any such map $\alpha : G \to G$ may be written as

$$\alpha(x) = xg(y) = L_x \circ g(y),$$

where $g : G \to G$ is a homomorphism and $x \in G$ is fixed. In the same way a **right-affine map** is a map $\beta(y) = g(y)x$. Evidently, these concepts generalize the notion of an affine map on a vector space, seen as an abelian group, where the homomorphisms are linear maps and left translations are the same as right translations.

Affine maps are denoted by their components (g, y), with an additional subscript l or r whenever necessary to distinguish between left and right.

The set of affine maps contains the identity map (given by (id, 1)) and is closed under composition. In fact, a simple computation shows that

1. $(g, y)_l \circ (h, z)_l = (g \circ h, yg(z))_l$,
2. $(g, y)_r \circ (h, z)_r = (g \circ h, g(z)y)_r$.

These expressions show that an affine map (g, x) is bijective if and only if the homomorphism g is an automorphism. In this case, the inverse is also an affine map, given by $(g, x)^{-1} = \left(g^{-1}, g^{-1}\left(x^{-1}\right)\right)$ (both in the left and right case).

In view of these comments, the **left-affine group** $\mathrm{Aff}_l G$ of a given group G is defined as the Cartesian product $\mathrm{Aut}G \times G$ endowed with the product of condition (1) above, hence defined by composing affine maps on the left. This product, in fact, satisfies the axioms of a group since map composition is associative. The **right-affine group** $\mathrm{Aff}_r G$ is defined similarly, via the product in (2) above. When distinguishing structures on the left or on the right is not necessary, both groups will be denoted indistinctly by $\mathrm{Aff}G$.

Either affine group naturally contains the group $\mathrm{Aut}G$ (which is isomorphic to the subgroup $\mathrm{Aut}G \times \{1\}$) and the group G (which is isomorphic to $\{1\} \times G$). Furthermore, the conjugation

$$(g, 1)(1, x)(g, 1)^{-1} = (g, 1)(1, x)\left(g^{-1}, 1\right) = (1, g(x))$$

(valid for both to the left- and right-affine groups) shows that the subgroup $\{1\} \times G$ is normal in $\mathrm{Aff}G$.

Suppose now that G is a connected Lie group. Then $\mathrm{Aut}G$ is a Lie group and its action on G is differentiable. The product in the affine group (left or right) is defined by an expression involving the products in G and $\mathrm{Aut}G$ together with the action of $\mathrm{Aut}G$ on G. Hence the product in the affine group is differentiable, turning the (left- or right-)affine group into a Lie group.

Proposition 9.7 *If G is a connected Lie group, then* Aff$G$ *is Lie group.*

The next objective will be to determine the Lie algebra $\mathfrak{aff}G$ of Aff$G$. The vector space underlying $\mathfrak{aff}G$ is certainly the product $\mathfrak{aut}G \times \mathfrak{g}$ of the Lie algebras of Aut$G$ and G.

The bracket, however, is not the direct product. To compute it note first that $\mathfrak{aut}G \times \{0\}$ and $\{0\} \times \mathfrak{g}$ are subalgebras of $\mathfrak{aff}G$, since these are the Lie algebras of the Lie subgroups Aut$G \times \{1\}$ and $\{1\} \times G$, respectively. This means that on $\mathfrak{aff}G = \mathfrak{aut}G \times \mathfrak{g}$ one can compute the brackets $[(X, 0), (Y, 0)] = ([X, Y], 0)$ and $[(0, X), (0, Y)] = (0, [X, Y])$.

There remains then to determine the bracket $[(X, 0), (0, Y)]$ with $X \in \mathfrak{aut}G$ and $Y \in \mathfrak{g}$. This is done by differentiating conjugation maps. The exponential maps on the subgroups Aut$G \times \{1\}$ and $\{1\} \times G$ are $\exp t\,(X, 0) = \left(e^{tX}, 1\right)$ and $\exp s\,(0, Y) = \left(1, e^{sY}\right)$. Hence,

$$C_{\exp t(X,0)}\left(e^{s(0,Y)}\right) = \left(e^{tX}, 1\right)\left(1, e^{sY}\right)\left(e^{-tX}, 1\right) = \left(1, e^{tX}\left(e^{sY}\right)\right),$$

where e^{tX} is seen as an automorphism of G (this conjugation holds for both the left- and the right-affine group). Therefore

$$\mathrm{Ad}\left(e^{t(X,0)}\right)(0, Y) = \frac{d}{ds} C_{e^{t(X,0)}}\left(e^{s(0,Y)}\right)_{|s=0} = \left(0, d\left(e^{tX}\right)_1 (Y)\right).$$

Now, $d\left(e^{tX}\right)_1$ is a 1-parameter group in Aut$\mathfrak{g}$. There exists a derivation $D \in \mathrm{Der}\mathfrak{g}$ such that $d\left(e^{tX}\right)_1 = e^{tD}$. Substituting this exponential in the expression above, one arrives at the desired bracket

$$[(X, 0), (0, Y)] = \mathrm{ad}\,((X, 0))\,(0, Y) = \frac{d}{dt}\mathrm{Ad}\left(e^{t(X,0)}\right)(0, Y)_{|t=0} = (0, DY).$$

The derivation D in the formula is an element of $\mathfrak{aut}G$ seen as a subalgebra of Der$\mathfrak{g}$, since it comes from a 1-parameter group of automorphisms of G.

Summarizing, the Lie algebra of Aff$G$ (both left and right) is defined as follows.

Proposition 9.8 *The Lie bracket on* $\mathfrak{aff}G = \mathfrak{aut}G \times \mathfrak{g}$ *is given by*

$$[(D_1, X_1), (D_2, X_2)] = ([D_1, D_2], D_1X_2 - D_2X_1 + [X_1, X_2]),$$

where $D_i \in \mathfrak{aut}G \subset \mathrm{Der}\mathfrak{g}$ *and* $X_i \in \mathfrak{g}$.

One final remark is that if $H \subset \mathrm{Aut}G$ is a Lie subgroup, then $H \times G$ is a Lie subgroup of Aff$G$, as is easily checked.

9.3 Semi-Direct Products

The semi-direct product of two groups is a construction that generalizes the direct product and is widely used in the description of simply connected Lie groups. The ingredients of this construction are two groups G and H and a differentiable homomorphism $\tau : G \to \mathrm{Aut}H$. A pair $(g, h) \in G \times H$ defines two affine maps on H, a left map and a right map, explicitly given by $x \mapsto h\tau(g)(x)$ and $x \mapsto \tau(g)(x)h$, $x \in H$. By composing these affine maps one obtains two group structures on $G \times H$, referred to as the **semi-direct product** (on the left or on the right) of G and H by the homomorphism τ. The products are concretely given by

1. $(g_1, h_1)(g_2, h_2) = (g_1 g_2, h_1 \tau(g_1)(h_2))$ (on the left) and
2. $(g_1, h_1)(g_2, h_2) = (g_1 g_2, \tau(g_1)(h_2) h_1)$ (on the right).

In both cases the neutral element is $(1, 1)$ and the inverse is

$$(g, h)^{-1} = \left(g^{-1}, \tau\left(g^{-1}\right)\left(h^{-1}\right)\right).$$

The semi-direct product is denoted by $G \times_\tau H$ (or $G \times_\tau^l H$ and $G \times_\tau^r H$ when distinguishing left product from right product becomes necessary). Since τ is a differentiable homomorphism the products above are differentiable, and therefore the semi-direct product of Lie groups is a Lie group.

A particular case is the affine group $\mathrm{Aff}H$, which is the semi-direct product $\mathrm{Aut}H \times_{\mathrm{id}} H$. Another special case comes from choosing τ constant, equal to id. The semi-direct product reduces then to the direct product $G \times H$ of G and H (as a left product. The right semi-direct product becomes the direct product $G \times \widehat{H}$, where $\widehat{H}$ is the group defined by the multiplication $(h_1, h_2) \mapsto h_2 h_1$).

Any semi-direct product $G \times_\tau H$ contains copies of its components. The subset $G \times \{1\} \subset G \times_\tau H$ is a subgroup isomorphic to G while $\{1\} \times H$ is a subgroup isomorphic to H that is additionally normal in $G \times_\tau H$. But in general $G \times \{1\}$ is not normal. In fact, a simple computation involving conjugations shows that $G \times \{1\}$ is normal in the semi-direct product if and only if $\tau = \mathrm{id}$, i.e. when the product is direct. The subgroups $G \times \{1\}$ and $\{1\} \times H$ are simply denoted by G and H, respectively. They are Lie subgroups since closed.

The Lie algebra of a semi-direct product $G \times_\tau H$ is given by the semi-direct product of the Lie algebras, as defined below.

Definition 9.9 Let $\mathfrak{g}$ and $\mathfrak{h}$ be Lie algebras and $\rho : \mathfrak{g} \to \mathrm{Der}\mathfrak{h}$ a homomorphism of Lie algebras. The semi-direct product $\mathfrak{g} \times_\rho \mathfrak{h}$ is the Lie algebra on $\mathfrak{g} \times \mathfrak{h}$ endowed with bracket

$$[(X_1, Y_1), (X_2, Y_2)] = ([X_1, X_2], \rho(X_1) Y_2 - \rho(X_2) Y_1 + [Y_1, Y_2]).$$

A Lie algebra $\mathfrak{l}$ is isomorphic to a semi-direct product $\mathfrak{g} \times_\rho \mathfrak{h}$ if and only if $\mathfrak{l} = \mathfrak{g}_1 \oplus \mathfrak{h}_1$ (direct sum of vector spaces) and $\mathfrak{g}_1$ is a subalgebra isomorphic to $\mathfrak{g}$ and $\mathfrak{h}_1$ is an ideal isomorphic to $\mathfrak{h}$.

Given a semi-direct product $G \times_\tau H$ let $\mathfrak{g}$ and $\mathfrak{h}$ be the Lie algebras of G and H, respectively. The homomorphism $\tau : G \to \mathrm{Aut}H$ is differentiable and its differential $\rho = d\tau_1$ at the identity is a homomorphism $\rho : \mathfrak{g} \to \mathfrak{aut}H$ of the corresponding Lie algebras. Yet $\mathfrak{aut}H$ is a subalgebra of the algebra of derivations $\mathrm{Der}\mathfrak{h}$. It makes then sense to write the semi-direct product as $\mathfrak{g} \times_\rho \mathfrak{h}$ with $\rho = d\tau_1$.

Proposition 9.10 *The Lie algebra of $G \times_\tau H$ is $\mathfrak{g} \times_\rho \mathfrak{h}$ where $\rho = d\tau_1$.*

Proof The argument is similar to the computation made above for the bracket on the Lie algebra of the affine group $\mathrm{Aff}G$. Both $\mathfrak{g}$ and $\mathfrak{h}$ are Lie subalgebras, and a bracket of type $[(X, 0), (0, Y)]$ may be defined by differentiating the conjugation map

$$C_{\exp t(X,0)}\left(e^{s(0,Y)}\right) = \left(1, \tau\left(e^{tX}\right)\left(e^{sY}\right)\right).$$

The derivative generates the homomorphism ρ and the bracket formula appearing in the definition of semi-direct product of Lie algebras. □

The construction of the semi-direct product is very useful to obtain the simply connected Lie group associated with a given Lie algebra. This is because the simply connected group of a product $\mathfrak{g} \times_\rho \mathfrak{h}$ is the semi-direct product of the corresponding simply connected groups. In fact, let $\widetilde{G}$ and $\widetilde{H}$ denote the connected, simply connected Lie groups with Lie algebra $\mathfrak{g}$ and $\mathfrak{h}$, respectively. The group $\mathrm{Aut}\widetilde{H}$ is isomorphic to $\mathrm{Aut}\mathfrak{h}$, whose Lie algebra is $\mathrm{Der}\mathfrak{h}$. Since $\widetilde{G}$ is simply connected, the homomorphism $\rho : \mathfrak{g} \to \mathrm{Der}\mathfrak{h}$ extends to a homomorphism $\tau : \widetilde{G} \to \mathrm{Aut}\widetilde{H}$. This allows to build $\widetilde{G} \times_\tau \widetilde{H}$, whose Lie algebra is $\mathfrak{g} \times_\rho \mathfrak{h}$. Clearly $\widetilde{G} \times_\tau \widetilde{H}$ is simply connected, as Cartesian product of simply connected spaces. Therefore the only connected and simply connected Lie group with Lie algebra $\mathfrak{g} \times_\rho \mathfrak{h}$ is the semi-direct product $\widetilde{G} \times_\tau \widetilde{H}$.

The semi-direct product of Lie groups has great theoretical relevance to unfold the full theory, due to a result on Lie algebras known as **Levi's decomposition theorem**.[1] This result asserts that any Lie algebra of finite dimension splits as a semi-direct product of a semi-simple subalgebra by a solvable ideal.

Altogether, then, the problem of classifying connected and simply connected Lie groups is reduced to the two main classes of Lie algebras, namely the semi-simple and the solvable ones.

[1] See *Álgebras de Lie* [47, Ch. 5] and Varadarajan [53, Section 3.14].

9.4 Derived Groups and Lower Central Series

This section focuses on the properties of the derived group and the lower central series of a Lie group. The semi-direct product of Lie groups defined in the previous section will be used repeatedly.

The construction of the semi-direct product also provides information on normal subgroups of simply connected Lie groups, as is proved next.

Proposition 9.11 *Let G be a connected and simply connected Lie group and $H \subset G$ a connected normal Lie subgroup. Then, H is closed and G/H is simply connected.*

Proof The Lie algebra $\mathfrak{h}$ of H is an ideal in the Lie algebra $\mathfrak{g}$ of G. This allows to write the quotient Lie algebra $\mathfrak{q} = \mathfrak{g}/\mathfrak{h}$. Let Q be the connected and simply connected Lie group with Lie algebra $\mathfrak{q}$. The canonical homomorphism $\theta : \mathfrak{g} \to \mathfrak{g}/\mathfrak{h}$ extends to a homomorphism $\phi : G \to Q$ such that $\theta = d\phi_1$. The Lie algebra of the kernel $\ker\phi$ is equal to $\ker\theta = \mathfrak{h}$. Since H and the identity component of $\ker\phi$ have the same Lie algebra, they coincide as Lie groups. But $\ker\phi$ is a closed subgroup, and then so is $H = (\ker\phi)_0$.

Actually, $\ker\phi$ is connected because the natural fibration $G/(\ker\phi)_0 \to G/\ker\phi = Q$ is a covering map, so $(\ker\phi)_0 = \ker\phi$, because Q is simply connected. Hence $H = \ker\phi$ and $G/H = Q$, showing that G/H is simply connected and concluding the proof. □

It is possible to show that the normal subgroup H of the previous proposition is also simply connected.[2]The following proposition proves this in a particular case that applies to the derived groups of G.

Proposition 9.12 *Let G be a connected and simply connected Lie group and $H \subset G$ a connected normal Lie subgroup. Suppose that* $\dim G = \dim H + 1$. *Then H is simply connected.*

Proof Let $\mathfrak{h} \subset \mathfrak{g}$ be the Lie subalgebra of H and take $X \in \mathfrak{g} \setminus \mathfrak{h}$ so that $\mathfrak{g} = \mathbb{R}X \oplus \mathfrak{h}$. As $\mathfrak{h}$ is an ideal, this equality says that $\mathfrak{g}$ is isomorphic to the semi-direct product $\mathbb{R} \times_\theta \mathfrak{h}$ where $\theta : \mathbb{R} \to \mathrm{Der}\mathfrak{h}$ is given by $\theta(t) = \mathrm{ad}(tX)_{|\mathfrak{h}}$. Therefore G is isomorphic to the semi-direct product $\mathbb{R} \times \widetilde{H}$, where $\widetilde{H}$ is the simply connected Lie group with Lie algebra $\mathfrak{h}$. The differential of the isomorphism $G \approx \mathbb{R} \times \widetilde{H}$ maps the Lie algebra of H to the Lie algebra of $\{0\} \times \widetilde{H}$. This proves that H and $\widetilde{H}$ are isomorphic, because they are connected. Hence H is simply connected. □

These results on normal subgroups will be applied to the derived group of a Lie group.

Generally speaking, if G is a group its **derived group** G' is defined as the subgroup generated by the commutators $[x, y] = xyx^{-1}y^{-1}$, $x, y \in G$. Further

[2]See Varadarajan [53, Theorem 3.18.2] and Exercise 20 at the end of this chapter.

derived groups $G^{(k)}$ are defined inductively by $G^{(k)} = \left(G^{(k-1)}\right)'$, where $G^{(0)} = G$. All these subgroups are normal (because $u\,[g,h]\,u^{-1} = \left[ugu^{-1}, uhu^{-1}\right]$), and $G^{(k)}/G^{(k+1)}$ is an abelian group for any $k \geq 0$.

In a similar way one may define inductively the **derived Lie algebras** of a Lie algebra $\mathfrak{g}$ by starting with $\mathfrak{g}^{(0)} = \mathfrak{g}$, letting $\mathfrak{g}'$ be the subspace spanned by brackets $[X, Y]$, $X, Y \in \mathfrak{g}$, and then putting $\mathfrak{g}^{(k+1)} = \left(\mathfrak{g}^{(k)}\right)'$. These derived algebras are ideals in $\mathfrak{g}$ and for each $k \geq 0$ the quotient $\mathfrak{g}^{(k)}/\mathfrak{g}^{(k+1)}$ is an abelian Lie algebra.[3]

Proposition 9.13 *If G is a connected Lie group, then G' is a connected normal Lie subgroup.*

Proof For starters, note that G' is a path-connected subgroup (differentiably so). In fact, given $g, h \in G$ there exist differentiable paths g_t and h_t, $t \in [0,1]$, with $g_0 = h_0 = 1$, $g_1 = g$ and $h_1 = h$. Hence $g_t h_t g_t^{-1} h_t^{-1}$ is a differentiable path joining the identity 1 to the commutator $[g,h]$. Moreover, if $g = [g_1, h_1] \cdots [g_k, h_k]$ is an arbitrary element of G', there exists a path from 1 to g obtained by multiplying the paths between 1 and the commutators in the product. This shows that G' is path-connected. Finally, G' is a Lie subgroup by Theorem 6.19. □

The next step is to verify that the Lie algebra $\mathcal{L}\left(G'\right)$ of G' is the derived algebra $\mathfrak{g}'$. To this end consider the commutator

$$\alpha(t) = e^{\sqrt{t}X} e^{\sqrt{t}Y} e^{-\sqrt{t}X} e^{-\sqrt{t}Y} \qquad t \geq 0.$$

Its derivative (on the right) is $\alpha'(0+) = -[X,Y]$ (see Proposition 6.12). This means that any bracket between elements of $\mathfrak{g}$ is the derivative of a curve in G'. Hence $\mathfrak{g}' \subset \mathcal{L}\left(G'\right)$.

Proposition 9.14 *Let G be a connected Lie group. Then the Lie algebra of G' equals $\mathcal{L}\left(G'\right) = \mathfrak{g}'$, so $G' = \langle \exp \mathfrak{g}' \rangle$.*

Proof There only remains to prove the inclusion $\mathcal{L}\left(G'\right) \subset \mathfrak{g}'$. For this purpose suppose first that G is simply connected. In this case, Proposition 9.11 ensures that $\langle \exp \mathfrak{g}' \rangle$ is a closed subgroup. It follows that $G/\langle \exp \mathfrak{g}' \rangle$ is an abelian Lie group, since its Lie algebra $\mathfrak{g}/\mathfrak{g}'$ is abelian. This implies that the commutators of G are contained in $\langle \exp \mathfrak{g}' \rangle$, because if $p : G \to G/\langle \exp \mathfrak{g}' \rangle$ indicates the canonical homomorphism, then $p\,[g,h] = [p(g), p(h)] = 1$. Hence $G' \subset \langle \exp \mathfrak{g}' \rangle$, so $\mathcal{L}\left(G'\right) \subset \mathfrak{g}'$.

Now for a general connected group $G = \widetilde{G}/\Gamma$ with $\widetilde{G}$ simply connected and Γ a discrete central subgroup, let $\pi : \widetilde{G} \to G$ be the canonical homomorphism. Then $\pi\left(\widetilde{G}'\right) = G'$ as π is onto. The argument in the first paragraph tells that $\mathcal{L}\left(\widetilde{G}'\right) = \mathfrak{g}'$, and since $d\pi_1$ is an isomorphism it follows that $\mathcal{L}\left(G'\right) = \mathfrak{g}'$, concluding the proof. □

[3] See *Álgebras de Lie* [47, Chapters 1, 2] and Varadarajan [53, Section 3.7].

For simply connected groups G, Propositions 9.11 and 9.12 provide additional features of the derived group G'.

Proposition 9.15 *Let G be a connected and simply connected Lie group. Then the derived group G' is closed and simply connected, and the quotient G/G' is simply connected.*

Proof Proposition 9.11 implies G' is closed and G/G' is simply connected. That G' is simply connected comes from applying Proposition 9.12 repeatedly. The point is that any subspace V with $\mathfrak{g}' \subset V \subset \mathfrak{g}$ is an ideal in $\mathfrak{g}$ because brackets live in $\mathfrak{g}'$. Hence there exist ideals $\mathfrak{g}_1, \ldots, \mathfrak{g}_k$, $k = \dim \mathfrak{g} - \dim \mathfrak{g}'$ such that

$$\mathfrak{g} = \mathfrak{g}_1 \supset \cdots \supset \mathfrak{g}_k \supset \mathfrak{g}'$$

and $\dim \mathfrak{g}_i = \dim \mathfrak{g}_{i+1} + 1$. The subgroups $G_i = \langle \exp \mathfrak{g}_i \rangle$ are normal. By Proposition 9.12 G_1 is simply connected. An induction argument shows that the subgroups G_i, as well as G', are simply connected. □

Higher derived groups $G^{(k)}$ are introduced inductively, so any $G^{(k)}$ is the derived group of the previously defined $G^{(k-1)}$. Therefore the aforementioned properties of G' pass, by induction, to $G^{(k)}$, $k \geq 2$.

Corollary 9.16 *Let G be a connected Lie group. Then the derived groups $G^{(k)}$ are the connected Lie subgroups $G^{(k)} = \langle \exp \mathfrak{g}^{(k)} \rangle$. If, furthermore, G is simply connected, then $G^{(k)}$, $k \geq 0$, is closed and simply connected. The quotients $G^{(k)}/G^{(i)}$, $k \leq i$, are simply connected as well.*

Proof The only thing that is still missing is proving $G^{(k)}/G^{(i)}$ is simply connected if $k < i + 1$. But this follows directly from Proposition 9.11. □

Corollary 9.17 *Let G be a connected Lie group. Then its **derived series***

$$G = G^{(0)} \supset \cdots \supset G^{(k)} \supset \cdots$$

stabilizes, i.e. there exists an index k_0 such that $G^{(k)} = G^{(k_0)}$ for any $k \geq k_0$.

Proof The derived series

$$\mathfrak{g} = \mathfrak{g}^{(0)} \supset \cdots \supset \mathfrak{g}^{(k)} \supset \cdots$$

stabilizes since $\dim \mathfrak{g}^{(k+1)} < \dim \mathfrak{g}^{(k)}$ if $\mathfrak{g}^{(k+1)} \neq \mathfrak{g}^{(k)}$. The same must then happen for the derived groups $G^{(k)} = \langle \exp \mathfrak{g}^{(k)} \rangle$. □

The corollary is not true in general. An example can be found in Exercise 22 at the end of the chapter.

The **lower central series**

$$G = G^1 \supset G^2 \supset \cdots \supset G^k \supset \cdots .$$

can be treated in a similar manner. The above is defined inductively: $G^1 = G$, $G^2 = G'$, and $G^{k+1} = \left[G, G^k\right]$ is the subgroup generated by commutators $[x, y] = xyx^{-1}y^{-1}$, $x \in G$, $y \in G^k$. The groups G^k are normal, and for each $k \geq 0$ the quotient G^k/G^{k+1} is an abelian group.

In the same way the **lower central series**

$$\mathfrak{g} = \mathfrak{g}^1 \supset \mathfrak{g}^2 \supset \cdots \supset \mathfrak{g}^k \supset \cdots$$

of a Lie algebra $\mathfrak{g}$ is defined by $\mathfrak{g}^1 = \mathfrak{g}$, $\mathfrak{g}^2 = \mathfrak{g}'$, and $\mathfrak{g}^{k+1} = \left[\mathfrak{g}, \mathfrak{g}^k\right]$ is the subspace spanned by brackets $[X, Y]$, $X \in \mathfrak{g}$ and $Y \in \mathfrak{g}^k$. Each $\mathfrak{g}^k$ is an ideal in $\mathfrak{g}$, and the quotients $\mathfrak{g}^k/\mathfrak{g}^{k+1}$ are abelian Lie algebras.[4]

The groups G^k are path-connected since they are spanned by commutators $[g, h]$, which are joined to the identity by differentiable paths. Hence they are Lie groups. When G is simply connected it follows from Proposition 9.11 that each G^k is closed and the quotient G/G^k is simply connected.

The goal now is to show that the Lie algebra $\mathcal{L}\left(G^k\right)$ of G^k equals $\mathfrak{g}^k$. This was already proved for $k = 2$ because $G^2 = G'$ and $\mathfrak{g}^2 = \mathfrak{g}'$. Induction on k shows $\mathfrak{g}^k \subset \mathcal{L}\left(G^k\right)$ in the same way used for the derived algebra. Take $X \in \mathfrak{g}$, $Y \in \mathfrak{g}^{k-1}$ and the commutator

$$\alpha(t) = e^{\sqrt{t}X} e^{\sqrt{t}Y} e^{-\sqrt{t}X} e^{-\sqrt{t}Y} \qquad t \geq 0.$$

Its right derivative $\alpha'(0+) = -[X, Y]$, so $[X, Y] \in \mathcal{L}\left(G^k\right)$ if $X \in \mathfrak{g}$ and $Y \in \mathfrak{g}^{k-1}$. Therefore $\mathfrak{g}^k \subset \mathcal{L}\left(G^k\right)$.

To prove the opposite inclusion, it is enough to consider a simply connected group. This is because if $\pi : \widetilde{G} \to G = \widetilde{G}/\Gamma$ is the canonical projection, then $\pi\left(\widetilde{G}^k\right) = G^k$, hence the Lie algebras of $\widetilde{G}^k$ and G^k coincide.

Proposition 9.18 *The Lie algebra $\mathcal{L}\left(G^k\right)$ of G^k is equal to $\mathfrak{g}^k$, that is, $G^k = \langle \exp \mathfrak{g}^k \rangle$.*

Proof There remains to prove inclusion $G^k \subset \langle \exp \mathfrak{g}^k \rangle$, by induction on k. Take G simply connected (see the comment above). Proposition 9.11 implies $\langle \exp \mathfrak{g}^k \rangle$ is closed and $G/\langle \exp \mathfrak{g}^k \rangle$ is a Lie group with Lie algebra $\mathfrak{g}/\mathfrak{g}^k$. Denote by $p : G \to G/\langle \exp \mathfrak{g}^k \rangle$ the canonical projection.

Assuming the result true for $k-1$, one must prove that if $g \in G$ and $h \in G^{k-1}$ then $ghg^{-1}h^{-1} \in \langle \exp \mathfrak{g}^k \rangle$, that is, $p\left(ghg^{-1}h^{-1}\right) = 1$.

Now, $p\left(\langle \exp \mathfrak{g}^{k-1} \rangle\right)$ is contained in the center of $G/\langle \exp \mathfrak{g}^k \rangle$ because $\mathfrak{g}^{k-1}/\mathfrak{g}^k \subset \mathfrak{z}\left(\mathfrak{g}/\mathfrak{g}^k\right)$. By induction hypothesis $G^{k-1} = \langle \exp \mathfrak{g}^{k-1} \rangle$, so $p\left(G^{k-1}\right)$ is contained in the center of $G/\langle \exp \mathfrak{g}^k \rangle$. This means $p\left(ghg^{-1}h^{-1}\right) = 1$ if $h \in G^{k-1}$, showing that $G^k \subset \langle \exp \mathfrak{g}^k \rangle$ and ending the proof. □

[4] See *Álgebras de Lie* [47, Chapters 1, 2] and Varadarajan [53, Section 3.5].

Just like the derived series, also the lower central series of G stabilizes, since this occurs at the Lie algebra level. Moreover, the lower central series of $\mathfrak{g}$ is part of a finite sequence of ideals

$$\mathfrak{g} = \mathfrak{g}_1 \supset \cdots \supset \mathfrak{g}_k = \{0\},$$

where $\dim \mathfrak{g}_i = \dim \mathfrak{g}_{i+1} + 1$. As in Proposition 9.15, therefore, it can be proved that if G is simply connected, then G^k, $k \geq 1$, is closed and simply connected and the quotient G/G^k is also simply connected.

9.5 Exercises

1. Let $\mathfrak{g}$ be the Heisenberg Lie algebra. This consists of matrices of the form

$$\begin{pmatrix} 0 & x & z \\ 0 & 0 & y \\ 0 & 0 & 0 \end{pmatrix} \qquad x, y, z \in \mathbb{R}.$$

   Determine the Lie algebras $\mathrm{Der}\mathfrak{g}$ of derivations and $\mathrm{ad}(\mathfrak{g})$ of inner derivations.
2. Let $\mathfrak{g}$ be a Lie algebra such that $\mathrm{Aut}\mathfrak{g} = \mathrm{Int}\mathfrak{g}$. Show that $\mathrm{Aut}G = \mathrm{Aut}\mathfrak{g}$ for any connected Lie group G with Lie algebra $\mathfrak{g}$.
3. Prove that if G is a connected Lie group, then the conjugation map $g \in G \mapsto C_g \in \mathrm{Aut}G$ is differentiable.
4. Let the Lie algebra $\mathfrak{g}$ be isomorphic to a semi-direct product. Otherwise put, there exist a subalgebra $\mathfrak{h}$ and an ideal $\mathfrak{n}$ such that $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{n}$. Let $\widetilde{G}$ be the connected and simply connected Lie group with Lie algebra $\mathfrak{g}$. Prove that the connected Lie subgroup $\langle \exp \mathfrak{n} \rangle$ is simply connected.
5. Let G be a connected and simply connected Lie group and $\mathfrak{h} \subset \mathfrak{g}$ an ideal in the Lie algebra $\mathfrak{g}$ of G. Prove that $\langle \exp \mathfrak{h} \rangle$ is closed.
6. Let $D : \mathbb{R}^n \to \mathbb{R}^n$ be a linear map without imaginary eigenvalues (in particular $\ker D = \{0\}$). Construct the semi-direct product $\mathfrak{h} = \mathbb{R} \times_\rho \mathbb{R}^n$ where $\rho : \mathbb{R} \to \mathfrak{gl}(n, \mathbb{R})$ is given by $\rho(t) = tD$. Prove that there is a unique connected Lie group with $\mathfrak{h}$.
7. Let G be a Lie group and denote by $\mathrm{End}G$ the semigroup of differentiable endomorphisms of G. Check that if G is connected and simply connected, then $\mathrm{End}G$ is isomorphic to the semigroup $\mathrm{End}\mathfrak{g}$ of endomorphisms of $\mathfrak{g}$. Describe $\mathrm{End}G$ in case $G = \widetilde{G}/\Gamma$ is not simply connected.
8. Let G be a connected and simply connected Lie group with Lie algebra $\mathfrak{g}$. A derivation $D \in \mathrm{Der}\mathfrak{g}$ defines a 1-parameter group $\exp tD \in \mathrm{Aut}G$ and hence a flow ϕ_t in G. In turn ϕ_t defines a vector field $\widetilde{D}$ by $\widetilde{D}(x) = \frac{d}{dt}\phi_t(x)$, called an **infinitesimal automorphism** of G. Show $\widetilde{D}$ satisfies the formula

$$\widetilde{D}(\exp X) = d(\exp)_X(DX).$$

9. Let $\widetilde{D}$ be an infinitesimal automorphism of a Lie group. Show that if X is an (left- or right-)invariant vector field, then $[\widetilde{D}, X]$ is also invariant (of the same type).
10. (Variation of parameters) Given a connected Lie group G with Lie algebra $\mathfrak{g}$, take an element $D \in \mathrm{Der}(\mathfrak{g})$ of the Lie algebra of $\mathrm{Aut}G$, that is, e^{tD}, $t \in \mathbb{R}$, and extend it to a 1-parameter group ϕ_t of automorphisms of G. Denote by $\widetilde{D}$ the vector field on G whose flow is ϕ_t (infinitesimal automorphism). Let X be a right invariant vector field in G and show that the solutions to the differential equation $\dot{x} = \widetilde{D}(x) + X(x)$ are given by $\phi_t(c(t))$, where $c'(t) = \left((\phi_{-t})_* X\right)(c(t))$. Generalize this fact to the case where X depends of t (as in Section 5.5). Take in particular $G = (\mathbb{R}^n, +)$ to obtain the classical formula of the variation of parameters.
11. Use the previous exercise to obtain an expression for the exponential map on the affine group $\mathrm{Aff}G$ in terms of the curve $c(t)$. Apply the result to a semi-direct product.
12. Let V be a vector space ($\dim V < \infty$) and A a linear operator on V. Construct the semi-direct product $\mathfrak{g} = \mathbb{R} \times_\theta V$ defined by the homomorphism $\theta : \mathbb{R} \to \mathrm{Der}V = \mathrm{End}V$, $\theta(t) = tA$. Let G be a connected Lie group with Lie algebra $\mathfrak{g}$. Prove that the exponential map of G is onto if 0 is the only imaginary eigenvalue A.
13. Given a connected Lie group G, let μ and ν be the left invariant Haar measures on G and $\mathrm{Aut}G$, respectively. Define $\Delta : \mathrm{Aut}G \to \mathbb{R}$ by $\Delta(g) = \det(dg_1)$. Show that $\Delta(g)\, \nu \times \mu$ is a Haar measure on $\mathrm{Aff}G = \mathrm{Aut}G \times G$. Let $P = H \times_\rho G$ be the semi-direct product defined by the homomorphism $\rho : H \in \mathrm{Aut}G$. Prove that $\Delta(\rho(h))\, \theta \times \mu$ is a left invariant Haar measure on P, where θ stands for some left invariant Haar measure on H. (Compare this with Exercise 29 in Chapter 5.)
14. Find an example of a connected Lie group G, but not simply connected, such that $\mathrm{Aut}G$ is not connected.
15. Let G be a Lie group and $\Gamma \subset G$ a discrete subgroup. Take $\phi \in \mathrm{Aut}G$ such that $\phi(\Gamma) = \Gamma$ and define the diffeomorphism $\overline{\phi}$ of G/Γ by $\overline{\phi}(g\Gamma) = \phi(g)\Gamma$. Suppose the only fixed point of ϕ in G is the identity, and prove that the fixed points of $\overline{\phi}$ in G/Γ are isolated.
    Apply the result to the case in which G/Γ is the torus $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$, and show that if 1 is not an eigenvalue of $g \in \mathrm{Sl}(n, \mathbb{Z})$, then the action of g on $\mathbb{T}^n$ has finitely many fixed points.
16. Let G be a Lie group with Lie algebra $\mathfrak{g}$ and tangent bundle $TG \approx G \times \mathfrak{g}$. Denote by $p : G \times G \to G$ the product in G. Show that the differential $dp : TG \times TG \to TG$ defines a Lie group structure on TG isomorphic to the semi-direct product $G \times_{\mathrm{Ad}} \mathfrak{g}$ (in this product $\mathfrak{g}$ is viewed as an abelian group).

17. Let G be a connected and nonabelian Lie group with Lie algebra $\mathfrak{g}$. Suppose that the only ideals in $\mathfrak{g}$ are $\{0\}$ and $\mathfrak{g}$, and that the center is trivial $Z(G) = \{1\}$. Prove the following statements:

    a. G is simple as an abstract group, i.e. the only normal subgroups of G are $\{1\}$ and G itself. (Hint: at some point you will need to apply Exercise 32, Chapter 9.)
    b. Every automorphism of $\mathfrak{g}$ extends to an automorphism of G and $\mathrm{Aut}G$ is isomorphic to $\mathrm{Aut}\mathfrak{g}$.

18. Let G be a connected and simply connected Lie group and $H \subset G$ a connected Lie subgroup. Suppose that the closure $\overline{H}$ of H is a normal subgroup, and prove that $H = \overline{H}$ i.e. H is closed.
19. Let G be a connected and simply connected Lie group and $H \subset G$ a connected Lie subgroup. Show that H is not dense in G.
20. Let G be a connected and simply connected Lie group with Lie algebra $\mathfrak{g}$ and $H = \langle \exp \mathfrak{h} \rangle$ a connected normal subgroup where $\mathfrak{h}$ is an ideal. The purpose of this exercise is to prove H is simply connected, thus complementing Proposition 9.11.

    a. Suppose $\mathfrak{h}$ is a maximal ideal, that is, if $\mathfrak{h}_1$ is an ideal containing $\mathfrak{h}$, then either $\mathfrak{h}_1 = \mathfrak{h}$ or $\mathfrak{h}_1 = \mathfrak{g}$. Prove that there exists a subalgebra $\mathfrak{l}$ such that $\mathfrak{g} = \mathfrak{l} \oplus \mathfrak{h}$. (Hint: $\mathfrak{g}/\mathfrak{h}$ only has the trivial ideals $\{0\}$ and $\mathfrak{g}/\mathfrak{h}$, hence either $\dim \mathfrak{g}/\mathfrak{h} = 1$ or $\mathfrak{g}/\mathfrak{h}$ is simple. In the first case argue as in Proposition 9.12. In the second case apply the Levi decomposition theorem.)
    b. Show that there exist subalgebras $\mathfrak{l}_1, \ldots, \mathfrak{l}_k$ such that $\mathfrak{g} = \mathfrak{l}_1 \oplus \cdots \oplus \mathfrak{l}_k \oplus \mathfrak{h}$ and $\mathfrak{l}_j \oplus \cdots \oplus \mathfrak{l}_k \oplus \mathfrak{h}$ is an ideal of $\mathfrak{g}$ for any $j = 1, \ldots, k$.
    c. Construct G by forming successive semi-direct products, and conclude $H = \langle \exp \mathfrak{h} \rangle$ is simply connected.

21. Give an example of a connected Lie group G whose derived group is not closed. (Hint: take the semi-direct product of Lie algebras $\mathbb{R} \times_\theta \mathbb{R}^3$ where $\theta(1) : \mathbb{R}^3 \to \mathbb{R}^3$ is a linear map mapping the first coordinate axis to a line contained in the plane spanned by the other two axes and having irrational slope.)
22. Find an example of a group G whose derived series $G = G^{(0)} \supset \cdots \supset G^{(k)} \supset \cdots$ does not stabilize: $G^{(k+1)} \neq G^{(k)}$ for all $k \geq 0$. Do the same for the lower central series $G = G^0 \supset \cdots \supset G^k \supset \cdots$. (Hint: consider "lower triangular matrices" in an infinite dimensional space with basis $\{e_1, e_2, \ldots\}$.)

Chapter 10
Solvable and Nilpotent Groups



The results in the previous chapter regarding the derived series $G^{(n)}$ and the lower central series G^n can be applied to study solvable Lie groups (those for which $G^{(n)} = \{1\}$ for some n) and nilpotent Lie groups, where $G^n = \{1\}$ for some n. It will be shown here that a connected Lie group G is solvable if and only if its Lie algebra is solvable. The same result holds for connected nilpotent groups. Furthermore, the simply connected Lie groups in these classes can be constructed by successive semi-direct products of simply connected abelian groups. It follows that a solvable (in particular, nilpotent) connected and simply connected Lie group is diffeomorphic to some Euclidean space $\mathbb{R}^n$.

10.1 Solvable Groups

The derived groups $G^{(k)}$ of a Lie group G are defined inductively by setting $G^{(0)} = G$ and letting $G^{(k)} = \left(G^{(k-1)}\right)'$ be the subgroup generated by commutators $[x, y] = xyx^{-1}y^{-1}$, $x, y \in G^{(k)}$. These are normal subgroups and, for each $k \geq 0$, $G^{(k)}/G^{(k+1)}$ is an abelian group. The decreasing sequence of groups

$$G = G^{(0)} \supset \cdots \supset G^{(k)} \supset \cdots$$

is called the **derived series** of G.

As we saw in Section 9.4, Chapter 9, if G is a connected Lie group then every $G^{(k)}$ is a connected Lie subgroup. The corresponding Lie algebras $\mathcal{L}\left(G^{(k)}\right)$, called **derived Lie algebras**, $\mathfrak{g}^{(k)}$ are defined, inductively, by $\mathfrak{g}^{(0)} = \mathfrak{g}$ and $\mathfrak{g}^{(k)} = [\mathfrak{g}^{(k-1)}, \mathfrak{g}^{(k-1)}]$. The derived series of $\mathfrak{g}$ is the decreasing sequence of ideals

$$\mathfrak{g} = \mathfrak{g}^{(0)} \supset \cdots \supset \mathfrak{g}^{(k)} \supset \cdots .$$

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A group G is said to be **solvable** if its derived series ends with the trivial group, meaning $G^{(k)} = \{1\}$ for some $k \geq 0$ (in this case, $G^{(i)} = \{1\}$ if $i \geq k$). Similarly, a Lie algebra $\mathfrak{g}$ is called **solvable** if $\mathfrak{g}^{(k)} = \{0\}$ for some $k \geq 0$.

The fact that $\mathcal{L}\left(G^{(k)}\right) = \mathfrak{g}^{(k)}$ means $G^{(k)} = \langle \exp \mathfrak{g}^{(k)} \rangle$ and shows straightforward that $G^{(k)} = \{1\}$ if and only if $\mathfrak{g}^{(k)} = \{0\}$. Hence, we have the following criterion for connected Lie groups to be solvable.

Proposition 10.1 *A connected Lie group G is solvable if and only if its Lie algebra $\mathfrak{g}$ is solvable.*

A typical example of solvable Lie algebra is the algebra of triangular matrices:

$$\begin{pmatrix} a_1 & \cdots & * \\ \vdots & \ddots & \vdots \\ 0 & \cdots & a_n \end{pmatrix}_{n \times n}.$$

This is the Lie algebra of the group T of triangular matrices whose diagonal entries are positive. The manifold underlying T is $\mathbb{R}_+^n \times \mathbb{R}^N$, $N = n\,(n-1)\,/2$, a Euclidean space.

This example illustrates a property enjoyed by every connected and simply connected solvable Lie group, for we will prove below that these groups are diffeomorphic to Euclidean spaces. The proof requires the following facts about solvable Lie algebras.

A **Jordan–Hölder decomposition** of a Lie algebra is a sequence of subalgebras $\mathfrak{g} = \mathfrak{g}_0 \supset \mathfrak{g}_1 \supset \cdots \supset \mathfrak{g}_k = \{0\}$ such that $\mathfrak{g}_{i+1}$ is an ideal of $\mathfrak{g}_i$, for each $i = 1, \ldots, k$, and the quotient space $\mathfrak{g}_i/\mathfrak{g}_{i+1}$ only contains the trivial ideals ($\mathfrak{g}_i/\mathfrak{g}_{i+1}$ is either a simple Lie algebra or $\dim \mathfrak{g}_i/\mathfrak{g}_{i+1} = 1$).

The next proposition shows that solvable Lie algebras admit Jordan–Hölder decompositions in which the above quotients have dimension 1. (As a matter of fact, the converse is true as well. Such a decomposition only occurs in solvable Lie algebras.)

Proposition 10.2 *Let $\mathfrak{g}$ be a solvable Lie algebra. Then there exists a sequence of subalgebras*

$$\mathfrak{g} = \mathfrak{g}_0 \supset \mathfrak{g}_1 \supset \cdots \supset \mathfrak{g}_k = \{0\}$$

such that $\mathfrak{g}_{i+1}$ is an ideal in $\mathfrak{g}_i$ and $\dim \mathfrak{g}_i = \dim \mathfrak{g}_{i+1} + 1$, for $i = 0, \ldots, k-1$.

Proof Start with the derived series $\mathfrak{g} = \mathfrak{g}^{(0)} \supset \mathfrak{g}' \supset \cdots \supset \mathfrak{g}^{(k)} = \{0\}$, where each term is an ideal in $\mathfrak{g}$. The inclusion $\mathfrak{g}^{(i)} \supset \mathfrak{g}^{(i+1)}$ between successive terms of the series can be refined by inserting vector subspaces

$$\mathfrak{g}^{(i)} \supset V_1 \supset \cdots \supset V_l \supset \mathfrak{g}^{(i+1)},$$

in such a way that the dimensions increase by 1 at each step. Since $[\mathfrak{g}^{(i)}, \mathfrak{g}^{(i)}] \subset \mathfrak{g}^{(i+1)}$, it follows that $[V_j, V_r] \subset \mathfrak{g}^{(i+1)}$. In particular, $[V_j, V_{j+1}] \subset \mathfrak{g}^{(i+1)} \subset V_{j+1}$, so V_{j+1} is an ideal in V_j. This concludes the proof. □

This proposition actually shows that the derived series

$$\mathfrak{g} = \mathfrak{g}^{(0)} \supset \cdots \supset \mathfrak{g}^{(s)} \supset \{0\}$$

can be fit into a Jordan–Hölder decomposition. A little additional work shows that any ideal of $\mathfrak{g}$ belongs to such a decomposition.

Proposition 10.3 *Let $\mathfrak{g}$ be a solvable Lie algebra and $\mathfrak{h} \subset \mathfrak{g}$ an ideal. There exists a Jordan–Hölder decomposition*

$$\mathfrak{g} = \mathfrak{g}_0 \supset \mathfrak{g}_1 \supset \cdots \supset \mathfrak{g}_k = \{0\},$$

such that $\mathfrak{h} = \mathfrak{g}_i$ for some i.

Proof The given subalgebra $\mathfrak{h}$ is solvable since $\mathfrak{h}^{(i)} \subset \mathfrak{g}^{(i)}$. Hence there exists a decomposition

$$\mathfrak{h} = \mathfrak{h}_0 \supset \mathfrak{h}_1 \supset \cdots \supset \mathfrak{h}_s = \{0\}$$

with $\dim(\mathfrak{h}_i/\mathfrak{h}_{i+1}) = 1$. On the other hand also $\mathfrak{g}/\mathfrak{h}$ is solvable (because $(\mathfrak{g}/\mathfrak{h})^{(i)} = \pi\left(\mathfrak{g}^{(i)}\right)$).[1] Hence there exists a sequence

$$\mathfrak{g}/\mathfrak{h} = \mathfrak{l}_0 \supset \mathfrak{l}_1 \supset \cdots \supset \mathfrak{l}_r = \{0\}$$

with $\dim(\mathfrak{l}_i/\mathfrak{l}_{i+1}) = 1$. Let $\pi : \mathfrak{g} \to \mathfrak{g}/\mathfrak{h}$ denote the canonical projection. Then $\pi^{-1}(\mathfrak{l}_{i+1})$ has codimension 1 in $\pi^{-1}(\mathfrak{l}_i)$, so

$$\mathfrak{g} = \pi^{-1}(\mathfrak{l}_0) \supset \cdots \supset \mathfrak{h} = \pi^{-1}(\mathfrak{l}_r) \supset \mathfrak{h}_1 \supset \cdots \supset \mathfrak{h}_s = \{0\}$$

is the desired decomposition. □

Returning to Lie groups, now we can prove that connected and simply connected solvable Lie groups are diffeomorphic to Euclidean spaces.

Proposition 10.4 *Let G be a connected and simply connected solvable Lie group with Lie algebra $\mathfrak{g}$. Take a Jordan–Hölder decomposition*

$$\mathfrak{g} = \mathfrak{g}_0 \supset \mathfrak{g}_1 \supset \cdots \supset \mathfrak{g}_k \supset \mathfrak{g}_{k+1} = \{0\}$$

[1] See *Álgebras de Lie* [47, Ch. 1], and Varadarajan [53, Section 3.7].

with $\dim(\mathfrak{g}_i/\mathfrak{g}_{i+1}) = 1$. *Then each of the connected subgroups* $\langle \exp \mathfrak{g}_i \rangle$ *is closed and diffeomorphic to a Euclidean space (*$\mathbb{R}^n$ *for some n). In particular,* $G = \langle \exp \mathfrak{g}_0 \rangle$ *is a Euclidean space.*

Proof Let G_i indicate the connected and simply connected Lie group with Lie algebra $\mathfrak{g}_i$. The claim is proved by reconstructing G by successive semi-direct products $\mathbb{R} \times_s G_{i+1} \approx G_i$. The construction is such that $\langle \exp \mathfrak{g}_i \rangle$ becomes isomorphic to G_i, for each i.

The isomorphism $\mathbb{R} \times_s G_{i+1} \approx G_i$ is defined as in Proposition 9.12, since $\mathfrak{g}_i = \langle X \rangle \oplus \mathfrak{g}_{i+1}$ and $X \in \mathfrak{g}_i \setminus \mathfrak{g}_{i+1}$. In order to use induction note that $G_k \approx \mathbb{R}$, as $\dim \mathfrak{g}_k = 1$. Therefore,

$$G_{k-1} \approx \mathbb{R} \times_s \mathbb{R},$$

where the second factor is the subgroup with Lie algebra $\mathfrak{g}_k$. In the same way, G_{k-2} is isomorphic to $\mathbb{R} \times_s (\mathbb{R} \times_s \mathbb{R})$ and so on and so forth.

From the sequence of semi-direct products, we end up with $G \approx G_0$, whose underlying manifold is $\mathbb{R}^{k+1}$. □

Corollary 10.5 *Let G be a connected and simply connected solvable Lie group with Lie algebra* $\mathfrak{g}$*, and* $H \subset G$ *a connected normal Lie subgroup. Then H is closed and diffeomorphic to some Euclidean space. The quotient* G/H*, too, is a Euclidean space.*

Proof The Lie algebra $\mathfrak{h}$ of H is an ideal, and $H = \langle \exp \mathfrak{h} \rangle$. By Proposition 10.3, the ideal $\mathfrak{h}$ belongs in some Jordan–Hölder decomposition of $\mathfrak{g}$. Hence the previous proposition ensures that H is Euclidean. On the other hand by Proposition 9.11, Chapter 9, it follows that H is closed and G/H simply connected. Therefore the solvable group G/H is also diffeomorphic to a Euclidean space. □

The good properties of connected normal subgroups stated in this corollary also apply to subgroups that are not normal, as shown in the next result.

Proposition 10.6 *Let G be a connected and simply connected solvable Lie group. Any connected subgroup* $H \subset G$ *is closed and simply connected, so H is diffeomorphic to a Euclidean space.*

Proof Suppose first that H is closed. In this case, the proof that H simply connected is by induction on the dimension of G. If $\dim G = 1$, there is nothing to prove since $G \approx \mathbb{R}$ and $H = \{1\}$ or $H = G$.

By induction hypothesis, the statement holds for the derived group G' since $\dim G' < \dim G$, and by previous results, G' is simply connected. The quotient G/G' is isomorphic to $\mathbb{R}^n$ since it is abelian and simply connected. Let $\pi : G \to G/G'$ be the canonical homomorphism. Then $\pi(H) = H/H \cap G'$ is isomorphic to a connected subgroup of $\mathbb{R}^n$, so a vector subspace. That is, $H/H \cap G'$ is simply connected, guaranteeing $H \cap G'$ is connected (since $H/\left(H \cap G'\right)_0 \to H/H \cap G'$ is a covering map). The induction hypothesis then implies $H \cap G'$ is simply connected.

Therefore, $H/H \cap G'$ and $H \cap G'$ are simply connected, which in turn implies H is simply connected (see Exercise 16 in Chapter 7). This concludes the proof that H is simply connected when closed and connected.

Finally, if H is connected, then its closure $\overline{H}$ is connected and closed, hence simply connected. But H is normal in $\overline{H}$. By Proposition 9.11 in Chapter 9, H is a closed subgroup of $\overline{H}$, i.e. $H = \overline{H}$, as required. □

10.2 Nilpotent Groups

Given a group G the **lower central series**

$$G = G^1 \supset G^2 \supset \cdots \supset G^k \supset \cdots$$

is defined inductively by $G^1 = G$, $G^2 = G'$, and the general term $G^{k+1} = \left[G, G^k\right]$ is the subgroup generated by commutators $[x, y] = xyx^{-1}y^{-1}$, $x \in G$, $y \in G^k$. The subgroups G^k are normal, and G^k/G^{k+1} is abelian for all $k \geq 0$.

In a similar way, the lower central series

$$\mathfrak{g} = \mathfrak{g}^1 \supset \mathfrak{g}^2 \supset \cdots \supset \mathfrak{g}^k \supset \cdots$$

of a Lie algebra $\mathfrak{g}$ is defined by setting $\mathfrak{g}^1 = \mathfrak{g}$, $\mathfrak{g}^2 = \mathfrak{g}'$, and letting $\mathfrak{g}^{k+1} = \left[\mathfrak{g}, \mathfrak{g}^k\right]$ be the subspace spanned by the brackets $[X, Y]$, for $X \in \mathfrak{g}$ and $Y \in \mathfrak{g}^k$. Each $\mathfrak{g}^k$ is an ideal in $\mathfrak{g}$, and the quotients $\mathfrak{g}^k/\mathfrak{g}^{k+1}$ are abelian Lie algebras.[2]

A group G is called **nilpotent** if its lower central series terminates in the trivial group, i.e. $G^k = \{1\}$ for some $k \geq 0$, and hence $G^i = \{1\}$ if $i \geq k$. Analogously, a Lie algebra $\mathfrak{g}$ is said **nilpotent** if $\mathfrak{g}^k = \{0\}$ for some $k \geq 0$.

If G is a connected Lie group, then, by Proposition 9.18, Chapter 9, the Lie algebra $\mathcal{L}\left(G^k\right)$ of G^k is $\mathfrak{g}^k$, which means $G^k = \langle \exp \mathfrak{g}^k \rangle$. This immediately shows that $G^k = \{1\}$ if and only if $\mathfrak{g}^k = \{0\}$ and gives the following criterion for the nilpotency of connected Lie groups.

Proposition 10.7 *A connected Lie group G is nilpotent if and only if its Lie algebra $\mathfrak{g}$ is nilpotent.*

The lower central series contains the derived series, in the sense that $G^{(k)} \subset G^{k+1}$ and $\mathfrak{g}^{(k)} \subset \mathfrak{g}^{k+1}$. That is why nilpotent groups (as well as Lie algebras) are also solvable, and the properties of connected solvable groups described in the previous section hold in particular for connected nilpotent groups. Consequently, a connected and simply connected nilpotent Lie group is diffeomorphic to Euclidean space, as well as any of its connected subgroups.

[2] See *Álgebras de Lie* [47, Ch. 1, 2], and Varadarajan [53, Section 3.5].

In the nilpotent case, this picture is improved by the fact that the exponential map $\exp : \mathfrak{g} \to G$ is a diffeomorphism. This is not the case for solvable groups, in general (see an example below).

Theorem 10.8 *Let G be a connected and simply connected nilpotent Lie group with Lie algebra $\mathfrak{g}$. Then the exponential map $\exp : \mathfrak{g} \to G$ is a diffeomorphism.*

In other words, in connected simply connected nilpotent Lie groups, the exponential map is a global coordinate system of the first kind.

To prove Theorem 10.8, the following elementary properties of nilpotent Lie algebras will be used:[3]

1. If $\mathfrak{g}$ is nilpotent, then $\operatorname{ad}(X)$, $X \in \mathfrak{g}$, is a nilpotent linear map. That is because $\operatorname{ad}(X)(Y) = [X, Y] \in \mathfrak{g}^2$, $\operatorname{ad}(X)^2(Y) \in \mathfrak{g}^3$ etc, so $\operatorname{ad}(X)^k = 0$ if $\mathfrak{g}^{k+1} = \{0\}$.
2. The center $\mathfrak{z}(\mathfrak{g})$ of a nilpotent Lie algebra $\mathfrak{g}$ is different from $\{0\}$. In fact if $k+1$ is the first exponent for which $\mathfrak{g}^{k+1} = \{0\}$, then $\mathfrak{g}^k \neq \{0\}$ is contained in $\mathfrak{z}(\mathfrak{g})$ since $\left[\mathfrak{g}, \mathfrak{g}^k\right] = \mathfrak{g}^{k+1} = \{0\}$.
3. Let $\mathfrak{g}$ be a nilpotent Lie algebra. Then every subalgebra of $\mathfrak{g}$ is nilpotent. If $\mathfrak{h} \subset \mathfrak{g}$ is an ideal, then $\mathfrak{g}/\mathfrak{h}$ is nilpotent.

The first step in the proof of Theorem 10.8 is to verify that exp is a local diffeomorphism. The formula for the differential of exp, proved in Chapter 8, reads

$$d(\exp)_X = dL_{e^X} \circ T_X = dL_{e^X} \circ \frac{e^{\operatorname{ad}(X)} - 1}{\operatorname{ad}(X)}.$$

The only eigenvalue $\operatorname{ad}(X)$, $X \in \mathfrak{g}$, is 0 since this linear map is nilpotent. Hence 1 is the only eigenvalue of $T_X = \dfrac{e^{\operatorname{ad}(X)} - 1}{\operatorname{ad}(X)}$ because the map $f(t) = \dfrac{e^t - 1}{t}$ satisfies $1 = f(0)$. This implies that T_X is injective, so $d(\exp)_X$ is also injective. Hence $d(\exp)_X$ is an isomorphism for every $X \in \mathfrak{g}$ since the domain and range have the same dimension. Altogether, exp is a local diffeomorphism.

So to show that exp is diffeomorphism, there remains to check that it is a bijection.

The proof that exp is bijective goes by induction on the dimension of G. First, if $\dim G = 1$ and G is connected and simply connected, then G, as well as $\mathfrak{g}$, is equal to $\mathbb{R}$ and exp is the identity map.

To proceed with induction let $k \geq 1$ be such that $\mathfrak{g}^k \neq \{0\}$ and $g^{k+1} = \{0\}$. We may assume $k \geq 2$, for otherwise $\mathfrak{g}$ is abelian and the exponential is clearly a diffeomorphism. In case $k \geq 2$, $\mathfrak{g}/\mathfrak{g}^k$ is a nilpotent algebra with $0 < \dim\left(\mathfrak{g}/\mathfrak{g}^k\right) < \dim \mathfrak{g}$. As was proved in Chapter 9, $G^k = \langle \exp \mathfrak{g}^k \rangle$ is closed and simply connected and the quotient G/G^k is a connected and simply connected nilpotent Lie group. Denote by $\pi : G^k \to G/G^k$ the canonical homomorphism.

[3] Seea *Álgebras de Lie* [47, Ch. 1, 2], and Varadarajan [53, Section 3.5].

By induction hypothesis, the exponential maps on G^k and G/G^k are onto. Hence given $g \in G$, there exists $\overline{X} = d\pi_1(X) \in \mathfrak{g}/\mathfrak{g}^k$, $X \in \mathfrak{g}$, such that $e^{\overline{X}} = \pi(g)$. Then

$$\pi\left(e^X\right) = e^{\overline{X}} = \pi(g),$$

i.e. there exists $h \in G^k$ such that $g = e^X h$. By the surjectivity of the exponential map on G^k, there exists $Y \in \mathfrak{g}^k$ such that $h = e^Y$. Therefore,

$$g = e^X e^Y = e^{X+Y},$$

because $[X, Y] = 0$. Since $g \in G$ was arbitrary, this shows that $\exp : \mathfrak{g} \to G$ is onto.

To prove injectivity, take $X, Y \in \mathfrak{g}$ such that $e^X = e^Y$. Then,

$$e^{d\pi_1(X)} = \pi\left(e^X\right) = \pi\left(e^Y\right) = e^{d\pi_1(X)}.$$

By applying again the inductive hypothesis, it follows that the exponential map is 1-1 on G^k. Hence $d\pi_1(X) = d\pi_1(Y)$, which means that there exists $Z \in \mathfrak{g}^k$ such that $X = Y + Z$. However, Z commutes with X and Y, so

$$e^X = e^{Y+Z} = e^Y e^Z,$$

which implies that $e^Z = 1$. Then $Z = 0$ since the exponential map is 1-1 on G^k. So we have $X = Y$, and the exponential map on G is injective. This ends the proof of Theorem 10.8.

Theorem 10.8 has the following consequences.

Corollary 10.9 *Let G be a connected nilpotent Lie group. Then* $\exp : \mathfrak{g} \to G$ *is a covering map. The group G is simply connected if and only if the exponential map is one-to-one.*

Proof Write $G = \widetilde{G}/\Gamma$ with $\widetilde{G}$ simply connected and Γ a discrete central subgroup. The canonical homomorphism $\pi : \widetilde{G} \to G$ is a covering map and $\exp_G = \pi \circ \exp_{\widetilde{G}}$, showing that $\exp_G$ is a covering since $\exp_{\widetilde{G}}$ is a diffeomorphism. □

Corollary 10.10 *Let G be a connected nilpotent Lie group with Lie algebra $\mathfrak{g}$. Any connected subgroup of G has the form $H = \exp \mathfrak{h}$, where $\mathfrak{h} \subset \mathfrak{g}$ is a subalgebra. If G is further simply connected, any connected subgroup* $\exp \mathfrak{h}$ *is closed and simply connected.*

Proof If H is connected, then the exponential map in H is onto since H is nilpotent. And if G is simply connected, the exponential map is 1-1. Hence $H = \exp \mathfrak{h}$ is simply connected. As $\exp : \mathfrak{g} \to G$ is a diffeomorphism and $\mathfrak{h} \subset \mathfrak{g}$ is closed, moreover, it follows that $H = \exp \mathfrak{h}$ is closed. □

Example Let $\mathfrak{n} \subset \mathfrak{gl}(n, \mathbb{R})$ be the Lie algebra of upper triangular matrices with zeroes along the diagonal. This nilpotent Lie algebra is the Lie algebra of the group N of upper triangular matrices with 1s on the diagonal. Since N is connected, it follows that $N = \langle \exp \mathfrak{n} \rangle$ equals $\exp \mathfrak{n}$. Moreover, N is simply connected, and hence, its connected subgroups are simply connected and closed by Corollaries 10.9 and 10.10. □

Proposition 10.11 *Let G be a connected nilpotent Lie group with Lie algebra $\mathfrak{g}$. If $\mathfrak{z}(\mathfrak{g})$ denotes the center of $\mathfrak{g}$, then the center of G is*

$$Z(G) = \exp \mathfrak{z}(\mathfrak{g}).$$

Proof The inclusion $\exp \mathfrak{z}(\mathfrak{g}) \subseteq Z(G)$ holds because G is connected. For the reverse inclusion, write $G = \widetilde{G}/\Gamma$ with $\widetilde{G}$ simply connected and Γ a discrete central subgroup. Take $g\Gamma \in Z(G)$ ($g \in \widetilde{G}$) and let $X \in \mathfrak{g}$ be such that $g = e^X$. Then for every $h \in G$, $he^X h^{-1} = e^X \Gamma$. Put equivalently, there exists $\gamma_h \in \Gamma$ with $e^{\mathrm{Ad}(h)X} = e^X \gamma_h$. This relation, applied to $h = e^{tY}$ with $Y \in \mathfrak{g}$ and $t \in \mathbb{R}$, defines the continuous curve $t \mapsto \gamma_t = e^{\mathrm{Ad}(e^{tY})X} e^{-X}$ that is contained in Γ and satisfies

$$e^{\mathrm{Ad}(e^{tY})X} = e^X \gamma_t.$$

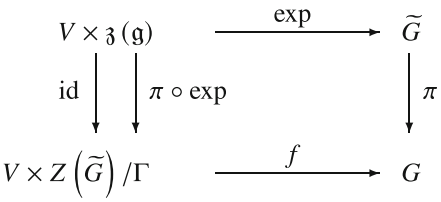
Since Γ is discrete, γ_t is constant and equals to $\gamma_0 = 1$. Hence $\mathrm{Ad}\left(e^{tY}\right) X = X$ by the injectivity of the exponential map on $\widetilde{G}$, and so

$$e^{t\,\mathrm{ad}(Y)} X = X.$$

By taking the derivative with respect to t above, it follows that $\mathrm{ad}(Y) X = 0$, which means $X \in \mathfrak{z}(\mathfrak{g})$. This proves that $g = e^X$ for $X \in \mathfrak{z}(\mathfrak{g})$ provided $g \in Z(G)$, concluding the proof. □

Corollary 10.12 *Let $G = \widetilde{G}/\Gamma$ be a connected nilpotent Lie group with $\widetilde{G}$ simply connected. Then G is diffeomorphic to a "cylinder" $\mathbb{R}^n \times Z\left(\widetilde{G}\right)/\Gamma$, where $n = \dim \widetilde{G} - \dim Z\left(\widetilde{G}\right)$.*

Proof By the above corollary, $Z\left(\widetilde{G}\right) = \exp \mathfrak{z}(\mathfrak{g})$. Take a complementary subspace $V \subset \mathfrak{g}$ to the Lie algebra center: $\mathfrak{g} = \mathfrak{z}(\mathfrak{g}) \oplus V$. The map $f : V \times Z\left(\widetilde{G}\right)/\Gamma \to G$ given by $f(X, g\Gamma) = \pi\left(e^X g\right)$ is the desired diffeomorphism. In fact, f fits in the commutative diagram



where the upper arrow exp is $(X, Y) \mapsto e^X e^Y = e^{X+Y}$. The diagram explains that f a is local diffeomorphism since the exponential maps are diffeomorphisms and the projections (denoted by π) are covering maps. Consequently, also f is onto. Finally, taking $f(X_1, g_1\Gamma) = f(X_2, g_2\Gamma)$, there exists $z = e^Z \in Z\left(\widetilde{G}\right)$ such that $e^{X_1} = e^{X_2} e^Z$, which implies $X_1 = X_2$ and hence $g_1\Gamma = g_2\Gamma$. □

Corollary 10.13 *If a connected nilpotent Lie group is compact, then it is abelian.*

Proof The proof is straightforward from the previous corollary, since G compact forces $n = \dim \widetilde{G} - \dim Z\left(\widetilde{G}\right) = 0$. □

The corollary will be obtained by a different method in Chapter 11, where compact Lie groups are investigated in detail.

A nilpotent Lie algebra $\mathfrak{g}$ may be viewed as the underlying manifold of a connected and simply connected Lie group, whose Lie algebra is $\mathfrak{g}$itself. In fact, if G is such a group, then $\exp : \mathfrak{g} \to G$ is a diffeomorphism and hence the product on G can be transferred to a product $c : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$ making $\mathfrak{g}$ a Lie group isomorphic to G. This product is given by

$$e^X e^Y = e^{c(X,Y)}.$$

The Baker–Campbell–Hausdorff (BCH) formula, proved in Chapter 8, gives an expression for $c(X, Y)$ in terms of the bracket of $\mathfrak{g}$. In case, $\mathfrak{g}$ is nilpotent, this formula is a polynomial in X and Y. It is defined for every pair (X, Y) and differentiable. This means that $\mathfrak{g}$, endowed with the product given by BCH, is a Lie group with Lie algebra $\mathfrak{g}$. This is an efficient method to build the connected and simply connected Lie group with nilpotent Lie algebra $\mathfrak{g}$.

For instance, if $\mathfrak{g}^3 = \{0\}$, the BCH formula becomes

$$c(X, Y) = X + Y + \frac{1}{2}[X, Y].$$

This is a product that defines on $\mathfrak{g}$ a Lie group with Lie algebra $\mathfrak{g}$.

Following this construction of nilpotent Lie groups, the next proposition shows that the Haar measure on a connected nilpotent Lie group is essentially given by the Lebesgue measure on its Lie algebra.

Proposition 10.14 *Let G be a connected and simply connected nilpotent Lie group with Lie algebra $\mathfrak{g}$, so that $\exp : \mathfrak{g} \to G$ is a diffeomorphism. Let dX denote the Lebesgue measure on $\mathfrak{g}$ and set $\mu = \exp_*(dX)$. Then μ is a (left and right) Haar measure on G, and integration on G passes onto $\mathfrak{g}$:*

$$\int_G f(g)\,\mu(dg) = \int_{\mathfrak{g}} f(\exp X)\,dX.$$

Proof The formula for the differential of the exponential map reads $d\exp_X = dL_{\exp X} \circ T_X$, where $T_X = \left(e^{\mathrm{ad}(X)} - 1\right)/\mathrm{ad}(X)$. Since $\mathfrak{g}$ is Nilpotent, each $\mathrm{ad}(X)$

is nilpotent and hence has 0 as unique eigenvalue. Then 1 is the only eigenvalue of T_X, implying $\det T_X = 1$.

Let ν be a left invariant volume form on G whose value at the identity is denoted by ν_0. If $X \in \mathfrak{g}$, then $\left(d \exp_X\right)^* \nu_{\exp X}$ is a volume form on $\mathfrak{g}$. The relation $d \exp_X = dL_{\exp X} \circ T_X$ shows that

$$\begin{aligned}\left(d \exp_X\right)^* \nu_{\exp X} &= T_X^* \left(\left(dL_{\exp X}\right)^* \nu_{\exp X}\right) = T_X^* \nu_0 \\ &= (\det T_X)\, \nu_0.\end{aligned}$$

Yet $\det T_X = 1$, and so $\left(d \exp_X\right)^* \nu_{\exp X} = \nu_0$ is independent of X. It follows that the measure on $\mathfrak{g}$ defined by the volume form $\exp^* \nu$ is the Lebesgue measure (possibly renormalized). □

This section closes with an following example explaining that the exponential map of a connected and simply connected solvable Lie group is not a diffeomorphism, in general.

Example Consider an $n \times n$ real matrix A having at least one nonzero imaginary eigenvalue. Let $\mathfrak{g}$ be the solvable Lie algebra of $(n+1) \times (n+1)$ matrices of the form

$$\begin{pmatrix} tA & v \\ 0 & 0 \end{pmatrix}$$

with v being a $n \times 1$ column matrix and $t \in \mathbb{R}$. The eigenvalues of

$$\operatorname{ad} \begin{pmatrix} tA & 0 \\ 0 & 0 \end{pmatrix}$$

are 0 and $t\lambda$, where λ is an eigenvalue of A. Then there exists $X \in \mathfrak{g}$ such that $\operatorname{ad}(X)$ has a nonzero eigenvalue belonging in $2\pi i \mathbb{Z}$. This implies that on any Lie group G with Lie algebra $\mathfrak{g}$, the exponential map $\exp : \mathfrak{g} \to G$ has singular points and hence is not a diffeomorphism. □

10.3 Exercises

1. Show that a connected and simply connected solvable Lie group G admits a global coordinate system of the second kind. That is, there exists a basis $\{X_1, \ldots, X_n\}$ of the Lie algebra $\mathfrak{g}$ of G such that the map $\phi(t_1, \ldots, t_n) = e^{t_1 X_1} \cdots e^{t_n X_n}$ is a diffeomorphism.
2. Show that a connected nilpotent Lie group is unimodular. Find examples of connected solvable Lie groups that are not unimodular.
3. Given a solvable Lie group G, let

$$(t_1, \ldots, t_n) \in \mathbb{R}^n \longmapsto e^{t_1 X_1} \cdots e^{t_n X_n} \in G$$

be a global coordinate system of the second kind. Find the expression of the Haar measure of G in this frame system. (Hint: apply Exercise 13, Chapter 9.)

4. Give an example of a connected solvable Lie group whose exponential map is not onto. Then do the same replacing "onto" by "1-1."
5. Given a connected and simply connected nilpotent group G, find the expression of the Haar measure of G in the coordinate system of the first kind defined by the exponential map $\exp : \mathfrak{g} \to G$. Generalize to the case where G is connected but not simply connected.
6. Given a group G and a normal subgroup $H \subset G$, show that G is solvable if and only if H and G/H are solvable. Find an example showing that this property does not hold for nilpotent groups.
7. Let G be a solvable but not nilpotent Lie group. Show that G is not compact.
8. Consider the formal series with real coefficients $\exp x = \sum_{k\geq 0} \frac{1}{k!} x^k$ and $\log(1+x) = \sum_{k\geq 1} \frac{(-1)^{k+1}}{k} x^k$ in the variable x. In the logarithm replace x with $\exp x - 1$ and verify that

$$\log(\exp x) = \log(1 + (\exp x - 1)) = x.$$

Use this identity to show that if A and B are $n \times n$ nilpotent matrices, then $e^A = e^B$ if and only if $A = B$.

9. Let $\mathcal{N}$ be the set of nilpotent linear maps on $\mathfrak{gl}(n, \mathbb{R})$.
   (a) Show that if $X \in \mathcal{N}$ then $g = e^X$ is unipotent, i.e. $g - 1$ is nilpotent. Also prove that $X \in \mathcal{N}$ if e^{tX} is unipotent for every $t \in \mathbb{R}$.
   (b) Let $\mathcal{U} \in \mathrm{Gl}(n, \mathbb{R})$ be the set of unipotent matrices. Prove that $\exp : \mathcal{N} \to \mathcal{U}$ is a bijection. (Hint: use the power series expansions of the exponential and logarithm.)
   (c) Give an example of a non-nilpotent matrix X such that e^X is unipotent.
10. As a particular case of Exercise 8, let $A \in \mathfrak{gl}(n, \mathbb{R})$ be a nilpotent linear map and T_A the power series of the function $f(x) = \frac{e^x - 1}{x}$ evaluated at A. Show that $e^A = 1$ implies $AT_A = 0$, and that $A = 0$ if $e^A = 1$.
11. Let $\mathfrak{g} \subset \mathfrak{gl}(n, \mathbb{R})$ be a Lie algebra, and suppose there exists a basis β of $\mathbb{R}^n$ in which any $A \in \mathfrak{g}$ has upper triangular matrix with zeroes on the diagonal. Prove that $G = \langle \exp \mathfrak{g} \rangle$ is a simply connected nilpotent Lie group.
12. Let G be a connected and simply connected nilpotent Lie group with Lie algebra $\mathfrak{g}$. Given a derivation $D \in \mathrm{Der}(\mathfrak{g})$ define the infinitesimal automorphism $\widetilde{D}$ as in Exercise 8, Chapter 9. Show that, in the global coordinate system of the first kind given by the exponential, $\widetilde{D}(X) = D(X)$ for every $X \in \mathfrak{g}$.
13. Given a semi-direct product $\mathfrak{p} = \mathfrak{g} \times_\rho \mathfrak{h}$ of Lie algebras show that $\mathfrak{p}$ is solvable if $\mathfrak{g}$ and $\mathfrak{h}$ are solvable.

14. Give examples of nilpotent Lie algebras $\mathfrak{g}, \mathfrak{h}$ whose semi-direct product $\mathfrak{p} = \mathfrak{g} \times_\rho \mathfrak{h}$ is not nilpotent.
15. Let G be the group of upper triangular $n \times n$ matrices whose diagonal entries are strictly positive. Given a partition $n_1 + \cdots + n_k = n$ of n, let H be the group of block-diagonal matrices where blocks have sizes n_i, $i = 1, \ldots, k$. Show that any g decomposes uniquely as a product $g = g_1 g_2$, where $g_1 \in H$ and g_2, upper triangular, has identity matrices as diagonal blocks.

Chapter 11
Compact Groups



In this chapter the simply connected groups which are universal coverings of compact groups are studied. It is proved that the Lie algebra $\mathfrak{g}$ of a compact group G decomposes in the direct sum $\mathfrak{g} = \mathfrak{z}(\mathfrak{g}) \oplus \mathfrak{k}$, where $\mathfrak{z}(\mathfrak{g})$ is the center of $\mathfrak{g}$ and $\mathfrak{k}$ is a semi-simple algebra. The simply connected group associated with $\mathfrak{g}$ is the direct product of the simply connected groups of $\mathfrak{z}(\mathfrak{g})$ and $\mathfrak{k}$. The latter is a compact group, a result of the Weyl theorem, which is proved here.

11.1 Compact Lie Algebras

In Chapter 3 it was proved, using integrals with respect to the Haar measure, that if $\rho : G \to \mathrm{Gl}(V)$ is a finite dimensional real representation of a Hausdorff compact group G, then there exists an inner product $(\cdot, \cdot)$ on V which is invariant by $\rho(g)$, $g \in G$ (see Proposition 4.1).

In particular, if G is a compact Lie group, then there exists an inner product $(\cdot, \cdot)$ on its Lie algebra $\mathfrak{g}$ with respect to which $\mathrm{Ad}(g)$ is an isometry for every $g \in G$, that is,

$$(\mathrm{Ad}(g)\, Y, \mathrm{Ad}(g)\, Z) = (Y, Z) \qquad Y, Z \in \mathfrak{g}.$$

The formula $\mathrm{Ad}(\exp tX) = \exp(t\,\mathrm{ad}(X))$, $t \in \mathbb{R}$, ensures that $\exp(t\,\mathrm{ad}(X))$ is an isometry of $(\cdot, \cdot)$ for every $X \in \mathfrak{g}$. This implies that $\mathrm{ad}(X)$ is skew-symmetric with respect to this inner product or, as it is said in Lie algebra theory, the inner product is invariant by the adjoint representation of $\mathfrak{g}$.

These properties ensure that in the case of a compact group, $\mathrm{ad}(X)$, $X \in \mathfrak{g}$, and $\mathrm{Ad}(g)$, $g \in G$, are semi-simple linear maps, that is, their complexifications are diagonalizable.

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These facts justify the introduction of the following class of Lie algebras, whose definition is purely algebraic.

Definition 11.1 A real Lie algebra $\mathfrak{g}$ is called **compact** if there exists on $\mathfrak{g}$ an **invariant inner product** $(\cdot, \cdot)$, that is,

$$(\operatorname{ad}(X)\, Y, Z) + (Y, \operatorname{ad}(X)\, Z) = 0 \qquad X, Y, Z \in \mathfrak{g}.$$

The Lie algebras of compact Lie groups are compact. Conversely, it will be proved along this chapter that a compact Lie algebra is the Lie algebra of some compact group. The relation between both classes of algebras and Lie groups is not completely closed because there are noncompact Lie groups with compact Lie algebras. For instance, an abelian Lie algebra is compact, since any inner product is invariant. Even though they are the Lie algebras of compact tori, they are also the Lie algebras of noncompact groups such as the simply connected groups diffeomorphic to $\mathbb{R}^n$.

This example of the abelian Lie algebra is essentially unique because a compact Lie algebra $\mathfrak{g}$ decomposes in the direct sum of its center $\mathfrak{z}(\mathfrak{g})$ and a semi-simple ideal $\mathfrak{k}$. This ideal is also a compact algebra and it follows from the Weyl theorem that the simply connected group with Lie algebra $\mathfrak{k}$ is compact.

Before proceeding, it must be noted that if $\mathfrak{h} \subset \mathfrak{g}$ is a subalgebra of the compact algebra $\mathfrak{g}$, then $\mathfrak{h}$ is also compact because the restriction to $\mathfrak{h}$ of the inner product on $\mathfrak{g}$ is an invariant inner product on $\mathfrak{h}$.

The first step in the description of $\mathfrak{g}$ is to obtain its decomposition in a direct sum of ideals.

In general, let $\rho : \mathfrak{h} \to \mathfrak{gl}(V)$ be a representation of a Lie algebra in a real vector space V such that, for every $X \in \mathfrak{g}$, $\rho(X)$ is skew-symmetric with respect to an inner product $(\cdot, \cdot)$. Then, if $W \subset V$ is an invariant subspace, the same happens with its orthogonal complement $W^{\perp}$ because, if $u \in W$, $v \in W^{\perp}$ and $X \in \mathfrak{h}$, then

$$(\rho(X)\, v, u) = -(v, \rho(X)\, u) = 0,$$

since $\rho(X)\, u \in W$. This gives the decomposition $V = W \oplus W^{\perp}$ in invariant subspaces W and $W^{\perp}$. Decomposing the subspaces successively yields a decomposition

$$V = V_1 \oplus \cdots \oplus V_n$$

in invariant subspaces which are irreducible by the representation ρ. (Compare with the proof of Proposition 4.2 of Chapter 3.)

This decomposition can be applied to the adjoint representation of a compact Lie algebra. In this case, an invariant subspace is an ideal of the algebra.

To deal with irreducibility, the following concepts of Lie algebras are used: A Lie algebra $\mathfrak{g}$ is **simple** if $\dim \mathfrak{g} > 1$ and its adjoint representation is irreducible, that is, if the only ideals of $\mathfrak{g}$ are the trivial ones, $\{0\}$ and $\mathfrak{g}$. The Lie algebra $\mathfrak{g}$ is **semi-simple** if $\mathfrak{g} = \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_n$, where each $\mathfrak{g}_i$ is a simple ideal (that is, $\mathfrak{g}_i$ is a

simple Lie algebra). These ideals are called the simple components of $\mathfrak{g}$. It is clear that the center $\mathfrak{z}(\mathfrak{g})$ of a semi-simple Lie algebra is zero, as follows directly from the definitions by the following arguments:

1. If $\mathfrak{g}$ is simple, then $\mathfrak{z}(\mathfrak{g}) = \{0\}$ or $\mathfrak{z}(\mathfrak{g}) = \mathfrak{g}$, since $\mathfrak{z}(\mathfrak{g})$ is an ideal. But if $\mathfrak{z}(\mathfrak{g}) = \mathfrak{g}$, then $\mathfrak{g}$ is abelian and as $\dim \mathfrak{g} > 1$, any of its subspaces would be an ideal. Hence, $\mathfrak{z}(\mathfrak{g}) = \{0\}$.
2. If $\mathfrak{g} = \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_n$, then the simple components commute with each other because if $\mathfrak{a}$ and $\mathfrak{b}$ are ideals with $\mathfrak{a} \cap \mathfrak{b} = \{0\}$, then $[X, Y] = 0$ if $X \in \mathfrak{a}$ and $Y \in \mathfrak{b}$, since $[X, Y] \in \mathfrak{a} \cap \mathfrak{b}$. Now, write $Z \in \mathfrak{z}(\mathfrak{g})$ in the form $Z = Z_1 + \cdots + Z_n$, given by the decomposition of $\mathfrak{g}$. Take $X \in \mathfrak{g}_j$.Then $[X, Z] = [X, Z_j] = 0$, hence $Z_j \in \mathfrak{z}\left(\mathfrak{g}_j\right) = \{0\}$. That is, $Z_j = 0$ for any index j and so $Z = 0$.

Another property to be used in the proof below is that, if $\mathfrak{g}$ is semi-simple, then its derived algebra $\mathfrak{g}'$ coincides with $\mathfrak{g}$. Indeed, if $\mathfrak{g}$ is simple, then $\mathfrak{g}'$ is an ideal $\neq \{0\}$ (as $\mathfrak{g}$ is not abelian), and hence $\mathfrak{g}' = \mathfrak{g}$. In the semi-simple case, this argument ensures that each simple component is contained in the subspace spanned by the images of the adjoint maps. Hence the semi-simple algebra, which is the direct sum of the simple components, also satisfies this property.

Having these concepts in mind the decomposition of a compact Lie algebra is given in the next theorem.

Theorem 11.2 *A compact Lie algebra* $\mathfrak{g}$ *has the decomposition*

$$\mathfrak{g} = \mathfrak{z}(\mathfrak{g}) \oplus \mathfrak{u},$$

where $\mathfrak{z}(\mathfrak{g})$ *is the center of* $\mathfrak{g}$ *and* $\mathfrak{u}$ *is a semi-simple ideal. This decomposition is unique since* $\mathfrak{u} = \mathfrak{z}(\mathfrak{u})^{\perp} = \mathfrak{g}'$.

Proof Take a decomposition of $\mathfrak{g}$ into invariant subspaces which are irreducible by the adjoint representation and write this decomposition as

$$\mathfrak{g} = \mathfrak{i}_1 \oplus \cdots \oplus \mathfrak{i}_m \oplus \mathfrak{u}_1 \oplus \cdots \oplus \mathfrak{u}_n, \tag{11.1}$$

in such a way that $\dim \mathfrak{i}_j = 1$ and $\dim \mathfrak{u}_j > 1$. Each component is an ideal of $\mathfrak{g}$. They commute with each other as they are ideals with zero intersection. Moreover, the ideals $\mathfrak{u}_j$ with $\dim \mathfrak{u}_j > 1$ are simple because an ideal $\mathfrak{a} \subset \mathfrak{u}_j$ of $\mathfrak{u}_j$ is also an ideal of $\mathfrak{g}$, since $\mathfrak{u}_j$ commutes with the other components.

Write $\mathfrak{i} = \mathfrak{i}_1 \oplus \cdots \oplus \mathfrak{i}_m$ and $\mathfrak{u} = \mathfrak{u}_1 \oplus \cdots \oplus \mathfrak{u}_n$. Then $\mathfrak{u}$ is semi-simple as it is a sum of simple ideals. The ideal $\mathfrak{i}$ is abelian because $\dim \mathfrak{i}_j = 1$. As $\mathfrak{i}$ commutes with $\mathfrak{u}$, it follows that $\mathfrak{i} \subset \mathfrak{z}(\mathfrak{g})$. Actually, this inclusion is an equality since otherwise $\mathfrak{z}(\mathfrak{g}) \cap \mathfrak{u} \neq \{0\}$. But $\mathfrak{z}(\mathfrak{g}) \cap \mathfrak{u} \subset \mathfrak{z}(\mathfrak{u}) = \{0\}$ because $\mathfrak{u}$ is semi-simple.

The uniqueness is a consequence of the fact that if $\mathfrak{u}$ is a semi-simple subalgebra that complements $\mathfrak{z}(\mathfrak{g})$, then $\mathfrak{u} = \mathfrak{z}(\mathfrak{g})^{\perp}$. To see this, it suffices to verify that $\mathfrak{u} \subset \mathfrak{z}(\mathfrak{g})^{\perp}$ due to the dimensions of the subspaces. This, in turn, is proved by showing that if $X \in \mathfrak{u}$, then the image of $\operatorname{ad}(X)$ is orthogonal to $\mathfrak{z}(\mathfrak{g})$, as $\mathfrak{u}$ is spanned by the

images of $\operatorname{ad}(X)$, $X \in \mathfrak{u}$. But this follows immediately from the invariance of the inner product since if $Y = \operatorname{ad}(X)(W)$, $W \in \mathfrak{u}$, then, for every $Z \in \mathfrak{z}(\mathfrak{g})$, it holds

$$(Z, Y) = (Z, \operatorname{ad}(X)(W)) = -(\operatorname{ad}(X)(Z), W) = 0,$$

concluding the proof. □

In the previous theorem the semi-simple ideal $\mathfrak{u}$, as well as its simple components, are compact Lie algebras. This is so because every subalgebra of a compact Lie algebra is also compact. It follows from this observation that a compact Lie algebra $\mathfrak{g}$ is semi-simple if and only if $\mathfrak{z}(\mathfrak{g}) = \{0\}$.

The invariant inner product $(\cdot, \cdot)$ on a compact Lie algebra is not unique. For instance, $a(\cdot, \cdot)$, $a > 0$, is also invariant or, still, a direct sum of invariant inner products on the components of decomposition (11.1) is also an invariant inner product, so that the set of invariant inner products is a cone of dimension equal to the number of components in (11.1).

There is, however, a natural choice based on the Cartan–Killing form of $\mathfrak{g}$, which is defined by

$$\mathcal{K}_{\mathfrak{g}}(X, Y) = \operatorname{tr}(\operatorname{ad}(X) \operatorname{ad}(Y)) \qquad X, Y \in \mathfrak{g}.$$

Proposition 11.3 *Let $\mathfrak{g}$ be a compact Lie algebra. Then its Cartan–Killing form $\mathcal{K}_{\mathfrak{g}}(\cdot, \cdot)$ is negative semidefinite. Moreover, for $X \in \mathfrak{g}$, $\mathcal{K}_{\mathfrak{g}}(X, X) = 0$ if and only if $X \in \mathfrak{z}(\mathfrak{g})$. Therefore, $\mathfrak{g}$ is semi-simple if and only if $\mathcal{K}_{\mathfrak{g}}(\cdot, \cdot)$ is negative definite.*[1]

Proof If $X \in \mathfrak{g}$, then $\operatorname{ad}(X)$ is skew-symmetric with respect to an inner product, therefore its eigenvalues are purely imaginary. Let $ia_1, \ldots, ia_n \in i\mathbb{R}$ be the eigenvalues of $\operatorname{ad}(X)$. Then,

$$\mathcal{K}_{\mathfrak{g}}(X, X) = \operatorname{tr}\left(\operatorname{ad}(X)^2\right) = -\left(a_1^2 + \cdots + a_k^2\right) \leq 0,$$

showing that $\mathcal{K}_{\mathfrak{g}}(\cdot, \cdot)$ is negative semidefinite. It also follows from this expression that $\mathcal{K}_{\mathfrak{g}}(X, X) = 0$ if and only if the eigenvalues of $\operatorname{ad}(X)$ are all zero. But as it is skew-symmetric, $\operatorname{ad}(X)$ is a semi-simple linear map, hence $\operatorname{ad}(X) = 0$ if and only if its eigenvalues are zero. Therefore, $\mathcal{K}_{\mathfrak{g}}(X, X) = 0$ if and only if $X \in \mathfrak{z}(\mathfrak{g})$.

Finally, if $\mathfrak{g}$ is semi-simple, then $\mathfrak{z}(\mathfrak{g}) = \{0\}$. Hence, $\mathcal{K}_{\mathfrak{g}}(X, X) = 0$ implies that $X = 0$, showing that the Cartan–Killing form is negative definite. Conversely, if $\mathcal{K}_{\mathfrak{g}}(\cdot, \cdot)$ is negative definite, then $\mathfrak{z}(\mathfrak{g}) = \{0\}$ and $\mathfrak{g}$ is semi-simple. □

In general, for an arbitrary Lie algebra the Cartan–Killing form is invariant by the adjoint representation. The above proposition then ensures that $-\mathcal{K}_{\mathfrak{g}}(\cdot, \cdot)$ is

[1] One of the Cartan criteria generalizes this last statement by showing that a Lie algebra is semi-simple if and only if its Cartan–Killing form is not degenerate. See *Álgebras de Lie* [47], Chapter 3 and Varadarajan [53], Section 3.9.

an invariant inner product on a compact semi-simple Lie algebra $\mathfrak{g}$. Among the invariant inner products, the negative of the Cartan–Killing form is a natural choice as it is intrinsically defined from the Lie algebra. For compact Lie algebras in general it is possible to take the Cartan–Killing form on the semi-simple component and extend it with an arbitrary inner product on the center.

Turning to the Lie groups with compact Lie algebras, the first step is to look at the group of automorphisms $\mathrm{Aut}\mathfrak{g}$ of a compact Lie algebra $\mathfrak{g}$.

Proposition 11.4 *Let $\mathfrak{g}$ be a compact semi-simple Lie algebra. Then the group $\mathrm{Aut}\mathfrak{g}$ of automorphisms of $\mathfrak{g}$ is compact.*

Proof The group $\mathrm{Aut}\mathfrak{g}$ is a closed subgroup of $\mathrm{Gl}(\mathfrak{g})$. Moreover, if ϕ is an automorphism of $\mathfrak{g}$, then $\mathrm{ad}(\phi X) = \phi \mathrm{ad}(X)\phi^{-1}$ and this implies that ϕ is an isometry of the Cartan–Killing form. As $\mathcal{K}_{\mathfrak{g}}(\cdot,\cdot)$ is negative definite, its group of isometries is compact, hence $\mathrm{Aut}\mathfrak{g}$ is compact. □

Consequently, the connected component of the identity $\mathrm{Aut}_0\mathfrak{g}$ of $\mathrm{Aut}\mathfrak{g}$ is a compact Lie group. The Lie algebra of these groups is $\mathrm{Der}\mathfrak{g}$, the algebra of derivations of $\mathfrak{g}$. In case $\mathfrak{g}$ is semi-simple, $\mathrm{Der}\mathfrak{g}$ is isomorphic to $\mathfrak{g}$ by the adjoint representation. Indeed ad is injective since $\ker \mathrm{ad} = \mathfrak{z}(\mathfrak{g}) = \{0\}$. On the other hand, a Lie algebra theorem ensures that if $\mathfrak{g}$ is semi-simple (not necessarily compact), then every derivation of $\mathfrak{g}$ is an inner derivation, that is, if $D \in \mathrm{Der}\mathfrak{g}$, then there exists $X \in \mathfrak{g}$ such that $D = \mathrm{ad}(X)$. This shows the surjectivity of ad, showing that $\mathrm{ad} : \mathfrak{g} \to \mathrm{Der}\mathfrak{g}$ is an isomorphism.

Therefore, $\mathrm{Aut}_0\mathfrak{g}$ is a connected compact group with Lie algebra (isomorphic to) $\mathfrak{g}$. Conversely, if $\mathrm{Aut}_0\mathfrak{g}$ is compact, then $\mathfrak{g}$ is a compact Lie algebra because it is the Lie algebra of a compact group.

Actually, if $\mathfrak{g}$ is a semi-simple Lie algebra (not necessarily compact), then $\mathrm{Aut}_0\mathfrak{g}$ is the smallest Lie group with Lie algebra $\mathfrak{g}$. Indeed, if G is a connected Lie group with Lie algebra $\mathfrak{g}$, then $\mathrm{Ad} : G \to \mathrm{Aut}_0\mathfrak{g}$ is a local diffeomorphism, since $\mathrm{ad} : \mathfrak{g} \to \mathrm{Der}\mathfrak{g}$ is an isomorphism. This implies that Ad is surjective, because its image is an open subgroup of $\mathrm{Aut}_0\mathfrak{g}$. Hence $\mathrm{Aut}_0\mathfrak{g} \approx G/Z(G)$ and therefore $\mathrm{Aut}_0\mathfrak{g}$ is the quotient of any connected Lie group G with Lie algebra $\mathfrak{g}$. These facts are included in the following proposition for future reference.

Proposition 11.5 *Let $\mathfrak{g}$ be a semi-simple Lie algebra, then $\mathrm{Aut}_0\mathfrak{g} = G/Z(G)$ for every connected Lie group with Lie algebra $\mathfrak{g}$ and the center of $\mathrm{Aut}_0\mathfrak{g}$ is trivial. Moreover, $\mathrm{Aut}_0\mathfrak{g}$ is compact if and only if $\mathfrak{g}$ is a compact Lie algebra.*

Proof It remains to show that $Z(\mathrm{Aut}_0\mathfrak{g}) = \{1\}$. So let $\widetilde{G}$ be the connected and simply connected Lie group with Lie algebra $\mathfrak{g}$ and denote by $p : \widetilde{G} \to \widetilde{G}/Z\left(\widetilde{G}\right) = \mathrm{Aut}_0\mathfrak{g}$ the canonical projection. The subgroup $p^{-1}(Z(\mathrm{Aut}_0\mathfrak{g}))$ is discrete and normal, so that $p^{-1}(Z(\mathrm{Aut}_0\mathfrak{g})) \subset Z\left(\widetilde{G}\right)$ and therefore $Z(\mathrm{Aut}_0\mathfrak{g}) = \{1\}$. □

The previous paragraphs show that if $\mathfrak{g}$ is a semi-simple compact Lie algebra, then there exists at least one compact Lie group G with Lie algebra $\mathfrak{g}$, namely the group $\mathrm{Aut}_0\mathfrak{g}$. In the next section it is proved that a Lie group is compact if its Lie

algebra is compact and semi-simple. This information, together with Theorem 11.2, provides the final result that the universal covering of a connected compact Lie group has the form $\mathbb{R}^n \times G$, where G is a simply connected compact Lie group whose Lie algebra is semi-simple.

11.2 Finite Fundamental Group

Let $\mathfrak{g}$ be a compact semi-simple Lie algebra and denote by $\widetilde{G}$ the connected and simply connected Lie group with Lie algebra $\mathfrak{g}$. The objective of this section is to present a first proof of the Weyl theorem, which ensures that $\widetilde{G}$ is compact. By this theorem, every connected Lie group with Lie algebra $\mathfrak{g}$ is compact. The claim that $\widetilde{G}$ is compact is equivalent to saying that the fundamental group of $\mathrm{Aut}_0\mathfrak{g}$ is finite (hence the title of this section). This is so because $\mathrm{Aut}_0\mathfrak{g}$ is a compact group whose Lie algebra is $\mathrm{Der}\mathfrak{g} \approx \mathfrak{g}$, as seen in the previous section.

Later, another proof of the Weyl theorem will be presented which relies on the algebraic and geometric structure of compact Lie algebras and Lie groups.[2] Although more engaging, that proof has the advantage of furnishing the fundamental group of $\mathrm{Aut}_0\mathfrak{g}$. The proof presented here is of existential kind and has an analytic approach. It is based on the following theorem about extensions of homomorphisms.

Theorem 11.6 *Suppose that L is a connected Lie group and $\Gamma \subset L$ a discrete central subgroup such that L/Γ is compact. Let $\theta : \Gamma \to \mathbb{R}_+$ be a homomorphism with values in the multiplicative group of reals. Then there exists a differentiable homomorphism $\phi : L \to \mathbb{R}_+$ which extends θ, that is, $\phi(\gamma) = \theta(\gamma)$ if $\gamma \in \Gamma$.*

Before this extension theorem is proved, it will be used to prove the Weyl theorem.

The idea is that if G is a connected Lie group whose Lie algebra $\mathfrak{g}$ is semi-simple, then the only homomorphism ϕ of G with values in $\mathbb{R}_+$ is the trivial one, $\phi \equiv 1$ (see Lemma 11.9 below). On the other hand, if $\mathfrak{g}$ is a compact semi-simple Lie algebra and $\mathrm{Aut}_0\mathfrak{g} = \widetilde{G}/\Gamma$, then the compactness of $\mathrm{Aut}_0\mathfrak{g}$ allows to show that the abelian group Γ is finitely generated. In this way, if Γ were infinite, there would exist a nontrivial homomorphism $\theta : \Gamma \to \mathbb{R}_+$. By the extension theorem θ would be the restriction of a nontrivial homomorphism $\phi : \widetilde{G} \to \mathbb{R}_+$, contradicting the fact that $\mathfrak{g}$ is semi-simple. To carry out this proof, the following lemmas, proved in sequence, are required.

Lemma 11.7 *Let G be a Lie group and $H \subset G$ a closed subgroup, such that G/H is compact. Then, there exists a compact set with nonempty interior C, such that $1 \in C^\circ$ and $G = C^\circ H$. Moreover, it is possible to take C such that $C^{-1} = C$.*

[2] A third proof, using Riemannian geometry, is pointed out at the end of this chapter. A fourth proof, arguing with curves, can be found in Zelobenko [61].

Proof Let V be a compact neighborhood of $1 \in G$, such that $V^{-1} = V$. Then, for each $x = gH \in G/H$, the set $Vx = \{hx : h \in V\}$ is a neighborhood of x on G/H. The sets Vx cover G/H and, as G/H is compact, there exist $x_1 = g_1 H, \ldots, x_n = g_n H$, such that $G/H = Vx_1 \cup \cdots \cup Vx_n$. Let $C = V \cup Vg_1 \cup \cdots \cup Vg_n$. Then, by construction, $1 \in V^\circ \subset C^\circ$ and if $g \in G$, then $gH \in Vx_i$ for some i, which implies that $g \in Vg_i H$. Therefore, taking the union of C with $g_1^{-1}V^{-1} \cup \cdots \cup g_n^{-1}V^{-1}$ yields a new compact set which satisfies the desired properties. □

Lemma 11.8 *Let $\mathfrak{g}$ be a semi-simple compact Lie algebra and denote by $\widetilde{G}$ the connected and simply connected group with Lie algebra $\mathfrak{g}$. Write $\mathrm{Aut}_0\mathfrak{g} = \widetilde{G}/\Gamma$, with Γ isomorphic to the fundamental group $\pi_1(\mathrm{Aut}_0\mathfrak{g})$. Then Γ is finitely generated.*

Proof Let $C = C^{-1}$ be the symmetric compact set of the previous lemma with $1 \in C^\circ$ and

$$\widetilde{G} = C^\circ\Gamma = \bigcup_{\gamma\in\Gamma} C^\circ\gamma.$$

The set C^2 is compact and since the open sets $C^\circ\gamma$, $\gamma \in \Gamma$, cover C^2, there exists a finite set $\{\gamma_1, \ldots, \gamma_n\} \subset \Gamma$ such that

$$C^2 \subset C^\circ\gamma_1 \cup \cdots \cup C^\circ\gamma_n. \tag{11.2}$$

Let Γ_1 be the subgroup of Γ generated by $\{\gamma_1, \ldots, \gamma_n\}$.

The lemma is a consequence of the equality $\Gamma = \left(C^2 \cap \Gamma\right)\Gamma_1$. Indeed, $C^2 \cap \Gamma$ is finite, because C^2 is compact and Γ is discrete. As Γ_1 is finitely generated, this equality proves the lemma.

To prove that $\Gamma = \left(C^2 \cap \Gamma\right)\Gamma_1$, let $p : \widetilde{G} \to \widetilde{G}/\Gamma_1$ be the canonical homomorphism. Then, $p\left(C^2\right)$ has nonempty interior in the group $\widetilde{G}/\Gamma_1$, as $C^\circ \neq \emptyset$. Moreover, $p\left(C^2\right)$ is a subgroup. Indeed, if $g, h \in C^2$, then, by (11.2), there exist $c_1, c_2 \in C^\circ$ and $\gamma_1, \gamma_2 \in \Gamma_1$ such that $g = c_1\gamma_1$ and $h = c_2\gamma_2$. As Γ is a central subgroup, $gh^{-1} = c_1c_2^{-1}\gamma_1\gamma_2^{-1}$ and therefore

$$p(g)\,p(h)^{-1} = p\left(gh^{-1}\right) = p\left(c_1c_2^{-1}\right) \in p\left(C^2\right),$$

as $C^{-1} = C$. Consequently, $p\left(C^2\right)$ is an open subgroup of the connected group $\widetilde{G}/\Gamma_1$, so that $p\left(C^2\right) = \widetilde{G}/\Gamma_1$.

This means that, for every $g \in \widetilde{G}$, there exists $\gamma \in \Gamma_1$ such that $g\gamma \in C^2$. In particular, if $g \in \Gamma$, then $g\gamma \in C^2 \cap \Gamma$, which shows that $\Gamma \subset \left(C^2 \cap \Gamma\right)\Gamma_1$ and hence $\Gamma = \left(C^2 \cap \Gamma\right)\Gamma_1$, concluding the proof. □

Lemma 11.9 *Suppose that G is a connected Lie group whose Lie algebra $\mathfrak{g}$ is semi-simple (compact or not). Then, the only differentiable homomorphism $\phi : G \to \mathbb{R}_+$ is the trivial homomorphism $\phi(g) = 1$ for every $g \in G$.*

Proof It suffices to show that the infinitesimal homomorphism $d\phi_1 : \mathfrak{g} \to \mathbb{R}$ is identically zero, as G is connected. Since $\mathbb{R}$ is an abelian algebra, $d\phi_1$ is zero in a bracket, that is,

$$d\phi_1[X, Y] = [d\phi_1 X, d\phi_1 Y] = 0.$$

But $\mathfrak{g}$ is semi-simple, and thus $\mathfrak{g}' = \mathfrak{g}$, where $\mathfrak{g}'$ is the derived algebra. This means that $\mathfrak{g}$ is generated by the brackets $[X, Y]$, $X, Y \in \mathfrak{g}$. Hence, $d\phi_1$ is identically zero, concluding the proof. □

It is now possible to formally state and prove the **Weyl theorem** of the finiteness of the fundamental group.

Theorem 11.10 *Let $\mathfrak{g}$ be a compact semi-simple Lie algebra and denote by $\widetilde{G}$ the simply connected group with Lie algebra $\mathfrak{g}$. Then, $\widetilde{G}$ is compact.*

Proof Let $\Gamma \approx \pi_1(\mathrm{Aut}_0\mathfrak{g})$ be the discrete central subgroup such that $\mathrm{Aut}_0\mathfrak{g} = \widetilde{G}/\Gamma$. It must be proved that Γ is finite, since $\mathrm{Aut}_0\mathfrak{g}$ is compact. By Lemma 11.8, the abelian group Γ is finitely generated and therefore isomorphic to $\mathbb{Z}^k \times \mathbb{Z}_{m_1} \times \cdots \times \mathbb{Z}_{m_n}$. Suppose by contradiction that Γ is not finite, so that $k \geq 1$. Then there exists a nontrivial homomorphism $\theta : \Gamma \to \mathbb{R}_+$ of the form $\theta(\gamma) = e^{p(\gamma)}$, where p is a projection into one of the components of $\mathbb{Z}^k$. By Theorem 11.6, there exists a homomorphism $\phi : \widetilde{G} \to \mathbb{R}_+$ which extends θ. This homomorphism is nontrivial, which is a contradiction by Lemma 11.9. □

11.2.1 Extension Theorem

In this section it is proved Theorem 11.6, which ensures that a homomorphism θ of a discrete central subgroup $\Gamma \subset L$ extends to L if L/Γ is compact.

The first step is the following lemma, which extends θ to a continuous function satisfying the homomorphism property when elements of Γ are involved.

Lemma 11.11 *There exists a continuous function $f : L \to \mathbb{R}_+$ such that:*

1. $f(\gamma) = \theta(\gamma)$ *if* $\gamma \in \Gamma$.
2. $f(x) > 0$ *for every* $x \in L$ *and* $f(1) = 1$.
3. $f(x\gamma) = f(x)\theta(\gamma)$ *for every* $x \in L$.

Proof Let $p : L \to L/\Gamma$ be the canonical homomorphism and take a compact set $C \subset L$ such that $p(C) = L/\Gamma$ given by Lemma 11.7. Then there exists a continuous function with compact support $g : L \to \mathbb{R}$ such that $g(x) = 1$ if $x \in C$ and $g(y) \geq 0$ for every $y \in L$.

Define the function $f_0 : L \to \mathbb{R}$ by

$$f_0(x) = \sum_{\gamma \in \Gamma} g(x\gamma)\,\theta\left(\gamma^{-1}\right). \tag{11.3}$$

This function is well defined and is continuous. Indeed, denote by K the support of g and take a compact set with nonempty interior $U \subset L$. Then, for $x \in U$ and $\gamma \in \Gamma$, $g(x\gamma) = 0$ unless $x\gamma \in K$, that is, $\gamma \in x^{-1}K$. Hence, for $x \in U$, the sum in (11.3) extends to $\Gamma_U = U^{-1}K \cap \Gamma$. But this set is finite because Γ is discrete and $U^{-1}K$ is compact. Hence the restriction of f_0 to U is a finite sum of continuous functions and therefore f_0 is continuous in U. It follows that f_0 is continuous, since U is arbitrary.

This function is strictly positive since, given $x \in L$, there exists $\gamma \in \Gamma$ such that $x\gamma \in C$ and $g(x\gamma) = 1$, by the choices of C and g. Thus it is possible to define the function $f(x) = f_0(x)/f(1)$ that satisfies the second property stated. The third property comes from

$$\begin{aligned} f(x\gamma_0) &= \sum_{\gamma \in \Gamma} g(x\gamma_0\gamma)\,\theta\left(\gamma^{-1}\right) = \sum_{\gamma \in \Gamma} g(x\gamma_0\gamma)\,\theta\left(\gamma^{-1}\gamma_0^{-1}\right)\theta(\gamma_0) \\ &= \left(\sum_{\gamma \in \Gamma} g(x\gamma)\,\theta\left(\gamma^{-1}\right)\right)\theta(\gamma_0) = f(x)\,\theta(\gamma_0). \end{aligned}$$

Finally, $f(\gamma) = f(1\gamma) = f(1)\,\theta(\gamma) = \theta(\gamma)$ if $\gamma \in \Gamma$. □

The idea now is to define a new function $h > 0$ such that $h(x\gamma) = h(x)$, $\gamma \in \Gamma$, and which satisfies

$$h(xy)\,h(x)^{-1}\,h(y)^{-1} = f(xy)\,f(x)^{-1}\,f(y)^{-1}.$$

Once this function h is obtained, the desired homomorphism will be $\phi(x) = f(x)\,h(x)^{-1}$.

To obtain h, consider the continuous function $F : L \times L \to \mathbb{R}$ given by

$$F(x, y) = \log f(xy) - \log f(x) - \log f(y).$$

Then, for every $\gamma \in \Gamma$, it holds

$$\begin{aligned} F(x\gamma, y\gamma) &= \log f\left(xy\gamma^2\right) - \log f(x\gamma) - \log f(y\gamma) \\ &= F(x, y), \end{aligned}$$

as $\Gamma \subset Z(L)$. This means that there exists a continuous function $F_0 : (L/\Gamma) \times (L/\Gamma) \to \mathbb{R}$ such that $F(x, y) = F_0(px, py)$, where $p : L \to L/\Gamma$ is the canonical projection.

A straightforward computation using its definition shows that the function F satisfies the identity

$$F(xy, u) - F(y, u) = F(x, yu) - F(x, y), \tag{11.4}$$

which is also satisfied by F_0. Moreover, $F(1, y) = F(x, 1) = 0$, as $f(1) = 1$.

Lemma 11.12 *There exists a continuous function* $a : L/\Gamma \to \mathbb{R}$ *such that* $F_0(x, y) = a(xy) - a(x) - a(y)$ *and* $a(1) = 0$.

Proof Define

$$b(x) = \int_{L/\Gamma} F_0(x, u)\, \mu(du),$$

where μ is the Haar measure of L/Γ normalized by $\mu(L/\Gamma) = 1$. The integral exists, as F_0 is continuous and L/Γ is compact. Then, $b(xy) - b(x) - b(y)$ is given by the following integral,

$$\int_{L/\Gamma} (F_0(xy, u) - F_0(x, u) - F_0(y, u))\, \mu(du),$$

which, by the identity (11.4), is rewritten as

$$\int_{L/\Gamma} (F_0(x, yu) - F_0(x, y) - F_0(x, u))\, \mu(du).$$

The integrals of the first and third integrands cancel each other, as μ is left invariant. Hence,

$$b(xy) - b(x) - b(y) = -\int_{L/\Gamma} F_0(x, y)\, \mu(du) = -F_0(x, y).$$

It follows that $a = -b$ is the desired function, given that $a(1) = \int_{L/\Gamma} F_0(1, u)\, \mu(du) = 0$ because $F(1, u) = 0$. □

Turning to the definition of the extension ϕ, let $h : L \to \mathbb{R}_+$ be defined by

$$h(x) = \exp a(px),$$

where $a : L/\Gamma \to \mathbb{R}$ is given by the previous lemma. Then $h(x\gamma) = h(x)$ if $x \in L$ and $\gamma \in \Gamma$. From the property of the function a it follows that

$$\begin{aligned} h(xy)\, h(x)^{-1}\, h(y)^{-1} &= \exp(a(p(xy)) - a(p(x)) - a(p(y))) \\ &= \exp F_0(px, py) = \exp F(x, y). \end{aligned}$$

But $F(x, y) = \log f(xy) - \log f(x) - \log f(y)$, hence

$$h(xy)\, h(x)^{-1}\, h(y)^{-1} = f(xy)\, f(x)^{-1}\, f(y)^{-1},$$

that is, $f(xy)/h(xy) = (f(x)/h(x))(f(y)/h(y))$. Hence $\phi(x) = f(x)/h(x)$ is a homomorphism which extends θ since, if $\gamma \in \Gamma$, then

$$\phi(\gamma) = f(\gamma)/h(\gamma) = f(\gamma) = \theta(\gamma),$$

as $h(\gamma) = e^{a(1)} = 0$, concluding the proof of the extension theorem.

11.3 Compact and Complex Lie Algebras

The objective of this section is to describe the structure of a semi-simple compact Lie algebra $\mathfrak{u}$ from its complexification $\mathfrak{g} = \mathfrak{u}_{\mathbb{C}}$, which is also a semi-simple Lie algebra. The point is that the eigenvalues of $\operatorname{ad}(X)$, $X \in \mathfrak{u}$, are purely imaginary and therefore the eigenspaces of $\operatorname{ad}(X)$ are not contained in $\mathfrak{u}$ but in $\mathfrak{g}$. The brackets between the eigenspaces describe the Lie algebra structure of $\mathfrak{g}$ and hence of $\mathfrak{u}$ through the **Weyl unitary trick**, which establishes a bijection between the compact semi-simple Lie algebras and the complex Lie algebras. The name unitary comes from the example $\mathfrak{su}(n)$, which is the Lie algebra of the special unitary group $\mathrm{SU}(n)$. This example will appear along the exposition.[3]

11.3.1 Weyl Unitary Trick

To describe the compact semi-simple Lie algebras, it suffices to obtain the simple algebras. The simple compact Lie algebras are obtained by the Weyl construction, which involves the complexification $\mathfrak{g} = \mathfrak{u}_{\mathbb{C}}$ of the compact algebra $\mathfrak{u}$. From the theory of real Lie algebras, it is known that $\mathfrak{g}$ is a complex simple Lie algebra. (In principle, the complexification of a simple real Lie algebra is semi-simple by the well known Cartan criterion for semi-simple algebras.[4]The simple algebras whose complexifications are not simple are the realifications of the complex simple Lie algebras. These Lie algebras are not compact. Hence the complexification of a compact simple Lie algebra is also simple.)

[3]The Lie algebras $\mathfrak{so}(n)$ are not as good as $\mathfrak{su}(n)$ as guiding examples because their Cartan subalgebras are not given—in the natural representation—as algebras of diagonal matrices, as is the case with $\mathfrak{su}(n)$.

[4]See *Álgebras de Lie* [47], Chapter 3 and Varadarajan [53], Section 3.9.

As it is a simple complex algebra, $\mathfrak{g} = \mathfrak{u}_{\mathbb{C}}$ is one of the Lie algebras of the Cartan–Killing classification. This classification is encoded by the Dynkin diagrams, which are reproduced below in Subsection 11.3.2.

On the other hand, the Lie algebra $\mathfrak{u}$ is a compact real form of $\mathfrak{g}$. One of the central results related to the Weyl unitary trick is that two compact real forms of the complex simple Lie algebra $\mathfrak{g}$ are obtained from each other by an automorphism of $\mathfrak{g}$ and, hence, are isomorphic. This means that the classification of the simple complex Lie algebras also classifies the simple compact algebras: To each Dynkin diagram there corresponds a unique equivalence class of simple compact Lie algebras and, conversely, every simple compact algebra is obtained from a Dynkin diagram.

Before describing the general Weyl construction, which establishes a bijection between compact and complex simple Lie algebras, it is convenient to see the example of the Lie algebra $\mathfrak{su}(n)$ which models this construction.

Example The Lie algebra of skew-Hermitian matrices

$$\mathfrak{su}(n) = \{A \in \mathfrak{gl}(n, \mathbb{C}) : A + \overline{A}^T = 0\}$$

is the Lie algebra of the group SU (n). It is simple and compact. The complexification of $\mathfrak{su}(n)$ is (isomorphic to) $\mathfrak{sl}(n, \mathbb{C})$, since a matrix X can be written as

$$X = \frac{X - \overline{X}^T}{2} + \frac{X + \overline{X}^T}{2} \in \mathfrak{su}(n) + i\mathfrak{su}(n).$$

The diagonal elements of a skew-Hermitian matrix are purely imaginary and $\mathfrak{su}(n)$ contains the algebra of diagonal matrices

$$\mathfrak{t} = \{\mathrm{diag}\{ix_1, \ldots, ix_n\} : x_j \in \mathbb{R},\ x_1 + \cdots + x_n = 0\}.$$

This subalgebra is a maximal abelian algebra, as a nondiagonal matrix does not commute with some diagonal matrix.

The complexification $\mathfrak{h} = \mathfrak{t}_{\mathbb{C}}$ of $\mathfrak{t}$ is the algebra of diagonal matrices in $\mathfrak{sl}(n, \mathbb{C})$, which is also a maximal abelian algebra. If $H \in \mathfrak{h}$ is the diagonal matrix $H = \mathrm{diag}\{a_1, \ldots, a_n\}$, then the eigenvalues of $\mathrm{ad}(H)$ are 0 and $\alpha_{jk}(H) = a_j - a_k$, $j \neq k$. The eigenspace associated with the eigenvalue $a_j - a_k$ contains the 1-dimensional subspace $\mathfrak{g}_{jk}$ spanned by the basic matrix $E_{jk} = \left(\delta_{rj}\delta_{sk}\right)_{r,s}$. These eigenspaces, common to $\mathrm{ad}(H)$, $H \in \mathfrak{h}$, decompose $\mathfrak{g} = \mathfrak{sl}(n, \mathbb{C})$ as

$$\mathfrak{g} = \mathfrak{h} \oplus \sum_{j \neq k} \mathfrak{g}_{jk}.$$

The subalgebras $\mathfrak{t}$ and $\mathfrak{h}$ are Cartan subalgebras of $\mathfrak{su}(n)$ and $\mathfrak{sl}(n, \mathbb{C})$, respectively (see definition below). In the terminology of complex Lie algebras, the linear functionals α_{jk}, $j \neq k$, are called roots of $\mathfrak{h}$, while the subspaces $\mathfrak{g}_{jk}$ are the corresponding root spaces.

The subspace $\mathfrak{h}_{\mathbb{R}}$ of real diagonal matrices is defined from the roots α_{jk} as the set of elements $H \in \mathfrak{h}$ such that $\alpha_{jk}(H) \in \mathbb{R}$ for every root α_{jk}. With this, the Cartan subalgebra $\mathfrak{t} = i\mathfrak{h}_{\mathbb{R}}$ of $\mathfrak{su}(n)$ is obtained from the roots. The Lie algebra $\mathfrak{su}(n)$ is generated by $\mathfrak{t}$ and

$$\{E_{jk} - E_{kj}, i\left(E_{jk} + E_{kj}\right)\},$$

where E_{jk}, $j \neq k$, are the generators of the root spaces $\mathfrak{g}_{jk}$. □

Given a complex Lie algebra $\mathfrak{g}$, the starting point for the **Weyl construction** of a compact real form is a Cartan subalgebra.

In general, if $\mathfrak{g}$ is a Lie algebra, then it is said that $\mathfrak{h} \subset \mathfrak{g}$ is a **Cartan subalgebra** if $\mathfrak{h}$ is nilpotent and coincides with its normalizer in $\mathfrak{g}$, that is, if $[X, \mathfrak{h}] \subset \mathfrak{h}$, then $X \in \mathfrak{h}$. In case $\mathfrak{g}$ is a complex semi-simple Lie algebra, a subalgebra $\mathfrak{h} \subset \mathfrak{g}$ is a Cartan subalgebra if and only if it is abelian, the adjoints of its elements $\mathrm{ad}(H)$, $H \in \mathfrak{h}$, are diagonalizable and $\mathfrak{h}$ is maximal with these two properties.[5]

A root of the Cartan subalgebra $\mathfrak{h}$ is a linear functional $\alpha \in \mathfrak{h}^*$, such that the root space

$$\mathfrak{g}_\alpha = \{X \in \mathfrak{g} : \forall H \in \mathfrak{h},\ [H, X] = \alpha(H)\, X\}$$

is not $\{0\}$. Let $\Pi \in \mathfrak{h}^*$ be the set of roots. Then it can be proved that $\dim_{\mathbb{C}} \mathfrak{g}_\alpha = 1$ and

$$\mathfrak{g} = \mathfrak{h} \oplus \sum_{\alpha \in \Pi} \mathfrak{g}_\alpha. \tag{11.5}$$

This decomposition in a direct sum is called **root space decomposition** of $\mathfrak{g}$.

From the decomposition (11.5) one obtains a **Weyl basis**, which is formed by a basis of $\mathfrak{h}$ and by elements $X_\alpha \in \mathfrak{g}_\alpha$ such that $\mathcal{K}_{\mathfrak{g}}(X_\alpha, X_{-\alpha}) = 1$ and $[X_\alpha, X_\beta] = m_{\alpha,\beta} X_{\alpha+\beta}$ with $m_{\alpha,\beta} \in \mathbb{R}$. The existence of bases satisfying these conditions is proved in Lie algebra theory.[6]

As the last ingredient, let $\mathfrak{h}_{\mathbb{R}}$ be the (real) subspace of $\mathfrak{h}$ of the elements $H \in \mathfrak{h}$ such that $\alpha(H) \in \mathbb{R}$ for every root $\alpha \in \Pi$. This subspace is spanned over $\mathbb{R}$ by $H_\alpha \in \mathfrak{h}$ defined by $\alpha(\cdot) = \mathcal{K}_{\mathfrak{g}}(H_\alpha, \cdot)$, $\alpha \in \Pi$.

[5] See *Álgebras de Lie* [47], Chapters 4, 6 and Helgason [20], Section III.3.

[6] See *Álgebras de Lie* [47], Chapter 12 and Helgason [20], Section III.6.

Theorem 11.13 *Given a Weyl basis, the subspace* $\mathfrak{u}$ *spanned over* $\mathbb{R}$ *by*

$$i\mathfrak{h}_{\mathbb{R}} \qquad A_\alpha = X_\alpha - X_{-\alpha} \qquad iS_\alpha = i\,(X_\alpha + X_{-\alpha})$$

is a ***compact real form*** *of* $\mathfrak{g}$. *Two compact real forms of* $\mathfrak{g}$ *are isomorphic by an inner automorphism of* $\mathfrak{g}$. *Conversely, every compact semi-simple Lie algebra is the compact real form of a complex Lie algebra* $\mathfrak{g}$.

Two compact semi-simple algebras $\mathfrak{u}_1$ *and* $\mathfrak{u}_2$ *are isomorphic if and only if their complexifications* $(\mathfrak{u}_1)_{\mathbb{C}}$ *and* $(\mathfrak{u}_2)_{\mathbb{C}}$ *are isomorphic.*

The structure of the compact real form $\mathfrak{u}$ is given by the decomposition (11.5). For each root $\alpha \in \Pi$, let $\mathfrak{u}_\alpha$ be the real vector space spanned by A_α and iS_α. Then, $\dim_{\mathbb{R}} \mathfrak{u}_\alpha = 2$ and

$$\mathfrak{u} = \mathfrak{t} \oplus \sum_{\alpha\in\Pi} \mathfrak{u}_\alpha. \tag{11.6}$$

The brackets between the generators of this decomposition are given by

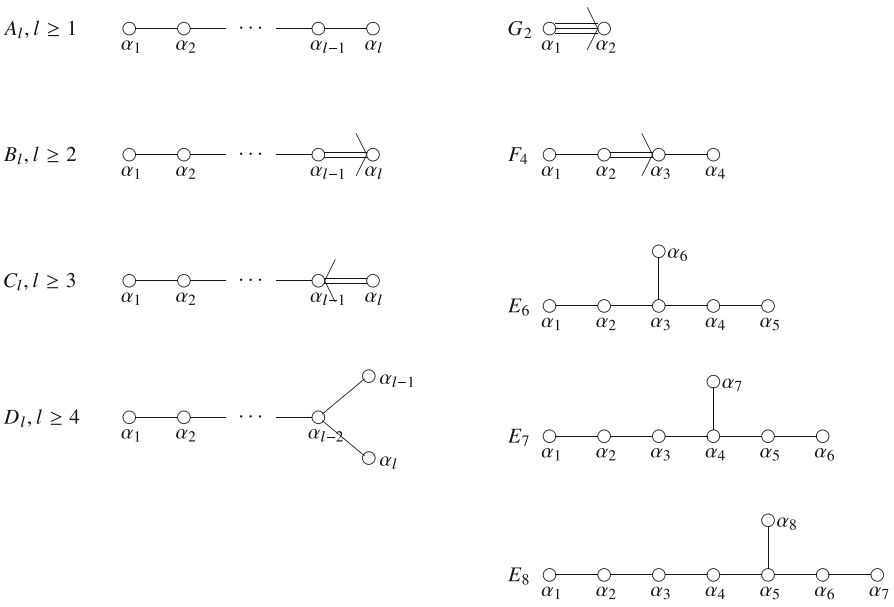
- $[iH_\alpha, A_\beta] = \beta(H_\alpha)S_\beta$.
- $[iH_\alpha, S_\beta] = -\beta(H_\alpha)A_\beta$.
- $[A_\alpha, A_\beta] = m_{\alpha,\beta}A_{\alpha+\beta} + m_{-\alpha,\beta}A_{\alpha-\beta}$.
- $[S_\alpha, S_\beta] = -m_{\alpha,\beta}A_{\alpha+\beta} - m_{\alpha,-\beta}A_{\alpha-\beta}$.
- $[A_\alpha, S_\beta] = m_{\alpha,\beta}S_{\alpha+\beta} + m_{\alpha,-\beta}S_{\alpha-\beta}$.
- $[A_\alpha, S_{-\alpha}] = 2iH_\alpha$.

Example In the case of the Lie algebra $\mathfrak{sl}(n,\mathbb{C})$, the proof of uniqueness (up to conjugation) of the compact real form can be carried out directly in the following manner. Let $\mathfrak{u} \subset \mathfrak{sl}(n,\mathbb{C})$ be a compact real form. Then $\mathfrak{u}$ is a simple algebra since $\mathfrak{sl}(n,\mathbb{C})$ is simple. Therefore, by the Weyl theorem (proved above) the connected subgroup of $\mathrm{Sl}(n,\mathbb{C})$ given by $U = \langle \exp\mathfrak{u}\rangle$ is compact. Hence, there exists a Hermitian metric $\mathcal{H}$ on $\mathbb{C}^n$ that is U-invariant. This implies that there exists $g \in \mathrm{Sl}(n,\mathbb{C})$ such that $gUg^{-1} \subset \mathrm{SU}(n)$ and therefore $g\mathfrak{u}g^{-1} \subset \mathfrak{su}(n)$. But then the equality holds since $\dim\mathfrak{u} = \dim\mathfrak{su}(n)$, as both algebras are real forms of $\mathfrak{sl}(n,\mathbb{C})$. □

11.3.2 Dynkin Diagrams

Theorem 11.13 provides the classification of simple compact Lie algebras from their complexifications. The latter are classified by the **Dynkin diagrams**, listed below.

In these diagrams, the series A_l ($l \geq 1$), B_l ($l \geq 2$), C_l ($l \geq 3$), and D_l ($l \geq 4$) represent the classical Lie algebras, which are listed below with their compact real forms $\mathfrak{u}$.



A_l The complex Lie algebra A_l is $\mathfrak{sl}(l+1,\mathbb{C})$, $n = l+1$, whose compact real form is $\mathfrak{u} = \mathfrak{su}(n)$.

B_l The complex Lie algebra is the algebra of skew-symmetric complex matrices in odd dimension $\mathfrak{so}(2l+1,\mathbb{C})$, with compact real form $\mathfrak{u} = \mathfrak{so}(2l+1,\mathbb{R})$. Such realizations hold for $l \geq 2$, that is, for $\mathfrak{so}(n)$ with $n \geq 5$. The Lie algebra $\mathfrak{so}(3)$ is isomorphic to $\mathfrak{su}(2)$, given in A_1.

C_l Is realized by the symplectic algebra $\mathfrak{sp}(l,\mathbb{C}) = \{A \in M_{2l\times 2l}(\mathbb{C}) : AJ + JA^T = 0\}$, where

$$J = \begin{pmatrix} 0_{l\times l} & -1 \\ 1 & 0_{l\times l} \end{pmatrix}.$$

This Lie algebra is formed by $2l \times 2l$ complex symplectic matrices, which are matrices of type

$$\begin{pmatrix} A & B \\ C & -A^T \end{pmatrix} \qquad B - B^T = C - C^T = 0.$$

The compact real form is given by the skew-Hermitian symplectic matrices, that is, matrices of the form

$$\begin{pmatrix} A & B \\ -\overline{B} & \overline{A} \end{pmatrix}.$$

This real form is denoted $\mathfrak{sp}(l)$ and is isomorphic to the algebra of $l \times l$ skew-Hermitian quaternionic matrices, that is, $Q + \overline{Q}^T = 0$. The isomorphism is given by

$$Q = A + jB \mapsto \begin{pmatrix} A & -\overline{B} \\ B & \overline{A} \end{pmatrix}.$$

D_l This series covers the skew-symmetric matrices in even dimension, $\mathfrak{so}(2l, \mathbb{C})$ with compact real form $\mathfrak{so}(2l, \mathbb{R})$, for $l \geq 4$. In low dimensions, the following isomorphisms hold: (i) $\mathfrak{so}(6) \approx \mathfrak{su}(4)$; (ii) $\mathfrak{so}(4) \approx \mathfrak{so}(3) \oplus \mathfrak{so}(3)$, which is not simple; and (iii) $\mathfrak{so}(2) \approx \mathbb{R}$, which is abelian.

11.3.3 Cartan Subalgebras and Regular Elements

As in the case of complex semi-simple Lie algebras, the Cartan subalgebras of a compact Lie algebra $\mathfrak{u}$ play a central role in its description. The main properties of the Cartan subalgebras of $\mathfrak{u}$ are collected in the following items:

1. $\mathfrak{t} \subset \mathfrak{u}$ is a Cartan subalgebra if and only if $\mathfrak{t}$ is a maximal abelian subalgebra.
2. The center $\mathfrak{z}(\mathfrak{u})$ of $\mathfrak{u}$ is contained in every Cartan subalgebra, as follows from the characterization of the previous item.
3. Given two Cartan subalgebras $\mathfrak{t}_1$ and $\mathfrak{t}_2$, there exists an inner automorphism $g \in \mathrm{Aut}_0\mathfrak{u}$, such that $\mathfrak{t}_2 = g(\mathfrak{t}_1)$.
4. The Lie algebra $\mathfrak{u}$ is the union of its Cartan subalgebras. More precisely, if $\mathfrak{t}$ is a fixed Cartan subalgebra, then

$$\mathfrak{u} = \bigcup_{g \in \mathrm{Aut}_0\mathfrak{u}} g(\mathfrak{t}).$$

   The terms $g(\mathfrak{t})$ of this union are Cartan subalgebras.
5. All Cartan subalgebras of $\mathfrak{u}$ have the same dimension. This common dimension is called **rank** of $\mathfrak{u}$.
6. An element $X \in \mathfrak{u}$ is a **regular element** of $\mathfrak{u}$ if $\dim \ker \mathrm{ad}(X)$ is minimal among the dimensions of the centralizers of elements of $\mathfrak{u}$. Then, $\mathfrak{t} \subset \mathfrak{u}$ is a Cartan subalgebra if and only if $\mathfrak{t} = \ker \mathrm{ad}(X)$ for some regular element $X \in \mathfrak{t}$. So that the rank of $\mathfrak{u}$ coincides with the dimension of $\ker \mathrm{ad}(X)$ for every regular element $X \in \mathfrak{u}$.

These properties are consequences of the theory of Cartan subalgebras for complex semi-simple algebras, combined with the Weyl unitary trick. However, they can be proved directly by exploiting the existence of the invariant inner product, as is done in the sequel.

The first step to prove the above properties is to show that the Cartan subalgebras are maximal abelian.

Proposition 11.14 *Let* $\mathfrak{u}$ *be a compact Lie algebra and* $\mathfrak{t} \subset \mathfrak{u}$ *a Cartan subalgebra. Then* $\mathfrak{t}$ *is maximal abelian.*

Proof Take an arbitrary $X \in \mathfrak{t}$. To verify that $\mathfrak{t}$ is abelian it must be checked that $\mathfrak{t} \subset \mathfrak{z}(X) = \ker \operatorname{ad}(X)$. Since it is a subalgebra, $\mathfrak{t}$ is invariant by $\operatorname{ad}(X)$. But, by definition, $\mathfrak{t}$ is a nilpotent subalgebra, therefore the restriction $\operatorname{ad}(X)_{|\mathfrak{t}}$ is a nilpotent linear map. But $\operatorname{ad}(X)$ is semi-simple, as well as the restriction $\operatorname{ad}(X)_{|\mathfrak{t}}$, since $\operatorname{ad}(X)$ is skew-symmetric with respect to the invariant inner product. This way, $\operatorname{ad}(X)_{|\mathfrak{t}}$ is both semi-simple and nilpotent. This implies that $\operatorname{ad}(X)_{|\mathfrak{t}} = 0$, that is, $\mathfrak{t} \subset \ker \operatorname{ad}(X)$, showing that $\mathfrak{t}$ is abelian.

The maximality comes from the fact that $\mathfrak{t}$ is its own normalizer. Indeed, if $\mathfrak{t}'$ is an abelian algebra with $\mathfrak{t} \subset \mathfrak{t}'$, then the elements of $\mathfrak{t}'$ normalize $\mathfrak{t}$, hence $\mathfrak{t} = \mathfrak{t}'$. □

The following lemma of linear algebra will be used to prove the converse of the previous proposition.

Lemma 11.15 *If* $T : V \to V$ *is a skew-symmetric linear map with respect to the inner product* $(\cdot, \cdot)$*, then* $\ker T^2 = \ker T$.

Proof It is clear that $\ker T \subset \ker T^2$. On the other hand, take $v \in \ker T^2$. Then $\left(T^2 v, w\right) = 0$ for every $w \in V$. In particular, $\left(T^2 v, v\right) = 0$ and as T is skew-symmetric, it follows that $(Tv, Tv) = 0$. Hence, $Tv = 0$, showing that $v \in \ker T$.

□

Proposition 11.16 *Let* $\mathfrak{u}$ *be a compact Lie algebra and suppose that* $\mathfrak{t} \subset \mathfrak{u}$ *is a maximal abelian subalgebra. Then* $\mathfrak{t}$ *is a Cartan subalgebra.*

Proof By assumption, $\mathfrak{t}$ is abelian, so it suffices to show that it coincides with its own normalizer. To see that, observe that if $X \notin \mathfrak{t}$, then there exists $Y \in \mathfrak{t}$ such that $[X, Y] \neq 0$ since otherwise the subspace spanned by $\mathfrak{t}$ and X would be an abelian subalgebra containing $\mathfrak{t}$ properly, contradicting the maximality hypothesis.

Now, suppose by contradiction that there exists $X \notin \mathfrak{t}$ such that $[X, \mathfrak{t}] \subset \mathfrak{t}$ and take $Y \in \mathfrak{t}$ with $[X, Y] \neq 0$. Then $[Y, [Y, X]] = 0$, since $[Y, X] \in \mathfrak{t}$, which is abelian. In other words, $X \in \ker \operatorname{ad}(Y)^2$. It follows by the previous lemma that $X \in \ker \operatorname{ad}(Y)$, that is, $[Y, X] = 0$, which contradicts the choices of X e Y. □

Corollary 11.17 *Let* $\mathfrak{u}$ *be a compact Lie algebra. Then, for every* $X \in \mathfrak{u}$*, there exists a Cartan subalgebra* $\mathfrak{t} \subset \mathfrak{u}$ *such that* $X \in \mathfrak{t}$*. That is,* $\mathfrak{u}$ *is the union of its Cartan subalgebras.*

Proof Indeed, it is enough to take a maximal abelian subalgebra $\mathfrak{t}$ containing X. The existence of $\mathfrak{t}$ is easily proved by considering the set of all abelian subalgebras containing X. This set is nonempty (as it contains the subspace spanned by X), and by the previous proposition, among these algebras, those with maximal dimension are Cartan subalgebras. □

The following propositions establish relations between the Cartan subalgebras and the regular elements.

Proposition 11.18 *Let* $\mathfrak{t}$ *be a Cartan subalgebra. Then there exists* $X_0 \in \mathfrak{t}$ *such that* $\mathfrak{t} = \mathfrak{z}(X_0) = \ker \operatorname{ad}(X_0)$.

Proof The adjoints $\operatorname{ad}(X)$, $X \in \mathfrak{t}$, commute to each other, their complexifications in $\mathfrak{g} = \mathfrak{u}_{\mathbb{C}}$ are diagonalizable and their eigenvalues are purely imaginary. Therefore, these adjoints are simultaneously diagonalizable. This means that there exists a set R of linear functionals $\alpha : \mathfrak{t} \to \mathbb{R}$ and subspaces $\mathfrak{g}_\alpha \subset \mathfrak{g}$, such that

$$\mathfrak{g}_\alpha = \{Y \in \mathfrak{g} : \forall X \in \mathfrak{t},\ \operatorname{ad}(X)\, Y = i\alpha(X)\, Y\}$$

and $\mathfrak{g} = \sum_{\alpha \in R} \mathfrak{g}_\alpha$. (The linear functionals $\alpha \neq 0$ form the set Π of roots of the Cartan subalgebra.) The zero linear functional is one of the elements of R, as 0 is an eigenvalue of each adjoint $\operatorname{ad}(X)$. Since $\mathfrak{t}$ is abelian, the subspace $\mathfrak{g}_0$ contains $\mathfrak{t}$ and hence $\mathfrak{t} \subset \mathfrak{g}_0 \cap \mathfrak{u}$. Actually, there is the equality $\mathfrak{t} = \mathfrak{g}_0 \cap \mathfrak{u}$, since the elements of $\mathfrak{g}_0 \cap \mathfrak{u}$ normalize $\mathfrak{t}$, which is a Cartan subalgebra.

The fact that R is a finite set ensures that there exists $X_0 \in \mathfrak{t}$ such that $\alpha(X_0) \neq 0$ for every $\alpha \in R$, $\alpha \neq 0$. This X_0 satisfies the statement as the kernel of $\operatorname{ad}(X_0)$ is $\mathfrak{g}_0 \cap \mathfrak{u} = \mathfrak{t}$. □

It will be proved that X_0 in the above proposition is regular in $\mathfrak{u}$. Before that, it must be verified that, for a regular element X, the kernel of $\operatorname{ad}(X)$ is a Cartan subalgebra.

Proposition 11.19 *Let* $X \in \mathfrak{u}$ *be a regular element. Then*

$$\mathfrak{z}(X) = \ker \operatorname{ad}(X)$$

is the only Cartan subalgebra containing X. *Moreover, if a Cartan subalgebra* $\mathfrak{t}$ *contains a regular element* X, *then* $\mathfrak{t} = \mathfrak{z}(X)$.

Proof By Corollary 11.17, there exists a Cartan subalgebra $\mathfrak{t}$ such that $X \in \mathfrak{t}$. The fact that $\mathfrak{t}$ is abelian implies that $\mathfrak{t} \subset \mathfrak{z}(X)$. This inclusion is an equality since, by the previous proposition, there exists an element $X_0 \in \mathfrak{t}$ such that $\mathfrak{z}(X_0) = \mathfrak{t} \subset \mathfrak{z}(X)$. As X is regular, $\dim \mathfrak{z}(X) \leq \dim \mathfrak{z}(X_0) = \dim \mathfrak{t}$, hence $\mathfrak{t} = \mathfrak{z}(X) = \ker \operatorname{ad}(X)$. It follows from this equality that $\mathfrak{z}(X)$ is the only Cartan subalgebra containing the regular element X, concluding the proof. □

The conclusion of this discussion about Cartan subalgebras and regular elements will be obtained from the next proposition. It shows not only that any element $X \in \mathfrak{u}$ belongs to a Cartan subalgebra (as in Corollary 11.17), but that X belongs to a conjugate of Cartan subalgebra fixed in advance.

Proposition 11.20 *Let* $\mathfrak{u}$ *be a compact Lie algebra and* $\mathfrak{t}$ *a Cartan subalgebra of* $\mathfrak{u}$. *Suppose that* $\mathfrak{t}$ *contains a regular element* H. *Then, for every* $X \in \mathfrak{u}$, *there exists* $g \in \operatorname{Aut}_0 \mathfrak{u}$ *such that* $gX \in \mathfrak{t}$.

Proof Consider the function

$$g \in \mathrm{Aut}_0\mathfrak{u} \longmapsto (gX, H) \in \mathbb{R},$$

where $(\cdot, \cdot)$ denotes an invariant inner product.

This function is differentiable and, as $\mathrm{Aut}_0\mathfrak{u}$ is compact, it has a minimum at some $g_0 \in \mathrm{Aut}_0\mathfrak{u}$. Hence, for any $Y \in \mathfrak{u}$, the function $f : \mathbb{R} \to \mathbb{R}$ given by

$$f(t) = \left(e^{t\mathrm{ad}(Y)} g_0 X, H\right)$$

assumes a minimum at $t = 0$.

As $\exp(t\mathrm{ad}(Y))$ is an isometry of $(\cdot, \cdot)$, it holds

$$f(t) = \left(g_0 X, e^{-t\mathrm{ad}(Y)} H\right).$$

Taking derivatives yields

$$f'(0) = (g_0 X, [Y, H]) = 0,$$

which is the same as

$$([H, g_0 X], Y) = 0,$$

since $\mathrm{ad}(H)$ is skew-symmetric. As $Y \in \mathfrak{u}$ is arbitrary, this implies that $[H, g_0 X] = 0$, that is, $g_0 X \in \mathfrak{z}(H) = \mathfrak{t}$, concluding the proof. □

Combining this proposition and the previous ones there is the following description of the Cartan subalgebras of $\mathfrak{u}$ and their conjugations.

Theorem 11.21 *Let $\mathfrak{u}$ be a compact Lie algebra and $\mathfrak{t} \subset \mathfrak{u}$ a Cartan subalgebra. Then, the following properties hold:*

1. $\mathfrak{t} = \mathfrak{z}(X_0) = \ker \mathrm{ad}(X_0)$ *for a regular element* $X_0 \in \mathfrak{t}$.
2. *If $\mathfrak{t}_1$ is a Cartan subalgebra, then there exists $g \in \mathrm{Aut}_0\mathfrak{u}$ such that* $\mathfrak{t}_1 = g(\mathfrak{t})$.
3. $\mathfrak{u} = \bigcup_{g \in \mathrm{Aut}_0\mathfrak{u}} g(\mathfrak{t})$.

Proof Choose a regular element $X \in \mathfrak{u}$; then, by Proposition 11.19, $\mathfrak{z}(X) = \ker \mathrm{ad}(X)$ is a Cartan subalgebra containing X. Choose also $X_0 \in \mathfrak{t}$ such that $\mathfrak{t} = \ker \mathrm{ad}(X_0)$, as ensured by Proposition 11.18. By Proposition 11.20 above, there exists $g \in \mathrm{Aut}_0\mathfrak{u}$ such that $gX_0 \in \mathfrak{z}(X)$. Hence, $\mathfrak{z}(X) \subset \ker \mathrm{ad}(gX_0)$. However, the equality $\mathrm{ad}(gX_0) = g\mathrm{ad}(X_0)g^{-1}$ implies that

$$g(\mathfrak{t}) = g(\ker \mathrm{ad}(X_0)) = \ker \mathrm{ad}(gX_0),$$

hence $\mathfrak{z}(X) \subset g(\mathfrak{t})$. But both $\mathfrak{z}(X)$ and $g(\mathfrak{t})$ are Cartan subalgebras. Therefore, the inclusion $\mathfrak{z}(X) \subset g(\mathfrak{t})$ is in fact an equality. This means that

$$\dim \ker \operatorname{ad}(X_0) = \dim \mathfrak{t} = \dim g(\mathfrak{t}) = \dim \ker \operatorname{ad}(X).$$

As X is regular, it follows that X_0 is also regular. This means that the arbitrary Cartan subalgebra $\mathfrak{t}$ contains a regular element X_0 such that $\mathfrak{t} = \mathfrak{z}(X_0) = \ker \operatorname{ad}(X_0)$, which proves item (1).

For item (2), take $\mathfrak{t}_1 = \ker \operatorname{ad}(X_1)$. Again, by Proposition 11.20, there exists $g \in \operatorname{Aut}_0\mathfrak{u}$ such that $gX_1 \in \mathfrak{t}$. Then,

$$g(\mathfrak{t}_1) = g(\ker \operatorname{ad}(X_1)) = \ker \operatorname{ad}(gX_1) = \mathfrak{t},$$

showing the conjugation between the Cartan subalgebras.

To conclude, every element of $\mathfrak{u}$ belongs to a Cartan subalgebra, by Corollary 11.17, and these Cartan subalgebras have the form $g(\mathfrak{t})$, $g \in \operatorname{Aut}_0\mathfrak{u}$, which proves (3). □

Finally, it is proved that the Weyl construction yields a natural Cartan subalgebra of the compact real form.

Proposition 11.22 *Let $\mathfrak{u}$ be the compact real form given by the Weyl construction, as in Theorem 11.13. Then the subspace $\mathfrak{t} = i\mathfrak{h}_{\mathbb{R}}$ is a Cartan subalgebra of $\mathfrak{u}$.*

Proof By definition, a Cartan subalgebra is nilpotent and coincides with its normalizer. The first condition is satisfied by $\mathfrak{t}$, which is abelian, since $\mathfrak{h}$ is abelian and $\mathfrak{t} \subset \mathfrak{h}$. For the second condition, take $X \in \mathfrak{u}$ and write $X = H + \sum_\alpha X_\alpha$, as in decomposition (11.5). If in this decomposition $X_\alpha \neq 0$ for some root α, then there exists $H' \in \mathfrak{t}$ such that $[H', X_\alpha] \neq 0$ and, of course, $[H', X_\alpha] \in \mathfrak{u}_\alpha$. This means that, if $[X, \mathfrak{t}] \subset \mathfrak{t}$, then for every root α, $X_\alpha = 0$, that is, $X \in \mathfrak{t}$, showing that $\mathfrak{t}$ is its own normalizer. □

Example A Cartan subalgebra of $\mathfrak{su}(n)$ is the algebra of diagonal matrices with purely imaginary eigenvalues

$$\mathfrak{t} = \{\operatorname{diag}\{ix_1, \ldots, ix_n\} : x_j \in \mathbb{R},\ x_1 + \cdots + x_n = 0\}.$$

The inner automorphisms of $\mathfrak{su}(n)$ are given by conjugations by elements of $\mathrm{SU}(n)$. Therefore, the Cartan subalgebras of $\mathfrak{su}(n)$ are the subalgebras $\mathfrak{t}$ conjugated by elements of $\mathrm{SU}(n)$. This means that, given an orthonormal basis of $\mathbb{C}^n$, the set of elements of $\mathfrak{su}(n)$ which are diagonal with respect to this basis is a Cartan subalgebra and, conversely, every Cartan subalgebra is given in this way from an orthonormal basis.

The assertion that $X \in \mathfrak{su}(n)$ belongs to a Cartan subalgebra translates into the fact, well known in linear algebra, that skew-Hermitian matrices are diagonalizable on orthonormal bases. The same assertions hold for the reducible algebra $\mathfrak{u}(n) = \mathfrak{z} \oplus \mathfrak{su}(n)$, where the center $\mathfrak{z}$ is the algebra of scalar matrices with purely imaginary eigenvalues. □

11.4 Maximal Tori

Let U be a connected compact Lie group with Lie algebra $\mathfrak{u}$. A **maximal torus** in U is an abelian connected compact subgroup which is maximal (with respect to inclusion) with these properties. The maximal tori play, at the level of the compact Lie group, a role analogous to the Cartan subalgebras. As proved along this section, the analogy is complete, since every element of U belongs to some maximal torus, that is, U is the union of its maximal tori, which are pairwise conjugate to each other. Moreover, the maximal tori are in bijection with the Cartan subalgebras, as shows the following result.

Proposition 11.23 *If $T \subset U$ is a maximal torus, then its Lie algebra $\mathfrak{t}$ is a Cartan subalgebra. Conversely, if $\mathfrak{t} \subset \mathfrak{u}$ is a Cartan subalgebra, then $\exp \mathfrak{t} = \langle \exp \mathfrak{t} \rangle$ is a maximal torus.*

Proof By Proposition 11.14, it must be proved that $\mathfrak{t}$ is maximal abelian. Let $\mathfrak{s} \supset \mathfrak{t}$ be an abelian subalgebra. Then $\langle \exp \mathfrak{s} \rangle$ is a connected abelian group, as well as its closure $\overline{T} = \overline{\langle \exp \mathfrak{s} \rangle}$, which is a torus. As $T \subset \overline{T}$, it follows that $T = \overline{T}$ and hence that $\mathfrak{s} = \mathfrak{t}$.

On the other hand, if $\mathfrak{t} \subset \mathfrak{u}$ is a Cartan subalgebra, then $\mathfrak{t}$ is abelian and hence $\langle \exp \mathfrak{t} \rangle$ is connected and abelian, as well as its closure $\overline{\langle \exp \mathfrak{t} \rangle}$. The Lie algebra $\mathfrak{s}$ of $\overline{\langle \exp \mathfrak{t} \rangle}$ is abelian and contains $\mathfrak{t}$. By Proposition 11.14, $\mathfrak{s} = \mathfrak{t}$, which implies that $\overline{\langle \exp \mathfrak{t} \rangle} = \langle \exp \mathfrak{t} \rangle$, that is, $\langle \exp \mathfrak{t} \rangle$ is a torus, which is maximal in the same way as the Lie algebra $\mathfrak{t}$ is maximal. Finally, $\exp \mathfrak{t} = \langle \exp \mathfrak{t} \rangle$, since $\mathfrak{t}$ is abelian. □

A consequence of this proposition is that if $\mathfrak{z}(\mathfrak{u})$ is the center of $\mathfrak{u}$, then the subgroup $\langle \exp \mathfrak{z}(\mathfrak{u}) \rangle = \exp \mathfrak{z}(\mathfrak{u})$ is contained in every maximal torus, since $\mathfrak{z}$ is contained in every Cartan subalgebra. This group is in fact a torus.

Proposition 11.24 *Let $\mathfrak{z}(\mathfrak{u})$ be the center of $\mathfrak{u}$. Then $\langle \exp \mathfrak{z}(\mathfrak{u}) \rangle = \exp \mathfrak{z}(\mathfrak{u})$ is compact and connected and hence a torus.*

Proof First, $\langle \exp \mathfrak{z}(\mathfrak{u}) \rangle = \exp \mathfrak{z}(\mathfrak{u})$, since $\mathfrak{z}(\mathfrak{u})$ is an abelian Lie algebra. Take the closure $Z = \overline{\exp \mathfrak{z}(\mathfrak{u})}$, which is a compact connected abelian subgroup and moreover is contained in the center of U. Therefore, the Lie algebra $\mathfrak{z}$ of Z is contained in the center $\mathfrak{z}(\mathfrak{u})$ of $\mathfrak{u}$. Hence $\mathfrak{z} = \mathfrak{z}(\mathfrak{u})$ and it follows that $\exp \mathfrak{z}(\mathfrak{u}) = Z$ is compact, concluding the proof. □

If $T = \exp \mathfrak{t}$ is a maximal torus and $u \in U$, then uTu^{-1} is also a maximal torus. Proposition 11.23 combined with Theorem 11.21 show that any maximal torus is obtained in this way by conjugation.

Proposition 11.25 *Let $T = \exp \mathfrak{t}$ and $T_1 = \exp \mathfrak{t}_1$ be maximal tori. Then there exists $u \in U$ such that $T_1 = uTu^{-1}$.*

Proof By Theorem 11.21, there exists $g \in \mathrm{Aut}_0\mathfrak{u}$ such that $\mathfrak{t}_1 = g(\mathfrak{t})$. But the adjoint representation $\mathrm{Ad} : U \to \mathrm{Aut}_0\mathfrak{u}$ is surjective. Therefore, there exists $u \in U$ such that $\mathrm{Ad}(u) = g$ and hence $\mathfrak{t}_1 = \mathrm{Ad}(u)\,\mathfrak{t}$, ensuring that $T_1 = uTu^{-1}$. □

It thus follows that the image of the exponential $\exp \mathfrak{u}$ coincides with the union of the maximal tori of U, since every element of $\mathfrak{u}$ belongs to a Cartan subalgebra. Hence $\exp \mathfrak{u}$ is a compact subset of U because if $T \subset U$ is a maximal torus, then the map $U \times T \to U$, $(u, t) \mapsto utu^{-1}$ is continuous and is defined on the compact set $U \times T$. Its image, which is compact, is the union of the tori of U, which coincides with $\exp \mathfrak{u}$.

The objective now is to show that the exponential map is surjective or, what is the same, that U is the union of its maximal tori.

To this purpose it is enough to consider the case in which $\mathfrak{u}$ is semi-simple. This is so because if $\mathfrak{u} = \mathfrak{z}(\mathfrak{u}) \oplus \mathfrak{k}$ with $\mathfrak{k}$ semi-simple, then, by Proposition 11.24 above, the group $Z = \langle \exp \mathfrak{z}(\mathfrak{u}) \rangle = \exp \mathfrak{z}(\mathfrak{u})$ is closed. The quotient U/Z is compact and semi-simple, since its Lie algebra $\mathfrak{u}/\mathfrak{z}(\mathfrak{u})$ is isomorphic to $\mathfrak{k}$. Once the surjectivity of the exponential on U/Z is proved the surjectivity on U follows. Indeed, let $\pi : U \to U/Z$ be the canonical homomorphism and take $g \in U$. Then there exists $X \in \mathfrak{k}$ such that $\pi\left(e^X\right) = \pi(g)$, which means that $g = e^X z$ with $z = e^Y \in \exp \mathfrak{z}(\mathfrak{u})$. As $[X, Y] = 0$, it follows that $g = e^X e^Y = e^{X+Y}$ and from this one concludes the surjectivity of the exponential map on the reducible group U from the surjectivity on the semi-simple group U/Z.

The next lemma will be used in the proof of the semi-simple case.

Lemma 11.26 *Given $u \in U$, denote by $Z(u)$ its centralizer, whose Lie algebra*

$$\mathfrak{z}(u) = \{X \in \mathfrak{u} : \operatorname{Ad}(u) X = X\}$$

is the eigenspace of $\operatorname{Ad}(u)$ *associated with the eigenvalue* 1. *Let $\mathfrak{e}$ be the orthogonal complement of $\mathfrak{z}(u)$ in $\mathfrak{u}$ (with respect to the invariant inner product). Then* $\operatorname{Ad}(u)\, \mathfrak{e} = \mathfrak{e}$ *and the restriction* $(\operatorname{Ad}(u) - \mathrm{id})_{|\mathfrak{e}}$ *is invertible.*

Proof The invariance of $\mathfrak{e}$ follows from the fact that $\operatorname{Ad}(u)$ is an isometry and $\operatorname{Ad}(u)\, \mathfrak{z}(u) = \mathfrak{z}(u)$. The centralizer $\mathfrak{z}(u)$ is the eigenspace associated with the eigenvalue 1 of $\operatorname{Ad}(u)$, so that 1 is not an eigenvalue of its restriction to $\mathfrak{e}$. □

Theorem 11.27 *Let U be a compact group and take a maximal torus $T = \exp \mathfrak{t} \subset U$. Then*

$$U = \bigcup_{g \in U} g T g^{-1},$$

that is, every element of U is conjugate to an element of T.

Proof As mentioned above, there is no loss of generality in assuming that U is semi-simple. The proof is by induction on the dimension of U. The only compact semi-simple Lie algebra with minimal dimension is $\mathfrak{su}(2)$ with $\dim \mathfrak{su}(2) = 3$. The exponential map is surjective for every Lie group with Lie algebra $\mathfrak{su}(2)$, since this occurs in SU (2), which is simply connected. With this, the induction procedure starts from dimension 3.

The induction hypothesis says that any proper subgroup $K \subset U$ is the union of its maximal tori even though K is not semi-simple, by the comments above.

For $A \subset U$ and $K \subset U$, write

$$A^K = \bigcup_{g \in K} gAg^{-1}$$

and $A^\times = A \setminus Z(U)$, where $Z(U)$ is the center of U. Then $\left(A^U\right)^\times = \left(A^\times\right)^U$, as the elements of $Z(U)$ are fixed by conjugations.

The set $U^\times$ is open, connected and dense in U, as $\dim U \geq 3$ and $Z(U)$ is finite. Moreover, since T^U is compact and $U^\times$ is dense, it suffices to prove that $U^\times = \left(T^U\right)^\times = \left(T^\times\right)^U \subset T^U$. This will be proved by showing that $\left(T^\times\right)^U$ is open and closed in the connected set U^x.

The set $\left(T^\times\right)^U$ is closed in $U^\times$, since $\left(T^\times\right)^U = T^U \cap U^\times$ and T^U is compact. To verify that it is open, it is enough to show that every $u \in T^\times$ belongs to the interior of $\left(T^\times\right)^U$, since in this case any $gug^{-1} \in \left(T^\times\right)^U$ is contained in the interior of $g\left(T^\times\right)^U g^{-1} = \left(T^\times\right)^U$.

Thus, take $u \in T^\times$ and let $Z(u)$ be its centralizer, which is a proper compact subgroup as $u \notin Z(U)$. The identity component $K = Z(u)_0$ contains T, which is a maximal torus of K. In particular $u \in K$.

The induction hypothesis can be applied to K, to conclude that $K = T^K$ and hence $K \subset T^U$.

Consider now the map $\psi : U \times K \to U$ given by $\psi(g,k) = gkg^{-1}$. Its differential at $(1,u)$ is given by

$$d\psi_{(1,u)}(X,Y) = (X + Y - \mathrm{Ad}(u)\, X)(u),$$

where X and Y are right invariant vector fields, $X \in \mathfrak{u}$ and $Y \in \mathfrak{k}$, the Lie algebra of K.

This differential is surjective. Indeed, taking $X = 0$, the vectors $d\psi_{(1,u)}(0,Y) = Y(u)$, $Y \in \mathfrak{k}$, cover the tangent space $T_u K$. On the other hand, taking $Y = 0$ one obtains the vectors $d\psi_{(1,u)}(X,0) = ((\mathrm{id} - \mathrm{Ad}(u))\, X)(u)$ which, by the previous lemma, cover the complement of $T_u K$ in $T_u U$, showing the surjectivity of $d\psi_{(1,u)}$.

Therefore, there exist open sets $A \subset U$, $B \subset K$ with $(1,u) \in A \times B$ such that $u = \psi(1,u)$ is in the interior of $\psi(A \times B)$. As $u \notin Z(U)$, it is possible to take $A \subset U^\times$, so that $\psi(A \times B) \subset \left(K^\times\right)^U \subset \left(T^\times\right)^U$. These arguments show that u is in the interior of $\left(T^\times\right)^U$, concluding the proof of the theorem. □

The result of the following corollary was widely discussed above.

Corollary 11.28 *Let U be a compact group with Lie algebra $\mathfrak{u}$. Then $\exp : \mathfrak{u} \to G$ is surjective.*

Corollary 11.29 *The center $Z(U)$ of U is contained in every maximal torus of U.*

Proof Given $z \in Z(U)$ there exists, by the theorem, a maximal torus T with $z \in T$. As $gzg^{-1} = z$, $g \in U$, it follows that $z \in gTg^{-1}$. □

Example A maximal torus of SU (n) is the subgroup of diagonal matrices

$$T = \{\text{diag}\{z_1, \ldots, z_n\} : |z_j| = 1,\ z_1 \cdots z_n = 1\}.$$

By conjugation, the maximal tori of SU (n) are the subgroups of diagonal matrices with respect to the different orthonormal bases of $\mathbb{C}^n$. The assertion that $g \in \text{SU}(n)$ belongs to a maximal torus translates into the fact, known in linear algebra, that the unitary matrices are diagonalizable by orthonormal bases. The center of SU (n) is formed by the scalar matrices $z \cdot \text{id}$ such that $z^n = 1$ and therefore is isomorphic to $\mathbb{Z}_n$.

In the reducible group U (n), the maximal tori are subgroups of diagonal matrices without the restriction of determinant 1. □

To conclude this section, it will be proved that the maximal tori are also maximal as abelian groups.

Theorem 11.30 *Let* $\mathfrak{u}$ *be a compact semi-simple Lie algebra,* $\mathfrak{t}$ *a Cartan subalgebra of* $\mathfrak{u}$ *and* $g \in \text{Aut}\mathfrak{u}$ *such that* $g(H) = H$ *for every* $H \in \mathfrak{t}$. *Then there exists* $H_g \in \mathfrak{t}$ *such that* $g = e^{\text{ad}(H_g)}$.

Proof Let $\mathfrak{h} = \mathfrak{t}_{\mathbb{C}}$ be the Cartan subalgebra of the complexification $\mathfrak{g} = \mathfrak{u}_{\mathbb{C}}$ of $\mathfrak{u}$. The automorphism g extends to an automorphism of $\mathfrak{g}$, also denoted by g, which fixes the elements of $\mathfrak{h}$.

Let Π be the set of roots of $\mathfrak{h}$, so that

$$\mathfrak{g} = \mathfrak{h} \oplus \sum_{\alpha \in \Pi} \mathfrak{g}_\alpha.$$

If $H \in \mathfrak{h}$, then $g(H) = H$ and therefore $g\,\text{ad}(H)\,g^{-1} = \text{ad}(gH) = \text{ad}(H)$ and it follows that $g(\mathfrak{g}_\alpha) = \mathfrak{g}_\alpha$. Hence, for each $\alpha \in \Pi$, there exists $z_\alpha \in \mathbb{C}$ such that $g(X_\alpha) = z_\alpha X_\alpha$, where X_α is a generator of $\mathfrak{g}_\alpha$ (which is a 1-dimensional subspace). The eigenvalues z_α satisfy $|z_\alpha| = 1$, as $g \in \text{Aut}\mathfrak{u}$, which is a compact group.

Write $z_\alpha = e^{i\theta_\alpha}$ with $\theta_\alpha \in \mathbb{R}$. Then $\theta_{\alpha+\beta} = \theta_\alpha + \theta_\beta (\text{mod}\, 2\pi)$ for every pair of roots α and β such that $\alpha + \beta \in \Pi$. Indeed, $g[X_\alpha, X_\beta] = e^{i\theta_{\alpha+\beta}}[X_\alpha, X_\beta]$, since $[X_\alpha, X_\beta] \in \mathfrak{g}_{\alpha+\beta}$. But

$$g[X_\alpha, X_\beta] = [gX_\alpha, gX_\beta] = [e^{i\theta_\alpha}X_\alpha, e^{i\theta_\beta}X_\beta] = e^{i(\theta_\alpha+\theta_\beta)}[X_\alpha, X_\beta],$$

which shows that $e^{i\theta_{\alpha+\beta}} = e^{i(\theta_\alpha+\theta_\beta)}$, that is, $\theta_{\alpha+\beta} = \theta_\alpha + \theta_\beta (\text{mod}\, 2\pi)$.

Let now $\Sigma = \{\alpha_1, \ldots, \alpha_l\} \subset \Pi$ be a simple system of roots. Then a root $\alpha \in \Pi$ is written as

$$\alpha = n_1\alpha_1 + \cdots + n_l\alpha_l$$

with $n_j \in \mathbb{Z}$. From the equality $\theta_{\alpha+\beta} = \theta_\alpha + \theta_\beta (\mathrm{mod}\, 2\pi)$ it follows that

$$\theta_\alpha = n_1\theta_{\alpha_1} + \cdots + n_l\theta_{\alpha_l} \,(\mathrm{mod}\, 2\pi)\,. \tag{11.7}$$

Finally, define $H_g \in \mathfrak{t}$ such that $\alpha_j\left(H_g\right) = i\theta_{\alpha_j}$, for the simple roots $\alpha_j \in \Sigma$. The existence of H_g comes from the fact that Σ is a basis and the roots assume purely imaginary values in $\mathfrak{t}$. For a root $\alpha = n_1\alpha_1 + \cdots + n_l\alpha_l$ it holds

$$\alpha\left(H_g\right) = i\left(n_1\theta_{\alpha_1} + \cdots + n_l\theta_{\alpha_l}\right) = i\theta_\alpha \,(\mathrm{mod}\, 2\pi)\,.$$

Therefore, if $X_\alpha \in \mathfrak{g}_\alpha$, then

$$e^{\mathrm{ad}(H_g)}X_\alpha = e^{\alpha(H_g)}X_\alpha = e^{i\theta_\alpha}X_\alpha = g\left(X_\alpha\right),$$

showing that $g = e^{\mathrm{ad}(H_g)}$ and concluding the proof. □

It follows from this theorem that the maximal tori are also maximal as abelian groups not necessarily connected.

Corollary 11.31 *Let U be a compact group, $T \subset U$ a maximal torus, and $S \subset U$ an abelian subgroup, such that $T \subset S$. Then $T = S$.*

Proof Write $\mathfrak{u} = \mathfrak{z}(\mathfrak{u}) \oplus \mathfrak{k}$ with $\mathfrak{k}$ semi-simple, so that the Lie algebra of T is $\mathfrak{z}(\mathfrak{u}) \oplus \mathfrak{t}$, with $\mathfrak{t}$ a Cartan subalgebra of $\mathfrak{k}$. If $k \in S$, then $\mathrm{Ad}(k)$ is an automorphism of $\mathfrak{u}$ which restricts to an automorphism of $\mathfrak{k}$, such that $\mathrm{Ad}(k)\,X = X$ for every $X \in \mathfrak{t}$. By Theorem 11.30, $\mathrm{Ad}(k) = e^{\mathrm{ad}(H)}$ for some $H \in \mathfrak{t}$. Hence $k = e^H z$ for some $z \in Z(U)$. Since both e^H and z belong to T, it follows that $k \in T$, that is, $S \subset T$. □

11.5 Center and Roots

In this section, $\mathfrak{u}$ is a compact semi-simple Lie algebra. Let U be a compact Lie group with Lie algebra $\mathfrak{u}$ and $T = \exp\mathfrak{t}$ a maximal torus of U. As happens with abelian groups in general, the exponential map $\exp : \mathfrak{t} \to T$ is a homomorphism whose kernel L_U is a lattice of the additive group of $\mathfrak{t}$, that is, a discrete subgroup isomorphic to $\mathbb{Z}^{\dim\mathfrak{t}}$.

As U runs through the locally isomorphic groups with Lie algebra $\mathfrak{u}$, the Cartan subalgebra $\mathfrak{t}$ remains unchanged, while the lattice L_U changes. An analysis of this lattice from the algebraic properties of $\mathfrak{u}$ and $\mathfrak{t}$ (basically the roots of the complexified Cartan subalgebra $\mathfrak{h} = \mathfrak{t}_{\mathbb{C}}$) allows a description of the center $Z(U)$ of U, as the center is contained in the maximal torus T. This description provides another proof of the Weyl theorem of the compactness of the simply connected group with Lie algebra $\mathfrak{u}$.

It can be assumed without loss of generality that $\mathfrak{u}$ is the real compact form of the complex semi-simple algebra $\mathfrak{g}$ obtained by the Weyl construction of Theorem 11.13. The notations are the same of Section 11.3. In particular, $\mathfrak{t} = i\mathfrak{h}_{\mathbb{R}}$, where $\mathfrak{h}$ is a Cartan subalgebra of $\mathfrak{g}$. The roots of $\mathfrak{h}$ are denoted by Π.

For the adjoint group $\mathrm{Aut}_0\mathfrak{u}$, the lattice $L_{\mathrm{Aut}_0\mathfrak{u}}$ is easily described by the roots in Π.

Proposition 11.32 *Let $U = \mathrm{Aut}_0\mathfrak{u}$ be the adjoint group. Then, for $H \in \mathfrak{h}_{\mathbb{R}}$, $\exp iH = 1$ if and only if $\alpha(H) \in 2\pi\mathbb{Z}$ for every root $\alpha \in \Pi$, that is, the lattice in $\mathfrak{t}$ given by the kernel of $\exp : \mathfrak{t} \to U$ is*

$$L_0 = \{iH \in i\mathfrak{h}_{\mathbb{R}} : \forall \alpha \in \Pi,\ \alpha(H) \in 2\pi\mathbb{Z}\}. \tag{11.8}$$

Proof The eigenvalues of $\mathrm{ad}(iH)$, $H \in \mathfrak{h}_{\mathbb{R}}$, are 0 and $i\alpha(H)$ with $\alpha \in \Pi$. Therefore, the eigenvalues of $e^{\mathrm{ad}(iH)}$ are 1 and $e^{i\alpha(H)}$, $\alpha \in \Pi$. It follows that $e^{\mathrm{ad}(iH)} = 1$ if and only if $\alpha(H) \in 2\pi\mathbb{Z}$ for every root $\alpha \in \Pi$. □

The lattice L_0 for the adjoint group given in (11.8) is the biggest lattice in $\mathfrak{t}$, defined from the compact groups U with Lie algebra $\mathfrak{u}$. Indeed, the following proposition holds.

Proposition 11.33 *For the compact group U with Lie algebra $\mathfrak{u}$, let*

$$L_U = \{iH \in i\mathfrak{h}_{\mathbb{R}} : \exp_U iH = 1\}$$

be the corresponding lattice. Then $L_U \subset L_0$ and the center $Z(U)$ is isomorphic to L_0/L_U. The isomorphism associates $iH \in L_0$ with $e^{iH} \in Z(U)$.

Proof If $iH \in L_U$, then $\mathrm{Ad}\left(e^{iH}\right) = e^{\mathrm{ad}(iH)} = 1$ and therefore $iH \in L_0$, that is, $L_U \subset L_0$. Given $iH_1, iH_2 \in L_0$, $e^{iH_1} = e^{iH_2}$ if and only if $iH_1 - iH_2 \in L_U$. Hence the map $iH \to e^{iH}$ can be factored to define an injective homomorphism on L_0/L_U which assumes values in $Z(U)$, since $\mathrm{Ad}\left(e^{iH}\right) = 1$ if $iH \in L_0$. This homomorphism is surjective. Indeed, if $z \in Z(U)$, then there exists $iH \in \mathfrak{t}$ such that $z = e^{iH}$, as the center is contained in every maximal torus. Since $\mathrm{Ad}(z) = 1$, it follows that $iH \in L_0$. □

In other words, L_0 is an upper bound (with respect to inclusion) for the lattices $L_U \subset \mathfrak{t}$ defined by the locally isomorphic groups with Lie algebra $\mathfrak{u}$.

It is also possible to obtain a lower bound for the lattices L_U based on the geometry of the set of roots and, ultimately, in the fact that $\mathrm{SU}(2)$ is simply connected. To get such a lower bound, the following concepts and facts related to the roots of a complex semi-simple Lie algebra are introduced.

Let $\mathcal{K}_{\mathfrak{g}}(\cdot,\cdot)$ be the Cartan–Killing form of $\mathfrak{g} = \mathfrak{u}_{\mathbb{C}}$. Its restriction to the Cartan subalgebra $\mathfrak{h}$ is non-degenerate and is an inner product on $\mathfrak{h}_{\mathbb{R}}$, denoted by $\langle\cdot,\cdot\rangle$. This real subspace is spanned by H_α, $\alpha \in \Pi$, where H_α is defined by $\alpha(\cdot) = \mathcal{K}_{\mathfrak{g}}(H_\alpha, \cdot)$. For $\alpha \in \Pi$, define the "co-root"

$$H_\alpha^\vee = \frac{2}{\langle H_\alpha, H_\alpha \rangle} H_\alpha$$

and denote by $\mathfrak{g}(\alpha)$ the subspace spanned by H_α, $\mathfrak{g}_\alpha$ and $\mathfrak{g}_{-\alpha}$ (recalling that, if $\alpha \in \Pi$, then $-\alpha \in \Pi$). The subspace $\mathfrak{g}(\alpha)$ is a complex subalgebra of $\mathfrak{g}$ isomorphic to $\mathfrak{sl}(2, \mathbb{C})$. To obtain an isomorphism, take $X_\alpha \in \mathfrak{g}_\alpha$ and $Y_\alpha \in \mathfrak{g}_{-\alpha}$ such that $\mathcal{K}_\mathfrak{g}(X_\alpha, Y_\alpha) = 1$. Then $\{X_\alpha, H_\alpha^\vee, Y_\alpha\}$ is a basis of $\mathfrak{g}(\alpha)$ and the isomorphism is given by

$$X_\alpha \longleftrightarrow \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \qquad H_\alpha^\vee \longleftrightarrow H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \qquad Y_\alpha \longleftrightarrow \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

(The choice of $H_\alpha^\vee$ comes from the fact that this is the only multiple of H_α such that $\alpha\left(H_\alpha^\vee\right) = 2$, since ± 2 are the eigenvalues of $\operatorname{ad}(H)$ in $\mathfrak{sl}(2, \mathbb{C})$.[7])

The additive subgroup generated by $2\pi i H_\alpha^\vee$, $\alpha \in \Pi$, is a lattice. Indeed, by the Killing formula, for every pair of roots $\alpha, \beta \in \Pi$, the Killing number $\beta\left(H_\alpha^\vee\right) \in \mathbb{Z}$. Hence $2\pi i H_\alpha^\vee \in L_0$ for every root $\alpha \in \Pi$. On the other hand, the set of roots Π spans the dual $\mathfrak{h}^*$ of $\mathfrak{h}$, so that $2\pi i H_\alpha^\vee$, $\alpha \in \Pi$, spans $\mathfrak{t} = i\mathfrak{h}_\mathbb{R}$. This means that $2\pi i H_\alpha^\vee$, $\alpha \in \Pi$, generates a lattice of $\mathfrak{t}$. This lattice is denoted by $L_{\min}$. It is contained in L_U for every compact U, as it will be verified in the sequel.

Proposition 11.34 *Let U be a compact Lie group with Lie algebra $\mathfrak{u}$. Then, for every root $\alpha \in \Pi$, it holds that $\exp_U i H_\alpha^\vee = 1$. Hence the lattice $L_{\min}$ generated by $2\pi i H_\alpha^\vee$, $\alpha \in \Pi$, is contained in L_U.*

Proof The isomorphism between $\mathfrak{g}(\alpha)$ and $\mathfrak{sl}(2, \mathbb{C})$ restricts to a homomorphism $\theta : \mathfrak{su}(2) \to \mathfrak{u}$ whose image is spanned by $iH_\alpha^\vee$, $X_\alpha - Y_\alpha$ and $i(X_\alpha + Y_\alpha)$ and satisfies

$$iH_\alpha^\vee = \theta(iH) = \theta \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}.$$

By the fact that $\mathrm{SU}(2)$ is simply connected, θ extends to a homomorphism $\phi : \mathrm{SU}(2) \to U$. Then

$$e^{2\pi i H_\alpha^\vee} = \phi\left(e^{2\pi i H}\right) = 1,$$

which shows that $2\pi i H_\alpha^\vee \in L_U$ for every root $\alpha \in \Pi$. □

In short, for any compact group U with Lie algebra $\mathfrak{u}$, the lattice L_U has lower and upper bounds given by

$$L_{\min} \subset L_U \subset L_0. \tag{11.9}$$

[7] See *Álgebras de Lie* [47], Chapter 5 and Varadarajan [53], Chapter 4.

Since $Z(U) = L_0/L_U$, its order is bounded by the order of $L_0/L_{\min}$, which is finite, as shows the following fact about lattices in $\mathbb{R}^n$.

Lemma 11.35 *Let $L_1 \subset L_2$ be lattices of $\mathbb{R}^n$. Then the index of L_1 in L_2 is finite. More precisely, suppose that L_1 is generated over $\mathbb{Z}$ by the basis $\{v_1, \ldots, v_n\}$ and L_2 is generated by the basis $\{w_1, \ldots, w_n\}$. Let g be the linear map such that $gw_j = v_j$. Then $|L_2/L_1| = \det g$.*

Proof If $L \subset \mathbb{R}^n$ is a lattice generated by the basis $\{f_1, \ldots, f_n\}$, then the quotient $\mathbb{R}^n/L$ is a torus. The Lebesgue measure on $\mathbb{R}^n$ induces a Haar measure on this torus, whose volume is the volume of the parallelepiped generated by $\{f_1, \ldots, f_n\}$.

The canonical projection $\mathbb{R}^n/L_2 \to \mathbb{R}^n/L_1$ is a covering with $m = |L_2/L_1|$ leaves. Hence, the volume of $\mathbb{R}^n/L_2$ with respect to the Lebesgue measure is m times the volume of $\mathbb{R}^n/L_1$.

Now, if g is as in the statement, then the volume of the parallelepiped generated by $\{v_1, \ldots, v_n\}$ is equal to $\det g$ times the volume of the parallelepiped generated by $\{w_1, \ldots, w_n\}$. Therefore, the volume of $\mathbb{R}^n/L_1$ is equal to $\det g$ times the volume of $\mathbb{R}^n/L_2$, which shows that $|L_2/L_1| = \det g$. □

The information gathered so far about the lattices L_U is already enough to give an alternate proof of the **Weyl theorem** .

Theorem 11.36 *If $\mathfrak{u}$ is a compact semi-simple Lie algebra, then the fundamental group of* $\mathrm{Aut}_0\mathfrak{u}$ *is finite.*

Proof Write $\mathrm{Aut}_0\mathfrak{u} = \widetilde{U}/\Gamma$ with $\widetilde{U}$ simply connected. By the Lemma 11.8, the fundamental group Γ is finitely generated and therefore is isomorphic to $\mathbb{Z}^k \times \mathbb{Z}_{m_1} \times \cdots \times \mathbb{Z}_{m_n}$. Suppose by contradiction that Γ is not finite, that is, $k \geq 1$. Then, for every integer $N > 0$, there exists a subgroup $L \subset \Gamma$ such that Γ/L is finite and has order $|\Gamma/L| > N$. Taking $N > |L_0/L_{\min}|$ yields a contradiction. Indeed, the center of $\widetilde{U}/L$ is isomorphic to Γ/L, which is finite, and it follows that $\widetilde{U}/L$ is compact. By the inclusions (11.9) it would follow that $|\Gamma/L| \leq |L_0/L_{\min}|$, contradicting the choice of L. □

The compactness of the universal covering group $\widetilde{U}$ allows to write the inclusions (11.9)

$$L_{\min} \subset L_{\widetilde{U}} \subset L_0,$$

which ensures that the order of the fundamental group of $\mathrm{Aut}_0\mathfrak{u}$ (that is, the center of $\widetilde{U}$) is at most the order of $L_0/L_{\min}$. Indeed, it will be proved in the sequel that $L_{\widetilde{U}} = L_{\min}$, which leads to the conclusion that the fundamental group of $\mathrm{Aut}_0\mathfrak{u}$ is isomorphic to $L_0/L_{\min}$ and therefore can be determined algebraically.

The proof that $L_{\widetilde{U}} = L_{\min}$ uses representations of $\mathfrak{g} = \mathfrak{u}_{\mathbb{C}}$. Given $iH \in L_0 \backslash L_{\min}$, there exists a finite dimensional faithful representation $\rho : \mathfrak{g} \to \mathfrak{gl}(V)$ such that $e^{i\rho(H)} \neq 1$ (see below). Hence, $iH \notin L_U$ if $U = \langle \exp \rho(\mathfrak{u}) \rangle$, which is a group with Lie algebra (isomorphic to) $\mathfrak{u}$. It then follows that $iH \notin L_{\widetilde{U}}$ since, for every group U with Lie algebra $\mathfrak{u}$, it holds that $L_{\widetilde{U}} \subset L_U$ by the definition of the lattice L_U.

The results and notations about representations required to prove the above assertion are listed below:[8]

1. The restriction $\langle \cdot, \cdot \rangle$ to $\mathfrak{h}$ of the Cartan–Killing form $\mathcal{K}_{\mathfrak{g}}(\cdot, \cdot)$ of $\mathfrak{g}$ defines the non-degenerate form on $\mathfrak{h}^*$ by

$$\langle \alpha, \beta \rangle = \alpha\left(H_\beta\right) = \beta\left(H_\alpha\right) = \langle H_\alpha, H_\beta \rangle,$$

   where $\alpha(\cdot) = \langle H_\alpha, \cdot \rangle$.
2. If Π is the set of roots of $\mathfrak{h}$, then a simple root system

$$\Sigma = \{\alpha_1, \ldots, \alpha_l\} \subset \Pi$$

   is a basis of $\mathfrak{h}^*$ such that, for every $\alpha \in \Pi$, $\alpha = n_1\alpha_1 + \cdots + n_l\alpha_l$, where the coefficients n_i are integer numbers having the same sign.
3. Given a root $\alpha \in \Pi$, the co-root is defined by

$$\alpha^\vee = \frac{2}{\langle \alpha, \alpha \rangle}\alpha.$$

   (It is known that $\langle \alpha, \alpha \rangle > 0$.) The set $\Pi^\vee = \{\alpha^\vee : \alpha \in \Pi\}$ is a root system such that $\Sigma^\vee = \{\alpha^\vee : \alpha \in \Sigma\}$ is a simple root system for $\Pi^\vee$.
4. The **Cartan matrix** of Π, or rather of Σ, is defined by

$$C(\Sigma) = \left(\frac{2\langle \alpha_i, \alpha_j \rangle}{\langle \alpha_i, \alpha_i \rangle}\right)_{i,j}.$$

   The Cartan matrix $C\left(\Sigma^\vee\right)$ is defined the same way. A simple computation shows that $C\left(\Sigma^\vee\right) = C(\Sigma)^T$.
5. The set of fundamental weights $\Omega = \{\omega_1, \ldots, \omega_l\}$ is the dual basis of $\Sigma^\vee$ and is defined by the relations

$$\langle \omega_j, \alpha_i^\vee \rangle = \frac{2\langle \omega_j, \alpha_i \rangle}{\langle \alpha_i, \alpha_i \rangle} = \delta_{ij}. \tag{11.10}$$

With these notations, the lattice $L_{\min}$ is generated (over $\mathbb{Z}$) by H_α, $\alpha \in \Sigma^\vee$. Indeed, by definition $L_{\min}$ is generated by $H_\alpha^\vee = H_{\alpha^\vee}$, $\alpha \in \Pi$, that is, $L_{\min}$ is generated by H_β, $\beta \in \Pi^\vee$. But the elements of $\Pi^\vee$ are linear combinations with integer coefficients of $\Sigma^\vee$. It follows that $\Sigma^\vee$ generates $L_{\min}$. This entails the following characterization of $L_{\min}$.

[8] See *Álgebras de Lie* [47], Chapters 6, 9, 11, Varadarajan [53], Chapter 4 and Helgason [20], Chapter III.

Proposition 11.37 *Let Ω be the set of fundamental weights. Then*

$$L_{\min} = \{iH \in i\mathfrak{h}_{\mathbb{R}} : \forall \omega \in \Omega,\ \omega(H) \in 2\pi\mathbb{Z}\}.$$

Proof By the definition of Ω in (11.10), its dual basis in $\mathfrak{h}$ is H_β, $\beta \in \Sigma^\vee$. Hence, $H \in \mathfrak{h}$ is written

$$H = \omega_1(H)\, H_{\alpha_1^\vee} + \cdots + \omega_l(H)\, H_{\alpha_l^\vee},$$

and it is seen that H is in the lattice $L_{\min}$ generated by $\{H_{\alpha_1^\vee}, \ldots, H_{\alpha_l^\vee}\}$ if and only if $\omega_i(H) \in \mathbb{Z}$, $i = 1, \ldots, l$. □

The following result about representations of semi-simple Lie algebras will be used in the sequel.[9]

Proposition 11.38 *Let $\omega = n_1\omega_1 + \cdots + n_l\omega_l$ be a linear combination of Ω with integer coefficients ≥ 0. Then there exists a finite dimensional (faithful) irreducible representation $\rho : \mathfrak{g} \to \mathfrak{gl}(V)$ such that, for every $H \in \mathfrak{h}$, $\omega(H)$ is an eigenvalue of $\rho(H)$.*

It is now possible to describe the lattice for $\widetilde{U}$ and hence the fundamental group of $\mathrm{Aut}_0\mathfrak{u}$.

Theorem 11.39 *Let $\mathfrak{u}$ be a compact semi-simple Lie algebra and $\widetilde{U}$ its simply connected group. Then*

$$\begin{aligned} L_{\widetilde{U}} &= L_{\min} = 2\pi i\mathbb{Z} \cdot \Sigma^\vee \\ &= \{iH \in i\mathfrak{h}_{\mathbb{R}} : \forall \omega \in \Omega,\ \omega(H) \in 2\pi\mathbb{Z}\}. \end{aligned}$$

Hence the fundamental group $\pi_1(\mathrm{Aut}_0\mathfrak{u}) \approx L_0/L_{\min}$.

Proof For every group U with Lie algebra $\mathfrak{u}$,

$$L_{\min} \subset L_{\widetilde{U}} \subset L_U \subset L_0.$$

It must be proved that, given $iH \in L_0 \setminus L_{\min}$, there exists a group U such that $iH \notin L_U$. By Proposition 11.37, if $iH \in L_0 \setminus L_{\min}$, then there exists a fundamental weight $\omega \in \Omega$ such that $\omega(H)$ is not an integer multiple of 2π. Let $\rho : \mathfrak{g} \to \mathfrak{gl}(V)$ be the representation given by Proposition 11.38, such that $\omega(H)$ is an eigenvalue of $\rho(H)$. The group $U = \langle \exp \rho(\mathfrak{u}) \rangle$ has Lie algebra $\rho(\mathfrak{u})$ isomorphic to $\mathfrak{u}$ and contains $e^{i\rho(H)}$. The fact that $\omega(H)$ is an eigenvalue of $\rho(H)$ implies that $e^{i\omega(H)} \neq 1$ is an eigenvalue of $e^{i\rho(H)}$, which is thus different from 1. Hence $iH \notin L_U$, which concludes the proof. □

[9] See the theorem of highest weight representation in *Álgebras de Lie* [47], Chapter 11 and Varadarajan [53], Section 4.6.

The order $L_0 \setminus L_{\min}$ can be computed from Lemma 11.35 once a generator set of L_0 is obtained.

Proposition 11.40 *Denote by $\Omega^\vee$ the set of fundamental weights for the root system $\Pi^\vee$. Then the lattice*

$$L_0 = \{iH : \forall \alpha \in \Pi,\ \alpha(H) \in 2\pi\mathbb{Z}\}$$

is generated by $2\pi i H_\beta$ with $\beta \in \Omega^\vee$.

Proof Let $B = \{H_1, \ldots, H_l\}$ be the dual basis of $\Sigma = \{\alpha_1, \ldots, \alpha_l\} \subset \Pi$. Every $H \in \mathfrak{h}$ is written

$$H = \alpha_1(H) H_1 + \cdots + \alpha_l(H) H_l,$$

which implies that L_0 is generated by $2\pi i B$. It must be checked that $B = \{H_\beta : \beta \in \Omega^\vee\}$. To see this, observe that $\left(\alpha^\vee\right)^\vee = \alpha$ and therefore $\left(\Pi^\vee\right)^\vee = \Pi$ and $\left(\Sigma^\vee\right)^\vee = \Sigma$. From the definition of the fundamental weights it follows that $\Omega^\vee = \{\beta_1, \ldots, \beta_l\}$ is given by the relations $\langle \alpha_j, \beta_k \rangle = \delta_{jk}$, that is, $H_j = H_{\beta_j}$, concluding the proof. □

A consequence of this proposition and of Lemma 11.35 is that the index of $L_{\min}$ in L_0 is the determinant of the linear map g that maps the generators $2\pi i H_\beta$, $\beta \in \Omega^\vee$, of L_0 into the generators $2\pi i H_\alpha$, $\alpha \in \Sigma^\vee$ of $L_{\min}$. The matrix of this linear map is precisely the Cartan matrix $C\left(\Sigma^\vee\right) = C(\Sigma)^T$.

Indeed, writing $\Sigma^\vee = \{\gamma_1, \ldots, \gamma_l\}$ and $\Omega^\vee = \{\beta_1, \ldots, \beta_l\}$, each β_j is obtained through the relations

$$\frac{2\langle \gamma_i, \beta_j \rangle}{\langle \gamma_i, \gamma_i \rangle} = \delta_{ij} \qquad i = 1, \ldots, l.$$

Taking coordinates with respect to $\Sigma^\vee$, the matrix $\left[\beta_j\right]$ of coordinates of β_j satisfies the linear system

$$C\left(\Sigma^\vee\right)\left[\beta_j\right] = c_j,$$

where c_j is the column matrix with entry 1 in the j-th position and 0 elsewhere. Therefore, the coordinates of $\{\beta_1, \ldots, \beta_l\}$ with respect to $\Sigma^\vee$ are the columns of the matrix $C\left(\Sigma^\vee\right)^{-1}$. It follows that if g is the linear map defined by $g\left(\beta_j\right) = \gamma_j$, then its matrix on the basis $\Sigma^\vee$ is $C\left(\Sigma^\vee\right) = C(\Sigma)^T$.

Summarizing this discussion the following proposition furnishes the fundamental group of $\mathrm{Aut}_0\mathfrak{u}$ in terms of the Cartan matrix.

Proposition 11.41 *Let $\mathfrak{u}$ be a compact semi-simple Lie algebra and denote by $C(\Sigma)$ the Cartan matrix of its complexification. Then the order of the fundamental group $\pi_1(\mathrm{Aut}_0\mathfrak{u})$ is $\det C(\Sigma)$.*

The determinant $\det C(\Sigma)$ is known as the **connectivity index** of the semi-simple algebra. It is directly computed from its Dynkin diagram and, as was seen, coincides with the order of the fundamental group of $\mathrm{Aut}_0\mathfrak{u}$. The table below shows the indices for the diagrams in the classification, which correspond to simple algebras. This table is easily obtained from the Cartan matrices associated with the Dynkin diagrams.[10]

Σ	A_l	B_l	C_l	D_l	E_6	E_7	E_8	G_2	F_4
$\det C(\Sigma)$	$l+1$	2	2	4	3	2	1	1	1

The following observations are read from the table of the connectivity indices:

1. The connectivity index of A_l, which corresponds to $\mathfrak{su}(n)$, $n = l+1$, is n, reflecting the fact that $\mathrm{SU}(n)$ is simply connected and its center $Z(\mathrm{SU}(n))$ has order n, as $Z(\mathrm{SU}(n)) \approx \mathbb{Z}_n$. Indeed, both results, $\det C(\Sigma) = n$ and $Z(\mathrm{SU}(n)) \approx \mathbb{Z}_n$, which can be obtained algebraically, show in an indirect way that $\mathrm{SU}(n)$ is simply connected.
2. The class B_l, $l \geq 2$, corresponds to the algebras $\mathfrak{so}(n)$ with $n = 2l+1$. For odd values of n, $n \geq 5$, the center of the group $\mathrm{SO}(n)$ is trivial. Therefore, the connectivity index of B_l, $l \geq 2$, shows that $\mathrm{SO}(n)$, $n = 2l+1$, is the adjoint group of $\mathfrak{so}(n)$. The universal covering group is a double covering of $\mathrm{SO}(n)$. This covering group is denoted by $\mathrm{Spin}(n)$.
3. The compact Lie algebras associated with C_l, $l \geq 3$, are given by $\mathfrak{sp}(l)$, which is given by the quaternionic skew-Hermitian matrices. A Lie group with Lie algebra $\mathfrak{sp}(l)$ is the group $\mathrm{Sp}(l)$ of quaternionic unitary matrices. This group is connected (see Chapter 7, Exercise 17). The center of $\mathrm{Sp}(l)$ is $\mathbb{Z}_2$ (given by $\{\pm 1\}$), which confirms that the connectivity index of C_l is 2, as shows the table.
4. The class D_l, $l \geq 4$, corresponds to the algebras $\mathfrak{so}(n)$ with $n = 2l$. For even values of n, $n \geq 8$, the group $\mathrm{SO}(n)$ has center $\{\pm 1\}$. Therefore, the connectivity index of D_l, $l \geq 4$, shows that $\pi_1(\mathrm{SO}(n))$, $n = 2l$, has order 2, that is, $\pi_1(\mathrm{SO}(n)) = \mathbb{Z}_2$ (compare with Exercise 18 of Chapter 7). As in the odd case, the universal covering group of $\mathrm{SO}(n)$ is a double covering, which is denoted by $\mathrm{Spin}(n)$.
   The fundamental group of the adjoint group $\mathrm{Aut}_0(\mathfrak{so}(2l))$ has order 4. It is $\mathbb{Z}_4$ if l is odd and $\mathbb{Z}_2 \times \mathbb{Z}_2$ if l is even (the Exercise 12 at the end of this chapter points out a proof of this fact).
5. The connectivity indices of the exceptional algebras show that in the cases E_8, G_2, and F_4 the adjoint group is the only compact Lie group with the respective Lie algebras. In the other cases there exist the adjoint group and the universal covering group, which is a triple covering for E_6 and a double covering for E_7.

[10] See *Álgebras de Lie* [47], Chapters 6, 7, 11 and Varadarajan [53], Section 4.5.

11.6 Riemannian Geometry

The objective of this section is to comment without details on the proofs of the surjectivity of the exponential map and of the Weyl theorem from general theorems of Riemannian geometry. The concepts and notations are the usual ones adopted in most textbooks about Riemannian geometry.

Let U be a compact Lie group with Lie algebra $\mathfrak{u}$. An inner product $(\cdot,\cdot)$ on $\mathfrak{u}$, invariant by $\mathrm{Ad}\,(u)$, $u \in U$, defines on U a Riemannian metric (also denoted by $(\cdot,\cdot)$) that is invariant by left translations,

$$(v, w)_u = \left(dL_{u^{-1}}(v), dL_{u^{-1}}(w)\right), v, w \in T_u U.$$

By definition, this metric is left invariant (the left translations are isometries). Since the inner product $(\cdot,\cdot)$ is invariant by $\mathrm{Ad}\,(u)$, $u \in U$, it follows that the Riemannian metric is invariant also by right translations, that is, it is bi-invariant.[11]

The Levi-Civita connection ∇ of $(\cdot,\cdot)$ is also bi-invariant, that is, $L_{u*}\nabla_X Y = \nabla_{L_{u*}X} L_{u*}Y$ and $R_{u*}\nabla_X Y = \nabla_{R_{u*}X} R_{u*}Y$ for $u \in U$ and X, Y vector fields. The connection is computed by the formula

$$\begin{aligned} 2(\nabla_X Y, Z) = {} & X(Y, Z) + Y(X, Z) - Z(X, Y) \\ & - ([X, Z], Y) - ([Y, Z], X) + ([X, Y], Z), \end{aligned}$$

where X, Y, and Z are vector fields. Taking (left or right) invariant vector fields, the first three terms are zero because (X, Y) is constant if X and Y are invariant vector fields. By the invariance of the inner product, at the identity the three last terms reduce to

$$-([X, Z], Y) - ([Y, Z], X) + ([X, Y], Z) = ([X, Y], Z).$$

Therefore, if X and Y are invariant vector fields, then

$$\nabla_X Y = \frac{1}{2}[X, Y]. \tag{11.11}$$

On the other hand, the curvature is given by

$$\begin{aligned} R(X, Y)Z &= \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z \\ &= \frac{1}{4}[X, [Y, Z]] - \frac{1}{4}[Y, [X, Z]] \frac{1}{2}[[X, Y], Z] \end{aligned}$$

if X, Y, and Z are invariant vector fields. Applying the Jacobi identity yields

[11] See Chapter 14 for more details about the construction of the bi-invariant metric.

$$R(X,Y)Z = -\frac{1}{4}[[X,Y],Z]. \tag{11.12}$$

The geodesics of $(\cdot,\cdot)$ are easily obtained from formula (11.11). Indeed, if X is an invariant vector field, then $\nabla_X X = 0$. Hence, the trajectories of X are geodesics. In other words, for every $X \in \mathfrak{g}$, the curves $e^{tX}u$ and ue^{tX} are geodesics starting at $u \in U$. (Observe that $ue^{tX} = e^{t\mathrm{Ad}(u)X}u$, so that the sets of curves $e^{tX}u$ and ue^{tY}, with X and Y running through $\mathfrak{u}$, coincide.) In particular, the geodesics starting at the identity 1 are the 1-parameter groups e^{tX}. This means that the exponential map, based at $1 \in U$, defined by the Riemannian metric, coincides with the exponential map of U.

This equality between the exponential maps yields the surjectivity of the exponential $\exp : \mathfrak{u} \to U$ by an application of the Hopf–Rinow theorem.[12] This theorem ensures that any two points on a geodesically complete Riemannian manifold are connected by a geodesic. The Riemannian metric $(\cdot,\cdot)$ is geodesically complete in the sense that its geodesics are defined for every $t \in \mathbb{R}$, as happens to any compact Riemannian manifold.

The Weyl theorem of the finite fundamental group (for compact semi-simple groups) follows from the Bonnet-Myers theorem.[13]This theorem ensures that if the Ricci curvature

$$\mathrm{Ric}(v) = \mathrm{tr}(w \mapsto R(w,v)v)$$

of a Riemannian manifold M satisfies $\mathrm{Ric}(v) > c > 0$ for every tangent vector v with $\|v\| = 1$, then the universal covering of M is compact. For a compact semi-simple group this condition is satisfied by a bi-invariant metric. Indeed, by the curvature formula (11.12), it follows that the Ricci curvature is given for $X \in \mathfrak{u}$, with $\|X\| = 1$, by

$$\begin{aligned}\mathrm{Ric}(X) &= -\frac{1}{4}\sum_{i=1}^{n}([[Y_i,X],X],Y_i)\\ &= -\frac{1}{4}\sum_{i=1}^{n}\left(\mathrm{ad}(X)^2 Y_i, Y_i\right),\end{aligned}$$

where Y_i is an orthonormal basis of $\mathfrak{u}$. That is, $\mathrm{Ric}(X) = -\frac{1}{4}\mathrm{tr}\left(\mathrm{ad}(X)^2\right)$. As the Cartan–Killing form of $\mathfrak{u}$ is negative definite, $\mathrm{tr}\left(\mathrm{ad}(X)^2\right) < 0$ if $X \neq 0$, which shows that $\mathrm{Ric}(X) > 0$ and assumes a minimum $c > 0$ when $\|X\| = 1$.

Therefore, if $\mathfrak{u}$ is semi-simple then U admits a Riemannian metric which satisfies the conditions of the Bonnet-Myers theorem, showing that its universal covering group is compact.

[12] See Carmo [7].

[13] See Carmo [7].

11.7 Exercises

1. Show that the Lie algebra $\mathfrak{su}(n)$ is semi-simple using the fact that $\mathrm{SU}(n)$ is compact and simply connected.
2. Use the description of the compact Lie algebras to show that the spheres S^2 and S^4 do not admit Lie group structures.
3. Let S^3 be the 3-dimensional sphere and suppose that a differentiable map $p : S^3 \times S^3 \to S^3$ satisfies the group axioms. Show that p is the product on $\mathrm{SU}(2)$ or, what is the same, the product on the unit sphere of quaternions.
4. Let K be a compact Lie group. Show that the set of finite order elements $x \in K$ is dense in K. Show also that for every $n \in \mathbb{N}$, there exists $x \in K$ whose order is finite and greater than n.
5. Let $\mathfrak{u}$ be a compact Lie algebra and U a connected Lie group with Lie algebra $\mathfrak{u}$. Show that U is the direct product of a connected compact group by a simply connected abelian group. (Write $\exp \mathfrak{t}$ as a cylinder, where $\mathfrak{t}$ is a Cartan subalgebra.)
6. Let θ be an automorphism of a semi-simple Lie algebra $\mathfrak{g}$. Show that $\phi : \mathrm{Aut}_0(\mathfrak{g}) \to \mathrm{Aut}_0(\mathfrak{g})$ given by $\phi(g) = \theta g \theta^{-1}$ is an automorphism of the adjoint group $\mathrm{Aut}_0(\mathfrak{g})$ which extends θ.
7. Let U be a compact Lie group with Lie algebra $\mathfrak{u}$ and $\Gamma \subset U$ a discrete (and hence finite) subgroup. Take a bi-invariant volume form ν on U whose Haar measure μ_ν is normalized by $\mu_\nu(U) = 1$. This volume form defines a differential form on $\mathfrak{u}$ which is invariant by the adjoint representation, which in turn defines an invariant form $\overline{\nu}$ on U/Γ. Denote by $\mu_{\overline{\nu}}$ the Haar measure on U/Γ defined by $\overline{\nu}$. Show that $\mu_{\overline{\nu}}(U/\Gamma) = 1/|\Gamma|$. Use this to justify the arguments of the proof of Lemma 11.35.
8. Let K be a connected compact Lie group and $T \subset K$ a maximal torus. Show that T contains the center $Z(K)$ of K.
9. Let U be a compact Lie group with Lie algebra $\mathfrak{u}$. Take a Cartan subalgebra $\mathfrak{t} \subset \mathfrak{u}$ and describe geometrically the set of the elements $X \in \mathfrak{t}$ which are not singularities of the exponential map.
10. Let G be a solvable Lie group and $H \subset G$ a connected compact subgroup. Show that H is a torus.
11. Let $L_1 \subset L_2 \subset \mathbb{R}^n$ be lattices of $\mathbb{R}^n$ generated by the bases $\beta_v = \{v_1, \ldots, v_n\}$ and $\beta_w = \{w_1, \ldots, w_n\}$, respectively. Let g be the linear map such that $g w_j = v_j$, and denote by $M = [g]_{\beta_w}$ the matrix of g on basis β_w. Show that the entries of M are integer numbers and that, if $k \in \mathbb{N}$ is such that the matrix kM^{-1} has integer entries, then, for every $x \in L_2$, $kx \in L_1$. Show also that if some entry of kM^{-1} is not an integer, then there exists $x \in L_2$ such that $kx \notin L_1$.
12. Use the previous exercise to show that the fundamental group of $\mathrm{Aut}(\mathfrak{so}(2l))$, $l \geq 4$, is $\mathbb{Z}_4$ if l is odd and $\mathbb{Z}_2 \times \mathbb{Z}_2$ if l is even.

Chapter 12
Noncompact Semi-Simple Groups



This chapter addresses noncompact semi-simple Lie groups. The starting point is the construction of the Cartan and Iwasawa decompositions of the Lie groups. Either decomposition shows that the differentiable manifold underlying a noncompact semi-simple and connected Lie group G is the product of some Euclidean space with a connected Lie group K whose Lie algebra is compact. Due to these splittings, the question of describing the universal covering $\widetilde{G}$ of G reduces to determining the universal covering of K, which has been done earlier.

This reduction of a semi-simple group to its "compact part" extends to any Lie group by way of the Levi decomposition theorem for Lie algebras. According to this result, a simply connected Lie group is the semi-direct product of a semi-simple Lie group and a solvable group. The latter is a Euclidean space, as established in Chapter 10.

12.1 Cartan Decompositions

Let $\mathfrak{g}$ be a real, semi-simple Lie algebra. Its complexification $\mathfrak{g}_{\mathbb{C}}$ is also semi-simple. This follows from Cartan's criterion, whereby a Lie algebra $\mathfrak{l}$ is semi-simple if and only if its Cartan–Killing form $\mathcal{K}_{\mathfrak{l}}(X, Y) = \operatorname{tr}(\operatorname{ad}(X)\operatorname{ad}(Y))$ is non-degenerate.[1] The Cartan–Killing forms $\mathcal{K}_{\mathfrak{g}}$ and $\mathcal{K}_{\mathfrak{g}_{\mathbb{C}}}$ are both non-degenerate, since $\mathcal{K}_{\mathfrak{g}}$ is the restriction to $\mathfrak{g}$ of $\mathcal{K}_{\mathfrak{g}_{\mathbb{C}}}$. Therefore $\mathfrak{g}$ is semi-simple if and only if $\mathfrak{g}_{\mathbb{C}}$ is semi-simple.

[1] See *Álgebras de Lie* [47, Ch. 3] and Knapp [33, Ch. I].

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12.1.1 Cartan Decomposition of a Lie Algebra

A **Cartan decomposition** of $\mathfrak{g}$ is a direct sum

$$\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{s}$$

with $\mathfrak{k} = \mathfrak{g} \cap \mathfrak{u}$ and $\mathfrak{s} = \mathfrak{g} \cap i\mathfrak{u}$, where $\mathfrak{u}$ is a compact real form of $\mathfrak{g}_{\mathbb{C}}$. The existence of Cartan decompositions is warranted using the conjugations of $\mathfrak{g}_{\mathbb{C}}$. Let σ be the conjugation map of $\mathfrak{g}_{\mathbb{C}}$ with respect to $\mathfrak{g}$ (that is, $\sigma(X + iY) = X - iY$ if $X, Y \in \mathfrak{g}$). It can be proved that there exists a compact real form $\mathfrak{u}$ of $\mathfrak{g}_{\mathbb{C}}$ whose conjugation commutes with σ.[2] The commutativity implies $\tau(\mathfrak{g}) = \mathfrak{g}$ and $\sigma(\mathfrak{u}) = \mathfrak{u}$, since $\mathfrak{g}$ is the fixed-point set of σ and $\mathfrak{u}$ that of τ. Therefore any $X \in \mathfrak{g}$ can be written as

$$X = \frac{X + \tau X}{2} + \frac{X - \tau X}{2},$$

which describes the direct sum $\mathfrak{g} = (\mathfrak{g} \cap \mathfrak{u}) \oplus (\mathfrak{g} \cap i\mathfrak{u})$ explicitly.

The Cartan decomposition is not unique, though, because different compact real forms may provide direct sums $\mathfrak{g} = (\mathfrak{g} \cap \mathfrak{u}) \oplus (\mathfrak{g} \cap i\mathfrak{u})$. However, no generality is lost in fixing one of them, since two Cartan decompositions are obtained from one another by an inner automorphism of $\mathfrak{g}$. Hence, results obtained from different decompositions are equivalent.

Here is a list of properties of Cartan decompositions:

1. on $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{s}$, the bracket is given by

$$[\mathfrak{k}, \mathfrak{k}] \subset \mathfrak{k} \qquad [\mathfrak{k}, \mathfrak{s}] \subset \mathfrak{s} \qquad [\mathfrak{s}, \mathfrak{s}] \subset \mathfrak{k}.$$

   In particular, $\mathfrak{k}$ is Lie subalgebra. These relations follow directly from $\mathfrak{k} = \mathfrak{g} \cap \mathfrak{u}$ and $\mathfrak{s} = \mathfrak{g} \cap i\mathfrak{u}$ (since $\mathfrak{u}$ is a Lie algebra), $[\mathfrak{u}, i\mathfrak{u}] \subset i\mathfrak{u}$ and $[i\mathfrak{u}, i\mathfrak{u}] \subset \mathfrak{u}$.
2. The restriction $\theta = \tau_{|\mathfrak{g}}$ is an involutive automorphism of $\mathfrak{g}$, that is, $\theta^2 = \mathrm{id}$, called Cartan involution. Its eigenspace, associated with the eigenvalue 1, is $\mathfrak{k}$, while $\mathfrak{s}$ is the -1-eigenspace.
3. The Cartan–Killing form $\mathcal{K}_{\mathfrak{g}}(X, Y)$ of $\mathfrak{g}$ is negative-definite on $\mathfrak{k}$ and positive-definite on $\mathfrak{s}$. This is because $\mathfrak{u}$ is a compact semi-simple algebra, so its Cartan–Killing form $\mathcal{K}_{\mathfrak{u}}(\cdot, \cdot)$ is negative-definite. The restrictions of $\mathcal{K}_{\mathfrak{g}_{\mathbb{C}}}(\cdot, \cdot)$ to $\mathfrak{u}$ and $\mathfrak{g}$ produce the respective Cartan–Killing forms $\mathcal{K}_{\mathfrak{u}}(\cdot, \cdot)$ and $\mathcal{K}_{\mathfrak{g}}(\cdot, \cdot)$. On $\mathfrak{g} \cap \mathfrak{u}$, therefore, $\mathcal{K}_{\mathfrak{g}}(\cdot, \cdot)$ coincides with $\mathcal{K}_{\mathfrak{u}}(\cdot, \cdot)$ and is negative-definite. Likewise, $\mathcal{K}_{\mathfrak{g}}(\cdot, \cdot)$ is positive-definite on $\mathfrak{g} \cap i\mathfrak{u}$.
4. If $\mathfrak{g} = \mathfrak{k}_1 \oplus \mathfrak{s}_1 = \mathfrak{k}_2 \oplus \mathfrak{s}_2$ are two Cartan decompositions, there exists $g \in \mathrm{Aut}_0\mathfrak{g}$ such that $g\mathfrak{k}_1 = \mathfrak{k}_2$ and $g\mathfrak{s}_1 = \mathfrak{s}_2$.

[2] See *Álgebras de Lie* [47, Theorem 12.18] and Helgason [20, Theorem III.7.1].

5. If $X \in \mathfrak{k}$ and $Y \in \mathfrak{s}$, then $\mathcal{K}_{\mathfrak{g}}(X, Y) = 0$, because $\mathcal{K}_{\mathfrak{g}}(X, Y) = \mathcal{K}_{\mathfrak{g}}(\theta X, Y) = \mathcal{K}_{\mathfrak{g}}(X, \theta Y)$, as θ is an automorphism. Therefore $\mathcal{K}_{\mathfrak{g}}(X, Y) = \mathcal{K}_{\mathfrak{g}}(X, \theta Y) = -\mathcal{K}_{\mathfrak{g}}(X, Y)$, which implies $\mathcal{K}_{\mathfrak{g}}(X, Y) = 0$.
6. The bilinear form $B_\theta(X, Y) = -\mathcal{K}_{\mathfrak{g}}(X, \theta Y)$ is an inner product. In fact, $\mathcal{K}_{\mathfrak{g}}$ is negative-definite on $\mathfrak{k}$ and positive-definite on $\mathfrak{s}$, and these two subspaces are orthogonal under $\mathcal{K}_{\mathfrak{g}}$.
7. If $X \in \mathfrak{k}$ and $Z, W \in \mathfrak{g}$, then

$$\begin{aligned} B_\theta([X, Z], W) &= -\mathcal{K}_{\mathfrak{g}}([X, Z], \theta W) = \mathcal{K}_{\mathfrak{g}}(Z, \theta[X, W]) \\ &= -B_\theta(Z, [X, W]). \end{aligned}$$

   That is to say, $\operatorname{ad}(X)$ is skew-symmetric with respect to B_θ.
8. If $Y \in \mathfrak{s}$ and $Z, W \in \mathfrak{g}$, then

$$\begin{aligned} B_\theta([Y, Z], W) &= -\mathcal{K}_{\mathfrak{g}}([Y, Z], \theta W) = -\mathcal{K}_{\mathfrak{g}}(Z, \theta[Y, W]) \\ &= B_\theta(Z, [Y, W]). \end{aligned}$$

   In other words, $\operatorname{ad}(Y)$ is symmetric with respect to B_θ.
9. $\mathfrak{k}$ is a compact Lie algebra, being a subalgebra of the compact algebra $\mathfrak{u}$. It is actually a **maximal compact** subalgebra,: it is not properly contained in any compact subalgebra. In fact, if $\mathfrak{k} \subset \mathfrak{l}$ and $\mathfrak{k} \neq \mathfrak{l}$, given $Z \in \mathfrak{l} \setminus \mathfrak{k}$ there exist $X \in \mathfrak{k}$ and $0 \neq Y \in \mathfrak{s}$ such that $Z = X + Y$. Then $Y \in \mathfrak{l}$ and $\mathcal{K}_{\mathfrak{g}}(Y, Y) > 0$, and $\mathfrak{l}$ is not a compact Lie algebra. (Actually, after developing the theory further, it is possible to show that $\mathfrak{k}$ is not properly contained in any proper subalgebra, compact or not.)
10. If $\mathfrak{g}$ is the real space underlying a complex semi-simple Lie algebra, any Cartan decomposition of $\mathfrak{g}$ has the form $\mathfrak{g} = \mathfrak{u} \oplus i\mathfrak{u}$, where $\mathfrak{u}$ is a compact real form of the complex Lie algebra $\mathfrak{g}$.
11. The isomorphism $\operatorname{ad} : \mathfrak{g} \to \operatorname{Der}\mathfrak{g}$, together with the Cartan involution θ, defines an automorphism $\overline{\theta}$ of $\operatorname{Der}\mathfrak{g}$ by the commutative diagram

$$\begin{array}{ccc} \mathfrak{g} & \xrightarrow{\theta} & \mathfrak{g} \\ \downarrow & & \downarrow \\ \operatorname{Der}\mathfrak{g} & \xrightarrow{\overline{\theta}} & \operatorname{Der}\mathfrak{g} \end{array}$$

   That is, $\overline{\theta}(\operatorname{ad}(X)) = \operatorname{ad}(\theta X)$. Since θ is an automorphism, formula $\operatorname{ad}(\theta X) = \theta \circ \operatorname{ad}(X) \circ \theta^{-1}$ holds, and $\overline{\theta}$ is the conjugation by θ on $\operatorname{Der}\mathfrak{g}$. This conjugation is given by $\overline{\theta}(\operatorname{ad}(X)) = -\operatorname{ad}(X)^T$, where the transpose is taken with respect to the inner product B_θ. In fact,

$$B_\theta([X, Y], Z) = -\mathcal{K}_{\mathfrak{g}}([X, Y], \theta Z) = \mathcal{K}_{\mathfrak{g}}(Y, [X, \theta Z]),$$

since $\operatorname{ad}(X)$ is skew-symmetric for the Cartan–Killing form $\mathcal{K}_{\mathfrak{g}}$. Then

$$B_\theta([X,Y],Z) = \mathcal{K}_{\mathfrak{g}}(Y,\theta[\theta X,Z]) = -B_\theta(Y,[\theta X,Z]),$$

meaning $\operatorname{ad}(\theta X) = -\operatorname{ad}(X)^T$.

Example Consider $\mathfrak{g} = \mathfrak{sl}(n,\mathbb{R})$, $\mathfrak{g}_{\mathbb{C}} = \mathfrak{sl}(n,\mathbb{C})$, and $\mathfrak{u} = \mathfrak{su}(n)$. Then $\mathfrak{k} = \mathfrak{g} \cap \mathfrak{u} = \mathfrak{so}(n)$, while $\mathfrak{s} = \mathfrak{g} \cap i\mathfrak{u}$ is the space of symmetric matrices with zero trace. In this case, $\tau(Z) = -\overline{Z}^T$ and $\theta(X) = -X^T$. The inner product reads $B_\theta(X,Y) = \operatorname{tr}(\operatorname{ad}(X)\operatorname{ad}(\theta Y))$. It can be proved that B_θ is a multiple of the inner product $\operatorname{tr}\left(XY^T\right)$. This Cartan decomposition surmises the presence of an inner product in $\mathbb{R}^n$ (to compute transpose matrices). Different inner products will define other Cartan decompositions. □

The Cartan decomposition of $\mathfrak{sl}(n,\mathbb{R})$ into symmetric and skew-symmetric matrices is typical of linear semi-simple Lie algebras. For these algebras, it is possible to describe a Cartan decomposition whereby the matrices in $\mathfrak{k}$ are skew-symmetric and the elements in $\mathfrak{s}$ are symmetric. Another manifestation of this is given by the symplectic algebra.

Example Let $\mathfrak{sp}(n,\mathbb{R}) = \{A \in \mathfrak{gl}(2n,\mathbb{R}) : AJ + JA^T = 0\}$ be the Lie algebra of real symplectic matrices, where

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

is written in block form with $n \times n$ blocks. The elements of $\mathfrak{sp}(n,\mathbb{R})$ are real matrices of type

$$\begin{pmatrix} A & B \\ C & -A^T \end{pmatrix} \qquad \text{with} \quad B - B^T = C - C^T = 0.$$

A Cartan decomposition consists of

$$\mathfrak{k} = \{\begin{pmatrix} A & B \\ -B & A \end{pmatrix} : \; A + A^T = B - B^T = 0\},$$

which is formed by skew matrices, and the set

$$\mathfrak{s} = \{\begin{pmatrix} A & B \\ B & -A \end{pmatrix} : \; A - A^T = B - B^T = 0\}$$

of symmetric matrices in $\mathfrak{sp}(n,\mathbb{R})$. These subspaces give a Cartan decomposition of $\mathfrak{sp}(n,\mathbb{R})$. □

12.1.2 Global Cartan Decomposition

The objective is to construct the Cartan decomposition at the Lie group level. This says that a noncompact semi-simple Lie group splits as $G = KS = SK$, where $K = \exp \mathfrak{k}$, $S = \exp \mathfrak{s}$, and $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{s}$ are Cartan decompositions of the Lie algebra $\mathfrak{g}$ of G. These decompositions give diffeomorphisms between G and $K \times S$, which means that $g \in G$ can be written in a unique way as $g = sk$ or $g = s'k'$, with $s, s' \in S$, $k = k' \in K$, so that the maps $g \mapsto k$, $g \mapsto s$, etc. are differentiable.

The Cartan decomposition is found first for the adjoint group $\mathrm{Aut}_0\mathfrak{g}$ of inner automorphisms of $\mathfrak{g}$, which is a Lie group with Lie algebra $\mathfrak{g}$. Fix a Cartan decomposition $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{s}$, and let θ be the corresponding Cartan involution, then define the inner product $B_\theta = -\mathcal{K}_\mathfrak{g}(X, \theta Y)$.

Let $\theta_0(g) = \theta g \theta^{-1}$ be the conjugation by θ. If g is an automorphism of $\mathfrak{g}$, then $\theta g \theta^{-1}$ is also an automorphism, so θ_0 defines an automorphism of the adjoint group $\mathrm{Aut}_0\mathfrak{g}$. The relation $\theta \mathrm{ad}(X) \theta^{-1} = \mathrm{ad}(\theta X)$ shows that θ_0 is an extension of θ viewed as an automorphism of $\mathrm{Der}\mathfrak{g} \approx \mathfrak{g}$. This extension is given by $\theta_0(g) = \left(g^{-1}\right)^T$, where transposes are taken with respect to the inner product B_θ. This follows from a computation similar to the one made above for $\mathrm{ad}(X)$. In fact, if $g \in \mathrm{Aut}_0\mathfrak{g}$ and $X, Y \in \mathfrak{g}$, then

$$B_\theta(gX, Y) = -\mathcal{K}_\mathfrak{g}(gX, \theta Y) = -\mathcal{K}_\mathfrak{g}\left(X, g^{-1}\theta Y\right),$$

since g is an isometry for the Cartan–Killing form $\mathcal{K}_\mathfrak{g}$. Hence,

$$B_\theta(gX, Y) = -\mathcal{K}_\mathfrak{g}\left(X, \theta\left(\theta^{-1}g^{-1}\theta\right)Y\right) = B_\theta\left(X, \left(\theta^{-1}g^{-1}\theta\right)Y\right),$$

that is, $\theta^{-1}g^{-1}\theta = g^T$, and so $\theta_0(g) = \theta g \theta^{-1} = \left(g^{-1}\right)^T$. It is clear that $\theta_0^2 = 1$, implying that g^T is an automorphism for any automorphism g.

Consider now

$$K_{\mathrm{ad}} = \{g \in \mathrm{Aut}_0\mathfrak{g} : \theta_0(g) = g\}.$$

This is a closed subgroup with Lie algebra $\mathrm{ad}(\mathfrak{k})$, because $\mathrm{ad}(\mathfrak{k})$ are the fixed points of θ. Moreover, an automorphism g belongs to K_{ad} if and only if $\left(g^{-1}\right)^T = g$, which means $K_{\mathrm{ad}} = \mathrm{SO}(\mathfrak{g}, B_\theta) \cap \mathrm{Aut}_0\mathfrak{g}$, where $\mathrm{SO}(\mathfrak{g}, B_\theta)$ is the isometry group of B_θ. Therefore K_{ad} is compact. (Later K_{ad} will be proved to be connected, ensuring that $K_{\mathrm{ad}} = \langle \exp \mathrm{ad}(\mathfrak{k}) \rangle = \exp \mathrm{ad}(\mathfrak{k})$.)

The group K_{ad} globalizes, so-to-speak, the skew-symmetric part in the Cartan decomposition. As for the symmetric part, suppose $g \in \mathrm{Aut}_0\mathfrak{g}$ is symmetric and positive-definite with respect to B_θ. Then g is diagonalizable, with positive real eigenvalues, and there exists $A \in \mathfrak{gl}(\mathfrak{g})$ symmetric (with respect to B_θ) such that $g = e^A$.

Lemma 12.1 *The map A such that $g = e^A$ is a derivation of $\mathfrak{g}$, and $e^{tA} \in \mathrm{Aut}_0\mathfrak{g}$ for all $t \in \mathbb{R}$. Furthermore A is an inner derivation, that is, $A = \mathrm{ad}(X)$ with $X \in \mathfrak{s}$.*

Proof Let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of g, and $\{X_1, \ldots, X_n\}$ a basis of $\mathfrak{g}$ diagonalizing g, with $gX_j = \lambda_i X_j$, $j = 1, \ldots, n$. Then $AX_j = a_i X_j$, where $a_j = \log \lambda_j$. Let c^l_{jk} be the structure constants of $\{X_1, \ldots, X_n\}$

$$[X_j, X_k] = \sum_{l=1}^{n} c^l_{jk} X_l. \tag{12.1}$$

Applying g to the above relation and using that g is an automorphism yield

$$\lambda_j \lambda_k [X_j, X_k] = \sum_{l=1}^{n} \lambda_l c^l_{jk} X_l \,.$$

Replacing the bracket with the linear combination (12.1) gives

$$c^l_{jk} \lambda_j \lambda_k = c^l_{jk} \lambda_l$$

for all j, k, l. Cancelling terms and subsequently multiplying by the structure constants, one obtains

$$c^l_{jk} e^{ta_j} e^{ta_k} = c^l_{jk} e^{ta_l} \,,$$

which immediately shows e^{tA} is an automorphism for all $t \in \mathbb{R}$. Therefore A is a derivation.

The derivation A is inner, since $\mathfrak{g}$ is semi-simple. Properties (7) and (8) of the Cartan decomposition, stated above, show that $X \in \mathfrak{s}$, since $\mathrm{ad}(X)$ is symmetric with respect to B_θ. □

Now let $S = \exp \mathfrak{s}$ be the image of the differentiable map $\exp : \mathfrak{s} \to \mathrm{Aut}_0\mathfrak{g}$. The map is an immersion, since if $X \in \mathfrak{s}$, then $d(\exp)_X = dE_{e^X} \circ T_X$ with

$$T_X = \frac{e^{\mathrm{ad}(X)} - 1}{\mathrm{ad}(X)} = \sum_{k \geq 0} \frac{1}{k!} \mathrm{ad}(X)^k \,,$$

and the above is an isomorphism since the eigenvalues of $\mathrm{ad}(X)$ are real (see Chapter 8). Therefore the restriction of $d(\exp)_X$ to $\mathfrak{s}$ is injective. Furthermore, $\exp : \mathfrak{s} \to \mathrm{Aut}_0\mathfrak{g}$ is injective, since if $X \in \mathfrak{s}$, then e^X is symmetric and positive-definite with respect to B_θ. In this way, X can be recovered from e^X by taking the logarithms of the eigenvalues, as in the proof of the previous lemma.

In conclusion, $S = \exp \mathfrak{s}$ is an immersed submanifold of $\mathrm{Aut}_0\mathfrak{g}$ diffeomorphic to $\mathfrak{s}$. The tangent space $T_{e^X} S$ to S at e^X, for $X \in \mathfrak{s}$, is given by $dE_{e^X} \circ T_X(\mathfrak{s})$.

The following lemma, which will be used to prove the injectivity of the Cartan decomposition, explains that $T_X(\mathfrak{s})$ is transversal to $\mathfrak{k}$.

Lemma 12.2 *If $X \in \mathfrak{s}$, then $\mathfrak{k} \cap T_X(\mathfrak{s}) = \{0\}$.*

Proof If $Y \in \mathfrak{s}$, then $T_X(Y) = \sum_{j\geq 0} \frac{1}{j!} \mathrm{ad}(X)^j(Y)$. In the series, the terms with even j belong to $\mathfrak{s}$, while those of odd degree are in $\mathfrak{k}$, as follows from the brackets $[\mathfrak{k}, \mathfrak{s}] \subset \mathfrak{s}$ and $[\mathfrak{s}, \mathfrak{s}] \subset \mathfrak{k}$. The claim is that the sum $E_X(Y)$ of even-degree terms is not zero unless $T_X(Y) = 0$.

Since T_X is the power series of the function $f(x) = \frac{e^x - 1}{x}$ evaluated at $\mathrm{ad}(X)$, the even-term sum E_X is given by the power series of $f(x) + f(-x) = \frac{\sinh(x)}{x}$. This function is strictly positive, so 0 is not an eigenvalue of $E_X = \frac{\sinh(\mathrm{ad}(X))}{\mathrm{ad}(X)}$. Hence E_X is injective, and therefore $E_X(Y) = 0$ implies $T_X(Y) = Y = 0$, concluding the proof. □

From these lemmas, it is possible to infer the Cartan decomposition of the adjoint group $\mathrm{Aut}_0\mathfrak{g}$.

Theorem 12.3 *The map $\phi : \mathfrak{s} \times K_{\mathrm{ad}} \to \mathrm{Aut}_0\mathfrak{g}$ defined by $\phi(X, k) = \exp(\mathrm{ad}(X))\, k$ is a diffeomorphism, and $\mathrm{Aut}_0\mathfrak{g} = SK_{\mathrm{ad}}$, where $S = \exp \mathfrak{s}$. Furthermore, K_{ad} is connected.*

Proof First, ϕ is onto, because if $g \in \mathrm{Aut}_0\mathfrak{g}$, then $g^T = \theta\left(g^{-1}\right)$, and gg^T belongs to $\mathrm{Aut}_0\mathfrak{g}$. As gg^T is positive-definite, Lemma 12.1 ensures that $s = \sqrt{gg^T} \in \exp \mathrm{ad}(\mathfrak{s})$. Take $k = s^{-1}g$, which belongs to $\mathrm{Aut}_0\mathfrak{g}$. Then,

$$kk^T = s^{-1}gg^T s^{-1} = s^{-1}s^2 s^{-1} = 1,$$

so $k \in K_{\mathrm{ad}} = \mathrm{SO}(\mathfrak{g}, B_\theta) \cap \mathrm{Aut}_0\mathfrak{g}$. As $g = ks$, this shows that ϕ is surjective.

The injectivity comes from the following observation. If $g = sk$, $s \in S$, $k \in K_{\mathrm{ad}}$, then $gg^T = skk^T s^T = s^2 \in \mathrm{Aut}_0\mathfrak{g}$, and so $s = \sqrt{gg^T}$. That implies the S-component is unique. Hence the K_{ad}-component is also unique.

To conclude the argument proving ϕ is a diffeomorphism, there remains to show that the differential of ϕ is an isomorphism at every point. Take $(X, k) \in \mathfrak{s} \times K$, $Y \in \mathfrak{s}$, and $A \in \mathfrak{k}$, viewed as a right invariant vector field on K. Then,

$$\begin{aligned} d\phi_{(X,k)}(Y, A(k)) &= e^X \cdot (T_X(Y) + A) \cdot k \\ &= \left(dE_{e^X}\right)_1 \circ (dD_k)_1 (T_X(Y) + A). \end{aligned}$$

This differential vanishes if and only if $T_X(Y) + A = 0$, i.e. $T_X(Y) = -A$. By Lemma 12.2, this happens only if $T_X(Y) = 0$, which in turn implies $Y = 0$, since T_X is isomorphism, because $X \in \mathfrak{s}$. Therefore $A = Y = 0$, showing that $d\phi_{(X,k)}$ is an isomorphism. As ϕ is bijective and a local diffeomorphism, ϕ is indeed a diffeomorphism.

At last, from the diffeomorphism $S \times K_{\text{ad}} \approx SK_{\text{ad}} = \text{Aut}_0\mathfrak{g}$, it follows that $\text{Aut}_0\mathfrak{g}/K_{\text{ad}}$ is diffeomorphic to S, which is simply connected. Therefore, K_{ad} is connected. □

The Cartan decomposition of an arbitrary group is easily obtained from the decomposition of the group $\text{Aut}_0\mathfrak{g}$ by applying the adjoint representation.

Theorem 12.4 *Let G be a semi-simple, connected Lie group and $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{s}$ be a Cartan decomposition of its Lie algebra. Write $K = \langle \exp \mathfrak{k} \rangle$ and $S = \exp \mathfrak{s}$. Then*

1. *$G = SK = KS$, and any $g \in G$ can be uniquely written as $g = sk$ or $g = ks$, for some $k \in K$ and $s \in S$.*
2. *S is an embedded submanifold of G diffeomorphic to $\mathfrak{s}$ under the embedding $\exp : \mathfrak{s} \to S$.*
3. *The maps $K \times S \to G$ given by $(k, s) \mapsto ks$ and $(k, s) \mapsto sk$ are diffeomorphisms.*
4. *The center $Z(G)$ of G is contained in K.*
5. *$K = \exp \mathfrak{k}$, and K is compact if and only if $Z(G)$ is finite.*

Proof The adjoint representation $\text{Ad} : G \to \text{Aut}_0\mathfrak{g}$ is a covering, with kernel $Z(G)$. The preimage $\widehat{K} = \text{Ad}^{-1}(K_{\text{ad}})$ is a closed subgroup whose Lie algebra is $\mathfrak{k}$ and that contains $Z(G)$.

The Cartan decomposition applies to $\widehat{K}$: the map $\phi : \mathfrak{s} \times \widehat{K} \to G$, $\phi(X, k) = \exp(\text{ad}(X))\, k$ is a diffeomorphism. To prove this statement, let us distinguish the exponential maps of G and $\text{Aut}_0\mathfrak{g}$ by writing exp and $\exp_{\text{ad}}$, respectively.

Given $g \in G$, $\text{Ad}(g) = sk$ with $s = \exp_{\text{ad}} Y \in \exp_{\text{ad}} \mathfrak{s}$ and $k \in K_{\text{ad}}$ by the Cartan decomposition of $\text{Aut}_0\mathfrak{g}$. Now, $\text{Ad} \circ \exp = \exp_{\text{ad}}$, so $\text{Ad}(g) = \text{Ad}(\exp Y)\, k$. In other words, $\text{Ad}\left((\exp Y)^{-1} g\right) = k \in K_{\text{ad}}$. Therefore $(\exp Y)^{-1} g \in K$ and then $g = (\exp Y)\, k_1$ with $k_1 \in K$. That is, the map ϕ is onto.

The injectivity is proved in a similar fashion. If $g = (\exp X_1)\, k_1 = (\exp X_2)\, k_2$, then $\left(\exp_{\text{ad}} X_1\right) \text{Ad}(k_1) = \left(\exp_{\text{ad}} X_2\right) \text{Ad}(k_2)$, which implies $\exp_{\text{ad}} X_1 = \exp_{\text{ad}} X_2$ and $\text{Ad}(k_1) = \text{Ad}(k_2)$. Therefore $X_1 = X_2$ (by the uniqueness of the decomposition in the adjoint group), so $\exp X_1 = \exp X_2$, and hence $k_1 = k_2$.

The proof that ϕ is local diffeomorphism is the same as in Theorem 12.3, because $\mathfrak{k}$ is the Lie algebra of $\widehat{K}$ and the property of T_X stated in Lemma 12.2 depends only on the Lie algebra.

Applying the inverse map of G gives us the decomposition $G = KS$ and the diffeomorphism $(X, k) \mapsto k \exp(\text{ad}(X))$.

As in Theorem 12.3, $G/\widehat{K}$ is diffeomorphic to S, which is simply connected. Consequently, $\widehat{K}$ is connected, and therefore $\widehat{K} = K = \langle \exp \mathfrak{k} \rangle = \exp \mathfrak{k}$.

Now, by construction, $K_{\text{ad}} = \widehat{K}/Z(G) = K/Z(G)$. Since K_{ad} is compact, it follows that K is compact if and only if $Z(G)$ is finite, proving the last claim and hence the theorem. □

The diffeomorphisms $G \approx K \times S \approx K \times \exp \mathfrak{s}$ immediately show that G is simply connected if and only if K is simply connected. Theorem 12.4 (5), together with Weyl's theorem on compact groups, tells that for any G, $Z(G)$ is finite if $\mathfrak{k}$

is semi-simple, i.e. $\mathfrak{z}(\mathfrak{k}) = \{0\}$. On the other hand, if $\mathfrak{z}(\mathfrak{k}) \neq \{0\}$ and K is simply connected, then K is not compact. Hence $Z(G)$ is infinite if $\mathfrak{z}(\mathfrak{k}) \neq \{0\}$ and G is simply connected.

Another consequence of the global Cartan decomposition is that the differentiable manifold G is diffeomorphic to the Cartesian product of a compact, connected Lie group and a Euclidean space. This is so, even though K is not compact, since the differentiable manifold is the product of a compact connected Lie group and a simply connected abelian group (see Chapter 11, Exercise 5).

12.2 Iwasawa Decompositions

The Iwasawa decomposition of a noncompact semi-simple Lie group G supplies us with yet another diffeomorphism between G and a product $K \times E$, where K is a Lie group with compact Lie algebra and E a Euclidean space. In contrast to Cartan decompositions, in an Iwasawa decomposition the Euclidean component is a simply connected, solvable subgroup.

12.2.1 Iwasawa Decomposition of a Lie Algebra

Real semi-simple Lie algebras admit root space decompositions in analogy to complex Lie algebras. But contrary to the complex case, the eigenspace decomposition of $\mathrm{ad}(H)$, with H in a Cartan subalgebra, does not work because there might be complex eigenvalues that are not real.

What takes the place of Cartan subalgebras are the maximal abelian subalgebras $\mathfrak{a} \subset \mathfrak{s}$ in the symmetric component of a fixed Cartan decomposition $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{s}$. For such a subalgebra $\mathfrak{a}$ and an element $H \in \mathfrak{a}$, the map $\mathrm{ad}(H))$ is diagonalizable with real eigenvalues since it is a symmetric linear map for the inner product B_θ. As $\mathfrak{a}$ is abelian, the maps $\mathrm{ad}(H)$, $H \in \mathfrak{a}$, are simultaneously diagonalizable, and if $\alpha \in \mathfrak{a}^*$, then the "root space"

$$\mathfrak{g}_\alpha = \{X \in \mathfrak{g} : \forall H \in \mathfrak{a},\ \mathrm{ad}(H)X = \alpha(H)X\},$$

if non-null, is a common eigenspace for $\mathrm{ad}(H)$, $H \in \mathfrak{a}$. In this situation, α is called a (restricted) root of $\mathfrak{a}$ whenever $\alpha \neq 0$, and one has a direct sum

$$\mathfrak{g} = \mathfrak{g}_0 \oplus \sum_\alpha \mathfrak{g}_\alpha, \tag{12.2}$$

with α running through the set of roots. As a matter of fact, the subspace $\mathfrak{g}_0$ is the centralizer of $\mathfrak{a}$, whence a subalgebra. The above splitting can be made more

precise by taking Cartan decompositions of the elements of $\mathfrak{g}_0$, as the following lemma proves.

Lemma 12.5 *Let $\mathfrak{g}_0$ be as in (12.2) and define $\mathfrak{m} = \mathfrak{g}_0 \cap \mathfrak{k}$. Then $\mathfrak{g}_0 = \mathfrak{m} \oplus \mathfrak{a}$, and the Lie algebra $\mathfrak{g}$ decomposes as*

$$\mathfrak{g} = \mathfrak{m} \oplus \mathfrak{a} \oplus \sum_{\alpha} \mathfrak{g}_\alpha. \tag{12.3}$$

Proof Take $X + Y \in \mathfrak{g}_0$ with $X \in \mathfrak{k}$ and $Y \in \mathfrak{s}$. If $H \in \mathfrak{a}$, then $0 = [H, X + Y] = [H, X] + [H, Y]$, with $[H, X] \in \mathfrak{s}$ and $[H, Y] \in \mathfrak{k}$. Therefore $[H, X] = [H, Y] = 0$. Equivalently, the elements $X \in \mathfrak{m} = \mathfrak{g}_0 \cap \mathfrak{k}$ and $Y \in \mathfrak{s}$ centralize $\mathfrak{a}$. Since $\mathfrak{a}$ is maximal abelian, it follows that $Y \in \mathfrak{a}$, which concludes the proof. □

Decompositions (12.2) and (12.3) subsume the choice of a maximal abelian subalgebra $\mathfrak{a}$. This choice, although arbitrary, does not lose generality, since two abelian subalgebras $\mathfrak{a}_1, \mathfrak{a}_2 \subset \mathfrak{s}$ are conjugate through an element of K_{ad}. Justifying this fact is similar to proving that Cartan subalgebras in compact algebras are conjugated (see Proposition 11.20). The proof relies on the concept of regular element in $\mathfrak{a}$.

An element $H \in \mathfrak{a}$ is called **regular real** if $\alpha(H) \neq 0$ for every root α. The set of regular real elements is open and dense in $\mathfrak{a}$, since there is a finite number of roots. If $H \in \mathfrak{a}$ is regular real, then the dimension of $\ker \text{ad}(H)$ is the smallest among the dimensions of the kernels of the elements of $\mathfrak{a}$, and the centralizer $\mathfrak{z}(H)$ coincides with $\mathfrak{g}_0 = \mathfrak{m} \oplus \mathfrak{a}$. Therefore if $X \in \mathfrak{s}$ satisfies $[H, X] = 0$ for some regular real element, then $X \in \mathfrak{a}$.

Now the bracket $[\mathfrak{k}, \mathfrak{s}] \subset \mathfrak{s}$ shows that the subgroup K_{ad} leaves the subspace $\mathfrak{s}$ invariant, in the sense that $k(\mathfrak{s}) = \mathfrak{s}$ if $k \in K_{\text{ad}}$. This implies that if $\mathfrak{a} \subset \mathfrak{s}$ is maximal abelian and $k \in K_{\text{ad}}$, then $k(\mathfrak{a}) \subset \mathfrak{s}$ is also a maximal abelian subalgebra. The following proposition shows that all maximal abelian subalgebras are obtained in this way, by conjugating elements of K_{ad}.

Proposition 12.6 *Let $\mathfrak{a} \subset \mathfrak{s}$ be a maximal abelian subalgebra. For every $X \in \mathfrak{s}$, then, there exists $k \in K_{\text{ad}}$ such that $gX \in \mathfrak{a}$.*

Proof The argument is similar to that of Proposition 11.20. Let $H \in \mathfrak{a}$ be a regular real element and define the function $f : K_{\text{ad}} \to \mathbb{R}$ by $f(k) = \mathcal{K}_{\mathfrak{g}}(kX, H)$, where $\mathcal{K}_{\mathfrak{g}}$ is the Cartan–Killing form of $\mathfrak{g}$ (an inner product on $\mathfrak{s}$). By compactness, this function assumes a minimum value $f(k_0)$. As in Proposition 11.20, then, $k_0 X$ commutes with H. But the latter is regular real, so $k_0 X \in \mathfrak{a}$. □

Corollary 12.7 *If $\mathfrak{a}, \mathfrak{a}_1 \subset \mathfrak{s}$ are maximal abelian, there exists $k \in K_{\text{ad}}$ such that $k\mathfrak{a}_1 = \mathfrak{a}$.*

Proof Take a regular real element $X \in \mathfrak{a}_1$ and let $k \in K_{\text{ad}}$ be such that $k(X) \in \mathfrak{a}$. If $Y \in \mathfrak{a}$, then $\left[k^{-1}(Y), X\right] = [Y, k(X)] = 0$, and so $k^{-1}(Y) \in \mathfrak{a}_1$. This shows that $k^{-1}(\mathfrak{a}) \subset \mathfrak{a}_1$. Swapping roles, there exists $u \in K$ such that $u(\mathfrak{a}_1) \subset \mathfrak{a}$. Hence $\dim \mathfrak{a}_i = \dim \mathfrak{a}$, which implies $k^{-1}(\mathfrak{a}) = \mathfrak{a}_1$. □

The common dimension of any maximal abelian algebra $\mathfrak{a} \subset \mathfrak{s}$ is called the **real rank** of $\mathfrak{g}$. In general, the real rank differs from the rank, which is the common dimension of all Cartan subalgebras. A maximal abelian algebra $\mathfrak{a}$ is contained in a Cartan subalgebra, but the inclusion may be proper.

The starting point to define an Iwasawa decomposition on $\mathfrak{g}$ is to choose a Cartan decomposition $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{s}$, a maximal abelian algebra $\mathfrak{a} \subset \mathfrak{s}$ and a regular real element $H \in \mathfrak{a}$. Using the splitting of $\mathfrak{g}$ into the root spaces of $\mathfrak{a}$, define

$$\mathfrak{n} = \mathfrak{n}_H^+ = \sum_{\alpha(H)>0} \mathfrak{g}_\alpha,$$

which is the sum of the eigenspaces of $\operatorname{ad}(H)$ with eigenvalues > 0. With these choices, the **Iwasawa decomposition** is, by definition,

$$\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{a} \oplus \mathfrak{n}. \tag{12.4}$$

Theorem 12.8 *The Iwasawa decomposition (12.4) is a direct sum.*

Proof First of all, $\mathfrak{k} \cap \mathfrak{a} = \{0\}$ because $\mathfrak{a} \subset \mathfrak{s}$ and $\mathfrak{a} \cap \mathfrak{n}_H^+ = \{0\}$, since $\mathfrak{a} \subset \ker \operatorname{ad}(H)$ and $\mathfrak{n}_H^+$ is the sum of the $\operatorname{ad}(H)$-eigenspaces with strictly positive eigenvalues. To check the intersection $\mathfrak{k} \cap \mathfrak{n}_H^+ = \{0\}$, let

$$\mathfrak{n}_H^- = \sum_{\alpha(H)<0} \mathfrak{g}_\alpha$$

be the sum of the eigenspaces of $\operatorname{ad}(H)$ with negative eigenvalues. If θ is the Cartan involution, then $\theta\left(\mathfrak{n}_H^+\right) = \mathfrak{n}_H^-$ and $\theta\left(\mathfrak{n}_H^-\right) = \mathfrak{n}_H^+$. In fact, if $\operatorname{ad}(H) X = [H, X] = \lambda X$ then $\theta[H, X] = \lambda \theta X$. Since $\theta H = -H$, it follows

$$[H, \theta X] = (-\lambda)\, \theta X.$$

Hence θ permutes the eigenspaces of $\operatorname{ad}(H)$, by flipping the sign of the eigenvalue. That said, take $X \in \mathfrak{k} \cap \mathfrak{n}_H^+$. Then $\theta X = X$, because $X \in \mathfrak{k}$, and hence $\theta X = X \in \mathfrak{n}_H^+ \cap \mathfrak{n}_H^- = \{0\}$, showing that $\mathfrak{k} \cap \mathfrak{n}_H^+ = \{0\}$.

Therefore the sum $\mathfrak{k} \oplus \mathfrak{a} \oplus \mathfrak{n}_H^+$ on the right side of (12.4) is indeed direct. There remains to see that the sum equals $\mathfrak{g}$.

Since $H \in \mathfrak{a}$ is regular real, the kernel of $\operatorname{ad}(H)$ is $\mathfrak{g}_0 = \mathfrak{m} \oplus \mathfrak{a}$, so $\mathfrak{g} = \mathfrak{n}_H^- \oplus \mathfrak{m} \oplus \mathfrak{a} \oplus \mathfrak{n}_H^+$. Now the claim is that $\mathfrak{n}_H^- \oplus \mathfrak{m} \subset \mathfrak{k} \oplus \mathfrak{a} \oplus \mathfrak{n}_H^+$. Take $X = Y + Z \in \mathfrak{n}_H^- \oplus \mathfrak{m}$. Then $\theta(X) = \theta(Y) + Z$ because $\theta(Z) = Z$ and

$$\begin{aligned} X &= X + \theta X - \theta X \\ &= Y + \theta Y + Z - \theta Y. \end{aligned}$$

The right hand side lives in $\mathfrak{k} \oplus \mathfrak{n}_H^+$, since $Y + \theta Y + Z \in \mathfrak{k}$ and $\theta Y \in \mathfrak{n}_H^+$. The claim is therefore proved. □

The component $\mathfrak{k}$ of an Iwasawa decomposition $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{a} \oplus \mathfrak{n}$ is a compact subalgebra, while $\mathfrak{a}$ is abelian. The following proposition deals with the component $\mathfrak{n}$ and the sum $\mathfrak{a} \oplus \mathfrak{n}$.

Proposition 12.9 *The component $\mathfrak{n} = \mathfrak{n}_H^+$ of the Iwasawa decomposition is a nilpotent subalgebra, and $\mathfrak{a} \oplus \mathfrak{n}$ is a solvable subalgebra.*

Proof Let X and Y be eigenvectors of $\operatorname{ad}(H)$ with respective eigenvalues λ and μ. Then,

$$\begin{aligned}[H, [X, Y]] &= [[H, X], Y] + [X, [H, Y]] \\ &= (\lambda + \mu)[X, Y].\end{aligned}$$

Therefore either $[X, Y] = 0$ or $[X, Y]$ is an eigenvector with eigenvalue $\lambda + \mu$. Let $\mathfrak{g} = V_1 \oplus \cdots \oplus V_N$ be the decomposition of $\mathfrak{g}$ into $\operatorname{ad}(H)$-eigenspaces with eigenvalues $\lambda_1 > \cdots > \lambda_N$ (an eigenspace V_i equals either $\mathfrak{g}_0$ or a sum of root spaces).

Take a basis $B = \{X_1, \ldots, X_N\}$ of $\mathfrak{g}$ consisting of bases of the subspaces V_i, ordered accordingly. Let X be an eigenvector of $\operatorname{ad}(H)$ with eigenvalue $\lambda > 0$. The matrix of $\operatorname{ad}(X)$ in the basis B is upper triangular with zero diagonal entries, since $[X, X_i]$ either vanishes or is an eigenvector with eigenvalue $\lambda + \lambda_i$. Taking linear combinations, the adjoints $\operatorname{ad}(Y)$, $Y \in \mathfrak{n}$, can be triangularized simultaneously in the basis B. Therefore $\operatorname{ad}(\mathfrak{n})$ is nilpotent, as contained in a nilpotent Lie algebra. But ad is injective, so $\mathfrak{n}$ is nilpotent.

The sum $\mathfrak{a} \oplus \mathfrak{n}$ is a subalgebra, since $\mathfrak{a}$ normalizes $\mathfrak{n}$ (i.e. $[\mathfrak{a}, \mathfrak{n}] \subset \mathfrak{n}$). It is also solvable, because $\mathfrak{n}$ is nilpotent and $(\mathfrak{a} \oplus \mathfrak{n})/\mathfrak{n} \approx \mathfrak{a}$ abelian. □

Eventually, it should be emphasized that the Iwasawa decomposition $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{a} \oplus \mathfrak{n}$ is not a semi-direct product, since none of the summands is an ideal in $\mathfrak{g}$.

Example Consider the algebra $\mathfrak{sl}(n, \mathbb{R})$. An Iwasawa decomposition is made of $\mathfrak{k} = \mathfrak{so}(n)$, the algebra $\mathfrak{a}$ of traceless diagonal matrices and the Lie algebra $\mathfrak{n}$ of upper triangular matrices with null diagonal. The splitting arises by picking a regular element $H = \operatorname{diag}\{a_1, \ldots, a_n\}$ with $a_1 > \cdots > a_n$. □

12.2.2 Global Iwasawa Decomposition

The global Iwasawa decomposition of a semi-simple, connected, and noncompact Lie group G is the product $G = KAN$ of three closed subgroups, where $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{a} \oplus \mathfrak{n}$ is an Iwasawa decomposition of the Lie algebra $\mathfrak{g}$ of G, $K = \exp \mathfrak{k}$, $A = \exp \mathfrak{a}$, and $N = \exp \mathfrak{n}$.

The subgroup K is the same appearing in the Cartan decomposition and is closed (and compact if $Z(G)$ is finite). The abelian subgroup $A = \exp \mathfrak{a}$ is closed, because $S = \exp \mathfrak{s}$ is closed in G, $\mathfrak{a}$ is closed in $\mathfrak{s}$, and $\exp : \mathfrak{s} \to S$ is a diffeomorphism. This also shows that $\exp : \mathfrak{a} \to A$ is a diffeomorphism. The other subgroups, associated with the algebras $\mathfrak{n}$ and $\mathfrak{a} \oplus \mathfrak{n}$, are treated in the following lemma regarding the adjoint group $\mathrm{Aut}_0\mathfrak{g}$.

Lemma 12.10 *Let* $N_{\mathrm{ad}} = \langle \exp \mathrm{ad}\,(\mathfrak{n}) \rangle$ *and* $A_{\mathrm{ad}} = \exp \mathrm{ad}\,(\mathfrak{a})$. *Then* N_{ad} *and* $A_{\mathrm{ad}}N_{\mathrm{ad}}$ *are closed and simply connected. The Lie algebra of* $A_{\mathrm{ad}}N_{\mathrm{ad}}$ *is* $\mathrm{ad}\,(\mathfrak{a} \oplus \mathfrak{n})$. *The intersections* $K_{\mathrm{ad}} \cap A_{\mathrm{ad}}N_{\mathrm{ad}} = \{1\}$ *and* $A_{\mathrm{ad}} \cap N_{\mathrm{ad}} = \{1\}$ *are trivial.*

Proof In the proof of Proposition 12.9, we constructed a basis B for $\mathfrak{g}$ such that, if $X \in \mathfrak{n}$, the matrix of $\mathrm{ad}\,(X)$ is upper triangular with zeroes on the diagonal. Therefore N is a connected subgroup of the nilpotent, closed and simply connected subgroup $\widehat{N}$ made by the linear maps of $\mathfrak{g}$, whose matrices in the basis B are upper triangular with 1s on the diagonal. By Corollaries 10.9 and 10.10, N_{ad} is closed and simply connected (see also the example after Corollary 10.10).

The product $A_{\mathrm{ad}}N_{\mathrm{ad}}$ is a subgroup, because A_{ad} normalizes N_{ad}, as follows from $[\mathfrak{a}, \mathfrak{n}] \subset \mathfrak{n}$. With respect to the basis B mentioned above, the elements of A_{ad} are diagonal, while those of N_{ad} are upper triangular. Therefore if $a_k n_k \in A_{\mathrm{ad}}N_{\mathrm{ad}}$ is a convergent sequence, the diagonal part a_k converges and $\lim a_k \in A_{\mathrm{ad}}$, which is closed. Hence n_k converges and $\lim n_k \in N_{\mathrm{ad}}$, so $\lim a_k n_k \in A_{\mathrm{ad}}N_{\mathrm{ad}}$, which shows that $A_{\mathrm{ad}}N_{\mathrm{ad}}$ is closed.

There remains to check that the group $A_{\mathrm{ad}}N_{\mathrm{ad}}$ is simply connected. To do this, define the semi-direct product $A_{\mathrm{ad}} \times_\tau N_{\mathrm{ad}}$, where $\tau : A_{\mathrm{ad}} \to \mathrm{Aut}N_{\mathrm{ad}}$ is defined by $\tau(a)(n) = ana^{-1}$. The map $\phi : A_{\mathrm{ad}} \times_\tau N_{\mathrm{ad}} \to A_{\mathrm{ad}}N_{\mathrm{ad}}$, $\phi(a, n) = an$, is a differentiable homomorphism. It is surjective by definition. To show it is 1-1, suppose that $an = a_1 n_1$. Then $a_1^{-1}a = n_1 n^{-1} \in A_{\mathrm{ad}} \cap N_{\mathrm{ad}}$, and this intersection reduces to $\{1\}$ since the elements in A_{ad} are diagonal with respect to B, while the elements of N_{ad} are upper triangular with diagonal entries 1. Hence ϕ is an isomorphism. As $A_{\mathrm{ad}} \times_\tau N_{\mathrm{ad}}$ is simply connected, one concludes that $A_{\mathrm{ad}}N_{\mathrm{ad}}$ is simply connected. It follows from this isomorphism that the Lie algebra of $A_{\mathrm{ad}}N_{\mathrm{ad}}$ is $\mathrm{ad}\,(\mathfrak{a} \oplus \mathfrak{n})$.

Finally, an element $g \in K_{\mathrm{ad}} \cap A_{\mathrm{ad}}N_{\mathrm{ad}}$ is an isometry for the inner product B_θ. At the same time, its eigenvalues are real > 0. This is only possible if $g = 1$, and hence $K_{\mathrm{ad}} \cap A_{\mathrm{ad}}N_{\mathrm{ad}} = \{1\}$. □

The fact that $A_{\mathrm{ad}}N_{\mathrm{ad}}$ is a closed subgroup guarantees the quotient $\mathrm{Aut}_0\mathfrak{g}/A_{\mathrm{ad}}N_{\mathrm{ad}}$ is a differentiable manifold. This has an interesting consequence from the point of view of the Iwasawa decomposition. It allows to prove the surjectivity of the decomposition, using the compactness of K_{ad} together with the fact that $\mathfrak{a} \oplus \mathfrak{n}$ is complementary to $\mathfrak{k}$ in $\mathfrak{g}$.

Lemma 12.11 *The action of the group* K_{ad} *on* $\mathrm{Aut}_0\mathfrak{g}/A_{\mathrm{ad}}N_{\mathrm{ad}}$ *is transitive: for every* $g \in \mathrm{Aut}_0\mathfrak{g}$, *there exists* $k \in K_{\mathrm{ad}}$ *such that* $gA_{\mathrm{ad}}N_{\mathrm{ad}} = kA_{\mathrm{ad}}N_{\mathrm{ad}}$. *This means that* $\mathrm{Aut}_0\mathfrak{g} = K_{\mathrm{ad}}A_{\mathrm{ad}}N_{\mathrm{ad}}$.

Proof Denote by $\pi : \mathrm{Aut}_0\mathfrak{g} \to \mathrm{Aut}_0\mathfrak{g}/A_{\mathrm{ad}}N_{\mathrm{ad}}$ the canonical projection and let $x_0 = 1 \cdot A_{\mathrm{ad}}N_{\mathrm{ad}}$ be the origin of $\mathrm{Aut}_0\mathfrak{g}/A_{\mathrm{ad}}N_{\mathrm{ad}}$. The orbit

$$K_{\mathrm{ad}} \cdot x_0 = \{kA_{\mathrm{ad}}N_{\mathrm{ad}} : k \in K_{\mathrm{ad}}\} = \pi\left(K_{\mathrm{ad}}\right)$$

is compact, because K_{ad} is compact.

The orbit $K_{\mathrm{ad}} \cdot x_0$ is a submanifold of $\mathrm{Aut}_0\mathfrak{g}/A_{\mathrm{ad}}N_{\mathrm{ad}}$, whose tangent space at x_0 equals $d\pi_1\left(\mathfrak{k}\right)$ (see Theorem 13.8). As $\mathfrak{k}$ is a complement to $\mathfrak{a} \oplus \mathfrak{n}$ in $\mathfrak{g}$, the dimension of $K_{\mathrm{ad}} \cdot x_0$ coincides with the dimension of $\mathrm{Aut}_0\mathfrak{g}/A_{\mathrm{ad}}N_{\mathrm{ad}}$. Therefore $K_{\mathrm{ad}} \cdot x_0$ is open and closed in the connected manifold $\mathrm{Aut}_0\mathfrak{g}/A_{\mathrm{ad}}N_{\mathrm{ad}}$, whence $K_{\mathrm{ad}} \cdot x_0 = \mathrm{Aut}_0\mathfrak{g}/A_{\mathrm{ad}}N_{\mathrm{ad}}$.

The fact that every coset of $A_{\mathrm{ad}}N_{\mathrm{ad}}$ contains an element of K_{ad} means that $\mathrm{Aut}_0\mathfrak{g} = K_{\mathrm{ad}}A_{\mathrm{ad}}N_{\mathrm{ad}}$. □

We are now in the position to state and prove the global Iwasawa decomposition.

Theorem 12.12 *Let G be a connected semi-simple Lie group and $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{a} \oplus \mathfrak{n}$ an Iwasawa decomposition of the Lie algebra $\mathfrak{g}$ of G. Then $G = KAN$, where $K = \exp\mathfrak{k}$, $A = \exp\mathfrak{a}$ and $N = \exp\mathfrak{n}$. The map $\phi : K \times A \times N \to KAN$, $\phi\left(k, a, n\right) = kan$, is a diffeomorphism. The groups A, N, and AN are simply connected and diffeomorphic to Euclidean spaces.*

Proof The images, under the adjoint representation, of the Iwasawa components of G are the components of $\mathrm{Aut}_0\mathfrak{g}$:

$$\mathrm{Ad}\left(K\right) = K_{\mathrm{ad}} \quad \mathrm{Ad}\left(A\right) = A_{\mathrm{ad}} \quad \mathrm{Ad}\left(N\right) = N_{\mathrm{ad}}.$$

The restriction of Ad to each of these groups is a covering space, modulo the center $Z\left(G\right)$. The groups A_{ad} and N_{ad} are simply connected, and hence $\mathrm{Ad} : A \to A_{\mathrm{ad}}$ and $\mathrm{Ad} : N \to N_{\mathrm{ad}}$ are isomorphisms, which shows that A and N are simply connected and diffeomorphic to Euclidean spaces.

The map ϕ is surjective, because if $g \in G$, then $\mathrm{Ad}\left(g\right) = \mathrm{Ad}\left(k\right)\mathrm{Ad}\left(a\right)\mathrm{Ad}\left(n\right)$, $k \in K$, $a \in A$, and $n \in N$, as follows from the previous lemma, which ensures surjectivity of the Iwasawa decomposition on $\mathrm{Aut}_0\mathfrak{g}$. Therefore, $g = kanz$ with $z \in Z\left(G\right)$, i.e. $g = \left(kz\right)an \in KAN$, since $Z\left(G\right) \subset K$.

As for injectivity, suppose $kan = k_1a_1n_1$. Then,

$$\mathrm{Ad}\left(k_1^{-1}k\right) = \mathrm{Ad}\left(a_1n_1\left(an\right)^{-1}\right) \in K_{\mathrm{ad}} \cap A_{\mathrm{ad}}N_{\mathrm{ad}} = \{1\},$$

and $\mathrm{Ad}\left(an\right) = \mathrm{Ad}\left(a_1n_1\right)$. This implies $\mathrm{Ad}\left(a_1^{-1}a\right) = \mathrm{Ad}\left(n_1^{-1}n\right) \in A_{\mathrm{ad}} \cap N_{\mathrm{ad}} = \{1\}$, that is, $\mathrm{Ad}\left(a\right) = \mathrm{Ad}\left(a_1\right)$ and $\mathrm{Ad}\left(n\right) = \mathrm{Ad}\left(n_1\right)$. But Ad is 1-1 on A and N, so $a = a_1$ and $n = n_1$. This forces $k = k_1$, showing the injectivity of the decomposition.

The last thing to prove is that ϕ is local diffeomorphism. For this, take invariant vector fields $X \in \mathfrak{k}$, $Y \in \mathfrak{a}$, and $Z \in \mathfrak{n}$ with X left invariant, Y bi-invariant (on A), and Z right invariant. Then,

$$d\phi_{(k,a,n)}\,(X, Y, Z) = k\,(X + Y + \mathrm{Ad}\,(a)\,Z)\,an.$$

This differential vanishes if and only if $X + Y + \mathrm{Ad}\,(a)\,Z = 0$, and since $\mathrm{Ad}\,(a)\,Z \in \mathfrak{n}$, necessarily $X = Y = Z = 0$. So the differential at any point is injective. As the dimensions $\dim\,(K \times A \times N) = \dim G$ are equal, ϕ is indeed a local diffeomorphism. □

Example For the linear group $\mathrm{Sl}\,(n, \mathbb{R})$, the Iwasawa decomposition can be interpreted in terms of Gram–Schmidt orthonormalization. Consider the Iwasawa decomposition $\mathfrak{sl}\,(n, \mathbb{R}) = \mathfrak{so}\,(n) \oplus \mathfrak{a} \oplus \mathfrak{n}$, where $\mathfrak{a}$ is the algebra of diagonal matrices and $\mathfrak{n}$ that of upper triangular matrices. The Iwasawa decomposition $\mathrm{Sl}\,(n, \mathbb{R}) = \mathrm{SO}\,(n)\,AN$ is given by the group A of diagonal matrices of determinant 1 with positive entries, while N is the group of upper triangular matrices with 1s on the diagonal. Take a matrix $g \in \mathrm{Sl}\,(n, \mathbb{R})$. Its decomposition $g = kan \in \mathrm{SO}\,(n)\,AN$ is obtained by making the columns of g orthonormal. The process consists in right multiplying g by an upper triangular matrix $(an)^{-1}$, which produces an orthogonal matrix k. □

12.3 Classification

Let $\mathfrak{g}$ be a real, simple Lie algebra. Its complexification $\mathfrak{g}_{\mathbb{C}}$ is also simple. If $\mathfrak{g}_{\mathbb{C}}$ is Simple, then $\mathfrak{g}$ is simple because $\mathfrak{i}_{\mathbb{C}} \subset \mathfrak{g}_{\mathbb{C}}$ is an ideal in $\mathfrak{g}_{\mathbb{C}}$ if $\mathfrak{i} \subset \mathfrak{g}$ is an ideal. Conversely, if $\mathfrak{g}$ is simple, then $\mathfrak{g}_{\mathbb{C}}$ may not be simple. For this reason, real Lie algebras are distinguished into two classes, depending on whether $\mathfrak{g}_{\mathbb{C}}$ is simple or not. When $\mathfrak{g}_{\mathbb{C}}$ is not simple, $\mathfrak{g}$ is a real form of a complexified Lie algebra. In this case, $\mathfrak{g}_{\mathbb{C}}$ is the sum of two copies of $\mathfrak{g}$ viewed as a complex Lie algebra.[3] The classification of these real forms is given by the classification of complex Lie algebras via Dynkin diagrams, discussed in Chapter 11.

The classification of simple real algebras whose complexifications are simple is quite engaging. It can be achieved using Satake[4] or Vogan[5] diagrams. The following tables lay out this classification. The first table contains the so-called classical algebras, which are Lie algebras of matrices. The column $\mathfrak{k}$ lists the maximal compact subalgebras (which appear in the Cartan decomposition). The second table lists the exceptional algebras. They are named according to their complexification (for example, E_6^{-14} is a real form of the complex Lie algebra E_6). The superscript accounts for the difference $\dim \mathfrak{s} - \dim \mathfrak{k}$.

[3] See *Álgebras de Lie* [47, Ch. 12] and Knapp [33, Section VI.9].

[4] See *Álgebras de Lie* [47, Ch. 14] and Warner [57, Section 1.1].

[5] See Knapp [33, Chapter VI].

Classical noncompact simple Lie algebras

$\mathfrak{g}_{\mathbb{C}}$	$\mathfrak{g}$	$\mathfrak{k}$	Rank	Real rank
$\mathfrak{sl}(n,\mathbb{C})$	$\mathfrak{sl}(n,\mathbb{R})$	$\mathfrak{so}(n,\mathbb{R})$	$n-1$	$n-1$
$\mathfrak{sl}(2n,\mathbb{C})$	$\mathfrak{sl}(n,\mathbb{H})$ $\mathfrak{su}^*(2n)$	$\mathfrak{sp}(n)$	$2n-1$	$n-1$
$\mathfrak{sl}(n,\mathbb{C})$	$\mathfrak{su}(p,q)$ $p+q=n$	$\mathfrak{su}(p)\oplus\mathfrak{su}(q)\oplus\mathbb{R}$	$n-1$	$\min\{p,q\}$
$\mathfrak{so}(2n+1,\mathbb{C})$	$\mathfrak{so}(p,q)$ $p+q=2n+1$	$\mathfrak{so}(p)\oplus\mathfrak{so}(q)$	n	$\min\{p,q\}$
$\mathfrak{so}(2n,\mathbb{C})$	$\mathfrak{so}(p,q)$ $p+q=2n$	$\mathfrak{so}(p)\oplus\mathfrak{so}(q)$	n	$\min\{p,q\}$
$\mathfrak{so}(2n,\mathbb{C})$	$\mathfrak{so}^*(2n)$	$\mathfrak{u}(n)$	n	$n/2$ or $(n-1)/2$
$\mathfrak{sp}(n,\mathbb{C})$	$\mathfrak{sp}(n,\mathbb{R})$	$\mathfrak{u}(n)$	n	n
$\mathfrak{sp}(n,\mathbb{C})$	$\mathfrak{sp}(p,q)$ $p+q=n$	$\mathfrak{sp}(p)\oplus\mathfrak{sp}(q)\oplus\mathbb{R}$	n	$\min\{p,q\}$

Exceptional noncompact simple Lie algebras

$\mathfrak{g}$	$\mathfrak{k}$	Real rank	Dimension
G_2^2	$\mathfrak{su}(2)\oplus\mathfrak{su}(2)$	2	14
F_4^4	$\mathfrak{su}(2)\oplus\mathfrak{sp}(3)$	4	52
F_4^{-20}	$\mathfrak{so}(9)$	1	52
E_6^6	$\mathfrak{sp}(4)$	6	78
E_6^2	$\mathfrak{su}(2)\oplus\mathfrak{su}(6)$	4	78
E_6^{-14}	$\mathfrak{so}(10)\oplus\mathbb{R}$	2	78
E_6^{-26}	F_4	2	78
E_7^7	$\mathfrak{su}(8)$	7	133
E_7^{-5}	$\mathfrak{su}(2)\oplus\mathfrak{so}(12)$	4	133
E_7^{-25}	$E_6\oplus\mathbb{R}$	3	133
E_8^8	$\mathfrak{so}(16)$	8	248
E_8^{-24}	$\mathfrak{su}(2)\oplus E_7$	4	248

12.4 Exercises

1. Let G be a noncompact, semi-simple, connected Lie group with Cartan decomposition $G = KS$ and Iwasawa decomposition $G = KAN$. Prove the following statements:

   a. the exponential map of AN is surjective;
   b. the manifold G/AN is diffeomorphic to K;
   c. the action of AN on G/K is transitive;
   d. the action of K on G/AN is transitive.

2. Let $\mathfrak{g}$ be a noncompact semi-simple Lie algebra. Show that if $X \in \mathfrak{g}$ is such that $\mathrm{ad}(X)$ is nilpotent, then the adjoint orbit of X (that is, $\{g(X) : g \in \mathrm{Aut}_0\mathfrak{g}\}$)

is not closed. (Hint: apply the Jacobson–Borel–Morozov theorem proved in Knapp [33, Section X.2].)

3. Let $\mathfrak{g}$ be a noncompact semi-simple Lie algebra and $X \in \mathfrak{g}$ a vector such that $\operatorname{ad}(X)$ is a semi-simple linear map. Show that there is a Cartan subalgebra $\mathfrak{h} \subset \mathfrak{g}$ such that $X \in \mathfrak{h}$. (Hint: define $\phi : \mathfrak{g} = \ker \operatorname{ad}(X) \oplus \operatorname{Im} \operatorname{ad}(X) \to \mathfrak{g}$ by $\phi(Y + Z) = e^{\operatorname{ad}(Z)}(Y + X)$. Show that the image of ϕ contains a regular element, and conclude that $\ker \operatorname{ad}(X)$ contains a regular element.)
4. Show that if $\mathfrak{g}$ is a complex Lie algebra, then the fundamental group of the adjoint group $\operatorname{Aut}_0\mathfrak{g}$ is finite, and hence any connected group G with Lie algebra $\mathfrak{g}$ has finite center.
5. Let $\mathfrak{g}_{\mathbb{C}}$ be a complex semi-simple Lie algebra and $\mathfrak{g}$ a real form of $\mathfrak{g}_{\mathbb{C}}$. Denote by $G_{\mathbb{C}}$ the connected and simply connected Lie group with Lie algebra $\mathfrak{g}_{\mathbb{C}}$, and set $G = \langle \exp \mathfrak{g} \rangle \subset G_{\mathbb{C}}$. Show that G has finite center. (Take a compact real form $\mathfrak{u}$ of $\mathfrak{g}_{\mathbb{C}}$ such that $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{s}$, with $\mathfrak{k} = \mathfrak{g} \cap \mathfrak{u}$ and $\mathfrak{s} = \mathfrak{g} \cap i\mathfrak{u}$, is a Cartan decomposition of $\mathfrak{g}$. Show that the component $K = \exp \mathfrak{k}$ of the Cartan decomposition of G is the subgroup of fixed points of the automorphism θ of $G_{\mathbb{C}}$ that extends the conjugation of $\mathfrak{g}_{\mathbb{C}}$ with respect to $\mathfrak{g}$. Conclude K is compact.)
6. Let us call a Lie group G with Lie algebra $\mathfrak{g}$ **complexifiable** if there is a Lie group $G_{\mathbb{C}}$ whose Lie algebra is the complexification $\mathfrak{g}_{\mathbb{C}}$ of $\mathfrak{g}$ and such that G is isomorphic to $\langle \exp \mathfrak{g} \rangle \subset G_{\mathbb{C}}$. Show that a complexifiable group has finite center. (Hint: use the previous exercise.)
7. Consider a noncompact, semi-simple, and connected linear group $G \subset \operatorname{Gl}(n, \mathbb{R})$. Show that its center $Z(G)$ is finite.
8. Let G be a noncompact and connected semi-simple group. Show that if G has finite center, then any connected semi-simple subgroup $G_1 \subset G$ has finite center. (Hint: use the previous exercise to verify that $\operatorname{Ad}(G_1) \subset \operatorname{Ad}(G)$ has finite center, and show that $G_1 \to \operatorname{Ad}(G_1)$ is a finite covering.)
9. Show that the universal covering $\widetilde{\operatorname{Sl}(2, \mathbb{R})}$ of $\operatorname{Sl}(2, \mathbb{R})$ does not admit a faithful representation of finite dimension.
10. Find the real simple Lie algebras $\mathfrak{g}$ whose simply connected group $\widetilde{G}$ has infinite center.
11. Let G be a connected and simply connected Lie group. Show that the differentiable manifold underlying G is diffeomorphic to the product of a connected and compact group and a Euclidean space.
12. Let G be a connected Lie group. Prove there is an integer n such that the map $(X_1, \ldots, X_n) \mapsto e^{X_1} \cdots e^{X_n}$ is onto. Show that if G is semi-simple, then $n = 2$ works. Find a noncompact semi-simple Lie group whose exponential map is not onto.

Part IV
Transformation Groups

Overview

This part considers the differentiable actions of Lie groups in differentiable manifolds and some of their applications in the study of differential geometry in homogeneous spaces.

The basic facts about differentiable actions $G \times M \to M$ of a Lie group G are developed in Chapter 13. The first step is to put in evidence the Lie algebra $\mathfrak{g}$ of G. This is done by defining, for each $X \in \mathfrak{g}$, a vector field $\widetilde{X}$ whose flow is given by the action of e^{tX}. The map $X \mapsto \widetilde{X}$ is a homomorphism of Lie algebras, that is, $[\widetilde{X}, \widetilde{Y}] = \widetilde{[X, Y]}$, where in the left hand side it is the Lie bracket of vector fields in M and in the right hand side is the bracket of right invariant vector fields in G. (For actions on the left, right invariant vector fields are replaced by left invariant.) The homomorphism $X \mapsto \widetilde{X}$ is called the infinitesimal action of the Lie algebra $\mathfrak{g}$ associated with the action of the Lie group G.

The infinitesimal action gives rise to a distribution Δ in M defined by $\Delta(x) = \{\widetilde{X}(x) : X \in \mathfrak{g}\}$. This distribution is integrable and its connected maximal integral manifolds are the orbits of the action of G if the group is connected. In case the group is not connected, its orbits are unions of integral manifolds. This fact has the consequence that the orbits of the action of G are quasi-regular manifolds and are conveniently packed by the adapted charts of the distribution.

A natural question is whether an infinitesimal action of a Lie algebra $\mathfrak{g}$ comes from a global action of some Lie group. The affirmative answer to this question is given by the Lie–Palais theorem, which ensures the existence of an action of the simply connected Lie group with Lie algebra $\mathfrak{g}$ in case the vector fields of the infinitesimal action are complete (what happens when the manifold M is compact).

Still in Chapter 13, there is a section where the concepts of principal bundles (whose fibers are Lie groups) and their associated bundles (whose fibers are spaces where Lie groups act) are introduced. These concepts are later on applied to look at natural fibrations between quotient spaces, such as $G \to G/H$, which has the structure of a principal fiber bundle with fiber H.

Chapter 14, which complements this part, is intended to make a (brief) introduction to a vast area of differential geometry that studies invariant geometric structures in homogeneous spaces. Four types of invariant geometries were chosen: complex and almost complex structures, differential forms, Riemannian metrics, and symplectic forms. The basic principle is that the analysis of any invariant geometric structure is reduced to an algebraic study of what occurs at a single point.

Regarding the almost complex structures, a discussion about their integrability is presented, based on the Nijenhuis tensor. Some examples are given. One of them is that of complex groups reaching the conclusion that a Lie group is complex if and only if its Lie algebra is complex. About the differential forms, a theorem due to Chevalley and Eilenberg is proved. It states that the de Rham cohomology of a homogeneous space of a compact Lie group coincides with the cohomology of the invariant differential forms. This reduces the computation of cohomology to algebraic issues involving the Lie algebra of the Lie group.

As to Riemannian manifolds, the text is more parsimonious, since the literature on the subject is wide and easily accessible (see Helgason's classical book [20]). In symplectic geometry, one of the central questions is the construction of the symplectic Kirillov–Kostant–Souriaux form in the coadjoint orbits of a Lie algebra. This symplectic form enters into the discussion of the moment maps of Hamiltonian actions for which the emphasis is put on its equivariance with respect to the coadjoint action. This equivariance is analyzed on the basis of the cohomology of group representations.

Chapter 13
Lie Group Actions



This chapter discusses the differentiable actions of Lie groups and their orbits. The model for orbits are quotient spaces G/H. When H is closed, the quotient G/H admits a structure of differentiable manifold, which was built in Chapter 6. One of the present objectives is then to verify that an orbit $G \cdot x$ is an immersed submanifold diffeomorphic to the quotient space G/G_x, where G_x is the isotropy subgroup at x, which is a closed subgroup. In this direction, a convenient point of view is to look at orbits as maximal integral manifolds of a singular distribution (see Appendix B). This approach provides the additional information that they are quasi-regular immersed submanifolds.

13.1 Group Actions

The various concepts and results that were developed in the study of topological group actions continue to apply to Lie groups. In particular, an action of the Lie group G is a map $\phi : G \times M \to M, \phi(g, x) = gx$ such that the partial map $g \mapsto \phi_g$, $\phi_g(x) = \phi(g, x)$ is a homomorphism from G to the group of the invertible maps of M. The **action is differentiable** if ϕ is a differentiable map.

In this case, the partial maps $\phi_g : M \to M$ and $\phi_x : G \to M$, $\phi_g(x) = \phi_x(g) = \phi(g, x)$ are differentiable for all $g \in G$ and $x \in M$. The relation $(\phi_g)^{-1} = \phi_{g^{-1}}$ implies the maps ϕ_g are diffeomorphisms. That is to say, the homomorphism $g \mapsto \phi_g$ takes values in the group $\mathrm{Diff}M$ of diffeomorphisms of M. As before, a diffeomorphism ϕ_g is simply denoted by g.

When an action is differentiable, the isotropy subgroup $G_x = \{g \in G : gx = x\}$ is closed, and therefore a Lie subgroup of G. There is a natural one-to-one correspondence between the orbit $G \cdot x$ and the homogeneous space G/G_x. In the sequel, it will be proved that $G \cdot x$ is a submanifold of M and the bijection with

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G/G_x is actually a diffeomorphism when G/G_x is endowed with the differentiable quotient structure.

The Lie algebra $\mathfrak{g}_x$ of the isotropy group G_x is called **isotropy algebra** at x. It consists of vectors $X \in \mathfrak{g}$ such that $e^{tX} \in G_x$ for all $t \in \mathbb{R}$:

$$\mathfrak{g}_x = \{X \in \mathfrak{g} : \forall t \in \mathbb{R},\ e^{tX}x = x\}.$$

For any $g \in G$ and $x \in M$, one has $G_{gx} = gG_xg^{-1}$, which implies that $\mathfrak{g}_{gx} = \mathrm{Ad}\,(g)\,(\mathfrak{g}_x)$.

As is usual in Lie group theory, a basic technique when studying group actions is to introduce the corresponding infinitesimal object.

Definition 13.1 Let $\mathfrak{g}$ be a Lie algebra and M a C^∞ manifold. Denote by $\Gamma(TM)$ the Lie algebra of vector fields on M endowed with the usual Lie bracket. An **infinitesimal action** of $\mathfrak{g}$ on M is a homomorphism $\mathfrak{g} \to \Gamma(TM)$.

A differentiable action of G on M induces an infinitesimal action of $\mathfrak{g}$ in the following way. Given $X \in \mathfrak{g}$ and $x \in M$, the curve in M defined by $t \mapsto e^{tX}x$ is differentiable. Its derivative at the origin

$$\widetilde{X}(x) = \frac{d}{dt}\left(e^{tX}x\right)_{|t=0} = \frac{d}{dt}\phi_x\left(e^{tX}\right)_{|t=0} = (d\phi_x)_1(X)$$

is a tangent vector to $x \in M$. Therefore $x \in M \mapsto \widetilde{X}(x) \in T_xM$ defines a vector field on M.

Its flow $\widetilde{X}_t$ is e^{tX} (or rather, $\phi_{e^{tX}}$). In fact, for any $x \in M$ the curve $t \mapsto e^{tX}x$ is a trajectory of $\widetilde{X}$, because

$$\frac{d}{dt}\left(e^{tX}x\right) = \frac{d}{ds}\left(e^{(t+s)X}x\right)_{|s=0} = \widetilde{X}\left(\exp(tX)\right).$$

Consequently, the vector fields $\widetilde{X}$ are complete, since their flows e^{tX} are defined for all $t \in \mathbb{R}$, hence globally. Moreover, the right invariant field $X \in \mathfrak{g}$ and the associated field $\widetilde{X}$ are ϕ_x-related for every $x \in M$, because $\phi_x \circ E_{e^{tX}} = \phi_{e^{tX}} \circ \phi_x$, i.e. ϕ_x intertwines the flows of X and $\widetilde{X}$.

From this last comment, it follows that the map $X \in \mathfrak{g} \mapsto \widetilde{X} \in \Gamma(TM)$ is an infinitesimal action (viewing X as a right invariant field). In fact, since the fields $X, Y \in \mathfrak{g}$ are ϕ_x-related, their brackets are also ϕ_x -related, so

$$[\widetilde{X}, \widetilde{Y}](x) = (d\phi_x)_1[X, Y] = \widetilde{[X, Y]}(x).$$

Proposition 13.2 *The map $X \in \mathfrak{g} \mapsto \widetilde{X} \in \Gamma(TM)$ is a homomorphism if $\mathfrak{g}$ is the Lie algebra of right invariant vector fields on G.*

Thus far actions of G on M were implicitly left actions, so it should be pointed out that in the above proposition one takes the bracket of right invariant vector fields.

If G acts on M on the right, instead, the same map $X \mapsto \widetilde{X}$ defines a homomorphism of the Lie algebra of left invariant fields.

The rule for translating the fields $\widetilde{X}$ by elements of G is given by the following formula, which is used quite often.

Proposition 13.3 *If $g \in G$ and $X \in \mathfrak{g}$, then $g_* \widetilde{X} = \widetilde{(\mathrm{Ad}\,(g)\, X)}$, i.e.*

$$(dg)_{g^{-1}x} \left(\widetilde{X}\,(gx) \right) = \widetilde{(\mathrm{Ad}\,(g)\, X)}\,(x)\,.$$

Proof It is enough to check that the translations under g of the trajectories $\widetilde{X}$ are trajectories of $\widetilde{\mathrm{Ad}\,(g)\, X}$. Since the flow is $\widetilde{X}_t = e^{tX}$, the translation of the trajectory of $\widetilde{X}$ through $g^{-1}x$ is

$$g e^{tX} g^{-1} x = e^{t \mathrm{Ad}(g) X} x,$$

and the right side is the trajectory of $\widetilde{\mathrm{Ad}\,(g)\, X}$ starting at x. □

In the simplified notation described in Section 5.1, the formula of the above proposition can be written as

$$g \widetilde{X}\,(x) = \widetilde{g X g^{-1}}\,(gx)\,.$$

It is obtained by multiplying by g^{-1} and g.

As a particular case of infinitesimal action, consider the action of G on G/H, where H is a closed subgroup. In this scenario, if $x_0 = 1 \cdot H$ is the origin of G/H, then ϕ_{x_0} is the canonical projection $\pi : G \to G/H$. Hence the following description of $\widetilde{X}$ comes about.

Proposition 13.4 *Let G be a Lie group and H a closed subgroup. Denote by $\pi : G \to G/H$ the canonical projection and suppose X is a right invariant field on G. Then $\widetilde{X}$ is the projection of X, that is, X and $\widetilde{X}$ are π-related.*

In terms of the fields $\widetilde{X}$, $X \in \mathfrak{g}$, the isotropy algebra is given by

$$\mathfrak{g}_x = \{X \in \mathfrak{g} : \widetilde{X}\,(x) = 0\},$$

because $X \in \mathfrak{g}_x$ if and only if $e^{tX} x = x$ for all $t \in \mathbb{R}$, that is, if x is a singularity of the vector field $\widetilde{X}$.

Examples

1. Consider the canonical action of $\mathrm{Gl}\,(n, \mathbb{R})$ on $\mathbb{R}^n$, $(g, x) \mapsto gx$. Given a matrix $A \in \mathfrak{gl}\,(n, \mathbb{R})$,

$$\widetilde{A}\,(x) = \frac{d}{dt} \left(e^{tA} x \right)_{|t=0} = Ax.$$

So, the vector field $\widetilde{A}$ is the linear vector field on $\mathbb{R}^n$ defined by A. The homomorphism property here means that the Lie bracket of $\widetilde{A}$ and $\widetilde{B}$ is the linear field defined by the matrix $BA - AB$.

2. The previous example generalizes to group representations. If $\rho : G \to \mathrm{Gl}(n, \mathbb{R})$ is a (differentiable) representation of G, the map $G \times \mathbb{R}^n \to \mathbb{R}^n$, defined by $(g, x) \mapsto \rho(g)\, x$, is an action of G on $\mathbb{R}^n$. The corresponding infinitesimal action reads

$$\widetilde{X}(x) = d\rho_1(X)\, x$$

for $X \in \mathfrak{g}$, the Lie algebra of G. This action corresponds to the infinitesimal representation of $\mathfrak{g}$.

3. A Lie group G acts on itself by left translations. Since the flow of a right invariant field is given by left translations, it follows that $\widetilde{X}$ is the right invariant field corresponding to $X \in \mathfrak{g}$.
4. Let $\phi : G \to H$ be a differentiable homomorphism. Then G acts on H on the left by $(g, h) \in G \times H \mapsto \phi(g)\, h$. As in the previous example, $\widetilde{X}$ is the right invariant field on H defined by $d\phi_1(X) \in \mathfrak{h}$ (the Lie algebra of H).
5. The linear group $G = \mathrm{Gl}(n, \mathbb{R})$ acts on the sphere of radius 1 (defined by the standard inner product). The action is in terms of the 1-1 correspondence between the sphere and the set of half-lines in $\mathbb{R}^n$ emanating from the origin.
   If r is a half-line starting at the origin and $g \in \mathrm{Gl}(n, \mathbb{R})$, then gr is a half-line, which defines the action $(g, r) \mapsto gr$ on the set of half-lines and, therefore, on S^{n-1}. For $x \in S^{n-1}$ and $g \in \mathrm{Gl}(n, \mathbb{R})$, the action is denoted by $g * x$. By definition, $g * x$ is the intersection with S^{n-1} of the ray spanned by gx:

$$g * x = \frac{gx}{\|gx\|},$$

   where $\|gx\|$ denotes the Euclidean norm. This action is differentiable. The corresponding infinitesimal action is given by

$$\widetilde{A}(x) = \frac{d}{dt}\left(e^{tA} * x\right)_{|t=0} = \frac{d}{dt}\left(\frac{e^{tA}x}{\left|e^{tA}x\right|}\right)_{|t=0}.$$

   Computing the derivative yields

$$\widetilde{A}(x) = Ax - \langle Ax, x\rangle x,$$

   which is the tangent vector to S^{n-1} at x obtained by projecting Ax along the half-line spanned by x.
6. A minor adaptation in the previous example provides an action of $\mathrm{Gl}(n, \mathbb{R})$ on the projective space $\mathbb{P}^{n-1}$, i.e. the space of one-dimensional subspaces of $\mathbb{R}^n$. Replacing half-lines by the corresponding lines does the trick. When one identifies the tangent space to $\mathbb{P}^{n-1}$ at $[x]$ with the tangent space to S^{n-1} at $x \in S^{n-1}$, the expression of $\widetilde{A}$ one gets coincides with the one presented earlier.

□

13.1.1 Orbits

The infinitesimal action on M induced by the action of a Lie group G is applied, among other things, to the study of the orbits of G. The reason is that orbits can be obtained as maximal integral manifolds of the distribution defined by the infinitesimal action.

Let $\phi : G \times M \to M$ be a differentiable action and $\theta : \mathfrak{g} \to \Gamma(TM), \theta(X) = \widetilde{X}$, the corresponding infinitesimal action. For $x \in M$, define the subspace $\Delta_{\mathfrak{g}}(x) \subset T_x M$ by

$$\Delta_{\mathfrak{g}}(x) = \{\widetilde{X}(x) \in T_x M : X \in \mathfrak{g}\}.$$

The map $x \mapsto \Delta_{\mathfrak{g}}(x)$ is a distribution on M. By its very definition, $\Delta_{\mathfrak{g}}$ is a differentiable distribution, because it is generated by the vector fields $\widetilde{X}$, $X \in \mathfrak{g}$. In general, the dimension of $\Delta_{\mathfrak{g}}$ will not be constant. For instance, for the standard action of $\mathrm{Gl}(n, \mathbb{R})$ on $\mathbb{R}^n$, $\Delta_{\mathfrak{g}}$ reduces to $\{0\}$ at the origin, while $\Delta_{\mathfrak{g}}(x)$ is the entire tangent space if $x \neq 0$. What usually happens is $\dim \Delta_{\mathfrak{g}}(x) = \dim \mathfrak{g} - \dim \mathfrak{g}_x$, since $\mathfrak{g}_x$ is the kernel of the linear map $X \in \mathfrak{g} \to \widetilde{X}(x) \in T_x M$.

Proposition 13.5 *The distribution $\Delta_{\mathfrak{g}}$ is invariant under the action of G, that is, $g_* \Delta_{\mathfrak{g}} = \Delta_{\mathfrak{g}}$. More explicitly,*

$$dg_x \Delta_{\mathfrak{g}}(x) = \Delta_{\mathfrak{g}}(gx)$$

for all $g \in G$.

Proof The translation of a vector field $\widetilde{X}$ by an element $g \in G$ is given by the formula $g_* \widetilde{X} = \widetilde{\mathrm{Ad}(g) X}$. This implies that for every $g \in G$ and $x \in X$,

$$dg_x \left(\widetilde{X}(x)\right) = \widetilde{\mathrm{Ad}(g) X}(gx).$$

Since the distribution $\Delta_{\mathfrak{g}}$ is spanned by the vector fields $\widetilde{X}$, $X \in \mathfrak{g}$, one has $dg_x \Delta_{\mathfrak{g}}(x) \subset \Delta_{\mathfrak{g}}(gx)$. The opposite inclusion is obtained in the same way, translating by g^{-1} instead of g. □

This proposition shows that $\Delta_{\mathfrak{g}}$ is a characteristic distribution (see Definition B.8). By Theorem B.9, therefore:

Proposition 13.6 *The distribution $\Delta_{\mathfrak{g}}$ is integrable.*

The integral manifolds of this distribution are the orbits of the G-action. To see this, let $I_{\mathfrak{g}}(x)$ be the maximal integral manifold of $\Delta_{\mathfrak{g}}$ passing through x. Since the distribution is G-invariant (Proposition 13.5), for each $g \in G$ the set $g I_{\mathfrak{g}}(x)$ is an integral manifold of the distribution. It is clear that $gx \in g I_{\mathfrak{g}}(x)$, so $g I_{\mathfrak{g}}(x) \subset I_{\mathfrak{g}}(gx)$. This inclusion is not proper, for otherwise $g^{-1}\left(I_{\mathfrak{g}}(gx)\right)$ would be an integral manifold of $\Delta_{\mathfrak{g}}$ containing $I_{\mathfrak{g}}(x)$. Consequently,

$$gI_{\mathfrak{g}}(x) = I_{\mathfrak{g}}(gx) \qquad g \in G, x \in M. \tag{13.1}$$

The next lemma, in particular, shows that if G is connected its elements preserve the maximal connected integral manifolds of $\Delta_{\mathfrak{g}}$.

Lemma 13.7 *Retaining the previous notations, suppose G is connected. Then $gI_{\mathfrak{g}}(x) = I_{\mathfrak{g}}(gx) = I_{\mathfrak{g}}(x)$ for every $g \in G$ and $x \in M$. That is, maximal connected integral manifolds of $\Delta_{\mathfrak{g}}$ are G-invariant.*

Proof Given $g \in G$, write $g = e^{X_k} \cdots e^{X_1}$ and define the path $\alpha : [0, k] \to M$ by

$$\alpha(t) = e^{(t-i+1)X_i} e^{X_{i-1}} \cdots e^{X_1} x \qquad t \in [i-1, i].$$

This curve is a concatenation of trajectories of the vector fields $\widetilde{X}_i$, which are tangent to $\Delta_{\mathfrak{g}}$. Then the curve is contained in a single maximal connected integral manifold of $\Delta_{\mathfrak{g}}$. Its starting point is x and the end point is gx. Hence $I_{\mathfrak{g}}(gx) = I_{\mathfrak{g}}(x)$, proving the lemma. □

The invariance of the lemma defines, in the connected case, an action $G \times I_{\mathfrak{g}}(x) \to I_{\mathfrak{g}}(x)$ on each maximal connected integral manifold. All these actions are differentiable since integral manifolds are quasi-regular. Furthermore, the orbit $G \cdot x$ is contained in $I_{\mathfrak{g}}(x)$. In fact, the following result shows that $G \cdot x = I_{\mathfrak{g}}(x)$, which characterizes G-orbits in terms of the infinitesimal action and the distribution defined by it.

Theorem 13.8 *Suppose G is connected. Then for every $x \in M$ the orbit $G \cdot x$ coincides with the maximal connected integral manifold $I_{\mathfrak{g}}(x)$ of $\Delta_{\mathfrak{g}}$ through x.*

Proof The idea is to prove that G-orbits are open sets in maximal integral manifolds. Let $\{X_1, \ldots, X_k\}$ be a basis of $\mathfrak{g}$ and take $y \in I_{\mathfrak{g}}(x)$. Define the map

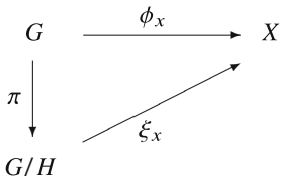
$$\rho(t_1, \ldots, t_k) = e^{t_1 X_1} \cdots e^{t_k X_k} y.$$

Its image is contained in the orbit $G \cdot y$. Furthermore, the image of its differential at the origin is spanned by the partial derivatives $\widetilde{X}_i(y)$, which in turn generate the tangent space $\Delta_{\mathfrak{g}}(y)$ to $I_{\mathfrak{g}}(x)$. By the open-mapping theorem, therefore, y belongs to the interior (with respect to the intrinsic topology of $I_{\mathfrak{g}}(x)$) of the image of ρ. This implies that $y \in (G \cdot y)^{\circ}$, and hence the $G \cdot y$ are open subsets of $I_{\mathfrak{g}}(x)$. But the complement of an orbit is a union of orbits, so $G \cdot x$ is open, closed, and nonempty inside the connected set $I_{\mathfrak{g}}(x)$, showing that $G \cdot x = I_{\mathfrak{g}}(x)$. □

The orbits of a nonconnected group in general are unions of maximal integral manifolds of $\Delta_{\mathfrak{g}}$. In fact, the orbit $G_0 \cdot x$ of the identity component is $I_{\mathfrak{g}}(x)$. So,

$$G \cdot x = \bigcup_{g \in G} gG_0 \cdot x = \bigcup_{g \in G} gI_{\mathfrak{g}}(x) = \bigcup_{g \in G} I_{\mathfrak{g}}(gx).$$

The next objective is to identify an orbit $G \cdot x$ with the homogeneous space G/G_x. As seen in Chapter 2, the map $\xi_x : G/G_x \to G \cdot x$, $\xi_x\,(gG_x) = gx$, as per the diagram



is bijective. It is continuous and differentiable for the quotient structure, since $\xi_x \circ \pi = \phi_x$, where $\phi_x\,(g) = gx$ is the partial map of the action $\phi : G \times M \to M$. Since $G \cdot x$ is a quasi-regular submanifold, ϕ_x is differentiable with values in $G \cdot x$. By Theorem 6.22, it follows that ξ_x is differentiable as well.

Proposition 13.9 *The map $\xi_x : G/G_x \to G \cdot x$ is a diffeomorphism.*

Proof Since ξ_x is differentiable and bijective, it suffices to check that it is a local diffeomorphism. Equivalently, the differential of ξ_x is bijective at every point. Since $\xi_x \circ \pi = \phi_x$, the image of the differential of ξ at $gG_x \in G/G_x$ coincides with the image of the differential $(d\phi_x)_g$, which is the set of vectors $\widetilde{X}\,(gx)$. So the image of $d\,(\xi_x)_{gG_x}$ is $\Delta_{\mathfrak{g}}\,(gx)$, which means that the differential is onto, and therefore injective, because the dimensions of G/G_x and $G \cdot x$ are equal, to $\dim G - \dim G_x$. □

One particular case covered by this proposition is that of transitive actions, in which case there is a single orbit, namely the whole manifold M. If so, $\psi_x : G/G_x \to M$ is a diffeomorphism and the tangent spaces T_zM, $z \in M$, coincide with $\Delta_{\mathfrak{g}}\,(z)$. This means that every tangent vector $v \in T_zM$ is of the form $v = \widetilde{X}\,(z)$ for some $X \in \mathfrak{g}$.

Finally, the following result applies to transitive actions of the connected component at the identity.

Proposition 13.10 *Suppose that the action of G on M is transitive and let C be a connected component of M. Then the restriction of the action of G to its identity component G_0 is a transitive action of G_0 on C.*

Proof First of all, the restriction to G_0 defines actions on the connected components of M, because if one takes $g \in G_0$ then $g\,(C)$ is contained in a connected component of M by the continuity of the action. Since $g\,(C)$ and $1\,(C)$ are necessarily in the same connected component, $g\,(C) \subset C$.

Now, for every $x \in C$ the tangent spaces to the orbits Gx and G_0x coincide with the distribution $\Delta_{\mathfrak{g}}\,(x)$. Since the action of G is transitive, $\Delta_{\mathfrak{g}}\,(x) = T_xM = T_xC$. Therefore the orbit G_0x is an open submanifold. The complement of G_0x is open because it is a union of open orbits. Hence G_0x is open and closed in the connected set C. This proves $G_0x = C$, i.e. the action is transitive. □

13.2 Lie–Palais Theorem

A natural question is whether the infinitesimal actions of Lie algebras come from actions of Lie groups. Put otherwise, if $\mathfrak{g}$ is a finite dimensional real Lie algebra and $\theta : \mathfrak{g} \to \Gamma(TM)$ an infinitesimal action of $\mathfrak{g}$, are there a Lie group G with Lie algebra $\mathfrak{g}$ and an action $\phi : G \times M \to M$ such that θ is the infinitesimal action corresponding to ϕ? A necessary condition for this to happen is that the vector fields $\theta(X)$, $X \in \mathfrak{g}$, are complete, since any vector field $\widetilde{X}$ obtained from a group action is complete.

The **Lie–Palais theorem**[1] says that the completeness of the vector fields $\theta(X)$, $X \in \mathfrak{g}$, is sufficient to integrate the infinitesimal action θ to an action of a Lie group G. In particular, on a compact manifold M, every infinitesimal action comes from a global action.

To construct an action $\phi : G \times M \to M$ that globalizes θ, one idea that comes to mind is based on the following observation: if ψ_t^X denotes the flow of the vector field $\theta(X)$ then

$$\psi_t^X(x) = \phi\left(e^{tX}, x\right)$$

for all $t \in \mathbb{R}$ and $x \in M$. Hence the putative action should be defined by

$$\phi(g, x) = \phi\left(e^{X_1} \cdots e^{X_n}, x\right) = \psi_1^{X_1} \circ \cdots \circ \psi_1^{X_n}(x)$$

if $g = e^{X_1} \cdots e^{X_n} \in G$.

The disadvantage with such direct definition of ϕ is that it is hard to prove that it is well defined, i.e. that the right hand side does not depend on how $g \in G$ is written as a product of exponentials. Due to this snag, one usually adopts another tactic, similar to the construction of Lie group homomorphisms extending Lie algebra homomorphisms (see Chapter 7). By this approach, the maps $\phi_x : G \to M$, $\phi_x(g) = \phi(g, x)$, are constructed through their graphs in $G \times M$. These, in turn, are given by integral manifolds of an integrable distribution.

Keeping this in mind, let G be the connected and simply connected group with Lie algebra $\mathfrak{g}$. Given an infinitesimal action $\theta : \mathfrak{g} \to \Gamma(TM)$, define in $G \times M$ the distribution

$$\Delta_\theta(g, x) = \{\left(X^d(g), \theta(X)(x)\right) \in T_{(g,x)}G \times M : X \in \mathfrak{g}\}.$$

This distribution satisfies the following properties:

1. $\dim \Delta_\theta(g, x) = \dim G$ for all $(g, x) \in G \times M$, since the linear map $X \in \mathfrak{g} \mapsto \left(X^d(g), \theta(X)(x)\right) \in \Delta_\theta(g, x)$ is an isomorphism.

[1] See Palais [43].

2. Δ_θ is differentiable. In fact, if $\{X_1, \ldots, X_n\}$ is a basis of $\mathfrak{g}$, the vector fields $\left(X_1^d, \theta\left(X_1\right)\right), \ldots, \left(X_n^d, \theta\left(X_n\right)\right)$ form a global parametrization of Δ_θ.
3. Δ_θ is integrable, as follows from the Frobenius theorem, or more precisely Corollary B.15. In fact, if $X, Y \in \mathfrak{g}$ then

$$[\left(X^d, \theta\left(X\right)\right), \left(Y^d, \theta\left(Y\right)\right)] = \left([X, Y]^d, \theta[X, Y]\right),$$

because θ is a homomorphism. Hence the global parametrization of the previous item is closed under the bracket, as required in Corollary B.15. Alternatively, if $\{X_1, \ldots, X_n\}$ is a basis of $\mathfrak{g}$ and ψ_t^i denotes the flow of $\theta\left(X_i\right)$, it is possible to show that the map

$$\left(t_1, \ldots, t_n\right) \mapsto \left(e^{t_1 X_1} \cdots e^{t_n X_n} g, \psi_{t_1}^1 \circ \cdots \circ \psi_{t_1}^1 (x)\right)$$

defines an integral manifold of Δ_θ passing through (g, x), provided $\left(t_1, \ldots, t_n\right)$ is small enough (compared with the proof of Theorem 13.8).

Let $I_\theta (g, x)$ be the maximal connected integral manifold of Δ_θ, which contains (g, x). When the vector fields $\theta\left(X\right)$ are complete, the integral manifold has good properties in regard to the projection $p : G \times M \to G$, as the following lemmas show.

Lemma 13.11 *The restriction to an integral manifold $I_\theta (g, x)$ of the projection $p : G \times M \to G$ is a local diffeomorphism. If the vector fields $\theta\left(X\right)$, $X \in \mathfrak{g}$ are complete, the projection is surjective.*

Proof The image of the tangent vector $\left(X^d\left(g\right), \theta\left(X\right)(x)\right)$ under the differential of p, restricted to $\Delta_\theta (g, x)$, equals $X^d\left(g\right)$. This map is onto and hence an isomorphism between $\Delta_\theta (g, x)$ and $T_g G$. It follows that p is a local diffeomorphism. The surjectivity of p comes from the fact that the trajectories of the vector fields $\left(X^d, \theta\left(X\right)\right)$ are entirely contained in the integral manifolds of Δ_θ. Such a trajectory has the form $\left(e^{tX} g, \psi_t (x)\right)$, where ψ_t is the flow of $\theta\left(X\right)$. Since $\theta\left(X\right)$ is complete, it follows that $p\left(I_\theta (g, x)\right)$ contains $e^{tX} g$ for all $X \in \mathfrak{g}$. Taking successive concatenations of trajectories of the vector fields $(Y, \theta\left(Y\right))$ shows that $p\left(I_\theta (g, x)\right)$ contains arbitrary products of the type $e^{X_1} \cdots e^{X_n} g$, and therefore $p\left(I_\theta (g, x)\right) = G$. □

The fact that $p : I_\theta (g, x) \to G$ is a local diffeomorphism ensures that if $(h, y) \in I_\theta (g, x)$, then there exist open connected subsets $A_y \subset I_\theta (g, x)$ and $B_y \subset G$, with $(h, y) \in A_y$ and $h \in B_y$, such that $p : A_y \to B_y$ is a diffeomorphism. In general, the sets B_y depend on y. However, with the assumption that the vector fields $\theta\left(X\right)$ are complete, it is possible to find sets A_y, B_y, such that B_y is constant as a function of y, and hence prove the desired covering property.

Lemma 13.12 *Suppose the vector fields $\theta(X)$, $X \in \mathfrak{g}$ are complete. Then for every maximal connected integral manifold $I_\theta(g,x)$ of $\Delta_\theta(g,x)$, the projection $p : I_\theta(g,x) \to G$ is a covering map.*

Proof Take open connected sets $V \subset \mathfrak{g}$ and $U \in G$, containing the respective origins, such that $\exp : V \to U$ is diffeomorphism. Assume that $V = -V$. For $X \in \mathfrak{g}$, denote by ψ_t^X the flow of the vector field $\theta(X)$. The hypothesis that $\theta(X)$ is complete ensures that $\psi_{t=1}^X(y)$ is well defined for all $y \in M$.

Now let $h \in G$ and $y \in M$ be such that $(h, y) \in I_\theta(g,x)$, and define the set

$$A_y = \{\left(e^X h, \psi_1^X(y)\right) : X \in V\}.$$

This is contained in $I_\theta(g,x)$, because the vector fields $\left(X^d, \theta(X)\right)$ are tangent to Δ_θ. Clearly, A_y is the image of the differentiable map $f_{h,y} : V \to I_\theta(g,x)$ given by $f_{h,y}(X) = \left(e^X h, \psi_1^X(y)\right)$. This map satisfies $p \circ f_{h,y} = D_h \circ \exp$, and by the chain rule $f_{h,y}$ is a local diffeomorphism. Therefore each A_y is a connected open subset of $I_\theta(g,x)$. Their projections $p\left(A_y\right)$ are equal to $B = Uh$. Each restriction $p_y : A_y \to Uh$ is injective, since $X = Y$ if $e^X h = e^Y h$, and hence $\psi_1^X(y) = \psi_1^Y(y)$. Hence p_y is diffeomorphism, because p is a local diffeomorphism and p_y is additionally onto. The sets A_y will be employed to show that $p : I_\theta(g,x) \to G$ is covering map. These sets satisfy the following properties:

1. A_y is connected, since diffeomorphic to Uh.
2. $A_{y_1} \cap A_{y_2} = \emptyset$ if $y_1 \neq y_2$. In fact, if

$$\left(e^X h, \psi_1^X(y_1)\right) = \left(e^Y h, \psi_1^Y(y_2)\right) \in A_{y_1} \cap A_{y_2},$$

   then $e^X h = e^Y h$ and so $X = Y$. Hence $\psi_1^X(y_1) = \psi_1^X(y_2)$, that is, $y_1 = y_2$.
3. $p^{-1}(Uh) = \bigcup_{y \in p^{-1}\{h\}} A_y$. In fact, $A_y \subset p^{-1}(Uh)$ for all $y \in p^{-1}\{h\}$ by construction of A_y. On the other hand, the elements of $p^{-1}(Uh)$ are of the form $\left(e^X h, z\right)$ with $X \in V$ and $z \in M$, since $\left(e^X h, z\right) \in I_\theta(g,x)$. Take $\left(e^X h, z\right) \in p^{-1}(Uh)$, $X \in V$. Then the trajectory of the vector field $(X, \theta(X))$ starting at $\left(e^X h, z\right)$ remains in $I_\theta(g,x)$. For $t \in [-1, 0]$, the first coordinate of this trajectory is $e^{(1+t)}h$. Hence the trajectory does not leave $p^{-1}(Uh)$. For $t = -1$, the trajectory assumes the value $\left(e^{-X}e^X h, \psi_{-1}^X(z)\right) = \left(h, \psi_{-1}^X(z)\right)$, implying $\left(e^X h, z\right) \in A_y$ with $y = \psi_{-1}^X(z) \in p^{-1}\{h\}$. Consequently, the reverse inclusion $p^{-1}(Uh) \subset \bigcup_{y \in p^{-1}\{h\}} A_y$ also holds.

These properties show that p is a covering map, concluding the proof of the lemma. □

When the group G is simply connected, the covering maps $p : I_\theta(g,x) \to G$ are bijective. But p is a local diffeomorphism, hence each projection $p : I_\theta(g,x) \subset$

$G \times M \to G$ is a diffeomorphism. Therefore $I_\theta(g, x)$ is the graph of a differentiable map $G \to M$.

Denote by $\phi_x : G \to M$ the differentiable map whose graph is the integral manifold $I_\theta(1, x)$. The following proposition provides an expression for ϕ_x in terms of exponentials in G and of the flows ψ_t^X of the vector fields $\theta(X)$. This expression allows to show that $\phi(g, x) = \phi_x(g)$ is the desired global action of the simply connected group G, which integrates the infinitesimal action θ.

Proposition 13.13 *Take $x \in M$ and $X_1, \ldots, X_n \in \mathfrak{g}$. Then*

$$\phi_x\left(e^{X_1} \cdots e^{X_n}\right) = \psi_1^{X_1} \circ \cdots \circ \psi_1^{X_n}(x). \tag{13.2}$$

Proof The trajectories of the vector field $\left(X^d, \theta(X)\right)$ remain in the maximal integral manifolds. The trajectory beginning at (g, y) is given by $\left(e^{tX}, \psi_t^X(y)\right)$. Concatenating trajectories from $(1, x)$, one sees that

$$\left(e^{X_1} \cdots e^{X_n}, \psi_1^{X_1} \circ \cdots \circ \psi_1^{X_n}(x)\right)$$

belongs to the integral manifold $I_\theta(1, x)$. Hence, by definition of ϕ_x, the second coordinate is the value of ϕ_x on the first coordinate, which proves (13.2). □

Now it is possible to state and complete the proof of the Lie–Palais theorem, which integrates an infinitesimal action of a Lie algebra to a global action of Lie group.

Theorem 13.14 *Let $\mathfrak{g}$ be a real Lie algebra with $\dim \mathfrak{g} < \infty$ and G the corresponding connected and simply connected Lie group. Let $\theta : \mathfrak{g} \to \Gamma(TM)$ be an infinitesimal action of $\mathfrak{g}$ and suppose that the vector fields $\theta(X)$ are complete. Then there exists a differentiable action $\phi : G \times M \to M$ corresponding to θ.*

Proof Define $\phi(g, x) = \phi_x(g)$, where the graph of $\phi_x : G \to M$ is the integral manifold $I_\theta(1, x)$. This defines an action of G on M since:

1. if $x \in M$, then $\phi(1, x) = x$, as $(1, x)$ is the only element of $I_\theta(1, x)$ that projects to 1;
2. $\phi(g, \phi(h, x)) = \phi(gh, x)$ for $g, h \in G$. In fact, if $g = e^{X_1} \cdots e^{X_n}$ and $h = e^{Y_1} \cdots e^{Y_n}$, by formula (13.2) it follows that

$$\begin{aligned}
\phi(g, \phi(h, x)) &= \phi\left(g, \psi_1^{Y_1} \circ \cdots \circ \psi_1^{Y_m}(x)\right) \\
&= \psi_1^{X_1} \circ \cdots \circ \psi_1^{X_n} \circ \psi_1^{Y_1} \circ \cdots \circ \psi_1^{Y_m}(x) \\
&= \phi(gh, x).
\end{aligned}$$

The differentiability of the action ϕ is a consequence of the differentiable dependence of solutions on parameters, see formula (13.2), after taking a frame system of the second kind around any $g \in G$. □

The action of G defines a homomorphism $\Theta : G \to \mathrm{Diff}(M)$ with values in the diffeomorphism group $\mathrm{Diff}(M)$. By (13.2), it is given by

$$\Theta\left(e^{X_1} \cdots e^{X_n}\right) = \psi_1^{X_1} \circ \cdots \circ \psi_1^{X_n},$$

hence the image of Θ is the subgroup $\mathrm{Diff}(\theta)$ of $\mathrm{Diff}(M)$ generated by the flows ψ^X of the vector fields $\theta(X)$, $X \in \mathfrak{g}$. That is,

$$\mathrm{Diff}(\theta) = \{\psi_{t_1}^{X_1} \circ \cdots \circ \psi_{t_n}^{X_n} : X_i \in \mathfrak{g},\ t_i \in \mathbb{R}\}.$$

Therefore $\mathrm{Diff}(\theta)$ is isomorphic to $G/\ker\Theta$, which is a Lie group since $\ker\Theta$ is closed in G. The Lie algebra of $\ker\Theta$ is $\ker\theta$, since $\Theta\left(e^{tX}\right) = \psi_1^{tX} = \psi_t^X =$ id if and only if $\theta(X) = 0$. Hence the Lie algebra of $\mathrm{Diff}(\theta)$ is isomorphic to $\mathfrak{g}/\ker\theta$. The latter is isomorphic to the image of θ, i.e. the Lie algebra of vector fields $\{\theta(X) : X \in \mathfrak{g}\}$.

These remarks apply in particular to a finite dimensional Lie algebra of vector fields whose infinitesimal action is the inclusion.

Corollary 13.15 *Let $\mathfrak{g}$ be a finite dimensional Lie algebra of vector fields of a manifold M, such that every $X \in \mathfrak{g}$ is complete. Denote by $\mathrm{Diff}(\mathfrak{g})$ the diffeomorphism group of M generated by the flows ψ^X of the elements of $\mathfrak{g}$:*

$$\mathrm{Diff}(\mathfrak{g}) = \{\psi_{t_1}^{X_1} \circ \cdots \circ \psi_{t_n}^{X_n} : X_i \in \mathfrak{g},\ t_i \in \mathbb{R}\}.$$

Then $\mathrm{Diff}(\mathfrak{g})$ is a Lie group whose Lie algebra is isomorphic to $\mathfrak{g}$.

Finally, the next example illustrates an infinitesimal action that can be integrated to a local action that cannot be global since the vector fields are not complete.

Example The image of the map

$$x \in \mathbb{R} \longmapsto \begin{pmatrix} x \\ 1 \end{pmatrix} \in \mathbb{R}^2$$

is the horizontal line r containing $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$. The set of one-dimensional subspaces that cross r is open and dense in the projective line $\mathbb{P}^1$. Hence the above map embeds $\mathbb{R}$ as an open dense subset of $\mathbb{P}^1$. The restriction of the standard action of $\mathrm{Gl}(2, \mathbb{R})$ to this open dense set defines a local action of $\mathrm{Gl}(2, \mathbb{R})$ on $\mathbb{R}$ by rational maps (Möbius functions). In fact, let $g * p$, for $g \in \mathrm{Gl}(2, \mathbb{R})$ and $p \in \mathbb{P}^1$, denote the action on the

projective line. If p is the subspace generated by $\begin{pmatrix} x \\ 1 \end{pmatrix}$ and $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then $g * p$ is the subspace spanned by

$$\begin{pmatrix} ax + b \\ cx + d \end{pmatrix}.$$

Provided $cx + d \neq 0$, this vector spans the same subspace as

$$\begin{pmatrix} (ax + b) / (cx + d) \\ 1 \end{pmatrix}.$$

Using the notation

$$g * x = \frac{ax + b}{cx + d},$$

the map $\phi(g, x) = g * x$ defines a local action of $\mathrm{Gl}(2, \mathbb{R})$ on $\mathbb{R}$. It is clear that ϕ is not defined on the entire $\mathrm{Gl}(2, \mathbb{R}) \times \mathbb{R}$. Nevertheless, $g * (h * x) = (gh) * x$ for the values at which it is defined. In any case, ϕ is defined on neighborhoods of $(1, x)$, for all $x \in \mathbb{R}$, which allows to define vector fields

$$\widetilde{A}(x) = \frac{d}{dt}\left(e^{tA} * x\right)_{|t=0} = d(\phi_x)_1(A),$$

where ϕ_x is the partial map $\phi_x(g) = \phi(g, x)$. Since ϕ is the restriction of a global action, $A \mapsto \widetilde{A}$ defines an infinitesimal action of $\mathfrak{gl}(2, \mathbb{R})$ on $\mathbb{R}$. To compute $\widetilde{A}$ write

$$A = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \qquad e^{tA} = \begin{pmatrix} a_t & b_t \\ c_t & d_t \end{pmatrix}.$$

Then

$$\widetilde{A}(x) = \frac{d}{dt}\left(\frac{a_t x + b_t}{c_t x + d_t}\right)_{|t=0}.$$

But $a_0 = d_0 = 1$ and $c_0 = d_0 = 0$, so

$$\widetilde{A}(x) = \beta + (\alpha - \delta)x - \gamma x^2.$$

These vector fields are associated with **Riccati differential equations** and, in general, they are not complete. The map $\theta(A) = \widetilde{A}$ defines an infinitesimal action of $\mathfrak{gl}(2, \mathbb{R})$ that does not integrate to a global action. □

13.2.1 Families of Vector Fields

The purpose of this subsection is to strengthen the Lie–Palais theorem by achieving the same integrability result but with fewer assumptions. The gist is that the completeness of a generating set of vector fields is enough.

Theorem 13.16 *Let $\mathfrak{g}$ be a finite dimensional Lie algebra and $\theta : \mathfrak{g} \to \Gamma(TM)$ a homomorphism from $\mathfrak{g}$ to the Lie algebra of C^∞-vector fields on M (infinitesimal action). Suppose there exists a subset $D \subset \mathfrak{g}$ that generates $\mathfrak{g}$ (as a Lie algebra) and such that the vector fields $\theta(X)$ are complete if $X \in D$. Then θ integrates to an action $\phi : G \times M \to M$ of a Lie group G with Lie algebra $\mathfrak{g}$. A posteriori, $\theta(Y)$ is complete for every $Y \in \mathfrak{g}$.*

The theorem is proved by suitably modifying the above proof of the Lie–Palais theorem. The main changes occur where coordinate systems of the second kind on G are used. These systems are given by products of exponentials of arbitrary elements of $\mathfrak{g}$. The following proposition, itself a special case of a theorem of Chow, allows to take products of elements of $\mathbb{R}D$ instead.

Proposition 13.17 *Let G be a connected Lie group with Lie algebra $\mathfrak{g}$ and suppose $D \subset \mathfrak{g}$ generates $\mathfrak{g}$ (as a Lie algebra). Then*

1. *For every pair $g, h \in G$, there exist $Y_1, \ldots, Y_k \in D$ (possibly not all distinct) and $t_i \in \mathbb{R}$ such that*

$$g = e^{t_1 Y_1} \cdots e^{t_k Y_k} h.$$

   In particular, every $g \in G$ is a product of exponentials of elements of $\mathbb{R}D$.
2. *The $Y_1, \ldots, Y_k \in D$ and the $t_1, \ldots, t_k \in \mathbb{R}$ can be chosen so that the map $\rho : \mathbb{R}^k \to G$ defined by*

$$\rho(s_1, \ldots, s_k) = e^{s_1 Y_1} \cdots e^{s_k Y_k} h$$

   has rank $\dim \mathfrak{g}$ *at $(t_1, \ldots, t_k)$ (Hence, $k \geq \dim \mathfrak{g}$.)*

The proof follows the steps indicated in Exercise 33, Chapter 6.

Just as in the proof of Theorem 13.14, the first step to prove Theorem 13.16 is to define the distribution Δ_θ on $G \times M$. The definition of Δ_θ, as well as the proof of its integrability as an application of the Frobenius theorem, requires the completeness of the vector fields (see item (3) in the discussion preceding Lemma 13.11).

In Lemma 13.11, completeness is used to guarantee that the projection p restricted to a maximal integral manifold is onto. Here the same argument works by taking vector fields $\left(X^d, \theta(X)\right)$ with X in $\mathbb{R}D$ instead of arbitrary vectors $X \in \mathfrak{g}$. In fact, by the assumptions made on D every element of G can be written as a product of exponentials in $\mathbb{R}D$, and the trajectories of $\left(X^d, \theta(X)\right)$, $X \in D$, can be extended as one pleases.

Completeness is necessary also to show that the projection p on G is a covering map. Specifically, it is crucial for the definition of the subsets A_y, B_y in the proof of Lemma 13.12. At present, these subsets must be introduced using flows of vector fields defined by elements of D, instead of arbitrary elements in $\mathfrak{g}$.

For the new definition of A_y, take $Y_1, \ldots, Y_k \in D$, $t_1, \ldots, t_k \in \mathbb{R}$ and $\rho(s_1, \ldots, s_k) = e^{s_1 Y_1} \cdots e^{s_k Y_k}$ as in Proposition 13.17, in such a way that $e^{t_1 Y_1} \cdots e^{t_k Y_k} = 1$ and ρ has rank $\dim \mathfrak{g}$ at $(t_1, \ldots, t_k)$. With this choice, there exists an open neighborhood $W \subset \mathbb{R}^k$ of $(t_1, \ldots, t_k) \in W$ such that ρ is a submersion on W. Hence there exist an open set $V \subset \mathbb{R}^n$ ($n = \dim \mathfrak{g}$) and an immersion $\phi : V \to W$ such that $\rho \circ \phi$ is a diffeomorphism (by the local form of submersions). The map $\rho \circ \phi : V \to G$ plays the role of the exponential in the previous proof.

Let $\psi_s^{Y_i}$, $i = 1, \ldots, k$, be the flows of $Y_1, \ldots, Y_k$ and put $x = \rho(t_1, \ldots, t_k)$. Define maps $f_{h,y} : V \to I_\theta(g, x)$, $h \in G$ and $y \in M$, by

$$f_{h,y}(u_1, \ldots, u_n) = \left(\rho \circ \phi(u_1, \ldots, u_n)\, h,\, \psi_{s_1}^{Y_1} \circ \cdots \circ \psi_{s_k}^{Y_k}(y)\right), \tag{13.3}$$

where $(s_1, \ldots, s_k) = \phi(u_1, \ldots, u_n)$. As in the proof of Lemma 13.12, these maps are well defined. They are local diffeomorphisms since $p \circ f_{h,y} = R_h \circ \rho$ is a diffeomorphism and p a local diffeomorphism.

With all this in place, the new subsets A_y are defined by $A_y = f_{h,y}(V)$. Any such is open since each $f_{h,y}$ is a local diffeomorphism. Its projection $B_y = p\left(f_{h,y}(V)\right) = \rho(W)\, h$ is open in G and does not depend on y, only on h. If V is taken to be connected, then A_y and B_y are connected as well.

The sets A_y and B_y satisfy the three properties in the proof of Lemma 13.12, which permits to conclude that $p : I_\theta(g, x) \to G$ is a covering map. In fact,

1. property (1) holds by construction (starting with a connected V).
2. For property (2), suppose that

$$f_{h,y_1}(u_1, \ldots, u_n) = f_{h,y_2}(\overline{u}_1, \ldots, \overline{u}_n).$$

A glance at the first coordinates in (13.3) tells that $u_i = \overline{u}_i$, so the corresponding s_i are also equal. Therefore,

$$\psi_{s_1}^{Y_1} \circ \cdots \circ \psi_{s_k}^{Y_k}(y_1) = \psi_{s_1}^{Y_1} \circ \cdots \circ \psi_{s_k}^{Y_k}(y_2),$$

showing that $y_1 = y_2$ as $\psi_{s_1}^{Y_1} \circ \cdots \circ \psi_{s_k}^{Y_k}$ is diffeomorphism. Hence $A_{y_1} \cap A_{y_2} = \emptyset$ if $y_1 \neq y_2$.

3. For property (3), define $U = \rho \circ \phi(V) = \rho(W)$. Take $(l, z) \in p^{-1}(U)$, that is, $l = p(z) = e^{s_1 Y_1} \cdots e^{s_k Y_k} h$, where $(s_1, \ldots, s_k) = \phi(u_1, \ldots, u_n)$ and $(u_1, \ldots, u_n) \in V$. Let $\eta_t^{Y_i}$ be the flow of the complete vector field $\left(Y_i^d, \theta(Y_i)\right)$:

$$\eta_t^{Y_i}(h, y) = \left(e^{tY_i} h,\, \psi_t^{Y_i}(y)\right).$$

Then $\eta_{-s_k}^{Y_k} \circ \cdots \circ \eta_{-s_1}^{Y_1}(z) = (h, y)$ where $y = \psi_{-s_k}^{Y_k} \circ \cdots \circ \psi_{-s_1}^{Y_1}(z)$. Hence $(l, z) = f_{h,y}(u_1, \ldots, u_n)$, showing that $(l, z) \in A_y$.

Therefore Lemma 13.12 holds under the assumption that the vector fields in a generating set D are complete. The same was checked above for Lemma 13.11, thus proving that $I_\theta(g, x)$ is the graph of a map $\phi_x : G \to M$. Proposition 13.13 proves the key equality ensuring that $\phi(g, x) = \phi_x(g)$ is indeed an action. Here the same proof works by taking $Y_1, \ldots, Y_k \in D$ in place of $X_1, \ldots, X_n \in \mathfrak{g}$. Finally, the proof of Theorem 13.14 follows verbatim, where in the second part one takes only elements of D and properly applies Proposition 13.13 combined with Proposition 13.17 (1).

This concludes the proof of Lie–Palais theorem under the weaker assumptions of Theorem 13.16.

Just as in Corollary 13.15, one can start from a finite dimensional Lie algebra of vector fields on M. Regarding Lie algebras of vector fields, the following completeness result for vector fields is an immediate consequence of Theorem 13.16.

Corollary 13.18 *Let D be a family of complete vector fields on a manifold M, and assume that the Lie algebra $\mathfrak{g}$ generated by D is finite dimensional. Then every vector field on $\mathfrak{g}$ is complete. Hence all iterated Lie brackets of vector fields on D are complete.*

In this corollary, the assumption that $\mathfrak{g}$ is finite dimensional is essential, since in general the brackets of complete vector fields may not be complete. An example of this situation is presented below. Recall that incomplete trajectories of a vector field X on a manifold M are unbounded, in the sense that if $\mathbf{x} : (\alpha, \omega) \to M$ is a maximal solution of $\dot{x} = X(x)$ in which, say $\omega < \infty$, then $\lim_{t \to \omega} \mathbf{x}(t) = \infty$. That is to say, $\mathbf{x}(t)$ eventually exits any compact subset of M.

Example On the real line $\mathbb{R}$ the vector fields

$$X = \cos x \frac{d}{dx} \qquad \text{and} \qquad Y = x^n \sin x \frac{d}{dx}$$

with $n \geq 2$ are complete, since their singular sets are bounded from above and below. Hence the trajectory of either vector field is bounded, implying completeness. The Lie bracket

$$\begin{aligned}\left[\cos x \frac{d}{dx}, x^n \sin x \frac{d}{dx}\right] &= x^n \left[\cos x \frac{d}{dx}, \sin x \frac{d}{dx}\right] + n x^{n-1} \cos x \sin x \frac{d}{dx} \\ &= x^n \frac{d}{dx} + \frac{n}{2} x^{n-1} \sin 2x \frac{d}{dx} \\ &= x^n \left(1 + \frac{n}{2x} \sin 2x\right) \frac{d}{dx}\end{aligned}$$

is an incomplete vector field. To see this, observe first that

$$1 + \frac{n}{2x} \sin 2x > 1 - \frac{n}{2x} > 1 - \frac{1}{2} = \frac{1}{2}$$

when $x > n$, so in this range $a(x) = x^n \left(1 + \frac{n}{2x} \sin 2x\right) > \frac{x^n}{2}$. This implies that the set of singularities of $[X, Y]$ is bounded above. Call m be the maximum among the singularities. The solutions of $\frac{dx}{dt} = a(x)$ in the unbounded interval $(m, +\infty)$ are found by rewriting the equation as $\frac{dx}{a(x)} = dt$ and integrating both sides. A primitive $g(x)$ of $1/a(x)$ is strictly increasing on $(m, +\infty)$. Hence there is an interval $J = (\alpha, \omega)$ such that $\phi : (m, +\infty) \to J$ is a diffeomorphism, where $\omega = \lim_{x \to +\infty} \phi(x)$. The solution with initial condition $x(0) = x_0$ is given by

$$x(t) = g^{-1}(g(x_0) + t) \qquad t \in -g(x_0) + J.$$

Such a solution tends to $+\infty$ in finite time if J is bounded from above ($\omega < +\infty$). At present,

$$\frac{1}{x^n \left(1 + \frac{n}{x} \cos^2 x\right)} < \frac{2}{x^n}$$

if $x > n$, and hence

$$\int_n^{+\infty} \frac{dx}{x^n \left(1 + \frac{n}{x} \cos^2 x\right)} < \int_n^{+\infty} \frac{2dx}{x^n} < +\infty$$

since $n \geq 2$. This implies $\omega < +\infty$, and hence $[X, Y] = x^n \left(1 + \frac{n}{x} \cos^2 x\right) \frac{d}{dx}$ is not complete. □

13.3 Bundles

This section discusses the notions of a principal bundle and the bundles associated to it. These concepts crop up naturally when considering maps between different homogeneous spaces.

13.3.1 Principal Bundles

A **principal bundle** $P(M, G)$ (often denoted simply by $P \to M$) consists of two topological spaces, the total space P and the base M, plus the structure group G. These objects are related as follows:

1. The group G acts freely on P on the right by the action $D : (p, g) \mapsto pa$, $p \in P$, $g \in G$. (That is, if $pg = p$ for some p then $g = 1$.)
2. The orbit space of the action is M. This means that there exists a surjective map

$$\pi : P \longrightarrow M,$$

whose preimages $\pi^{-1}\{x\}$, $x \in M$, are the G-orbits.
3. P is locally trivial, meaning that for every $x \in M$ there exist a neighborhood U of x and a bijective map

$$\psi : \pi^{-1}(U) \longrightarrow U \times G,$$

called local trivialization, of the form

$$\psi(p) = (\pi(p), \phi(p)),$$

where $\phi : \pi^{-1}(U) \to G$ is a map that satisfies

$$\phi(pg) = \phi(p)\, g \tag{13.4}$$

for all $p \in \pi^{-1}(U)$ and $g \in G$.

The bundle $P \to M$ is said to be a topological bundle if the maps involved in the definition are continuous (and homeomorphisms when bijective). The principal bundle is of class C^k, $k \geq 1$, if the spaces are C^k-manifolds (in particular G must be a Lie group) and the maps involved C^k (and diffeomorphisms if bijections). In this case, the projection $\pi : P \to M$ becomes a submersion because, under the local trivialization ψ, it can be identified with the projection on the first coordinate $U \times G \to U$.

The fibers of the principal bundle are denoted by $P_x = \pi^{-1}\{x\}$, $x \in M$, or $P_p = \pi^{-1}\{\pi(p)\}$, $p \in P$.

Examples

1. The product $M \times G$ is a principal bundle with structure group G and right action $D_h(x, g) = (x, g)\, h = (x, gh)$. In particular, a group G can be understood as a principal bundle where the base reduces to a point $M = \{x\}$. This product is called **trivial bundle**.
2. Let M be a differentiable manifold with tangent bundle TM. The **frame bundle** is the set BM of all bases of TM. An element $p \in BM$ is then a basis

$$\{f_1, \ldots, f_n\} \tag{13.5}$$

of some tangent space T_xM, $x \in M$. Equivalently, $p \in BM$ can be seen as an invertible linear map (a frame) $p : \mathbb{R}^n \to T_xM$, $x \in M$. Given the map p, the set

$$\{p(e_1), \ldots, p(e_n)\},$$

where $\{e_1, \dots e_n\}$ is the canonical basis of $\mathbb{R}^n$, a basis of $T_x M$. Vice versa, the basis (13.5) determines $p : \mathbb{R}^n \to T_x M$ by

$$p\,(x_1, \dots, x_n) = x_1 f_1 + \cdots + x_n f_n\,.$$

The projection $BM \to M$ associates to $p : \mathbb{R}^n \to T_x M$ the point $x \in M$ in such a way that the fiber BM_x is the set of frames of $T_x M$.
The group $\mathrm{Gl}\,(n, \mathbb{R})$ acts on BM on the right by

$$(p, g) \mapsto pg = p \circ g,$$

with $p \in BM$ and $g \in \mathrm{Gl}\,(n, \mathbb{R})$. The action is free since the elements of BM are invertible linear maps ($p \circ g = p$ if and only if $g = 1$). The action is also transitive on the fibers, because given a linear map $p : \mathbb{R}^n \to T_x M$, the other maps are of the form $q = p \circ g$ for some $g \in \mathrm{Gl}\,(n, \mathbb{R})$. This construction defines BM as a principal bundle with structure group $\mathrm{Gl}\,(n, \mathbb{R})$ and base M.
The condition expressing local triviality is obtained by taking charts of M. These give, for every $x \in M$, a neighborhood U and vector fields $X_1, \dots, X_n$ on U (the coordinate vector fields) that are linearly independent at every point of U. These vector fields define sections of BM, which trivialize it locally. (See the discussion below, which relates local trivializations to sections.)

3. Denote by $\mathrm{B}_k\,(n)$ the set of injective linear maps

$$p : \mathbb{R}^k \longrightarrow \mathbb{R}^n.$$

(The elements of $\mathrm{B}_k\,(n)$ are identified with sets of k linearly independent elements of $\mathbb{R}^n$, or to real $n \times k$ matrices of rank k.)
The group $\mathrm{GL}\,(k, \mathbb{R})$ acts on $\mathrm{B}_k\,(n)$ by right multiplication of matrices. The action is free since the elements of $\mathrm{B}_k\,(n)$ are 1-1 linear maps.
The set of orbits for this right action is the Grassmannian $\mathrm{Gr}_k\,(n)$ of k-dimensional subspaces of $\mathbb{R}^n$. In fact, the images of linear maps on $\mathrm{B}_k\,(n)$ are subspaces of dimension k in $\mathbb{R}^n$. This gives a map

$$p \in \mathrm{B}_k\,(n) \longmapsto \mathrm{im}\,p \in \mathrm{Gr}_k\,(n)\,,$$

whose fibers coincide with the $\mathrm{Gl}\,(k, \mathbb{R})$-orbits. This is because the images of two elements p and $q = pa$ in the same orbit coincide. Conversely, if the images of p, q coincide, then it makes sense to write $p^{-1}q$, where p^{-1} denotes the inverse of p as a map from $\mathbb{R}^k$ onto its image. Then $p^{-1}q \in \mathrm{Gl}\,(k, \mathbb{R})$ and, as $q = p\left(p^{-1}q\right)$, two elements with the same image belong in the same $\mathrm{Gl}\,(k, \mathbb{R})$-orbit.
This construction defines the principal bundle $\mathrm{B}_k\,(n)\,(\mathrm{Gr}_k\,(n)\,, \mathrm{Gl}\,(k, \mathbb{R}))$ with structure group $\mathrm{Gl}\,(k, \mathbb{R})$.
That this bundle is locally trivial can be seen either directly, by constructing local sections, or indirectly, by looking at it as an associated bundle to the bundle

$\mathrm{Gl}(n,\mathbb{R})$ $(\mathrm{Gr}_k(n), P)$ obtained from the transitive action of $\mathrm{Gl}(n,\mathbb{R})$ on $\mathrm{Gr}_k(n)$ (see Section 13.4).
When $k = 1$, the Grassmannian is the projective space $\mathbb{P}^{n-1}$. Then $\mathrm{B}_k(n)$ becomes $\mathbb{R}^n \setminus \{0\}$ and the projection $\mathbb{R}^n \setminus \{0\} \longrightarrow \mathbb{P}^{n-1}$ associates $v \in \mathbb{R}^n \setminus \{0\}$ to the subspace spanned by v.

4. Here is a modification of the previous example. Instead of all bases of a k-subspace, consider only orthonormal bases (with respect to a fixed inner product on $\mathbb{R}^n$). This gives the **Stiefel manifold** $\mathrm{St}_k(n)$, the space of orthonormal sets in $\mathbb{R}^n$ with k elements. Equivalently, $p \in \mathrm{St}_k(n)$ is a linear isometry

$$p : \mathbb{R}^k \longrightarrow \mathbb{R}^n$$

for the standard inner products of $\mathbb{R}^k$ and $\mathbb{R}^n$. Alternatively, $p \in \mathrm{St}_k(n)$ may be realized as an $n \times k$ matrix. The condition of being an isometry becomes

$$p^T p = 1,$$

where p^T is the transpose and 1 the $k \times k$ identity matrix. The projection

$$\mathrm{St}_k(n) \longrightarrow \mathrm{Gr}_k(n)$$

given by the image of an element defines a principal bundle with structure group $\mathrm{O}(k)$.
In case $k = 1$, $\mathrm{St}_k(n)$ is the sphere S^{n-1} and the projection

$$S^{n-1} \longrightarrow \mathbb{P}^{n-1}$$

is the antipodal identification on the sphere.

5. Let M be a manifold and $\widetilde{M}$ its universal covering. The canonical covering map $\widetilde{M} \to M$ defines a principal bundle, whose structure group is the fundamental group of M. □

A **morphism** between two principal bundles $P(M,G)$ and $Q(N,H)$ is a map $\phi : P \to Q$ such that there exists a homomorphism $\theta : G \to H$ satisfying $\phi(pa) = \phi(p)\theta(a)$, for $p \in Q$ and $a \in G$. This condition for ϕ ensures that the image of a fiber of P is contained in a fiber of Q. Therefore ϕ induces a map $f : M \to N$ between the bundle bases, given by $f(x) = \pi(\phi(p))$ for any $p \in P_x$, where $\pi : Q \to N$ is the projection of Q. The bundles P and Q are isomorphic if θ is an isomorphism and ϕ bijective. In that case, $\phi^{-1} : Q \to P$ together with θ^{-1} defines a morphism between $Q(N,H)$ and $P(M,G)$. In the particular case where $G = H$ and $\theta = \mathrm{id}$, the morphism is called an endomorphism (and automorphism if invertible). If ϕ and θ are injective, then by definition the image of ϕ is a **principal subbundle** of Q. Now, if $M = N$, $G \subset H$ and $\theta : G \hookrightarrow H$ is the inclusion, then P is called a G-**reduction** of Q.

The condition of local triviality in the definition of a principal bundle $P \to M$ means that P is in practice a well-organized collection of groups (or Lie groups, in the differentiable case).

Local triviality can be looked at from another point of view. This is akin to what happens in the theory of manifolds, where many properties are derived from the features of transition functions (for instance, a manifold M is of class C^k if $\alpha_1 \alpha_2^{-1}$ is C^k for any pair of charts α_1 and α_2). Along this line of thought, a principal bundle may be defined as a manifold where transition functions belong in a certain class. This idea is more or less implicit in the discussion to follow.

Let $\psi : \pi^{-1}(U) \longrightarrow U \times G$ be a local trivialization as from the definition, and call $\phi : \pi^{-1}(U) \to G$ the second component of ψ. The set

$$\psi^{-1}\{(x, 1) : x \in M\}$$

is contained in $\pi^{-1}(U)$, and since the first component of ψ is the projection on M, it also meets each fiber $\pi^{-1}\{x\}$, $x \in U$, at a single point. Call this point $\sigma(x)$. Thus $\sigma : U \to P$ is a **local section** of P, i.e. it satisfies $\pi(\sigma(x)) = x$. Moreover, $\phi(\sigma(x)) = 1$ by definition of ϕ, and because of (13.4) ϕ is determined by σ by

$$\phi(\sigma(x)\, a) = \phi(\sigma(x))\, a = a$$

for any $a \in G$. Hence $\sigma(x)\, a$ covers the whole fiber over x as a runs through G. Since ψ is completely determined by ϕ, it is also determined by σ. Explicitly,

$$\psi(\sigma(x)\, a) = (x, \phi(\sigma(x)\, a)) = (x, a), \tag{13.6}$$

which shows that the existence of the local section implies the existence of the trivialization.

Let $\psi_1 : \pi^{-1}(U_1) \to U_1 \times G$ and $\psi_2 : \pi^{-1}(U_2) \to U_2 \times G$ be two local trivializations such that $U_1 \cap U_2 \neq \emptyset$. Call ϕ_1, ϕ_2 their second components and σ_1, σ_2 the corresponding sections.

Since $\sigma_1(x)$ and $\sigma_2(x)$ belong to the same fiber, for each $x \in U_1 \cap U_2$ there exists $\theta(x) \in G$ such that

$$\sigma_2(x) = \sigma_1(x)\, \theta(x).$$

This sets up a function $\theta : U_1 \cap U_2 \to G$ describing the the coordinate change $\psi_1 \psi_2^{-1}$. In fact, by (13.6),

$$\psi_2^{-1}(x, a) = \sigma_2(x)\, a,$$

and so

$$\psi_1 \psi_2^{-1}(x, a) = \psi_1(\sigma_2(x)\, a) = \psi_1(\sigma_1(x)\, \theta(x)\, a) = (x, \theta(x)\, a).$$

In words, the coordinate change is the left multiplication by $\theta(x)$. For this reason, θ is called **transition function** between the trivializations ψ_1 and ψ_2 (in this order).

The transition function provides the change of coordinates between two trivializations, but not the trivializations themselves. Starting from transition functions, one can reconstruct the whole bundle, with the recipe outlined below.

Let ψ_3 be a third trivialization with domain U_3 that intersects $U_1 \cap U_2$. Indicate by θ_{ij} the transition function from ψ_i to ψ_j. Then

- $\psi_1 \psi_2^{-1}(x, a) = (x, \theta_{12}(x)\, a)$;
- $\psi_2 \psi_3^{-1}(x, a) = (x, \theta_{23}(x)\, a)$;
- $\psi_3 \psi_1^{-1}(x, a) = (x, \theta_{31}(x)\, a)$.

Composing the first two lines above produces the third:

$$\psi_1 \psi_2^{-1} \psi_2 \psi_3^{-1} = \psi_1 \psi_2^{-1}(x, \theta_{23}(x)\, a) = (x, \theta_{12}\theta_{23}(x)\, a)\,,$$

and comparing which to ψ_3 yields

$$\theta_{31}(x) = \theta_{12}(x)\, \theta_{23}(x)\,. \tag{13.7}$$

Altogether, the maps θ_{ij} that define the transition functions of a given bundle obey condition (13.7), called the cocycle property. The converse statement shall not proven here, but goes like this.[2]

Theorem 13.19 *Let M be a manifold and G a Lie group. Suppose there exists a family of maps $\theta : U \to G$, as U varies among open subsets in M, whose domains cover M and such that every triple whose domains intersect satisfies* (13.7)*. Then there exists a unique (up to isomorphism) principal bundle P with structure group G whose trivializations have G-valued transition functions.*

13.3.2 Associated Bundles

The ingredients that enter the definition of an associated bundle are a principal bundle $\pi : P \to M$ and a left action of the structure group G on a space F.

The group G acts on the right on the product $P \times F$ by $g(p, v) = \left(pg, g^{-1}v\right)$, for $g \in G$ and $(p, v) \in P \times F$. This action determines an equivalence relation on $P \times F$, whereby $(p, v) \sim (q, w)$ if and only if there is an element $g \in G$ such that $q = pg$ and $w = g^{-1}v$. The equivalence class of the pair $(p, v) \in P \times F$ is denoted by $p \cdot v$ or $[p, v]$.

[2] See e.g. Kobayashi–Nomizu [34, Proposition I.5.2] in the differentiable context, or Husemoller [29, Theorem 5.3.2] for topological bundles.

The set E of equivalence classes of $\sim$ is called **associated bundle** of P with **typical fiber** F and base M. The associated bundle is denoted by $E = P \times_G F$. The following observations justify the terminology employed:

1. If $(p, v) \sim (q, w)$, then p and q are on the same fiber of P. Therefore, the map $\pi_E : P \times_G F \to M$ defined by $\pi_E (p \cdot v) = \pi (p)$ is well defined, which makes $E = P \times_G F$ a bundle over M.
   The fibers of $E \to M$ are denoted by $E_x = \pi^{-1}\{x\}$, $x \in M$, or $E_\xi = \pi_E^{-1}\{\pi_E (\xi)\}$, $\xi = p \cdot v \in E$.
2. Given $p \in P$ the pairs (p, v) and (p, w) are equivalent if and only if $v = w$. In fact, $(p, v) \sim (p, w)$ if there exists $a \in G$ such that $p = pa$ and $w = a^{-1}v$. Since the G-action on P is free, it follows that $a = 1$ and hence $w = v$. In other words, for a given $p \in P$ an equivalence class $p \cdot v \in P \times_G F$ is determined by a unique $v \in F$.
3. Each $p \in P$ gives rise to a bijection

$$v \in F \longmapsto p \cdot v \in E_x \qquad x = \pi (p) . \tag{13.8}$$

   By the previous item, in fact, this map is 1-1. On the other hand, an element of E_x has the form $q \cdot w$ with $q \in P_p$. Then $q = pa$, for $a \in G$, which implies $q \cdot w = pa \cdot w = paa^{-1} \cdot aw = p \cdot aw$ has the form $p \cdot v$, showing that (13.8) is onto as well.
   It is customary to use the same letter p to indicate this bijection, which justifies the notation $p \cdot v$ for the coset of (p, v).

The idea behind (13.8) is that the bundle P in effect parametrizes the collection of fibers of the associated bundle $E \to M$. More precisely, each element $p \in P$ parametrizes the fiber E_x, where $x = \pi (p)$, by the typical fiber F. Two elements p and q in the same fiber of Q give different parametrizations, obtained from each other by letting G act on F. In fact, if $q = pa$ for some $a \in G$, then $q \cdot v = pa \cdot v = p \cdot av$. Therefore the bijection defined by q is obtained from the bijection defined by p by composing with the right action of $a \in G$.

Associated bundles admit local trivializations inherited from the trivializations of the principal bundle. In fact, let $\chi : U \to P$ be a local section of P. The map $\psi_\chi : U \times F \to \pi_E^{-1} (U)$ defined by $(x, v) \mapsto \chi (x) \cdot v$ is a bijection that trivializes the associated bundle over U. If χ_1 is another local section, then $\chi_1 (x) = \chi (x) a (x)$ for some $a (x) \in G$ if both χ, χ_1 are defined at $x \in M$. Hence $\chi_1 (x) \cdot v = \chi (x) \cdot av$, and if ψ_{χ_1} is the trivialization corresponding to χ_1, then ψ_{χ_1} and ψ_χ are related by

$$\psi_\chi^{-1} \circ \psi_{\chi_1} (x, v) = (x, av) . \tag{13.9}$$

This map permutes the fibers, and the maps between fibers come from the action of G.

In the context of differentiable bundles, it is not difficult to construct a differentiable structure on an associated bundle, using these local trivializations:

Proposition 13.20 *Let $\pi : P \to M$ be a differentiable principal bundle and suppose the action of G on F is differentiable. Then $P \times_G F$ is a differentiable manifold such that the projection $\pi_E : P \times_G F \to M$ is a submersion. Moreover, the fibers E_x are closed and embedded submanifolds, and the parametrizations $v \in F \mapsto p \cdot v \in E_x$, for $x = \pi(p)$, are diffeomorphisms.*

Proof Take local sections $\chi : U \to P$, so the trivializations described above change under the differentiable maps $(x, v) \mapsto (x, a(x)\, v)$. Therefore these trivializations provide a differentiable atlas for $P \times_G F$.

The projection π_E is a submersion because in a local chart of $\pi^{-1}(U) \approx U \times F$ it identifies with the projection on the first component. Hence the fibers are closed embedded submanifolds. Finally, taking local charts of E as products of type $U \times F$, it can be shown that the parametrizations by elements of P are diffeomorphisms.

□

Examples

1. Consider a differentiable manifold M with $\dim M = n$. The frame bundle BM was constructed earlier as the collection of frames of the tangent bundle TM, and its structure group is $\mathrm{Gl}(n, \mathbb{R})$. Conversely, TM can be recovered from BM by identifying it as the associated bundle $BM \times_{\mathrm{Gl}(n,\mathbb{R})} \mathbb{R}^n$ coming from the standard linear action of $\mathrm{Gl}(n, \mathbb{R})$ on $\mathbb{R}^n$. In fact, there is an almost tautological bijection between TM and $BM \times_{\mathrm{Gl}(n,\mathbb{R})} \mathbb{R}^n$, defined by associating to the coset of $(p, v) \in BM \times \mathbb{R}^n$ the tangent vector $p(v) \in T_x M$, $x = \pi(p)$ (where $p : \mathbb{R}^n \to T_x M$ comes from the definition of BM). This map is well defined by the fact that the associated bundle was constructed from the standard $\mathrm{Gl}(n, \mathbb{R})$-action on $\mathbb{R}^n$. In fact, if (p, v) and $(q, w) = \left(pa, a^{-1}v\right)$ belong to the same equivalence class, then $q(w) = pa\left(a^{-1}v\right) = pv$.
2. The previous construction of TM generalizes to vector bundles. Let $P(M, G)$ be a principal bundle and $\rho : G \to \mathrm{Gl}(V)$ a representation of G on a vector space V. Then G acts on the left on V. The ensuing associated bundle $E = P \times_\rho V$ is a **vector bundle**, because it is given by a map $\pi : E \to M$ satisfying two properties: (i) each fiber has the structure of a vector space (coming from the bijections $v \mapsto p \cdot v$, $p \in P$); (ii) there are local trivializations $U \times V \to \pi^{-1}(U)$, which are transformed into one another by linear maps between the fibers, as follows from formula (13.9).
   If $\dim V < \infty$ and P is a differentiable bundle, then $P \times_\rho V$ is a differentiable manifold. The above construction also works for much more general representations than finite dimensional ones.
   Any vector bundle $E \to M$ (satisfying the three conditions above) can be constructed as an associated bundle. This is done by defining the frame bundle BE of $E \to M$ in analogy to BM, using linear isomorphisms $p : \mathbb{R}^k \to E_x$, $k = \dim E_x$. The vector bundle $E \to M$ is then recovered as the associated bundle of BE.
3. The tensor bundles of a differentiable manifold M can be obtained as associated bundles of BM. For instance, the cotangent bundle T^*M is the associated bundle

$BM \times_{\rho^*} (\mathbb{R}^n)^*$ obtained via the dual standard representation ρ^*, whereby if $g \in \mathrm{Gl}(n, \mathbb{R})$ and $\alpha \in (\mathbb{R}^n)^*$ is a linear functional, then $\rho^*(g)(\alpha) = \alpha \circ g^{-1}$. □

Two particular instances of associated bundles deserve special attention. They will be presented in the following two propositions.

Proposition 13.21 *Let $P(M, G)$ be a principal bundle and G/H a homogeneous space of G. The subgroup H acts on the left on P, and denote by P/H the set of orbits. Then P/H is identified with the associated bundle $P \times_G G/H$ coming from the canonical action of G on G/H.*

Proof Denote by $x_0 = 1H$ the origin of G/H. An element of P/H is a right orbit pH, for some $p \in P$. Define a map sending $pH \in P/H$ to the coset $p \cdot x_0 \in P \times_G G/H$. This function is

1. well defined, for if $q = ph \in pH$ then $q \cdot x_0 = ph \cdot x_0 = p \cdot hx_0 = p \cdot x_0$;
2. one-to-one, since $q \cdot x_0 = p \cdot x_0$ implies $q = pg$ and $x_0 = g^{-1}x_0$. The latter means $g^{-1} \in H$, so $g \in H$. From the former, it follows that $qH = pH$;
3. onto, because if $q \cdot x \in P \times_G G/H$, there exists $g \in G$ such that $g^{-1}x = x_0$. This implies $p \cdot x_0 = q \cdot x$ if $p = qg^{-1}$, showing that $q \cdot x$ belongs to the range of the map.

To sum up, $pH \mapsto p \cdot x_0$ is a bijection, identifying P/H with $P \times_G G/H$. □

The identification given in Proposition 13.21 has in local coordinates a very simple expression. If $U \times G \approx \pi^{-1}(U)$ is a local trivialization of P, the right action of H on $U \times G$ is defined, and given by $(z, g)h \mapsto (z, gh)$, for $z \in U$, $g \in G$, and $h \in H$ (see 13.4)). The set of orbits in $\pi^{-1}(U)$ then identifies with $U \times G/H$. On the other hand, the elements of $\pi_E^{-1}(U)$ can be written as $(z, 1) \cdot x$ with $z \in U$ and $x \in G/H$ (since $(z, 1) \in U \times G$ is identified with an element of $\pi^{-1}(U)$). In the trivial bundle $U \times G$, the bijection between P/H and $P \times_G G/H$ reads

$$(z, gH) \in (U \times G)/H \longmapsto (z, 1) \cdot gH \in (U \times G) \times_G G/H.$$

Through this local description, it follows immediately that the identification $P/H \approx P \times_G G/H$ is a diffeomorphism when all bundles are differentiable.

Proposition 13.22 *Consider a principal bundle $P(M, G)$ and a homomorphism $P(M, G)$ of (Lie) groups. Suppose G acts on the left on H by $(g, h) \mapsto \phi(g)h$ and denote by $P \times_\phi H$ the associated bundle of this action. Then $P \times_\phi H$ is a principal bundle with structure group H.*

Proof Define the right action of H on $P \times_\phi H$ by $(p \cdot h)h_1 = p \cdot (hh_1)$. This is free if $p \cdot (hh_1) = p \cdot h$ then $p = pg$ for some $g \in G$ and $hh_1 = \phi(g)^{-1}h$. The first fact implies $g = 1$. Substituting this in the second relation gives $hh_1 = h$, and so $h_1 = 1$. Furthermore, the action is transitive on the fibers, because $p : h \in H \mapsto p \cdot h$ is a bijection between H and the fiber. To conclude that this right action defines

$P \times_\phi H \to M$ as a principal bundle it suffices to verify the local trivialization condition. But the latter follows from the trivializations of the associated bundles, in general. □

A particular case of the proposition above is when $G = H$ and $\phi = \mathrm{id}$. Then $P \times_{\mathrm{id}} G$ is isomorphic to P under the mapping that assigns the element $pg \in P$ to the coset $p \cdot g \in P \times_{\mathrm{id}} G$. Said otherwise, P can be seen as an associated bundle of itself.

A **section** of an associated bundle $\pi : E = P \times_G F \to M$ is a map $\sigma : M \to E$ such that $\pi \circ \sigma = \mathrm{id}$. Sections can be defined by F-valued equivariant maps defined on the total space P. In fact, given a section σ, let $f_\sigma : P \to F$ be

$$f_\sigma(p) = p^{-1}(\sigma(\pi(p))),$$

where $p^{-1} : E_{\pi(p)} \to F$ is the inverse of the bijection defined by $p \in P$ between the typical fiber F and the fiber $E_{\pi(p)}$ of E over $\pi(p)$. This map is equivariant, i.e. it satisfies

$$f_\sigma(pa) = a^{-1} \cdot f(p), \tag{13.10}$$

as $(pa)^{-1} = a^{-1} \circ p^{-1}$.

Conversely, let $f : P \to F$ be equivariant and define $\widetilde{f} : P \to P \times_G F$ by $\widetilde{f}(p) = p \cdot f(p)$. If $a \in G$, then

$$\widetilde{f}(pa) = pa \cdot f(pa) = pa \cdot a^{-1} f(p),$$

because f is equivariant. Hence $\widetilde{f}(pa) = \widetilde{f}(p)$, that is, $\widetilde{f}$ is constant on the fibers of P. This allows to define the map $\sigma_f : M \to P \times_G F$ by

$$\sigma_f(x) = p \cdot f(p)$$

for any $p \in P_x$, which is a section since $p \cdot f(p)$ lives in the fiber over x.

Summarizing, there exists a 1-1 correspondence between sections of the associated bundle $P \times_G F \to M$ and equivariant maps $P \to F$. The bijection maps $\sigma \mapsto f_\sigma$, and its inverse is $f \mapsto \sigma_f$ since, by the given definitions, $\sigma_{f_\sigma} = \sigma$ and $f_{\sigma_f} = f$.

The function f_σ is called the **equivariant function** associated to the section σ.

In a local trivialization, the correspondence $\sigma \mapsto f_\sigma$ is described as follows. Let $\chi : U \to P$ be a local section over $U \subset M$. Then $(x, g) \in U \times G \mapsto \chi(x)\, g \in P$ is a local trivialization of P over U, and the associated bundle $P \times_G F$ admits local trivialization

$$(x, v) \in U \times F \mapsto \chi(x) \cdot v \in P \times_G F.$$

If $\sigma : M \to P \times_G F$ is a section, its restriction to U is given, via the identification of the bundle with $U \times F$, by $\sigma(x) = (x, \tau(x))$, with $\tau : U \to F$. That is, the local trivialization associates $\sigma(x) = (x, \tau(x))$ to $\chi(x) \cdot \tau(x) \in P \times_G F$, so $f_\sigma(\chi(x)) = \tau(x)$. As f_σ is equivariant, the local form of f_σ is

$$f_\sigma(x, g) = f_\sigma(\chi(x)\, g) = g^{-1} f_\sigma(\chi(x)) = g^{-1}\tau(x). \tag{13.11}$$

From this expression, it follows immediately that on differentiable bundles f_σ is differentiable if and only if τ is differentiable. Therefore $\sigma \mapsto f_\sigma$ preserves differentiability.

Proposition 13.23 *Let $P \to M$ and $P \times_G F \to M$ be differentiable bundles. Then a section $\sigma : M \to P \times_G F$ is differentiable if and only if its equivariant function $f_\sigma : P \to F$ is differentiable.*

In general, an associated bundle may not admit sections. For example, a principal bundle $P \to M$, viewed as associated bundle of itself, only admits sections if it is globally trivial. The following proposition relates the existence of sections in $P \times_G G/H$ to H-reductions of P.

Proposition 13.24 *Let P be a principal bundle with structure group G, $H \subset G$ a subgroup, σ a section of $P \times_G G/H$. Then $f_\sigma^{-1}\{o\}$ is right invariant under H, where $o = 1H$ is the origin of G/H. If, moreover, P and σ are differentiable and H is closed, then $f_\sigma^{-1}\{o\}$ is a differentiable H-reduction of P.*

Proof Take $h \in H$. As f_σ is equivariant, $f_\sigma(ph) = h^{-1} f_\sigma(p)$. Hence, if $p \in f_\sigma^{-1}\{o\}$, then $f_\sigma(ph) = h^{-1}(o) = o$, showing that $f_\sigma^{-1}\{o\}$ is invariant under H. To prove the differentiable case, pick a local trivialization $(x, g) \in U \times G \mapsto \chi(x)\, g \in P$ given by a section $\chi : U \subset M \to P$. By (13.11), f_σ is given on $U \times G$ by

$$f_\sigma(x, g) = g^{-1}\tau(x)$$

if $\sigma(x) = (x, \tau(x))$. Hence $f_\sigma^{-1}\{o\} = \{(x, g) : \tau(x) = g(o)\}$. Fix $x_0 \in U$ and take a differentiable section $\xi : V \subset G/H \to G$ such that $V \subset G/H$ is open and $\tau(x_0) \in V$ (for the existence of this section, see Proposition 13.25 below). This section satisfies $p \circ \xi = \mathrm{id}$, where $p : G \to G/H$ is the canonical projection. This means that $\xi(y)(o) = y$ for all $y \in V$. Therefore $\xi(\tau(x))(o) = \tau(x)$, which implies $(x, \xi(\tau(x))) \in f_\sigma^{-1}\{o\}$. That is to say, $\xi \circ \tau$ is a differentiable local section defined on the open set $\tau^{-1}(V)$ that contains x_0. Since x_0 is arbitrary, $f_\sigma^{-1}\{o\}$ is indeed a differentiable bundle. □

13.4 Homogeneous Spaces and Bundles

Let G be a group. A subgroup $H \subset G$ acts on the right on G freely, and the orbits are the cosets gH and the orbit space is G/H. When G is a Lie group and H a

closed subgroup, by virtue of the earlier construction of the differentiable structure on G/H, one can prove that the canonical projection $\pi : G \to G/H$ defines a differentiable principal bundle.

Proposition 13.25 *Let G be a Lie group and $H \subset G$ a closed subgroup. Then $G \to G/H$ is a principal bundle with structure group H.*

Proof There remains to verify local triviality. To this end, recall the notations of Theorem 6.22. Charts on G/H were constructed by restricting π to sets of the form ge^V, where V is a neighborhood of 0 in the Lie algebra $\mathfrak{g}$ of G. If V_g denotes the image of such chart, the elements of V_g have the form lH with $l = ge^Y$, $Y \in V$. Hence the map $lH \mapsto ge^Y$ is a differentiable section of $G \to G/H$, concluding the proof. □

Let G be a group and $H_1 \subset H_2$ two subgroups of G. There is a natural surjective map $G/H_1 \to G/H_2$ associating to the coset gH_1 the coset gH_2 (the latter is the unique coset containing gH_1). This map is in fact the projection of an associated bundle, as the following construction shows.

Proposition 13.26 *Let G be a Lie group and $H_1 \subset H_2$ closed subgroups of G. Then G/H_1 is a bundle over G/H_2 with canonical projection $G/H_1 \to G/H_2$ given by $gH_1 \mapsto gH_2$. If, furthermore, H_1 is normal in H_2, then $G/H_1 \to G/H_2$ is a principal bundle with structure group H_2/H_1.*

Proof By Proposition 13.21, the associated bundle $G \times_{H_2} H_2/H_1$ is identified with the quotient of the right action of H_1 on G, i.e. G/H_1. The same proposition implies that the image of $gH_1 \in G/H_1$ under $\pi : G/H_1 \to G/H_2$ coincides with the projection of g on G/H_2, that is, $\pi(gH_1) = gH_2$. Finally, suppose H_1 is normal in H_2. Then the action of H_2 on H_2/H_1 comes from the canonical homomorphism $H_2 \to H_2/H_1$, and the last claim descends from Proposition 13.22. □

13.5 Exercises

1. Use formula $g_*\widetilde{X} = \widetilde{(\mathrm{Ad}(g)\, X)}$ to show, directly from the definition of Lie bracket, that the map $X \mapsto \widetilde{X}$ is a homomorphism of Lie algebras, i.e. $\widetilde{[X, Y]} = [\widetilde{X}, \widetilde{Y}]$.
2. Let G be a connected Lie group, H a closed subgroup, and K a compact subgroup. Suppose that $\dim K - \dim(K \cap H) = \dim G/H$. Show that K acts transitively on G/H.
3. Given a Lie group G and two subgroups $H, L \subset G$ with H closed, show that L has an open orbit in G/H if and only if there exists $g \in G$ such that $\mathfrak{g} = \mathfrak{h} + \mathrm{Ad}(g)\, \mathfrak{l}$, where $\mathfrak{g}$, $\mathfrak{h}$, and $\mathfrak{l}$ are the Lie algebras of G, H, and L, respectively.
4. Let G be a connected Lie group and $H, K \subset G$ two subgroups such that H is closed and K is compact. Denote by $\mathfrak{g}, \mathfrak{h}, \mathfrak{k}$ the respective Lie algebras. Show

that $G = HK$ if $\mathfrak{g} = \mathfrak{h} + \mathfrak{k}$. Do the same assuming that $\mathfrak{g} = \mathfrak{h} + \mathrm{Ad}(g)\,\mathfrak{k}$ for some $g \in G$.

5. Retaining the notations of the previous exercise, suppose $H \cap K = \{1\}$ and $\mathfrak{g} = \mathfrak{h} \oplus \mathrm{Ad}(g)\,\mathfrak{k}$ for all $g \in G$. Show that the map $(h, k) \in H \times K \mapsto hk \in G$ is a diffeomorphism.
6. Use the previous exercise to prove that $\mathrm{Sl}(n, \mathbb{R}) = T\,\mathrm{SO}(n) = \mathrm{SO}(n)\,T$, where T is the subgroup of upper triangular matrices with positive diagonal entries. Interpret decomposition $\mathrm{Sl}(n, \mathbb{R}) = \mathrm{SO}(n)\,T$ by applying the Gram–Schmidt orthonormalization process to the columns of a matrix.
7. Let G be a connected Lie group and $H \subset G$ a closed subgroup. Show that if G/H is simply connected then H is connected.
8. Let G be a connected and noncompact semi-simple group with Iwasawa decomposition $G = KAN$. Show that the action of K on G/AN is transitive.
9. Prove that the distribution $\Delta_{\mathfrak{g}}$, defined by a given G-action, is integrable, by showing that the map $\xi_x : G/H_x \to M$, $\xi_x(gH_x) = gx$, is an immersion and defines an integral manifold of $\Delta_{\mathfrak{g}}$ passing through x.
10. A flag in $\mathbb{R}^n$ is a family of subspaces $f = (V_1 \subset \cdots \subset V_k)$ of $\mathbb{R}^n$. Given a finite sequence of integers $\mathbf{r} = \{r_1, \ldots, r_k\}$ with $0 < r_1 \leq \cdots \leq r_k \leq n$, denote by $\mathbb{F}^n(\mathbf{r})$ the set of flags $f = (V_1 \subset \cdots \subset V_k)$ with $\dim V_i = r_i$.
    Show that $\mathrm{Gl}(n, \mathbb{R})$ acts transitively on $\mathbb{F}^n(\mathbf{r})$, by establishing a bijection between $\mathbb{F}^n(\mathbf{r})$ and the homogeneous space $\mathrm{Gl}(n, \mathbb{R})/Q$, where Q is some isotropy subgroup. Determine Q and show that Q is closed. Conclude that $\mathbb{F}^n(\mathbf{r})$ is a differentiable manifold.
    Prove that the subgroups $\mathrm{Sl}(n, \mathbb{R})$ and $\mathrm{SO}(n)$ act transitively on $\mathbb{F}^n(\mathbf{r})$ and write $\mathbb{F}^n(\mathbf{r})$ as homogeneous spaces $\mathrm{Sl}(n, \mathbb{R})/P$ and $\mathrm{SO}(n)/M$. Conclude that $\mathbb{F}^n(\mathbf{r})$ is compact. (Suggestion: for $\mathrm{SO}(n)$ use exercise 2.)
11. Solve the previous exercise for complex flags, i.e. those formed by subspaces of $\mathbb{C}^n$. Replace $\mathrm{Gl}(n, \mathbb{R})$ with $\mathrm{Gl}(n, \mathbb{C})$, $\mathrm{Sl}(n, \mathbb{R})$ with $\mathrm{Sl}(n, \mathbb{C})$ and $\mathrm{SO}(n)$ with $\mathrm{SU}(n)$.
12. Use transitive actions of groups to construct topologies and differentiable structures on the following sets:

    a. the set of bases of $\mathbb{R}^n$;
    b. the set of ordered bases of $\mathbb{R}^n$;
    c. the set of orthonormal bases of $\mathbb{R}^n$ (with respect to a fixed inner product);
    d. the set of inner products on $\mathbb{R}^n$;
    e. the set of linear complex structures on $\mathbb{R}^{2n}$ (these are linear maps $J : \mathbb{R}^{2n} \to \mathbb{R}^{2n}$ such that $J^2 = -\mathrm{id}$);
    f. the set of symplectic forms on $\mathbb{R}^{2n}$ (i.e. bilinear, skew-symmetric, and non-degenerated forms);
    g. the set of quadratic forms on $\mathbb{R}^n$ of given signature;
    h. the set of elements conjugated to an element x in a Lie group G (i.e. $\{gxg^{-1} : g \in G\}$).

13. Let β be an ordered basis of $\mathbb{C}^n$. The Borel subalgebra of $\mathfrak{sl}(n, \mathbb{C})$ defined by β is the subalgebra $\mathfrak{b}_\beta$ of linear maps that, in the basis β, are upper triangular. Denote by $\mathbb{B} = \{\mathfrak{b}_\beta : \beta \text{ is a basis}\}$ the set of Borel subalgebras. Show that $\mathrm{Sl}(n, \mathbb{C})$ acts transitively on $\mathbb{B}$ and verify that, as a homogeneous space, $\mathbb{B}$ coincides with $\mathbb{F}^n_{\mathbb{C}}(\mathbf{r})$, where $\mathbf{r} = (1, 2, \ldots, n-1)$.
14. Let G be a Lie group with Lie algebra $\mathfrak{g}$. Two subalgebras $\mathfrak{h}_1, \mathfrak{h}_2 \subset \mathfrak{g}$ are said to be G-conjugated if there exists $g \in G$ such that $\mathrm{Ad}(g)\,\mathfrak{h}_1 = \mathfrak{h}_2$. Construct a differentiable structure on the set of subalgebras G-conjugated to a given Lie subalgebra $\mathfrak{h} \subset \mathfrak{g}$.
15. Given a Lie group G and a closed subgroup $H \subset G$, suppose G/H is compact. Denote by $\mathfrak{h}$ the Lie algebra of H and show that the set of G-conjugated subalgebras to $\mathfrak{h}$ (see previous exercise) is compact.
16. Let $G \times M \to M$ be a differentiable action. Given $x_0 \in M$, let G_{x_0} be the isotropy group. The **isotropy representation** of G_{x_0} is the homomorphism $g \in G_{x_0} \mapsto dg_{x_0} \in \mathrm{Gl}\left(T_{x_0}M\right)$. Suppose G compact and M connected. Also assume that the orbit $G \cdot x_0$ has dimension > 0 and the isotropy representation is irreducible. Show M is compact.
17. Let $G \times M \to M$ be a differentiable action of the Lie group on the manifold M. Denote by $\mathfrak{g}$ the Lie algebra of G and take a path $A : (a, b) \subset \mathbb{R} \to \mathfrak{g}$. This curve defines a time-dependent differential equation $\dot{x} = \widetilde{A(t)}(x)$ on M. Show that the solutions to this equation are $\phi(t, s)(x)$, where $\phi(t, s) \in G$ is the solution of $\dot{g} = A(t)\,g$, $g \in G$, with initial condition $\phi(s, s) = 1$. Then prove that these solutions extend to the whole interval (a, b).
18. Let G be a Lie group, $G \times M \to M$ a differentiable action of G, and $F \subset M$ a closed subset. Define

$$\mathfrak{g}_F = \{X \in \mathfrak{g} : \forall t \in \mathbb{R},\ \exp(tX) \cdot F \subset F\}$$

(that is, $e^{tX}x \in F$ if $x \in F$). Show that $\mathfrak{g}_F$ is a Lie subalgebra. (Hint: use the limits of Section 6.4.)
19. Describe the orbits of the adjoint and coadjoint representations of the Heisenberg group, that is, the group of 3×3 matrices of the form

$$\begin{pmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix}.$$

20. Describe the orbits of the adjoint representation of the group $\mathrm{Sl}(2, \mathbb{R})$.
21. Let G be a Lie group with Lie algebra $\mathfrak{g}$ and denote by $\mathfrak{g}^*$ the dual of $\mathfrak{g}$. Consider the coadjoint representation of G on $\mathfrak{g}^*$. Take $\alpha \in \mathfrak{g}^*$ and verify that the isotropy algebra of the orbit $G \cdot \alpha$ of α is

$$\mathfrak{g}_\alpha = \{X \in \mathfrak{g} : \alpha \circ \mathrm{ad}(X) = 0\}.$$

Fix $\alpha \in \mathfrak{g}^*$ and define the skew-symmetric bilinear form $\omega_\alpha(X, Y) = \alpha[X, Y]$, $X, Y \in \mathfrak{g}$. Supposing $\mathfrak{e} \subset \mathfrak{g}$ is a complement to $\mathfrak{g}_\alpha$, i.e. $\mathfrak{g} = \mathfrak{e} \oplus \mathfrak{g}_\alpha$, prove the restriction of ω_α to $\mathfrak{e}$ is non-degenerate (that is, if $\omega_\alpha(X, Y) = 0$ for all $Y \in \mathfrak{e}$ then $X = 0$). Conclude that the orbits of the coadjoint representation have even dimension.

22. Let G be a Lie group and $\phi : G \to G$ an automorphism of G. Show that the fixed-point set $H = \{x \in G : \phi(x) = x\}$ is a Lie subgroup of G.
Vice versa, assume ϕ is involutive: $\phi^2 = \mathrm{id}$. Show that the map $xH \in G/H \mapsto x\phi\left(x^{-1}\right) \in G$ is an injective immersion.
23. Given a Lie group G, a closed subgroup H, and an automorphism θ of G, suppose $\theta(H) = H$. Prove θ defines a diffeomorphism $\widetilde{\theta}$ of G/H by $\widetilde{\theta}(gH) = \theta(g)H$. Let X be an element of the Lie algebra $\mathfrak{g}$ of G. Show that if $\widetilde{X}$ is the vector field on G/H induced by X, then $\widetilde{\theta}_* \widetilde{X} = \widetilde{d\theta_1(X)}$.
24. Let $G \times M \to M$ be an analytic action of the (analytic) Lie group G on the connected analytic manifold M. Denote by k the maximum dimension of the orbits of G. Show that the set of points $x \in M$ such that $\dim(G \cdot x) = k$ is an open and dense subset of M.
25. Show that if $G \times M \to M$ is a differentiable action of the Lie group G on the differentiable manifold M, the function $x \mapsto \dim(G \cdot x)$ is lower semicontinuous: $\{x \in M : \dim(G \cdot x) > b\}$ is open for all $b \in \mathbb{R}$.
26. Given a differentiable action $G \times M \to M$ of the Lie group G on the differentiable manifold M, let $G \cdot x$ be an orbit of G. Verify that the closure $\overline{G \cdot x}$ is a G-invariant set and hence a union of G-orbits. Show that, for every $y \in \overline{G \cdot x}$, the orbit $G \cdot y$ satisfies $\dim(G \cdot y) \leq \dim(G \cdot x)$. Find examples of actions where $\dim(G \cdot y) = \dim(G \cdot x)$ for some $y \in \overline{G \cdot x}$. Also give examples where $\dim(G \cdot y) < \dim(G \cdot x)$ for all $y \in \overline{G \cdot x}$.
27. Let $G \times M \to M$ be a differentiable action of the Lie group G on the differentiable manifold M. Define on M the equivalence relation given by G-orbits: $x \sim y$ if and only if $y = gx$ for some $g \in G$. Assume that the action is free, and construct on the orbit space $M/\sim$ a differentiable structure whose topology is the quotient topology and such that the canonical projection $M \to M/\sim$ is a submersion.
28. Given a differentiable action $\phi : G \times M \to M$ with G and M connected, let $\widetilde{G}$ and $\widetilde{M}$ be the universal coverings. Use Theorem 13.14 to construct an action $\widetilde{\phi} : \widetilde{G} \times \widetilde{M} \to \widetilde{M}$ such that the covering map $\pi : \widetilde{M} \to M$ is equivariant. Show that the orbits of $\widetilde{\phi}$ are coverings of the orbits of ϕ. Do the same for an arbitrary covering $\widehat{M}$ of M. Prove that if the action on M is transitive the same occurs on any covering.
29. Let G be a compact semi-simple Lie group and $H \subset G$ a closed subgroup. Show that the fundamental group $\pi_1(G/H)$ is finite.
30. Given a Lie group G and $\Gamma \subset G$ a normal discrete subgroup, consider the quotient group G/Γ and the canonical projection $\pi : G \to G/\Gamma$. Show that if $\Gamma_1 \subset G/\Gamma$ is a discrete subgroup of G/Γ, then $\pi^{-1}(\Gamma_1)$ is a discrete subgroup of G.

31. Given a Lie group G and a closed subgroup $\Gamma \subset G$, suppose $\rho : \Gamma \to H$ is a differentiable homomorphism into the Lie group H. Take the principal bundle $G \to G/\Gamma$ and construct, as in Proposition 13.22, the principal bundle $G \times_\rho H$ over G/Γ. Show that if ρ extends to a differentiable homomorphism $G \to H$, then $G \times_\rho H$ is a trivial bundle.
32. A **complete parallelism** on a manifold M is a set of vector fields $\{X_1, \ldots, X_n\}$, $n = \dim M$, such that $\{X_1(x), \ldots, X_n(x)\}$ is a basis of $T_x M$ for all $x \in M$ and $[X_i, X_j] = \sum_k c_{ij}^k X_k$ with c_{ij}^k constants. Show that if M admits a parallelism $\{X_1, \ldots, X_n\}$ such that the vector fields $\sum_k a_i X_i$, $a_i \in \mathbb{R}$, are complete, then there is a Lie group structure on M such that $\{X_1, \ldots, X_n\}$ are right invariant. (Hint: use the Lie–Palais theorem.)

Chapter 14
Invariant Geometry



This chapter discusses a number of invariant geometric structures on homogeneous spaces. The structures of concern will be defined in terms of tensors (complex structures, differential forms, Riemannian metrics, and symplectic forms). The basic principle is that on a homogeneous space $M = G/H$ (H closed), invariant structures are determined by their values at the origin $x_0 = 1 \cdot H$. What happens is that $h(x_0) = x_0$ for any $h \in H$, implying that the differential dh_{x_0} is a linear map of the tangent space $T_{x_0}M$ to itself. This differential therefore defines a map $\rho : H \to \mathrm{Gl}\left(T_{x_0}M\right)$, which is a group homomorphism by the chain rule. Hence ρ is a representation of H on the tangent space $T_{x_0}M$, called the **isotropy representation** of G/H.

An invariant geometric structure on G/H given by a tensor defines a tensor on $T_{x_0}M$ that is invariant under the isotropy representation. Vice versa, given an isotropy-invariant tensor on $T_{x_0}M$, one can define a G-invariant tensor on G/H. Instead of focusing on general principles, this procedure will be worked out on a case-by-case basis in the examples considered in the chapter.

14.1 Complex Manifolds

A **complex manifold** M is defined by an atlas of $\mathbb{C}^n$-valued charts and coordinate changes given by holomorphic maps between open subsets of $\mathbb{C}^n$. (Recall that a map $f : U \subset \mathbb{C}^n \to \mathbb{C}^n$ is said holomorphic if it is differentiable over $\mathbb{C}$. Equivalently, f is differentiable as a map of $\mathbb{R}^{2n}$, and its differential df_x at each point is complex-linear, i.e. it commutes with the multiplication by $i = \sqrt{-1}$.)

Complex manifolds are in particular real manifolds if $\mathbb{C}^n$ is interpreted as $\mathbb{R}^{2n}$. If M is a complex manifold, then each tangent space T_xM is (has the structure of) a complex vector space.

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Complex manifolds can be seen as real manifolds endowed with an additional structure given by an almost complex structure, which extends the concept of complex manifold.

Before getting to almost complex structures on M, let V be a real vector space. A linear map $J : V \to V$ is a complex structure on V if $J^2 = -\mathrm{id}$. If V admits a complex structure, then $\dim V$ is even since the eigenvalues of J are $\pm i$ and always appear in pairs.

Definition 14.1 Let M be a differentiable manifold. An **almost complex structure** on a differentiable manifold M is a tensor J whose value J_x at any $x \in M$ is a linear map $J_x : T_x M \to T_x M$ satisfying $J_x^2 = -\mathrm{id}$.

The structure J is differentiable if, for every differentiable vector field X, the vector field JX is also differentiable.

It should be noted that if M admits an almost complex structure, then $\dim M$ must be even, since its tangent spaces have even dimensions.

If M is a complex manifold, then the tangent spaces $T_x M$ are vector spaces over $\mathbb{C}$. The multiplication by i on each $T_x M$ defines an almost complex structure on M. Complex manifolds are therefore almost complex. The characterization of almost complex manifolds that are complex is the content of the **Newlander–Nirenberg theorem**. This result is stated in terms of the **Nijenhuis tensor** of J,

$$N_J(X, Y) = J[X, Y] - [JX, Y] - [X, JY] - J[JX, JY], \tag{14.1}$$

where X and Y are vector fields on M. This is the statement of Newlander–Nirenberg.

Theorem 14.2 *Let M be an almost complex manifold with structure J. Then M is a complex manifold if and only if $N_J = 0$. In this case, the multiplication by i in $T_x M$ coincides with J_x.*

An almost complex structure J is said to be **integrable** if its Nijenhuis tensor vanishes identically. The Newlander–Nirenberg theorem is therefore saying that integrable almost complex structures are exactly those that come from complex structures.

In the world of almost complex manifolds, a **holomorphic map** between manifolds M and N with almost complex structures J^M and J^N is a differentiable map $f : M \to N$, such that $df_x \circ J_x^M = J_{f(x)}^N \circ df_x$ for every $x \in M$. That is, the differentials df_x are complex-linear maps.

Now let $M = G/H$ be a homogeneous space, where G is a Lie group and H be a closed subgroup. An almost complex structure J on G/H is called ***G*-invariant** if the elements of G are holomorphic maps with respect to J. In other words,

$$dg_x \circ J_x = J_{g(x)} \circ dg_x$$

for every $g \in G$ and $x \in G/H$. An invariant almost complex structure J is completely determined by its value at the origin $x_0 = 1 \cdot H$, because if $x \in G/H$ is

given by $x = gx_0$, then $J_x = dg_{x_0} \circ J_{x_0} \circ \left(dg_{x_0}\right)^{-1}$. Moreover, the almost complex structure J_{x_0} on the vector space $T_{x_0}M$ is invariant under the isotropy representation, in the sense that $J_{x_0} = dh_{x_0} \circ J_{x_0} \circ \left(dh_{x_0}\right)^{-1}$ if $h \in H$.

Conversely, let J_0 be a complex structure on the vector space $T_{x_0}M$, i.e. $J_0 : T_{x_0}M \to T_{x_0}M$ satisfies $J_0^2 = -\mathrm{id}$. Suppose J_0 is invariant by the isotropy representation, which means $J_0 = dh_{x_0} \circ J_0 \circ \left(dh_{x_0}\right)^{-1}$ for $h \in H$. Then,

$$J_x = dg_{x_0} \circ J_0 \circ \left(dg_{x_0}\right)^{-1} \qquad x = gx_0 \in G/H \tag{14.2}$$

defines a G-invariant almost complex structure on G/H. This is because the expression for J_x in (14.2) does not depend on the particular $g \in G$ for which $x = gx_0$, since J_0 is isotropy-invariant. An almost complex structure defined by this procedure is C^∞.

The Lie algebras of G and H will be denoted by $\mathfrak{g}$ and $\mathfrak{h}$, respectively, in the sequel.

Example Suppose H is compact. The Lie algebra of G splits as $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$, where $\mathfrak{m}$ is an $\mathrm{Ad}\,(H)$-invariant subspace (for instance, $\mathfrak{m}$ may be the orthogonal complement of $\mathfrak{h}$ with respect to some invariant inner product). The map $X \in \mathfrak{m} \mapsto \widetilde{X}\,(x_0) \in T_{x_0}G/H$ is an isomorphism. The formula $g_*\widetilde{X} = \widetilde{\mathrm{Ad}\,(g)\,X}$, with $g \in G$ and $X \in \mathfrak{g}$, implies that the isotropy representation is equivalent to the adjoint representation of H on $\mathfrak{m}$.

Hence almost complex structures on G/H are in 1-1 correspondence with complex structures $J_0 : \mathfrak{m} \to \mathfrak{m}$ such that $\mathrm{Ad}\,(h)\,J_0 = J_0\mathrm{Ad}\,(h)$ for all $h \in H$.

The complex structures that might commute with $\mathrm{Ad}\,(H)$ are detected through the decomposition of $\mathfrak{m}$ in invariant irreducible subspaces. Suppose J_0 is a complex structure on $\mathfrak{m}$ that commutes with $\mathrm{Ad}\,(h)$, $h \in H$. If $W \subset \mathfrak{m}$ is an $\mathrm{Ad}\,(H)$-invariant subspace, then J_0W is also invariant. In fact, if $h \in H$, then

$$\mathrm{Ad}\,(h)\,J_0W = J_0\mathrm{Ad}\,(h)\,W = J_0W.$$

Furthermore, if W is irreducible, then J_0W is irreducible as well, because invariance of $U \subset J_0W$ implies that $J_0U \subset J_0^2W = W$ is invariant. As J_0 and $\mathrm{Ad}\,(h)$, for $h \in H$, commute, the representations of H on U and on J_0U are equivalent, and J_0 itself is an equivalence between these representations.

These observations apply in particular to the situation where

$$\mathfrak{m} = \mathfrak{m}_1 \oplus \cdots \oplus \mathfrak{m}_s$$

is a decomposition of $\mathfrak{m}$ into irreducible, invariant subspaces whose H-representations are inequivalent. In this case, an invariant complex structure J_0 must satisfy $J_0\mathfrak{m}_i = \mathfrak{m}_i$ for every $i = 1, \ldots, s$, i.e. there exist invariant complex structures on $\mathfrak{m}$ if and only if each $\mathfrak{m}_i$ admits one such. In the sequel, concrete examples of homogeneous spaces will be described in which this situation occurs.

The observations made thus far also apply to the case where H is semi-simple, whose representations split into invariant and irreducible subspaces, just as in the compact case. □

If J is an invariant almost complex structure on G/H, then the identity

$$g_*[X, Y] = [g_*X, g_*Y]$$

shows that $g_*N_J(X, Y) = N_J(g_*X, g_*Y)$ if $g \in G$.

Hence the Nijenhuis tensor N_J vanishes identically if and only if it is zero at the origin x_0 of G/H. Therefore the following criterion applies for an invariant almost complex structure to be complex.

Proposition 14.3 *An almost complex structure on the homogeneous space G/H is complex if and only if the Nijenhuis tensor N_J is zero at the origin $x_0 = 1 \cdot H$ of G/H.*

In a nutshell, both the construction of invariant almost complex structures and the criteria for deciding whether they are complex or not are reduced to analyzing what happens at the origin of the homogeneous space. This depends only on the isotropy representation and, in many instances, it is carried out by purely algebraic calculations.

Before presenting examples, a comment is in order regarding the right hand side of (14.1) that defines the Nijenhuis tensor. This tensor involves Lie brackets of vector fields, which in principle depend on the values of the vector fields on a whole neighborhood of a point. However, this dependence disappears due to the interaction of the four brackets in the expression, to the effect that $N_J(X, Y)$ depends only on the values of X and Y at a given point. Therefore, N_J is indeed a tensor (see Exercise 3 at the end of the chapter).

So there is a certain amount of freedom in the choice of the vector fields when computing the Nijenhuis tensor. On a homogeneous space G/H, a natural choice is to pick the vector fields $\widetilde{X}$, for $X \in \mathfrak{g}$, as these generate the tangent spaces at each point.

Example Take $G = \mathrm{SU}(n)$ and let $H = T \subset \mathrm{SU}(n)$ be the maximal torus consisting of diagonal matrices in $\mathrm{SU}(n)$. The quotient $\mathrm{SU}(n)/T$ can be identified with a flag manifold, parametrizing flags $(W_1 \subset \cdots \subset W_n)$ in $\mathbb{C}^n$ with $\dim W_j = j$ (see Chapter 13, Exercise 11).

The Lie algebra $\mathfrak{t}$ of T consists of diagonal matrices with trace 0 and purely imaginary entries. The adjoint representation of T decomposes $\mathfrak{su}(n)$ in invariant subspaces,

$$\mathfrak{su}(n) = \mathfrak{h} \oplus \sum_{j<k} V_{jk},$$

where V_{jk} is the subspace of matrices in $\mathfrak{su}(n)$ with nonzero entries only at positions j, k and k, j. The subspace $V = \sum_{j<k} V_{jk}$ identifies with the tangent space of G/H at the origin.

A subspace V_{jk} has dimension 2 and is spanned by

$$A_{jk} = E_{jk} - E_{kj} \qquad S_{jk} = i\left(E_{jk} + E_{kj}\right),$$

where E_{jk} are the basic matrices. The adjoint representations ρ_{jk} on V_{jk} of T and $\mathfrak{t}$ are irreducible. In the basis $\{A_{jk}, S_{jk}\}$, the representation of $\mathfrak{t}$ is given by the matrix

$$\rho_{jk}(X) = \begin{pmatrix} 0 & -\alpha_{jk}(X) \\ \alpha_{jk}(X) & 0 \end{pmatrix},$$

where $X = \operatorname{diag}\{ia_1, \ldots, ia_n\}$ and $\alpha_{jk}(X) = a_j - a_k$. It follows from this expression that the representation on $V_{j,k}$ is not equivalent to the representation on V_{rs} whenever $(j,k) \neq (r,s)$. As discussed in the above example, therefore, T-invariant complex structures on V are sums of T-invariant complex structures on each V_{jk}.

On the other hand, an easy calculation with 2×2 matrices shows that each V_{jk} admits only two invariant complex structures, namely $\pm J_{jk}$, where

$$J_{jk} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

with matrix written in the basis $\{A_{jk}, S_{jk}\}$. Consequently, an $\operatorname{Ad}(T)$-invariant complex structure on V is a direct sum of the types

$$J = \sum_{j<k} \delta_{jk} J_{jk} \qquad \delta_{jk} = \pm 1.$$

There are 2^N such structures, where $N = \frac{n(n-1)}{2}$ is the number of subspaces V_{jk}.

The Nijenhuis tensor at the origin is the skew-symmetric bilinear map on V with values in V. To compute it explicitly, it is better to work in the space $V_{\mathbb{C}}$ generated by E_{jk}, $j \neq k$ over $\mathbb{C}$, i.e. the complexification of V. Both J and N_J complexify to $V_{\mathbb{C}}$, and for $j < k$, one has

$$J_{jk} E_{jk} = i E_{jk} \qquad J_{jk} E_{kj} = -i E_{kj}.$$

For $j \neq k$, therefore, J is given by $J E_{jk} = i\varepsilon_{jk} E_{jk}$ and is determined by the possible signs $\varepsilon_{jk} = \pm 1$, $j \neq k$ with the restriction that $\varepsilon_{kj} = -\varepsilon_{jk}$. The Nijenhuis tensor satisfies $N_J\left(E_{jk}, E_{rs}\right) = 0$ when $\left[E_{jk}, E_{rs}\right] = 0$, i.e. when $\{j,k\} \cap \{r,s\} = \emptyset$. On the other hand,

$$N_J\left(E_{jk}, E_{ks}\right) = i\left(\varepsilon_{js}\left(1 + \varepsilon_{jk}\varepsilon_{ks}\right) - \varepsilon_{jk} - \varepsilon_{ks}\right) E_{js},$$

whose right hand side is different from zero if and only if $\varepsilon_{jk} = \varepsilon_{ks} \neq \varepsilon_{js}$.

For example, if J is such that $\varepsilon_{jk} = +1$ when $j < k$, then J is integrable. Similarly, suppose w is a permutation of $\{1, \ldots, n\}$. Then the complex structure determined by $\varepsilon_{jk} = +1$ if $w(j) < w(k)$ is integrable. It can be checked that these are the only integrable almost complex structures. □

The description of complex structures on $\mathrm{SU}(n)/T$ in the previous example extends to the so-called complex generalized flag manifolds, i.e. quotients of compact Lie groups by their maximal tori. Here, too, there exist finitely many integrable almost complex structures, which are in 1-1 correspondence with the Weyl chambers of the Lie algebra $\mathfrak{t}$ of the torus.[1]

14.1.1 Complex Lie Groups

A complex Lie group can be defined in the same way as a Lie group: a group whose underlying set is a complex manifold, and such that the product is holomorphic with respect to the given complex structure.

Recall that a complex manifold is a real manifold with an additional structure. From this perspective, a complex Lie group can be defined as a real Lie group endowed with an integrable almost complex structure such that the product is a holomorphic map.

This approach to complex groups befits the above discussions on invariant almost complex and complex structures on homogeneous spaces. To see this, start by taking a complex Lie group G and let J be the complex structure given by multiplication by i on each tangent space. Since the product p is holomorphic, left translations L_g and right translations R_g ($g \in G$) are holomorphic maps. This means $J_{gh} \circ d\left(L_g\right)_h = d\left(L_g\right)_h \circ J_h$ and $J_{hg} \circ d\left(R_g\right)_h = d\left(R_g\right)_h \circ J_h$ for any $g, h \in G$. In other words, J is a bi-invariant complex structure, i.e. both left and right invariant.

The next step would be to consider bi-invariant almost complex structures on G. Bi-invariance appears through the action $(g, h)\, x = gxh^{-1}$ of $G \times G$ on G. This is transitive, and the isotropy group at the identity $1 \in G$ is the diagonal

$$\Delta_G = \{(g, g) \in G \times G : g \in G\},$$

so G identifies with $G \times G/\Delta_G$. The isotropy representation $\rho : \Delta_G \to \mathrm{Gl}(\mathfrak{g})$ is the adjoint representation $\rho(g, g) = \mathrm{Ad}(g) = \left(dC_g\right)_1$.

A bi-invariant almost complex structure J on G is therefore given by a complex structure $J_1 : \mathfrak{g} \to \mathfrak{g}$ that commutes with the adjoint maps:

$$J_1 \circ \mathrm{Ad}(g) = \mathrm{Ad}(g) \circ J_1. \tag{14.3}$$

[1] See San Martin–Negreiros [48].

At a point $g \in G$, the almost complex structure reads

$$J_g = d\left(L_g\right)_1 \circ J_1 \circ d\left(L_g\right)_1^{-1} = d\left(R_g\right)_1 \circ J_1 \circ d\left(R_g\right)_1^{-1}. \tag{14.4}$$

The question is when is an almost complex structure integrable, i.e. when does the Nijenhuis tensor vanish. But here the condition becomes trivial, since bi-invariance automatically implies integrability.

In fact, the commutate relation $J_1 \circ \mathrm{Ad}(g) = \mathrm{Ad}(g) \circ J_1$, $g \in G$, applied to $g = e^{tX}$, $X \in \mathfrak{g}$, forces (by differentiating in t)

$$J_1 \circ \mathrm{ad}(X) = \mathrm{ad}(X) \circ J_1 \tag{14.5}$$

for all $X \in \mathfrak{g}$. The meaning of this is that $\mathfrak{g}$ is a Lie algebra over $\mathbb{C}$, in the sense that if J_1 is interpreted as multiplication by i, (14.5) says that $i[X, Y] = [X, iY]$. Moreover,

$$i[X, Y] = -i[Y, X] = -[Y, iX] = [iX, Y],$$

so $\mathfrak{g}$ becomes a complex vector space with bracket $[\cdot, \cdot]$ bilinear over $\mathbb{C}$.

Keeping in mind the interpretation of J_1 as multiplication by i in $\mathfrak{g}$, the Nijenhuis tensor at the identity is

$$\begin{aligned} N_J(X, Y) &= J_1[X, Y] - [J_1X, Y] - [X, J_1Y] - J_1[J_1X, J_1Y] \\ &= i[X, Y] - [iX, Y] - [X, iY] - i[iX, iY] \end{aligned}$$

with $X, Y \in \mathfrak{g}$. Since the last term is zero, we have $(N_J)_1 = 0$ at the origin. By the invariance (for example, on the left) of J, N_J vanishes identically.

These observations lead up to the following characterization of complex groups.

Theorem 14.4 *Let G be a Lie group with Lie algebra $\mathfrak{g}$, and suppose there is a map $J_1 : \mathfrak{g} \to \mathfrak{g}$ such that $J_1^2 = -\mathrm{id}$ and $J_1 \circ \mathrm{Ad}(g) = \mathrm{Ad}(g) \circ J_1$ for every $g \in G$. Then there exists a complex structure on G turning it into a complex Lie group. In that case, $\mathfrak{g}$ is a complex Lie algebra. Furthermore, if $\mathfrak{g}$ is a complex Lie algebra and G is connected, then G is a complex Lie group.*

Proof There only remains to check the last statement. If $\mathfrak{g}$ is a complex Lie algebra, then there exists an endomorphism $J_1 : \mathfrak{g} \to \mathfrak{g}$ with $J_1^2 = -\mathrm{id}$ such that $J_1 \circ \mathrm{ad}(X) = \mathrm{ad}(X) \circ J_1$. By the usual procedure, it follows that $J_1 \circ \mathrm{Ad}\left(e^X\right) = \mathrm{Ad}\left(e^X\right) \circ J_1$ if $X \in \mathfrak{g}$. Hence, if G is connected, $J_1 \circ \mathrm{Ad}(g) = \mathrm{Ad}(g) \circ J_1$ for all $g \in G$, and hence G is a complex Lie group. □

Classical examples of complex Lie groups are the matrix groups $\mathrm{Gl}(n, \mathbb{C})$, $\mathrm{Sl}(n, \mathbb{C})$, $\mathrm{Sp}(n, \mathbb{C})$, and $\mathrm{SO}(n, \mathbb{C})$, whose Lie algebras are complex.

14.2 Differential Forms and de Rham Cohomology

Given a differentiable action of the Lie group G on the manifold M, a differential k-form α on M is G-invariant if $g^*\alpha = \alpha$, for all $g \in G$, where $g^*\alpha$ is the pullback, defined by $g^*\alpha\,(X_1, \ldots, X_k) = \alpha\,(g_*X_1, \ldots, g_*X_k)$.

When $M = G/H$ is a homogeneous space, an invariant differential form α is completely determined by its value α_{x_0} at the origin $x_0 = 1 \cdot H$ of G/H. The alternating k-form α_{x_0} is invariant under the isotropy representation in the sense that $\left(dh_{x_0}\right)^* \alpha_{x_0} = \alpha_{x_0}$ if $h \in H$. Equivalently,

$$\alpha_{x_0}(dh_{x_0}v_1, \ldots, dh_{x_0}v_k) = \alpha_{x_0}(v_1, \ldots, v_k),$$

where $v_1, \ldots, v_k \in T_{x_0}G/H$.

At an arbitrary point $x \in G/H$, the k-form α_x is given by $\alpha_x = \left(g^{-1}\right)^* \alpha_{x_0}$ if $x = gx_0$, so

$$\alpha_x(w_1, \ldots, w_k) = \alpha_{x_0}(\left(dg_{x_0}\right)^{-1} w_1, \ldots, \left(dg_{x_0}\right)^{-1} w_k)$$

if $w_1, \ldots, w_k \in T_xG/H$. This expression is independent of the particular $g \in G$ such that $x = gx_0$, because α_{x_0} is invariant by the isotropy representation.

The space of differential k-forms on G/H is denoted by $\wedge^k\,(G/H)$, while the subset of invariant k-forms is indicated by $\wedge^k_{\text{inv}}\,(G/H)$. The latter is a subspace of $\wedge^k\,(G/H)$ as $\alpha \mapsto g^*\alpha$ is linear. Moreover, $\wedge^k_{\text{inv}}\,(G/H)$ is a subalgebra of the exterior algebra, since $g^*\,(\alpha \wedge \beta) = g^*\alpha \wedge g^*\beta$.

One of the issues to discuss regarding a differential k-form α is its exterior derivative $d\alpha$, whose definition, on a differentiable manifold M, is

$$\begin{aligned}(d\alpha)(X_1, \ldots, X_{k+1}) = &\textstyle\sum_i (-1)^{i+1} X_i \alpha(X_1, \ldots, \widehat{X_i}, \ldots, X_{k+1}) \\ &+ \textstyle\sum_{i<j} (-1)^{i+j} \alpha([X_i, X_j], X_1, \ldots, \widehat{X_i}, \ldots, \widehat{X_j}, \ldots, X_{k+1}),\end{aligned} \tag{14.6}$$

where $X_1, \ldots, X_{k+1}$ are vector fields on M.

For an invariant differential form α on a homogeneous space G/H, formula (14.6) simplifies, since the terms of the first sum can be incorporated in the second sum.

Lemma 14.5 *Consider an invariant differential k-form α on G/H and $X, X_1, \ldots X_k \in \mathfrak{g}$. Then,*

$$\widetilde{X}\alpha(\widetilde{X}_1, \ldots, \widetilde{X}_k) = \alpha([\widetilde{X}, \widetilde{X}_1], \widetilde{X}_2, \ldots, \widetilde{X}_k) + \cdots + \alpha(\widetilde{X}_1, \widetilde{X}_2, \ldots, [\widetilde{X}, \widetilde{X}_k]),$$

where $\widetilde{X}$ is the vector field on G/H induced by $X \in \mathfrak{g}$.

Proof If $x \in G/H$, then by definition,

$$\begin{aligned}\widetilde{X}\alpha(\widetilde{X}_1,\dots,\widetilde{X}_k)(x) &= \frac{d}{dt}\alpha(\widetilde{X}_1,\dots,\widetilde{X}_k)\left(e^{tX}x\right)_{|t=0}\\ &= \frac{d}{dt}\alpha_{e^{tX}x}(\widetilde{X}_1\left(e^{tX}x\right),\dots,\widetilde{X}_k\left(e^{tX}x\right))_{|t=0}.\end{aligned}$$

As α is invariant, $\alpha_{e^{tX}x} = \left(e^{-tX}\right)^*\alpha_x$, and so

$$\begin{aligned}&\alpha_{e^{tX}x}(\widetilde{X}_1\left(e^{tX}x\right),\dots,\widetilde{X}_k\left(e^{tX}x\right))\\ &= \alpha_x(\left(de^{-tX}\right)_{e^{tX}x}\widetilde{X}_1\left(e^{tX}x\right),\dots,\left(de^{-tX}\right)_{e^{tX}x}\widetilde{X}_k\left(e^{tX}x\right)).\end{aligned}$$

Differentiating in t and taking in account that

$$\frac{d}{dt}\left(de^{-tX}\right)_{e^{tX}x}\widetilde{X}_i\left(e^{tX}x\right)_{|t=0} = [\widetilde{X},\widetilde{X}_i](x),$$

the claimed formula follows. □

Proposition 14.6 *Let α be an invariant differential k-form on G/H. Then,*

$$(d\alpha)(\widetilde{X}_1,\dots,\widetilde{X}_{k+1}) = \sum_{i<j}(-1)^{i+j}\alpha([\widetilde{X}_i,\widetilde{X}_j],\widetilde{X}_1,\dots,\widehat{\widetilde{X}_i},\dots,\widehat{\widetilde{X}_j},\dots,\widetilde{X}_{k+1}) \tag{14.7}$$

with $X_1,\dots,X_{k+1}\in\mathfrak{g}$.

Proof The previous lemma and formula (14.6) for the exterior differential show that $d\alpha(\widetilde{X}_1,\dots,\widetilde{X}_{k+1})$ is a sum of terms of type

$$\alpha([X_i,X_j],X_1,\dots,\widehat{X_i},\dots,\widehat{X_j},\dots,X_{k+1}) \qquad i<j.$$

There remains to show that the coefficient of the term corresponding to $i,j,i<j$, equals $(-1)^{i+j}$. This term appears once in the second sum of (14.6) with coefficient $(-1)^{i+j}$. By the previous lemma, it appears twice in the first sum of (14.6), where the derivatives along $\widetilde{X}_i$ and $\widetilde{X}_j$ are taken. The derivative along $\widetilde{X}_i$ contributes with

$$(-1)^{i+1}\alpha(\widetilde{X}_1,\dots,[\widetilde{X}_i,\widetilde{X}_j],\dots,\widetilde{X}_k) = (-1)^{i+j}\alpha([\widetilde{X}_i,\widetilde{X}_j],\widetilde{X}_1,\dots,\widehat{X}_j,\dots,\widetilde{X}_k),$$

because the bracket on the left side appears at position j, and α is skew. The derivative along $\widetilde{X}_j$ contributes by the same term but with coefficient $-(-1)^{i+j}$. Therefore the sum of the three terms has coefficient $(-1)^{i+j}$, concluding the proof. □

Formula (14.7) for the exterior derivative d is written for vector fields $\widetilde{X}$, $X\in\mathfrak{g}$, although d only depends on their value at a given point. At the origin x_0 of the homogeneous space G/H, this can be explained as follows. Take a subspace $\mathfrak{m}\subset\mathfrak{g}$

such that $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$, where $\mathfrak{h}$ is the Lie algebra of H. The tangent space $T_{x_0} G/H$ is identified with $\mathfrak{m}$ by the isomorphism $X \in \mathfrak{m} \mapsto \widetilde{X}(x_0) \in T_{x_0} G/H$. Denote by $\mathrm{pr}_{\mathfrak{m}}$ the projection on $\mathfrak{m}$ with respect to the decomposition $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$. If $X, Y \in \mathfrak{m}$, then $[\widetilde{X}, \widetilde{Y}](x_0) = \widetilde{Z}(x_0)$, where $Z = \mathrm{pr}_{\mathfrak{m}}[X, Y]$. Using this notation, expression (14.7) reads

$$(d\alpha)_{x_0}(\widetilde{X}_1, \ldots, \widetilde{X}_{k+1}) = \sum_{i<j} (-1)^{i+j} \alpha_{x_0}([X_i, X_j]_{\mathfrak{m}}, X_1, \ldots, \widehat{X_i}, \ldots, \widehat{X_j}, \ldots, X_{k+1}) \tag{14.8}$$

if $X_1, \ldots, X_{k+1} \in \mathfrak{m}$. The right hand side depends solely on the Lie algebra $\mathfrak{g}$ and defines a linear map $d : \wedge^k \mathfrak{m} \to \wedge^{k+1} \mathfrak{m}$.

Before we continue, observe that if α is an invariant differential form, then $d\alpha$ is also invariant, since $g^* d\alpha = d g^* \alpha = d\alpha$. This entails, in particular, that if $d\alpha$ vanishes at a point, then it is identically zero.

The exterior derivative d satisfies $d^2 = 0$, a fact that gives rise to the **de Rham cohomology** of differentiable manifolds. A differential form α is called closed if $d\alpha = 0$, and exact if $\alpha = d\beta$ for some form β. Since $d^2 = 0$, every exact form is closed. The kth de Rham cohomology group $\mathcal{H}^k_{\mathrm{dR}}(M)$ of a manifold M is, by definition, the space of closed k-forms modulo exact k-forms:

$$\mathcal{H}^k_{\mathrm{dR}}(M) = \ker d_k / \mathrm{im} d_{k-1},$$

where d_k is the exterior derivative restricted to k-forms. Two closed differential forms α and β are said to be cohomologous if there exists γ such that $\alpha - \beta = d\gamma$, and one writes $\alpha \sim \beta$. The cohomology class of α is denoted by $[\alpha]$.

Invariant forms, too, possess a further, analogous notion of cohomology. If α is a G-invariant differential form, then $d\alpha$ is also G-invariant. So given a differentiable action $G \times M \to M$, one can introduce the **invariant cohomology group** $\mathcal{H}^k_{\mathrm{inv}}(M, G)$ just like the quotient above, where now d_k is the exterior derivative restricted invariant forms. In the case of homogeneous spaces $M = G/H$, this cohomology will be denoted by $\mathcal{H}^k_{\mathrm{inv}}(G/H)$. Given invariant closed forms α and β, write $\alpha \sim_{\mathrm{inv}} \beta$ if $\alpha - \beta = d\gamma$ for some invariant form γ. The invariant cohomology class of α is denoted by $[\alpha]_{\mathrm{inv}}$.

Let α and β be invariant closed forms on G/H. It is evident from the definitions that $\alpha \sim \beta$ if $\alpha \sim_{\mathrm{inv}} \beta$. Hence $[\alpha]_{\mathrm{inv}} = [\beta]_{\mathrm{inv}}$ implies $[\alpha] = [\beta]$, which provides well-defined maps

$$\mathcal{H}^k_{\mathrm{inv}}(G/H) \to \mathcal{H}^k_{\mathrm{dR}}(G/H) \qquad [\alpha]_{\mathrm{inv}} \mapsto [\alpha]. \tag{14.9}$$

In general, these maps are neither injective nor surjective, whence the cohomology groups $\mathcal{H}^k_{\mathrm{inv}}(G/H)$ and $\mathcal{H}^k_{\mathrm{dR}}(G/H)$ may differ. A typical example is that of an abelian group $\mathbb{R}^n$.

Example Consider the homogeneous space $\mathbb{R}^n = \mathbb{R}^n/H$, $H = \{0\}$, coming from the action of the group on itself by translations. If $(x_1, \ldots, x_n)$ are coordinates on $\mathbb{R}^n$, then differential k-forms are written as

$$\alpha = \sum a_I(x)\, dx_{i_1} \wedge \cdots \wedge dx_{i_k}$$

with $1 \leq i_1 < \cdots < i_k \leq n$, where $I = (i_1, \ldots, i_k)$, and $a_I : \mathbb{R}^n \to \mathbb{R}$ is differentiable. The exterior derivative is given, in these coordinates, by $d\alpha = \sum da_I \wedge dx_{i_1} \wedge \cdots \wedge dx_{i_k}$. Since the 1-forms dx_i are translation-invariant, α is invariant if and only if every a_I is a constant function. So if α is invariant, then $d\alpha = 0$ and, therefore, distinct invariant forms cannot be cohomologous. This means that $\mathcal{H}^k_{\text{inv}}(\mathbb{R}^n) = \wedge^k_{\text{inv}}(\mathbb{R}^n)$, and the latter is not zero. On the other hand, it is known that $\mathcal{H}^k_{\text{dR}}(\mathbb{R}^n) = \{0\}$ for $k \geq 1$.[2] Eventually, $\mathcal{H}^k_{\text{inv}}(\mathbb{R}^n) \neq \mathcal{H}^k_{\text{dR}}(\mathbb{R}^n)$. □

That said, it will be proved below that the maps (14.9) are isomorphisms when G is compact, in which case $\mathcal{H}^k_{\text{inv}}(G/H) = \mathcal{H}^k_{\text{dR}}(G/H)$ for every $k \geq 1$.

Suppose from now on that G is compact. The reason why the above isomorphisms come about depends on the possibility of integrating differential forms with respect to the Haar measure on G and thus obtaining differential forms that are invariant under G.

This is how differential forms are integrated. Let α be a differential k-form on G/H. For any $g \in G$, $g^*\alpha$ is a new differential form on G/H whose value $(g^*\alpha)_x$ at $x \in G/H$ is a skew form in $\wedge^k T^*_x G/H$, $T^*_x G/H$ being the cotangent space at x. The map $g \in G \mapsto (g^*\alpha)_x \in \wedge^k T^*_x G/H$ assumes values in a vector space. It makes sense then to take its integral

$$(I\alpha)_x = \int_G \left(g^*\alpha\right)_x \mu(dg),$$

where μ is the Haar measure of G. The integral exists because the integrand is a continuous function of g, and G is compact. (If $v_1, \ldots, v_k \in T_x G/H$, then $(g^*\alpha)_x(v_1, \ldots, v_k) = \alpha_{gx}(dg_x(v_1), \ldots, dg_x(v_k))$, which depends continuously on g since α and the action $G \times G/H \to G/H$ are both differentiable.)

Fix the point $x \in G/H$. Then $(I\alpha)_x$ is an alternating form on $T_x G/H$, i.e. an element of $\wedge^k T^*_x G/H$. Hence $x \mapsto (I\alpha)_x$ defines a section of $\wedge^k T^* G/H$. This is differentiable, because if $X_1, \ldots, X_k$ are differentiable vector fields on G/H, then

$$I\alpha(X_1, \ldots, X_k)(x) = \int_G \alpha_{gx} dg_x(X_1(x)), \ldots, dg_x(X_k(x))\, \mu(dg).$$

The integrand is differentiable in the variables x and g, so the integral is differentiable as a function of x (see Chapter 5, Exercise 28).

In one word, $I\alpha$ is a k-form.

[2] This fact is known as Poincaré lemma, see Bott and Tu [5, Chapter 1].

Lemma 14.7 *$I\alpha$ is G-invariant.*

Proof Given $g, h \in G$ and $v \in T_x G/H$,

$$\left(h^* (I\alpha)\right)_x (v) = (I\alpha)_{hx} (dh_x (v)) = \int_G \left(g^*\alpha\right)_{hx} (dh_x (v)) \mu (dg) .$$

The last integrand is the same as

$$\alpha_{ghx} (dg_{hx} \circ dh_x (v)) = \alpha_{ghx} \left(d (gh)_x (v)\right) = \left((gh)^* \alpha\right)_x (v) ,$$

and so

$$\left(h^* (I\alpha)\right)_x (v) = \int_G \left(g^*\alpha\right)_x (v) \mu (dg) = (I\alpha)_x (v) ,$$

proving the invariance of $I\alpha$. □

Therefore $\alpha \mapsto I\alpha$ defines a map on the space of differential forms, whose image is the subspace of invariant forms, since $I\alpha = \alpha$ if α is invariant. From the definition, the linearity of I is straightforward. Another feature of I, which allows its use in cohomology, is that it commutes with the exterior derivative.

Lemma 14.8 *$I \circ d = d \circ I$.*

Proof Since $g^* d = dg^*$ for all $g \in G$, one can write

$$(Id\alpha)_x = \int_G \left(g^* d\alpha\right)_x \mu (dg) = \int_G \left(dg^*\alpha\right)_x \mu (dg) . \tag{14.10}$$

Formula (14.6), which defines the exterior derivative, computes the last integral above. Let $X_1, \ldots, X_{k+1}$ be vector fields on G/H. The value of $(dg^*\alpha)_x$ on these vector fields is given by two sums. A term in the first sum has form $X_i g^*\alpha(X_1, \ldots, \widehat{X_i}, \ldots, X_{k+1})$ and integrates to

$$\int_G X_i g^*\alpha(X_1, \ldots, \widehat{X_i}, \ldots, X_{k+1})\mu (dg) = X_i \int_G g^*\alpha(X_1, \ldots, \widehat{X_i}, \ldots, X_{k+1})\mu (dg) ,$$

since if a function $f (x, g)$ is differentiable, then $\int_G f (x, g) \mu (dg)$ is differentiable as well and

$$X \int_G f (x, g) \mu (dg) = \int_G Xf (x, g) \mu (dg)$$

for a vector field X. The second sum involves terms of type

$$g^*\alpha([X_i, X_j], X_1, \ldots, X_{k+1}),$$

whose integrals satisfy

$$\int_G g^*\alpha([X_i, X_j], X_1, \ldots, X_{k+1})\mu\,(dg) = \left(\int_G g^*\alpha\mu\,(dg)\right)([X_i, X_j], X_1, \ldots, X_{k+1}).$$

Adding all these terms, the last integral in (14.10) becomes the exterior derivative $d\,(I\alpha)$, concluding the proof. □

At this juncture, it can be proved that the inclusion map $\mathcal{H}^k_{\text{inv}}\,(G/H) \to \mathcal{H}^k_{\text{dR}}\,(G/H)$ is injective.

Proposition 14.9 *If G is compact, then the natural homomorphism*

$$[\alpha]_{\text{inv}} \in \mathcal{H}^k_{\text{inv}}\,(G/H) \mapsto [\alpha] \in \mathcal{H}^k_{\text{dR}}\,(G/H)$$

is one-to-one.

Proof Let α and β be invariant and closed differential forms, and suppose $[\alpha] = [\beta]$. Then there is a form γ such that $\alpha - \beta = d\gamma$. Applying I to both sides and recalling the previous lemma,

$$\alpha - \beta = I\,(\alpha - \beta) = I\,(d\gamma) = d\,(I\gamma)\,.$$

Hence $[\alpha]_{\text{inv}} = [\beta]_{\text{inv}}$ because $I\gamma$ is invariant. □

The surjectivity of the homomorphisms depends on finding, for each closed form β, a closed invariant form α such that $[\alpha] = [\beta]$. The candidate is the invariant form $\alpha = I\beta$, which is closed if β is closed since $d\alpha = d\,(I\beta) = I\,(d\beta) = 0$. Surjectivity will then hold as soon as one shows that any closed form β is cohomologous to $I\beta$. This will be proved with the additional hypothesis that G is connected.

Getting hold of the required invariant form α satisfying $[\alpha] = [\beta]$ requires a little C^1 singular homology. This homology theory is defined on the real vector space

$$C = \bigoplus_{k\geq 0} C_k,$$

sum of $\mathbb{R}$-vector spaces C_k generated by k-dimensional cycles of class C^1 on $M = G/H$. By definition, a k-dimensional cycle is a continuous map $\sigma : \Delta_k \to M$, where Δ_k is the k-dimensional simplex

$$\Delta_k = \{(x_1, \ldots, x_k) \in \mathbb{R}^k : x_i \geq 0,\ x_1 + \cdots + x_k \leq 1\}.$$

A cycle σ is said to be of class C^1 if it is the restriction to Δ_k of a C^1 map $\sigma : U \to M$ on an open set $U \subset \mathbb{R}^{k+1}$ containing Δ_{k+1}.[3]

[3]Restricting to C^1 cycles is necessary in order to apply Stokes' theorem. In the classical article [15], Eilenberg showed that, on a C^r manifold, the singular homology of cycles of class C^r coincides

The space of cycles C admits, in each degree, a boundary operator $\partial_k : C_k \to C_{k-1}$ mapping cycles $\sigma : \Delta_k \to M$ to $\partial_k \sigma = \sum_f \varepsilon_f \sigma_f$. The sum is taken over the faces f of Δ_k, which are $(k-1)$-simplexes. The coefficients $\varepsilon_f = \pm 1$ reflect the choice of orientation of the boundary of Δ_k, and σ_f is the restriction of σ to f.

The boundary operators satisfy $\partial_k \circ \partial_{k+1} = 0$, implying $\mathrm{im}\partial_{k+1} \subset \ker \partial_k$. Thus the singular homology of M is defined by

$$\mathcal{H}_k(M) = \ker \partial_k / \mathrm{im}\partial_{k+1}.$$

Two k-cycles, σ_1 and σ_2 are homologous if there exists $\tau \in C_{k+1}$ such that $\sigma_1 - \sigma_2 = \partial_{k+1}\tau$, i.e. $\sigma_1 - \sigma_2 \in \mathrm{im}\partial_{k+1}$. The homology class of $\sigma \in C$ is written as $[\sigma]$.

The explicit expression for the boundary operators ∂_k will not be given here. It should be noted, however, that they are defined so to achieve **Stokes' theorem**, which may be encapsulated as

$$\int_\sigma d\alpha = \int_{\partial_{k+1}\sigma} \alpha. \tag{14.11}$$

In the formula, α is a k-form and $\sigma : \Delta_{k+1} \to M$ a $(k+1)$-cycle of class C^1.

The reason for working with differentiable cycles, and the corresponding homology, is to make sense to the integral of a differential form with respect to a cycle and therefore have access to Stokes' theorem. For a cycle $\sigma : U \to M$ of class C^1 with $\Delta_k \subset U$, the integral is defined by taking $\sigma^*\alpha$, a volume form on $U \subset \mathbb{R}^k$. The integral of α over σ is defined as

$$\int_\sigma \beta = \int_{\Delta_k} f_{\sigma,\beta}(x)\, dx,$$

where dx is the Lebesgue on $\mathbb{R}^k$, and $f_{\sigma,\beta} : \Delta_k \to \mathbb{R}$ is the function

$$\sigma^*\beta = f_{\sigma,\beta}(x)\, dx_1 \wedge \cdots \wedge dx_k, \tag{14.12}$$

with $dx_1 \wedge \cdots \wedge dx_k$ being the volume form of $\mathbb{R}^k$ in the coordinate system $(x_1, \ldots, x_k)$.

Now, by Stokes' theorem, if α is a closed k-form

$$\int_{\partial_{k+1}\sigma} \alpha = 0$$

for any C^1 cycle σ of dimension $k+1$. Hence if τ_1 and τ_2 are homologous k-cycles then

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with the singular homology of continuous cycles, the latter being the customary framework among the singular homologies of topological spaces.

$$\int_{\tau_1} \alpha = \int_{\tau_2} \alpha.$$

By this relation, the integral of α passes to the quotient and defines a linear map $T_\alpha : \mathcal{H}_k(M) \to \mathbb{R}$,

$$T_\alpha[\sigma] = \int_\sigma \alpha.$$

The linear functional T_α actually depends only on the cohomology class of α. This is because if α and β are closed k-forms with $\alpha - \beta = d\gamma$ and σ is a k-cycle, then

$$\int_\sigma (\alpha - \beta) = \int_\sigma d\gamma = \int_{\partial_k \sigma} \gamma,$$

and the last integral is zero if $\partial_k \sigma = 0$. The converse of this statement is one of the main results in the theory of de Rham cohomology. It is stated in the next proposition, which will not be proved.

Proposition 14.10 *Let α and β be closed k-forms. Then $T_\alpha = T_\beta$ if and only if α and β are cohomologous.*

The proof of the surjectivity of the homomorphism $\mathcal{H}^k_{\text{inv}}(G/H) \to \mathcal{H}^k_{\text{dR}}(G/H)$ depends on two more lemmas.

The first lemma is related to the action of G on the singular homology $\mathcal{H}_k(G/H)$. If $g \in G$ and $\sigma : \Delta_k \to G/H$ is a cycle of class C^1, then $g\sigma = g \circ \sigma$ is also a cycle. The composition of $g \in G$ with cycles on G/H induces a linear map on $C = \bigoplus_{k \geq 0} C_k$, also denoted by g. It follows from the definition of boundary operator that g commutes with ∂_k, and maps are induced on the homology spaces $\mathcal{H}_k(G/H)$. These are usually denoted by $g_*[\sigma] = [g\sigma]$, $\sigma \in C$.

Lemma 14.11 *If G is connected, then $g_* = \mathrm{id}$ for all $g \in G$.*

Proof For any $g \in G$, there is a curve g_t, $t \in [0, 1]$, between $g_0 = 1$ and $g_1 = g$. This path defines a homotopy between the map $g : G/H \to G/H$ and the identity map. But homotopic maps induce the same map on singular homology, and so $g_* = \mathrm{id}$. □

Lemma 14.12 *Let β be a differential form and σ a cycle. Then,*

$$\begin{aligned} f_{\sigma, I\beta}(x) &= \int_G f_{\sigma, g^*\beta}(x)\, \mu(dg) \\ &= \int_G f_{g\sigma, \beta}(x)\, \mu(dg), \end{aligned}$$

where $f_{\sigma,\alpha}$ was defined in (14.12).

Proof Let $\{e_1, \ldots, e_k\}$ be the standard basis of $\mathbb{R}^k$. Then

$$f_{\sigma, I\beta}(x) = \sigma^*(I\beta)_x(e_1, \ldots, e_k) = I\beta_{\sigma(x)}(d\sigma_x e_1, \ldots, d\sigma_x e_k).$$

By definition of $I\beta$, the right hand side reads

$$\begin{aligned}\int_G \left(g^*\beta\right)_{\sigma(x)}(d\sigma_x e_1, \ldots, d\sigma_x e_k)\,\mu(dg) &= \int_G \sigma^*\left(g^*\beta\right)_{\sigma(x)}(e_1, \ldots, e_k)\,\mu(dg)\\ &= \int_G f_{\sigma, g^*\beta}(x)\,\mu(dg).\end{aligned}$$

This proves the first equality in the statement. The second one follows from $f_{\sigma, g^*\beta}(x) = f_{g\sigma,\beta}$ because $\sigma^*(g^*\beta) = (g\sigma)^*\beta$. □

It is now possible to prove that a differential form β is cohomologous to its average $\alpha = I\beta$ with respect to the Haar measure of G.

Proposition 14.13 *Let G be compact and connected. Given a differential form β, write, as before,*

$$I\beta = \int_G \left(g^*\beta\right)\mu(dg).$$

Then β are $I\beta$ are cohomologous.

Proof By Proposition 14.10, one is required to prove $T_{I\beta} = T_\beta$. If $[\sigma]$ is a singular homology class, then by definition

$$T_{I\beta}[\sigma] = \int_\sigma I\beta = \int_{\Delta_k} f_{\sigma, I\beta}(x)\,dx.$$

But by Lemma 14.12, $f_{\sigma, I\beta}(x) = \int_G f_{g\sigma,\beta}(x)\,\mu(dg)$, so

$$T_{I\beta}[\sigma] = \int_{\Delta_k}\int_G f_{g\sigma,\beta}(x)\,\mu(dg)\,dx.$$

Interchanging the order of integration (which is allowed by Fubini theorem), it follows that

$$\begin{aligned}T_{I\beta}[\sigma] &= \int_G \left(\int_{\Delta_k} f_{g\sigma,\beta}(x)\,dx\right)\mu(dg)\\ &= \int_G \left(\int_{g\sigma}\beta\right)\mu(dg)\\ &= \int_G T_\beta[g\sigma]\,\mu(dg).\end{aligned}$$

Now Lemma 14.11 ensures that $T_\beta[g\sigma] = T_\beta[\sigma]$, and hence

$$T_{I\beta}[\sigma] = \int_G T_\beta[\sigma]\,\mu(dg) = I_\beta[\sigma].$$

This ends the proof. □

As mentioned earlier, the fact that $I\beta \sim \beta$ implies that the natural homomorphism $\mathcal{H}^k_{\text{inv}}(G/H) \to \mathcal{H}^k_{\text{dR}}(G/H)$ is surjective since $[\beta]$ is the image of the invariant class $[I\beta]$. This proves the following theorem about the de Rham cohomology of homogeneous spaces of compact and connected Lie groups.

Theorem 14.14 *If G is compact and connected, then the homomorphism*

$$[\alpha]_{\text{inv}} \in \mathcal{H}^k_{\text{inv}}(G/H) \mapsto [\alpha] \in \mathcal{H}^k_{\text{dR}}(G/H)$$

is an isomorphism.

Formula (14.8) for the exterior derivative of invariant forms tells that $d : \wedge^k\mathfrak{m} \to \wedge^{k+1}\mathfrak{m}$ depends only on algebraic properties of the Lie algebra, as do the cohomology spaces $\mathcal{H}^k_{\text{inv}}(G/H)$. In particular, the latter is finite dimensional.

The following theorem summarizes the results of this section. It reduces the de Rham cohomology of G/H to an "algebraic" cohomology.

Theorem 14.15 *Let G be a compact and connected Lie group and $H \subset G$ a closed subgroup with Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$. Take an $\text{Ad}(H)$-invariant subspace $\mathfrak{m} \subset \mathfrak{g}$ such that $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$ and denote by $\wedge^k_{\text{inv}}\mathfrak{m}$ the space of $\text{Ad}(H)$-invariant k-forms on $\mathfrak{m}$. Define $d_k : \wedge^k_{\text{inv}}\mathfrak{m} \to \wedge^{k+1}_{\text{inv}}\mathfrak{m}$ by*

$$d_k\alpha(X_1, \ldots, X_{k+1}) = \sum_{i<j}(-1)^{i+j}\alpha([X_i, X_j]_{\mathfrak{m}}, X_1, \ldots, \widehat{X_i}, \ldots, \widehat{X_j}, \ldots, X_{k+1})$$

with $X_1, \ldots, X_{k+1} \in \mathfrak{m}$, where $[X_i, X_j]_{\mathfrak{m}}$ is the projection on $\mathfrak{m}$ of the bracket $[X_i, X_j]$. Then $\mathcal{H}^k_{\text{dR}}(G/H)$ is isomorphic to

$$\mathcal{H}^k_{\text{inv}}(G/H) = \ker d_k / rm Im\, d_{k-1}.$$

Example A compact and connected Lie group G can be seen as a homogeneous space in two ways, either as $G = G/\{1\}$ under left translations or as $G = G \times G/\Delta_G$ with action $(g,h)\,x = gxh^{-1}$, where $\Delta_G = \{(g,g) : g \in G\}$ is the diagonal. Theorem 14.14 applies to both cases and provides isomorphisms of the de Rham cohomology of G with the cohomology of left invariant forms (for $G = G/\{1\}$) and bi-invariant forms (for $G = G \times G/\Delta$). Therefore the left invariant and bi-invariant cohomology theories coincide.

The left invariant cohomology is defined on the alternating forms on Lie algebra $\mathfrak{g}$ of G by the exterior derivative (14.7). In terms of the cohomology of Lie algebra

representations,[4] this is the cohomology of the trivial $\mathfrak{g}$-representation. On the other hand, the bi-invariant cohomology is defined on the subspace of alternating forms on $\mathfrak{g}$ that are fixed by the adjoint representation, which is typically smaller than the space of all alternating forms. □

Example Consider the homogeneous space $S^n = \mathrm{SO}(n+1)/\mathrm{SO}(n)$, $n \geq 2$. The isotropy representation is equivalent to the canonical representation of $\mathrm{SO}(n)$ on $\mathbb{R}^n$. The latter admits invariant volume forms $a dx_1 \wedge \cdots \wedge dx_n$, for $a \in \mathbb{R}$, since $\det g = 1$ if $g \in \mathrm{SO}(n)$. To see that these are the only nonzero invariant forms, take the basis $\{\varepsilon_1, \ldots, \varepsilon_n\}$ of $(\mathbb{R}^n)^*$ dual to the standard basis. Then a basis for the space $\wedge^m (\mathbb{R}^n)^*$ of m-forms on $\mathbb{R}^n$ is

$$\varepsilon_I = \varepsilon_{i_1} \wedge \cdots \wedge \varepsilon_{i_m}$$

with multi-index $I = \{1 \leq i_1 < \cdots < i_m \leq n\}$. The $\mathrm{SO}(n)$-representation is defined by

$$g \cdot \varepsilon_I = g \cdot \varepsilon_{i_1} \wedge \cdots \wedge g \cdot \varepsilon_{i_m} \qquad g \in \mathrm{SO}(n),$$

where $g \cdot \varepsilon_i = \varepsilon_i \circ g^{-1}$ is the dual representation. The infinitesimal representation reads

$$X \cdot \varepsilon_I = \frac{d}{dt} e^{tX} \varepsilon_I = \sum_{j=1}^{m} X \cdot \varepsilon_{i_j} \qquad X \in \mathfrak{so}(n),$$

where $X \cdot \varepsilon_i = -\varepsilon_i \circ X$ and $X \in \mathfrak{so}(n)$. A form $\alpha \in \wedge^m (\mathbb{R}^n)^*$ is $\mathrm{SO}(n)$-invariant if and only if $X \cdot \alpha = 0$ for all $X \in \mathfrak{so}(n)$.

Take a matrix $A_{jk} = E_{kj} - E_{jk} \in \mathfrak{so}(n)$ with $j < k$. Then $A_{jk} \cdot \varepsilon_j = \varepsilon_k$, $A_{jk} \cdot \varepsilon_k = \varepsilon_j$, and A_{jk} is zero on the remaining basis elements. Hence $A_{jk} \cdot \varepsilon_I = 0$ if $j, k \notin I$ or $j, k \in I$. Moreover, given a multi-index I such that $j \in I$ and $k \notin I$, let $I_{(jk)}$ be the multi-index obtained from I by replacing j with k, placed in the correct position so as to maintain the increasing order of the indices. Then,

$$A_{jk} \cdot \varepsilon_I = (-1)^r \varepsilon_{I_{(jk)}} \qquad \text{and} \qquad A_{jk} \cdot \varepsilon_I = -(-1)^r \varepsilon_{I_{(jk)}},$$

where r is the number of indices $i \in I$ such that $j < i < k$.

Now let $\alpha = \sum_I a_I \varepsilon_I$ be such that $X \cdot \alpha = 0$, $X \in \mathfrak{so}(n)$, and take a multi-index I. If $m < n$, then there exist $j \in I$, $k \notin I$, such that

$$A_{jk} \cdot \alpha = (-1)^r \left(a_I \varepsilon_{I(jk)} - a_{I_{(jk)}} \varepsilon_I \right) + \cdots,$$

[4] See *Álgebras de Lie* [47, Chapter 5] and Varadarajan [53, Section 3.12].

where $\cdots$ indicates a linear combination of the remaining basis elements. As $A_{jk} \cdot \alpha = 0$, it follows that $a_I = 0$, and so $\alpha = 0$.

Based on these calculations, Theorem 14.14 shows that $\mathcal{H}^n(S^n) = \mathbb{R}$ and $\mathcal{H}^m(S^n) = \{0\}$ if $1 \leq m < n$.

The same arguments apply to the projective space

$$\mathbb{R}P^n = \mathrm{SO}(n)/\{\pm 1\} \times \mathrm{O}(n).$$

The isotropy representation ρ on $\mathbb{R}^n$ is now $\rho(\det g, g) = (\det g)\, g$. As before, there are no invariant m-forms if $1 \leq m < n$, so $\mathcal{H}^m(\mathbb{R}P^n) = \{0\}$ if $m < n$. Now consider $m = n$: if $\det g = -1$, then $\det(-g) = (-1)^n \det g = (-1)^{n+1}$, and the volume forms are invariant if and only if n is odd. Therefore $\mathcal{H}^n(\mathbb{R}P^n) = \mathbb{R}$ if n is odd, while all de Rham cohomology groups with n even vanish. □

14.3 Riemannian Manifolds

A Riemannian metric $m(\cdot, \cdot)$ on a manifold M is invariant by the action of a group G if the elements of the group are isometries of the metric:

$$m_{gx}(dg_x u, dg_x v) = m_x(u, v) \qquad u, v \in T_x M \tag{14.13}$$

for all $g \in G$ and $x \in M$. Then both the Levi-Civita connection ∇ and the curvature $R(\cdot, \cdot)$ of m are invariant under any $g \in G$.

When $M = G/H$ is a homogeneous space, an invariant metric is completely determined by its value m_{x_0} at the origin, which is an inner product on $T_{x_0} G/H$ invariant under the isotropy representation. Conversely, an isotropy-invariant inner product on $T_{x_0} G/H$ defines, through (14.13), a Riemannian metric on G/H that is invariant under translations.

A necessary condition for the existence of the inner product m_{x_0} is that the image $\rho(H)$ of H by the isotropy representation ρ is a subgroup of the orthogonal group, whence $\rho(H)$ must have compact closure. This condition is also sufficient in this case, since the closure of $\rho(H)$ admits an invariant inner product and the same happens with $\rho(H)$. In particular, if H is compact, then G/H admits invariant Riemannian metrics.

In presence of inner products invariant under the isotropy representation, their set (and hence the set of invariant metrics on G/H) is a cone. In fact, since the closure of $\rho(H)$ is compact, there is a decomposition

$$T_{x_0} G/H = V_1 \oplus \cdots \oplus V_s$$

in invariant and irreducible subspaces by $\overline{\rho(H)}$ (and therefore by $\rho(H)$), such that an invariant inner product is a sum of inner products on the components V_i. The set of such inner products is parametrized by $\mathbb{R}^s_+$.

The following examples illustrate the general construction.

Example If G (and hence H) is compact, then any homogeneous space G/H admits invariant Riemannian metrics. The number of such metrics depends on the decomposition of the isotropy representation. For example, in the case of the sphere $S^n = \mathrm{SO}(n+1)/\mathrm{SO}(n)$, the isotropy representation is irreducible, so there is essentially a unique invariant metric (the metric induced by the embedding $S^n \hookrightarrow \mathbb{R}^{n+1}$). On the flag manifold $\mathrm{SU}(n)/T$, cf. Section 14.1, there are $\frac{n(n-1)}{2}$ two-dimensional irreducible components. The set of invariant metrics is parametrized by the cone $\mathbb{R}^s_+$, $s = \frac{n(n-1)}{2}$. □

Example Any Lie group G admits Riemannian metrics that are left (or right) invariant. This is because the isotropy group at 1 for the action by (left or right) translations reduces to $\{1\}$. Hence any inner product on T_1G defines a left invariant Riemannian metric and (possibly other) right invariant Riemannian metrics.

On the other hand, if G admits a bi-invariant metric, then its Lie algebra $\mathfrak{g}$ is compact, since the isotropy representation at 1 for $G = G \times G/G$ is the adjoint representation. An inner product $\langle \cdot, \cdot \rangle$ on $\mathfrak{g}$ invariant by the adjoint representation of G is also invariant by the adjoint representation of $\mathfrak{g}$. The existence of such an inner product implies $\mathfrak{g}$ is a compact Lie algebra. Conversely, if $\mathfrak{g}$ is a compact Lie algebra and G is connected, then the invariant inner product on $\mathfrak{g}$ induces a bi-invariant metric on G. □

Example A class of examples of noncompact homogeneous spaces admitting invariant Riemannian metrics are the noncompact **symmetric spaces**. Let $\mathfrak{g}$ be a noncompact semi-simple Lie algebra with Cartan decomposition $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{s}$ and denote by G its adjoint group. As was proved in Chapter 12, the subgroup $K = \langle \exp \mathfrak{k} \rangle$ is compact and the exponential map $\exp : \mathfrak{s} \to S = \exp(\mathfrak{s})$ a diffeomorphism. The homogeneous space G/K is diffeomorphic to S and hence to $\mathfrak{s}$ and admits invariant metrics because K is compact. The tangent space to G/K at the origin is identified with $\mathfrak{s}$. The restriction to $\mathfrak{s}$ of the Cartan–Killing form of $\mathfrak{g}$ is an inner product, see Chapter 12, and so it defines a natural metric on G/K. If $\mathfrak{g}$ is simple, then the adjoint representation of K on $\mathfrak{s}$ is irreducible, and there is an essentially unique invariant metric, namely the Cartan–Killing form.

The case $\mathfrak{g} = \mathfrak{sl}(n, \mathbb{R})$ is special. Here S is the submanifold of positive-definite matrices (with determinant 1), and $\mathfrak{s}$ is the space of traceless symmetric matrices. The Cartan–Killing form on $\mathfrak{s}$ is a multiple of the inner product $\langle X, Y \rangle = \mathrm{tr}(XY)$. □

14.4 Symplectic Manifolds

A **symplectic form** ω on a vector space V ($\dim V < \infty$) is a skew-symmetric bilinear form that is non-degenerate: for any $x \in V$, there exists $y \in V$ such

that $\omega(x, y) \neq 0$. Listed below are a number of algebraic properties enjoyed by symplectic vector spaces.

1. The non-degeneracy of ω is equivalent to the requirement that the linear map $\iota : V \to V^*$, $\iota(x)(\cdot) = \omega(x, \cdot)$, is an isomorphism.
2. For $W \subset V$, let

$$W^{\perp} = \{y \in \mathbb{R}^n : \forall x \in W, \omega(x, y) = 0\}$$

   be the orthogonal subspace with respect to ω. Then $W^{\perp} = \iota^{-1}(W^{\circ})$, where $W^{\circ} \subset V^*$ is the annihilator of W. Since $\dim W + \dim W^{\circ} = \dim V$, it follows that $\dim W + \dim W^{\perp} = \dim V$.
3. A subspace $W \subset V$ is called isotropic if the restriction of ω to $W \times W$ is identically zero. Hence W is isotropic if and only if $W \subset W^{\perp}$. There exist nontrivial isotropic subspaces, since $\omega(x, x) = 0$ for any x, and therefore the subspace generated by any $x \in V$ is isotropic. Moreover, the dimension formula $\dim W + \dim W^{\perp} = \dim V$ implies that $\dim W \leq \dim V/2$ if W is an isotropic subspace.
4. If $W \subset V$ is an isotropic subspace such that $\operatorname{ddim} W < \dim V/2$, then W is properly contained in $W^{\perp}$. Hence, if $x \in W^{\perp} \setminus W$, then $U = W \oplus \langle x \rangle$ is also an isotropic subspace, with $\dim U = \dim W + 1$.
5. It follows from the previous item, by induction, that there exists an isotropic subspace $W \subset V$ with $\dim W = \dim V/2$, and hence $\dim V$ is always even. (An isotropic subspace with maximum dimension $(\dim V)/2$ is called a Lagrangian subspace.)
6. It is possible to write ω in a canonical form. Supposing $\dim V = 2n$, there is a basis $B = \{e_1, \ldots, e_n, f_1 \ldots, f_n\}$ such that

$$\omega(e_i, f_j) = \delta_{ij} \qquad \omega(e_i, e_j) = 0 \qquad \omega(f_i, f_j) = 0.$$

   The matrix of ω in this basis is

$$[\omega]_B = J = \begin{pmatrix} 0 & -1_{n\times n} \\ 1_{n\times n} & 0 \end{pmatrix}.$$

   Hence if $\{dx_1, \ldots, dx_n, dy_1, \ldots, dy_n\}$ denotes the dual basis to B,

$$\omega = dx_1 \wedge dy_1 + \cdots + dx_n \wedge dy_n. \tag{14.14}$$

A **symplectic form** on a differentiable manifold M is a closed 2-form ω, which is non-degenerate: for any $x \in M$ and $v \in T_x M$, there exists $w \in T_x M$ such that $\omega_x(v, w) \neq 0$. For each $x \in M$, the bilinear form ω_x is a symplectic form on the vector space $T_x M$, so the existence of a symplectic form on M implies $\dim M$ is even.

A manifold M endowed with a symplectic form is called a symplectic manifold. A classical result on symplectic manifolds is **Darboux's theorem**,[5] which ensures that ω is locally equivalent to the standard symplectic form (14.14) on a vector space. Precisely, around any $x \in M$, there exist coordinates $(x_1, \ldots, x_n, y_1, \ldots, y_n)$ such that ω equals (14.14), locally.

On a symplectic manifold (M, ω), the map $\iota_x(v) = \omega_x(v, \cdot)$, $x \in M$, is an isomorphism between the tangent space $T_x M$ and cotangent space $T_x^* M$. The ensuing diffeomorphism between the bundles TM and T^*M allows to define the fundamental object of symplectic geometry: Hamiltonian vector fields.

Let $f : M \to \mathbb{R}$ be a differentiable function (say C^k, $k \geq 1$) with differential df_x, $x \in M$. Then $\iota_x^{-1} df_x$ is a tangent vector at x, so there is a vector field X_f on M defined by

$$df_x(\cdot) = \omega\left(X_f(x), \cdot\right).$$

Such X_f is called a **Hamiltonian vector field** on M. The function f is called the Hamiltonian function (or energy function) of X_f. Observe that there is no uniqueness for Hamiltonian functions, because if f is a Hamiltonian function of the Hamiltonian vector field X and $c \in \mathbb{R}$ is a constant, then $f + c$ is also a Hamiltonian function of X.

Listed below are some properties of Hamiltonian vector fields, which will be used later in the study of group actions.

1. Let X be a Hamiltonian vector field with flow ϕ_t. Then the Lie derivative $L_X \omega = \frac{d}{dt}\left(\phi_t^* \omega\right)_{t=0}$ vanishes, implying that $\phi_t^* \omega = \omega$. In fact, Cartan's formula for the Lie derivative reads

$$L_X \omega = d i_X \omega + i_X d\omega,$$

   where $i_X \omega$ is the 1-form $i_X \omega(\cdot) = \omega(X, \cdot)$. In this formula, the last term is zero because $d\omega = 0$. On the other hand, $d(i_X \omega) = d(df) = 0$ if f is Hamiltonian function of X.
2. Conversely, suppose X is a vector field on (M, ω) such that $L_X \omega = 0$. Applying Cartan's formula shows that the 1-form $i_X \omega$ is closed. If it is further exact, $i_X \omega = df$, then X is Hamiltonian. This occurs, for instance, when the first de Rham cohomology group is trivial: $\mathcal{H}^1_{\mathrm{dR}}(M) = \{0\}$. In any case, M is locally diffeomorphic to $\mathbb{R}^n$, and therefore $i_X \omega$ is always locally exact provided it is closed. Hence there exist open sets in M on which X restricts to a Hamiltonian vector field. For this reason, a vector field X such that $L_X \omega = 0$ is called locally Hamiltonian.
3. Let X and Y be the Hamiltonian fields of the functions f_X and f_Y, respectively. The Lie bracket $[X, Y]$ is the Hamiltonian vector field of the function $\omega(X, Y)$,

[5] See Abraham and Marsden [1].

called the **Poisson bracket** of f_X, f_Y and indicated $\{f_X, f_Y\}$. The Poisson bracket is skew-symmetric and satisfies the Jacobi identity.[6]

A differentiable action $G \times M \to M$ of a Lie group G on a symplectic manifold (M, ω) is called a **symplectic action** if the elements of G preserve ω: $g^*\omega = \omega$ for all $g \in G$. In this case, the infinitesimal action of Lie algebra $\mathfrak{g}$ of G is given by locally Hamiltonian fields. This is because if $X \in \mathfrak{g}$, the flow of the induced field $\widetilde{X}$ is the exponential e^{tX}. Hence $L_X\omega = 0$ since $\left(e^{tX}\right)^* \omega = \omega$. Conversely, if the vector fields $\widetilde{X}$, $X \in \mathfrak{g}$ are locally Hamiltonian and G is connected, the action of G is symplectic. A **Hamiltonian action** is a symplectic action in which the vector fields $\widetilde{X}$, $X \in \mathfrak{g}$, are (globally) Hamiltonian.

Not every symplectic action is Hamiltonian (see an example below). However, it is worth noting that if G and M are connected and G acts on (M, ω) symplectically, then the action lifts to a Hamiltonian action on the universal covering space of M. In fact, let $p : \widetilde{G} \to G$ and $\pi : \widetilde{M} \to M$ denote the universal coverings. By the Lie–Palais theorem, the G-action on M lifts to a $\widetilde{G}$-action on $\widetilde{M}$ such that $\pi \circ g = p(g) \circ \pi$ for every $g \in \widetilde{G}$ (see Chapter 13, Exercise 28). The action of $\widetilde{G}$ preserves the symplectic form $\pi^*\omega$ on $\widetilde{M}$, and therefore this action is Hamiltonian because the de Rham cohomology of $\widetilde{M}$ is zero (Proposition 14.10).

Let us specialize M to a homogeneous space G/H. An invariant 2-form ω is completely determined by its value ω_0 at the origin x_0 of G/H, which is a skew-symmetric bilinear form on $T_{x_0}G/H$ that is invariant by the isotropy representation: $\left(dh_{x_0}\right)^* \omega_0 = \omega_0$ for all $h \in H$. The 2-form ω is then given at T_xG/H by

$$\omega_x = \left(dg_x^{-1}\right)^* \omega_0,$$

where $g \in G$ satisfies $gx_0 = x$. Since dg_x^{-1} is an isomorphism, it follows that if ω_0 is non-degenerate, each ω_x is also non-degenerate. At the same time, Proposition 14.6 gives the expression for the exterior derivative of an invariant form on a homogeneous space, which for 2-forms comes down to

$$(d\omega)(\widetilde{X}_1, \widetilde{X}_2, \widetilde{X}_3) = -\omega([\widetilde{X}_1, \widetilde{X}_2], \widetilde{X}_3) + \omega([\widetilde{X}_1, \widetilde{X}_3], \widetilde{X}_2) - \omega([\widetilde{X}_2, \widetilde{X}_3], \widetilde{X}_1). \tag{14.15}$$

The above equals the cyclic sum

$$-\left(\omega([\widetilde{X}_1, \widetilde{X}_2], \widetilde{X}_3) + \omega([\widetilde{X}_2, \widetilde{X}_3], \widetilde{X}_1) + \omega([\widetilde{X}_3, \widetilde{X}_2], \widetilde{X}_1)\right).$$

Looking at this from, the point of view of the algebraic formula (14.8) will give a condition for the exactness of ω. Let $\mathfrak{m}$ be a complement to the Lie algebra $\mathfrak{h}$ of H, and denote by $\mathrm{pr}_{\mathfrak{m}}$ the projection on $\mathfrak{m}$ in the decomposition $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$. For $X, Y \in \mathfrak{m}$ put $[X, Y]_{\mathfrak{m}} = \mathrm{pr}_{\mathfrak{m}}[X, Y]$. Then $d\omega = 0$ if and only if

[6] See Abraham and Marsden [1].

$$\omega_0([X_1, X_2]_{\mathfrak{m}}, X_3) + \omega_0([X_2, X_3]_{\mathfrak{m}}, X_1) + \omega_0([X_3, X_2]_{\mathfrak{m}}, X_1) = 0 \qquad (14.16)$$

for every triple (X_1, X_2, X_3) of elements of $\mathfrak{m}$. The following proposition summarizes this discussion.

Proposition 14.16 *Choose* $\mathfrak{m}$ *so that* $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$. *Then a symplectic form on* G/H *is determined by a skew-symmetric and non-degenerate bilinear form* ω_0 *on* $\mathfrak{m}$ *that satisfies (14.16).*

If ω is an invariant symplectic form on G/H, then the action of G on G/H is symplectic. It may not be Hamiltonian, as the following example shows.

Example Let G be the torus $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ acting on itself by translations. The Lie algebra of T^2 is the abelian algebra $\mathbb{R}^2$, which admits the standard symplectic form $\omega = dx \wedge dy$. The latter induces an invariant symplectic form on T^2 and hence a symplectic, non-Hamiltonian action. In fact, the invariant vector fields are the vector fields induced on T^2 by the constant fields $a\frac{\partial}{\partial x} + b\frac{\partial}{\partial y}$, for $a, b \in \mathbb{R}$, of $\mathbb{R}^2$. These vector fields are not Hamiltonian. To see this, take for example the vector field X induced by $\partial/\partial x$. The 1-form $\alpha = i_X\omega = \omega(\partial/\partial x, \cdot)$, restricted to the open set of T^2 represented by the square $(0, 1)^2 \subset \mathbb{R}^2$, is just dy. This 1-form is not exact, because if $\alpha = df$, then f must satisfy $f(x, y) = y + c$ at $(0, 1)^2$ for some constant c. But a function of type $y + c$ does not extend continuously to a function on T^2, showing that the action cannot be Hamiltonian. □

14.4.1 Coadjoint Representation

A fundamental class of examples of Hamiltonian homogeneous spaces are the orbits of the coadjoint representation Ad^* of G on the dual $\mathfrak{g}^*$ of the Lie algebra $\mathfrak{g}$. This representation is defined by $\mathrm{Ad}^*(g)\,\alpha = \alpha \circ \mathrm{Ad}\left(g^{-1}\right)$, $g \in G$ and $\alpha \in \mathfrak{g}^*$. Its infinitesimal representation $\mathrm{ad}^* : \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g}^*)$ is given by $\mathrm{ad}^*(X)\,\alpha = -\alpha \circ \mathrm{ad}(X)$. The group G acts on $\mathfrak{g}^*$ by the representation Ad^*. For this action, the vector fields $\widetilde{X}$ induced by elements $X \in \mathfrak{g}$ are given by $\widetilde{X} = \mathrm{ad}^*(X)$.

For $\alpha \in \mathfrak{g}^*$, let $G \cdot \alpha = \mathrm{Ad}^*(G)\,\alpha$ be the orbit under the coadjoint representation. This coadjoint orbit is identified with the homogeneous space G/Z_α, where Z_α is the closed subgroup

$$Z_\alpha = \{g \in G : \alpha \circ \mathrm{Ad}\left(g^{-1}\right) = \alpha\}.$$

The Lie algebra of Z_α is given by $\mathfrak{z}_\alpha = \{X \in \mathfrak{g} : \alpha \circ \mathrm{ad}(X) = 0\}$, that is,

$$\mathfrak{z}_\alpha = \{X \in \mathfrak{g} : \forall Y \in \mathfrak{g},\ \alpha[X, Y] = 0\}.$$

The tangent space to $G \cdot \alpha$ at α is

$$T_\alpha (G \cdot \alpha) = \{\mathrm{ad}^* (X)\, \alpha : X \in \mathfrak{g}\}.$$

The **Kirillov–Kostant–Souriau symplectic form** Ω (abbreviated KKS) on the adjoint orbit $G \cdot \alpha$ is

$$\Omega_\alpha \left(\widetilde{X} (\alpha), \widetilde{Y} (\alpha)\right) = \alpha [X, Y] \qquad X, Y \in \mathfrak{g}. \tag{14.17}$$

There are two equivalent ways of interpreting this expression as a symplectic form on the adjoint orbit $G \cdot \alpha$. The first is to think of Ω_α as a skew-symmetric bilinear form on $T_\alpha (G \cdot \alpha)$. As it is invariant by the isotropy representation, it defines a translation-invariant 2-form on $G \cdot \alpha$. The second interpretation is to define, for each $\beta \in G \cdot \alpha$, the form Ω_β as in (14.17) with β in place of α, and consider $\beta \mapsto \Omega_\beta$ as a 2-form on $G \cdot \alpha$. It will be proved below that both interpretations produce the same 2-form.

If Ω_α is viewed as bilinear form on $T_\alpha (G \cdot \alpha)$, the following items show that Ω_α defines a translation-invariant symplectic form at $G \cdot \alpha = G/Z_\alpha$.

1. Ω_α is well defined. In fact, take $X, X_1, Y, Y_1 \in \mathfrak{g}$ such that $\widetilde{X} (\alpha) = \widetilde{X}_1 (\alpha)$ and $\widetilde{Y} (\alpha) = \widetilde{Y}_1 (\alpha)$, so $\alpha \circ \mathrm{ad}\, (X) = \alpha \circ \mathrm{ad}\, (X_1)$ and $\alpha \circ \mathrm{ad}\, (Y) = \alpha \circ \mathrm{ad}\, (Y_1)$. Hence,

$$\begin{aligned} \Omega_\alpha \left(\widetilde{X}_1 (\alpha), \widetilde{Y}_1 (\alpha)\right) &= \alpha \circ \mathrm{ad}\, (X_1) (Y_1) = \alpha \circ \mathrm{ad}\, (X) (Y_1) \\ &= -\alpha \circ \mathrm{ad}\, (Y_1) (X) = -\alpha \circ \mathrm{ad}\, (Y) (X) \\ &= \Omega_\alpha \left(\widetilde{X} (\alpha), \widetilde{Y} (\alpha)\right). \end{aligned}$$

2. Ω_α is skew-symmetric.
3. Ω_α is non-degenerate. In fact, take $0 \neq v \in T_\alpha (G \cdot \alpha)$ and choose $X \in \mathfrak{g}$ such that $v = \widetilde{X} (\alpha)$. Then $\alpha \circ \mathrm{ad}\, (X) \neq 0$, that is, there exists $Y \in \mathfrak{g}$ such that $\alpha [X, Y] \neq 0$. This means $\Omega_\alpha \left(v, \widetilde{Y} (\alpha)\right) \neq 0$, proving non-degeneracy.
4. Ω_α is invariant by the isotropy representation. Take $X \in \mathfrak{g}$ and $h \in Z_\alpha$, so the isotropy representation is $dh_\alpha \left(\widetilde{X} (\alpha)\right) = \widetilde{\mathrm{Ad}\, (h)\, X} (\alpha)$. Then,

$$\begin{aligned} \Omega_\alpha \left(dh_\alpha \left(\widetilde{X} (\alpha)\right), dh_\alpha \left(\widetilde{Y} (\alpha)\right)\right) &= \alpha [\mathrm{Ad}\, (h)\, X, \mathrm{Ad}\, (h)\, Y] \\ &= \alpha \circ \mathrm{Ad}\, (h) [X, Y]. \end{aligned}$$

   But $\alpha \circ \mathrm{Ad}\, (h) = \alpha$, so the last term equals $\alpha [X, Y] = \Omega_\alpha \left(\widetilde{X} (\alpha), \widetilde{Y} (\alpha)\right)$, and invariance is granted.
5. The 2-form Ω on $G \cdot \alpha = G/Z_\alpha$ defined by Ω_α is closed. This is a consequence of the Jacobi identity: if $X, Y, Z \in \mathfrak{g}$, then by Proposition 14.6 (see also (14.15)),

$$\begin{aligned} (d\Omega)_\alpha (\widetilde{X} (\alpha), \widetilde{Y} (\alpha), \widetilde{Z} (\alpha)) &= -\Omega_\alpha([\widetilde{X}, \widetilde{Y}], \widetilde{Z}) - \Omega_\alpha([\widetilde{Y}, \widetilde{Z}], \widetilde{X}) - \Omega_\alpha([\widetilde{Z}, \widetilde{Y}], \widetilde{X}) \\ &= -\alpha ([X, Y], Z] + [Y, Z], X] + [Z, Y], X]) = 0. \end{aligned}$$

The Kirillov–Kostant–Souriau form Ω on an adjoint orbit is built on the basis of the above facts. On an element $\beta = \mathrm{Ad}^*(g)\,\alpha$ in the orbit, the translation Ω_β of Ω_α is

$$\begin{aligned}\Omega_\beta\left(\widetilde{X}(\beta), \widetilde{Y}(\beta)\right) &= \Omega_\alpha\left(dg_\beta^{-1}\widetilde{X}(\beta), dg_\beta^{-1}\widetilde{Y}(\beta)\right)\\ &= \Omega_\alpha\left(\widetilde{\mathrm{Ad}\left(g^{-1}\right)X}(\alpha), \widetilde{\mathrm{Ad}\left(g^{-1}\right)Y}(\alpha)\right)\\ &= \alpha\left[\mathrm{Ad}\left(g^{-1}\right)X, \mathrm{Ad}\left(g^{-1}\right)Y\right]\\ &= \alpha\circ\mathrm{Ad}\left(g^{-1}\right)[X,Y].\end{aligned}$$

Hence,

$$\Omega_\beta\left(\widetilde{X}(\beta), \widetilde{Y}(\beta)\right) = \beta[X,Y]$$

has the same expression as the one used to define Ω_α.

The Kirillov–Kostant–Souriau construction shows, in particular, that the orbits of the coadjoint representation have even dimension.

One of the essential properties of the KKS form Ω is that the action of G on coadjoint orbits is Hamiltonian. Hence the induced vector fields $\widetilde{X}$, $X \in \mathfrak{g}$, are Hamiltonian with respect to Ω, as shall be proved next.

Proposition 14.17 *Let $\mathcal{O} \subset \mathfrak{g}^*$ be a coadjoint orbit. Given $X \in \mathfrak{g}$, the vector field $\widetilde{X} = \mathrm{ad}^*(X)$ is Hamiltonian with Hamiltonian function $f_X : \mathcal{O} \to \mathbb{R}$ given by*

$$f_X(\alpha) = \alpha(X) \qquad \alpha \in \mathcal{O}.$$

Proof As f_X is the restriction of a linear map, its differential $(df_X)_\alpha$, $\alpha \in \mathcal{O}$, coincides with f_X. Explicitly, if $\widetilde{Y}(\alpha) = \mathrm{ad}^*(Y)(\alpha)$, $Y \in \mathfrak{g}$, is a tangent vector then $(df_X)_\alpha\left(\widetilde{Y}(\alpha)\right) = \left(\mathrm{ad}^* Y(\alpha)\right)(X)$. Hence,

$$\begin{aligned}(df_X)_\alpha\left(\widetilde{Y}(\alpha)\right) &= -\alpha\circ\mathrm{ad}(Y)(X)\\ &= \alpha[X,Y]\\ &= \Omega_\alpha\left(\widetilde{X}(\alpha), \widetilde{Y}(\alpha)\right),\end{aligned}$$

showing that $\widetilde{X}$ is the Hamiltonian vector field defined by f_X. □

Examples of coadjoint orbits and their KKS symplectic forms are presented below.

Example The Cartan–Killing form $\mathcal{K}(\cdot,\cdot)$ of a semi-simple Lie algebra $\mathfrak{g}$ is non-degenerate and defines an isomorphism $K : \mathfrak{g} \to \mathfrak{g}^*$ by $K(X)(\cdot) = \mathcal{K}(X,\cdot)$. This map swaps adjoint and coadjoint representations, i.e. $K\,\mathrm{Ad}(g) = \mathrm{Ad}^*(g)\,K$ for all

$g \in G$. This is a consequence of the Ad-invariance of $\mathcal{K}$. Therefore K maps adjoint orbits to coadjoint orbits diffeomorphically. Then the Kirillov–Kostant–Souriau form Ω of a coadjoint orbit can be transported to an adjoint orbit by the pullback $K^*\Omega$. Moreover, the action of G on adjoint orbits is Hamiltonian since the induced fields $\widetilde{X} = \mathrm{ad}(X)$ satisfy $K\,\mathrm{ad}(X) = \mathrm{ad}^*(X)\,K$, and so $\widetilde{X}$ is the Hamiltonian vector field of the "height function" $f_X(\cdot) = \mathcal{K}(X, \cdot)$. Summing up, in the semi-simple case, the adjoint representation behaves like the coadjoint representation. These comments extend to any Lie algebra admitting a non-degenerate invariant bilinear form, such as reductive Lie algebras. □

Example Consider a concrete semi-simple group, $G = \mathrm{Sl}(2, \mathbb{R})$ and $\mathfrak{g} = \mathfrak{sl}(2, \mathbb{R})$. Since $\mathrm{Ad}(g)\,X = gXg^{-1}$, the polynomial $\det X$ is constant on each adjoint orbit. There are three families of orbits: (1) those of nilpotent matrices contained in the double cone $\{X : \det X = 0\}$, of which there are exactly three, namely the origin and the two halves of the cone, (2) those of matrices with imaginary eigenvalues, each of which is a sheet of the two-sheeted hyperboloid $\det X = c > 0$, and (3) the one-sheeted hyperboloid $\det X = c < 0$.

In coordinates (x, y, z) for the basis made by

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

the polynomial reads $\det X = x^2 - y^2 - z^2$. The double cone has equation $x^2 = y^2 + z^2$, while the hyperboloids are $x^2 = y^2 + z^2 = c > 0$ (two-sheeted, inside the cone) and $x^2 = y^2 + z^2 = c < 0$ (one-sheeted, outside the cone). □

Example Let $\mathfrak{g}$ be the Heisenberg Lie algebra with basis $\{X, Y, Z\}$ such that $[X, Y] = Z$ and other independent brackets zero. Denote by $\{\alpha, \beta, \gamma\}$ the dual basis to $\{X, Y, Z\}$. The coadjoint orbits of elements $x\alpha + y\beta + z\gamma$ with $z = 0$ are zero-dimensional, whereas the other orbits have dimension 2.

On the other hand, the adjoint orbit of X has dimension 1, since the centralizer of X is the subalgebra generated by $\{X, Z\}$. This adjoint orbit does not admit symplectic forms. □

14.4.2 Moment Map

A useful tool in the study of Hamiltonian actions is the moment map, a function that allows to relate an arbitrary Hamiltonian action with the coadjoint representation.

Let (M, ω) be a symplectic manifold, G a Lie group with Lie algebra $\mathfrak{g}$, and $G \times M \to M$ an action. The basic hypothesis is that this action is Hamiltonian. To simplify, the exposition M will be assumed connected.

Definition 14.18 A moment map of a Hamiltonian action is a map $\mu : M \to \mathfrak{g}^*$ such that for every $X \in \mathfrak{g}$, the function $f_X(x) = \mu(x)(X)$ is a Hamiltonian function for $\widetilde{X}$, that is, $df_X = \iota_{\widetilde{X}}\omega$.

Example Let $M = \mathrm{Ad}^*(G)\alpha$ be a coadjoint orbit endowed with the KKS form Ω. A vector field $\widetilde{X} = \mathrm{ad}^*(X)$, $X \in \mathfrak{g}$, is Hamiltonian for the function $f_X(\beta) = \beta(X)$, for $\beta \in \mathrm{Ad}^*(G)\alpha$. Therefore the action of G on the orbit is Hamiltonian, and $\mu = \mathrm{id}$ is a moment map. □

Any Hamiltonian action admits moment maps. Take in fact a basis $\{X_1, \ldots, X_N\}$ of $\mathfrak{g}$ and let $\{\alpha_1, \ldots, \alpha_N\}$ be the dual basis of $\mathfrak{g}^*$. Then,

$$\mu(x) = f_{X_1}(x)\alpha_1 + \cdots + f_{X_N}(x)\alpha_N$$

is a moment map provided $f_{X_1}, \ldots, f_{X_N}$ are Hamiltonian functions for the induced vector fields $\widetilde{X}_1, \ldots, \widetilde{X}_N$.

There is no uniqueness since if μ is a moment map and $\alpha \in \mathfrak{g}^*$ is a constant, then $\overline{\mu} = \mu + \alpha$ is another moment map. Vice versa, if μ and $\overline{\mu}$ are moment maps, then $\mu - \overline{\mu}$ is constant because M is connected. In the connected case, the set of moment maps is an affine space modeled on $\mathfrak{g}^*$.

In what follows the main issue will be deciding whether the moment maps (or some of them) are equivariant with respect to the actions of G on M and the coadjoint representation. The question is therefore to know whether

$$\mu(gx) = \mathrm{Ad}^*(g)\,\mu(x)$$

holds for all $g \in G$ and $x \in M$. Not every Hamiltonian action admits an Ad^*-equivariant moment map. The actions admitting such a moment map are called **equivariant Hamiltonian actions** (or Ad^*-equivariant).

Before delving deeper into the existence of Ad^*-equivariant moment maps, here is a homomorphism-flavored property of these maps.

Proposition 14.19 *Suppose μ is an Ad^*-equivariant moment map. Then,*

$$\mu(x)[X, Y] = \omega_x\left(\widetilde{X}, \widetilde{Y}\right) = \{f_X, f_Y\}(x),$$

for all $x \in M$ and $X, Y \in \mathfrak{g}$, where $\{\cdot, \cdot\}$ is Poisson bracket and $f_X(x) = \mu(x)(X)$. This relation means that the map $\widehat{\mu} : X \mapsto f_X$ is a homomorphism between $\mathfrak{g}$ and the space of functions endowed with the Poisson bracket.

Proof The differential of μ is given by

$$d\mu_x\left(\widetilde{X}(x)\right)(Y) = (df_Y)_x\left(\widetilde{X}(x)\right) = -\omega_x\left(\widetilde{X}, \widetilde{Y}\right),$$

because f_Y is the Hamiltonian function of $\widetilde{Y}$. If μ is equivariant, the same happens to its differential, so

$$d\mu_x\left(\widetilde{X}(x)\right) = \frac{d}{dt}\mu\left(e^{tX}x\right)_{t=0} = \frac{d}{dt}\mathrm{Ad}^*\left(e^{tX}\right)\mu(x)_{t=0}$$
$$= \mathrm{ad}^*(X)\,\mu(x) = -\mu(x)\,\mathrm{ad}(X).$$

It follows that $-\mu(x)[X,Y] = -\omega_x\left(\widetilde{X},\widetilde{Y}\right)$ if μ is equivariant, making $\widehat{\mu}$ a homomorphism. □

As mentioned above, moment maps may not be Ad^*-equivariant. What is always true is that, given a moment map μ, there is an affine representation of G on $\mathfrak{g}^*$ whose underlying linear representation is Ad^* and such that μ is equivariant for that affine representation.

An **affine representation** of a group G on a vector space V is a homomorphism $A: G \to \mathrm{Aff}(V)$ with values in the affine group of V. As $\mathrm{Aff}(V) = \mathrm{Gl}(V) \times V$, the representation A is written as $A(g) = (\rho(g), v(g))$, $g \in G$, with $\rho: G \to \mathrm{Gl}(V)$ and $v: G \to V$. The homomorphism property of A amounts to

$$(\rho(gh), v(gh)) = A(gh) = (\rho(g), v(g))(\rho(h), v(h))$$
$$= (\rho(g)\rho(h), \rho(g)v(h) + v(g)).$$

Therefore ρ is a linear G-representation, and v must satisfy the condition

$$v(gh) = \rho(g)v(h) + v(g). \tag{14.18}$$

Conversely, suppose $\rho: G \to \mathrm{Gl}(V)$ is a linear Representation, and $v: G \to V$ satisfies (14.18). Then $A(g) = (\rho(g), v(g))$ is an affine representation.

To define the affine representation rendering a moment map equivariant, some preliminary results are needed.

Lemma 14.20 *Given a symplectic action of G on (M,ω) suppose f_X is a Hamiltonian function for the vector field $\widetilde{X}$. Then $f_X \circ g$ is a Hamiltonian function for* $\widetilde{\mathrm{Ad}\left(g^{-1}\right)X}$.

Proof If $v \in T_xM$, then $d(f_X \circ g)_x(v) = d(f_X)_{gx} \circ dg_x(v)$, so

$$d(f_X \circ g)_x(v) = \omega\left(\widetilde{X}(gx), dg_x(v)\right) = \left(g^*\omega\right)_x\left(\widetilde{\mathrm{Ad}\left(g^{-1}\right)X}(x), v\right).$$

As $g^*\omega = \omega$, it follows that $d(f_X \circ g)_x(v) = \omega_x\left(v, \widetilde{\mathrm{Ad}\left(g^{-1}\right)X}(x)\right)$, that is, $f_X \circ g$ is Hamiltonian for $\widetilde{\mathrm{Ad}\left(g^{-1}\right)X}$. □

Proposition 14.21 *Let μ be a moment map for the Hamiltonian action of G on M. Then for any $g \in G$, the expression*

$$\mu(gx) - \mathrm{Ad}^*(g)\,\mu(x)$$

is constant as a function of $x \in M$.

Proof If $X \in \mathfrak{g}$, the function $f_X(x) = \mu(x)(X)$ is Hamiltonian for $\widetilde{X}$. Hence,

$$\mathrm{Ad}^*(g)\,\mu(x)(X) = \mu(x)\left(\mathrm{Ad}\left(g^{-1}\right)X\right)$$

is Hamiltonian for $\widetilde{\mathrm{Ad}\left(g^{-1}\right)X}$.

By the previous lemma, on the other hand, $f_X \circ g$ is also Hamiltonian for $\widetilde{\mathrm{Ad}\left(g^{-1}\right)X}$. Hence the following two functions of $x \in M$, namely $f_X \circ g(x) = \mu(gx)(X)$ and $\mathrm{Ad}^*(g)\,\mu(x)(X)$, differ by a constant (under the assumption, M is connected). That is, $\left(\mu(gx) - \mathrm{Ad}^*(g)\,\mu(x)\right)(X)$ is constant as a function of $x \in M$ for all $X \in \mathfrak{g}$. Therefore $\mu(gx) - \mathrm{Ad}^*(g)\,\mu(x) \in \mathfrak{g}^*$ does not depend on $x \in M$. □

On the basis of the above proposition, write, for any $x \in M$,

$$c_\mu(g) = \mu(gx) - \mathrm{Ad}^*(g)\,\mu(x) \in \mathfrak{g}^*. \tag{14.19}$$

This defines a map $c_\mu : G \to \mathfrak{g}^*$. In the sequel, it will be proved that c_μ satisfies property (14.18) of the vector part of an affine representation, whose linear component is Ad^*.

Proposition 14.22 *Given a Hamiltonian action and a moment map μ, define $c_\mu : G \to \mathfrak{g}^*$ as in (14.19), for any $x \in M$. Then,*

$$c_\mu(gh) = \mathrm{Ad}^*(g)\,c_\mu(h) + c_\mu(g).$$

Proof The definition of c_μ yields

$$\begin{aligned} c_\mu(gh) &= \mu(ghx) - \mathrm{Ad}^*(gh)\,\mu(x) \\ &= \mu(g(hx)) - \mathrm{Ad}^*(g)\,\mu(hx) + \mathrm{Ad}^*(g)\,\mu(hx) - \mathrm{Ad}^*(gh)\,\mu(x). \end{aligned}$$

The first two terms on the right read $\mu(g(hx)) - \mathrm{Ad}^*(g)\,\mu(hx) = c_\mu(g)$. The third and fourth terms combine to give

$$\begin{aligned} \mathrm{Ad}^*(g)\,\mu(hx) - \mathrm{Ad}^*(gh)\,\mu(x) &= \mathrm{Ad}^*(g)\left(\mu(hx) - \mathrm{Ad}^*(h)\,\mu(x)\right) \\ &= \mathrm{Ad}^*(g)\,c_\mu(h). \end{aligned}$$

Then $c_\mu(gh) = c_\mu(g) + \mathrm{Ad}^*(g)\,c_\mu(h)$, proving the claim. □

As c_μ satisfies (14.18) with $\rho = \mathrm{Ad}^*$, it defines the following affine representation.

Corollary 14.23 *Given a moment map $\mu : M \to \mathfrak{g}^*$, define $A_\mu : G \to \mathrm{Aff}(\mathfrak{g}^*)$ by*

$$A_\mu(g) = \left(\mathrm{Ad}^*(g), c_\mu(g)\right).$$

Then A_μ is an affine representation.

An almost immediate consequence of the definition of c_μ is the following equivariance property of μ.

Corollary 14.24 *Retaining the previous notations, the moment map $\mu : M \to \mathfrak{g}^*$ is A_μ-equivariant:*

$$\mu(gx) = A_\mu(g)\,\mu(x) \qquad x \in M,\ g \in G.$$

Proof The definition of the affine representation says that

$$A_\mu(g)\,\mu(x) = \mathrm{Ad}^*(g)\,\mu(x) + c_\mu(g),$$

which is equal to $\mu(gx)$ because $c_\mu(g) = \mu(gx) - \mathrm{Ad}^*(g)\,\mu(x)$. □

This corollary shows that μ is Ad^*-equivariant if and only if $c_\mu = 0$. This remark is the starting point for investigating the existence of Ad^*-equivariant moment maps.

Recall that if $\overline{\mu}$ is a moment map, then there exists $\alpha \in \mathfrak{g}^*$ such that $\overline{\mu} = \mu + \alpha$. Now the definition of c_μ gives

$$c_{\overline{\mu}}(g) = c_\mu(g) + \alpha - \mathrm{Ad}^*(g)\,\alpha. \tag{14.20}$$

This relation proves the following criterion for the existence of an Ad^*-equivariant moment map.

Proposition 14.25 *Take a moment map for the Hamiltonian action of G on M. A necessary and sufficient condition for the existence of an Ad^*-equivariant moment map is that there exists $\alpha \in \mathfrak{g}^*$ such that*

$$c_\mu(g) = \mathrm{Ad}^*(g)\,\alpha - \alpha. \tag{14.21}$$

The condition may be interpreted in two somehow complementary ways. One is in terms of equivalence of affine representations, and the other goes through the cohomology of group representations. They are described below.

- **Equivalent affine representations:** two affine representations $A_1, A_2 : G \to \mathrm{Aff}(V)$ are said to be equivalent if there exists $T \in \mathrm{Aff}(V)$ such that

  $$A_1(g) = T A_2(g) T^{-1}$$

  for all $g \in G$. If $A_1(g) = (\rho_1(g), v_1(g))$, $A_2(g) = (\rho_2(g), v_2(g))$, and $T = (P, w)$, equivalence means $\rho_1(g) = P\rho_2(g) P^{-1}$ and

  $$v_1(g) = -P\rho_2(g) P^{-1} w + P v_2(g) + w.$$

  Suppose $\rho_1 = \rho_2 = \rho$ and $T = (\mathrm{id}, w)$. Then,

  $$v_1(g) = -\rho(g) w + v_2(g) + w,$$

i.e.

$$v_1(g) - v_2(g) = w - \rho(g)\, w.$$

That said, let $A_\mu = \left(\mathrm{Ad}^*, c_\mu\right)$ and $A_{\overline{\mu}} = \left(\mathrm{Ad}^*, c_{\overline{\mu}}\right)$ be the affine representations defined by the moment maps μ and $\overline{\mu} = \mu + \alpha$. As noted above, $c_{\overline{\mu}}(g) - c_\mu(g) = \alpha - \mathrm{Ad}^*(g)\,\alpha$, which means A_μ and $A_{\overline{\mu}}$ are equivalent under the equivalence $T = (\mathrm{id}, \alpha)$.

On the other hand, (14.21) is satisfied if and only if A_μ is equivalent to the linear representation $\left(\mathrm{Ad}^*, 0\right)$. In this case, there exists an Ad^*-equivariant moment map.

- **Cohomology of group representations:** let G be a group and $\rho : G \to \mathrm{Gl}(V)$ a linear G-representation on the vector space V. Define spaces $F^0 = F^0(G, V) = V$ and, for $q \geq 1$, $F^q = F^q(G, V)$ as the set of maps $c : G^q \to V$. Introduce linear maps $\delta_q : F^q \to F^{q+1}$ for $q \geq 0$ by $\delta_0 v(g) = v - \rho(g)\, v$ and

$$\delta_q c\left(g_1, \ldots, g_{q+1}\right) = \rho(g_1)\, c\left(g_2, \ldots, g_{q+1}\right) + \sum_{i=1}^{q-1} (-1)^i\, c\left(g_1, \ldots, g_i g_{i+1}, \ldots, g_{q+1}\right) + (-1)^{q+1}\, c\left(g_1, \ldots, g_q\right).$$

If $q \geq 1$, then $\delta_{q+1} \circ \delta_q = 0$. Call

$$\mathcal{H}^q(\rho) = \frac{\ker \delta_q}{\mathrm{im}\delta_{q-1}}$$

qth **cohomology group of the representation**. The elements of $\ker \delta_q$ are called cocycles and those in $\mathrm{im}\delta_q$ are the coboundaries.

When $q = 1$,

$$\delta_1 c(g, h) = \rho(g)\, c(h) - c(gh) + c(g).$$

Hence $c : G \to V$ is a 1-cocycle if and only if

$$c(gh) = \rho(g)\, c(h) + c(g).$$

The property of c_μ given in Proposition 14.22 says that c_μ is a 1-cocycle for the representation Ad^*. On the other hand, the relationship between c_μ and $c_{\overline{\mu}}$ in (14.20) tells

$$c_{\overline{\mu}}(g) - c_\mu(g) = \alpha - \mathrm{Ad}^*(g)\,\alpha = \delta_0 \alpha$$

for the representation Ad^*. This means that different moment maps yield cohomologous cocycles.

The criterion of Proposition 14.25, therefore, translates to the statement that there is an Ad^*-equivariant moment map if and only if the 1-cocycle c_μ is cohomologous to 0. In particular, the existence is guaranteed if $\mathcal{H}^q\left(\mathrm{Ad}^*\right) = \{0\}$.

These two interpretations of the condition in Proposition 14.25 provide the following criteria to have an Ad^*-equivariant moment map.

Proposition 14.26 *Let $\mu : M \to \mathfrak{g}^*$ be a moment map of the Hamiltonian action $G \times M \to M$. Then either of the following statements is equivalent to the existence of an Ad^*-equivariant moment map $\overline{\mu}$:*

1. *the affine representation $A_\mu = \left(\mathrm{Ad}^*, c_\mu\right)$ is equivalent to the linear representation $\left(\mathrm{Ad}^*, 0\right)$;*
2. *the cocycle c_μ is cohomologous to 0.*

With regard to the uniqueness of Ad^*-equivariant moment maps, the relation $c_{\overline{\mu}}(g) = c_\mu(g) + \alpha - \mathrm{Ad}^*(g)\,\alpha$ if $\overline{\mu} = \mu + \alpha$ says that μ and $\overline{\mu}$ are equivariant if and only if $\mathrm{Ad}^*(g)\,\alpha = \alpha$. Therefore if the set of Ad^*-equivariant moment maps not empty, then it is an affine space modeled on

$$\{\alpha \in \mathfrak{g}^* : \forall g \in G,\ \mathrm{Ad}^*(g)\,\alpha = \alpha\}.$$

The criteria established in the above proposition for the existence of Ad^*-equivariant moment maps will be applied below to the Hamiltonian actions of certain groups. First of all, the cohomological criterion applies to compact groups G due to the following lemma, which ensuring the vanishing of the first cohomology group of an arbitrary G-representation.

Lemma 14.27 *Let G be a compact and Hausdorff topological group and $\rho : G \to \mathrm{Gl}(V)$ a finite dimensional, continuous representation. Suppose that $c : G \to V$ is a continuous 1-cocycle for ρ. Then c is a coboundary.*

Proof Integrating the two sides of the cocycle equation $c(gh) = \rho(g)\,c(h) + c(g)$ yields

$$\int c(gh)\,\nu(dh) = \int \rho(g)\,c(h)\,\nu(dh) + c(g),$$

where ν is the Haar measure of G normalized by $\nu(G) = 1$. In the left, integral g may be dropped because ν is left invariant. Moreover, $\rho(g)$ comes out of the integral on the right, by linearity. Hence,

$$\int c(h)\,\nu(dh) = \rho(g)\left(\int c(h)\,\nu(dh)\right) + c(g). \tag{14.22}$$

Now put $w = \int c(h)\,\nu(dh) \in V$, so

$$c(g) = w - \rho(g)\,w = (\delta_0 w)(g),$$

which shows c is a coboundary. □

This lemma says that for continuous cocycles of a compact group the first cohomology group, for any linear representation, is trivial.

Proposition 14.28 *Let G be a compact Lie group. Then a Hamiltonian G-action admits an* Ad^**-equivariant moment map $\mu : M \to \mathfrak{g}^*$. If G is semi-simple, then this moment map is unique.*

Proof The existence is a consequence of the previous lemma combined with the cohomological criterion established above. In the semi-simple case uniqueness is due to the fact that if $\alpha \in \mathfrak{g}^*$ is such that $\mathrm{Ad}^*(g)\,\alpha = \alpha$ for all $g \in G$, then $\alpha = 0$. Take in fact $X \in \mathfrak{g}$ such that $\alpha(\cdot) = \mathcal{K}(X, \cdot)$, where $\mathcal{K}$ is the Cartan–Killing form of $\mathfrak{g}$. Then $\mathrm{Ad}(g)\,X = X$ for every $g \in G$, which implies $X \in \mathfrak{z}(\mathfrak{g}) = \{0\}$. □

This result even applies to noncompact semi-simple groups.

Proposition 14.29 *Let G be a connected semi-simple Lie group. Then a Hamiltonian action of G admits a unique* Ad^**-equivariant moment map.*

Proof Given a moment map μ, let $A_\mu = \left(\mathrm{Ad}^*, c_\mu\right)$ be its affine representation, and denote by $B_\mu : \mathfrak{g} \to \mathfrak{aff}(\mathfrak{g}^*)$ the corresponding infinitesimal representation, given by $B_\mu = \left(\mathrm{ad}^*, \left(dc_\mu\right)_1\right)$. It is known that the first cohomology group of a finite dimensional representation of a semi-simple Lie algebra is trivial.[7] Hence B_μ is equivalent to the linear representation $\left(\mathrm{ad}^*, 0\right)$. This implies that A_μ is equivalent to $\left(\mathrm{Ad}^*, 0\right)$ and, therefore, there is an equivariant moment map. □

Up to now, Hamiltonian actions were not assumed to be transitive. As it turns out, the transitive case has features close to those of coadjoint orbits.

Let $M = G/H$ be a homogeneous space with invariant symplectic form ω such that the G-action is Hamiltonian. Suppose μ is an Ad^*-equivariant moment map. If x_0 is the origin of G/H and $x = gx_0$, then $\mu(x) = \mu(gx_0) = \mathrm{Ad}^*(g)\,\mu(x_0)$. This shows that the image of μ is the coadjoint orbit $\mathrm{Ad}^*(G)\,\mu(x_0)$. The latter may be equipped with the Kirillov–Kostant–Souriau form Ω.

Proposition 14.30 *If the action of G on G/H is Hamiltonian with respect to the invariant symplectic form ω and μ is an* Ad^**-equivariant moment map, then $\mu^*\Omega = \omega$.*

Proof If $x \in G/H$ and $X \in \mathfrak{g}$, then $\mu\left(e^{tX}x\right) = \mathrm{Ad}^*\left(e^{tX}\right)\mu(x)$. Differentiating this relation in t and evaluating at $t = 0$ yield

$$d\mu_x\left(\widetilde{X}(x)\right) = \mathrm{ad}^*(X)\,\mu(x)\,.$$

Hence

$$\begin{aligned}\Omega_{\mu(x)}\left(d\mu_x\left(\widetilde{X}\right), d\mu_x\left(\widetilde{Y}\right)\right) &= \Omega_{\mu(x)}\left(\mathrm{ad}^*(X)\,\mu(x), \mathrm{ad}^*(X)\,\mu(x)\right)\\ &= \mu(x)\,[X, Y]\end{aligned}$$

[7] See *Álgebras de Lie* [47, Chapter 5] and Varadarajan [53, Section 3.12].

by definition of Ω. By Proposition 14.19, the last term equals $\omega_x\left(\widetilde{X}, \widetilde{Y}\right)$, showing that $\mu^*\Omega = \omega$. □

Now take a Hamiltonian action on G/H with Ad^* -equivariant moment map μ, let x_0 be the origin of G/H, and call

$$Z_{\mu(x_0)} = \{g \in G : \mathrm{Ad}^*(g)\,\mu(x_0) = \mu(x_0)\}$$

the isotropy group at $\mu(x_0)$, so that $\mathrm{Ad}^*(G)\,\mu(x_0) = G/Z_{\mu(x_0)}$. Since μ is equivariant, $H \subset Z_{\mu(x_0)}$ because if $gx_0 = x_0$, then $\mathrm{Ad}^*(g)\,\mu(x_0) = \mu(gx_0) = \mu(x_0)$.

The inclusion $H \subset Z_{\mu(x_0)}$ implies, in terms of the homogeneous spaces $\mathrm{Ad}^*(G)\,\mu(x_0) = G/Z_{\mu(x_0)}$ and $M = G/H$, that the moment map μ becomes the canonical projection $G/H \to G/Z_{\mu(x_0)}$ that associates with the coset gH and the coset $gZ_{\mu(x_0)}$. Hence μ is a submersion. In particular, $\dim G/Z_{\mu(x_0)} \leq \dim M$.

Actually, the dimensions are equal. This is because, by the previous proposition, $\mu^*\Omega = \omega$. Hence, any wedge product $\Omega^{k\wedge} = \Omega \wedge \cdots \wedge \Omega$ satisfies $\mu^*\Omega^{k\wedge} = (-1)^k\,\omega^{k\wedge}$. When $\dim M = 2k$, in particular, $\omega^{k\wedge} \neq 0$, and it is not possible to have $\dim G/Z_{\mu(x_0)} < 2k$, for this would force $\Omega^{k\wedge} = 0$, i.e. $0 = \mu^*\Omega^{k\wedge} = (-1)^k\,\omega^{k\wedge}$, contradicting the non-degeneracy of ω.

Together, $\dim G/Z_{\mu(x_0)} = \dim G/H$ and $H \subset Z_{\mu(x_0)}$ imply that the Lie algebras of H and $Z_{\mu(x_0)}$ coincide and therefore must be

$$\mathfrak{z}_{\mu(x_0)} = \{X \in \mathfrak{g} : \mathrm{ad}^*(X)\,\mu(x_0) = 0\}.$$

Moreover, the canonical projection $G/H \to G/Z_{\mu(x_0)}$ is a covering map. Therefore μ is a covering map, and M is a covering space of the adjoint orbit $\mathrm{Ad}^*(G)\,\mu(x_0)$. In short, an Ad^* -equivariant Hamiltonian action on a homogeneous space does not differ much from a coadjoint orbit.

To conclude the section, two classical examples in symplectic geometry are considered.

Example (Abelian Groups) If G is a connected abelian group, then $G = \mathbb{T}^m \times \mathbb{R}^n$. Its Lie algebra $\mathfrak{g}$, as well as the dual $\mathfrak{g}^*$, is identified with $\mathbb{R}^{m+n}$. The adjoint and coadjoint representations are trivial, $\mathrm{Ad}(g) = \mathrm{Ad}^*(g) = \mathrm{id}$, $g \in G$.

A map $c : G \to \mathfrak{g}^* = \mathbb{R}^{m+n}$ satisfies the cocycle property if and only if $c(gh) = c(g) + c(h)$. Write $g \in G$ as $g = (x, y) \in \mathbb{T}^m \times \mathbb{R}^n$ and define $c_1(x) = c(x, 0)$ and $c_2(y) = c(0, y)$, so that $c(x, y) = c_1(x) + c_2(y)$. The first component c_1 is a cocycle on the torus $\mathbb{T}^m$, which is a compact and abelian group. Hence $c_1 = 0$ (see (14.22)). Therefore a cocycle on G can be written as

$$c(x, y) = c_2(y),$$

where $c_2 : \mathbb{R}^n \to \mathbb{R}^{n+m}$ is a linear map.

The only coboundary is 0 because $\delta_0 w = w - \mathrm{Ad}^*(g) w = w - w = 0$. Hence the first cohomology group of G (for the coadjoint representation) is the space of linear maps $\mathbb{R}^n \to \mathbb{R}^{n+m}$.

Now take a Hamiltonian action $G \times M \to M$. Since the cocycles of distinct moment maps are cohomologous, and two cocycles are cohomologous if and only if they are equal, all moment maps for this action have the same cocycle. In particular, if a moment map is Ad^*-equivariant, then every one is equivariant. In this case, $\mu(gx) = \mathrm{Ad}^*(g)\mu(x) = \mu(x)$, that is, μ is constant on G-orbits.

The Ad^*-equivariant case occurs when the Hamiltonian functions Poisson-commute with one another. In fact, define, as in Proposition 14.19, $\widehat{\mu}(X)(\cdot) = \mu(\cdot)(X)$. When μ is Ad^*-equivariant, then $\{\widehat{\mu}(X), \widehat{\mu}(Y)\} = \widehat{\mu}[X, Y] = 0$. This means that there is a choice of Hamiltonian $\widehat{\mu}(X)$, $X \in \mathfrak{g}$, such that these functions commute in the Poisson algebra. (In general, $\{\widehat{\mu}(X), \widehat{\mu}(Y)\}$ is a Hamiltonian function for $[\widetilde{X}, \widetilde{Y}] = \widetilde{[X, Y]} = 0$, and this allows us to conclude $\{\widehat{\mu}(X), \widehat{\mu}(Y)\}$ is constant, but possibly nonzero.)

G-orbits in M are immersed submanifolds and at the same time homogeneous spaces G/H. Since G is abelian, G/H is also an abelian group and, like G, it is the product of a torus by a Euclidean space. When the action is Ad^*-equivariant, the tangent spaces to the orbits

$$T_x(Gx) = \{\widetilde{X}(x) \in T_x M : X \in \mathfrak{g}\}$$

are isotropy subspaces, since $\omega\left(\widetilde{X}(x), \widetilde{Y}(x)\right) = \{\widehat{\mu}(X), \widehat{\mu}(Y)\}(x) = 0$.

What is classically called a **completely integrable system** is a Hamiltonian vector field X on a manifold M of dimension $2n$, with Hamiltonian function f such that there exist functions $g_1, \ldots, g_{n-1}$ satisfying $\{g_i, f\} = \{g_i, g_j\} = 0$. The Hamiltonian vector fields associated with these functions generate an abelian Lie algebra of dimension n. If the vector fields are complete, by the Lie–Palais theorem, their flows define a Hamiltonian action of $\mathbb{R}^n$ on M, which is Ad^*-equivariant. □

Example (Cotangent Bundles[8]*)* The cotangent bundle $\pi : T^*M \to M$ of a manifold M admits a canonical symplectic form $\omega = d\lambda$, where λ is the so-called Liouville 1-form: $\lambda_\alpha(v) = \alpha(\pi_* v)$ for $\alpha \in T^*M$ and $v \in T_\alpha T^*M$. A diffeomorphism ϕ of M lifts to a diffeomorphism $\phi^\#$ of T^*M by

$$\phi^\#(\alpha) = \left(d\phi_x^{-1}\right)^* \alpha = \alpha \circ d\phi_x^{-1},$$

where $x = \pi(\alpha)$. Similarly, a vector field X on M with flow ϕ_t lifts to a vector field $X^\#$ on T^*M whose flow is $\phi_t^\#$.

The lifted vector field $X^\#$ is Hamiltonian for the canonical symplectic form with Hamiltonian $f_X : T^*M \mapsto \mathbb{R}$ given by $f_X(\alpha) = \alpha(X(x))$ if $x = \pi(\alpha)$.

[8] See Abraham and Marsden [1].

An action of a Lie group G on M lifts to an action on T^*M by $g \cdot \alpha = \left(dg_x^{-1}\right)^* \alpha = \alpha \circ dg_x^{-1}$ if $x = \pi(\alpha)$ and $g \in G$. The corresponding infinitesimal action associates with $X \in \mathfrak{g}$ the vector field $\widetilde{X}^{\#}$ lifting of $\widetilde{X}$.

A moment map for the action of G on T^*M is $\mu(\alpha)(X) = \alpha\left(\widetilde{X}(x)\right)$ if $\alpha \in T^*M$, $X \in \mathfrak{g}$, and $x = \pi(\alpha)$. It is Ad^*-equivariant because

$$\begin{aligned}\mu(g \cdot \alpha)(X) &= \alpha\left(dg_x^{-1}\widetilde{X}(gx)\right) = \alpha\left(\widetilde{\mathrm{Ad}(g)X}(x)\right)\\ &= \mu(\alpha)(\mathrm{Ad}(g)X)\\ &= \mathrm{Ad}^*(g)(\mu(\alpha))(X).\end{aligned}$$

□

14.5 Exercises

1. Let G be a connected Lie group and $H \subset G$ a closed subgroup. Prove G/H is orientable if and only if the isotropy representation satisfies $\det dh_{x_0} > 0$ for all $h \in H$. Show that if H is connected, then G/H is orientable.
2. Let G be a connected Lie group and $\Gamma \subset G$ a discrete subgroup. Show that G/Γ is orientable. Give an example of a non-orientable quotient G/H with G connected.
3. Given a manifold M with complex structure J, let X, X_1, Y, Y_1 be vector fields such that $X(x) = X_1(x)$ and $Y(x) = Y_1(x)$. Show that $N_J(X, Y) = N_J(X_1, Y_1)$, where N_J is the Nijenhuis tensor defined in (14.1). (That is, the Nijenhuis tensor is indeed a tensor.)
4. Let $\mathfrak{g}$ be a Lie algebra with bracket $[\cdot,\cdot]$. Given a complex structure $J : \mathfrak{g} \to \mathfrak{g}$, define a new bracket $[\cdot,\cdot]_J$ by

$$[X, Y]_J = \frac{1}{2}([X, Y] - [JX, JY]).$$

   Verify that $[\cdot,\cdot]_J$ is skew-symmetric, and show it satisfies the Jacobi identity if and only if the Nijenhuis tensor N_J of J vanishes identically (i.e. J defines a left invariant (or right invariant) complex structure on the connected Lie groups with Lie algebra $\mathfrak{g}$).
5. In the notation of the previous exercise, find a nonabelian Lie algebra $\mathfrak{g}$ and a complex structure J such that $[\cdot,\cdot]_J \equiv 0$.
6. Let G be a connected Lie group whose Lie algebra $\mathfrak{g}$ is semi-simple. Show that G does not admit any bi-invariant 1-form.
7. Suppose α is a k-form on G/H and $\sigma : \Delta_k \to G/H$ is a k-cycle of class C^1. Prove that

$$\int_{g\sigma} \alpha = \int_{\sigma} g^* \alpha$$

for any $g \in G$.

8. Let G be a Lie group with Lie algebra $\mathfrak{g}$ and denote by ω the Maurer–Cartan form obtained by right translations. Show that ω is right invariant. Take a basis $\{X_1, \ldots, X_n\}$ of $\mathfrak{g}$ and let c_{ij}^k be the structure constants, defined by $[X_i, X_j] = \sum_k c_{ij}^k X_k$. Write

$$\omega(\cdot) = \omega^1(\cdot) X_1 + \cdots + \omega^n(\cdot) X_n,$$

where each $\omega^i(\cdot)$ is a real 1-form. Prove that $d\omega^k = -\sum_k c_{ij}^k \omega^i \wedge \omega^j$.

Part V
Appendices

Appendix A
Vector Fields and Lie Brackets

A **vector field** on a differentiable manifold M is a map $X : M \to TM$ that satisfies $X(x) \in T_x M$ if $x \in M$.

A vector field X defines an (autonomous) ordinary differential equation on M by

$$\frac{dx}{dt} = X(x). \tag{A.1}$$

If X is differentiable (of class C^1), then for every $x_0 \in M$ there exists a unique maximal solution $x(t)$ to this differential equation with initial condition $x(0) = x_0$. The maximal solution passing through x_0 is denoted by $t \mapsto X_t(x_0)$. Its domain of definition is an open interval $(\alpha, \omega) \subset \mathbb{R}$ that contains 0.

If $t \in \mathbb{R}$ is fixed, the map $X_t : x \mapsto X_t(x)$ is a local diffeomorphism of M in the sense that its domain $\mathrm{dom}X_t$ is an open set of M and $X_t : \mathrm{dom}X_t \to X_t(\mathrm{dom}X_t)$ is a diffeomorphism. The domain $\mathrm{dom}X_t$ is the set of elements of M whose maximal solutions extend to t. This means the elements such that interval of definition (α, ω) contains t. The vector field is said to be **complete** if $\mathrm{dom}X_t = M$ for all $t \in \mathbb{R}$. Equivalently, X is complete if all maximal solutions are always defined, i.e. on $\mathbb{R} = (-\infty, +\infty)$.

The set of local diffeomorphisms X_t, $t \in \mathbb{R}$, is called the **flow** of X. Up to restricting domains, the flow satisfies the homomorphic-like property $X_{t+s} = X_t \circ X_s$, in the sense that if $X_s(x)$ and $X_t(X_s(x))$ are defined, then also $X_{t+s}(x)$ is defined and $X_{t+s}(x) = X_t(X_s(x))$. This is due to the uniqueness of solution (A.1) with the given initial condition, by noting that $\mathrm{dom}X_{t+s} = X_s(\mathrm{dom}X_s) \cap \mathrm{dom}X_t$. Said property implies that $X_t \circ X_s = X_s \circ X_t$ and $X_{-t} = (X_t)^{-1}$.

To sum up, the flow X_t of X satisfies the following properties, which characterize it:

1. $X_0 = \mathrm{id}$.
2. $\dfrac{d}{dt} X_t(x) = X(X_t(x))$.

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3. $X_{t+s} = X_t \circ X_s = X_s \circ X_t$.

A vector field X is obtained from its flow by the second condition above and is often called the **infinitesimal generator** of its flow.

Let $\phi : M \to N$ be a differentiable map. Two vector fields X on M and Y on N are said to be ϕ-**related** if $d\phi$ maps X to Y, that is, $d\phi_x (X(x)) = Y(\phi(x))$ for any $x \in M$. In this case, the image under ϕ of a path in X is a path in Y. In terms of flows, this reads

$$\phi \circ X_t = Y_t \circ \phi.$$

A vector field X on M might not admit ϕ-related vector fields to it on N. For instance, suppose ϕ is not injective. If $\phi(x) = \phi(y)$ and $d\phi_x (X(x)) \neq d\phi_y (X(y))$, then there is no vector field ϕ-related to X. However, if $\phi : M \to N$ is a diffeomorphism and X is a vector field on M, there exists a unique vector field on N, denoted by $\phi_* X$, that is, ϕ-related to X. It is defined by

$$(\phi_* X)(y) = d\phi_{\phi^{-1}(y)} \left(X \left(\phi^{-1}(y) \right) \right).$$

Definition A.1 The **Lie bracket** between vector fields X and Y on M is defined by

$$[X, Y](x) = \frac{d}{dt} \left(d(X_{-t})_{X_t(x)} (Y(X_t(x))) \right)_{|t=0}. \tag{A.2}$$

The Lie bracket preserves ϕ-related vector fields.

Proposition A.2 *Let $\phi : M \to N$ be a differentiable map and X_1, X_2 vector fields on M. Suppose that Y_1 and Y_2 are vector fields on N that are ϕ-related to X_1 and X_2, respectively. Then $[X_1, X_2]$ and $[Y_1, Y_2]$ are ϕ-related.*

Proof The claim follows directly from the definition of Lie bracket combined with the chain rule and the relation $\phi \circ X_t = Y_t \circ \phi$ if X and Y are ϕ-related. □

In particular, if ϕ is a diffeomorphism, then

$$\phi_*[X, Y] = [\phi_* X, \phi_* Y]. \tag{A.3}$$

The properties of Lie bracket descend from its expression in local coordinates, which is found below.

Let U be an open set of $\mathbb{R}^n$. A vector field X on U is identified with a differentiable map $X : U \subset \mathbb{R}^n \to \mathbb{R}^n$. As solutions to an ODE depend differentiably on the parameter, the flow X_t of a C^k vector field X is also of class C^k. In particular, the differential $d(X_t)_x$, for $k \geq 1$, exists, and x belongs in the domain of X_t. The differential satisfies a linear differential equation usually called the adjoint equation. The next lemmas spell out the adjoint equation.

Lemma A.3 $\dfrac{d}{dt}\left(d\left(X_t\right)_x\right)_{|t=0} = d\left(X\right)_x$.

Proof Given $v \in \mathbb{R}^n$, define the map

$$\beta\left(t, s\right) = X_t\left(x + sv\right),$$

whose domain is a neighborhood of $0 \in \mathbb{R}^2$. The differential $d\left(X_t\right)$ is given by the partial derivative

$$\frac{\partial \beta}{ds}\left(t, 0\right) = d\left(X_t\right)_x\left(v\right).$$

Hence,

$$\frac{d}{dt}\left(d\left(X_t\right)_x\right)_{|t=0} = \frac{\partial^2 \beta}{\partial t \partial s}\left(0, 0\right) = \frac{\partial^2 \beta}{\partial s \partial t}\left(0, 0\right).$$

But $\dfrac{\partial \beta}{dt}\left(0, s\right) = X\left(x + sv\right)$, and differentiating the latter in s provides the claimed formula. □

The formula for the derivative of $t \mapsto d\left(X_t\right)_x$ for arbitrary t is easily obtained from the derivative at $t = 0$:

$$\frac{d}{dt}\left(d\left(X_t\right)_x\right) = \frac{d}{ds}\left(d\left(X_{t+s}\right)_x\right)_{|s=0} = \frac{d}{ds}\left(d\left(X_s\right)_{X_t(x)} \circ d\left(X_t\right)_x\right)_{|s=0}.$$

Hence the above lemma implies the following.

Proposition A.4 $\dfrac{d}{dt}\left(d\left(X_t\right)_x\right) = d\left(X\right)_{X_t(x)} \circ d\left(X_t\right)_x$.

This formula for the derivative means that the curve $t \mapsto d\left(X_t\right)_x$ satisfies the differential equation

$$\frac{dg}{dt} = d\left(X\right)_{X_t(x)} \circ g$$

on the space of linear maps of $\mathbb{R}^n$. This equation is linear, and its coefficients are not constants unless $X_t\left(x\right) = x$ for all t, that is, if x is a singularity of X.

Proposition A.5 *Let $X, Y : U \subset \mathbb{R}^n \to \mathbb{R}^n$ be differentiable vector fields on the open set $U \subset \mathbb{R}^n$. Then,*

$$\left[X, Y\right]\left(x\right) = dY_x\left(X\left(x\right)\right) - dX_x\left(Y\left(x\right)\right) \tag{A.4}$$

for all $x \in U$.

Proof The translation $(X_{-t})_{X_t(x)}\,(Y\,(X_t\,(x)))$ in the definition of Lie bracket can be read, in local coordinates, as a product of an $n \times n$ matrix by an $n \times 1$ matrix. Its derivative at $t = 0$ comes from differentiating a product and has two terms, namely

$$\frac{d}{dt}\left(d\,(X_{-t})_{X_t(x)}\right)_{|t=0}\,(Y\,(x)) + \frac{d}{dt}\,(Y\,(X_t\,(x)))_{|t=0}\,. \tag{A.5}$$

The derivative of the second term is $dY_x\,(X\,(x))$. To compute the derivative of the first term, let us consider the matrix product $d\,(X_{-t})_{X_t(x)}\,d\,(X_t)_x = \mathrm{id}$. Its derivative at $t = 0$ equals

$$\frac{d}{dt}\left(d\,(X_{-t})_{X_t(x)}\right)_{|t=0} + \frac{d}{dt}\left(d\,(X_t)_x\right)_{|t=0} = 0.$$

Hence, by Lemma A.3,

$$\begin{aligned}\frac{d}{dt}\left(d\,(X_{-t})_{X_t(x)}\right)_{|t=0} &= -\frac{d}{dt}\left(d\,(X_t)_x\right)_{|t=0}\\ &= -dX_x.\end{aligned}$$

Then the first term in (A.5) becomes $-dX_x\,(Y\,(x))$. Putting together the two expressions gives $[X, Y]\,(x) = dY_x\,(X\,(x)) - dX_x\,(Y\,(x))$, as claimed. □

From the expression in local coordinates, it follows that $[X, Y]$ is a C^{k-1}-vector field if X and Y are of class C^k. From the same local formula, further properties of Lie brackets can be proved:

1. $\mathbb{R}$-bilinearity: if X, Y, Z are vector fields and $a \in \mathbb{R}$, then

$$[aX + Y, Z] = a[X, Z] + [Y, Z];$$

2. skew-symmetry: $[X, Y] = -[Y, X]$;
3. Jacobi identity: $[X, [Y, Z]] = [[X, Y], Z] + [Y, [X, Z]]$.

In particular, the vector space of C^∞-vector fields forms a Lie algebra when endowed with the Lie bracket of vector fields. Formula (A.3) shows that the map $X \mapsto \phi_* X$ is a homomorphism of Lie algebras of vector fields.

Still from the local expression of the Lie bracket, it follows that if X, Y are vector fields on M and $f : M \to \mathbb{R}$ is a differentiable function, then $[X, Y]f = X\,(Yf) - Y\,(Xf)$, where Xf indicates the derivative of f in the direction of X, i.e. $Xf\,(x) = df_x\,(X\,(x))$.

Notice that the Lie bracket $[X, Y]\,(x)$ depends on the values of X and Y on the neighborhood of x, and not just at the point x. In fact, if one of the vector fields is multiplied by a function $f : M \to \mathbb{R}$, the bracket does not get multiplied by f but rather satisfies

$$[X, fY] = f[X, Y] + (Xf)\,Y.$$

This means that the Lie bracket is not a tensor.

The following proposition provides an analytical interpretation of the Lie bracket of vector fields as the first term in the Taylor expansion of the commutator of their flows.

Proposition A.6 *Let $X, Y : U \to \mathbb{R}^n$ be differentiable vector fields defined on the open set $U \subset \mathbb{R}^n$. Fix $x \in U$ and consider the curve*

$$\alpha(t) = X_t \circ Y_t \circ X_{-t} \circ Y_{-t}(x)$$

defined on an open interval containing $0 \in \mathbb{R}$. Then $\alpha'(0) = 0$ and $\alpha''(0) = 2[Y, X](x)$.

Proof The computation of the first derivative of α is only based on the definition of X_t. Define curves $\alpha_1(t) = Y_t \circ X_{-t} \circ Y_{-t}(x)$ and $\alpha_2(t) = X_{-t} \circ Y_{-t}(x)$. Then,

$$\begin{aligned}\alpha'(t) = {} & X(\alpha(t)) + d(X_t)_{\alpha_1(t)}(Y(\alpha_1(t))) \\ & -d(X_t)_{\alpha_1(t)} \circ d(Y_t)_{\alpha_2(t)}(X(\alpha_2(t))) \\ & -d(X_t \circ Y_t \circ X_{-t})_{Y_{-t}(x)}(Y(Y_{-t}(x)))\,.\end{aligned}$$

Evaluating at $t = 0$ gives $\alpha'(0) = 0$. For the second derivative, the previous proposition is used. Taking derivatives of each term of $\alpha'(t)$ and evaluating at $t = 0$ yield:

1. $dX_x\left(\alpha'(0)\right) = 0$.
2. $\dfrac{d}{dt}\left(d(X_t)_x\right)(Y(x))_{|t=0} + 0 + d(Y)_x\left(\alpha_1'(0)\right)$.
3. $-\dfrac{d}{dt}\left(d(X_t)_x\right)(X(x))_{|t=0} - \dfrac{d}{dt}\left(d(Y_t)_x\right)(X(x))_{|t=0} - d(X)_x\left(\alpha_2'(0)\right)$.
4. $-\dfrac{d}{dt}\left(d(X_t)_x\right)(Y(x))_{|t=0} - \dfrac{d}{dt}\left(d(Y_t)_x\right)(Y(x))_{|t=0}$
   $+\dfrac{d}{dt}\left(d(X_t)_x\right)(Y(x))_{|t=0} + d(Y)_x((Y(x))$.

Similarly,

$$\alpha_1'(0) = Y(x) - X(x) - Y(x) = -X(x) \qquad \alpha_2'(0) = -X(x) - Y(x)\,.$$

Using these values and the lemma above, the second term becomes

$$d(X)_x(Y(x)) - d(Y)_x(X(x))\,.$$

The third term is given by

$$-d(X)_x(X(x)) - d(Y)_x(X(x)) + d(X)_x(X(x)) + d(X)_x(Y(x))\,,$$

while the fourth one is

$$-d\left(X\right)_x\left(Y\left(x\right)\right)-d\left(Y\right)_x\left(Y\left(x\right)\right)+d\left(X\right)_x\left(Y\left(x\right)\right)+d\left(Y\right)_x\left(Y\left(x\right)\right).$$

Adding up produces the desired result:

$$\alpha''\left(0\right)=2\left(d\left(X\right)_x\left(Y\left(x\right)\right)-d\left(Y\right)_x\left(X\left(x\right)\right)\right)=2[Y,X]\left(x\right). \qquad \square$$

Finally, here is a criterion for the commutativity of flows in terms of Lie brackets.

Proposition A.7 *Let X and Y be vector fields on M. The following statements are equivalent:*

1. *X and Y commute, $[X,Y]=0$;*
2. *$\left(X_t\right)_* Y=Y$ for every t;*
3. *$\left(Y_t\right)_* X=X$ for every t;*
4. *$X_t\circ Y_s=Y_s\circ X_t$ for every s,t.*

Proof Suppose $[X,Y]=0$. Fix $x\in M$ and consider the curve $\alpha\left(t\right)=d\left(X_{-t}\right)_{X_t(x)}\left(Y\left(X_t\left(x\right)\right)\right)\in T_xM$. t=The claim is α is constant equal to $\alpha\left(0\right)=Y\left(x\right)$. This will imply $Y\left(X_t\left(x\right)\right)=d\left(X_t\right)_x\left(Y\left(x\right)\right)$, which is the same as $\left(X_t\right)_* Y=Y$, because x is arbitrary. Evidently, $\alpha'\left(0\right)=[X,Y]\left(x\right)=0$ by definition of Lie bracket. For the other values of t, $\alpha'\left(t\right)=\dfrac{d}{ds}\alpha\left(t+s\right)_{|s=0}$ and

$$\begin{aligned}\frac{d}{ds}\alpha\left(t+s\right)_{|s=0}&=d\left(X_{-t}\right)_{X_t(x)}\frac{d}{ds}\left(d\left(X_{-s}\right)_{X_{-s}(X_t(x))}Y\left(X_s\left(X_t\left(x\right)\right)\right)\right)\\&=d\left(X_{-t}\right)_{X_t(x)}[X,Y]\left(X_t\left(x\right)\right)\\&=0.\end{aligned}$$

Hence $\alpha'\left(t\right)=0$, showing that $\left(X_t\right)_* Y=Y$. The same argument ensures that $\left(Y_t\right)_* X=X$ since the Lie bracket is skew.

Assume now $\left(X_t\right)_* Y=Y$. Then the flow X_t maps trajectories of Y to trajectories, so $X_t\circ Y_s=Y_s\circ X_t$. Finally, if the flows of X and Y commute, then the commutator of X_t and Y_t is constant. By Proposition A.6, it follows that $[X,Y]=0$. $\square$

A.1 Exercises

1. A vector field X on an open subset in $\mathbb{R}^n$ can be written in coordinates as

$$X=\sum_i a^i\frac{\partial}{\partial x^i},$$

where $\frac{\partial}{\partial x^i}$ are the constant vector fields in the coordinate directions. Using this notation, show that if $X = \sum_i a^i \frac{\partial}{\partial x^i}$ and $Y = \sum_i b^i \frac{\partial}{\partial x^i}$, then

$$[X, Y] = \sum_{i,j} \left(a^i \frac{\partial b^j}{\partial x^i} - b^i \frac{\partial a^j}{\partial x^i} \right) \frac{\partial}{\partial x^j}.$$

2. Given a vector field X on the manifold M, suppose that $x \in M$ is a singularity of X, that is, $X(x) = 0$. Given vector fields Y_1, Y_2, show that $[X, Y_1](x) = [X, Y_2](x)$ provided $Y_1(x) = Y_2(x)$. Conclude that it is possible to define—unambiguously—the linear map $d_X : T_x M \to T_x M$ by $d_X(v) = [X, Y](x)$, where Y is a vector field such that $Y(x) = v$.
   Verify that if M is an open set of $\mathbb{R}^n$, then d_X is equal to $-dX_x$ when the vector field is seen as a map $X : M \to \mathbb{R}^n$.
3. A linear vector field on $\mathbb{R}^n$ is defined by $X_A(x) = Ax$, where A is an $n \times n$ matrix. Show that $[X_A, X_B] = X_{BA-AB}$. Use this fact to show that if the matrices A, B commute, then

$$\sum_{k \geq 0} \frac{1}{k!} (A + B)^k = \left(\sum_{k \geq 0} \frac{1}{k!} A^k \right) \left(\sum_{k \geq 0} \frac{1}{k!} B^k \right).$$

4. On the one-dimensional manifold S^1, parametrized by the angle θ, consider the vector fields $X(\theta) = \cos\theta \frac{d}{d\theta}$ and $Y(\theta) = \sin\theta \frac{d}{d\theta}$. Show that the Lie subalgebra of vector fields generated by X and Y has dimension 3 and is isomorphic to $\mathfrak{sl}(2, \mathbb{R})$.

Appendix B
Integrability of Distributions

B.1 Immersions and Submanifolds

Let V and M be differentiable manifolds of class C^k, $k = 1, \ldots, \infty, \omega$. An **immersion** $f : V \to M$ is a differentiable map such that df_x is one-to-one for every $x \in V$. When f is an **injective immersion**, the set $L = f(V)$ is called an **immersed submanifold** of M. In this case, the map $V \to f(V)$ is a bijection that allows to transfer the differentiable structure of V to $f(V)$. The latter structure is called the **intrinsic differentiable structure**. The topology underlying this differentiable structure is the **intrinsic topology** of the submanifold.

There is another natural topology on the immersed submanifold $f(V) \subset M$, namely the induced, or subset, topology. Since the immersion f is a continuous map, induced open sets are open in the intrinsic topology. In general, though, the two topologies are different (see some examples below). When these topologies coincide, the immersed submanifold is called an **embedded submanifold** or a regular submanifold, and the immersion is said to be an **embedding** or a **regular immersion**.

The following proposition is a characterization of embedded submanifolds.

Proposition B.1 *A subset $N \subset M$ is an embedded submanifold if and only if the following condition is met: for every $x \in N$, there exist (i) a neighborhood W of x in M; (ii) neighborhoods $V \subset \mathbb{R}^k$ and $U \subset \mathbb{R}^p$ of the origins; and (iii) a diffeomorphism $\psi : V \times U \to W$, such that $\psi(V \times \{0\}) = W \cap N$.*

In studying the integrability of distributions, it is necessary to consider a class of immersions that is larger than embeddings. This is the class of quasi-regular immersed submanifolds, which are defined below.

Definition B.2 An immersed submanifold $L = f(V)$ is said to be **quasi-regular** (or **quasi-embedded**) if the following condition holds:

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- Let N be a locally connected topological space and $\phi : N \to M$ a continuous map. Suppose the image of ϕ is contained in L. Then $\phi : N \to L$ is continuous in the intrinsic topology of L.

If a continuous map $\phi : N \to M$ assumes its values in L, then $\phi : N \to L$ is continuous in the induced topology. In particular, if this topology coincides with the intrinsic topology, then the condition of quasi-regularity is satisfied. In other words, an embedded submanifold is quasi-regular.

The following examples help clarify the concept of quasi-regular submanifold.

Examples

1. Let $\phi : (0, +\infty) \to \mathbb{R}^2$ be the C^1 immersion defined by

$$\phi(t) = \begin{cases} (\cos(t - \pi/2), \sin(t - \pi/2)) & \text{if} \quad 0 < t \leq 2\pi \\ (t - 2\pi, -1) & \text{if} \quad t \geq 2\pi \end{cases}.$$



It is injective but not quasi-regular. In fact, for $\epsilon > 0$ small enough define $\psi : (-\epsilon, \epsilon) \to \mathbb{R}^2$ by $\psi(t) = (\cos(t - \pi/2), \sin(t - \pi/2))$. Then ψ is continuous as a map on $\mathbb{R}^2$, but not continuous in the intrinsic topology of the submanifold. In fact, if V is the open set (in the intrinsic topology) of the picture, then $\psi^{-1}(V)$ is an interval of type $(-\delta, 0]$, which is not open in $(-\epsilon, \epsilon)$.

2. Consider the two-dimensional torus $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ with canonical projection $\pi : \mathbb{R}^2 \to T^2$. If r is a line in $\mathbb{R}^2$, its projection $\pi(r)$ is an immersed submanifold of T^2. In particular, lines through the origin $r_\alpha = \{t(1, \alpha) : t \in \mathbb{R}\}$ define immersions $\pi(r_\alpha) \hookrightarrow T^2$. If α is rational, then $\pi(r_\alpha)$ is a closed curve on T^2 and the corresponding immersion is an embedding. But if α is irrational, then $\pi(r_\alpha)$ is dense in T^2. In this case, the immersion is not an embedding because, for example, an interval of type $\{t(1, \alpha) : t \in (-\varepsilon, \varepsilon)\}$ is open in the intrinsic topology but not in the induced one.
Nevertheless, the immersions $\pi(r_\alpha)$ are quasi-regular. To see this, take $\phi : N \to T^2$ continuous, with $\phi(N) \subset \pi(r_\alpha)$ and N locally connected. The continuity of ϕ in the intrinsic topology is a local matter. By taking suitable restrictions of ϕ to neighborhoods of N, one can assume $\phi(N)$ is contained in an open rectangle

$$\begin{aligned} R &= \pi\{(t - \alpha s, \alpha t + s) : t \in I_1, s \in I_2\} \\ &= \pi(I_1 \cdot (1, \alpha) + I_2 \cdot (-\alpha, 1)) \end{aligned}$$

with I_1, I_2 open intervals in $\mathbb{R}$. If I_1, I_2 are small enough, the map

$$(t, s) \in I_1 \times I_2 \mapsto \pi\,(t\,(1, \alpha) + s\,(-\alpha, 1)) \in \mathbb{T}^2$$

is a homeomorphism. Hence the projection p_2 onto the line generated by $(-\alpha, 1)$ is well defined on R. This projection is a continuous map, so $p_2 \circ \phi$ is a continuous map to $I_2\,(-\alpha, 1)$. But the intersection of $\pi\,(r_\alpha)$ with the rectangle R has at most countably many connected components. Each connected component is of the form

$$(\pi\,(r_\alpha) \cap \pi I_2\,(-\alpha, 1)) + \pi\,(I_1\,(1, \alpha))\,.$$

This means that $p_2 \circ \phi$ takes values in a countable set, and being continuous, it must then be constant. Therefore $\phi\,(N) \cap R$ is contained in an interval range of type $I_1\,(1, \alpha) + (-\alpha s_0, s_0)$. This being so, let $A \subset R$ be an intrinsic open set. Then $\phi^{-1}\,(A) = \phi^{-1}\,(B)$, where B is open in T^2, ensuring that ϕ is continuous with respect to the intrinsic topology. (Notice that the key point in this proof is the countability of the range of $p_2 \circ \phi$.) □

The continuity in the definition of a quasi-regular submanifold extends to differentiability with respect to the intrinsic differentiable structure of the submanifold, as shown in the following result.

Proposition B.3 *Let L be a quasi-regular submanifold and N a connected differentiable manifold. If a differentiable map $\phi : N \to M$ takes values in L, then it is differentiable with respect to the intrinsic differentiable structure of L.*

Proof Take $y \in N$ and write $x = \phi\,(y) \in L$. The local form of immersions guarantees that there is an intrinsic neighborhood U of $x \in L$ such that the inclusion of U in M is equivalent to the canonical inclusion of a neighborhood V of the origin $\mathbb{R}^k$ in the neighborhood $V \times W$ of the origin in $\mathbb{R}^k \times \mathbb{R}^l$.

Since L is quasi-regular $\phi^{-1}\,(U)$ is an open set in N containing y. It follows that the restriction of ϕ to $\phi^{-1}\,(U)$ is equivalent to a map $z \mapsto (f\,(z)\,, g\,(z)) \in V \times W$, which takes values in $V \times \{0\}$, i.e. such that $g\,(z) = 0$. Hence $z \mapsto f\,(z) \in V$ is equivalent to the restriction of ϕ to $\phi^{-1}\,(U)$, with V diffeomorphic to U in the intrinsic topology. Since f is differentiable, it follows that ϕ is differentiable. □

Corollary B.4 *In the situation of the above proposition, if $\phi : N \to M$ is an immersion, then $\phi : N \to L$ is also an immersion. In particular, if $d\dim N = \dim L$, then ϕ is a local diffeomorphism.*

Proof As $\phi : N \to L$ is differentiable, the image of its differential is contained in the tangent space to L, and therefore $\phi : N \to L$ is an immersion. □

Corollary B.5 *If L is a quasi-regular submanifold with $\dim L = k$, the set underlying L admits a unique structure of immersed submanifold of dimension k.*

Proof Apply the above corollary to the identity map id of L. □

B.2 Characteristic Distributions

Let M be a differentiable manifold. A distribution Δ on M is a map that associates to each point $x \in M$ a vector subspace $\Delta(x) \subset T_x M$ of the tangent space at x. A distribution Δ is **regular** or **non-singular** if $\dim \Delta(x)$ is constant as a function of x, and **singular** otherwise. The constant dimension of $\Delta(x)$ in the regular case is called the dimension of the distribution.

An **integral submanifold** of Δ is an immersion $\phi : N \to M$ such that

$$\forall x \in N, \phi_* (T_x N) = \Delta(\phi(x)).$$

Definition B.6 A distribution Δ on M is said to be **integrable** at $x \in M$ if there is an integral manifold of Δ containing x. The distribution is integrable if it is integrable at every $x \in M$.

A local vector field in M is a vector field defined on an open subset $U \subset M$, that is, a map $X : U \to TM$ such that $X(x) \in T_x M$ for all $x \in U$.

A local vector field X in M is said to be tangent to Δ if $X(x) \in \Delta(x)$ for every x in the domain of X.

Definition B.7 A distribution Δ in M is **differentiable** at $x \in M$ if there exist differentiable local vector fields $X^1, \ldots, X^k$ defined on a neighborhood of x such that:

1. $X^1, \ldots, X^k$ are tangent to Δ.
2. The set $\{X^1(x), \ldots, X^k(x)\}$ spans $\Delta(x)$.

A distribution is differentiable if it is differentiable at every $x \in M$. The set of fields $X^1, \ldots, X^k$ is called a parameterization of Δ centered at x.

In general, a parameterization $X^1, \ldots, X^k$ centered at x may not span the distribution at points $y \neq x$. If, however, the distribution is regular, then there exists a neighborhood U of x such that $\{X^1(y), \ldots, X^m(y)\}$, $y \in U$, is linearly independent if $\{X^1(x), \ldots, X^m(x)\}$ is linearly independent. (This fact can be verified locally, by taking vector fields on an open set in $\mathbb{R}^n$, $n = \dim M$. The $n \times m$ matrix whose columns are the components of the $X^1, \ldots, X^m$ has rank m at x. As the determinant is a continuous function, the matrix has rank m on a neighborhood of x. See Lemma B.13 below.)

A central concept in the study of integrability is that of a characteristic distribution, defined next.

Definition B.8 Let M be a manifold and Δ a distribution on M.

1. A local vector field X with domain $U \subset M$ preserves the distribution Δ if

$$X_{t*}(\Delta(x)) \subset \Delta(X_t(x)) \tag{B.1}$$

for all $x \in U$ and $t \in \mathbb{R}$ such that $X_t(x)$ is defined. In this case, Δ is said to be **invariant** by X. (The symbol $*$ reminds that this is a differential.)

2. A local vector field $X : U \to TM$ is characteristic of the distribution Δ if X preserves Δ and is tangent to Δ, i.e. $X(y) \in \Delta(y)$ for all y in the domain of X.
3. A distribution Δ is characteristic at $x \in M$ if there are Δ-characteristic local vector fields $X^1, \ldots, X^k$ such that

$$\Delta(x) = \operatorname{span}\{X^1(x), \ldots, X^k(x)\}.$$

The distribution is characteristic if this happens at every $x \in M$.

Note that inclusion (B.1) is actually an equality, since $X_{-t*}(\Delta(X_t(x))) \subset \Delta(x)$.

Having established these concepts, it is possible to state and prove the following criterion for the integrability of distributions.

Theorem B.9 *For a distribution Δ, the following statements are equivalent:*

1. *Δ is characteristic.*
2. *Δ is differentiable and integrable.*

Proof Suppose Δ is characteristic. Then it is differentiable by definition. Therefore the main point is to show Δ is integrable, i.e. every $x \in M$ is contained in an integral manifold of Δ.

Fix $x \in M$. Since Δ is characteristic at x, there are characteristic vector fields $X^1, \ldots, X^k$, defined on a neighborhood of x and linearly independent at x, such that

$$\Delta(x) = \operatorname{span}\{X^1(x), \ldots, X^k(x)\}.$$

For some neighborhood U of the origin in $\mathbb{R}^k$, the composition

$$\rho(\tau) = \rho(t_1, \ldots, t_k) = X^1_{t_1} \circ \cdots \circ X^k_{t_k}(x)$$

with $\tau = (t_1, \ldots, t_k) \in U$ makes sense and defines a differentiable map $\rho : U \to M$.

The integral manifold will be constructed by checking that ρ is an immersion when restricted to some neighborhood $V \subset U$ of 0. For this, the image of the differential $d\rho_\tau$ at τ must be computed. This image is spanned by the partial derivative $\dfrac{\partial \rho}{\partial t_i}(\tau)$. Using that the t-derivative of $X_t(x)$ equals $X(X_t(x))$, it follows

$$\frac{\partial \rho}{\partial t_i}(\tau) = X^1_{t_1 *} \circ \cdots \circ X^{i-1}_{t_{i-1} *}\left(X^i(z_i)\right), \tag{B.2}$$

where $z_i = X^i_{t_i} \circ \cdots \circ X^k_{t_k}(x)$. Evaluating this at $\tau = 0$ yields

$$\frac{\partial \rho}{\partial t_i}(0) = X^i(x).$$

Hence the image of $d\rho_0$ is equal to $\Delta(x)$ by the choice of the vector fields, and so ρ has maximal rank at 0. Therefore ρ restricted to a neighborhood V of 0 is an immersion, as required.

To see that the restriction of ρ to V is an integral manifold of Δ, note that

$$X^1_{t_1 *} \circ \cdots \circ X^k_{t_k *}(\Delta) = \Delta,$$

because each X^i is a characteristic vector field and hence preserves Δ.

Combining the above equality with the expression for the partial derivatives of ρ in (B.2) and the fact that $X^i(z_i) \in \Delta(z_i)$ (because X^i is tangent to Δ), it follows that if $\tau \in U$, then $\dfrac{\partial \rho}{\partial t_i}(\tau) \in \Delta(\rho(\tau))$, and hence $\operatorname{im}(d\rho_\tau) \subset \Delta(\rho(\tau))$. This inclusion is in fact an equality when $\tau \in V$, because x and $\rho(\tau)$ are, respectively, the initial and end points of a curve arising as the concatenation of trajectories of characteristic fields of Δ. Since the dimension of Δ does not change along the trajectories of the characteristic fields, $\dim \Delta(\rho(\tau)) = \dim \Delta(x)$, and this dimension equals $\dim(\operatorname{im}(d\rho_\tau))$ because ρ is an immersion in V. Therefore ρ is an integral manifold, concluding the proof that Δ is integrable.

The converse is a consequence of Lemma B.17 below, which ensures that an integrable distribution is invariant by its tangent vector fields. Hence if Δ is differentiable and integrable, the vector fields of its parameterizations are characteristic, showing that the distribution is characteristic if it is differentiable and integrable.

□

The Frobenius theorem gives a sufficient condition for a differentiable and regular distribution to be integrable. The condition is expressed in terms of involutivity according to the following definition.

Definition B.10 A distribution Δ is **involutive** if the following holds:

- Let $X : U \to TM$ and $Y : V \to TM$ be local vector fields with $U \cap V \neq \emptyset$ and such that X and Y are tangent to Δ. Then $[X, Y] : U \cap V \to TM$ is a local vector field tangent to Δ.

With this concept, it is possible to state the Frobenius theorem.

Theorem B.11 *Let Δ be a differentiable distribution of constant dimension. Suppose that Δ is involutive. Then Δ is integrable.*

The gist of the proof is showing that a regular and involutive distribution is characteristic. For this, the following lemmas are required.

Lemma B.12 *Let $U \subset \mathbb{R}^n$ be an open set and A a differentiable map defined on U with values in the space of $m \times k$ matrices, with $k \leq m$. Suppose that* $\operatorname{rank} A(y) = k$ *for every $y \in U$. Let the maps $a : U \to \mathbb{R}^k$ and $b : U \to \mathbb{R}^m$ be such that b is differentiable and*

$$A(y)\,a(y) = b(y)$$

for every $y \in U$. Then $a : U \to \mathbb{R}^k$ is differentiable as well.

Proof Take $y_0 \in U$. Since the rank of $A(y_0)$ is k, up to a permutation of the rows of A one may assume

$$A = \begin{pmatrix} B(y) \\ C(y) \end{pmatrix},$$

where $B(y)$ is a $k \times k$ matrix with $\det B(y_0) \neq 0$. Writing A in this way, the continuity of $\det B(y_0)$ ensures that $B(y)$ is invertible for y belonging in some neighborhood U_1 of y_0. The inverse $B(y)^{-1}$ is differentiable because it has the form $(1/\det B(y))\, Q(B(y))$ for some polynomial Q in the entries of $B(y)$. On the other hand, for all $y \in U_1$,

$$a(y) = \begin{pmatrix} B(y)^{-1} & 0 \end{pmatrix} \begin{pmatrix} B(y) \\ C(y) \end{pmatrix} a(y) = \begin{pmatrix} B(y)^{-1} & 0 \end{pmatrix} b(y).$$

This shows that a is differentiable on U_1. Since y_0 is arbitrary, differentiability holds everywhere on U. □

Lemma B.13 *Let Z and $Y^1, \ldots, Y^k$ be differentiable vector fields on the open set $U \subset \mathbb{R}^n$ such that*

$$Z(y) = \sum_{j=1}^{k} a_j(y)\, Y^j(y).$$

Suppose $Y^1(y_0), \ldots, Y^k(y_0)$ are linearly independent at $y_0 \in U$. Then the coefficients a_j are differentiable on some neighborhood of y_0.

Proof Let $A(y)$, $y \in U$, be the $n \times k$ matrix whose columns are the vectors $Y^1(y), \ldots, Y^k(y) \in \mathbb{R}^n$. Viewed as a linear transformation $\mathbb{R}^k \to \mathbb{R}^n$, $A(y)$ is given by

$$A(y)(u_1, \ldots, u_k) = u_1 Y^1(y) + \cdots + u_k Y^k(y).$$

If $b(y)$ denotes the column vector of the components of Z and a the column vector of the coefficients a_j, then $A(y)\,a(y) = b(y)$. Since the Y^j are linearly independent at y_0, the matrix $A(y_0)$ has rank k. This rank persists for $A(y)$, for any y in a neighborhood of y_0, by continuity of the determinant. The previous lemma then shows that a is differentiable on this neighborhood of y_0. □

The next result will be used in the proof of the Frobenius theorem but has independent interest. It shows in particular that in regular differentiable distributions tangent vector fields are characteristic.

Theorem B.14 *Let Δ be an involutive distribution on M and U an open set of M. Suppose there are vector fields X and $Y^1, \ldots, Y^k$ on U such that:*

1. *X and Y^j, $j = 1, \ldots, k$, are tangent to Δ;*
2. *$\{Y^1(x), \ldots, Y^k(x)\}$ spans $\Delta(x)$ for all $x \in U$.*

Take $x \in U$ and let J be an interval such that $X_t(x) \in U$ for every $t \in J$. Then,

$$\Delta(X_t(x)) = X_{t*}\Delta(x) \tag{B.3}$$

for every $t \in J$.

Proof Inclusion (B.3) is equivalent to

$$X_{-t*}\Delta(X_t(x)) = \Delta(x).$$

To prove the latter, define functions v_i with values in T_xM by

$$v_i(t) = (X_{-t})_* \left(Y^i(X_t(x))\right), \quad i = 1, \ldots, k.$$

It suffices to check that $v_i(t) \in \Delta(x)$, $t \in J$. In turn, for that it is enough to show that for every linear functional $\lambda : T_xM \to \mathbb{R}$ such that $\Delta(x) \subset \ker\lambda$,

$$\lambda(v_i(t)) = 0 \quad t \in J.$$

That vanishing, in fact, ensures that $v_i(t)$ lies in the intersection of the kernels of linear functionals that are zero on $\Delta(x)$. Pick a functional λ that is zero on $\Delta(x)$ and set

$$w_i(t) = \lambda(v_i(t)) \qquad w = (w_1, \ldots, w_k).$$

By the definition of Lie bracket of vector fields (see the proof of Proposition A.7),

$$v_i'(t) = X_{-t*}\left([X, Y^i](X_t(x))\right).$$

By Lemma B.13, however, and the fact that the distribution is involutive,

$$[X, Y^j](y) = \sum_{j=1}^{k} b_{ij}(y)\, Y^j(y)$$

with b_{ij} differentiable functions on U. Write $a_{ij}(t) = b_{ij}(X_t(x))$, so that

$$v_i'(t) = X_{-t*}\left(\sum_{j=1}^{k} a_{ij}(t)\, Y^j(X_t(x))\right)$$
$$= \sum_{j=1}^{k} a_{ij}(t)\, v_j(t).$$

Now λ is linear, so $w_i'(t) = \lambda\left(v_i'(t)\right)$ and w satisfies the differential equation

$$w_i'(t) = \sum_{j=1}^{k} a_{ij}(t)\, w_j(t).$$

This is a linear ODE with continuous coefficients, and therefore it admits a unique solution with given initial condition $w(0)$. This solution is defined on the whole interval J. However, $w_i(0) = 0$ because $v_i(0) = Y^i(x) \in \Delta(x)$. Hence $w_i(t) = 0$ for all $t \in J$, which concludes the proof. □

With these preparatory results in place, the proof of the Frobenius theorem is easy.

Proof of the Frobenius Theorem Given $x \in M$, take vector fields $Y^1, \ldots, Y^k$ on a neighborhood U of x such that $\{Y^1(y), \ldots, Y^k(y)\}$ is a basis of $\Delta(y)$ for all $y \in U$. By Lemma B.13 and Theorem B.14, the vector fields $Y^1, \ldots, Y^k$ are characteristic for Δ. This shows that a regular and involutive distribution is characteristic, and therefore integrable. □

To apply Frobenius theorem, it is not usually necessary to verify the involutivity of all tangent vector fields. The following consequence of Lemma B.13 says it suffices to compute the brackets of a basis of Δ only.

Corollary B.15 *Let Δ be a regular distribution on M. Suppose that for every $x \in M$ there are tangent vector fields $X_1, \ldots, X_k$ to Δ defined around x, such that the set $\{X_1(x), \ldots, X_k(x)\}$ spans $\Delta(x)$ and the brackets $[X_i, X_j]$ are tangent to Δ. Then Δ is integrable.*

Proof If the vector fields span $\Delta(x)$, then they generate $\Delta(y)$ at any y in a neighborhood of x, so Δ is differentiable. By Lemma B.13, furthermore, if Y and Z are tangent to Δ, then $Y = \sum a_i X_i$ and $Z = \sum b_i X_i$ on a neighborhood x, with differentiable coefficients. Therefore,

$$[Y, Z] = \sum_{i,j} a_i b_j [X_i, X_j] + \sum_{i,j} \left(X_i b_j - X_i a_j\right) X_j,$$

showing that Δ is involutive. □

B.3 Maximal Integral Manifolds

The integrability results proved above, especially the Frobenius theorem, are existence theorems and have a local character. Global results, as well as the uniqueness of integral manifolds, are typically obtained by applying Zorn's Lemma, which allows to extend integral manifolds.

In this context, the concept of **maximal integral manifold** plays a central role. A maximal integral manifold is a connected integral submanifold L of Δ that is not properly contained in any other connected integral submanifold. Below, it will be proved that characteristic distributions admit maximal integral manifolds. For this purpose, some lemmas are required.

Lemma B.16 *Let $N \hookrightarrow M$ be an immersion, and suppose the vector field X of M is tangent to N, that is, $X(x) \in T_x N$ if $x \in N$.*

Then for every $x \in N$, there exist a neighborhood $V \subset N$ of x and a number $\epsilon > 0$ such that if $y \in V$ then $X_t(y) \in N$ provided $|t| < \epsilon$. Moreover, $X_t : V \to N$, $|t| < \epsilon$, is a diffeomorphism onto an open subset of N.

Proof Due to the local form of immersions, one may assume, without loss of generality, that M is a product $V \times W \subset \mathbb{R}^k \times \mathbb{R}^l$ with V and W open neighborhoods of the origin and that $N = V \times \{0\}$. Then take a small enough neighborhood of the origin $V_1 \times W_1 \subset V \times W$ and $\epsilon > 0$ so that $X_t(V_1 \times W_1) \subset V \times W$.

Since X is tangent to N, it defines by restriction a vector field $\bar{X}$ on $V \times \{0\}$. A trajectory α of $\bar{X}$ satisfies

$$\alpha'(t) = \bar{X}(\alpha(t)) = X(\alpha(t)),$$

and hence it is also a trajectory of X. The uniqueness of solutions ensures then that the trajectories of X starting in $V \times \{0\}$ stay in $V \times \{0\}$. If $|t| < \epsilon$, therefore $X_t(V_1 \times \{0\}) \subset V \times \{0\}$, proving the lemma. □

Lemma B.17 *Suppose Δ is an integrable distribution and X a local vector field tangent to Δ. Then X preserves Δ.*

Proof Let U be the domain of X, and take $y \in U$ and L an integral manifold of Δ through y. By the above lemma, there are $\epsilon > 0$ and $V \subset L \cap U$ (depending on y and L) such that X_s is a diffeomorphism of V on an open subset of L, for $|s| < \epsilon$. Therefore X_s maps tangent spaces of L to tangent spaces, and so

$$X_{s*}\Delta(z) = \Delta(X_s(z))$$

if $z \in V$. In particular, this equality holds when $z = y$ and $|s| < \epsilon$.

Now take $x \in \mathrm{dom} X_t$. For $t > 0$, define

$$m = \sup\{s \in [0, t] : \forall \sigma \in [0, s], X_{\sigma *}(\Delta(x)) \subset \Delta(X_\sigma(x))\}.$$

Applying the first part of the proof to $y = X_m(x)$, one concludes that $m = t$, and the claim is proved. □

Lemma B.18 *Let Δ be a characteristic distribution and N_1, N_2 integral manifolds of Δ. Then $N_1 \cap N_2$ is an open submanifold of both N_1 and N_2.*

Proof Assume $N_1 \cap N_2 \neq \emptyset$ and take $x \in N_1 \cap N_2$. Let $X^1, \ldots, X^k$ be characteristic vector fields of Δ defined on a neighborhood of x

$$\Delta(x) = \text{span}\{X^1(x), \ldots, X^k(x)\}.$$

Set

$$\rho(t_1, \ldots, t_k) = X^1_{t_1} \circ \cdots \circ X^k_{t_k}(x).$$

As in the proof of Theorem B.9, the map $\rho : U \to M$ is an immersion for some open set U containing the origin of $\mathbb{R}^k$. By Lemma B.16, if U is small enough, then $\rho(U) \subset N_1 \cap N_2$, and the maps $\rho : U \to N_1$ and $\rho : U \to N_2$ are immersions. Since the dimensions of U, N_1, and N_2 are equal, it can be assumed, by shrinking U if necessary, that these immersions are embeddings. Therefore $\rho(U)$ is an open submanifold of both N_1 and N_2, making the intersection $N_1 \cap N_2$ open in the two integral manifolds. □

Theorem B.19 *Let Δ be a characteristic distribution. Then each $x \in M$ is contained in a unique maximal integral manifold $I(x)$ of Δ. Moreover, if $N \subset M$ is a connected integral manifold of Δ with $x \in N$, then N is an open submanifold of $I(x)$.*

Proof Denote by $\mathcal{F}$ the set of integral submanifolds of Δ. For $x \in M$, set

$$\mathcal{F}_x = \{N \in \mathcal{F} : x \in N\}.$$

As Δ is integrable $\mathcal{F}_x \neq \emptyset$, and it is also partially ordered by the relation $N_1 \prec N_2$ if $N_1 \subset N_2$.

Let $\mathcal{H}$ be a chain of $\mathcal{F}_x$, i.e. a totally ordered subset of $\mathcal{F}_x$. To apply Zorn's Lemma, it must be proved that $\mathcal{H}$ has an upper bound in $\mathcal{F}_x$. The claim is that such an upper bound is

$$\tilde{N} = \bigcup_{n \in \mathcal{H}} N.$$

In fact, define a manifold structure on $\tilde{N}$ as follows. Take $y \in \tilde{N}$. Then there is $N \in \mathcal{H}$ containing y. Since N is a submanifold, its charts around y define charts of N. Doing this for all $y \in \tilde{N}$ defines a set of charts whose domains cover $\tilde{N}$. Two charts defined in this way are related differentiably, because if $y \in N_1 \cap N_2$ then either $N_1 \prec N_2$ or $N_2 \prec N_1$ (recall that $\mathcal{H}$ is totally ordered) and one of them is an

open submanifold of the other. Either way there is an atlas on $\tilde{N}$ such that N is an open submanifold of $\tilde{N}$, for all $N \in \mathcal{H}$. With this structure, $\tilde{N}$ becomes an integral submanifold of Δ containing x, and hence $\tilde{N} \in \mathcal{F}_x$.

Since every totally ordered subset of $\mathcal{F}_x$ admits an upper bound, Zorn's Lemma guarantees that $\mathcal{F}_x$ admits maximal elements. But a maximal element in $\mathcal{F}_x$ is a maximal integral manifold that contains x.

To check uniqueness, suppose $x \in N_1 \cap N_2$, with N_1 and N_2 maximal integral manifolds. By local uniqueness (Theorem B.18), $N_1 \cap N_2$ is an open submanifold of both N_1 and N_2 and therefore $N_1 \cup N_2$ is a connected integral submanifold containing N_1, N_2. Hence $N_1 = N_2 = N_1 \cup N_2$ by the maximality of N_1 and N_2.

The last statement follows from the fact that any connected integral manifold is contained in a maximal connected integral manifold as an open subset, by the previous lemma. □

The uniqueness of maximal integral manifolds ensures that any two such are either disjoint or equal (this property does not apply to arbitrary integral manifolds, only maximal ones). Consequently, maximal integral manifolds are the equivalence classes of the equivalence relation $x \sim_\Delta y$ if x and y belong to the same maximal integral manifold of Δ.

B.4 Adapted Charts

Adapted charts (also known as tubular neighborhoods) show that the integral manifolds of an integrable distribution are nicely arranged inside of it.

Definition B.20 Let Δ be an integrable distribution on M. An **adapted chart** (or adapted coordinate system) centered at x is a diffeomorphism $\psi : U \times V \to W$, where $U \subset \mathbb{R}^k$ and $V \subset \mathbb{R}^{n-k}$ are open sets containing the origin, and W is open in M and contains x, which meets the following conditions:

1. $\psi(0, 0) = x$.
2. $\dim \Delta(x) = k$.
3. For every $z \in V$, the set $\psi(U \times \{z\})$ is contained in a maximal connected integral manifold of Δ.
4. The map $\psi_0 : U \to \psi(U \times \{y\})$, $\psi_0(x) = \psi(0, y)$ is an integral manifold of Δ.

The open set W is called the domain of the chart.

An adapted chart is also known as a local **tubular neighborhood** (of the integral manifold through x).

The following proposition shows that characteristic distributions admit adapted charts centered at any point of M.

Proposition B.21 *Let Δ be a characteristic distribution. Then for every $x \in M$, there exists an adapted chart $\psi : U \times V \to M$ with $\psi(0,0) = x$.*

Proof Fix $x \in M$ and define the immersion

$$\rho : (t_1, \ldots, t_k) \longmapsto X^1_{t_1} \circ \cdots \circ X^k_{t_k}(x)$$

on some neighborhood V of the origin of $\mathbb{R}^k$, where $X^1, \ldots, X^k$ are characteristic vector fields that span $\Delta(x)$. The image of ρ is an open set of the maximal integral manifold through x. To construct the adapted chart, let $n = \dim M$. Then there is an immersion

$$\phi : W \longrightarrow M$$

with W a neighborhood of the origin in $\mathbb{R}^{n-k}$ such that $\phi(0) = x$ and ϕ is transversal to ρ at x:

$$\operatorname{im}(d\phi_0) \cap \operatorname{im}(d\rho_0) = \{0\} \qquad \text{and} \qquad \operatorname{im}(d\phi_0) \oplus \operatorname{im}(d\rho_0) = T_x M.$$

Let $\psi : V \times W \to M$ be defined by

$$\psi(v, w) = X^1_{t_1} \circ \cdots \circ X^k_{t_k}(\phi(w))$$

for $v = (t_1, \ldots, t_k) \in V$. Then some restriction of ψ is an adapted chart. In fact, it is clear that $\psi(0,0) = x$. On the other hand, the value of $d\psi_{(0,0)}$ on $(v, w) \in \mathbb{R}^k \times \mathbb{R}^{n-k}$ is

$$d\psi_{(0,0)}(v, w) = d\rho_0(v) + d\phi_0(w),$$

which shows that $d\psi_{(0,0)}$ is an isomorphism. Therefore there are neighborhoods $V_1 \subset V$ and $W_1 \subset W$ such that the restriction of ψ to $V_1 \times W_1$ is a diffeomorphism.

Now the points $\psi(v, w)$, $\psi(0, w)$, and $\phi(w)$ are in the same integral manifold because $\psi(v, w)$ is obtained from $\phi(w)$ by successive concatenations of trajectories of vector fields tangent to Δ, which by Lemma B.16 do not leave the integral manifolds. Fix w, so Lemma B.16 says the map

$$(t_1, \ldots, t_k) \longmapsto X^1_{t_1} \circ \cdots \circ X^k_{t_k}(\phi(w))$$

is differentiable in the intrinsic structure of $I(\psi(0, w))$. Since this map coincides with ρ if $w = 0$, it follows that ψ (or rather, a restriction of it) is an adapted chart. □

The existence of adapted charts makes it possible to prove various features of integral manifolds of an integrable distribution. One such is that connected integral manifolds are quasi-regular submanifolds, as will be proved in the following section.

Another useful property is related to vector fields tangent to the distribution:

Proposition B.22 *Let x_t be a C^1 curve tangent to the integrable distribution Δ. Then x_t is entirely contained in a maximal integral manifold of Δ. In particular, if a vector field X is tangent to Δ then its trajectories are contained in connected maximal integral manifolds.*

Proof Denote by $I(x)$ the maximal integral manifold through x_0 and suppose x_t is defined on the interval (α, ω). Put

$$m = \sup\{t \in (\alpha, \omega) : \forall s \in [0, t], X_s(x) \in I(x)\}.$$

The claim is $m = \omega$. Assuming by contradiction $m < \omega$, take an adapted chart $\psi : V \times W \to M$ at x_m and consider the curve $y_t = \psi^{-1} x_t$ in $V \times W$. Since x_t is tangent to the distribution, it follows that y_t is tangent to $V \times \{0\}$. Therefore if z_t denotes the projection of y_t on the second component, the derivative of z_t vanishes and hence z_t is constant. This implies y_t is contained in $V \times \{0\}$ against the hypothesis that m is the supremum. □

It should be emphasized that the above property of trajectories is valid only for maximal integral manifolds, and not for arbitrary integral manifolds.

B.5 Integral Manifolds Are Quasi-Regular

The existence of adapted charts for integrable distributions permits to prove that maximal integral manifolds in integrable distributions are **quasi-regular submanifolds**.

Before that, a few remarks regarding the topology of a manifold and its submanifolds are in order. For a differentiable manifold M, the following conditions are equivalent:

1. M is **paracompact**, that is, every open cover of M admits a locally finite refinement.
2. Each connected component of M is a countable union of compact subsets.
3. Connected components of M are completely separable: they admit a countable basis of open neighborhoods (one also says they satisfy the second countability axiom).
4. M is metrizable.

Apropos submanifolds,

Proposition B.23 *Let $L \subset M$ be a connected, immersed submanifold in the paracompact manifold M. Then L is also paracompact for the intrinsic topology, and therefore it admits a countable basis of open neighborhoods.*

Proof Since M is paracompact, it admits a Riemannian metric. One such metric induces a Riemannian metric on N since N is an immersed submanifold. As all connected Riemannian manifolds are metrizable, so is N. Consequently, N is paracompact, due to a theorem of Stone's that warrants paracompactness for metric spaces. □

The complete separability of connected immersed submanifolds is central in the following proof that maximal integral manifolds are quasi-regular.

Proposition B.24 *Let Δ be an integrable, differentiable distribution in the paracompact manifold M. Then its maximal integral manifolds are quasi-regular immersed submanifolds.*

Proof Given a maximal integral manifold $I \subset M$, the claim is that a continuous map $\phi : N \to M$, with N locally connected and $\phi(N) \subset I$, is continuous in the intrinsic topology. In other words, it must be checked that given $x \in N$, $\phi^{-1}(V)$ is a neighborhood of x for every intrinsic neighborhood V of $\phi(x)$. To do this, it suffices to take V in the domain W of an adapted chart centered at $\phi(x)$, because if $\phi^{-1}(V \cap W)$ is a neighborhood of x then $\phi^{-1}(V) \supset \phi^{-1}(V \cap W)$ is also a neighborhood. Without loss of generality, N can be taken connected, since if U is a connected neighborhood of x such that $\phi^{-1}(V) \cap U$ is a neighborhood of x then $\phi^{-1}(V)$ is a neighborhood of x as well.

Therefore let $\phi : N \to W$ be a continuous map with N connected, and suppose W is the domain of an adapted chart $\psi : U \times V \to W$ centered at $\phi(x)$ such that $\phi(N) \subset I \cap W$.

Denote by $p : W \to \psi(\{0\} \times V)$ the projection equivalent to the projection $U \times V \to V$. This map is continuous, so $p \circ \phi$ is also continuous. The image of $p \circ \phi$ is at most countable, for it coincides with $I \cap \psi(\{0\} \times V)$ since ϕ assumes values in I, and this set is at most countable (if not, I would contain an uncountable numbers of pairwise-disjoint open sets, contradicting the previous proposition).

Therefore $p \circ \phi$ is constant since N is connected. It follows that $\phi(N)$ is contained in the connected component of $I \cap W$ containing $\phi(x)$. But this component is $\psi(U \times \{0\})$, and its intrinsic topology coincides with the induced topology of $\psi(U \times V)$. This implies that ϕ is continuous for the intrinsic topology of I, concluding the proof. □

(Compare the above proof with Example (2) of Section B.1.)

The key argument in the previous proof is the countability of the intersection $I \cap \psi(\{0\} \times V)$, which descends from the complete separability of the integral manifold I. The same statement therefore holds in the following more general situation.

Corollary B.25 *Suppose N is a countable union of maximal integral manifolds of Δ. Then N is quasi-regular.*

The proof of the above proposition shows that maximal connected integral manifold I intersects the image of the adapted chart $\psi : U \times V \to W$ in a set of type $\psi(E \times V)$, where $E \subset U$ is countable at most. If $U \neq \{0\}$, this set has empty

interior in M since $E \times V$ has empty interior in $U \times V$. Since around every point of I there is an adapted chart, I has empty interior in M. The following consequence of the proof of the Proposition B.24 is then straightforward.

Corollary B.26 *Let I be a maximal connected integral manifold of a differentiable and integrable distribution Δ in a paracompact manifold M. Then I has empty interior in M if* $\dim I < \dim M$.

B.6 Exercises

1. Consider the following separation property for a subset $D \subset \mathbb{R}$: for every $x, y \in D$, there exists $z \in \mathbb{R} \setminus D$ lying between x and y. Verify that countable subsets satisfy this property. Show that if D satisfies the property and $f : N \to \mathbb{R}$ is a continuous function with N connected and $f(N) \subset D$, then f is constant. Prove that if N is connected and the image $f(N)$ of a continuous function $f : N \to \mathbb{R}^n$ is at most countable, then f is constant.
2. Show that the trajectories of a vector field on a differentiable manifold are quasi-regular submanifolds (of dimension 0 or 1). (Hint: use the tubular-neighborhood theorem for differential equations.)
3. Let Δ be a characteristic distribution on the manifold M. Suppose $\mathcal{F}$ is a family of vector fields on M such that for any $x \in M$ the subspace $\mathcal{F}(x) = \text{span}\{X(x) : X \in \mathcal{F}\}$ equals $\Delta(x)$. Take x, y in the same integral manifold of Δ. Show that there are vector fields $X^1, \ldots, X^k \in \mathcal{F}$ and $t_1, \ldots, t_k \in \mathbb{R}$ such that $y = X^1_{t_1} \circ \cdots \circ X^k_{t_k}(x)$. (It is enough to assume that the vector fields are local. The hypothesis on $\mathcal{F}(x)$ ensures that the domains of the local vector fields on $\mathcal{F}$ cover M.)
4. Let $\Gamma(TM)$ be the Lie algebra of (C^∞) vector fields on the manifold M (endowed with the Lie bracket). Let $\mathfrak{g}$ be a finite-dimensional Lie subalgebra of $\Gamma(TM)$ and define the distribution $\Delta_\mathfrak{g}(x) = \{X(x) : X \in \mathfrak{g}\}$. Show that $\Delta_\mathfrak{g}$ is integrable.
   The $\mathfrak{g}$-orbit of $x \in M$ is defined as
   $$O_\mathfrak{g}(x) = \{X^1_{t_1} \circ \cdots \circ X^k_{t_k}(x) \in M : k \geq 1,\ X^i \in \mathfrak{g}\}$$
   (where the t_i are chosen so that the flows and compositions are defined). Prove $O_\mathfrak{g}(x)$ is a maximal integral manifold of $\Delta_\mathfrak{g}$ for every $x \in M$.
5. Let L be a maximal integral submanifold of a differentiable and integrable distribution. Show that if L is locally closed, then it is closed.

References

1. ABRAHAN, R. and MARSDEN, J. E. *Foundations of mechanics*. Addison-Wesley, 1978.
2. BOREL, A. *Linear algebraic groups*. Benjamin Inc., 1969.
3. ————. "Hermann Weyl and Lie groups". *Hermann Weyl, 1885–1985, Centenary lectures delivered by C. N. Yang, R. Penrose, A. Borel, at the ETH Zürich.* Springer-Verlag, 1986.
4. ————. *Semisimple groups and Riemannian symmetric spaces*. Hindustan Book Agency, 1998.
5. BOTT, R. and TU, L. W. *Differential forms in algebraic topology*. Springer-Verlag, 1982.
6. BOURBAKI, N. *Elements de mathématique. Groupes et algèbres de Lie*. Hermann, 1972.
7. CARMO, M. P. Riemannian Geometry, Boston, Mass.: Birkhäuser, 1992.
8. CARTAN, E. *Notice sur les travaux scientifiques*. Gauthier-Villars, 1974 (Collection Discours de la Méthode).
9. CHEVALLEY, C. *Theory of Lie groups*, vol. 1. Princeton University Press, 1946.
10. ————. *Théorie des groupes de Lie*, vol. 2. Hermann, 1951.
11. ————. *The algebraic theory of spinors*. Columbia University Press, 1954.
12. CHEVALLEY, C. and EILENBERG, S. "Cohomology theory of Lie groups and Lie algebras". *Trans. Amer. Math. Soc.*, **63**, 1948, pp. 85–124.
13. DIEUDONNE, J. *Éléments d'analyse*, vol. 3. Gauthier-Villars, 1970.
14. DUISTERMAAT, J. J. and KOLK J. A. C. *Lie groups*. Springer-Verlag, 2000.
15. EILENBERG, S. "Singular homology in differentiable manifolds". *Ann. of Math.*, **48**, 1947, pp. 670–681.
16. FRITZSCHE, B. "Sophus Lie: A sketch of his life and work". *J. of Lie Theory*, **9**, 1999, pp. 1–38.
17. FULTON, W. and HARRIS, J. *Representation theory. A first course*. Springer-Verlag, 1991 (Graduate Texts in Mathematics).
18. HALMOS, P. R. *Measure theory*. Springer-Verlag, 1974.
19. HAWKINS, T. "Wilhelm Killing and the structure of Lie algebras". *Archive for History of Exact Sciences*, **26**, 1982, pp. 127–192.
20. HELGASON, S. *Differential geometry, Lie groups and symmetric spaces*. Academic Press, 1978.
21. HILGERT, J. and LAWSON, J. *Lie groups, convex cones and semigroups*. Oxford University Press, 1989.
22. HILGERT, J. and NEEB K.-H. *Lie semigroups and their applications*. Springer-Verlag, 1993 (Lecture Notes in Mathematics, 1.552).
23. HOCHSCHILD, G. *The structure of Lie groups*. Holden-Day Inc., 1965.

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https://doi.org/10.1007/978-3-030-61824-7

24. ————. *Basic theory of algebraic groups and Lie algebras*. Springer-Verlag, 1981 (Graduate Texts in Mathematics).
25. HOFMANN, K. H. and MORRIS, S. A. *The structure of compact groups, a primer for the student – A handbook for the expert*. Walter de Gruyter, 2006 (De Gruyter Studies in Mathematics, 25).
26. HUMPHREYS, J. E. *Introduction to Lie algebras and representation theory*. Springer-Verlag, 1972.
27. ————. *Reflection groups and Coxeter groups*. Cambridge University Press, 1990 (Cambridge Studies in Advanced Mathematics, 29).
28. ————. *Linear algebraic groups*. Springer-Verlag, 1998 (Graduate Texts in Mathematics, 21).
29. HUSEMOLLER, D. *Fibre bundles*. Springer-Verlag, 1966.
30. IWAHORI, N. "On real irreducible representations of Lie algebras". *Nagoya Math. J.*, **14**, 1959, pp. 59–83.
31. JACOBSON, N. *Lie algebras*. Interscience, 1962.
32. KAC, V. *Infinite dimensional Lie algebras*. Cambridge University Press, 1985.
33. KNAPP, A. W. *Lie groups: Beyond an introduction*. Birkhäuser, 1996.
34. KOBAYASHI S. and NOMIZU, K. *Foundations of differential geometry*. Interscience, 1963.
35. KOSZUL, J. L. "Homologie et cohomologie des algébres de Lie". *Bull. Soc. Math. France*, **78**, 1950, pp. 65–127.
36. LANG, S. $Sl_2(\mathbb{R})$. Addison-Wesley, 1975.
37. LIMA, E. L. *Grupo fundamental e espaços de recobrimento*. Impa, 1977 (Coleção Projeto Euclides).
38. LINS NETO, A. and CAMACHO, C. *Teoria geométrica das folheações*. Impa, 1977 (Coleção Projeto Euclides).
39. MATSUSHIMA, Y. *Groupes de Lie: Notes d'un cours donée en 1965–1966*. Université Grenoble, 1966.
40. MONTGOMERY, D. and ZIPPIN, L. *Topological transformation groups*. Interscience, 1955.
41. MOSTOW, G. D. "Some new decomposition theorems for semi-simple groups". *Memoirs Amer. Math. Soc.*, **14**, 1955, pp. 31–54.
42. ONISHCHIK, A. L. and VINBERG, E. B. *Lie groups and algebraic groups*. Springer-Verlag, 1990.
43. PALAIS, R. "A global formulation of the Lie theory of transitive groups". *Memoirs of AMS*, **22**, 1957.
44. PONTRYAGIN, L. S. *Topological groups*. Princeton University Press, 1958.
45. PORTEAUS, I. R. *Clifford algebras and the classical groups*. Cambridge University Press, 1995 (Cambridge Studies in Advanced Mathematics, 50).
46. SAMELSON, H. *Notes on Lie algebras*. Van Nostrand, 1969 (Van Nostrand Mathematical Studies).
47. SAN MARTIN, L. A. B. *Álgebras de Lie*. 2. ed. Editora da Unicamp, 2010.
48. SAN MARTIN, L. A. B. and NEGREIROS, C. J. C. "Invariant almost Hermitian structures on flag manifolds". *Advances in Math.*, **178**, 2003, pp. 277–310.
49. SATAKE I. *Classification theory of semi-simples algebraic groups*. Marcel Dekker, 1971.
50. SERRE, J.-P. *Algèbres de Lie semi-simples complexes*. Benjamin Inc., 1966.
51. SINGER, I. M. and STERNBERG, S. "On the infinite groups of Lie and Cartan, I". *Annales Institut Fourier*, **15**, 1965, pp. 1–114.
52. STILLWELL, J. *Naive Lie theory*. Springer-Verlag, 2010 (Undergraduate Texts in Mathematics).
53. VARADARAJAN, V. S. *Lie groups, Lie algebras and their representations*. Prentice-Hall Inc., 1974.
54. ————. *Harmonic analysis on real reductive groups*. Springer-Verlag, 1977 (Lecture Notes in Mathematics, 576).
55. ————. *An introduction to harmonic analysis on semisimple Lie groups*. Cambridge University Press, 1989 (Cambridge Studies in Advanced Mathematics, 16).

56. WALLACH, N. R. *Real reductive groups*. Academic Press, 1988.
57. WARNER, G. *Harmonic analysis on semi-simple Lie groups*. Springer-Verlag, 1972.
58. WEYL, H. *The classical groups*. Princeton University Press, 1946.
59. WUSSING, H. *The genesis of the abstract group concept:A contribution to the history of the origin of abstract group theory*. The MIT Press, 1984.
60. YANG, C. T. "Hilbert's fifth problem and related problems on transformation groups". *Mathematical developments arising from Hilbert problems*. American Mathematical Society, Providence R. I., 1976, pp. 142–146 (Proceedings Symposia in Pure Math., vol. 28).
61. ZELOBENKO, D. P. *Compact Lie groups and their representations*. American Mathematical Society, 1973 (Translations of Mathematical Monographs, 40).

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